

User Guide | PUBLIC
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Core Tool

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THE BEST RUN SAP

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1 Core Tool

SAP Convergent Charging 2023 is built upon a powerful, flexible pricing modeling, and charging solution. SAP CC drives a powerful set of systems and user applications to profitably rate and charge millions of service consumption events in prepaid or postpaid environments. SAP Convergent Charging's modular design includes an implementation framework for fast and low-cost integration (low TCO) to quickly deploy solutions that enable you to get a rapid return on your investment (rapid ROI).

SAP Convergent Charging

SAP Convergent Charging (SAP CC) is your solution for pricing, charging, and refilling processes. You can:

- Use the panel of tools to streamline the way you create pricing element with reusable charges, charge plans, and refill plans
- Use the intuitive GUI that enables "visual programming" to customize your pricing logic (price algorithms)

To help you get the most out of SAP Convergent Charging, SAP SE has set up a Help Portal which features a regularly updated knowledge base, the latest release notes, and expert advice from our SAP Support Team.

Purpose of This Primary Help

As an IT administrator or an IT consultant, this primary help describes **how to use your SAP CC Core Tool user interface** to configure and manage your data and system. It explains:

- Functions and limitations [page 10] of the user interface
- How to work with SAP Convergent Charging's concepts [page 12] and objects (data, users, others) [page 27]
- Tasks you have to perform and complete with Core Tool and associated procedures during both the implementation project phase and the production operation phase

For more information about help, refer to Help Resources [page 40].

> ⚠ Caution
>
> This product documentation and user assistance relates to the Data Model (DM) based on charge plans and allowance plans in pricing catalogs, and on provider contracts and allowances for the end customers of the subscription- and usage-based services.
>
> If **you work with offers and subscriptions from the former Data Model (DM)** of SAP Convergent Charging, refer to our dedicated product documentation and user assistance. See SAP CC 2023 Primary Help for Core Tool (with Offers and Subscriptions) and select your software version.
>
> In your Core Tool user interface, verify in the preferences [page 56] that the *Working Object Set* option is set to *Provider Contracts and Subscriptions*.

Target Audience

- Pricing Specialist
- Implementation consultants
- Project Team Member (Implementation, Integration)
- Power User
- Administrators (systems, applications, users)

Reader	Responsibilities	Knowledgeable
Pricing Specialist	This primary help is intended for SAP CC pricing specialists (power users, IT consultants) who create and manage the master data for a service provider and its end-customer services to monetize: reusable charges, charge plans [page 343], and refill plans. This master data is grouped in a pricing catalog [page 109] to configure the pricing logic for the use (usage-based charges, periodic or on-off fees) of the marketable end-customer services. **Note** For test purposes, consultants and project team members can manage master data related to the end customers for a service provider. This customer master data includes subscriber accounts [page 511] and provider contracts [page 573]. During the *Implementation Project Phase*, SAP Convergent Charging is integrated with a CRM application or a provisioning system that is responsible for the creation and maintenance of this customer master data See the SAP CC Administration Guide in help resources.	They should have knowledge of: • Core Tool • Pricing and rating concepts • Charging concepts • Refilling concepts
Application Administrator	This primary help is intended for application administrators who create business data [page 32] such as currencies, public holidays.	
User Administrator	This primary help is intended for user administrators who create SAP CC users [page 640].	They should have knowledge of: • Core Tool • Security concepts for SAP CC • Security policy for the company • Security-related settings in SAP CC
System Administrator	This primary help is intended for system administrators who manage the system with the Core Tool user interface: • Audits [page 654] • Generate License Auditing File [page 97]	

Related Information

Getting Started [page 35]
Core Tool Functions and Limitations [page 10]

1.1 What's New?

What's New in Core Tool

This section highlights the main changes in this *SAP Product Document and User Assistance Deliverable*. They relate both to **new or changed features** in the SAP Convergent Charging (SAP CC) software and to **documentation enhancements** for enriching your SAP user experience continuously.

What's New in SAP CC 2023 FPS 2?

As of SAP CC 2023 FPS 2, the Core Tool user interface provides you with the following features:

- Decimal numbers with up to 16 decimal places are supported.

What's New in SAP CC 2023 FPS 1?

As of SAP CC 2032 FPS 1, the Core Tool user interface provides you with the following features:

Consider the following changes:

- A new role called *Pricing Specialist for Price Data* is available for SAP CC users. Refer to the User Roles [page 641] section.
  - User admistrators can assign this role to SAP CC users who need to view or edit some price data in a subset of mapping tables or range tables.
  - Users granted this new role can use the Core Tool user interface to change their password. To work with mapping or range tables, they need to use the Cockpit user interface. Restricted actions are allowed depending on configured permissions.

What's New in SAP CC 2023 (in FPS 0)?

2023 is a new major release of the SAP software product SAP Convergent Charging (SAP CC).

Consider the following changes:

- A new Visibility Level named `External (Mandatory unmodifiable)` is available for the parameters of a charge plan.
- Rounding rules for the Arithmetic Computation [page 789] component are customizable for all the operations.

Related Information

What's New in Feature Package Stack 2

1.2 Core Tool Functions and Limitations

Core Tool Functions

In an SAP Convergent Charging environment, the main functions of the *Core Tool* user interface are the following:

- Configuration during the implementation phase:
  - Configurations of business data or technical data available for the complete system
  - Configurations of the master data related to the pricing of a customer service: You set up the pricing logic of its customer services
  - Configurations of the master data related to end customers of a service provider in a B2B environment with few subscriber data
  - Dummy configurations of the master data related to the end customers of a service provider for test or demo purposes
- Operations during the production operation phase:
  - Master data (pricing catalogs) and business data maintenance
  - Technical data maintenance (users)
  - Rerating operations
  - Auditing

> **Note**
>
> Core Tool provides you with some additional tools [page 59] to facilitate your:
>
> - Business operations
> - Data maintenance
> - Audit operations

For more information on data objects (business data, master data, transactional data, and so on), see Data Objects in Core Tool [page 27].

Core Tool Limitations

In case of an installation scenario integrated with SAP S/4HANA Cloud, the Prepaid Accounts Management feature is automatically deactivated within SAP CC, which impacts the following elements:

- Main window:
  - *New* and *Open* menus no more give the possibility to create refill plans, refill record classes, refill item classes and refill logic objects
  - *Open From File* and *Import* menus no more give the possibility to open or import objects relating to prepaid accounts
- *Allowance Plan* window: Prepaid-related fields are no more available within charged item mappings
- *Catalog* window: Refill Plan, Refill Record Class, Refill Item Class and Refill Logic categories are no more available
- *Charge component* window: Switching operator is no more available
- *Charged Item Class* window: Prepaid-related fields are no more available
- *Charge Plan* window:
  - Prepaid account assignments are no more available
  - Prepaid-related fields are no more available within charged item mappings and response item mappings
- *Choose Object* window (in case of *New* and *Open* actions):
  - *Refill plans*, *refill record classes*, *refill item classes* and *refill logic* objects are no more available
  - *Refilling* category is no more available
- *Internal Component Reference* component window:
  - Settings related to prepaid accounts are no more available
  - Overspending radio button has been converted to a checkbox for credit-limit purpose only
- *Provider Contract* window: The *Activate Refill Plan* action is no more available
- *Subscriber Account* window: Settings related to prepaid accounts are no more available

Related Information

Configuration of Business Data Step by Step [page 107]
Business Data [page 32]
Master Data [page 28]
Catalogs [page 109]
Rerating [page 59]
Audits [page 654]

1.3 Concept Overview

This section explains the main concepts necessary to be familiar with SAP Convergent Charging (SAP CC) and its Core Tool user interface:

- Business objects used to implement the pricing policy and evolution for the subsription- and usage-based services to monetize
- Processes step by step
- Rating (dynamic pricing) and charging (multiaccount determination and amount splitting)
- Account and balance management [page 13]
- Charging information flow [page 17]
- Session-based charging [page 20]
  - Session-Based Charging Limitations [page 22]
- Spending status monitoring [page 25]

> → Recommendation
>
> Consult the Application Help for more information and user assistance.

1.3.1 Configuring the Master Data

In SAP Convergent Charging 2023, the *SAP Master Data* includes:

- **Master data** related to the *pricing configuration* of an end-customer service to monetize. In your *Core Tool* user interface, the business configuration activities consist in setting up this data for SAP Convergent Charging.
- **Customer master data**, which are master data related to the *end customers* and *subscribers* of the service provider company and can represent very high volume of data to store and manage in SAP Convergent Charging.

Related Information

Master Data [page 28]
Customer Master Data [page 30]

1.3.2 Rating Context

The rating (dynamic pricing) function of SAP Convergent Charging 2021 uses a set of data to complete its tasks at runtime. All of this data, for which the values vary over time, is referred to as the *rating context*. Data that is part of a rating context is also referred to, in more general terms, as *rating context properties*.

The properties of a chargeable item or the following sources provide the results of usage used to calculate prices:

- Counters [page 297] values from provider contracts
- Parameters [page 303] values from provider contracts, charge plans, or charges
- Translation Tables [page 132] or Tier Tables [page 147] rows
- Provider contract properties such as dates, for example

> ⓘ Note
>
> When you are at the stage of adjusting and finalizing your charge plan, you can view the rating context in detail. You can also manipulate it in the user interface by performing a trial run [page 335] on the price plan (see).

1.3.3 Account and Balance Management (ABM)

> ⚠ Caution
>
> As of SAP Convergent Charging 5.0 SP 5, the behavior of this feature is modified in case of an integrated *SAP Solution* with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

> → Recommendation
>
> As of release 5.0, SAP Convergent Charging provides you with the *Allowance Management* feature to model time-limited service allowances.
>
> See Allowance Management in the SAP CC 2023 Application Help.

The *Account Balance Management (ABM)* feature adds prepaid accounts [page 514] and credit control features to SAP Convergent Charging and:

- Holds subscriber accounts.
- Controls the addition or deduction of monetary or non-monetary amounts from accounts.
- Performs credit reservation on accounts.
- Manages counters applicable for accounts.

The benefits of the account balance management are the following:

- Offer a single platform to manage balances for all subscribers, independent of their payment method.
- Support subscriber groups with sets of interrelated prepaid/postpaid balances, for example, for families or corporate customers.

- Enable flexible configuration of spending caps, allowances, credit limits, and service-specific wallets.
- Manage dynamic charging where the balances to be charged are determined per transaction at runtime.
- Ensure real-time online operations with high performance and high availability.

Subscriber Accounts

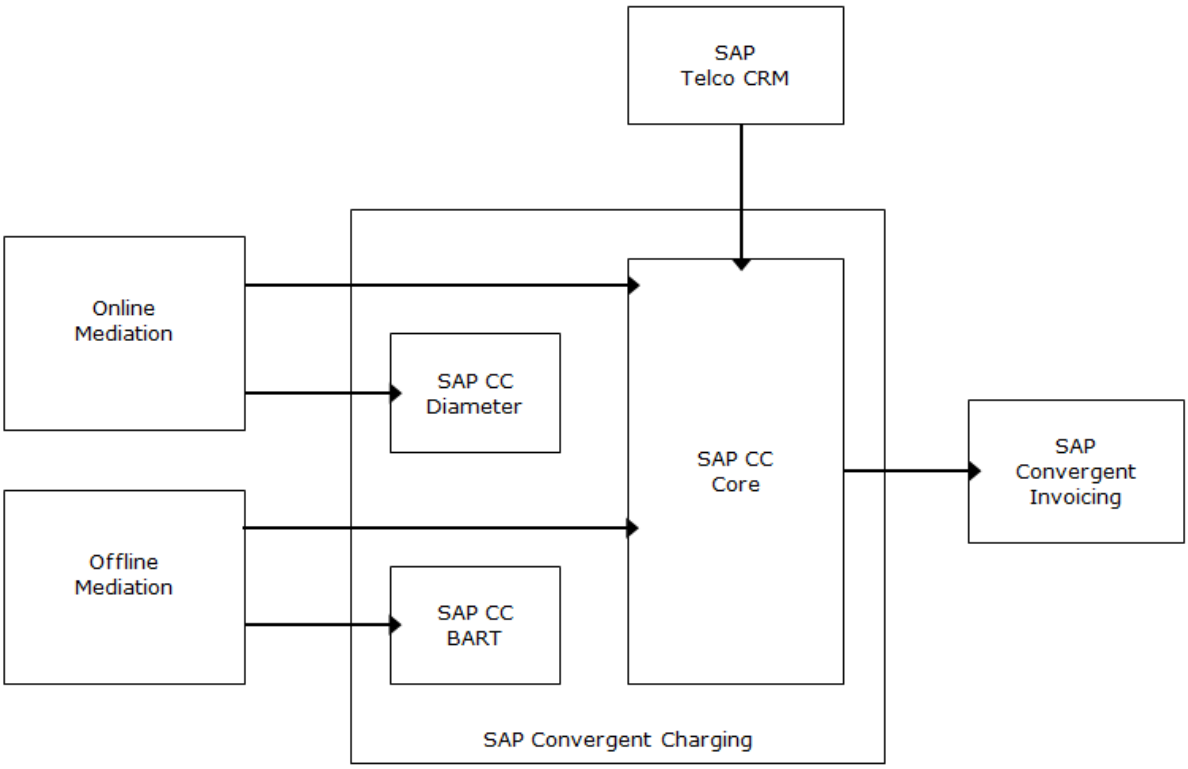
SAP Convergent Charging, the subscriber account [page 511] contains all the accounts which are debited or credited when a service is consumed. The accounts are the following:

- Prepaid Accounts [page 514]
- External Accounts [page 520] (postpaid)

Charging Policy and Multiaccount Determination

The charging policy identifies the accounts that will be debited or credited with the priced amount calculated with the price plan. The configuration is mainly based on information set in the SAP Master Data for SAP Convergent Charging:

- Master data related to the pricing configuration for the usage-based service delivered by a service provider company:
  - In reusable charges: Charging Plans [page 305]
  - In charge plan: Customized Charge: Account Assignments [page 385], charge plans (see expected account assignments)
- Master data related to the end customers who consumes the services: provider contracts (see Account Assignments [page 593])



> ⓘ Note
>
> **Online and Offline Charging**
>
> A distinction is usually made between online and offline charging:
>
> - **Online charging**: An online charging layer applies real-time technology to manage the multiple services. It enables an exchange of information between the charging layer and the network element or the applications in real time; the information exchanged can affect the delivery of service for the end user. The interaction is to handle authorization, rating, credit control, and the balance management of subscriber accounts. In SAP Convergent Charging 2021, the online charging function is used for prepaid services.
> - **Offline charging**: An offline charging function is necessary to handle postpaid services. The charging information does not affect the service rendered in real time. The final result of this charging mechanism is the forwarding of rated call-detail record files to the billing domain. The offline charging function is applied to users who pay for their services periodically, for example, at the end of the month.

Charging Process in a Business Scenario

SAP CC processes any service usage event (named *chargeable item*) coming from an external mediation system in the following order: *Identification*, *rating (dynamic pricing)* and *account determination*.

1. **Identification**: The properties (technical identifier for the end customer (see service user identifier), service identifier, usage date) contained in the chargeable item allow to identify a business agreement (Provider Contracts [page 573]) and its charges through an access [page 634].
2. **Rating and dependent fees**: For each charge, the price plan (Price Plans [page 299]) is carried out, and the related transactions are created with the calculated price amount. Dependent fees are computed. If some dependent charges (Charges [page 290]) exist, the steps "rating" and "account determination" are repeated for each dependent charge.
3. **Account determination and customer data updates (balances, counters)**: The charging plan (Charging Plans [page 305]) is carried out through charging references which are links to accounts from the subscriber account. During this process, persistent counters are updated, the prepaid account balances are debited or credited, and charged transactions are created to debit or credit the balances of external postpaid accounts. Account and balance notifications are managed and alerts are sent to external systems.
4. **Result data**: When all the charges are processed, all the charged transactions are sent to the external billing system or saved to files. Different formats are possible: transactions, charged items, billable items (SAP Convergent Invoicing).

> ⓘ Note
>
> About alerts and the rejection of the charging process:
>
> - An alert is triggered if one of the balances goes below a specified threshold.
> - In the case of a prepaid account, the charging process is rejected if the charged account is locked or closed (states of a prepaid account (Prepaid Accounts [page 514])).
> - The charging process is rejected when only one balance threshold is reached or if an incompatible currency is used.

Refilling Policy

The *refilling policy* identifies the prepaid account that will be credited with a refill (money, units, and so on). The refilling policy is mainly based on the Refill Plans [page 421] and their Refill Logic [page 395].

Refilling Process in a Business Scenario

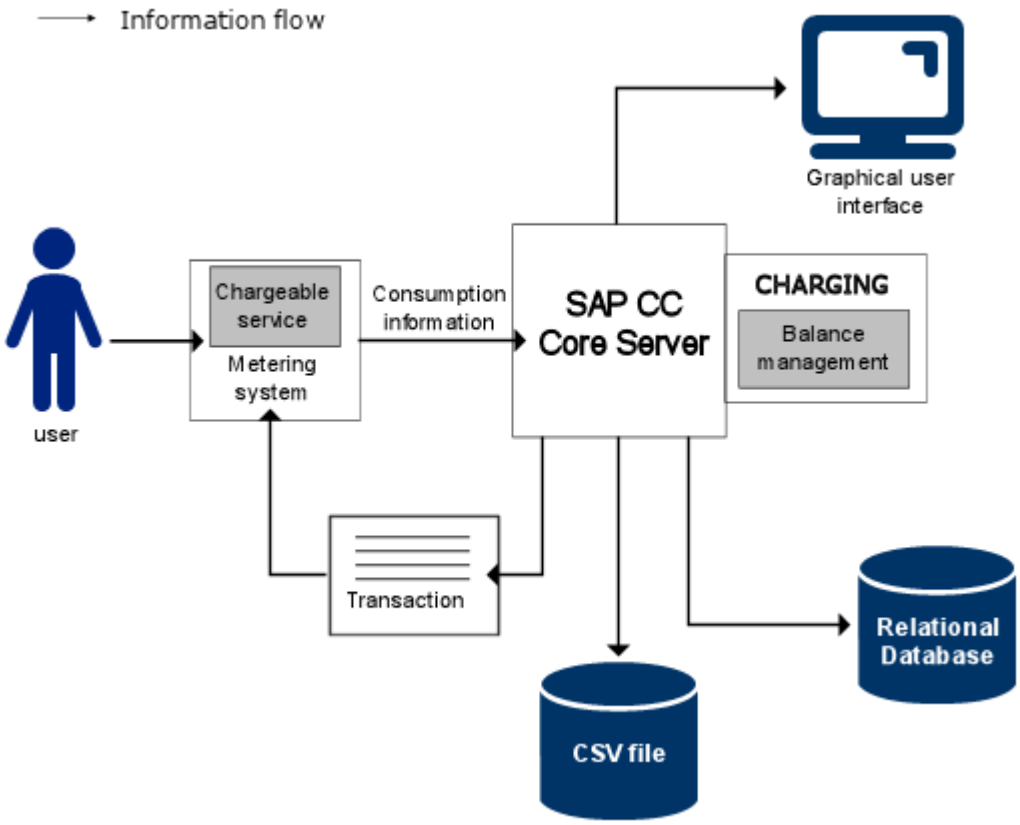
Any refill event (refill item class) coming from an external system is processed in the following order:

1. **Identification**: The properties (for example, account identifier, MSISDN, URL, and SIP) contained in the refill item class allow to identify the prepaid account.
2. **Refill**: The refill plan and its refill logic is carried out by the SAP CC system to compute the credit to add to the prepaid account and updates the prepaid balance and the counters of the related services accordingly. Refill records (or resulting transactions) are generated and stored in files.

1.3.4 Charging Information Flow

Charging information flows through SAP Convergent Charging (SAP CC) in the following way:

1. As an overall prerequisite, a party subscribes to the commercial product of a service provider. For example, in an external provisioning system such as SAP CRM. Agreed pricing and charging terms are saved in a subscriber account and in a provider contract.
2. The subscriber or an agreed end customer consumes a usage-based service whom pricing configuration is managed by SAP CC. An external system outside SAP Convergent Charging can meter the service consumptions.
3. A ticket (aka chargeable item), which is information about the consumption, arrives at SAP CC Core Server. The system identifies the service consumer and the existence of a provider contract and a belonging subscriber account.
4. The SAP CC Core Server system calculates (rates) dynamically the price amount of the consumed service according to data set in the identified provider contract: parameters, counter values, and charges of the activated charge plans.
5. The system determines then debits or credits prepaid accounts or external (postpaid) accounts with the amounts calculated with the executed price plans. The balance management system controls the addition or deduction of monetary or non-monetary amounts from accounts. The system also considers the existing service allowances that are defined in the provider contract.
6. One or more charged items (aka transactions) are produced, which are returned to the external system, your billing system, or saved to a CSV file. Counters in the provider contract are updated into the back-end database system.



Objects such as chargeable item classes, charges, and charge plans are saved into the back-end database system. As an SAP CC pricing specialist, you can use the graphical user interface (SAP CC Core Tool) to configure the pricing objects that are master data related to the service provider. You can perform trial runs [page 335] of price plans to test your pricing logic.

1.3.5 Basic Charging Concepts

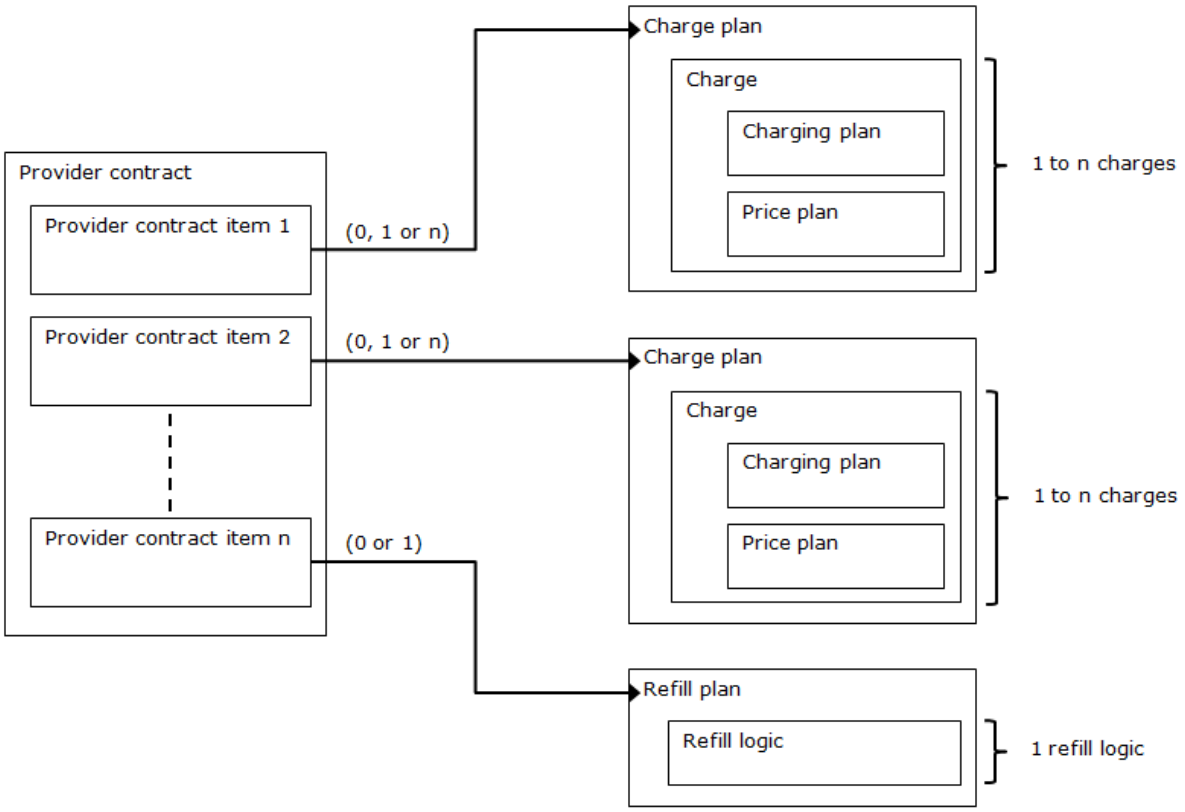
The concept of the charge plan [page 343] is fundamental in SAP CC SAP Convergent Charging. You configure the pricing for the subscription- and usage-based services to monetize by designing and implementing a pricing catalog. In this catalog, you build up a charge plan by listing all the charges that the end customers can subscribe to. Then you combine several charge plans and one refill plan at most in an external provisioning system. For instance, you can combine charge plans to build a commercial product in a CRM application. Each charge contained in the selected charge plans are activated when a provider contract is signed with an end customer.

A charge plan is a reusable pricing block which is made up of customized Charges [page 290]. A charge contains one or more Chargeable Item Classes [page 241][1] linked to a price plan [page 299]. A price plan defines how to calculate how much a customer must pay to consume a service. This information is used to dynamically rate the price amounts and other non-monetary amounts at runtime.

A provider contract [page 573] is the contract between a service provider company and a subscriber. It is composed of one or several contract items which reference a charge plan or a refill plan. It is created and maintained by an external provisioning system and synchronized with SAP Convergent Charging through a dedicated technical interface (Web Services).

Accesses [page 634] specify under which identifier the subscriber consumes the usage-based service. When SAP Convergent Charging is integrated with SAP CRM, the accesses are automatically created and maintained by SAP CRM. Note that a subscriber can access a service through various identifiers.

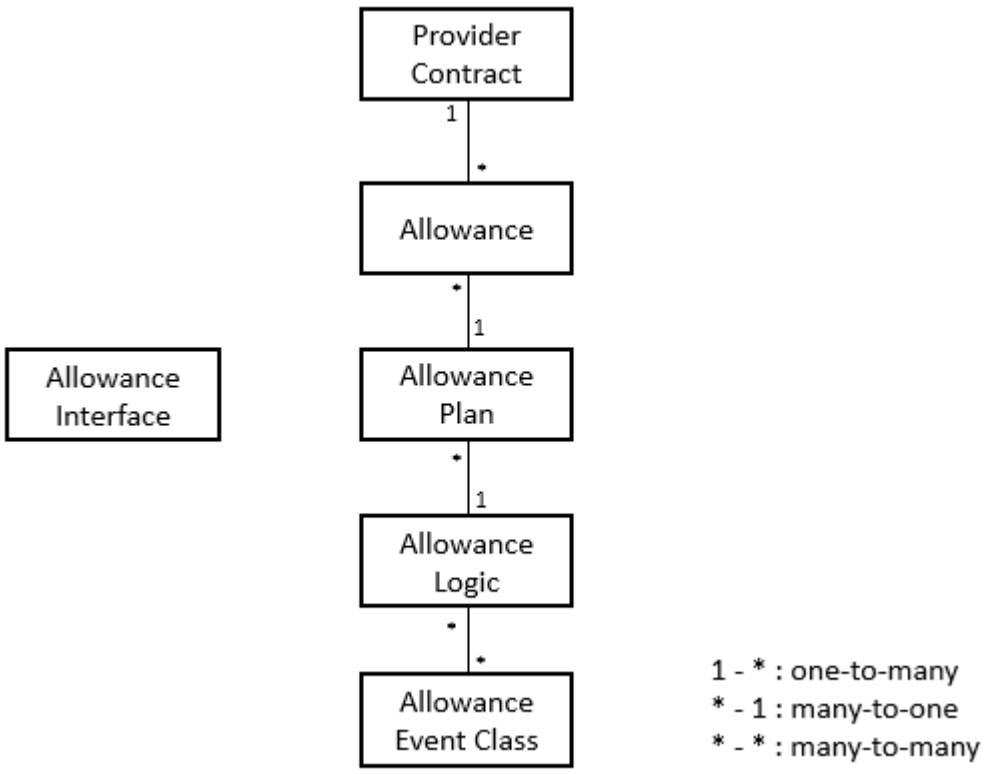
[1] Only if your price plan contains usage rate components. Chargeable item classes are not mandatory if your price plan only contains recurring or one-shot rates.



Additionally to that, the pricing catalogs may contain an allowance plan [page 466] defines the behavior of an "allowance of service consumption" (credit bucket, spending limit, promotion) when the charging and refilling processes are executed by the system.

Allowances [page 575] in provider contracts represent these granted service usages that are limited in quantity or in time. Each allowance refers to an allowance plan at the most, and the allowance plan contains one to several customized allowance logic objects. A provider contract may contain zero or several allowances.

The following illustration shows the relations between the objects relating to allowances:



Related Information

Provider Contracts [page 573]
Allowances [page 575]
Allowance Plans [page 466]
Allowance Logic [page 449]
Allowance Event Classes [page 441]
Allowance Interface [page 91]

1.3.6 Session-Based Charging and Reservation

In SAP Convergent Charging 2021, three types of online charging are provided:

- Event-based charging with or without credit control
- Session-based charging (with credit control and quota reservation)
- Blank charging for advice of charge functions

Offline charging of past events is supported in a postpaid environment.

Session-Based Charging

> → Remember
>
> SAP SE strongly recommends that you read the conditions and limitations on the session-based charging. Refer to the Session-Based Charging Limitations [page 22] topic.

The session-based charging is more suitable to handle a wide variety of rateable instances such as:

- Rating of volume: Video download, for example
- Rating of time: Telephone calls, for example

Concurrent Services in Real Time

The session-based charging is performed in real time and is able to manage the reservation of units for many services at the same time. For example, while calling on a mobile phone, a client can download a video. The units include the balance of a prepaid account and the counters in a provider contract).

Counter Reservation and Credit Control

A session-based charging is provided with counter reservation and credit control. The credit control is a mechanism which directly interacts in real time with a balance and controls or monitors the charges related to service usage. The credit control does the following:

- Checks if credit is available on the balance
- Reserves credit
- Deducts the reserved credit from the balance when the service is completed
- Refunds the balance with the reserved credit not used

How Does a Session-Based Charging Operate?

The session-based charging is well adapted for telephone services for two reasons:

- The amount of resources that a subscriber may consume cannot be a priori known.
- Other services debit the same balances and counters at the same time; therefore, counters and monetary balance must be reserved in advance.

In this case, a certain amount of monetary or non-monetary units is reserved on the subscriber's balance. Permission to use an amount of resources that matches the unit reservation is then allowed. When the granted units have been used or when a new, not yet authorized chargeable event occurs, a new request is sent for unit allocation. When resource usage has been executed, the actual amount of resource usage (used units) must be known so that reserved amounts can be credited to the subscriber's balance, ensuring that the correct amount gets debited.

1.3.6.1 Session-Based Charging Limitations

> ⚠ Caution
>
> SAP SE recommends that you read the following limitations on the Session-Based Charging and Reservation [page 20].

About Information on Reservations and Confirmations

A session-based charging is usually a succession of reservations and confirmations. The information relating to a reservation or a confirmation is available in a price plan [page 299] only through a custom property (aka user-defined property) that you must create and define in a chargeable item class [page 241] for the usage-based service to monetize.

This "reservation vs confirmation" information is required in the incoming chargeable items that relate to the service consumptions.

The downstream system (mediation) that massively sends chargeable items to the SAP CC system must include this "reservation/confirmation" information in these chargeable items for beeing charged for the service usages.

About Transient Counters

A transient counter [page 298] **is unique to a charging session**. The values of a transient counter are updated in memory both at the time of reservation and confirmation. If you want a transient counter to be incremented at a specified time (reservation or confirmation), you must create a dedicated property in a chargeable item class (refer to the previous paragraph *About information on reservations and confirmations*)About Information on Reservations and Confirmations [page 22]. Then, in the price plan settings, you add some instructions to test the property and update the transient counter at the specified time according to the test result.

> ✤ Example
>
> For example, when the counter is used to add up periods of time (sessions) to calculate the session duration, the counter must not be incremented when a quota is reserved but only when it is confirmed.
>
> The SAP CC pricing specialist must then configure the price plan so that the transient counter is incremented only at the time of confirmation. This requires that the information relating to the reservation/confirmation comes from a custom property of the incoming chargeable items.

About Persistent Counters

Unlike transient counters, persistent counters [page 297] **can be shared by several charging sessions**. For example, the persistent counters allow sessions to share free units. At the time of reservation, persistent

counters are not updated in the database, but counter reservations are stored in the database. Persistent counters are updated in the database at the time of confirmation.

When using persistent counters in a session-based charging environment, pay attention to the following:

- In a *Usage Rate* branch, use the operations such as + or – and avoid the operations such as set, reset, *, or /.
- In a *Recurring Rate* or *One-Shot Rate* branch, the operations such as set or reset can be used and possibly mixed with operations such as *, /, +, or –.

A persistent counter has the following values:

- **A current value**: This value is displayed in the counters of a decision tree, a price plan [page 299], or in a charged item (see Charged Item Classes [page 254]) when viewing a persistent counter. The current value includes the modifications resulting from the ongoing reservations.
- **A reservation-free value**: This value is displayed in the counters of a provider contract when viewing the provider contract [page 573] or its service allowances [page 575]. The reservation-free value does not include the modifications resulting from the ongoing reservations.

> ⓘ Note
>
> You can also access the reservation-free values through the *Property Introducer Operator* component. For more information, see the primary help of logic components: Working with Logic Components [page 671].

About Loyalty Points and Free Minutes

It is recommended to be careful when using session operations to establish a tariff giving loyalty points or free minutes. In the current session of a session-based charging, adding free units or loyalty points to a persistent counter must be done at the time of confirmation to prevent the counter from being used by other sessions. The most appropriate time is at the confirmation of the last consumption quota (last confirmation of the session). If free units or loyalty points are added to the persistent counter at the time of reservation, other sessions may use the persistent counter. For example, a persistent counter which is reserved by a first session, and then reserved and modified meanwhile by a second session, can cause a balance to go below zero.

About Reserving a Persistent Counter

Because a persistent counter may be reserved simultaneously by several sessions, its value may continuously change. For example, the value of a persistent counter at the time of reservation may be different from its value at the time of confirmation. Therefore, a price plan calculating a price must accurately test a persistent counter in order to avoid an amount of reservation less than the related amount of confirmation (see the example below). Testing a persistent counter of free units can be carried out by using a splitter component to split the value of the counter into free or not free units. This solution allows getting a correct price.

> ✤ Example
>
> As shown in this example, a session management is not always able to allow accurate tests on a persistent counter.

In the chargeable item class "Telephone services", each call is priced at $0.25 per minute (VIP tariff) or priced $0.50 per minute (normal tariff). In addition, in the related price plan, each national call adds 10 points to a persistent counter "C" of loyalty points while each international call subtracts 10 points from the counter "C". As a customer, John Smith has accumulated more than 200 loyalty points, and his account balance is $1.50.

1. Session 1 (national call) starts, and then reserves 2 minutes: pre.C = 195 (*), C = 205, VIP tariff, price = $0.50
2. Session 2 (international call) starts, and then reserves 2 minutes: pre.C = 205, C = 195, normal tariff, price = $1
3. Session 2 (international call) stops, and then confirms 2 minutes: pre.C = 195, C = 185 (195 – 10), normal tariff, price = $1
4. Session 1 (national call) stops, and then confirms 2 minutes: pre.C = 185, C = 195 (185 + 10), normal tariff, price = $1

(*): “pre.C” contains the value of the counter before rating while the counter “C” contains the value of the counter after rating.

This example shows that the context of the confirmation differs from the context of the reservation. This means that testing a persistent counter value shared by multiple sessions will not provide a constant result. Because the value of a persistent counter can change between the time of reservation and the time of confirmation, the result of a test branch in a price plan may change. Therefore, SAP SE strongly recommends to use a splitter component for managing free units and testing persistent counter values at the time of confirmation when the current counter value does not change.

About Emptying the Free Units of a Persistent Counter

At the time of reservation, if a session (*) empties the free units of a persistent counter without totally confirming the consumption of a service, another session running at the same time cannot use the free units of the counter, and thus, will entirely be charged. Therefore, a client will pay for the consumption of a service even if free units are still available. This exceptional case however occurs only when several sessions are simultaneously open and a persistent counter is emptied.

(*) In price plans including persistent counters of free units

Example

The persistent counter "C" contains 60 seconds. Session 1 starts and reserves 120 seconds. Session 1 empties the persistent counter "C" because the first minute is free. Session 2 starts, reserves 120 seconds, and cannot use the free units of the empty counter "C". Session 2 then stops and confirms 2 minutes, which are debited from the customer’s balance. A little while later, session 1 stops, and 30 seconds are only confirmed. At the end of both sessions, 30 seconds of free units are still available, and the respective customers’ balances have already been debited.

About Reserved, Confirmed Amounts, and Balances

As for any other rating operation which reserves, and then confirms a consumption quota, a balance may be less than the value of 0 if the related price plan does not respect the following rule: In the same context of

rating, and for the same duration, the confirmed amount must not be greater than the reserved amount. For example, if 5 minutes are reserved at the price of $2.50, the confirmed amount must not be greater than $12.50.

About Transactions

Each confirmation in a session-based charging generates a rated transaction that is assigned to the relevant accounts. The sessionID[2] field of transactions allows finding all the transactions generated by the same charging session.

1.3.7 Spending Status Monitoring

> ⓘ Note
>
> The spending status monitoring enables a registered and connected policy or monitoring system to make policy decisions that are based on information about an end customer and the service usage context. SAP CC converts some counter values in provider contracts as the form of spending statuses and manages the reporting of these indicators when they change.

In online environment, the spending status monitoring enables to implement policy control in your system landscape. Downstream systems (policy, monitoring) can make policy decisions based on end customer experience and service spending information reported by SAP Convergent Charging 2021.

The spending status monitoring adds a monitoring plan and spending status reporting features to SAP Convergent Charging 2021.

The benefits of the spending status monitoring enables a flexible configuration in master data through:

- The easy data configuration in pricing catalog (master data): Basic objects to configure monitoring plans
- The fine configuration in customer master data: Configuration in Contract Items [page 592] in provider contracts

> ⓘ Note
>
> - The spending status monitoring feature does not support the counter sharing between provider contracts (enterprise counter pooling). Do not share a pool of counters with parent and linked provider contracts.
> - The offline charging operations do not influence the spending status reporting. Counter changes relating to batch charging operations do not trigger new spending status reports.

[2] Provided by an external system.

Spending Status Reporting

The SAP CC systems can detect changes in the business context of the end customers and sends spending status reports to the registered policy or monitoring system.

Implementation

Refer to the SAP CC Administration Guide (see in Help Resources [page 40]) for more information about the implementation of this feature in your system landscape:

- When SAP CC is connected to SAP CRM.
- When SAP CC must be interfaced with another CRM or provisioning system.

Application Example

Example

A service provider needs to focus on the quality of services it delivers to increase perceived quality and performances and reduce churn. It must implement a fair usage management function in the system landscape to be able to act on traffic control and service usage (consumption).

An end customer subscribes to a mobile data plan allowing for "unlimited data usage".

To protect itself from customer abuse in network usage, the operator adds a clause to the contract small prints.

It explains that data usage is unlimited with regular quality of service up to a consumption threshold (2.5GB per month). Once the customer has used these 2.5GB in a month, service is continued but throttled, downgraded in quality of service.

The service provider implements policy control in its system landscape to ensure fair usage and to optimize the network bandwidth allocation with predefined policy rules.

A policy system registers to the SAP CC system and monitor fair usage of end customers.

To model this business case in SAP CC, the service provider configures in the pricing catalog (see Catalogs [page 109]) of this customer service:

- The necessary entry in the counter name dictionary: 'DATA_USAGE' is a counter name that will be used in the charge plans and monitoring plans
- The necessary reusable charge and charge plan
- The necessary Spending Status Description [page 483]: 'Data fair usage'
- The necessary range table class (see Range Table Classes [page 199]): 'RTC dedicated to data monitoring'
- The necessary range tables (see Range Tables [page 213]): 'RT dedicated to data monitoring'
- The necessary monitoring plan (see Monitoring Plans [page 491]): 'Monitoring plan dedicated to data usage'

Additional Information

See the Application Help about the spending status monitoring feature and related processes.

Related Information

Spending Statuses [page 598]
Configuration of Business Data Step by Step [page 107]
Spending Status Description [page 483]
Monitoring Plans [page 491]
Setting Up a Monitoring Plan in a Provider Contract [page 628]

1.3.8 Data Objects in Core Tool

SAP Convergent Charging 2023 extends your *core data* with other data useful for managing the pricing and charging of end-customer service's use: Master data, customer master data, business data. You configure this data with th Core Tool user interface.

For example, as a pricing specialist, you use the Core Tool user interface to configure some master data and business data. When you start working with this user interface, you usually create some pricing elements such as chargeable item classes [page 241] and reusable charges [page 290], and pricing macros [page 121]. You can use objects to configure more complex objects such as charge plans.

With the Core Tool user interface, you can configure some *technical data* as a system administrator. In an SAP environment, this data includes SAP CC users and audits for user operations.

Master Data

Master data is the set of data which is commonly used across the entire organization and all business processes. SAP Convergent Charging defines data objects that relate to the implementation of the pricing policy for chargeable services or products. As a pricing specialist, you configure this master data in a pricing catalog with the Core Tool user interface.

More information here: Master Data [page 28]

Customer Master Data

Customer master data (or master data for end customers) is a subset of the master data that relates to the end customers of the service provider.

- In business-to-customer (B2C) environment, this data represents very high volumes in the back-end database system.
- In a B2B environment, you may configure this data with the Core Tool user interface by creating a Subscriber Accounts [page 511] and some Provider Contracts [page 573].

More information here: Customer Master Data [page 30]

Business Data

Business data is a master data that is configured for a complete system. As an application administrator, you must configure this data.

More information here: Business Data [page 32]

Technical Data

With Core Tool, administrators (system or user) configure and manage some objects for technical features of SAP Convergent Charging.

More information here: Technical Data [page 34]

1.3.8.1 Master Data

In SAP Convergent Charging (SAP CC), the **SAP Master Data** includes:

- Master data related to the **pricing configuration settings for an end-customer service to monetize**
- Customer Master Data [page 30], which is **master data related to the end customers and subscribers** of the service provider and can represent very high volume of data to store and manage in SAP CC

The following master data allows you to configure the pricing logic of a chargeable service or product.

Pricing Catalogs [page 109]

To help you organize your work more efficiently, you create pricing catalogs for storing all the objects (pricing elements) that are needed for the pricing of a chargeable service or product that are offered by the service/product provider company.

To implement the pricing policy of an end-customer service in SAP CC, you configure the pricing logic in a catalog.

Chargeable Item Classes [page 241]

You need to define chargeable item classes that represent services which can be rated and charged according to their usage. As of release 4.0, the chargeable item classes are independent objects which are no longer subsumed within chargeable item packages.

Charged Item Classes [page 254]

You create charged item classes to retrieve all the necessary information included in the resulting transactions of the charging process. The SAP CC system generates this information in charged items.

Aggregation Policies [page 273]

You can merge multiple charged items into a single one through the aggregation policies, and thus decrease the number of consumption items which are sent to the SAP Convergent Invoicing in the SAP ERP system.

Reusable Charges [page 290]

You create reusable charges that define the price plan and the charging plan to be applied to the usage of a service by end customers.

Charge Plans [page 343]

To integrate with a CRM system or an external provisioning system, you create *charge plans* which represent the way that you sell services. You can combine charge plans to create commercial products.

Refill Item Classes [page 387]

You need to define *refill item classes* that describe the refill events of usage type coming from an external system such as SAP ERP/FI-CA or a Voucher Management System.

Refill Record Classes [page 414]

You create *refill record classes* to retrieve all the necessary information included in the resulting transactions of the refilling process. The system generates this information in refill records. You need to define refill item classes that describe the refill events also called refill items coming from an external system such as SAP ERP or a Voucher Management System.

Refill Logic [page 395]

You create the refill logic that defines how to compute the different credits (money, units, and so on) of a refill.

Refill Plans [page 421]

To integrate with a CRM system or another connected provisioning system, you create refill plans to credit the prepaid accounts of end customers.

Allowance Interface [page 91]

You configure the allowance interface to list and describe the common parameters and counters of all the allowances.

Allowance Event Classes [page 441]

You create the allowance event classes to define and describe the type of an allowance event and its set of properties. The allowance event class is similar to the chargeable item class or the refill item class.

Allowance Logic [page 449]

You create the allowance logic that is a decision tree defining the rules and conditions of use of a contract allowance based on this logic.

Allowance Plans [page 466]

You create an allowance plan that includes the allowance logic to define the behavior of an allowance when the charging and refilling processes are executed.

Monitoring Plans [page 491]

You create a monitoring plan to define and generate spending statuses that you can monitor through a policy system (monitoring system).

Pricing Macros [page 121]

You can create pricing macros that are small reusable modules for applying calculation formulas within the charges.

Translation Tables [page 132]

To assign specific output values to input values, you can create translation tables used within the charges and refill logic.

Tier Tables [page 147]

You need to define refill item classes that describe the refill events also called refill items coming from an external system such as SAPTo assign a combination of output values to numeric input intervals, you can create tier tables used within the charges and refill logic.

Mapping Tables [page 179] and Range Tables [page 213]

To assign specific output values to input values, you can create mapping/range tables used within the charges and refill logic. You can import a price table to use recurring charges defined in a CRM application.

> → Tip
>
> In SAP Convergent Charging, the Billable Item Mapping [page 263], Consumption Item Mapping [page 281], Currencies [page 81], and Public Holidays [page 73] are not part of the master data because they are available for the complete system and cannot be configured for each pricing catalog of a service provider. They are part of the business data [page 32].

Related Information

Customer Master Data [page 30]
Business Data [page 32]
Data Objects in Core Tool [page 27]
Configuration of Business Data Step by Step [page 107]

1.3.8.2 Customer Master Data

Customer master data is part of the *SAP Master Data* for the subscription- and usage-based service pricing. This high volume data relates to the end customers and subscribers of a provider contract company.

Subscriber Accounts

The subscriber accounts are automatically created in SAP CC Core Tool.

Subscriber Mapping Tables and Subscriber Range Tables

You can create Subscriber Mapping Tables [page 542] or Subscriber Range Tables [page 557] to store customer related data. Subscriber mapping/range tables are business data tables attached to a subscriber account and used in the provider contracts relating to the same subscriber account.

Provider Contracts

You can create provider contracts [page 573] that contain the terms of the agreements signed by service providers and end customers or subscribers.

Accesses

At provider contract level, a user access [page 634] specifies the *technical identifier* under which an end customer consumes a usage-based service.

Related Information

Master Data [page 28]
Business Data [page 32]

1.3.8.3 Data Tables

To configure the master data, SAP Convergent Charging manages several business data tables to store and use your pricing data, business data, and custom data. These data tables include business information that allows you to implement the marketing and pricing policy of an end-customer service for configuring the pricing logic in SAP Convergent Charging:

Data tables	Are used in	Can be redefined in charge plans	Can be redefined in provider contract
Mapping Tables [page 179]	• Charges • Refill Logic • Allowance Logic	Yes	Yes
Subscriber Mapping Tables [page 542]	N/A	No	Yes (through the relating Subscriber Accounts [page 511])
Range Tables [page 213]	• Charge • Refill Logic • Allowance Logic • Monitoring Plan	Yes	Yes
Subscriber Range Tables [page 557]	N/A	No	Yes (through the relating Subscriber Accounts [page 511])
Tier Tables [page 147]	• Charge • Refill Logic • Allowance Logic	No	No
Translation Tables [page 132]	• Charge • Refill Logic • Allowance Logic	No	No

1.3.8.4 Business Data

SAP Convergent Charging (SAP CC) defines some business data that is **configured for a complete SAP CC system** in your SAP system landscape.

The application administrator (or a system administrator) configures and maintains this data. The pricing specialist uses this data when configuring the pricing logic in the *Core Tool* user interface. The SAP CC systems use this data when processing the charging or refilling operations.

The following business data allows you to configure the pricing logic in the Core Tool user interface. Note that the application administrator needs to use some other SAP CC tools to manage some other business data.

Billable Item Mapping [page 263] and Consumption Item Mapping [page 281]

Regarding the billable item mapping, you map billable item classes (BIT classes) defined in the SAP ERP/FI-CA reference system onto Charged Item Classes [page 254] (or Refill Record Classes [page 414]) to define which information you need from the resulting business transactions of the charging and refilling processes.

Regarding the consumption item mapping, you map Chargeable Item Classes [page 241] onto consumption item classes defined in the SAP ERP/FI-CA reference system so that the SAP CC system can convert chargeable items to consumption items and load them to the relevant SAP ERP/FI-CA systems for temporary data storage.

> **Note**
>
> This business data is only relevant when SAP Convergent Charging is interfaced with SAP Convergent Invoicing in an integrated *SAP Solution* scenario with SAP S/4HANA or SAP ERP.

Counter Name Dictionary [page 86]

You set up a dictionary of counter names to facilitate the configuration of Charge Plans [page 343], Refill Plans [page 421], and Monitoring Plans [page 491] in the pricing catalog of a service provider.

Currencies [page 81]

You manage the currencies necessary to configure the master data and the customer master data.

Customer Management Areas [page 87]

In an *SAP Solution* scenario, you set up the pairs of SAP ERP/FI-CA systems and SAP CRM systems that maintain the customer master data in the SAP CC system and that manage the customer business billing/invoicing operations (consumption items, billable items).

Public Holidays [page 73]

You manage the public holidays necessary to configure the master data and the customer master data.

Spending Status Description [page 483]

You set up the spending status descriptions necessary to implement the spending status monitoring function.

VAT Rates and VAT Rules

As a pricing specialist, you use customized VAT rates and rules when configuring the master data and the customer master data.

See: Customized Charge: Tax [page 374]

See also: Tax Settings [page 523] (in subscriber accounts)

As an application administrator, you configure and maintain this business data with a dedicated user interface (see SAP CC Setup ToolSAP CC 2023 Setup Tool).

Transaction/Service Types

In the environment of US Telecommunications taxes, both the Core Tool user interface and the Core Server system use the predefined definitions of the transaction types and service types (by Avalara AvaTax for Communications (AFC)) to configure the SAP Master Data (pricing catalogs, subscriber accounts) and the system and to calculate US Telco taxes.

See: Customized Charge: Tax [page 374]

See also: Tax Settings [page 523] (in subscriber accounts)

As an application administrator, you configure and maintain this business data with a dedicated user interface (see SAP CC Setup ToolSAP CC 2023 Setup Tool).

Related Information

Master Data [page 28]

1.3.8.5 Technical Data

With the Core Tool user interface, you can configure and maintain some technical data. In an SAP environment, this data includes SAP CC users and audits for user operations.

Technical Data

SAP CC Users



As a user administrator (or system administrator) for SAP Convergent Charging, you can assign a logon (user ID or user name), a password, and one or more roles to each SAP user (individual or service).

More information here: Users [page 640]

Audits



The SAP CC system can save user operations to the back-end database for audit purposes. As a system administrator, you can create filters to search for audited user operations in your database. You complete some technical or business audits.

More information here: Audits [page 654]

1.3.9 Transactional Data

In SAP Convergent Charging (SAP CC), a **transactional data** is a data that is created during an event or operation and dependent on another data such as master data or business data.

For example, transactional data relates to the following:

- Incoming data:
  - *Chargeable items (CIs)* that are **usage/consumption events** for a chargeable service (digital service, customer service, or usage-based service).
    It contains some service consumption information that is necessary to rate (dynamic price) and charge the calculated fees to an account (in SAP CC or outside SAP CC) or decrease an allowance.
    It must contain a valid technical identifier of the service consumer (subscriber, end customer), a technical identifier for the consumed service, and the consumption date and time information. To be able to configure the service pricing, an SAP CC pricing specialist needs to set up a chargeable item class [page 241] that specifies the data structure.

    > **Note**
    >
    > To match with provider contracts, the system checks these technical IDs with the technical data that is set in the contract items. See Provider Contract>Charge Plan: Technical Data [page 617].

In *SAP Telecommunications industry*, call detail records (CDRs) contain such information.

- Resulting output data:
  - Rated transactions
  - Charged transactions (former releases of SAP CC)
  - Charged items (CITs)
  - Billable items (aka BITs) for SAP Convergent Invoicing in SAP S/4HANA or SAP ERP
  - Reffil records
  - Accounting operations
  - Charged chargeale items
  - Consumption items (for SAP Convergent Invoicing)
  - Notifications (business, technical)

1.4 Getting Started

If this is your first time using the SAP CC Core Tool user interface, get started by working through these steps:

1. Familiarize yourself with the graphical environment of the Core Tool workspace [page 42] and how to launch [page 35] the user interface.
2. Learn basic concepts (pricing, rating, charging) [page 18].
3. Learn Account and Balance Management (ABM) [page 13] features.
4. Learn the objects you will work with (Data Objects in Core Tool [page 27]).
5. Read through the steps involved in creating a charge plan in Configuration of Master Data Step by Step [page 104].

Related Information

Exploring the Core Tool Workspace [page 42]

1.4.1 Launching Core Tool and Logging On

Once the required files have been installed on your workstation by your IT administrator, you can launch the Core Tool user interface of SAP Convergent Charging (SAP CC).

Prerequisites

- Before you begin, make sure that you have the user ID and password given by your SAP CC user administrator.

- Verify that:
  - You have the HCI URLs of the dispatcher instances in the running Core Server system to connect to. You need these addresses to log on to the Core Tool user interface.
  - The HCI URLs of the dispatcher and updater instances must be directly accessible from the Core Tool host.

> **Note**
>
> If the *Connection* dialog box is not displayed after step 2 in the following procedure, refer to your SAP CC system administrator for troubleshooting.

Launch Core Tool

This procedure is valid for a Microsoft Windows environment.

1. Using *Windows Explorer*, open the `SAP Convergent Charging` directory on the Core Tool host.
2. Open the `bin` directory and double-click the `core_tool.bat` script file.
   The *Connection* dialog box appears.
3. In the *Connection* dialog box, type your user ID in the *Login* field. The corresponding user must be an individual SAP CC user.
4. Type your secret password in the *Password* field.
5. Type one or more machine, IPv4, or IPv6 addresses in the *Server URL* field. You can type as many addresses as you have dispatcher instances in your SAP CC system.
   Separate each address by a comma. For example, **`http://machine1:9100, http://machine2:9100, http://machine3:9100`**.
   Contact your SAP CC system administrator to know the HCI URLs of the dispatcher instances in the remote Core Server system.
6. Click *OK*.
7. Core Tool temporarilly connects to a dispatcher instance, is authenticated with your user credentials, releases the connection, and finally connects to an updater instance (thas was indicated by the dispatcher instance).
8. Core Tool is ready and displays its main window [page 43].

   You can work with it and perform your daily or exceptional tasks. A user session [page 37] starts if user session management has been enabled in the connected SAP system.

   > **Note**
   >
   > In case of logon issues, one of the following secondary windows is displayed:
   > - The error window that indicates a communication issue. See Troubleshooting Connection Issues to Core Server [page 828].
   > - The list of previously failed logon authentication attempts so you can verify the issue.
   > - The window used to change your password [page 644] if the password management policy is enabled and if it applies to you.

Failed Login Authentication Attempts (Window)

This window displays a list of all the previously failed login attempts. Each login attempt is displayed with the following information:

- The date and time relating to each failed attempt of the user name
- The IPv4 or IPv6 address of the computer which is the cause of the failed authentication

> ⓘ Note
>
> This window provides only information about the connected user.

Related Information

Exploring the Core Tool Workspace [page 42]
Core Tool Functions and Limitations [page 10]
Disconnecting from Core Tool [page 39]

1.4.2 Configuring Data

Procedures

- Configure the Business Data [page 32] or technical data available for the complete system
- Configure the Configuration of Master Data Step by Step [page 104]master data

1.4.3 User Sessions

> ⓘ Note
>
> This topic is relevant only if the *user session management* feature is enabled in your SAP CC system.

The user session begins when an SAP CC user [page 640] accesses the *Core Tool* user interface. The session ends when this SAP user logs out or quits the user interface or when the system administrator terminates the session.

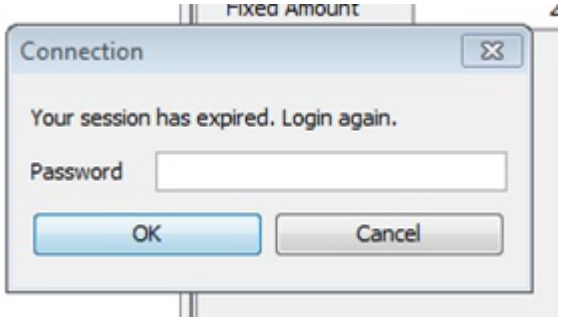
The following characteristics of user sessions are configured at system level:

- The maximum period during which the session remains inactive

- The maximum number of sessions allowed for the same user ID on *Core Tool*

Maximum Inactivity Period

When the session is inactive for a period longer than the period configured at system level, the session expires and you have to authenticate.



A new session starts and you are allowed to continue working with the *Core Tool* user interface.

> ⚠ Caution
>
> When the session expires, objects opened in edit mode (see Display/Edit Mode [page 46] are not protected against concurrent edition any longer (see Concurrent Edition Management [page 48]).

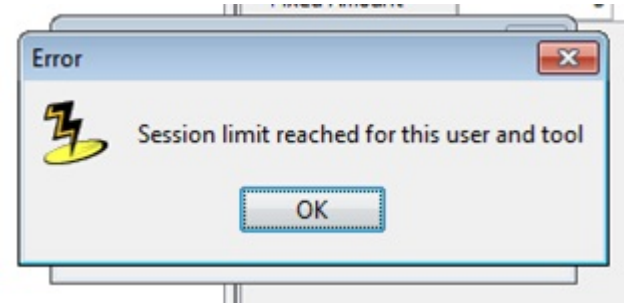
The following actions in the *Core Tool* prevent the session from being inactive and expiring:

- Opening an object
- Saving an object
- Clicking the root of an object tree

On the contrary, modifying a field without saving the object does not prevent the session from expiring. The session is considered as inactive.

Maximum Number of Sessions

If you try to open more sessions on *Core Tool* than the maximum number of sessions authorized, the following error displays:



You are not allowed to connect to another instance of *Core Tool* until one of your sessions ends.

How to Close a User Session?

You can close your session by Disconnecting from Core Tool [page 39].

If the Core Tool user interface you were connected to has failed, contact your system administrator. The system administrator is allowed to close your session by using Admin+.

Related Information

Users [page 640]

1.4.4 Disconnecting from Core Tool

Use this procedure to quit the SAP CC *Core Tool* user interface.

To disconnect from Core Tool

In the main window, do one of the following:

- From the menu bar, select *File* > *Exit*
- On the title bar, double-click on  , and then click *Close*.

The window closes and you are disconnected from the distant SAP CC system. Your user session ends.

1.4.5 Getting Help

Related Information

Help Resources [page 40]
Typographical Conventions [page 42]

1.4.5.1 Help Resources

To help you get the most out of SAP Convergent Charging, SAP SE provides you with an extensive list of resources you can use for support and guidance.

Integrated SAP Solution Scenarios

SAP Convergent Charging is integrated with different software components of the SAP Business Suite: SAP CRM, SAP ERP/FI-CA, and SAP Convergent Invoicing in SAP ERP.

The primary help provides you with specific information and recommendations for configuring data in SAP Convergent Charging for the integrated *SAP Solution* scenarios.

Consult the SAP Help Portal

On SAP Help Portal, you find extensive documentation from the SAP Library on SAP products and solutions, along with additional information about the documentation, information design at SAP, educations services, and links to the SAP Community Network (SCN).

> ⓘ Note
>
> SAP Convergent Charging is part of SAP Billing and Revenue Innovation Management. When your SAP CC system is connected to the SAP CRM system or to SAP Convergent Invoicing in the SAP ERP system, the following SAP libraries are relevant on SAP Help Portal at:
>
> - SAP EHP 4 for SAP CRM 7.0 > Application Help >SAP Library > SAP CRM for Industries > Telecommunications > Functions for Provider Sales and Order Management
>   - Products > Cross Catalog Mapping
>   - Business Agreement
>   - Provider Order and Provider Contract
>   - Sharing Contract
>
> and at:
>
> - SAP EHP 8 for SAP ERP 6.0 > Application Help > SAP Library > Industries in SAP ERP > SAP Telecommunications > Contract Accounts Receivable and Payable > SAP Convergent Invoicing
> -
>   - Master Data
>   - Rating
>   - Billing
>   - Discounts and Charges on Billable Items
>   - Partner Settlement
>   - Invoicing

Access the Reference Guides of SAP Convergent Charging

These guides are available in PDF format and provided on the DVD-ROM with the release of this product. They contain comprehensive information about each functional aspect of SAP Convergent Charging. The guides are also available on SAP Help Portal for online consultation or download.

- Getting Started Guide
- Administration Guide
- Installation and Maintenance Guide

To find a key word once a guide is open, press CTRL + F (Windows users), type the word in the *Find* field, and then click *Find Next*.

Access the Configuration Guide of SAP Convergent Charging

If you need more information about configuring and maintaining the master data for a service provider, we recommend that you consult the SAP Convergent Charging Administration Guide on SAP Help Portal.

Access the Primary help for Subscription-Based Business

In case of a system landscape with customer master data including subscriptions, you can set up the SAP CC Core Tool user interface to work with all kinds of data related to the charging process based either on subscriptions or provider contracts.

Consult the Primary Help of Core Tool [page 12] and the Logic Component Reference [page 671] Reference on the SAP site.

Access the Application Help Documentation

The Application Help document is provided in the Software Development Kit (SDK) of SAP Convergent Charging. A version is available in SAP Help Portal for online consultation.

Access the technical Documentation

The Web Services Documentation, the SAP CC Java/XML API References, and the SAP CC Error Code Reference documents are provided in the Software Development Kit (SDK) of SAP Convergent Charging. A version is available on SAP Help Portal for online consultation,

- Web Services Documentation
- Core APIs

- Error Code Reference

> → Tip
>
> You can also download these product documents on SAP Help Portal.

Access the SAP Notes

SAP Notes and SAP Knowledge Base Articles (KBA) help you find answers to questions that arise from your daily work with SAP Convergent Charging:

- Searching for 'cockpit' in SAP Notes and Knowledge Base Articles (KBAs)

1.4.5.2 Typographical Conventions

The following typographical conventions are used in the primary help of SAP Convergent Charging 2021:

- Text or a button that appears on the graphical user interface is written like *this*.
- System commands, names of folders and files, and codes used in APIs are written in `Code` font.
- Menu choices are shown in this format: *menu name* > *submenu name* > *menu itme name*.
- Text to be typed into a field is shown as follows: **`userinput`**.
- Blue text indicates an active hyperlink to another part of the primary help or a pop-up window that provides you with extra information on a task.
- UI names or features are displayed in purple.

1.4.6 Exploring the Core Tool Workspace

This section presents the elements of the graphical user interface of SAP CC 2023 Core Tool.

Main Window

The Main Window [page 43] integrates the SAP CC Core Tool functions into one workspace. It provides you with the same graphical interface whether your company uses one or several SAP Convergent Charging modules.

Objects and Icons

Inside the main window, you work with objects that you organize in a catalog for each service provider. Each object in SAP CC Core Tool opens in its own object window (see Objects Windows [page 45]) and can be recognized in the graphical user interface by distinctive Object Icons [page 51].

Because SAP CC Core Tool is an object-oriented program, icons are frequently used to represent the different objects and actions you can perform. Become familiar with the icons most commonly used on the tabs (Common Icons [page 53]) and with the windows of SAP CC Core Tool.

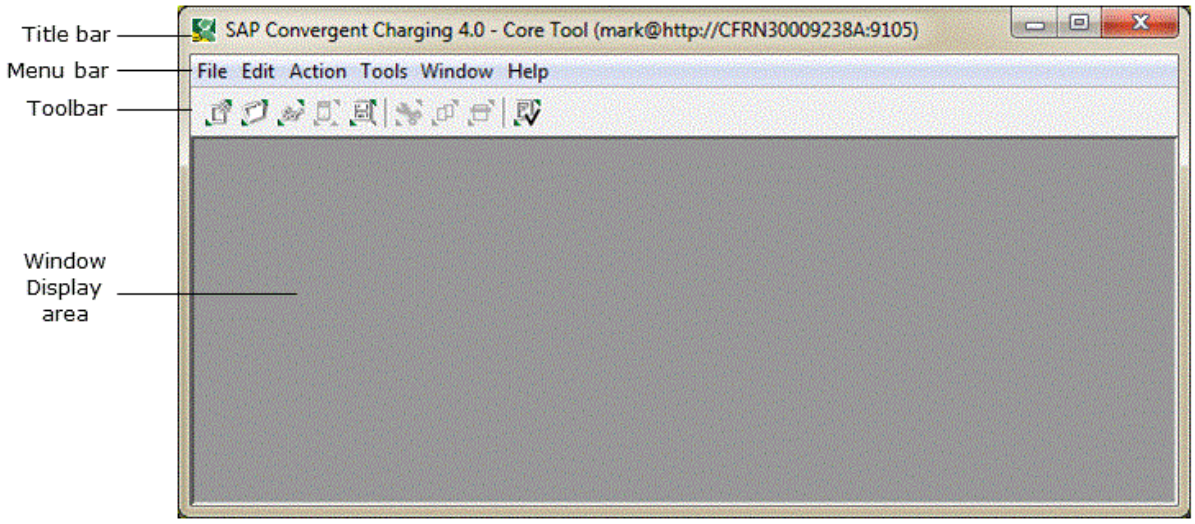
Related Information

Setting Up the Preferences [page 56]

1.4.6.1 Main Window

The main window appears once you have started [page 35] the *Core Tool* user interface for SAP Convergent Charging 2023.

Main Window



Title Bar

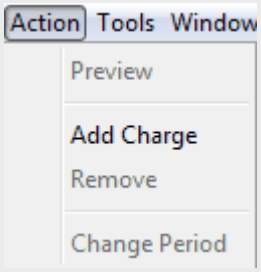
The *Title* bar displays the name of the software on the left and the *Minimize*, *Maximize*, and *Close* buttons on the right.

Menu Bar

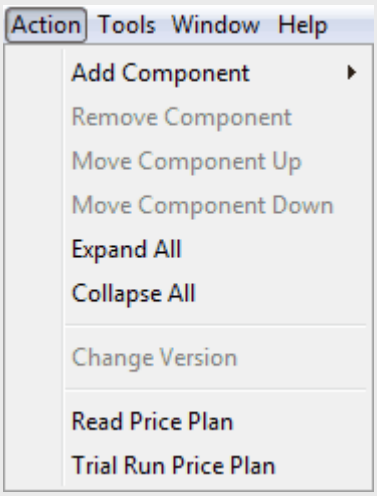
The *Menu* bar displays the different menus for manipulating objects in the graphical user interface. The submenu choices that are available depend on the type of object that is active in the window and the context of the work session. Unavailable choices are dimmed.

Example

Submenu choices in the *Action* menu differ depending on whether a charge plan or a charge is the active object.



Action menu for a charge plan



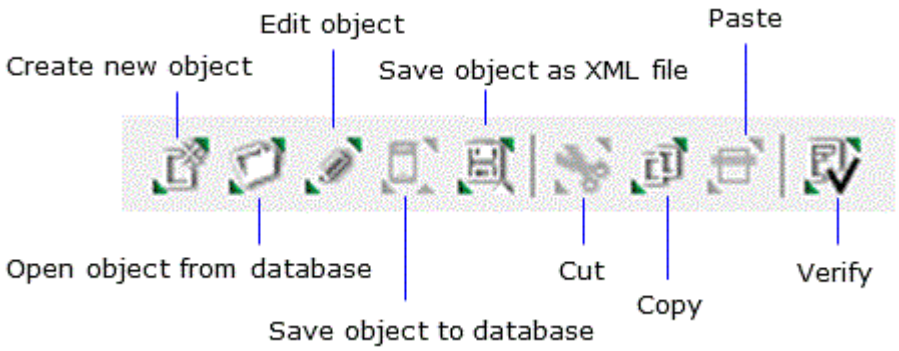
Action menu for a charge

See: Edit [page 55], Action, Tools [page 59], and Window [page 98]

Toolbar

The *Toolbar* buttons are shortcuts for certain menu commands. The toolbar is dynamic, meaning that the buttons which appear on it depend on the object window that is open. For buttons that are not shown below, rest the cursor over a button to read its description.

The following illustration shows the principal buttons of the toolbar:



Window Display Area

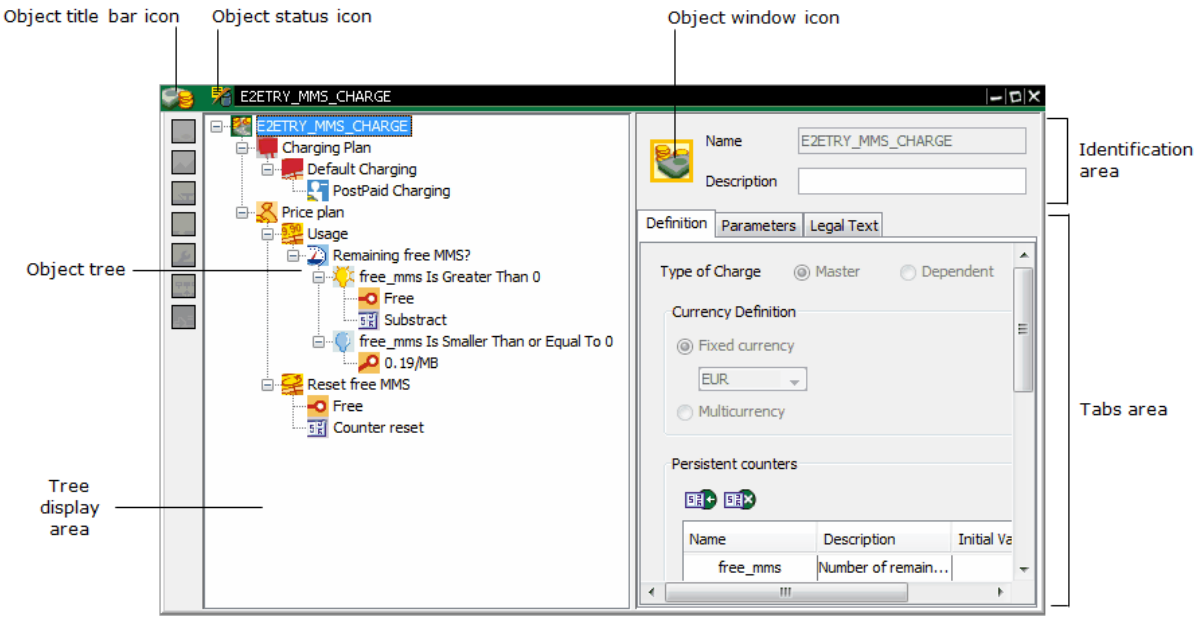
The *Window Display* area displays the different windows and dialog boxes that open during your work session.

1.4.6.2 Objects in the Core Tool Workspace

Objects Windows [page 45]
Display/Edit Mode [page 46]
Concurrent Edition Management [page 48]
Object Transport [page 48]
Object Export/Import [page 49]
Object Deletion [page 50]
Shortcuts for the Menus File and Edit [page 50]
Object Icons [page 51]
Object Status Icons [page 53]
Common Icons [page 53]

1.4.6.2.1 Objects Windows

The windows that you open to work with the subscriber accounts, chargeable item cass, charge, pricing macro, , charge plans, provider contracts objects contain common characteristics that are shown in the figure below:



> **Note**
>
> This window is in edit mode [page 46].

Two object icons [page 51] are shown on each object window: One on the far left of the title bar and one on the upper left of the right pane.

The object status icon [page 53] on the title bar shows you the status of the object you are working with:

- Being created
- Has been saved to the database
- Has been reopened and is in the process of being modified

You create the object as a tree structure in the tree display area. Each building block of an object tree is represented by an icon.

The identification area allows the object to be saved in the database under a unique name. You can choose to describe the object if you wish.

The tabs area contains one or more tabs that are available depending on how the object is structured. The tabs allow you to define and characterize the object and its elements in the tree display area.

1.4.6.2.2 Display/Edit Mode

In SAP CC Core Tool, main catalog objects open in the display mode. You have to switch to the edit mode if you need to modify an object by clicking the *Edit Mode* icon on the main toolbar.

If the object has been modified since you opened it, an *Information* dialog box opens to inform you that the object will be reloaded. Click *OK* to reload the object in the edit mode.

You come back to the display mode by clicking the *Display Mode* icon  on the main toolbar. You are asked to save your changes. If you click:

- *No*, your previous modifications are lost, the object is reloaded and the object windows is in the display mode.
- *Yes*, the object is saved and the object windows is in the display mode.
- *Cancel*, the object remains unchanged and the object windows is in the edit mode.

List of objects using the display or edit mode

- Aggregation policy
- Allowance event class
- Allowance logic
- Allowance plan
- Charge plan
- Chargeable item class
- Charged item class
- Charge
- Mapping table
- Monitoring plan
- Pricing macro
- Range table class
- Range table
- Refill item class
- Refill logic
- Refill plan
- Refill record class
- Tier table
- Translation table
-

Related Information

Concurrent Edition Management [page 48]

1.4.6.2.3 Concurrent Edition Management

> **Note**
>
> This topic is relevant:
>
> - Only if the User Sessions [page 37] management feature is enabled in your SAP CC system.
> - For the catalog objects using the Display/Edit Mode [page 46].

When user sessions are enabled in your system, the objects being edited in *Core Tool* cannot be modified or deleted at the same time in another *Core Tool*. If you therefore set an object in the edit mode, no other *Core Tool* user is allowed to edit this object.

> **Caution**
>
> When the session expires, objects opened in the edit mode are not protected against concurrent edition any longer.

> **Caution**
>
> Importing an object results in overwriting this object, even if it is being edited. Therefore, another user can modify an object being edited in *Core Tool* with the import operation (refer to Object Export/Import [page 49]).

If the object is already being edited by another Core Tool user, you are not allowed:

- To set the edit mode
- To delete the object

If you try to set the edit mode or to delete the object, a *Warning* dialog box opens to inform you who is editing the object and in which session. You can go and ask this user to switch the object you need to modify from the edit mode to the display mode. Otherwise, you can either wait for the session to expire or ask your administrator to delete this session if it is not used any longer.

> **Caution**
>
> SAP CC users who are not connected to *Core Tool* can modify an object being edited in *Core Tool*l. For instance, SAP CRM uses the SAP CC web service user to modify mapping tables.

1.4.6.2.4 Object Transport

SAP CC Core Tool makes it easy to transport a selected and consistent subset of data from a source SAP CC system to a target system.

The transport feature enables you to select a subset of data objects and versions that is consistent for the target system.

Related Information

Transporting Catalog and Data [page 109]

1.4.6.2.5 Object Export/Import

Transporting Objects by Using the Export/Import Operations

SAP CC Core Tool makes it easy to export and import a set of objects linked together.

For example, you could import the following master data:

- Pricing Macros [page 121] with their associated Translation Tables [page 132]
- Charges [page 290] with their associated Chargeable Item Classes [page 241], Translation Tables [page 132], Tier Tables [page 147], and Pricing Macros [page 121]
- Charge Plans [page 343] with Charges [page 290] and all the objects linked to the charges

> ⚠ Caution
>
> Export and import may lead to data inconsistencies. In a production or preproduction system landscape, use the transport operations that are based on transport requests [page 668] and change lists [page 660].
>
> To be sure that you transport a **consistent subset of data objects and object versions**, consider the transport dedicated feature and available operations [page 109]. See Transporting Catalog and Data [page 109].

Related Information

Object Transport [page 48]

1.4.6.2.6 Object Deletion

Deleting Objects

> ⚠ Caution
>
> There is no “undo” function in Core Tool. Before deleting an object, make sure that this action is necessary.

You can delete objects used at runtime for charging or refilling. You can only delete an object if you have first deleted all the objects that reference it. For example, you can only delete a charge plan once you have deleted all Provider Contracts [page 573] that refer to it.

1.4.6.2.7 Shortcuts for the Menus File and Edit

Shortcuts are available for the menus *File* and *Edit* of the *Menu* bar.

Shortcuts for the File menu

The list of available shortcuts are the following:

Shortcut	Corresponding Action
CTRL + N	Create a new object
CTRL + O	Open an existing object
CTRL + S	Save the currently edited object in the Core Database
CTRL + SHIFT + N	Save the currently edited object in a file
CTRL + K	Validate the currently edited object
CTRL + W	Close the currently edited object
F6	Switch between display mode and edit mode for the selected object

Shortcuts for the Edit menu

Shortcut	Corresponding Action
CTRL + C	Copy a tree structure from the selected node
CTRL + X	Cut a tree structure from the selected node
CTRL + V	Paste a tree structure under the selected node

1.4.6.2.8 Object Icons

Each object is represented by a specific icon, which allows to you recognize it in windows and tree structures of the Core Tool workspace.

Below, the list of object icons:

Object	Icon
Chargeable Item Classes [page 241]	
Charged Item Classes [page 254]	
Billable Item Mapping [page 263]	
Consumption Item Mapping [page 281]	
Charges [page 290]	
Charge Plans [page 343]	
Refill Plans [page 421]	
Allowance Event Classes [page 441]	
Allowance Plans [page 466]	
Allowance Logic [page 449]	

Object	Icon
Allowance Interface [page 91]	
Monitoring Plans [page 491]	
Subscriber Accounts [page 511]	
Provider Contracts [page 573]	
Accesses [page 634]	
Pricing Macros [page 121]	
Translation Tables [page 132]	
Tier Tables [page 147]	
Mapping Table Classes [page 167]	
Mapping Tables [page 179]	
Subscriber Mapping Tables [page 542]	
Range Table Classes [page 199]	
Range Tables [page 213]	
Subscriber Range Tables [page 557]	
Spending Status Description [page 483]	
Users [page 640]	
Audits [page 654]	
Change Lists [page 660]	

Related Information

Master Data [page 28]
Master Data [page 28]
Customer Master Data [page 30]

1.4.6.2.9 Object Status Icons

Object status refers to whether an object is being created, modified, or has been saved to the back-end database system.

An object status icon appears at the left of the title bar [page 43] of an object window:

-  *New*: Object is being created.
-  *Saved*: Object has been created and saved in the database.
-  *Being Modified*: Object has been retrieved from the database and is in the process of being modified.

Related Information

Objects Windows [page 45]

1.4.6.2.10 Common Icons

These icons are common to many of the windows and tabs of SAP Convergent Charging 2021. They represent the following:

Properties

 Default property

 User-defined property

 Generated (by the system at runtime) property

Dates and Times

Add a date or date/time

Tables and Lists

 Add a row

 Delete a row

 Move a row down one place in a list

 Move a row up one place in a list

Shared Items

 Counter

 Transient counter

 Add a counter

 Delete a counter

 Parameter

 Add a parameter

 Delete a parameter

1.5 Menus

In the Core Tool user interface of SAP Convergent Charging (SAP CC), use the menus to perform your tasks.

Edit [page 55]
In the menu bar, the *Edit* menu groups some basic editing functions, user interface preferences, and the verification tool.

Action [page 58]
In the menu bar, the *Action* menu groups some usefull contextual functions.

Tools [page 59]
In the menu bar, the *Tools* menu groups some usefull tools.

Window [page 98]

Related Information

Main Window [page 43]

1.5.1 Edit

In the menu bar, the *Edit* menu groups some basic editing functions, user interface preferences, and the verification tool.

In the menu bar, the *Edit* menu enables you to:

- Edit the objects (cut, copy, paste).
- Verify [page 58] these objects.
- Change your user interface preferences and options [page 55]. These settings are defined for the user interface and not for a particular SAP CC user.

  Copy [page 55]

  Preferences [page 55]

  Verify [page 58]

Related Information

Main Window [page 43]

1.5.1.1 Copy

Copying and Pasting a Catalog Object [page 117]

1.5.1.2 Preferences

The preferences represent some of the options of Core Tool and are saved in the local configuration file of the user interface. The preferences only relate to the installed Core Tool user interface and not to a specific SAP user.

Depending on how you (or the system administrator) install Core Tool, the preferences can be:

- Shared by several SAP users if Core Tool is installed in a public folder. In this case, we highly recommend to check the preferences every time you start the Core Tool program to ensure that no change occurred since your last logon.
- Specific to each SAP user if Core Tool is installed in a private folder. For example, the home folder of a user.

> → Tip
>
> For more information about the installation of the Core Tool user interfaces, refer to the SAP CC Installation and Maintenance Guide.
>
> For more information about the configuration of the Core Tool user interface, refer to the SAP CC Administration Guide.

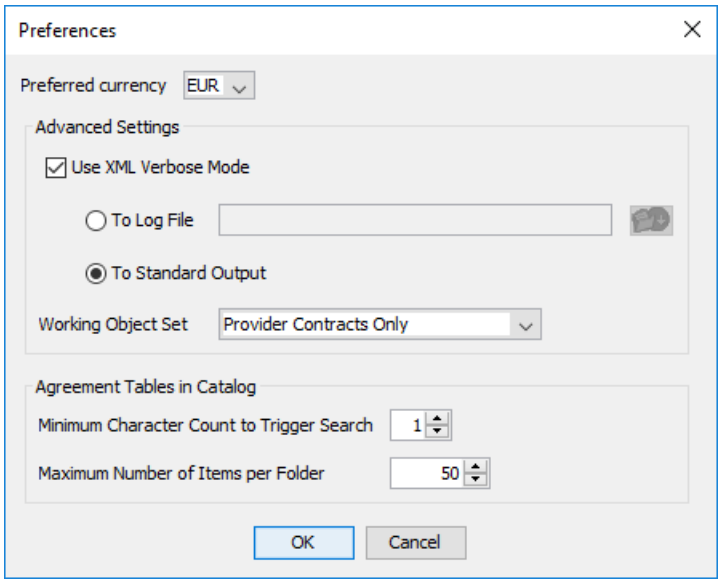
Tasks

- Setting Up the Preferences [page 56]

1.5.1.2.1 Setting Up the Preferences

You can change some options of your installed Core Tool user interface by using the *Preferences* window in the *Edit* menu.

> ⓘ Note
>
> Before changing an option, read the topic for the Edit menu [page 55].



Window: Preferences

The *Preferences* window allows you to change:

- The default currency displayed in any list of currencies of Core Tool
- The destination of the XML messages exchanged by your Core Tool user interface and the SAP CC system
- The subset of master data that SAP CC users can work with throughout Core Tool
- The settings for agreement tables search and display in the catalog view

Preferred Currency: Provides the list of currencies. Select the currency that you want to use as the default one in Core Tool.

Advanced Settings

- *Use XML Verbose Mode*: Select one of the following options to save the XML log messages
  - *To Log File*: Stores the exchanged XML messages to a log file. Type the folder path and the file name where the XML messages will be saved or click  to select a file.
  - *To Standard Output*: Sends the exchanged XML messages to the console of your application. For MS Windows operating system, the console is not visible by default.
- *Working Object Set*: Provides the list of master data subsets featured by SAP Convergent Charging. Select a master data subset that you want to work with in the SAP system landscape:
  - *Provider Contracts and Subscriptions*: You work with master data [page 28] dedicated to convergent charging services based both on either provider contracts or subscriptions. You use this mode for the user interface if your SAP CC system stores data for both data models (DMs) or if the charging process is only based on subscriptions.

    Your Core Tool user interface is able to manage all types of data.
    If you configure offers for the service pricing and subscriptions for the end customers of the service provider company, we recommend that you read our dedicated product documentation and user assistance. See SAP CC 2023 Primary Help for Core Tool (with Offers and Subscriptions) and select your software version.
  - *Provider Contracts Only*: You work with master data dedicated to convergent charging services based on charge plans and allowance plans (in service pricing catalogs) and provider contracts for end customers.

  As of release 2.0, SAP Convergent Charging supports tow data models (DMs).
  - New data model: reusable charges, charge plans, allowance plans (as of release 5.0), subscriber accounts, provider contracts with allowances
  - Former data model: reusable charges, offers, subscriber accounts, subscriptions

  Subscriber accounts are part of both data models.

  > ⓘ Note
  >
  > If you change the working object set, a message informs you that Core Tool must be restarted to apply the new configuration.

- *Agreement Tables in Catalog*: In an integrated *SAP Solution*, determines the triggering of a search and the display of agreement tables in the catalog view:
  - *Minimum Character Count to Trigger Search*: Choose the number of characters to type to trigger a search for agreement tables in the SAP CC system (Core Server) and its back-end database.

- *Maximum Number of Items per Folder*: Choose the number of subitems to be displayed in each item in the catalog view.

> → Recommendation
>
> For a matter of GUI memory performances, the default values are set to:
> - **1** for *Minimum Character Count to Trigger Search* (max. value is 99).
> - **26** for *Maximum Number of Items per Folder* (max. value is 999).
>
> It is preferable not to change these values.

> → Remember
>
> As the catalog view is not dynamically refreshed, you need to close and re-open it if you change these values to have a consistent number of displayed items.

1.5.1.3 Verify

The Core Tool user interface provides you with tools to verify your data objects. Depending on the currently displayed window, this *Verify* submenu command may be dimmed.

The verification operations are crucial in SAP CC. Core Tool verifies the entered information and provides you with some guidance.

It identifies and highlights issues in red. Correct the elements that are underlined in red.

> ⓘ Note
>
> Consult the primary help for more information about the possible verifications, object by object.

> → Tip
>
> Click the  *Verify* button on the main window toolbar [page 44].

1.5.2 Action

In the menu bar, the *Action* menu groups some usefull contextual functions.

Depending on the currently displayed window in the user interface, the *Action* menu groups some important operations for your activities.

1.5.3 Tools

In the menu bar, the *Tools* menu groups some usefull tools.

The SAP CC Core Tool user interface provides you with useful functions for:

- Business operations (administration)
  - Rerating [page 59] consumption related to provider contract, charge plans and so on
  - Refreshing [page 72] the rater instances of the connected SAP CC system in a test system landscape
- Data maintenance
  - Exporting [page 80] or importing [page 80] public holidays
  - Exporting [page 84], importing [page 84], or synchronizing [page 85] currencies
  - Editing [page 87] the counter dictionary
  - Working with bulk operations
- Auditing operation
  - Generating [page 97] a license auditing file

Tool access

From the menu bar in the main window, select *Tools* and the operation you need.

1.5.3.1 Rerating

> → Remember
>
> Rerating is no compliant with shared allowances [page 575]. When SAP Convergent Charging detects that a provider contract has created or used a shared allowance, the rerating process is rejected and an error message is displayed.

Business operation

During the production operation phase, as application administrator you may need to rerate data when exceptional mistakes or errors occur in price plans, translation tables, and other data objects, resulting in incorrect transactions stored in the third-party billing system. Core Tool allows rerating all service usage events occurred within a default period of 70 days[3] preceding the current date.

[3] The period may be different in your SAP CC system. Get the current value of the `COUNTER_SNAPSHOT_COUNT` system parameter with the *Admin+* user interface. Refer to the SAP CC Administration Guide for more information about the implementation of the rerating services in your system landscape.

> **Example**
>
> Pricing errors can lead to erroneous rating calculations. For example, this situation can arise from an incorrect price plan, incorrect content in translation tables, or incorrect use of translation tables or pricing macros. This results in incorrect transactions stored in the third-party billing system.

Prerequisites

The *Chargeable Items Recharging* feature must be enabled. Refer to the Application Help for further information about this feature.

Authorization

You must have the *Administrator* role in your SAP CC user [page 640].

Rerating process

You initialize (and perform) the rerating process according to the following variants and options:

In an integrated *SAP Solution* scenario:

- With SAP CC BART Server: The BART Server system of SAP CC performs this process to rerate subscriptions and provider contracts in batch. You trigger this process fromSAP CC Core Tool. BART Server works with SAP CC Core Server that works with SAP Convergent Invoicing in the SAP ERP/FI-CA system.
- With SAP Convergent Invoicing (in the SAP ERP/FI-CA system or SAP S/4HANA system) with consumption item management: SAP Convergent Invoicing in the SAP ERP system performs the rerating process. You can however trigger the rerating operations with the SAP CC Core Tool user interface or with SAP Convergent Invoicing. The rerating process only rerates provider contracts via the *Web Services* technical interface but does not rerate subscriptions.

In another scenario, SAP CC BART Server performs the rerating process and SAP CC Core Server must be interfaced with a third-party billing system. The implementation includes Java developments.

> **Note**
>
> During the installation, you must select the configuration file of the SAP CC Core Server system to choose one of these options. For more information, refer to the SAP CC Administration Guide.

Rerating process managed by BART Server

The rerating process consists of nine steps performed by:

- The Core Server system (step 1, 2, 3, 4, 6, 7, 9)

- The BART Server system (step 8)
- The external billing system (step 5)

The steps are the following:

1. Selecting the provider contract which must be rerated according to a search filter (for example, provider contract, translation table). If a parent contract or a linked contract [page 573] is selected, all the collection linked contracts and their parent contract are included in this selection.
2. Sending the selected provider contracts to the BART Server system.
3. Synchronizing the bulkloaders to make sure that the billable items are sent to the external billing system before the beginning of the rerating operation.
4. Checking that the state of the provider contracts can be restored from the counter snapshots. If the state of a provider contract cannot be restored, the provider contract cannot be rerated and is erroneous.
5. Checking that the billable items can be reversed at a specified date, or proposing another date after the specified date if the billable items have already been billed.
6. Restoring the selected provider contracts to the state they had at the time of the specified date.
7. Sending the request to the external billing system for reversing the billable items generated by the selected provider contracts during the rerating period.
8. Rating the usage events related to each provider contract.
9. Sending the rerated billable items to the external billing system.

When a translation table or a set of provider contracts is rerated, the related provider contracts are first restored, and then rerated by batch rating groups. The rerating process is completed when all the provider contracts have been successfully rerated.

> **Note**
>
> - At step 8, the provider contracts are grouped by batch rating groups.
> - The rerating process can only be stopped during step 6. Because the rerating process rates the batch rating groups of provider contracts, one after another, the rerating process stops after the current batch rating group is rerated.
> - The rerating process is only available if the SAP Convergent Charging solution is configured to work together with SAP Convergent Invoicing. This process can be adapted to third party billing systems which are different from SAP Convergent Invoicing, but this configuration requires specific developments. For further information, please refer to your SAP consultants.
> - For further information about the BART Server and the *BART Tool* components, refer to the dedicated topic.

Rerating process for SAP ERP/FI-CA

You can initialize the rerating process for SAP ERP/FI-CA by creating a rerating operation from the SAP CC Core Tool user interface. This operation enables SAP CC Core Server to send a list of provider contracts to SAP ERP/FI-CA for rerating purposes. When the operation is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server.

For more information about the rerating process, refer to the user assistance of SAP Convergent Invoicing (in the SAP ERP/FI-CA or SAP S/4HANA system. See Help Resources [page 40].

1.5.3.1.1 Working with Rerating Operations

Related Information

Rerating from Provider Contracts [page 62]
Rerating from a Charge Plan [page 63]
Rerating from a Translation Table [page 64]
Rerating from a Tier Table [page 65]
Rerating from a Pricing Macro [page 66]
Rerating from a Mapping Table [page 67]
Rerating from a Subscriber Mapping Table [page 68]
Rerating from a Range Table [page 69]
Rerating from a Subscriber Range Table [page 69]
Viewing the Current Rerating Session [page 70]

1.5.3.1.1.1 Rerating from Provider Contracts

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Provider Contracts* window, you do the following:

1. Set up the rerating date, and then the search filter to find contracts to rerate depending on your criteria.
2. Rerate the contracts found by the search filter.

From the menu bar in the main window, select *Tools* > *Rerating* > *From Provider Contracts*. The *Rerating from Provider Contracts* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[4] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, the rerating operation fails.

After you select a date, set up [page 609] the search filter for finding the contracts to be rerated.

[4] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the provider contracts are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of provider contracts sent and the total number of provider contracts to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

1.5.3.1.1.2 Rerating from a Charge Plan

> **Note**
>
> - This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide
> - Rerating is no compliant with a charge plan that uses shared allowances. If you try rerating this charge plan, the related dialog box informs you that the charge plan uses shared allowances, and you cannot start the rerating process because the *OK* button is not available. See the important note about rerating [page 59].

In the *Rerating from Charge Plan* window, you do the following:

1. Set up the rerating date, and then select a charge plan.
2. Rerate the contracts relating to the charge plan.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Charge Plan*. The *Rerating from Charge Plan* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[5] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Charge Plan: Click  to display the catalog of service providers. Select a service provider, and then select a charge plan.

[5] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the charge plan are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of charge plan sent and the total number of charge plan to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

1.5.3.1.1.3 Rerating from a Translation Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Translation Table* window, you do the following:

1. Set up the rerating date, and then select a translation table.
2. Rerate the contracts relating to the translation table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Translation Table*. The *Rerating from Translation Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[6] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Translation Table: Click to display the catalog of service providers. Select a service provider, and then select a translation table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the translation table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of translation table sent and the total number of translation table to send (for example, 3,000/5,000).

[6] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communications. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

1.5.3.1.1.4 Rerating from a Tier Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Tier Table* window, you do the following:

1. Set up the rerating date, and then select a tier table.
2. Rerate the contracts relating to the tier table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Tier Table*. The *Rerating from Tier Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[7] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Tier Table: Click  to display the catalog of service providers. Select a service provider, and then select a tier table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the tier table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of tier table sent and the total number of tier table to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communications. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

[7] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

1.5.3.1.1.5 Rerating from a Pricing Macro

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Pricing Macro* window, you do the following:

1. Set up the rerating date, and then select a pricing macro.
2. Rerate the contracts relating to the pricing macro.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Pricing Macro*. The *Rerating from Pricing Macro* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[8] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Pricing Macro: Click to display the catalog of service providers. Select a service provider, and then select a pricing macro.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the pricing macro are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of pricing macro sent and the total number of pricing macro to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

[8] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

1.5.3.1.1.6 Rerating from a Mapping Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Mapping Table* window, you do the following:

1. Set up the rerating date, and then select a mapping table.
2. Rerate the contracts relating to the mapping table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Mapping Table*. The *Rerating from Mapping Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[9] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Mapping Table: Click  to display the catalog of service providers. Select a service provider, and then select a mapping table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the mapping table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of mapping table sent and the total number of mapping table to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

[9] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

1.5.3.1.1.7 Rerating from a Subscriber Mapping Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Subscriber Mapping Table* window, you do the following:

1. Set up the rerating date, and then select a subscriber mapping table.
2. Rerate the contracts relating to the subscriber mapping table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Subscriber Mapping Table*. The *Rerating from Subscriber Mapping Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[10] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Subscriber Mapping Table: Click to display the catalog of service providers. Select a service provider, and then select a subscriber mapping table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the subscriber mapping table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of subscriber mapping table sent and the total number of subscriber mapping table to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with SAP Convergent Invoicing. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

[10] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

1.5.3.1.1.8 Rerating from a Range Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

In the *Rerating from Range Table* window, you do the following:

1. Set up the rerating date, and then select a range table.
2. Rerate the contracts relating to the range table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Range Table*. The *Rerating from Range Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[11] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Range Table: Click to display the catalog of service providers. Select a service provider, and then select a range table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the range table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of range table sent and the total number of range table to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communication with the server. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

1.5.3.1.1.9 Rerating from a Subscriber Range Table

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

[11] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

In the *Rerating from Subscriber Range Table* window, you do the following:

1. Set up the rerating date, and then select a subscriber range table.
2. Rerate the contracts relating to the subscriber range table.

From the menu bar in the main window, select *Tools* > *Rerating* > *From a Subscriber Range Table*. The *Rerating from Subscriber Range Table* window opens. You perform the operation in this window.

Rerate From: Click , and then select a date within the period of 40 days[12] preceding the current date. Before being rerated, all the contracts found by the search filter are restored to the state they had at the selected date.

Adapt to External System Date: Select this option if you want to rerate a contract from the rerating date provided by the external billing system. In a set of contracts to rerate, all or part of billable items of any contract may have already been invoiced. For each of these contracts, the external billing system therefore provides a new date from which the contract can be rerated. If you do not select this option and if the billing system cannot rerate a contract from the date that you have selected, then the rerating operation fails.

Subscriber Range Table: Click to display the catalog of service providers. Select a service provider, and then select a subscriber range table.

Buttons

OK: Click to send the selected contracts to SAP CC Core Server. The *Input* dialog box opens. Type the reason of the rerating operation, and then click *OK*. The *Operation Progress* dialog box opens to inform you that the subscriber range table are being sent to SAP Convergent Invoicing. A progress bar displays the instantaneous number of subscriber range table sent and the total number of subscriber range table to send (for example, 3,000/5,000). When the session is successfully completed, a message displays the rerating session identifier; otherwise, a message warns you that an error has occurred during the communications. Click *Close* to close the dialog box.

Cancel: Click to cancel all your changes, and then close the window.

1.5.3.1.1.10 Viewing the Current Rerating Session

> **Note**
>
> This menu is available only if you change the configuration of the SAP CC Core Server system. For more information, refer to the SAP CC Administration Guide

You can view the current rerating session. If no rerating session is in progress, a message displays that no rerating session is running. If a rerating session is in progress, the *Bulk Rerating* dialog box opens (see the process below if needed).

[12] The default period can be changed. For more information, refer to the SAP CC Administration Guide.

Bulk rerating

The *Bulk Rerating* dialog box displays information related to the rerating process in progress. This information is displayed in two different areas:

- The *Rerating Progress* area used to display information related to the different steps of the rerating process
- The *Subscription(s)/Contracts(s) Bulk Rerating Report* area used to display a global report containing details related to the rerating process

Rerating Progress

This area displays the following information:

- The lock code of the rerating process
- The different steps of the rerating process and their progress which display:
  - The number of charging contracts sent to the BART Server for rerating
  - The number of files of the bulk loader that are sent to the billing system
  - The number of charging contracts successfully locked
  - The number of days related to rerated CDRs (Consumption Detail Record)
  - The number of charging contracts successfully unlocked
- A progress bar which shows how the rerating process evolves over time

Subscription(s)/Contracts(s) Bulk Rerating Report

To display the global report of the rerating process, click the  button. The global report displays the status, the number, and the percentage of contracts relating to the rerating process. When the rating process is in progress, click  to refresh the list that displays the statuses of contracts and the progress rate.

The list of statuses is the following:

- *All*: No filtering
- *Restored*: Contracts for which counters were restored from a snapshot and which are ready to be rerated
- *Rerated*: Contracts which were successfully rerated
- *Adapted*: Contracts for which the billing system has suggested a date to reverse the charged items (this state is displayed if the user has chosen the *Adapt to External System Date* option)
- *Can be Adapted*: Contracts for which the billing system has suggested a date to reverse the charged items (this state is displayed even if the user has not chosen the *Adapt to External System Date* option)
- *Snapshot failure*: Contracts for which counters could not be restored from a snapshot
- *Billing failure*: Contracts for which an error occurred in the billing system during the rerating process
- *Unlock failure*: Contracts which could not be unlocked
- *Prepaid Account Failure*: Contracts for which the rerating process has been triggered by a provider contract linked to a prepaid account (which is not expected to trigger the rerating process)

Buttons

Cancel: Click to cancel the rerating process. Before charging contracts are sent to the BART Server, you can cancel the rerating process. When charging contracts are being sent to the BART Server, you cannot cancel the rerating process, but you can only stop it. If you click the Stop button, the rerating process ends only when all the charging contracts of the current batch rating group are rerated. Therefore, the charging contracts of the next batch rating groups are not rerated.

Stop: Refer to the Cancel button.

Close: Click to close the dialog box. You can reopen the dialog box by using the menu *Tools* > *Rerating* > *View Current Rating Session Status*.

From the menu bar in the main window, select *Tools* > *Rerating* > *View Current Rating Session Status*.

1.5.3.2 Refresh Rating Instances

By default, when you modify or create a data object in the pricing catalog [page 109] such as a charge plan, a reusable charge, or a pricing macro, a few minutes are needed to update the convergent charging services in the connected SAP CC system. An internal system scheduler ("Refresh Scheduler") or an external scheduler manages these data distribution and synchronization operations inside the SAP system between the different types of instances.

If you need to refresh your SAP system quickly when working in a development or test system landscape, use this command *Refresh Rating Instances* to force the data refreshment in the following instances:

- Rater instances
- Bulkloader instances

Once done, the rater and bulkloader instances are synchronized again and ready to process the requested charging operations.

> → Tip
>
> This command is also available in the SAP CC Admin+ user interface for administrators.

Manually Forcing the Refresh for the Rating Instances [page 73]

Related Information

Cached Structures Refreshing

1.5.3.2.1 Manually Forcing the Refresh for the Rating Instances

Refresh the Rater/Bulkloader Instances of the SAP CC System

1. In Core Tool, from the menu bar in the main window, select *Tools* > *Refresh the Rating instances*. The *Question* dialog box is displayed to ask if you want to refresh all the rating instances[13] in the connected SAP CC system.
2. Click *Yes*. A confirmation message is displayed to inform you that the rater instances (and the bulkloader instances) have been refreshed in the SAP CC system.
3. Click *OK*.

1.5.3.3 Public Holidays

SAP Convergent Charging provides you with function for setting up public holidays known by the SAP CC system. Public holidays information is used by the SAP CC system when processing the pricing logic configured in the price plan [page 299], the charging plan, [page 305] or the refill logic [page 395] during the charging and refilling processes.

> **Note**
>
> Public holidays information is part of the business data [page 32] configured in the complete SAP CC system. It is not included in the pricing catalog [page 109] of a service provider.

Transport

You can transport this business data between SAP CC systems:

- Export [page 80] public holidays
- Import [page 80] public holidays

1.5.3.3.1 Setting Up Some Public Holidays

During the project implementation phase or the production operation phase, you maintain the public holidays defined in the SAP CC systems.

[13] When SAP Convergent Charging (SAP CC) is connected to SAP Convergent Invoicing (in the SAP ERP/FI-CA or SAP S/4HANA system), the refresh operation forces also the update of the bulkloader instances in the SAP CC system.

To set up the public holidays:

- Export [page 80] the public holidays from the system to an XML file.
- Edit the exported XML file and change it. For details and examples on the XML syntax to be used, see XML File Format for the Public Holidays [page 74].
- Import [page 80] the public holidays.

1.5.3.3.1.1 XML File Format for the Public Holidays

XML Syntax for the public holidays

Root XML Element

The main XML element is `<publicHolidays>`. One and only one `<publicHolidays>` element is allowed in the XML file.

Public Holidays

Use the `<holiday>` XML element to define a public holiday.

A public holiday has:

- An identification code that is used to identify the holiday. It must be unique among the codes of public holidays.
- A description, possibly given in many different languages.
- One or several groups if public holidays are grouped, none otherwise. See below for details on how to set up the groups.
- A type selected among the four types described in the table below.

Type	Description	Examples	XML Samples

`<fixedDay>`	A holiday with a date which does not vary over years.
	In the XML file, you specify the month and the day corresponding to the date of the holiday.
		New Year's Day which always occurs on January 1.	`<holiday code="NewYearsDay">`
			`<description lang="en" label="New Year's Day" />`
			`<description lang="fr" label="Nouvel An" />`
			`<group code="US" />`
			`<group code="FR" />`
			`<group code="DE" />`
			`<group code="HR" />`
			`<group code="BE" />`
			`<group code="LU" />`
			`<group code="CH" />`
			`<fixedDay month="January" day="1" />`
			`</holiday>`

`<relativeDay>`	A holiday that always occurs the same weekday of a given month.
	In the XML file, you specify the weekday, the month of the holiday and the position of the weekday in this month.
		Thanksgiving which occurs the fourth Thursday of November in the United States.	`<holiday code="Thanksgiving">`
			`<description lang="en" label="Thanksgiving Day" />`
			`<description lang="fr" label="Thanksgiving" />`
			`<group code="US" />`
			`<relativeDay month="November" weekday="Thursday" position="fourth" />`
			`</holiday>`

Type	Description	Examples	XML Samples
<easterRelative>	A holiday related to Easter (for the Catholic Church). In the XML file, you specify an offset defining when the holiday occurs relatively to Easter Sunday which is used as a reference date.	Easter Sunday for the Catholic Church (offset to zero)	<holiday code="EasterSunday"> <description lang="en" label="Easter Sunday" /> <description lang="fr" label="Dimanche de Pâques" /> <group code="FR" /> <group code="HR" /> <easterRelativeDay offset="0" /> </holiday>
		Easter Monday which occurs one day after Easter Sunday (offset to one)	<holiday code="EasterMonday"> <description lang="en" label="Easter Monday" /> <description lang="fr" label="Lundi de Pâques" /> <group code="FR" /> <group code="DE" /> <group code="HR" /> <group code="BE" /> <group code="LU" /> <easterRelativeDay offset="1" /> </holiday>

Type	Description	Examples	XML Samples
		Good Friday which occurs two days before Easter Sunday (negative offset to -2)	`<holiday code="GoodFriday" > <description lang="en" label="Good Friday" /> <description lang="fr" label="Vendredi Saint" /> <group code="DE" /> <easterRelativeDay offset="-2" /> </holiday>`

Type	Description	Examples	XML Samples
<enumeratedDay>	A holiday occurring at a different date each year, this date being known in advance. In the XML file, you specify the list of dates when the holiday is planned.	Chinese New Year's Day	<holiday code="ChineseNewYear"> <description lang="en" label="Chinese New Year" /> <description lang="fr" label="Nouvel An Chinois" /> <enumeratedDay> <date value="1999-02-16" /> <date value="2000-01-05" /> <date value="2001-01-24" /> <date value="2002-01-12" /> <date value="2003-02-01" /> <date value="2004-01-22" /> <date value="2005-02-09" /> <date value="2006-01-29" /> <date value="2007-02-18" /> <date value="2008-02-07" /> <date value="2009-01-26" /> <date value="2010-02-14" /> <date value="2011-02-03" /> <date value="2012-01-23" /> <date value="2013-02-10" /> <date value="2014-01-31" /> <date value="2015-02-19" />

Type	Description	Examples	XML Samples
			<date value="2016-02-08" /> <date value="2017-01-28" /> <date value="2018-02-16" /> <date value="2019-02-05" /> <date value="2020-01-25" /> </enumeratedDay> </holiday>

Groups

You can define groups for public holidays with the `<group>` XML element.

The groups are used when you configure your price plans with the *Public Holidays Comparator* component and the *Public Holidays Splitter* component. They enable you to select in one operation all the public holidays belonging to the same group. See the primary help of logic components [page 671] for more information about the components.

A group has:

- A code, which is used for identifying the group. The code must be unique among the codes of all the groups.
- A description, possibly given in many different languages.

Example

The following XML sample enables to define a group for Belgium.

```
<groupDescription code="BE">
<description lang="en" label="Belgium" />
<description lang="fr" label="Belgique" />
</groupDescription>
```

The following XML sample shows how the group is used for Easter Monday.

```
<holiday code="EasterMonday">
<description lang="en" label="Easter Monday" />
<description lang="fr" label="Lundi de P&#xE2;ques" />
<group code="FR" />
<group code="DE" />
<group code="HR" />
<group code="BE" />
<group code="LU" />
<easterRelativeDay offset="1" />
</holiday>
```

1.5.3.3.2 Exporting Public Holidays

You can export to an XML file the list of public holidays used in the logic components *Public Holidays Splitter* and *Public Holidays Comparison*. For more information on using these components, click here [page 671].

You use this operation to set up [page 73] or transport public holidays.

To export a list of public holidays

1. From the menu bar in the main window, select *Tools* > *Public Holidays* > *Export*. The *Save* dialog box opens.
2. Select a directory to which you export the list of public holidays.
3. In the *File name* field, type the name of the file.
4. Click *Save*.

1.5.3.3.3 Importing Public Holidays

You can import a list of public holidays saved in an XML file. Once imported, the list of public holidays deletes and replaces[14] the one currently used in the components *Public Holidays Splitter* and *Public Holidays Comparison*. For more information on using these components, click Working with Logic Components [page 671].

You use this operation to set up [page 73] or transport public holidays.

To import a list of public holidays

1. From the menu bar in the main window, select *Tools* > *Public Holidays* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the public holidays file is saved, and then select the file.
3. Click *Open*. A confirmation dialog box opens.
4. Click *Yes* to import the list of public holidays.

[14] SAP SE strongly recommends that you consult your system administrator before importing a new list of public holidays.

Be careful when importing a new list of public holidays. After you import a new list, the old public holidays are replaced by the new ones with the exception of those already used in the charges set up before importing the new list. Besides, the old public holidays already used appear dimmed (unavailable) in the new list because not only they are no longer recognized by the SAP CC Core Server system but they are no longer active in the corresponding charges.

1.5.3.4 Currencies

SAP Convergent Charging assigns a currency to each computed amount.

You must configure currency information in the master data managed by SAP CC and related to the customer service pricing (see pricing catalog) and related to the end customers (see subscriber accounts, provider contracts) of the service provider.

Currencies are part of the business data configured in the complete SAP CC system. They are not included in the pricing catalog of a service provider.

SAP CC handles proprietary definitions for:

- Currencies (used to set up a reusable charge [page 311] for example)
- ISO 4217 Currency Codes [page 83]

> **Note**
>
> If SAP Convergent Charging is connected to SAP ERP/FI-CA in an integrated SAP solution, the currencies are defined in the SAP ERP/FI-CA reference system and replicated to the SAP CC system in background. You can only configure the ISO 4217 currency codes known by SAP CC.

Currency

A currency is identified by a unique code [15] and defined by:

- The precision which represents the number of decimal places
- A legible name
- A code which can be:
  - An ISO 4217 currency code defined by the International Standards Organization
  - A proprietary code that is used to group some non-ISO currencies like "PNT" (Point) does not correspond to standard currencies

> **Note**
>
> Relationship between major and minor currency units.
>
> The minor currency unit has usually a value that is 1/100 of the major unit and sometimes 1/1000. For example, the Japanese sen is 1/100th of a yen. However, some currencies have no minor currency unit.

> **Example**
>
> **Currencies defined in SAP CC**
>
> In the following examples, even if their precision differs, the currencies are equivalent because they have the same ISO code:

[15] You use currencies only based on ISO 4217 currency codes to configure the master data if SAP CC Diameter Server is part of the landscape as mediation system: Only numerical ISO 4217 three-digit codes are valid.

- The currency named “EUR”, based on the "EUR (978)" ISO code and is defined with two decimal precision
- The currency named “USD” is based on the "USD (840)" ISO code and is defined with two decimal precision
- Note the currency named “PNT”, based on a virtual "PNT" code and is defined with no decimal precision

Example

Currencies defined in SAP ERP/FI-CA and replicated in SAP CC

In the following examples, even if their precision differs, the currencies are equivalent because they have the same ISO code:

- The currency named “EUR” is based on the "EUR" ISO code and is defined with two decimal precision
- The currency named “EUR6”, based on the "EUR" ISO code and is defined with six decimal precision
- The currency named “USDN” is based on the "USD" ISO code and is defined with five decimal precision
- The currency named “USD6” ("CC-CI BIT amount USD (6 Dec.)"), based on the "USD" ISO code and is defined with six decimal precision

Unique code in SAP CC

To read and understand a currency easily in the SAP CC Core Tool, it is highly recommended to use only the ISO 4217 alphabetic codes when configuring a new currency because mixing alphabetic and numeric codes could lead to incompatibility problems.

Precision of low-level computations

To maximize their precision, the SAP CC system performs low-level amount computations with 12 or 32 decimal places (depending on the database precision). When stored in the business objects, the computed amounts are rounded according to the value of the TRANSACTION_ROUNDING_MODE system parameter and truncated according to the value of the TRANSACTION_PRECISION system parameter. The following rules are applied:

- If the TRANSACTION_PRECISION parameter is set to “-2”, the precision of the stored amounts corresponds to the precision of the database.
- If the TRANSACTION_PRECISION parameter is set to “-1”, the precision of the stored amounts corresponds to the precision of their currency.
- If the TRANSACTION_PRECISION parameter is set to a number between 0 and 6, the precision of the stored amounts corresponds to the value of this parameter. The amounts can then have 0 or up to 6 decimal places.

Example

If the TRANSACTION_ROUNDING_MODE parameter is set to NEAREST, the results are the following:

Computed amount	Stored amount TRANSACTION_PRECISION = 6	Stored amount TRANSACTION_PRECISION = -1
EUR 1.00	EUR 1.00	EUR 1.00

Computed amount	Stored amount TRANSACTION_PRECISION = 6	Stored amount TRANSACTION_PRECISION = -1
EUR 1.234567	EUR 1.234567	EUR 1.24
EUR6 1.00	EUR6 1.00	EUR6 1.000000
EUR3 1.234567	EUR3 1.234567	EUR3 1.235

For further information about these parameters, refer to the SAP CC 2023 System Parameter Reference and to the SAP CC 2023 Administration and Operation Guide.

Improve the currency process performance

The defined SAP currencies are stored in the back-end database of SAP Convergent Charging. To improve performance, the different client applications and server instances of the SAP CC system (Core Server) handle a cached structure named *"network currency book"* containing the defined SAP currencies. Through a DAL (Data Access Layer) dedicated to currencies, the active updater instance performs any change to the list of currencies. For refreshing data, the primary dispatcher instance is informed about any change to update its own currency book which is then forwarded to the other clients and server instances.

Transport

You can export [page 84] and import [page 84] the definitions of currencies and ISO 4217 currency codes.

If your SAP CC system is connected to SAP ERP/FI-CA system, you must manually synchronize currencies [page 85].

Maintenance

You must maintain this business data defined for an SAP CC system by exporting [page 84] and importing [page 84] this business data.

If SAP CC is connected to SAP ERP/FI-CA, you must manually synchronize currencies [page 85].

1.5.3.4.1 ISO 4217 Currency Codes

Every ISO 4217 currency must be configured and maintained in SAP CC. It is identified by a three-digit alphabetic code and defined by a three-digit numeric code, a minor currency unit, and a ratio.

1.5.3.4.2 Exporting Currencies

You can export in an XML file the list of currencies and ISO 4217 Currency Codes [page 83] stored into the back-end database system (SAP CC Core Database).

To export a list of currencies and a list of ISO codes

1. From the menu bar in the main window, select *Tools* > *Currencies* > *Export*. The *Save* dialog box opens.
2. Select a directory to which you want to export the lists of currencies and ISO 4217 currency codes.
3. In the *File name* field, type the name of the file.
4. Click *Save*.

1.5.3.4.3 Importing Currencies

You can import a list of currencies and a list of definitions of ISO currency codes saved in an XML file. Note that importing currencies:

- Is only available for users which are assigned the marketing role
- Can modify the existing currencies or add new ones but cannot remove any existing currency
- Modifies currencies stored in the back-end SAP CC Core Database system
- Updates the currencies known by all the instances of SAP CC Core Server
- Updates the connected SAP CC Diameter Server systems
- Updates the connected SAP CC BART Server systems
- Updates all the Java client application based on the SAP CC Java Libraries and that is connected to the SAP CC system (A mediation system or a provisioning system)

The other running Core Tool user interfaces are not refreshed.

> ⚠ Caution
>
> Currencies are used in the whole application and are shared by all the SAP CC users.
>
> Because importing currencies can modify existing currencies and change the computation of all the amounts, we strongly recommend that you consult your administrator before performing this operation.

> ⓘ Note
>
> - If SAP Convergent Charging is connected to the SAP ERP/FI-CA system which defines its own currencies, only the ISO 4217 currency codes are imported from the file. The list of currencies in the file are not taken into account when you import currencies.
> - If you want to maintain the currencies known by SAP CC, perform your changes in the SAP ERP/FI-CA reference system and synchronize both systems [page 85].

To import a list of currencies and a list of ISO codes

1. From the menu bar in the main window, select *Tools* > *Currencies* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the XML file containing the currencies are saved, and then select this file.
3. Click *Open*. A confirmation dialog box opens.
4. Click *Yes* to import the lists of currencies and ISO 4217 currency codes.

1.5.3.4.4 Synchronizing Currencies

You can synchronize the list of currencies defined in the Core Database of SAP Convergent Charging with a list of currencies defined in the SAP ERP/FI-CA reference system. Note that synchronizing currencies:

- Is only available for SAP CC users which are assigned the marketing role
- Is only available when SAP Convergent Charging is connected to the SAP ERP/FI-CA reference system
- Only takes into account the definitions of the currencies (the definitions of the ISO 4217 are not modified)
- Can add new currencies and modify or remove the existing currencies
- Modifies currencies stored both in the Core Database of SAP Convergent Charging and in the data caches used by the different running server instances

> ⚠ Caution
>
> Currencies are used in the whole application and shared by all the SAP CC users. Because synchronizing currencies can modify existing currencies, we strongly recommend that you consult your administrator before performing this operation. If a currency is removed from the list of currencies:
>
> - This currency is unknown to SAP Convergent Charging, and a message is displayed every time a business object tries to use this currency
> - You cannot use this currency when creating new business objects (master data)
> - The charging and refilling operations based on business objects which use this currency are not possible and fail

Tasks

You can synchronize currencies when:

- The currencies have been changed in the SAP ERP/FI-CA reference system
- The SAP CC user cannot see the appropriate currencies in the list when configuring master data in the Core Tool user interface

Prerequisite

The SAP CC system is connected to the SAP ERP/FI-CA system.

Procedure

To synchronize a list of currencies

1. From the menu bar in the main window, select *Tools* > *Currencies* > *Synchronize*.
   - When the synchronization of currencies is successful, an information dialog opens and displays the number of synchronized currencies
   - When the synchronization of currencies fails, an information dialog opens and displays the reason of failure
2. Export [page 84] the lists.
3. Verify that the list of ISO 4217 currency codes is up-to-date and complete regarding to the list of currencies. Add necessary ISO 4217 currency codes and import [page 84] the file.

1.5.3.5 Counter Name Dictionary

The application administrator configures and maintains this data in the system for all pricing catalogs.

The *counter name dictionary* defines a **set of counter names** to be used in charge plans [page 343] and refill plans [page 421] when declaring counters [page 353].

Each counter name must be unique in the dictionary. Each entry of the dictionary must describe the counter that can be declared in a plan.

This dictionary is part of the business data [page 32] configured in the complete SAP CC system. It is not included in the pricing catalog [page 109] of a service provider. It is shared by all pricing catalogs.

> ⓘ Note
>
> - In charge plans, the names defined in the counter name dictionary can be used only for counters whose visibility is set to external.
> - In monitoring plans [page 491], you use this dictionary to define which counter of contracts can be monitored and reported [page 507] to a policy or monitoring system.

When one or two charge plans and a refill plan use the same counter name, you can define this counter as shareable in the provider contract. Refer to the procedures to share counters with several contract items [page 596] or with multiple provider contracts [page 507].

> ⓘ Note

1.5.3.5.1 Editing the Counter Name Dictionary

Edit the Counter Name Dictionary

From the menu bar in the main window, select *Tools* > *Counter Dictionary* > *Edit*. The *Counter Dictionary* window opens. You are now able to manage the entries in the counter dictionary.

As an application administrator, you manage entries in the counter name dictionary that is defined for the complete system:

Use the following icon...	To...
	Add a new counter name to the counter name dictionary.
	Delete an entry from the name dictionary.
	Clear all the dictionary.

1.5.3.6 Customer Management Areas

> ⓘ Note
>
> This chapter is relevant only if you use SAP Convergent Charging with SAP Convergent Invoicing in SAP ERP/FI-CA in an integrated *SAP Solution* scenario. The system landscape includes one or more SAP ERP/FI-CA systems and one or more SAP CRM systems. For more information, refer to the SAP CC Administration Guide

SAP Convergent Charging defines logical system pairs that include an SAP ERP/FI-CA system and an SAP CRM system. A pair identifies the SAP systems connected to SAP CC that are responsible for the master data management and that are involved in the following business processes:

- Transactional data about service usage (billable items, consumption items)
- Customer master data provisioning and maintenance (subscriber accounts [page 511]), which is the business partner management function in an SAP Solution scenario
- ABM notifications about prepaid accounts [page 514] (balance amount alerts, expiration alerts), which is the business agreement management function in the SAP Solution

A customer management area defines:

- A unique area name in SAP CC whose name includes the SAP system ID (SID) of the SAP ERP/FI-CA system in the area in combination with the SAP Client number.
- A JCo or WS destination of the SAP ERP/FI-CA system that belongs to this area that manages master data in SAP CC

- A JCo destination of the SAP CRM system that belongs to this area and that receives ABM notifications

> **Note**
>
> The Core Tool user interface verifies that each area includes at least the information related to an SAP ERP system. When your system landscape includes one or more SAP CRM systems, each customer management area must include the necessary information relating to an SAP CRM system.

In an integrated *SAP Solution* scenario, each area defines the SAP ERP system that is responsible for creating and maintaining the subscriber accounts in the master data stored in the SAP CC system as part of the business partner management function in the system landscape. This information is stored in the subscriber accounts by each ERP system and is visible in the Definition [page 530] tab in Core Tool.

> **Note**
>
> - These settings are part of the business data [page 32] configured in the complete SAP CC system. It is not included in the pricing catalog [page 109] of a service provider.
> - The application administrator is responsible for these settings.

Default Area

One of the customer management areas is the default area. The SAP CC system considers this area when performing the business processes that relate to a subscriber account without any customer management area.

Authorization

You must have the *Administrator* role in your SAP CC user [page 640] to manage customer management areas.

Configuration

Setting up a customer management area.

Transport

You cannot transport this business data [page 32].

Maintenance

You must manually maintain this configuration. For example, you can add a new area, change an existing area, or delete an unused area. Once configured, the list of customer management areas cannot be emptied.

Related Information

Subscriber Account: Definition [page 530]

1.5.3.6.1 Opening the Customer Management Areas

Open the Customer Management Areas

- From the menu bar in the main window, select *Tools* > *Customer Management Areas* > *Edit*. The *Customer Management Areas* window opens.

View or manage the areas that are configured in SAP CC:

- Editing the Customer Management Areas [page 89]
- Deleting a Customer Management Area [page 91]

1.5.3.6.2 Editing the Customer Management Areas

As a system or application administrator, open [page 89] the customer management areas before editing them. The *Customer Management Areas* window must be open.

> → Remember
>
> The configuration and the customizing of a new customer management area requires to stop and restart some particular instances in the SAP CC Core Server system. Before editing the areas, refer to the SAP CC Administration Guide for more information about:
>
> - Enabling an SAP ERP/FI-CA in a new customer management area
> - Enabling an SAP CRM in a new customer management area

Customer Management Areas (Window)

Use the configuration table and its icons to create, change, and delete customer management areas. This procedure is partial and does not consider the necessary production operations with the SAP CC system to stop and restart the adequate instances of the system.

 : Click to add the definition of a customer management area to the table.

 : Click to remove an area definition from the table. See Deleting a Customer Management Area [page 91]. It is not possible to remove a row that is the default customer management area. First define another area than the default customer management area.

- *Name*: Type the name of the customer management area.
  SAP SE recommends that you use a syntax based on the SAP system ID (SID[16] or SAPID) of the targeted system in combination with the SAP Client number, separated by a underscore character. Do not use spaces. For example: ABC_123 or XYZ_191
- *ERP Destination*: Provides the list of JCo or WS destinations configured in the SAP CC system and that are available for the communications with the SAP S/4HANA system, SAP ERP system or, SAP S/4HANA Cloud Public Edition cloud system in your SAP system landscape. Select the appropriate JCo or WS destination.
- *CRM Destination*: Provides the list of JCo destinations configured in the SAP CC system and that are available for an SAP CRM application. Select the appropriate JCo or WS destination.

  > ### Note
  >
  > This column is not available in case of an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition.

- *Default Area*: Click to set up the corresponding area as the default customer management area.

> ### Note
>
> The Core Tool user interface verifies that each area includes at least the information related to an SAP ERP system.
>
> When your SAP system landscape includes one or more SAP CRM systems, each customer management area must include the necessary information relating to an SAP CRM system.

Next

Restart the dispatcher and the bulkloader instances in the Core Server system.

> ### Note
>
> Check if you must restart the rater instances in your Core Server system.
>
> If the system parameter JCO_CONNECTION_COUNT is set to 1 or more, then do restart all these rater instances.

Related Information

Creating JCo Destinations

[16] SAP System Identifier

1.5.3.6.3 Deleting a Customer Management Area

As a system or application administrator, you delete a customer management area that has never been used or you delete an obsolete area. In this second case, it means that an SAP ERP/FI-CA system is no more responsible for a subset of subscriber accounts [page 511] in customer master data [page 30] stored in SAP CC.

> **Note**
>
> You cannot delete a customer management area that is:
>
> - Referenced by a subscriber account [page 530]
> - The default customer management area in the SAP system landscape

Delete a Customer Management Area

- Search [page 537] and verify that no subscriber account refers to the customer management area you want to delete.
- Verify that there is no remaining data file in the output data folders. In each Working Directory of the SAP CC system, consider the subfolders that relate to the JCo destination associated to the obsolete customer management area. Use theAdmin+ user interface to get the value of the `POSTPAID_CIT_READER_ROOT_PATH` parameter.

> **Note**
>
> In a development or test system landscape, you can use the Message Client user interface with the `bl` command and the name of the obsolete customer management area.

- Verify that there is no remaining error file. Consult the SAP CC Administration Guide for more information.
- Open [page 89] the *Customer Management Areas* window.
- Edit [page 89] the areas and remove the relevant row in the configuration table.

1.5.3.7 Allowance Interface

In SAP Convergent Charging 2023, the *allowance interface* is a master data object listing and describing the common *parameters* and *counters* of all the allowances [page 575] that can be created in provider contracts.

The allowance interface allows you to **extend the default set of properties** for allowances in provider contracts:

- The unique identifier
- The activation date/time (aka Validity Start Date)
- The expiration date/time (aka Validity End Date)

Example

For example, you set up some parameters or counters in the allowance interface for:

- Sorting allowances in the *Allowance Event Sender*[17] component
- Computing properties in the *Allowance Property Introducer* component

When you add a counter or a parameter to the allowance interface, the SAP CC system adds it to the list of properties of each allowance, and you can view it in your Core Tool user interface in the Counters [page 470] or Parameters [page 471] tabs of any allowance plan whose creation date/time is later than that of the counter or parameter.

However, you cannot view this new counter or parameter in an allowance plan created before the new counter or parameter.

In SAP CC Core Tool:

- There is only one allowance interface that is shared by all pricing catalogs.
- The allowance interface is used to sort allowances in the *Allowance Event Sender* component.
- The allowance interface is used to compute properties in the *Allowance Property Introducer* component.

Note

- You cannot import or export the allowance interface.
- You cannot delete the allowance interface.
- The modified allowance interface replaces the previous version in the back-end database, but you cannot access the previous versions of the allowance interface that are saved in the database.
- The SAP CC user can redefine the default values of counters and parameters in the allowance interface. When the allowances are created, the default values of counters and parameters of the allowance interface can be redefined in the allowances if necessary.
- A parameter or a counter cannot be added to the allowance interface if one or more allowance plans already have a parameter or a counter with the same name.
- The persistent counters that are implemented by shared allowances [page 575] are not visible in the allowance interface.

Before Starting

Before setting up [page 96] the allowance interface, you must be familiar with the following:

- Basic Charging Concepts [page 18]
- Configuration of Master Data Step by Step [page 104]
- Provider Contracts [page 573]
- Allowances [page 575]
- Allowance Event Classes [page 441]
- Allowance Logic [page 449]
- Allowance Plans [page 466]

[17] Refer to the primary help relating to components [page 671]

Tasks

- Working with the Allowance Interface [page 95]
- Customizing the Allowance Interface for SAP CRM [page 94]

Related Information

Allowances [page 575]
Allowance Event Sender (Operator) [page 777]
Allowance Property Introducer (Operator) [page 784]

1.5.3.7.1 Example of an Allowance Interface

Example

An operator company offers services managed according to three classes of allowances:

- The "gold" allowances with unlimited authorization for using a service
- The "gift" allowances created by the end customer services
- The "normal" allowances bought by an end customer

The system must consume the services with the following constraints:

- Authorize the service if an end customer has a "gold" allowance for the service to consume.
- Consume the "gift" allowances before the "normal" allowances.
- First use the allowances with the lower remaining quantity of service ("gift" or "normal").

As an SAP CC pricing specialist, you translate this marketing and pricing policy by modeling and designing the management of allowances in SAP CC.

To use the allowances with the *Allowance Event Sender* component, the allowances need to provide:

- The type of the allowance (gold, gift, and normal) whose values are 2 for gold, 1 for gift, and 0 for normal.
- The remaining quantity of service.

As a pricing specialist for this this operator, you configure the allowance interface [page 91] to provide the type and the quantity of service. You set up [page 96] a parameter and a counter delcaration:

- A number parameter, named **`TYPE`**, representing the type of the allowance with the default value of 0.
- A counter, named **`REMAINING_QUANTITY`**, representing the remaining quantity with the default value of 0.

See also: Allowance Event Sender (Operator) [page 777]

1.5.3.7.2 Customizing the Allowance Interface for SAP CRM

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

To be displayed by the SAP CRM system, the allowances [page 575] and their allowance plans [page 466] must have a predefined parameter of string type and named `ALLOW_DEF_ID`. As a main parameter of the allowances, `ALLOW_DEF_ID` must be initialized when the allowance is created in a provider contract and must be used to customize the allowance.

To force the allowances to have the `ALLOW_DEF_ID` parameter, you customize the unique allowance interface [page 91] in the SAP CC system.

To configure the allowance interface in an SAP Solution scenario with SAP CRM and SAP ERP/FI-CA

1. Open [page 95] the allowance interface.
2. Add a new line, and then set up the following column values:

Column	Value
Name	**`ALLOW_DEF_ID`**
Type	*String*
Description	**`The ID of the allowance definition from which the allowance has been initialized`**
Default Value	*Not Set*

3. Save [page 97] the allowance interface by selecting the menu *File* > *Save*.

Next Step

Creating an Allowance Definition Table Class [page 170]

Related Information

Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

Setting Up the Allowance Interface [page 96]

1.5.3.7.3 Working with the Allowance Interface

Related Information

Opening the Allowance Interface [page 95]
Modifying the Allowance Interface [page 95]
Setting Up the Allowance Interface [page 96]
Saving the Allowance Interface [page 97]

1.5.3.7.3.1 Opening the Allowance Interface

To open the allowance interface

1. From the menu bar in the main window, select *Tools* > *Allowance Interface*. The *Allowance Interface* window opens.

1.5.3.7.3.2 Modifying the Allowance Interface

> **Note**
>
> "Allowance Interface", the name of the allowance interface is not modifiable. This name is assigned to the allowance interface during the installation of SAP CC.

To modify the allowance interface

1. Open [page 95] the allowance interface.
2. Set up [page 96] the allowance interface.

3. Verify the audit structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save [page 97] the allowance interface.

1.5.3.7.3.3 Setting Up the Allowance Interface

Open [page 95] the allowance interface before setting it up. The *Allowance Interface* window must be open.

> **Note**
>
> - You cannot change the type of a parameter already saved.
> - You cannot change the name of a parameter or a counter already saved.
> - You cannot change the name of the allowance interface because it is assigned to the allowance interface during the installation of SAP CC

Allowance Interface window

Name: Displays the name of the allowance interface.

Parameters tab

Use the table and its icons to create parameters.

 : Click to add a parameter row to the table.

 : Click to delete a parameter row from the table.

- *Name*: Type the name of the parameter.
- *Type*: Provides the list of parameter types. Select the parameter type: *Number*, *String*, and *Date*.
- *Description*: Type the description of the parameter.
- *Default Value*: Type the default value of the parameter.

Counters

Use the table and its icons to create counters.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

- *Name*: Type the name of the counter.
- *Description*: Type the description of the counter.
- *Default Value*: Type the default value of the counter.

1.5.3.7.3.4 Saving the Allowance Interface

Once you have modified the allowance interface, and then verified its structure, you can save the allowance interface to either the database or as an XML file on your hard disk.

> ⓘ Note
>
> After saving the allowance interface, you can no longer:
>
> - Modify a parameter name or a counter name
> - Change the type of a parameter

To save a new allowance interface to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. The *Allowance Interface Creation* dialog box opens to inform you that the allowance interface has been successfully modified.
3. Click *OK*. The allowance interface is saved in the database.

To save an allowance interface as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the allowance interface.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The allowance interface is saved as an XML file in your selected directory.

1.5.3.8 Generate License Auditing File

You can generate a license auditing file for the SAP Global License Auditing System (SAP GLAS).

> ⓘ Note
>
> You cannot generate a license auditing file if the SAP Convergent Charging license corresponds to a temporary license.

To generate a license auditing file for SAP GLAS:

1. From the menu bar in the main window, select *Tools* > *License auditing file* > *Generating a license auditing file*.
2. The *Save* dialog box opens.
3. Select a directory to which you want to generate the license auditing file.
4. In the *File name* field, type the name of the file.
5. Click *Save*.

Related Information

Viewing Metrics [page 103]

1.5.4 Window

In the menu bar, the *Window* menu allows opening the following work windows:

Viewing a Catalog [page 98]

Viewing Operations [page 99]

Viewing Object Change Logs [page 101]

Viewing Metrics [page 103]

Related Information

Viewing a Catalog [page 98]
Viewing Operations [page 99]
Viewing Object Change Logs [page 101]
Viewing Metrics [page 103]

1.5.4.1 Viewing a Catalog

The procedure is the same as the one for opening a pricing catalog [page 113].

1.5.4.2 Viewing Operations

From the menu bar in the main window, select *Window* > *View Operations*. The *Audited Operations* window opens.

The elements of the *Audited Operations* window allows you to search for audited user operations that are based on:

- The rules set in a previously created audit object (*Search From Audit* tab) as a custom filter
- Filter criteria (*Search From Filters* tab)
- Criteria (*Search by Criteria* tab)

Audited Operations (Window)

After you set up either the *Search From Audit* tab, the *Search From Filters* tab, or the *Search From Criteria* tab, you can start the search by clicking  . If there are audited operations that conform to the criteria, they appear in the results area.

Use the following icons to:

 : Clear the search results list from the results area.

 : Delete all the audited user operations that match the search criteria from the back-end database system. This button deletes the audited operation that are currently displayed and all the other audited operations that match the search criteria.

 : Scroll to the next audited user operations that match the criteria.

 : Scroll to the previous audited user operations that match the criteria.

 : Open the *Object Change Log* window to display the object change logs related to the selected audited operation.

Search From Audit (Tab)

Audit: Provides the list of existing audit objects (custom audit filters). Select an audit on which you want to base your search. Before searching for operations, click  to modify [page 659] the original audit if necessary. Make your modifications, save [page 658] them, and then close the audit window.

Period For Search: Choose the option to search for user operations by selecting either *last audited period* or *given dates*. If you select *given dates*, select these dates in the *From* and *To* fields by using the calendar  .

Search From Filters (Tab)

You set up this tab in the same way as you set up [page 656] the *Filters* tab of an audit object.

In the *From* and *To* fields, click the calendar to select the dates between which you want to find all the audited operations.

Search by Criteria (Tab)

To find audited operations, the *Search by Criteria* tab allows you to create a table containing a single or a combination of possible search criteria. To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> **Note**
>
> - If the criteria table consists of several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table comprises three fields. To set up a row, do the following:

1. In the first field, select a search criterion in the following list of identifiers:
   - *Reference*: Identifier of the audited operation
   - *Operation*: Type of the audited operation for every object supporting the fine-grained auditing feature
   - *User*: ID of the user who initiated the audited operation
   - *Operand*: Identifier (name, code, another reference) that characterizes the user operation and its action
   - *Error Message*: Error related to the execution of the audited operation
   - *Instance ID*: Identifier of the instance that has processed the user operation (for example, updater #1 or rater #1)
2. In the second field, select a search comparator such as *Equals*, *Starts With*, *Ends With*, *Contains*, *Is Like*, for example.
3. In the third field, select or type the string that you want to find in the database.
4. In the *From* and *To* fields, click the calendar to select the dates between which you want to find audited operations.

Related Information

Viewing Object Change Logs [page 101]

1.5.4.3 Viewing Object Change Logs

From the menu bar in the main window, select *Window* > *View Object Change Logs*. The *Object Change Logs* window opens.

To find object change logs, you perform the operation in two halves of the window:

- The top part of the window is the filter area, which allows you to find object change logs by using filter criteria. The information in the *Basic* and *Advanced* tabs is combined during the search operation.
- The bottom area of the window is the results area, which displays the search results that you perform.

You can find object change logs with or without using search criteria. If you do not use search criteria, click  to start the search. If you use search criteria, set up the *Basic* and *Advanced* tabs, and then click  to start the search. The object change logs are displayed in the results area. To display more columns of information in the results area, click  , and then select the types of information you want to display.

> **Note**
>
> - When the system is configured to record object snapshots, you can view and inspect these detailed object changes.
> - When the User Operation Auditing feature is enabled in the system, you can audit the SAP CC user and the user operation that are responsible for the data object change.
> - Refer to the SAP CC Application Help for more information and user assistance about the audit concept, features, and functions.

Basic tab

The *Basic* tab allows you to find object change logs depending on the criteria described below:

- *Object Type*: Contains the list of auditable objects.
- *Object ID Part 1*, *Object ID Part 2*, and *Object ID Part 3*: These three fields represent the different parts of the identifier of an object associated to a given change log. In the first field, select a search comparator such as *Equals*, *Starts With*, or *Contains*, for example. In the second field, type the string that you want to find in the database.
- *From* and *To*: Click the calendar  to select the dates between which you want to find change logs.

Advanced tab

To find object change logs, the *Advanced* tab allows you to create a table with one or more search criteria such as *Action*, *Object Reference*, or *User Operation Reference*. To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

Each row of the table consists of three fields. To set up a row, do the following:

- In the first field, select a search criterion in the list of provided criteria.
- In the second field, select a search comparator such as *Lower Than or Equal To*, *Greater Than*, *Lower Than* or *Equals*, for example.
- In the third field, type the string that you want to find in the database.

> ⓘ Note
>
> - If the criteria table consists of several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Results area

Use the following icons:

 : Start a new search whether you use search criteria or not.

 : Clear the results list.

 : Move to the next page. This button is available when the results list is large.

 : Move to the previous page. This button is available when the results list is large.

 : View the related user operation and the SAP CC user who triggered the operation. This button is available when an object change log is selected and when the system is configured to audit user operations.

 : Consult the object snapshot before change. This button is available only when an object snapshot exists.

 : Consult the object snapshot after change. This button is available only when an object snapshot exists.

> ⓘ Note
>
> Clicking the buttons  or  :
>
> - Either opens the editor of the object that is related to the snapshot.
> - Or asks for saving the object snapshot to a file when it is not possible to open the editor. This occurs when another object snapshot that is required for the display cannot be found in the back-end database system.
>
> You cannot change the objet in the editor, and some existing panes may not be available because of the existing dependencies between the related objects.
>
> To compare the object snapshots before and after data object changes, you can save a couple of snapshots to files.

Related Information

Viewing Operations [page 99]

1.5.4.4 Viewing Metrics

From the menu bar in the main window, select *Window* > *Metrics*. The *Metrics* window opens.

The elements of the *Metrics* window allows you to search for the number of transactions or subscriber accounts processed per day in SAP Convergent Charging. This information measures the usage of the SAP CC product in your system landscape.

Refer to the SAP CC Application Help for more information and user assistance about the *Product Usage Measurement* features.

Metrics window

To display the search results, you perform the operation in two halves of the window:

- The top part of the window is the filter area, which allows you to define the search rules.
- The bottom area of the window is the results area, which displays the search results.

Filter area

The filter area mainly consists of a table whose each row represents a search rule. You can combine as many rules as you want in the table. For example, you can search all the days during which the number of transactions is greater than 100 and the number of subscriber accounts is lower than 5.

 : Click to add a criterion to the table.

 : Click to remove the selected criterion

Metric: Select one of the following product sage metrics:

- Transaction
- Postpaid Transaction
- Prepaid Transaction
- Subscriber Account

Comparator: Select one of the following comparators:

- Equals
- Greater than
- Lower than
- Greater or Equals

- Lower or Equals

Value: Type the comparison value.

From and *To*: Click the calendars to display the dates. These fields allow you to select:

- An undefined period: *From* (empty field) *To* (empty field)
- A defined period: *From* April 8, 2010 12:00:00 AM *To* May 8, 2010 12:00:00 AM
- A period limited by a date in the future: *From* (empty field) *To* April 8, 2010 12:00:00 AM
- A period limited by a date in the past: *From* April 8, 2010 12:00:00 AM *To* (empty field)

Results area

Use the following icons to:

 : Clear the search results list from the results area.

 : Scroll to the next results matching the criteria.

 : Scroll to the previous results matching the criteria.

 : Save the results in a CSV file.

1.6 Configuration of Master Data Step by Step

Prerequisites

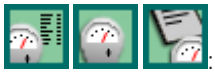
The business data [page 32] is configured in the SAP CC system.

First configure necessary business data with Core Tool (refer to Configuration of Business Data Step by Step [page 107]) and other user interfaces.

Configure the Master Data in a Pricing Catalog

The steps below outline the order in which you create and define new objects when you work with charge plans, refill plans, and provider contracts.

> ⓘ Note
>
> Some of the steps can be performed in an external provisioning system.
>
> Some steps may not apply to your business requirements and models.

1.  : As a pricing specialist, create a pricing catalog [page 112] for the chargeable service or product. You store all the pricing elements to this catalog.
2.  : In an integrated SAP Solution, import the price table [page 171] if you want to use recurring fees defined in SAP CRM (see Using SAP CRM Prices in Recurring Charges [page 319]).
3.  : Implement the allowance feature if you use Allowances [page 575] in your charge, in your refill logic, or in your allowance logic. You configure the creation of allowances through one or more *Create Allowance Operator* components.
   - Create an allowance event class [page 441].
     - Define the event properties [page 444] to use the allowance.
   - Create the Allowance Logic [page 449]
   - Set up [page 455] the allowance logic.
   - Create [page 468] an allowance plan:
     - Add [page 469] the allowance logic to the allowance plan.
     - Set up [page 469] the allowance plan.
     - Set up [page 477] the allowance logic in the allowance plan.
     - Release [page 469] the allowance plan.
   - Configure the creation of Allowances [page 575] in the allowance logic if you want to implement an allowance renewal policy, for example. Note that the related charge or the refill logic that participates to the product definition must has been subscribed and provisioned in the related Provider Contracts [page 573].
   - Configure the creation of allowances in a charge (step 5) or in the refill logic (step 10).
   - Customize this charge (step 8) or this refill logic (step 11) in a charge plan or in a refill plan to define the rules for creating allowances in provider contracts.

   You must first set up [page 585] this implementation if SAP Convergent Charging is connected to SAP ERP/FI-CA.
4.  Create Chargeable Item Classes [page 241].
   This step is optional if you do not use a *Usage Rate* component in your charge.
   You must first create the Consumption Item Mapping [page 281] if SAP Convergent Charging is connected to SAP ERP/FI-CA.
5.  : Create [page 309] a reusable charge.
   1. Build a price plan that will be used to fix the prices for the service/product uses.
   2. Build a charging plan that will be used to determine accounts to be charged.
6.  : Create [page 256] a charged item class.
   You must first create [page 266] the billable item mapping if SAP Convergent Charging is connected to SAP ERP/FI-CA.
7. Create the necessary counter names in the dictionary [page 87].
8.  : Create [page 350] a charge plan and customize charges for each chargeable service or product:
   1. Add master charges and dependent charges.

2. Set up the charge plan and define relationships between master and dependent charges.
3. Set up the customized charges.
4. Release the plan.

Option: If you want to implement top-up or refill services in a prepaid environment, you configure the data as described in steps 9 to 11.

9.  : Create [page 389] a refill item class.
   - Define the properties used to perform a refill plan.
10.  : Create [page 400] the refill logic.
    - Build the refill logic that will be used to refill a prepaid account.
11.  : Create [page 422] a refill plan.
    1. Add the refill logic.
    2. Set up the refill plan and the refill logic.
    3. Release the plan.

Option: If you want to implement spending status monitoring services in your system landscape, you configure the business data and master data as described in steps 12 to 15.

12.  : Create [page 486] the necessary spending status descriptions that must be jointly known by the policy or monitoring system and the SAP CC system.
13.  : Create dedicated range table class [page 206] that define the structure of range tables suitable for configuring monitoring plans.
14.  : Create the necessary range table [page 227] that define the spending thresholds of spending status labels.
15.  : Create a monitoring plan [page 495].
    1. Set up [page 496] the monitoring plan.
    2. Set up [page 496] the generation of necessary spending statuses.
    3. Release [page 504] the plan.

    This ends the configuration of the master data related to the service provider in the pricing catalog. This master data is used in a CRM application to combine charge plans and refill plans in commercial products that end customers can subscribe to.
    The master data related to the end customers (mostly subscriber accounts and provider contracts) is automatically created in SAP Convergent Charging when the system is connected to an external provisioning system or a CRM application. You can simulate these creations.
16.  : Create a subscriber account [page 527].
    1. Add accounts (prepaid, external postpaid).
    2. Set up these accounts.

    Option: You can create a subscriber mapping table [page 545] or a subscriber range tables [page 557] for this subscriber account.

17.  : Create a provider contract [page 599].
    1. Set up the provider contract.
    2. Add contract items to the provider contract.
    3. Set up these contract items.

    Option: You can create a subscriber mapping table [page 545] or a subscriber range table [page 557] for each provider contract relating to the new subscriber account.
18. Test and debug all your pricing settings in the development SAP system landscape.
19. Correct if necessary.
    - Rerate [page 59] if exceptional mistakes or errors occur in charge plans, provider contracts, or price plans, for example.

Next Steps

- Transport, test, and validate your pricing catalog and its elements in the appropriate system landscapes: test, QA, preproduction.
- Put your pricing settings into production by transporting the data to the productive systems.

Related Information

Configuration of Business Data Step by Step [page 107]

1.7 Configuration of Business Data Step by Step

After installing the SAP CC system landscape, the system administrator (Administrator role [page 640]) and the SAP CC pricing specialists or power users (Marketing role [page 640]) must configure some Business Data [page 32].

Prerequisites

The *SAP CC system administrator* has configured the necessary business data with the appropriate Admin+ and Setup Tool user interfaces. Before creating a pricing catalog [page 109] for implementing the pricing policy of a customer service, you can check this data. Ask for reports about:

- VAT taxes
- VAT rules

Procedure

In Core Tool, you set up the following business data whose some data is predefined during the installation of the SAP CC system.

> **Note**
>
> You can configure some business data when configuring the master data [page 104] for the first time.

Marketing Role	1. Manage the Currencies [page 81] necessary to configure the Master Data [page 28] of the pricing catalog and the customer master data. • In an integrated *SAP Solution* when SAP CC is connected to the ERP reference system, synchronize the currencies (Synchronizing Currencies [page 85]). • Otherwise check (see Exporting Currencies [page 84]) and change (see Importing Currencies [page 84]) the predefined currencies.
Marketing Role	2. Manage the Public Holidays [page 73] necessary to configure some pricing logic based on public holiday information relating to the customer service consumption or other operations: • Check (see Exporting Public Holidays [page 80]) and change (see Importing Public Holidays [page 80]) the predefined public holidays
Administrator Role	3. In an integrated *SAP Solution* with one ore more SAP ERP/FI-CA and CRM systems, manage the Customer Management Areas [page 87] necessary to configure master data. You set up the pairs of SAP ERP/FI-CA systems and SAP CRM systems that maintain the customer master data in the SAP CC system.
Marketing Role	4. You can configure the master data (refer to Configuration of Master Data Step by Step [page 104]) for a service provider.

Next Steps

Now the pricing specialists for SAP CC can configure the pricing catalogs the chargeable services or products to monetize.

Related Information

Configuration of Master Data Step by Step [page 104]

Configuration of Master Data Step by Step

1.8 Transporting Catalog and Data

Use these procedures to transport a consistent subset of pricing elements from a source SAP CC system to target SAP CC systems (as transport destinations).

Transport Data

1. Create a change list object [page 662] that includes the data objects to transport to the target systems.
2. Run or schedule a transport request [page 668].
3. When relevant, monitor your scheduled requests.
4. Verify the final execution status.
5. Troubleshoot [page 670] when necessary.

Transport Data by Using CTS

1. Create a change list object [page 662] that includes the data objects to transport.
2. Export the change list [page 666] containing the consistent master data to a file.
3. In SAP Solution Manager, use the exported file to trigger transport operations.

Related Information

Catalog Transport

1.9 Catalogs

A catalog is a storage structure used by SAP Convergent Charging to help you organize your work more efficiently.

In SAP Convergent Charging, a pricing catalog includes the *pricing configuration* for an end-customer service to monetize. A catalog contains all the pricing elements that configure the necessary pricing logic and charging logic for this chargeable service.

Depending on your modeling and design, you create a pricing catalog [page 112] for each service provider company that you represent, or for each end-customer service, or for a bundle of services that you must monetize.

In your SAP CC Core Tool user interface, each catalog consists of subfolders inside an *All Objects* folder. The *All Objects* folder stores all the pricing elements you create for a service provider such as chargeable item classes, reusable charges and pricing macros.

You can add folders to this basic structure but you cannot delete the basic structure itself.

Note that a catalog is made up of the following master data, whose availability or behavior depends (as of SAP CC 2023 FPS 2) on the installation scenario:

- Chargeable Item Class
- Charged Item Class
- Aggregation Policy
- Pricing Macro
- Translation Table
- Tier Table
- Mapping Table Class
- Mapping Table
- Range Table Class
- Range Table
- Charge
- Charge Plan
- Allowance Event Class
- Allowance Logic
- Allowance Plan
- Refill Item Class
- Refill Logic
- Refill Record Class
- Refill Plan
- Monitoring Plan

> **Note**
>
> - When an agreement table is created, it appears in a new subfolder located in the folder of the corresponding table type (range or mapping table).
> - If at least one agreement table exists, the subfolder *Agreement Tables* cannot be deleted.

Related Information

Working with Pricing Catalogs [page 112]

1.9.1 Catalog Owner

Catalog Owner

A **service provider** is an actor that provides some marketable value-added services to end customers. The service provider company monetizes the service usage and consumption. The company launches customer-focused pricing models and offers that differentiate from the competition. To price its end-customer services dynamically, the service provider owns a pricing catalog in *SAP Master Data*.

> ⓘ Note
>
> In SAP Convergent Charging 2021, the catalog owner information is a textual reference to:
>
> - The "contracting" service provider (such as the company name) that want to monetize one or multiple end-customer services
> - An end-customer service (such as the service name or a comprehensive abbreviation)
> - A bundle of end-customer services (such as the name of this bundle) that can be priced by using some common pricing elements from a unique catalog
>
> The exact meaning depends on your implementation requirements.

Modeling and Design

The SAP CC pricing specialist is an SAP CC individual user who represents the contracting service provider. This expert translates the service pricing policy and strategy to SAP Convergent Charging capabilities and features. He designs how to configure the SAP CC systems and data and performs the pricing configuration, validation, and maintenance.

> → Recommendation
>
> When modeling and designing the catalog owner information, use a **combination** of:
>
> - The *company name* of the service provider that you represent.
> - The name of the *marketable end-customer service* or the name of a service bundle that must be priced by using the corresponding pricing catalog.
>
> For example, configure your system with: *GAFA*, *GAFA Digital Services*, or *GAFA Duo Services*.
>
> Use the same information to define the name for the pricing catalog and facilitate the maintenance and troubleshooting.
>
> In a training system, SAP CC users can set up the catalog owners by using their user ID or family name for example.

Configuration

As an SAP CC pricing specialist, you set up the owner information when configuring a pricing catalog and its pricing elements such as a charge plan, a mapping table, or a pricing macro.

The SAP CC system uses this technical information to control the master data consistency. A pricing catalog and all its pricing elements must have the same owner information.

> ⓘ Note
>
> The catalog owner is not the SAP CC individual user (pricing specialist) who configures a pricing catalog and its elements.
>
> There is an exception. In a training system, a user can set up the owner information with a family name.

1.9.2 Working with Pricing Catalogs

As a pricing specialist, you manually configure and maintain your pricing catalogs and their elements (see Working with Catalog Objects [page 116]).

Note that you can also customize some folders (see Working with Catalog Folders [page 115]).

> → Remember
>
> A user administrator can assign an SAP CC user to a unique pricing catalog.

Related Information

Creating a Catalog [page 112]
Opening a Catalog [page 113]
Modifying a Catalog [page 114]
Deleting a Catalog [page 114]

1.9.2.1 Creating a Catalog

Use this procedure to create a pricing catalog.

Create a Pricing Catalog

1. From the menu bar in the main window, select *File* > *New* > *Catalog*. The *New Catalog* window opens.
2. In the *Owner* field, type the name of the service provider.
3. Click *OK*. The *Catalog* window opens.

> **Note**
>
> An SAP CC user associated to a pricing catalog cannot create a new catalog. If such a user tries to create a catalog, an error message is displayed to inform the user that creating a catalog cannot be performed.

Related Information

Working with Catalog Objects [page 116]

1.9.2.2 Opening a Catalog

At any point in your work session, you can check whether a pricing catalog has been created for a service provider and modify [page 114] it if necessary.

Open a Pricing Catalog

1. From the menu bar in the main window, select *Window* > *View Catalogs*. The *Catalog* window opens.
2. In the *Owner* drop-down menu, select the owner of the catalog that you want open.

> **Note**
>
> If the name of the service provider does not appear in the *Owner* drop-down menu, this means that no catalog has been created for the service provider. In this case, you can create [page 112] a pricing catalog for the service provider company and its chargeable services.

1.9.2.3 Modifying a Catalog

Modify Your Pricing Catalog

You can modify the pricing catalog structure only in the following ways:

- Add [page 115] new folders to the basic structure.
- Copy existing folder [page 115] or objects [page 117] (pricing elements) and paste them into new folders.
- Rename [page 116] new folders.
- Delete folders [page 116] or objects [page 120].
- Modify [page 117] objects.

> **Note**
>
> You cannot delete the default object folders (such as the *Chargeable Item Classes* and the *Charges* folders) from the *All Objects* folder.

1.9.2.4 Deleting a Catalog

Before you delete a pricing catalog, you must first delete all data objects contained in the catalog. Only empty catalogs can be deleted.

> **Caution**
>
> There is no “undo” function in your SAP CC Core Tool user interface. Before deleting an object, make sure that this action is necessary.

Delete a Pricing Catalog

1. From the menu bar in the main window, select *File* > *Delete* > *Catalog*. The *Choose Owner* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the catalog.
3. Click *OK*. A dialog box confirms that the catalog has been successfully deleted.

> **Note**
>
> If a warning message appears after clicking *OK* at step 3, you must delete the object or objects contained in the pricing catalog.

1.9.3 Working with Catalog Folders

Related Information

Adding a New Catalog Folder [page 115]
Copying and Pasting a Catalog Folder [page 115]
Renaming a Catalog Folder [page 116]
Deleting a Catalog Folder [page 116]

1.9.3.1 Adding a New Catalog Folder

You can add new folders to the basic catalog structure and add subfolders within those folders.

Add a Folder to a Catalog

1. Open [page 113] a pricing catalog.
2. In the *Owner* drop-down menu, select the owner of the catalog that you want to open.
3. Select the catalog or folder in which you want to add a new folder.
4. Right-click the item selected in the previous step and select *New Directory*. A folder called *New Directory* is added. If you want, you can rename [page 116] it.

1.9.3.2 Copying and Pasting a Catalog Folder

You can copy existing folders and paste them into new folders created [page 115] in the pricing catalog.

Copy and Paste a Catalog Folder

1. Open [page 113] a pricing catalog.
2. In the catalog, select the folder that you want to copy.
3. Right-click the item selected in the previous step, and then select *Copy*.
4. Select the destination folder for pasting the folder.
5. Right-click the folder selected in the previous step, and then select *Paste*. The folder appears in the destination folder.

> **Note**
>
> Using the same method, you can select *Cut* instead of *Copy* but only with folders outside the *All Objects* folder.

1.9.3.3 Renaming a Catalog Folder

You can only rename folders that have been created outside the *All Objects* folder.

Rename a Catalog Folder

1. Open [page 113] a pricing catalog.
2. In the catalog, select the folder that you want to rename.
3. Right-click the item selected in the previous step and select *Rename Directory*.
4. Type the new name for the folder.

1.9.3.4 Deleting a Catalog Folder

You can only delete folders that have been created outside the *All Objects* folder.

Delete a Catalog Folder

1. Open [page 113] a pricing catalog.
2. In the catalog, select the folder outside the *All Objects* folder that you want to delete.
3. Right-click the item selected in the previous step, and then select *Delete Directory*. The folder is deleted.

> **Note**
>
> Deleting a folder does not delete the objects that are included in the folder. The objects of the deleted folder are always available in the *All objects* folder.

1.9.4 Working with Catalog Objects

Related Information

Modifying a Catalog Object [page 117]
Copying and Pasting a Catalog Object [page 117]
Deleting the Versions of a Catalog Object [page 118]
Deleting a Catalog Object [page 120]
Exporting an Object from a Catalog [page 121]

1.9.4.1 Modifying a Catalog Object

You can change catalog elements that are displayed in the folders of your pricing catalog.

Modify a Catalog Object

1. Open [page 113] a pricing catalog for a chargeable service.
2. In the catalog, double-click the object that you want to modify. The related object window opens.
3. Switch to the editing mode [page 46].
4. Modify the object.
5. Save the object by selecting one of two methods:
   - To save and release the object to the SAP Convergent Charging back-end database, select *File* > *Save* > *In The Database*, choose an effective date and time, and then click *OK*.
   - To save the object as an XML file on your hard disk, select *File* > *Save* > *In A File*, name the file, and then click *Save*. You can use this text file for troubleshooting purposes.

1.9.4.2 Copying and Pasting a Catalog Object

You can copy existing objects and paste them into new folders created [page 115] in the catalog.

Copy and Paste a Catalog Object

1. Open [page 113] a pricing catalog.
2. In the catalog, select the object that you want to copy.
3. Right-click the item selected in the previous step, and then select *Copy*.
4. Select the destination folder for pasting the object.
5. Right-click the folder selected in the previous step, and then select *Paste*. The object appears in the destination folder.

> ⓘ Note
>
> Using the same method, you can select *Cut* instead of *Copy* but only with objects outside the *All Objects* folder.

> → Tip
>
> Use the same method to copy/paste an object from your user interface to a text editor. The pasted object is an XML element.

1.9.4.3 Deleting the Versions of a Catalog Object

When deleting one or more versions of a catalog object, you take the following decisions:

- Which versions you want to delete.
- Which version you want to extend.

> ⓘ Note
>
> - You can delete one or several consecutive versions at the same time.
> - You cannot delete all the versions of a catalog object.

When deleting one or more versions of a catalog object, you extend the version preceding or following the versions to delete. In the figures below, a pricing macro has three versions A, B, and C (Fig. 1) as follows:

- Version C: From the beginning of time to d3
- Version B: From d3 to d2
- Version A: From d2 to d1

If you delete version B (Fig. 2), you can extend version A to d3 (Fig. 3), or you can extend version C to d2 (Fig. 4).

Fig. 1

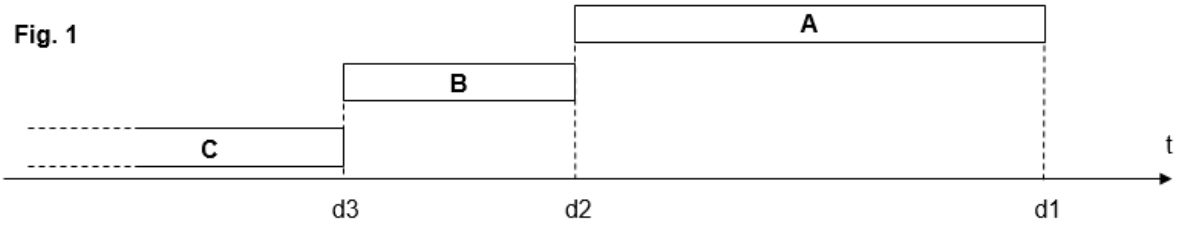


Fig. 2

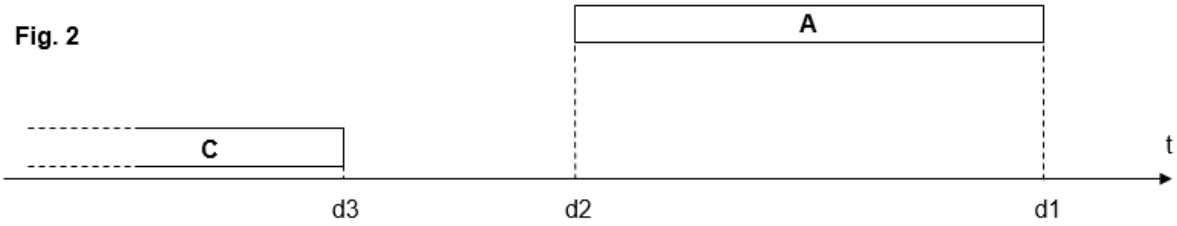


Fig. 3

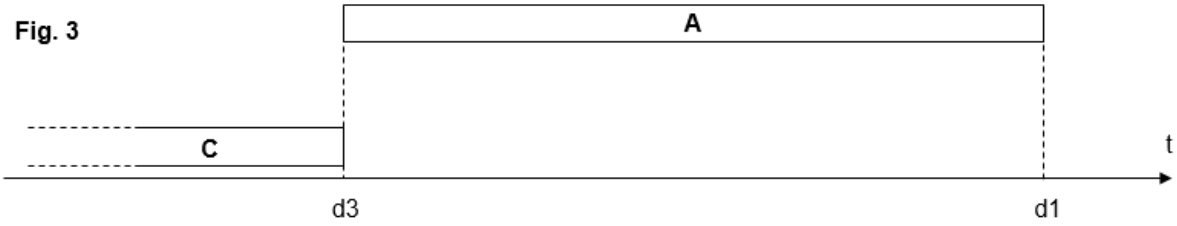
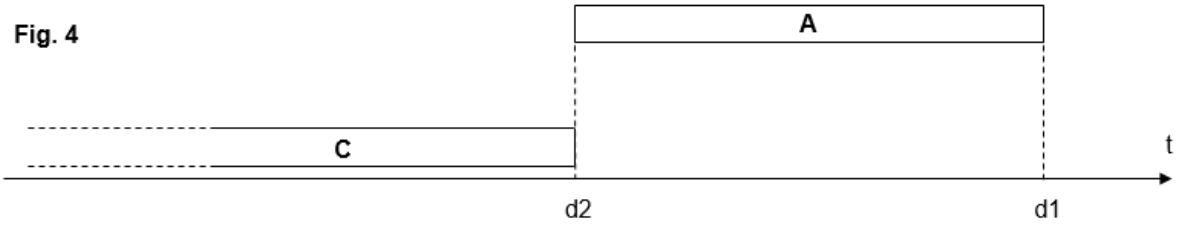


Fig. 4



Example

For example, you want to delete some versions of a pricing macro whose versions are ordered by date from the latest to the oldest:

Select versions to delete

Row	Version
1	From March 3, 2021 9:01:28 AM
2	Between February 3, 2021 9:00:49 AM and March 3, 2021 9:01:28 AM
3	Between January 3, 2021 9:00:04 AM and February 3, 2021 9:00:49 AM
4	Between December 3, 2020 8:59:19 AM and January 3, 2021 9:00:04 AM
5	Between November 3, 2020 8:58:30 AM and December 3, 2020 8:59:19 AM
6	Until November 3, 2020 8:58:30 AM

According to the previous table:

- If you delete the first version of the table, the second version (row 2) of the table is automatically extended.

- If you delete the last version of the table, the version (row 5) before the last one is automatically extended.
- If you delete consecutive versions of the table, you can extend the version that precedes or follows the versions to delete: If you delete three versions (rows 3, 4, and 5), you can extend two versions (row 2 or row 6).
- If you delete consecutive versions including the first version, the version that follows the versions to delete is extended: If you delete three versions (rows 1, 2, and 3), one version (row 4) is automatically extended.
- If you delete consecutive versions including the last version, the row that precedes the versions to delete is extended: If you delete three versions (rows 4, 5, and 6), one version (row 3) is automatically extended.

To delete the versions of a catalog object

1. Open [page 113] the catalog related to the object.
2. In the catalog, double-click the object whose versions you want to delete. The related object window opens.
3. Switch to the editing mode [page 46].
4. From the menu bar in the main window, select *Action* > *Delete Versions*[18]. The *Delete Versions* dialog box opens.
5. In the *Select Versions to Delete* table, select the versions that you want to delete. The *Version to Extend* table displays the versions to extend[19].
6. In the *Version to Extend* table, select the version that you want to extend. The selected version is in bold in the *Select Versions to Delete* table.
7. Click *Delete*. A message informs that you are about to delete the selected versions. The dialog box displays the number and the list of versions that you are to delete.
8. Click *Delete*. A message confirms that the versions have been successfully deleted.
9. Click *OK*[20].

1.9.4.4 Deleting a Catalog Object

You can only delete catalog objects that have been created outside the *All Objects* folder.

---

[18] If the menu is not available, this means that you have already modified the open object. Save or close the object, and then start again this procedure.

[19]
- If you select all the versions or no version in the *Select Versions to Delete* table, the *Version to Extend* table is empty and the *Delete* button is not available (dimmed).
- If you select the latest version or the oldest version in the *Select Versions to Delete* table, the version following the latest version or the version preceding the oldest version is in bold in this table, and this version is selected in the *Version to Extend* table.

[20] When you click *OK*, the window of the catalog object switches from the Edit mode to the Display Mode.

Delete a Catalog Object

1. Open [page 113] a pricing catalog.
2. In the catalog, select the object outside the *All Objects* folder that you want to delete.
3. Right-click the item selected in the previous step, and then select *Cut*. The object is deleted.

1.9.4.5 Exporting an Object from a Catalog

You can export objects to a directory of your hard disk, where they are saved as XML files.

Export a Catalog Object

1. From the menu bar in the main window, select *Window* > *View Catalog*. The *Catalog* window opens.
2. In the *Owner* drop-down menu, select the owner of the pricing catalog that you want to open.
3. Select the object that you want to export.
4. Right-click the item selected in the previous step and select *Export*. The *Save* dialog box opens.
5. Select a directory to export the object.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.10 Pricing Macros

When configuring the pricing logic for your chargeable service, service bundle, or product, SAP Convergent Charging provides you with an important reusable pricing element: the pricing macros as blocks of instructions and logic decisions.


A **pricing macro** in SAP Convergent Charging (SAP CC) is a reusable pricing element capable of retaining and applying a calculation formula. If your price plan, refill logic, or allowance logic has common calculation formula, you can manage and maintain the price plan or the refill logic more easily by creating pricing macros.

By using pricing macros, you simplify and optimize your pricing configurations. You gain flexibility.

A pricing macro looks like a price plan, and it is represented using the same multilevel tree structure. At runtime, the system executes this pricing macro. Using its own properties, the pricing macro calculates an amount and may generate output properties.

Once a pricing macro is created [page 123], you can then add [page 330] it to the price plan of a charge, to the refill logic of a refill plan, or to the allowance logic of an allowance plan. You use the logic component named macro introducer [page 804]. A pricing macro can be used in any charge or refill logic and may also be used within other pricing macros, resulting in cascading pricing macros.

Versions

You can modify and save different versions of a pricing macro: Changing the Version of a Pricing Macro [page 131].

Testing a Pricing Macro

Once you pricing macro is complete, SAP Convergent Charging provides you with the Rating Trial Runs [page 335] tool to test and verify your pricing macro and its logic configuration.

Transport

See: Transporting Catalog and Data [page 109]

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 0, pricing functions are introduced. Pricing functions are mapped to pricing macros managed by Cockpit. Even if pricing functions are visible in Core Tool, they cannot be edited by it. The Core Tool user interface manages exclusively pricing macros. Use Cockpit to manage pricing functions.

Related Information

Price Plans [page 299]
Refill Logic [page 395]
Allowance Logic [page 449]
Macro Introducer (Operator) [page 804]

1.10.1 Working with Pricing Macros

Related Information

Creating a Pricing Macro [page 123]
Opening an Existing Pricing Macro [page 127]
Modifying a Pricing Macro [page 128]

Setting Up a Pricing Macro [page 124]
Deleting a Pricing Macro [page 129]
Saving a Pricing Macro [page 126]
Exporting a Pricing Macro [page 130]
Importing a Pricing Macro [page 130]

1.10.1.1 Creating a Pricing Macro

A pricing macro is like a section of a price plan that you can add to any price plan or refill logic. You can choose to define properties that input information into the macro or output information from it. Like a price plan, a pricing macro is graphically represented by a decision tree. The elements of the *Pricing Macro* window enable you to create a pricing macro.

Create a Pricing Macro

1. From the menu bar in the main window, select *File* > *New* > *Pricing Macro*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. Click *OK*. The *New Pricing Macro* window opens.
4. Set up [page 124] the pricing macro.
5. Verify the pricing macro structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 126] the pricing macro.

> → Tip
>
> You can create a pricing macro by opening an existing pricing macro and copying it.

Create a Pricing Macro from an Existing Pricing Macro

1. From the menu bar in the main window, select *File* > *Open* > *Pricing Macro*. The *Open Pricing Macro* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. In the *Pricing Macros* folder of the catalog of the owner, select the pricing macro to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding pricing macro window opens.
5. Set up [page 124] the pricing macro.
6. Verify the pricing macro structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.

7. Save [page 126] the pricing macro.

Related Information

Setting Up a Pricing Macro [page 124]
Saving a Pricing Macro [page 126]

1.10.1.2 Setting Up a Pricing Macro

Before setting up a price macro, make sure one of the tasks listed below is done:

- Creating a Pricing Macro [page 123]
- Opening an Existing Pricing Macro [page 127]
- Modifying a Pricing Macro [page 128]

Like a price plan [page 299], a pricing macro is graphically represented by a decision tree, which can consist of price plan components using counters, parameters and pricing macros depending on the requirements of the plan.

> ⓘ Note
>
> Before inserting components into the pricing macro, you need to create properties that you will use in the components.

Set Up a Pricing Macro

1. In the left pane of the pricing macro window, click the pricing macro icon  .
2. In the right pane of the pricing macro window, identify the pricing macro.
   1. In the *Name* field, type the unique identifier of the pricing macro.
   2. In the *Description* field, describe the function of the pricing macro if necessary.
3. If necessary, add rating context properties [page 125] that contain the values that the pricing macro needs to make calculations.
   These values are passed to the pricing macro by the price plan or another pricing macro that contains the pricing macro that you are setting up.

   > → Tip
   >
   > If you want to work with counters, you can add some entries to these input properties.
4. If necessary, add generated properties [page 125] which are the calculation results of the pricing macro. These values are retrieved by the price plan or another pricing macro that contains the pricing macro that you are setting up.

5. Add and combine [page 330] logic components to build your business logic. Add functions, comparisons, splitters, or operators to the pricing macro, and then set up the components (*).

> **Note**
>
> (*) For more comprehensive details on using each component, refer to the primary help relating to components: Working with Logic Components [page 671]

Add a Rating Context Property

When required by your business needs, you define a rating context propety as input of the pricing macro:

1. In the *Rating Context Properties* area, click  to add a property. A new row is added to the configuration table.
2. In the *Name* column, type the property name or a counter name.
3. In the *Description* column, describe the function of the property if necessary.
4. In the *Type* column, select one of the following property types: *Number*, *String*, *Date*, *Counter*, *Multi-valued Number*, or *Multi-valued String*.

> **Note**
>
> - Multi-valued Number:
> Represents a list of numerical values. For example, the multi-valued number "ImageSize" contains the following list: 5000, 10000, 15000, 20000.
> - Multi-valued String:
> Represents a list of string values. For example, the multi-valued string "ImageType" contains the following list: image/jpeg:5000, image/gif:10000, image/eps:25000 and image/pic:8000.

5. If you want to remove the property, click  .

Add a Generated Property

When required by your business needs, you define a generated propety as output of the pricing macro:

1. In the *Generated Properties* area, click  to add a property. A new row is added to the configuration table.
2. In the *Name* column, type the property name.
3. In the *Description* column, describe the function of the property if necessary.
4. In the *Type* column, select one of the following property types: *Number*, *String*, *Date*, *Multi-valued Number* or *Multi-valued String*.
5. In the *Default Value* column, type a value corresponding to the property type.
6. If you want to remove the property, click  .

Related Information

Macro Introducer (Operator) [page 804]

1.10.1.3 Saving a Pricing Macro

Once you have created or modified a pricing macro, and then verified its structure, you are ready to save it. You can save the pricing macro to either the database or as an XML file on your hard disk.

Save a New Pricing Macro to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Pricing Macro Creation* dialog box opens to inform you that the pricing macro has been successfully saved and released.
4. Click *OK*.

Save a New Version of an Existing Pricing Macro to the Database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the pricing macro has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the pricing macro.
> 2. Click *Cancel* to come back to the pricing macro. See later in this chapter how to save the pricing macro as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the pricing macro.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date/time while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates/times by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates/times related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.

- *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.

4. Click *Save*. The *Pricing Macro Modification* dialog box opens to inform you that the modified pricing macro has been successfully saved and released.
5. Click *OK*.

Save a Pricing Macro as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the pricing macro.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The pricing macro is saved as an XML file in your selected directory.

1.10.1.4 Opening an Existing Pricing Macro

You can open:

- An existing pricing macro saved in the back-end database system
- A previous version of a pricing macro saved in the database
- An existing pricing macro saved in an XML file

Open an Existing Pricing Macro Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Pricing Macro*. The *Open Pricing Macro* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. In the *Pricing Macros* folder of the catalog of the owner, select the pricing macro, and then click *Open*. The pricing macro window opens in display mode.

Pricing macros are managed chronologically. After you have saved a pricing macro, you can access previous versions of it that exist in the database. However, if you modify and save a previous version of a pricing macro, all later versions of it are deleted.

Open a Previous Version of a Pricing Macro Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Pricing Macro*. The *Open Pricing Macro* dialog box opens.

2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. In the *Pricing Macros* folder of the catalog of the owner, select the pricing macro, and then click *Open*. The pricing macro window opens in display mode.
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the pricing macro from the list.
6. Click *View*.

Open an Existing Pricing Macro Saved in an XML File

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.10.1.5 Modifying a Pricing Macro

You can modify a pricing macro by adding or deleting rating components. The modified pricing macro replaces the previous version in the back-end database, though you can still access the previous version.

You cannot modify its name or its input and output properties once you have saved it.

Modify a Pricing Macro

1. *Open* a pricing macro.
2. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The pricing macro may be reloaded if it has been modified since you have opened it.
3. Set up [page 124] the pricing macro.
4. Verify the pricing macro structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Save [page 126] the pricing macro.

Related Information

Opening an Existing Pricing Macro [page 127]
Display/Edit Mode [page 46]
Setting Up a Pricing Macro [page 124]
Saving a Pricing Macro [page 126]

1.10.1.6 Deleting a Pricing Macro

Before you delete a pricing macro, you must first delete all objects which reference it.

Delete a Pricing Macro

1. From the menu bar in the main window, select *File* > *Delete* > *Pricing Macro*. The *Delete Pricing Macro* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. In the *Pricing Macros* folder of the catalog of the owner, select the pricing macro.
4. Click *Delete*. A deletion dialog box asks if you want to delete the pricing macro.
5. Confirm the deletion of the pricing macro by clicking *Yes*:
    1. If a dialog box confirms your choice, the pricing macro has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the pricing macro is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the pricing macro. The identifiers of these objects appear in the message.

1.10.1.7 Exporting a Pricing Macro

You can export pricing macros to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import the pricing macro back into SAP Convergent Charging 2021.

To export a pricing macro

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the pricing macro.
3. In the catalog of the owner, select the pricing macro.
4. Click *OK*. The Save dialog box opens.
5. Select a directory to export the pricing macro.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.10.1.8 Importing a Pricing Macro

You can import pricing macros in XML format from a directory in your database.

Several versions of a pricing macro can exist in a database, corresponding to their stages of modification. You can import all versions of an import into SAP CC *Core Tool*, or select only the versions you want.

To import a charge

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the charge is saved, and then select the charge file.
3. Click *Open*. The *Import Objects* dialog box opens.
4. Leave the default *All versions* option selected if you want to import all the versions of the imported objects, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
5. In the *Import* column, clear the checkboxes of objects you do not wish to import.
6. Click *Import*.

> **Note**
>
> SAP CC *Core Tool* can only import an object which has previously been exported and saved (*File* ▸ *Export*), rather than saved directly to a file (*File* ▸ *Save* ▸ *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.10.2 Using the Action Menu

Related Information

Changing the Version of a Pricing Macro [page 131]
Deleting the Versions of a Pricing Macro [page 132]
Reading a Pricing Macro [page 132]
Doing a Trial Run of a Pricing Macro [page 132]

1.10.2.1 Changing the Version of a Pricing Macro

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.10.2.2 Deleting the Versions of a Pricing Macro

The procedure is the same as Deleting the Versions of a Catalog Object [page 118].

1.10.2.3 Reading a Pricing Macro

Once you have created a pricing macro, you can generate a descriptive representation of it. This process is called reading a pricing macro.

To read a pricing macro

1. Open a pricing macro.
2. On the toolbar in the main window, click the *Read Pricing Macro* icon  . The *Read Pricing Macro* dialog box opens.
3. Check the pricing macro description for accuracy, and then click *Close*.

Related Information

Opening an Existing Pricing Macro [page 127]

1.10.2.4 Doing a Trial Run of a Pricing Macro

The procedure is the same as Doing a Trial Run of a Price Plan [page 335].

1.11 Translation Tables

> ⓘ Note
>
> You cannot redefine a translation table at Charge Plans [page 343], Refill Plans [page 421] or Allowance Plans [page 466] levels if your global charging services are based on Provider Contracts [page 573]. We recommend that you use Mapping Tables [page 179] and Subscriber Mapping Tables [page 542].

A **translation table** in SAP Convergent Charging (SAP CC) is a multicolumn data table. It allows you to create new properties that can be used in Price Plans [page 299] or Charging Plans [page 305], in the Refill Logic [page 395], or in the Allowance Logic [page 449].

A translation table is an n-dimensional table that consists of:

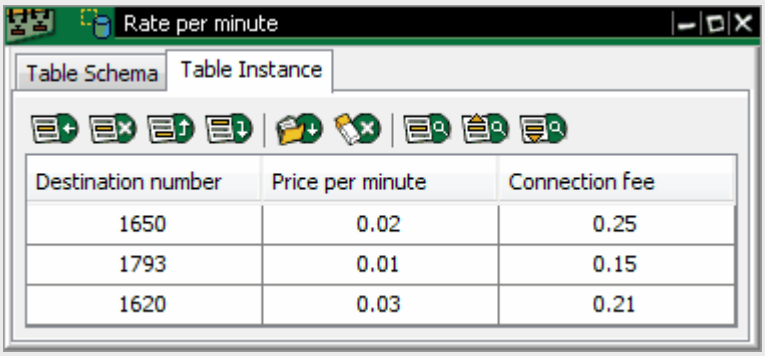
- A set of input columns (I1, I2, I3, ..., In) with n less than or equal to 5
- A set of output columns (O1, O2, O3, ..., Op) with p less than or equal to 15
- A set of rows

Each row of the table consists of a set of values of input columns (I1, I2, I3,..., In) to which corresponds a set of values of output columns (O1, O2, O3,..., Op).

After you have created a translation table, you can then add it to the price plan (see Adding a Component [page 330], the charging plan, the refill logic, or the allowance logic. You can use the same translation table more than once in any price plan, charging plan, or refill logic.

Example

The rate per minute for the U.S. during weekdays (out of holidays) depends on the destination. To calculate the consumption price in US dollars, you create a table with the following inputs and outputs: Destination number (input), Price per minute (output) and Connection fee (output). The following translation table determines the related price per minute and connection fee for a specified destination number:



Destination number	Price per minute	Connection fee
1650	0.02	0.25
1793	0.01	0.15
1620	0.03	0.21

Versions

You can modify and save different versions of a translation table.

See the task Changing the Version of a Translation Table [page 146].

→ Remember

Read the note about the order of the rows [page 135] in a translation table.

1.11.1 Working with Translation Tables

- Creating a Translation Table [page 134]
- Opening an Existing Translation Table [page 142]
- Changing the Version of a Translation Table [page 146]
- Modifying a Translation Table [page 143]
- Setting Up a Translation Table [page 135]
- Saving a Translation Table [page 140]
- Deleting a Translation Table [page 144]
- Deleting the Versions of a Translation Table [page 146]
- Exporting a Translation Table [page 145]
- Importing a Translation Table [page 145]

1.11.1.1 Creating a Translation Table

Use one of these procedures to create a translation table.

Create a Translation Table

1. From the menu bar in the main window, select *File* > *New* > *Translation Table*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. Click *OK*. The *New Translation Table* window opens.
4. Set up [page 135] the translation table.
5. Verify the translation table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 140] the new translation table.

> ⓘ Note
>
> Once you have saved a translation table, you can no longer modify its name or structure. You can only modify the *Table Instance* tab by adding content to it.

> → Tip
>
> You can create a translation table by opening an existing translation table and copying it.

Create a Translation Table from an Existing Translation Table

1. From the menu bar in the main window, select *File* > *Open* > *Translation Table*. The *Open Translation Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the *Translation Tables* folder of the catalog of the owner, select the translation table to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding translation table window opens.
5. *Modify* the translation table if necessary.
6. Verify the translation table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 140] the new translation table.

Related Information

Setting Up a Translation Table [page 135]
Modifying a Translation Table [page 143]
Saving a Translation Table [page 140]

1.11.1.2 Setting Up a Translation Table

Create, open, or modify a translation table before setting it up. A translation table window must be open.

> → Remember
>
> When a translation table is used in Price Plans [page 299], Charging Plans [page 305], Pricing Macros [page 121], or in the Refill Logic [page 395], a *Table Operator component* compares row by row the input columns of the translation table with price plan properties. The comparison stops when the price plan properties match at least one row of the input columns of the translation table; therefore, the order of the rows in the translation table is important.

> ⓘ Note
>
> Once you have saved a translation table, you can no longer modify its name or structure. You can only modify the *Table Instance* tab by adding content to it.

Translation Table (Window)

To set up a translation table, you must fill in or modify the *Table Schema* tab and the *Table Instance* tab.

Table Schema (Tab)

This tab allows you to create the structure of your translation table, which can consist of one to five input columns and one to fifteen output columns. Use the icons to fill in the tab.

Name: Type the unique identifier of the table.

Description: Describe the translation table if necessary.

Start and end dates can be specified for each row: If selected, this option adds the columns *Start Date* and *End Date* to the table in the *Table Instance* tab. This option is used in combination with the field *Reference Date* of a table operator component. You use this option as an additional criterion to sort the items of the table. You can, for example, fix the price of phone calls depending on prefixes and call dates.

Disallow duplicate input values for the same time: If selected, this option prevent you from adding two rows containing the same input values. An error message is displayed when two rows containing the same values are added to the table. If the option *Start and end dates can be specified for each row* is selected, two rows are considered as duplicated if they contain the same input values and if the periods delimited by *Start date* and *End date* overlap.

Example

Example of duplicate input values

I1	I2	I3	Start Date	End Date
John	Jane	Harry	null[21]	null
John	Jane	Harry	null	

I1	I2	I3	Start Date	End Date
John	null[22]	Harry	2007-01-01	2007-02-01
John	null	Harry	2007-01-15	2007-02-07

I1	I2	I3	Start Date	End Date
John	Peter	Harry	2007-01-01-10:48:05	null
John	Peter	Harry	null	2007-01-01-10:48-06[23]

Input Columns

- *Name*: type the input column name.
- *Description*: Describe the input column if necessary.

Output Columns

- *Name*: Type the output column name.
- *Description*: Describe the output column if necessary.
- *Type*: Provides the list of types. Select Number or String.
- *Default value*: Assign a value to the output column. When the translation table is

[21] Null represents an unspecified date in the past or the future.
[22] Null represents an empty cell.
[23] An overlap of 1 second is enough.

used in a price plan or a pricing macro, if no comparison is successful, a default value for each generated property is returned by the translation table.

Table Instance (Tab)

> **Note**
>
> Double-clicking a column sorts the data in that column, and the entire table is updated accordingly. Double-click a second time the column sorts the column (and the table) in reverse order.

Once you have created the structure of your translation table in the *Table Schema* tab, you can add one or more rows to the table, and then fill in the cells of each row with appropriate values. When adding a new row to the table, each cell of the input columns is assigned the default value "Any" that represents any value. You can replace "Any" by any value of your choice.

If you have selected the *Start and end dates can be specified for each row* option, two additional columns Start Date and End Date appear in the table. After you add a row, click the cell of each column to open a calendar, and then select a date if necessary. Depending on your choice, the item of the table is defined as follows:

- If only the cell of the *Start Date* column is empty: The corresponding item of the table is defined by a time period ranging from an unspecified date to the selected date.
- If only the cell of the *End Date* column is empty: The corresponding item of the table is defined by a time period ranging from the selected date to an unspecified date.
- If the cell of both columns is left blank: The corresponding item of the table is defined by a time period ranging from an unspecified date to an unspecified date.

Use the icons to fill in the data:

Using the Icons

Use the following icon...	To...
	Add a new row to the configuration table.
	Delete a row from the table.
	Add a new group to the table.
	Delete a group from the table.
	Move a row up.
	Move a row down.
	Import data into the table according to the CSV format.(*)
	Clear all the table.

Use the following icon...	To...
	Search for a numerical or string value in the table. The search stops when the first occurrence has been found. When the search is successful, the cell is highlighted. To find the next occurrence, use  or  . When you click  , a *Search in table* dialog box is displayed. In the *Search* field, type the value to search for or select a value in the list. If you want to do a case-sensitive search, click the *Match case* option. In the *Search in column(s)* check boxes, select the columns where you want to search for the numerical or string value.
	Search for the next occurrence towards the top of the table if the previous search with the icon  was successful. The search is performed from left to right, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the beginning of the table is reached, a message informs you that the value has not been found.
	Search for the next occurrence towards the bottom of the table if the previous search with the icon  was successful. The search is performed from right to left, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the end of the table is reached, a message informs you that the value has not been found.

> **Note**
>
> Instead of manually typing all of the data in the columns, you can import existing data from external .CSV type files by clicking  (refer to the process described below).

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns

Rule	Verifications
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

Related Information

Creating a Translation Table [page 134]
Opening an Existing Translation Table [page 142]
Modifying a Translation Table [page 143]

1.11.1.3 Saving a Translation Table

Once you have created or modified a translation table, and then verified its structure, you are ready to save it. You can save the translation table to either the database or as an XML file on your hard disk.

To save a new translation table to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Translation Table Creation* dialog box opens to inform you that the translation table has been successfully saved.
4. Click *OK*.

To save a new version of an existing translation table to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the translation table has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the translation table.
> 2. Click *Cancel* to come back to the translation table. See later in this chapter how to save the translation table as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the translation table.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Translation Table Modification* dialog box opens to inform you that the translation table has been successfully created.
5. Click *OK*.

> **Note**

To save a translation table as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the translation table.
3. In the File name field, type the name of the file.
4. Click *Save*. The translation table is saved as an XML file in your selected directory.

1.11.1.4 Opening an Existing Translation Table

You can open:

- An existing translation table saved in the database
- A previous version of a translation table saved in the database
- An existing translation table saved in an XML file

To open an existing translation table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Translation Table*. The *Open Translation Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the *Translation Tables* folder of the catalog of the owner, select the translation table. The translation table window opens in display mode.

Translation tables are managed chronologically. After you have saved a translation table, you can access the previous versions of it that exist in the database. However, if you modify and save a previous version of a translation table, all later versions are deleted.

To open a previous version of a translation table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Translation Table*. The *Open Translation Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the *Translation Table* folder of the catalog of the owner, select the translation table, and then click *Open*. The translation table window opens in display mode.
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the translation table from the list.
6. Click *View*.

To open an existing translation table saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.11.1.5 Modifying a Translation Table

You can modify a translation table by adding, modifying, or deleting content. When you save it in the database, you can modify its versions (see Changing the Version of a Translation Table [page 146]).

You cannot modify the name or the structure of a translation table once you have saved it.

> **Note**

To modify a translation table

1. From the menu bar in the main window, select *File* > *Open* > *Translation Table*. The *Open Translation Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the *Translation Tables* folder of the catalog of the owner, select the translation table.

4. Click *Open*. The translation table window opens in display mode.
5. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The translation table may be reloaded if it has been modified since you have opened it.
6. *Set up* the translation table.
7. Verify the translation table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. *Save* the translation table.

Related Information

Display/Edit Mode [page 46]
Setting Up a Translation Table [page 135]
Saving a Translation Table [page 140]

1.11.1.6 Deleting a Translation Table

Before you delete a translation table, you must first delete all objects which reference it.

To delete a translation table

1. From the menu bar in the main window, select *File* > *Delete* > *Translation Table*. The *Delete Translation Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the *Translation Tables* folder of the catalog of the owner, select the translation table.
4. Click *Delete*. A deletion dialog box asks if you want to delete the translation table
5. Confirm the deletion of the translation table by clicking *Yes*:
    1. If a dialog box confirms your choice, the translation table has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the translation table is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the objects that reference the translation table. The identifiers of these objects appear in the message.

1.11.1.7 Exporting a Translation Table

You can export translation tables to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import the translation table back into SAP Convergent Charging 2021.

To export a translation table

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the translation table.
3. In the catalog of the owner, select the translation table.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the translation table.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.11.1.8 Importing a Translation Table

You can import translation tables in XML format from a directory in your database.

Several versions of a translation table can exist in a database, corresponding to their stages of modification. You can import all versions of a translation table into SAP CC *Core Tool*, or select only the versions you want.

To import a translation table

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the translation table is saved, and then select the translation table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

> **Note**
>
> SAP CC *Core Tool* can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.11.2 Using the Action Menu

Related Information

Changing the Version of a Translation Table [page 146]
Deleting the Versions of a Translation Table [page 146]

1.11.2.1 Changing the Version of a Translation Table

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.11.2.2 Deleting the Versions of a Translation Table

The procedure is the same as Deleting the Versions of a Catalog Object [page 118]

1.12 Tier Tables

 A **tier table** in SAP Convergent Charging (SAP CC) is a table of correspondence for mapping an input column that contains value ranges to a set of output columns that contain string or number values. To compute final values, you can define some combinatorial rules to output columns.

A tier table is included in the pricing catalog [page 109] to implement subscription- and usage-based pricing models for your chargeable service or product.

> ### Example
>
> You can configure the pricing logic with tier tables for different industries. See Industry Examples of Tier Tables [page 148].

> ### Caution
>
> You cannot redefine any tier tables at charge plan level.

Use

> ### → Recommendation
>
> When you model and base your convergent charging services on provider contracts [page 573], SAP recommends that you use some range tables [page 213] and subscriber range tables [page 557] to design and configure the pricing logic that includes tiered or scaled pricing.

A tier table allows you to create new properties that can be used when designing the price plan, the charging plan, the refill logic, or the allowance logic.

A tier table is an n-dimensional data table that consists of:

- A *Group Name* column
- A *Range* column
- A set of one to five string or number output columns
- A set of rows

When creating a tier table, you use:

- The *Group Name* column to create groups of numerical intervals
- The *Range* column to represent the numerical intervals of each group
- The output columns to assign number or string values to each interval

> ### Note
>
> - You can create as many groups as you want.
> - You can create in each group ranging from -∞ to +∞ as many numerical intervals as you want.

- You can use negative or decimal numbers.
- You cannot create discontinuous or empty numerical intervals.
- You cannot delete or rename the default group.
- You must arrange the numerical intervals in increasing order.
- The numerical intervals cannot overlap.
- Each group of intervals is independent of the other ones.

After you have created a tier table, you can use it by adding a dedicated logic component [page 330] to the price plan, the charging plan, or the refill logic. You can use the same tier table more than once in any price plan, charging plan, or refill logic. You use the logic component: Tier Table Introducer (Operator) [page 822].

Redefinition

You cannot redefine any tier tables at charge plan level.

Versions

You can modify and save different versions of a tier table [page 166].

Related Information

Tier Table Introducer (Operator) [page 822]
Working with Tier Tables [page 151]

1.12.1 Industry Examples of Tier Tables

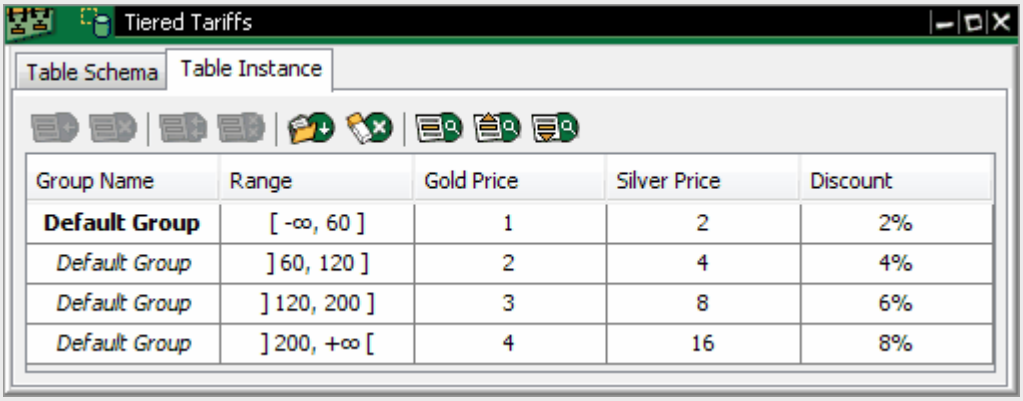
This topic presents some typical business use cases based on tier tables to model tiered pricing in the pricing logic:

Example	SAP Industry
Example of Tiered Pricing with Discounts [page 149]	*SAP for Media*
Transports Industry [page 150]	*SAP for Transportation and Logistics*

1.12.1.1 Example of Tiered Pricing with Discounts

Example

The marketing team of your company has defined the following tariffs, in which "Gold Price" and "Silver Price" are of number type and "Discount" is of string type:



Group Name	Range	Gold Price	Silver Price	Discount
Default Group	[-∞, 60]	1	2	2%
Default Group	] 60, 120]	2	4	4%
Default Group	] 120, 200]	3	8	6%
Default Group	] 200, +∞ [	4	16	8%

The marketing team hesitates about the final pricing and billing policy.

As a pricing specialist, you design and configure the pricing catalog for these usage-based services.

Tiered pricing is a pricing and billing strategy used to define a price per unit within a range. Tiered pricing works so that the price per unit decreases once each quantity within a tier has been consumed.

You determined four possible pricing configurations based on the computation mode:

- Single, non-linear
- Single, linear
- Cumulative, non-linear
- Cumulative, linear

In this example, the final computed prices for different weights are:

Computation Mode	Description	Example: 110 kg	Example: 300 kg
Single, non-linear	The property value determines a row of the tier table. The output column values of this row are assigned to the generated properties.	Gold Price = 1 Silver Price = 2	Gold Price = 4 Silver Price = 16

Computation Mode	Description	Example: 110 kg	Example: 300 kg
Single, linear	The property value determines a row of the tier table. For each output column, the output column value of this row is recalculated according to the formula "a(b-c)" where "a" is the output column value, "b" is the property value, and "c" is the lower boundary of the range including the property value. The resulting values are then assigned to the generated properties.	100=2(110-60) 200=4(110-60)	400=4(300-200) 1,600=16(300-200)
Cumulative, non-linear	The property value determines a row of the tier table. For each output column, the output column value of this row and the values of the output column of the previous rows are added together. The resulting values are then assigned to the generated properties	3=2+1 6=4+2	6=4+2 10=4+3+2+1 30=16+8+4+2
Cumulative, linear	The property value determines a row of the tier table. For each output column, the output column value of this row and the values of the output column of the previous rows are recalculated according to the formula "a(b-c)" (refer to the mode single, linear) and added together (*). The resulting values are then assigned to the generated properties.	160 = (1(60-0)) +(2(110-60)) 320 = (2(60-0)) +(4(110-60))	820 = (1(60-0)) +(2(120-60))+(3(200-120)) +(4(300-200)) 2,600 = (2(60-0)) +(4(120-50))+(8(200-120)) +(16(300-200))

1.12.1.2 Transports Industry

Example

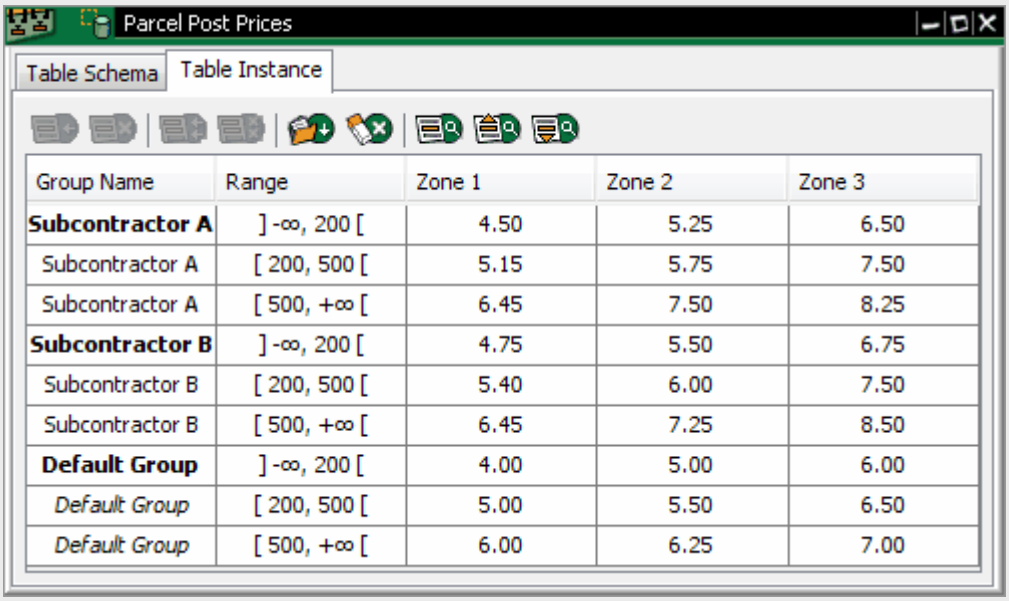
A company sells postal services which are delivered by subcontractors. The parcel postal tariffs are priced according to:

- The subcontractor
- The region of the world
- And the weight (in grams) of the parcel. There are three ranges (aka tiers):

-]-∞, 200[
- [200, 500[
- [500, +∞[

All prices are calculated in USD.

As a pricing specialist, you design and configure the following tier table:



Group Name	Range	Zone 1	Zone 2	Zone 3
Subcontractor A	] -∞, 200 [	4.50	5.25	6.50
Subcontractor A	[200, 500 [	5.15	5.75	7.50
Subcontractor A	[500, +∞ [	6.45	7.50	8.25
Subcontractor B	] -∞, 200 [	4.75	5.50	6.75
Subcontractor B	[200, 500 [	5.40	6.00	7.50
Subcontractor B	[500, +∞ [	6.45	7.25	8.50
Default Group	] -∞, 200 [	4.00	5.00	6.00
Default Group	[200, 500 [	5.00	5.50	6.50
Default Group	[500, +∞ [	6.00	6.25	7.00

In this example, if the subcontractor is unknown, the default group is used to determine the price of the postal service.

1.12.2 Working with Tier Tables

Related Information

Creating a Tier Table [page 152]
Opening an Existing Tier Table [page 162]
Changing the Version of a Tier Table [page 166]
Modifying a Tier Table [page 163]
Setting Up a Tier Table [page 153]
Saving a Tier Table [page 161]
Deleting a Tier Table [page 164]
Deleting the Versions of a Tier Table [page 167]
Exporting a Tier Table [page 165]
Importing a Tier Table [page 165]

1.12.2.1 Creating a Tier Table

Use one of these procedures to create a tier table.

Create a Tier Table

1. From the menu bar in the main window, select *File* > *New* > *Tier Table*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. Click *OK*. The *New Tier Table* window opens.
4. Set up [page 153] the tier table.
5. Verify the tier table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 161] the new tier table.

> ⓘ Note
>
> Once you have saved a tier table, you can no longer modify its name or structure. You can only modify the *Table Instance* tab by adding content to it. See Tab: Table Instance [page 157].

> → Tip
>
> You can create a tier table by opening an existing tier table and copying it.

Create a Tier Table from an Existing Tier Table

1. From the menu bar in the main window, select *File* > *Open* > *tier Table*. The *Open tier Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the *Tier Tables* folder of the catalog of the owner, select the tier table to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Tier Table* window opens.
5. Modify [page 163] the tier table if necessary.
6. Verify the tier table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 161] the tier table.

Related Information

Example of Tiered Pricing with Discounts [page 149]

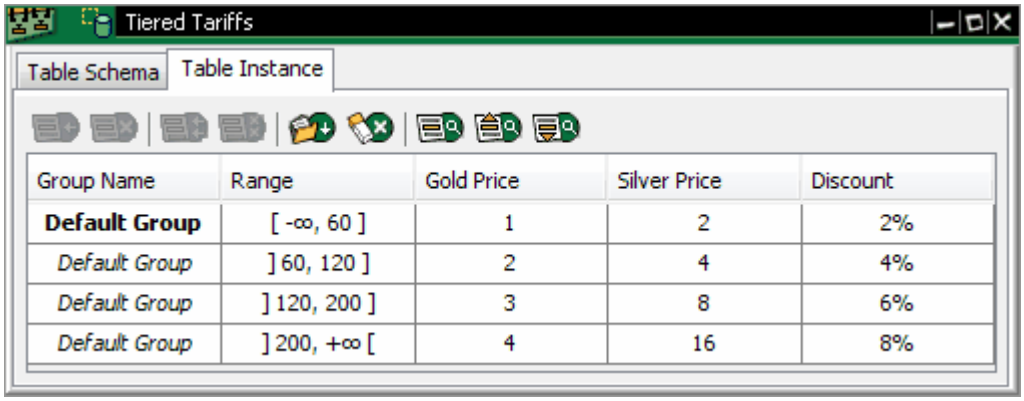
Setting Up a Tier Table [page 153]
Saving a Tier Table [page 161]
Modifying a Tier Table [page 163]

1.12.2.2 Setting Up a Tier Table

Create [page 134], open [page 142], or modify [page 143] a tier table before setting it up. A *Tier Table* window must be open.

> **Note**
>
> Once you have saved a tier table, you can no longer modify its name or structure. You can only modify the *Table Instance* tab by adding content to it.

All the examples described in this topic refer to the following illustration, in which "Gold Price" and "Silver Price" are of number type and "Discount" is of string type:



Window: Tier Table

To set up a tier table, you must fill in or modify the following tabs:

- Table Schema [page 153]
- Table Instance [page 157]

1.12.2.2.1 Tab: Table Schema

This tab allows you to define the data structure for your tier table, which can consist of one to five output columns. Use the icons to fill in the tab:

Use the Configuration Icons

Using the Icons

Use the following icon...	To...
	Add a new row to the configuration table.
	Delete a row from the table.
	Add a new group to the table.
	Delete a group from the table.
	Move a row up.
	Move a row down.
	Import data into the table according to the CSV format.(*)
	Clear all the table.
	Search for a numerical or string value in the table. The search stops when the first occurrence has been found. When the search is successful, the cell is highlighted. To find the next occurrence, use  or  . When you click  , a *Search in table* dialog box is displayed. In the *Search* field, type the value to search for or select a value in the list. If you want to do a case-sensitive search, click the *Match case* option. In the *Search in column(s)* check boxes, select the columns where you want to search for the numerical or string value.
	Search for the next occurrence towards the top of the table if the previous search with the icon  was successful. The search is performed from left to right, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the beginning of the table is reached, a message informs you that the value has not been found.
	Search for the next occurrence towards the bottom of the table if the previous search with the icon  was successful. The search is performed from right to left, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the end of the table is reached, a message informs you that the value has not been found.

> **Note**
>
> About tier tables and tier table operators.

You must use a Tier Table Introducer operator component [page 822] to insert a tier table in a price plan, a charging plan, or in the refill logic. You then choose a tier table and a property value.

The Tier Table Introducer operator component allows you to configure which numerical interval of the tier table includes the property value. When the numerical interval is determined, the output column values of the corresponding row are assigned to generated properties which can be used as new properties in the price plan, the charging plan, or in the refill logic.

For example, if the property value is equal to 140, the values of 3 (Gold Price) and 8 (Silver Price) and the string value "6%" (Discount) are assigned to the generated properties (Refer to the illustration above).

For more information about components, refer to the primary help relating to components: Working with Logic Components [page 671].

Table Schema (Tab)

Name: Type the unique identifier of the tier table.

Description: Describe the tier table if necessary.

> ### Note
>
> Whatever the computation mode, the output column values of string type are assigned to the corresponding generated properties if the output column values are related to the interval including the property value.
>
> For example, if the property value is equal to 110, the string value "4%" is assigned to the generated property. If the property value is equal to 300, the string value "8%" is assigned to the generated property (Refer to the illustration above).

Output Columns:

- *Name*: Type the output column name.
- *Description*: Describe the output column if necessary.
- *Type*: Provides the list of types. Select *Number* or *String*.
- *Computation Mode*: Provides the list of computation modes. This list is available only for the output columns of number type. The computation modes are the following:

Computation Mode	Description	Example
Single, non-linear	The property value determines a row of the tier table. The output column values of this row are assigned to the generated properties.	For example, if the property value is equal to 110 (range]60, 120]), the values of 1 (Gold Price) and of 2 (Silver Price) are assigned to the generated properties. If the property value is equal to 300 (range]200, +∞]), the values of 4 (Gold Price) and of 16 (Silver Price) are assigned to the generated properties.
Single, linear	The property value determines a row of the tier table. For each output column, the output column value of this row is recalculated according to the linear formula "A . (B - C)" Where: • "A" is the output column value • "B" is the property value • And "C" is the lower boundary of the range including the property value. The resulting values are then assigned to the generated properties.	For example, if the property value is 110 (range]60, 120]), the values of 100=2(110-60) and of 200=4(110-60) are assigned to the generated properties. If the property value is 300 (range]200, + ∞]), the values of 400=4(300-200) and of 1,600=16(300-200) are assigned to the generated properties.
Cumulative, non-linear	The property value determines a row of the tier table. For each output column, the output column value of this row and the values of the output column of the previous rows are added together. The resulting values are then assigned to the generated properties.	For example, if the property value is 110 (range]60, 120]), the values of 3=2+1 and of 6=4+2 are assigned to the generated properties. If the property value is 300 (range]200, + ∞]), the values of 10=4+3+2+1 and of 30=16+8+4+2 are assigned to the generated properties.

Computation Mode	Description	Example
Cumulative, linear	The property value determines a row of the tier table. For each output column, the output column value of this row and the values of the output column of the previous rows are recalculated according to the formula "a(b-c)" (refer to the mode single, linear) and added together[24]. The resulting values are then assigned to the generated properties.	For example, if the property value is 110 (range]60, 120]), the values of 160=(1(60-0))+(2(110-60)) and of 320=(2(60-0))+(4(110-60)) are assigned to the generated properties. If the property value is 300 (range]200, + ∞]), the values of 820=(1(60-0))+(2(120-60)) +(3(200-120))+(4(300-200)) and of 2600=(2(60-0))+(4(120-50)) +(8(200-120))+(16(300-200)) are assigned to the generated properties.

Related Information

Tier Table Introducer (Operator) [page 822]

1.12.2.2.2 Tab: Table Instance

Table Instance (Tab)

Once you have defined the structure of the tier table in the Table Schema tab [page 153], you can add one or more groups of rows to the table, and then fill in the cells of each row with appropriate values. A tier table contains a default group that you cannot rename or delete. You can only name or rename the groups that you create. Each group is independent of the other ones. For example, you can duplicate any group to which you assign a new name and new values.

To add a group to the table, click  , and to add a line to the group, click  .

To set up or modify an interval, click the interval in the *Range* column to display the *Range* dialog box, and then type your interval values. Use the Included option to include a value in an interval. Do not use this option when you select the values -∞ or +∞.

> → Remember
>
> Remember that the intervals must be arranged in increasing order. Pay attention to the characters that delimits the intervals[25] when including or excluding the values of intervals.

[24] Note that the computation is lightly different for the output column values of the previous rows. In the formula "a(b-c)", "b" refers to the upper boundary of the range while "c" refers to the lower boundary of the range. Furthermore, for a range such as [-∞, 60], the lower boundary "b" is equal to 0.

Use the icons to fill in the data:

Using the Icons

Use the following icon...	To...
	Add a new row to the configuration table.
	Delete a row from the table.
	Add a new group to the table.
	Delete a group from the table.
	Move a row up.
	Move a row down.
	Import data into the table according to the CSV format.(*)
	Clear all the table.
	Search for a numerical or string value in the table. The search stops when the first occurrence has been found. When the search is successful, the cell is highlighted. To find the next occurrence, use  or  . When you click  , a *Search in table* dialog box is displayed. In the *Search* field, type the value to search for or select a value in the list. If you want to do a case-sensitive search, click the *Match case* option. In the *Search in column(s)* check boxes, select the columns where you want to search for the numerical or string value.
	Search for the next occurrence towards the top of the table if the previous search with the icon  was successful. The search is performed from left to right, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the beginning of the table is reached, a message informs you that the value has not been found.
	Search for the next occurrence towards the bottom of the table if the previous search with the icon  was successful. The search is performed from right to left, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the end of the table is reached, a message informs you that the value has not been found.

[25] Open, closed, or mixed intervals

> **Note**
>
> Instead of manually typing all of the data in the columns, you can import existing data from external CSV files by clicking  (refer to the process described below).

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	> **Note** > Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values

Rule	Verifications
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

1.12.2.3 Saving a Tier Table

Once you have created or modified a tier table, and then verified its structure, you are ready to save it. You can save the tier table to either the database or as an XML file on your hard disk.

Save a New Tier Table to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Tier Table Creation* dialog box opens to inform you that the tier table has been successfully saved.
4. Click *OK*.

Save a New Version of an Existing Tier Table to the Database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the tier table has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the tier table.
> 2. Click *Cancel* to come back to the tier table. See later in this chapter how to save the tier table as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the tier table.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date/time while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates/times by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Tier Table Modification* dialog box opens to inform you that the tier table has been successfully created.

5. Click *OK*.

Save a Tier Table as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the tier table.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The tier table is saved as an XML file in your selected directory.

1.12.2.4 Opening an Existing Tier Table

You can open:

- An existing tier table saved in the back-end database system
- A previous version of a tier table saved in the back-end database
- An existing tier table saved in an XML file

Open an Existing Tier Table Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Tier Table*. The *Open tier Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the *Tier Tables* folder of the catalog of the owner, select the tier table. The *Tier Table* window opens in display mode.

Tier tables are managed chronologically. After you have saved a tier table, you can access the previous versions of it that exist in the database. However, if you modify and save a previous version of a tier table, all later versions are deleted.

Open a Previous Version of a Tier Table Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Tier Table*. The *Open Tier Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the *Tier Table* folder of the catalog of the owner, select the tier table, and then click *Open*. The *Tier Table* window opens in display mode.
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.

5. Select a version of the tier table from the list.
6. Click *View*.

Open an Existing Tier Table Saved in an XML File

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.12.2.5 Modifying a Tier Table

You can modify a tier table by adding, modifying, or deleting content. When you save it in the database, you can modify its versions [page 166].

You cannot modify the name or the data structure of a tier table once you have saved it.

To modify a tier table

1. From the menu bar in the main window, select *File* > *Open* > *Tier Table*. The *Open Tier Table* dialog box opens.

2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the *Tier Tables* folder of the catalog of the owner, select the tier table.
4. Click *Open*. The *Tier Table* window opens in display mode.
5. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The tier table may be reloaded if it has been modified since you have opened it.
6. Set up [page 153] the tier table.
7. Verify the tier table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. Save [page 161] the tier table.

Related Information

Display/Edit Mode [page 46]
Setting Up a Tier Table [page 153]
Saving a Tier Table [page 161]

1.12.2.6 Deleting a Tier Table

Before you delete a tier table, you must first delete all objects which reference it.

Delete a Tier Table

1. From the menu bar in the main window, select *File* > *Delete* > *Tier Table*. The *Delete Tier Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the *Tier Tables* folder of the catalog of the owner, select the tier table.
4. Click *Delete*. A deletion dialog box asks if you want to delete the tier table.
5. Confirm the deletion of the tier table by clicking *Yes*:
    1. If a dialog box confirms your choice, the tier table has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the tier table is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the objects that reference the tier table. The identifiers of these objects appear in the message.

1.12.2.7 Exporting a Tier Table

You can export tier tables to a directory of your hard disk, where they are saved as XML files.

> **Note**
>
> Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import [page 165] the tier table back into SAP Convergent Charging 2021.

Export a Tier Table

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the tier table.
3. In the catalog of the owner, select the tier table.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the tier table.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

Related Information

Importing a Tier Table [page 165]

1.12.2.8 Importing a Tier Table

You can import tier tables in XML format from a directory in your database.

Several versions of a tier table can exist in a database, corresponding to their stages of modification. You can import all versions of a tier table into SAP CC *Core Tool*, or select only the versions you want.

Import a Tier Table

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the tier table is saved, and then select the tier table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected, or do the following:

- To select only one version as of a certain date, select the *One version per object* option.
- To replace all versions by the imported version, select the *Replace all versions* option.
- To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date and time on which you save the imported version.

  Depending on this date and time, some existing versions will be replaced or not by the imported version.

6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

> **Note**
>
> SAP CC *Core Tool* can only import an object which has previously been exported [page 165] (*File* > *Export* >), rather than "directly saved" to a file (*File* > *Save* > *In a File* >).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.12.3 Using the Action Menu

Related Information

Changing the Version of a Tier Table [page 166]
Deleting the Versions of a Tier Table [page 167]

1.12.3.1 Changing the Version of a Tier Table

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

Related Information

Opening an Existing Tier Table [page 162]

1.12.3.2 Deleting the Versions of a Tier Table

The procedure is the same as Deleting the Versions of a Catalog Object [page 118]

1.13 Mapping Table Classes

 A **mapping table class** in SAP Convergent Charging (SAP CC) allows you to define a **common data structure for some mapping tables** (in your pricing catalog) and subscriber mapping tables (in customer master data). When you model and design some pricing logic based on these data lookup tables, a mapping table class is required.

A mapping table class defines the following characteristics for the corresponding mapping tables (and subscriber mapping tables):

- Number of input columns (I1, I2, I3, ..., In) with n less than or equal to 40
- Number of output columns (O1, O2, O3, ..., Op) with p less than or equal to 30

A mapping table class is included in the pricing catalog [page 109] for the provider company that monetizes a chargeable service or product.

See how to create [page 173], set up [page 174], save [page 176], and delete [page 177] a mapping table class.

> **Note**
>
> Once you have saved a mapping table class, you can no longer modify it with Core Tool. Therefore, you cannot have several versions of a mapping table class.

Use

You use mapping table classes when you want to base some pricing logic on data available in mapping tables or subscriber mapping tables. A corresponding class is expected.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

SAP Solution

In an integrated *SAP Solution* scenario, there are some specific mapping table classes and corresponding mapping tables. Refer to the integration guide that relates to your SAP system landscape.

Allowance Definition Table Class

If you want to use the concept of allowances in provider contracts, see Allowance Definition Table Class [page 169].

Price Table

If you want to configure periodic fees (see recurring rates [page 302]) in charge plans that are combined in SAP CRM, you must import a special mapping table class [page 171] named *"price table"*.

Refer to the topic Using SAP CRM Prices in Recurring Charges [page 319] for more information on how to use the price table for synchronizing recurring fees with SAP CRM prices.

Note that SAP CRM creates and maintains in SAP CC a mapping table for each commercial product.

Versions

SAP Convergent Charging 2023 does not manage several versions of a mapping table class.

Transport

You can use the Catalog and Data Transport feature to transport a consistent subset of data from a source system to target systems (as destinations). See Transporting Catalog and Data [page 109].

Otherwise, you can export [page 178] and import [page 178] some mapping table classes between different SAP CC systems.

Related Information

Working with Mapping Table Classes [page 172]

1.13.1 Allowance Definition Table Class

→ Remember

These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your integrated *SAP Solution*, the SAP CRM system manages the allowance definitions as mapping tables [page 179] stored in SAP CC. The structure of the allowance definition tables is thus defined by using mapping table classes [page 167] that must contain:

- One input column of string type and named **`ALLOW_DEF_ID`** (this column must contain the identifier of the allowance definition)
- One output column of string type and named **`ALLOW_DEF_DESC`**
- Some other main output columns that depend on your business case: **`BucketQuantity`**, **`Validity`**
- An additional information item of string type, named **`SAP_DATA_TYPE`** and set to **`AllowanceDefinition`**

As an SAP CC pricing specialist, you configure and maintain [page 170] this class.

Example

The following table shows an example of an allowance definition table class when you implement the allowance feature to model the management of end-customer service credits:

Input Column	Name	Type	Description
#1	**`ALLOW_DEF_ID`**	*String*	The identifier of the allowance definition

Output Column	Name	Type	Description
#1	**`ALLOW_DEF_DESC`**	*String*	A human readable short description of the allowance definition
#2	**`BucketQuantity`**	*Number*	The quantity of the allowance
#3	**`Validity`**	*Number*	The validity period in days of the allowance

This example shows that the purpose of an allowance definition is to define the quantity and the validity period of an allowance limited in quantity and in time.

Your Core Tool user interface provides you with a template to create an allowance definition table class in SAP CC.

Related Information

Creating an Allowance Definition Table Class [page 170]
Allowance Definition Tables [page 182]

1.13.1.1 Creating an Allowance Definition Table Class

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

Create an Ilowance Definition Table Class

1. Open the `allowance_definition_table_class.xml` XML file located in the installation directory of your Core Tool user interface in the templates folder.
2. Replace `${catalog}` by the name of your service pricing catalog.
3. Save and close the XML file.
4. Launch and log on [page 35] to the Core Tool user interface of SAP CC.
5. Open [page 177] the mapping table class from the XML file.
6. Rename the name of the displayed mapping table class.
7. Set up [page 174] the description of the mapping table class.
8. Declare necessary input and output columns.
9. Save [page 176] the mapping table class into the back-end database system.

Next Step

Creating an Allowance Definition Table [page 184]

Related Information

Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

1.13.2 Price Table

> → Remember
>
> These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your integrated *SAP Solution*, the **price table** is a specific and predefined mapping table class [page 167].

A price table specifies the data structure for price table instances [page 184] as mapping tables.

- Input column:
  - #1: type=*String* name=**`price.key`** description=**`The Price Table price key`**
- Output columns:
  - #1: type=*Currency* name=**`currency`** description=**`The Price Table currency`**
  - #2: type=*Number* name=**`amount`** description=**`Recurring charged amount`**

As an SAP CC pricing specialist, you configure and maintain [page 171] this particular mapping table class.

Related Information

Importing a Price Table as a Mapping Table Class [page 171]
Using SAP CRM Prices in Recurring Charges [page 319]
Price Table Instances [page 184]

1.13.2.1 Importing a Price Table as a Mapping Table Class

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

If you want to use recurring fees defined in an SAP CRM system, you must import the mapping table class called "price table [page 171]" from the installation directory to your back-end database system (Core Database). The format of the price table is XML.

Note that only one version of a mapping table class exists in the database because a mapping table class cannot be modified once it has been saved to the database. You can import the unique version of the price table with the *Core Tool* user interface.

Use the following procedures to customize the template of the price table and then import your price table as a new mapping table class [page 167].

Customize the Template of the Price Table

An XML template is provided to help you to create your price table.

1. Open the XML file `price_table.xml` from the installation directory of your *Core Tool* (`core_tool_install\templates`).
2. Replace `${owner}` by the name of your pricing catalog.
3. Save and then close the XML file.

Import the Price Table

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the price table is saved, and then select the modified `price_table.xml` file.
3. Click *Open*. The *Import Objects* dialog box opens.
4. Leave the default *All versions* option selected because there is only one version for the price table.
5. Click *Import*.

> **Note**
>
> - You can import a price table only once because one price table at most can be stored in the Core Database. If you want to change your price table, you must delete it and then import another price table.
> - The price table is a mapping table class. To open [page 177], export [page 178], delete [page 177] a price table, and so on, refer to the related topics listed in the mapping table class section.

Related Information

Using SAP CRM Prices in Recurring Charges [page 319]

1.13.3 Working with Mapping Table Classes

Specific tasks:

- Creating an Allowance Definition Table Class [page 170]
- Importing a Price Table as a Mapping Table Class [page 171]

Related Information

Creating a Mapping Table Class [page 173]

Setting Up a Mapping Table Class [page 174]
Opening an Existing Mapping Table Class [page 177]
Saving a Mapping Table Class [page 176]
Deleting a Mapping Table Class [page 177]
Exporting a Mapping Table Class [page 178]
Importing a Mapping Table Class [page 178]

1.13.3.1 Creating a Mapping Table Class

Use one of these procedures to create a mapping table class.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Create a Mapping Table Class

1. From the menu bar in the main window, select *File* > *New* > *Mapping Table Class*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table class.
3. Click *OK*. The *New Mapping Table Class* window opens.
4. Set up [page 174] the mapping table class.
5. Verify the mapping table class structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 176] the new mapping table class.

> ⓘ Note
>
> Once you have saved a mapping table class into the database, you can no longer modify it with the *Core Tool* user interface.

> → Tip
>
> You can create a mapping table class by opening an existing mapping table class and copying it.

Create a Mapping Table Class from an Existing Mapping Table Class

1. From the menu bar in the main window, select *File* > *Open* > *Mapping Table Class*. The *Open Mapping Table Class* dialog box opens.

2. In the *Owner* drop-down menu, select the owner of the mapping table class.
3. In the *Mapping Table Classes* folder of the catalog of the owner, select the mapping table class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding mapping table class window opens.
5. Modify the mapping table class if necessary.
6. Verify the mapping table class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 176] the new mapping table class.

Related Information

Setting Up a Mapping Table Class [page 174]
Saving a Mapping Table Class [page 176]

1.13.3.2 Setting Up a Mapping Table Class

Create [page 173] a mapping table class before setting it up. A *Mapping Table Class* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

The mapping table class allows you to define the structure of your Mapping Tables [page 179] (and Subscriber Mapping Tables [page 542]).

> ⓘ Note
>
> Once you have saved and released a mapping table class, you can no longer modify it with your Core Tool user interface. You can only create or modify mapping tables and subscriber mapping tables.

Window: Mapping Table Class

To set up a mapping table class, you must fill in or modify the *Table Schema* tab and the *Additional Information* tab.

Related Information

Setting Up a Mapping Table [page 188]

1.13.3.2.1 Tab: Definition

Name: Type the unique identifier of the mapping table class.

Description: Describe the mapping table class. This description is mandatory.

1.13.3.2.2 Tab: Table Schema

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

To set up your mapping table class, use the following icons to fill in the two configuration tables *Input Columns* and *Output Columns*:

 Add a new row to the configuration table. A row in this table, defines an input column or an output column in the mapping tables that are based on this class

 Delete a row from the table.

 Move a row up. The order in this table, specifies the order of the input columns (output columns) in the corresponding mapping tables.

 Move a row down.

> ⓘ Note
>
> In the table *Input Columns*, you can only add one input column of *Currency* type.

Input Columns

- *Name*: Type the input column name. This name must be unique within the set of input columns.
- *Description*: Describe the input column. This description is mandatory.
- *Type*: Provides the list of types. Select *String* or *Currency*.

Output Columns

- *Name*: Type the output column name. This name must be unique within the set of output columns.
- *Description*: Describe the output column. This description is mandatory.
- *Type*: Provides the list of types. Select Number or String.

1.13.3.2.3 Tab: Additional Information

Use the configuration table and its icons to define some additional information items.

 : Click to add a row to the configuration table.

 : Click to delete a row from the table.

Fill in each row that you add to the configuration table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date/Time*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information.

1.13.3.3 Saving a Mapping Table Class

Once you have created a mapping table class and then verified its configuration, you are ready to save it. You can save the mapping table class either into the back-end database or to an XML file on your hard disk.

> ⓘ Note
>
> Once you have saved a mapping table class into the database, you can no longer modify it with the *Core Tool* user interface.

Save and Release a Mapping Table Class into the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Mapping Table Class Creation* dialog box opens to inform you that the mapping table class has been successfully saved and released.
4. Click *OK*.

Save a Mapping Table Class to an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.

2. Select a directory to save the mapping table class.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The mapping table class is saved as an XML file in your selected directory.

1.13.3.4 Opening an Existing Mapping Table Class

You can open:

- An existing mapping table class saved in the database
- An existing mapping table class saved in an XML file

To open an existing mapping table class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Mapping Table Class*. The *Open Mapping Table Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table class.
3. In the *Mapping Table Classes* folder of the catalog of the owner, select the mapping table class. The *Mapping Table Class* window opens in display mode.

To open an existing mapping table class saved in an XML file

- Use the procedure for Importing a Mapping Table Class [page 178]

Related Information

Display/Edit Mode [page 46]

1.13.3.5 Deleting a Mapping Table Class

Before you delete a mapping table class, you must first delete all objects which reference it.

To delete a mapping table class

1. From the menu bar in the main window, select *File* > *Delete* > *Mapping Table Class*. The *Delete Mapping Table Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table class.
3. In the *Mapping Table Classes* folder of the catalog of the owner, select the mapping table class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the mapping table class
5. Confirm the deletion of the mapping table class by clicking *Yes*. A dialog box opens to inform you that the mapping table class has been successfully deleted.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the master data objects that reference the mapping table class. The identifiers of these objects appear in the message.

1.13.3.6 Exporting a Mapping Table Class

You can export mapping table classes to a directory of your hard disk, where they are saved as XML files. After exporting mapping table classes, you can import [page 178] them back into SAP Convergent Charging 2021.

To export a mapping table class

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table class.
3. In the catalog of the owner, select the mapping table class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the mapping table class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.13.3.7 Importing a Mapping Table Class

You can import mapping table classes in XML format from a directory to your database.

Note that there is only one version of a mapping table class in the database, as a mapping table class cannot be modified once it has been saved into the database. You can import the unique version of a mapping table class into SAP CC *Core Tool*.

To import a mapping table class

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the mapping table class is saved, and then select the mapping table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected, as there is only one version for each mapping table class.
6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

1.14 Mapping Tables

> ⚠ Caution
>
> **Do not modify or delete mapping tables with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Modifying or deleting mapping tables may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. To modify a mapping table, you must use this CRM or provisioning system in your SAP system landscape.
>
> - Consult the Application Help for more information about the product modeling process.
> - Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this master data in the pricing catalog of a the service provider for its subscription- and usage-based services.
>
> The manual procedures detailed in this section are relevant in:
>
> - A production system landscape with few contracts directly managed with the Core Tool user interface (no CRM interfaced with SAP CC)
> - A test, demo, or training system landscape

 A **mapping table** in SAP Convergent Charging (SAP CC) is a multicolumn table of business/technical data. This data table allows you to configure some logic based on data lookup and comparisons.

A mapping table allows you to create new *properties* (in the rating context) that you can use in the price plan, in the refill logic, or in the allowance logic.

The data table is composed of a set of rows. Each row of the mapping table consists of a set of values of input columns (I1, I2, I3, ..., In) to which corresponds a set of values of output columns (O1, O2, O3, ..., Op). For each row, you have to specify the period during which the row is valid. At runtime, SAP Convergent Charging uses mapping tables to compare a subset of information from rating context with data in the input columns in a selected mapping table and determines a subset of values from the output columns in the table. You must

however configure the pricing logic that defines the computation algorithm of the final values. See the logic component Mapping Table Introducer (Comparator) [page 744].

A mapping table is included in the pricing catalog of the service provider and is part of its master data. It is available for all the end customers of this service provider.

You define the data structure of your mapping table in a mapping table class [page 167].

Use

You use mapping tables to configure the pricing logic with Convergent Charging services based on provider contracts.

If the pricing is based on customer master data, you must implement the configuration of some subscriber mapping tables [page 542] that belong to subscriber accounts. In your plans, you set up a dedicated external parameter that contains the ID of the expected table. When a provider contract is created, the parameter gets a personalized value.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Mapping Table Step by Step

Before creating a mapping table, you must create [page 173], set up [page 174], and save [page 176] a mapping table class. You must perform the following steps to use mapping tables:

1. Create [page 187] a mapping table.
2. Set up [page 188] the mapping table.
3. Save [page 194] the mapping table.
4. Use the mapping table in the refill logic or in one or several price plans by setting up a dedicated logic component: Mapping Table Introducer (Comparator) [page 744].

   We recommend that you determine the appropriate customization level of the mapping table [page 185] that must be processed by the SAP CC system.
5. When relevant, transport [page 109] the mapping table and other consistent data to a target system.
6. Delete [page 197] the mapping table.

> → Tip
>
> You can use the same mapping table class to configure different mapping tables.

After you have created a mapping table, you can then add [page 330] it to the price plan of a charge [page 290] or to the refill logic. You can use the same mapping table more than once in any price plan or refill logic.

SAP Solution

In an integrated *SAP Solution* scenario, there are some specific mapping tables and classes.

Agreement Tables

See Agreement Tables [page 182].

Allowance Definition Tables

If you want to use the concept of allowances in provider contracts, see Allowance Definition Tables [page 182].

Price Table Instances

See Price Table Instances [page 184].

Integration

This master data may be created and maintained by an external system connected to the SAP CC system during the provisioning process.

> **Example**
>
> In an integrated *SAP Solution* scenario, SAP CRM manages some mapping tables stored in SAP Convergent Charging.

Versions

You cannot modify and save different versions of a mapping table.

Transport

You can use the Catalog and Data Transport feature to transport a consistent subset of data from a source system to target systems. See Transporting Catalog and Data [page 109].

Otherwise, you can:

- Export a mapping table [page 198] and export its class [page 178] to XML files
- Import a mapping table [page 198] and its class [page 178] from files to another SAP CC system.

Maintenance

The only mapping table which is available is the last modified one. You can modify and delete the rows of a mapping table, but you cannot change the name of a mapping table or the related mapping table class.

Related Information

Working with Mapping Tables [page 186]

1.14.1 Agreement Tables

In an integrated *SAP Solution* scenario with SAP SOM (SAP Subscription Order Management) in Convergent Invoicing, an agreement table is a particular mapping table or a particular range table stored in SAP CC that allows sharing data between several subscriber accounts.

> → Remember
>
> Agreement tables are part of customer master data. For technical and performance reasons, you are able to manage them like mapping or range tables.
>
> - To create an agreement table, use the procedures for creating a mapping table or a range table.
> - To search for an existing agreement table, use the procedures for finding a mapping table or a range table and by filtering the `Agreement ID` information.

1.14.2 Allowance Definition Tables

> → Remember
>
> These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In SAP Convergent Charging (SAP CC), *allowance definitions* are stored in the *allowance definition table* that is a specialized and dedicated mapping table [page 179] with:

- A predefined data structure:
  - 1 input column: `ALLOW_DEF_ID`
  - 3 output columns: `ALLOW_DEF_DESC`, `BucketQuantity`, and `Validity`
- An additional information item of string type, named **`SAP_DATA_TYPE`**, and set to **`AllowanceDefinition`**

→ Remember

In your integrated *SAP Solution* scenario, an allowance definition is used to define the allowance plan and respective parameters that will be used when an allowance is created for an active contract.

This mapping table relates to a specific class that is named allowance definition table class [page 169].

As a pricing specialist, you configure and maintain [page 184] these data lookup tables.

ⓘ Note

To be used by SAP CRM, the allowance definition table must be previously created in SAP CC.

See Also

- Allowance Definition Table Class [page 169]
- Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

Related Information

Example of Allowance Definition Table [page 183]
Creating an Allowance Definition Table [page 184]

1.14.2.1 Example of Allowance Definition Table

Example

The following table shows an example of an allowance definition table:

ALLOW_DEF_ID	*Start*	*End*	**ALLOW_DEF_DESC**	*BucketQuantity*	*Validity*
A01	201x	201x		100	30
A02	201x	201x		600	60
A03	201x	201x		1000	60

A01 defines a credit allowance limited in quantity and in time of 100 units valid for a period of 30 days. The allowance definition is valid for a period of one year.

→ Tip

Your Core Tool user interface provides you with a template to create [page 184] an allowance definition table in SAP Convergent Charging.

1.14.2.2 Creating an Allowance Definition Table

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

Create an Allowance Definition Table

1. Open the `allowance_definition_table.xml` XML file located in the installation directory of your Core Tool in the templates folder.
2. Replace `${catalog}` by the name of your service pricing catalog.
3. Replace `${mappingTableClass}` by the identifier of an allowance definition table class.
4. Save and close the XML file.
5. Launch and log on [page 35] to the Core Tool user interface of SAP CC.
6. Open [page 195] the mapping table from the XML file.
7. Change the name of the displayed mapping table.
8. Add allowance definitions as mapping table rows if necessary.
9. Save [page 194] the mapping table into the back-end database system.

Next Steps

- Creating an Allowance Event Class [page 443]
- Configuring the Creation of Allowances in Charge Plans [page 588]

Related Information

Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

1.14.3 Price Table Instances

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your integrated *SAP Solution*, we call **price table instance** a typical mapping table [page 179] that references a price table [page 171] as a mapping table class. Note that only the SAP CRM system creates price table instances.

> ⚠ Caution
>
> The SAP CRM system automatically creates and maintains the price table instances. Creating, modifying, or deleting price table instances may result in inconsistencies between these two systems. We strongly recommend that you do not modify or delete existing price table instances except under supervision of a support expert from your SAP Support Team.

> Example

Row ID	Validity Start Date/Time	Validity End Date/Time	price.key	currency	amount
..			..	.	.
...			...	..	.
.			.	.	.

Related Information

Using SAP CRM Prices in Recurring Charges [page 319]

1.14.4 Customized Use of Mapping Tables in the Charges and Refill Logic

You may want to create the charges or refill logic that can be used with different mapping tables during the charging or refilling operations.

The procedure below describes how to create a charge [page 290] containing a price plan with a Mapping Table Introducer as Comparator component [page 744] without specifying the name of the mapping table to be used. The idea is to use a parameter [page 303] as the name of a mapping table and to set up this parameter later on during:

- The manual configuration of a charge plan or refill plan in the pricing catalog
- Or the creation of a provider contract in customer master data

You can easily use different mapping tables or subscriber mapping tables.

> ⓘ Note
>
> You can use the same procedure in the refill logic [page 395].

Step 1: Configuration in the Pricing Catalog (Master Data Related to a Service Provider)

Configuration at Charge or Refill Logic Level

1. Create a charge with a logic component Mapping Table Introducer (Comparator) [page 744].
2. Add a parameter [page 312] with the type string to the charge.
3. Set up the visibility *External*.
4. In the *Mapping Table* field of the Mapping Table Introducer component, select this parameter as the name of the mapping table.

Configuration at Charge Plan Level

- Create a charge plan which contains the previously created charge. Note that the parameter is now assigned the type *Mapping Table ID*.
- Add a parameter [page 354] to the charge plan:
  1. Set up the type to *Mapping Table ID*.
  2. Set up the visibility *External* if you want to set the name of the mapping table at provider contract level or from a connected Customer Relationship Management application.
  3. Link this parameter to the parameter of the charge.

The parameter created in the charge plan can be used to set the name of the mapping table or to make the parameter available at provider contract level (parameter of a contract item).

Step 2: Configuration of the Customer Master Data

Configuration at Provider Contract Level

If the visibility of a parameter in a charge plan is set to *External*, a Customer Relationship Management (CRM) application or an external provisioning system can redefine the value of this parameter in a contract item [page 592] (this contract item enables to activate the charge plan in a provider contract) that subscribes to the charge plan in the pricing catalog.

The value of the parameter in the contract item can be:

- The name of a mapping table from the pricing catalog of the contracting service provider
- The name of a subscriber mapping table of the subscriber account

> **Note**
>
> You can use the same procedure to dynamically set a value in the mapping table. In that case, use a parameter with the special type *Mapping Table Key*.

1.14.5 Working with Mapping Tables

Related Information

Creating a Mapping Table [page 187]
Opening an Existing Mapping Table [page 195]
Modifying a Mapping Table [page 196]
Setting Up a Mapping Table [page 188]
Saving a Mapping Table [page 194]
Deleting a Mapping Table [page 197]
Exporting a Mapping Table [page 198]
Importing a Mapping Table [page 198]

1.14.5.1 Creating a Mapping Table

Prerequisites

Before creating a mapping table, you must design and create a mapping table class [page 173] that defines its data structure and suits your business requirements.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Create a Mapping Table

1. From the menu bar in the main window, select *File* > *New* > *Mapping Table*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. Click *OK*. The *New Mapping Table* window opens.
4. Set up [page 188] the mapping table.
5. Verify the mapping table structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 194] your new mapping table.

You can create a mapping table by opening an existing mapping table and copying it.

> ⓘ Note
>
> Once you have saved a mapping table, you can no longer modify its name or mapping table class. You can only modify the rows of the mapping table.

Create a Mapping Table from an Existing Mapping Table

1. From the menu bar in the main window, select *File* > *Open* > *Mapping table*. The *Open Mapping Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. In the *Mapping Tables* folder of the catalog of the owner, select the mapping table to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Mapping Table* window opens.
5. Modify [page 196] the mapping table if necessary.
6. Verify the mapping table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 194] your new mapping table.

Related Information

Setting Up a Mapping Table [page 188]
Saving a Mapping Table [page 194]
Modifying a Mapping Table [page 196]

1.14.5.2 Setting Up a Mapping Table

Create [page 187], open [page 195], or modify [page 196] a mapping table before setting it up. A *Mapping Table* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Window: Mapping Table

To set up a mapping table, you must:

1. Type the identification name of the mapping table, select the appropriate mapping table class [page 167] that defines its data structure, and fill in the Definition [page 189] tab.
2. Depending on the selected class and the business requirements, fill in or modify the following tabs:
   - Table Instance [page 189]
   - Additional Information [page 194]

> → Tip
>
> You can import already-existing data from a CSV file.

1.14.5.2.1 Tab: Definition

Fill in or modify the *Definition* tab:

Name: Type the unique identifier of this mapping table [page 179].

Description: Describe the mapping table.

Mapping Table Class: Select the mapping table class [page 167] that defines the data structure for your mapping table.

Agreement ID: Type the identification code of the related *master agreement*, if any. This ID is usefull in an integrated *SAP Solution* scenario with **SAP S/4HANA for Customer Management 2.0** as of release 2.0. When you set up this field, the mapping table is an *agreement table* and belongs to the identified master agreement. Consult Subscription Order Management in the Product Assistance for SAP S/4HANA for Customer Management 2.0 for more information and user assistance about:

- Master data
- Master agreement
- Available functions

See also the integration guides at help.sap.com/brim.

1.14.5.2.2 Tab: Table Instance

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

The *Table Instance* tab consists of two parts:

- The top part of the tab is the filter area, which allows you to find rows in the mapping table by using filter criteria.
- The bottom part of the tab is the result area, which displays the search results that you perform.

> → Tip
>
> Use the arrows ▲▼ if you want to hide or show the filter area or the result area.

Setting up the filter

> ⓘ Note
>
> When you add, remove, or change one or more rows in the result area, the filter area is no more available (dimmed) until you save your changes.

Use the following icons:

 : To start a new search. Every time you add, remove, or change one or more criteria, this button allows you to start a new search and display the results.

 : To reset the filter. After one or more searches, this button allows you to re-initialize the filter and display all the rows of the mapping table according to the number of results per page and the selected order.

 : To move to the previous page. This button is available when the result list is large.

 : To move to the next page. This button is available when the result list is large.

 : To add a criterion to the filter.

 : To remove a criterion from the filter.

Use the following fields:

Results per Page: Select the number of results that are displayed in the result area.

Identifier Sorted By: Select the order in which you want to display the results.

Found Rows: Displays the number of rows that are found by the search.

Total Number of Rows: Displays the total number of rows of the mapping table that is stored in the database.

Creating, modifying, or removing rows

Use the following icons:

 : To add a row to the mapping table. The new row is added after the selected row in the current page of the result area. Otherwise, the new row is added to the end of the table in the current page of the result area. When you click this button, the filter area is no more available (dimmed) until you save your changes.

 : To remove a row from the table. Select a row in the current page of the result area, and then click this button to remove the row from the table. When you click this button, the filter area is no more available (dimmed) until you save your changes.

 : To import data into the mapping table according to the CSV format (see *).

 : To export data of the mapping table into a CSV file (see **). This button is not available (dimmed):

- If you are modifying the mapping table
- If the mapping table contains no row

 : To remove the added and filtered rows from the mapping table. This button allows you to remove the rows that you have added to the mapping table and the rows that have been found by the last search. When you click this button, a message displays the number of added and filtered rows and asks if you want to remove the rows. The added and filtered rows are removed from the result area but are not deleted from the database until you save the mapping table.

Use the following fields:

Identifier: Type an identifier for the row. This identifier must be unique in the mapping table.

Input Columns: Type the input values for this row. You are allowed to enter duplicate input values. In that case, the selection of a row depends on the *Start Date* and the *End Date*. Note that the number of input columns is the same as the number of input columns defined in the related mapping table class.

Start Date: Click the cell of the *Start Date* column to open a calendar, and then select a date and a time. It is the start date/time of the validity period for this row in the mapping table.

End Date: Click the cell of the *End Date* column to open a calendar, and then select a date and a time. The *Start Date* and the *End Date* settings define the period when the row is valid.

Output Columns: Type the output values for this row. Note that the number of output columns is the same as the number of output columns defined in the related mapping table class.

> → Tip
>
> Instead of manually typing all of the data in the columns, you can import existing data from external CSV files by clicking  (refer to the process described below).

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	> ⓘ Note > Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.

Rule	Verifications
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.

5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

(**) Exporting Data to a CSV File

This procedure allows you to export data from mapping tables or subscriber mapping tables into a CSV file.

Export Data to a CSV File

1. Click the button  . The *CSV File Format Options* dialog box is displayed.
2. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
3. In the *Number Format Parameters* area, select the *Decimal Separator* type (*Point*, *Comma*) and the *Thousands Separator* type (*No*, *Quote*, *Comma*, *Space*, *Point*) to format numbers. If there is no *Thousands Separator*, select *No*.
4. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns (not editable for mapping tables).
5. Click *OK* to export data. The *Save* dialog box is displayed.
6. In the *Save in* field, select the directory where you want to export data.
7. In the *File name* field, type the name of the file, and then click *Save*. If the file already exists, a message asks you if you want to overwrite it.

Restrictions on input values and dates

You are not allowed to put the same input values, the same start date, and the same end date for two distinct rows.

Example

The following table shows you an example of duplicate input value:

Identifier	First input column	Start Date	End Date	First output column
Row 1	A	July, 18th	October, 4th	12
Row 2	A	July, 18th	October, 5th	19
Row 3	A	July, 17th	October, 4th	16

For example, an event occurs on the 20th of July with the input value A. You use the following rules to determine which row of the mapping table applies to the example.

Rule 1: When more than one period correspond to an input value (A) at a date (the 20th of July), the rows whose start date is closest to the reference date applies. In our example, Row 1 and Row 2 are selected because their start date (July, 18th) is closer to the reference date (July, 20th) than the start date of Row 3 (July, 17th).

Rule 2: If two rows have the same start date, the one that has the shortest period applies. In the previous example, Row 1 applies.

1.14.5.2.3 Tab: Additional Information

Use the configuration table and its icons to set up some additional information.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the configuration table:

Name: Type the name of the additional information item that characterizes your mapping table.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the SAP ERP system or the external provisioning system that is connected to SAP Convergent Charging 2021.

1.14.5.3 Saving a Mapping Table

Once you have created or modified a mapping table, and then verified its structure, you are ready to save it. You can save the mapping table either into the back-end database or to an XML file on your hard disk.

Save and Release a New Mapping Table into the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Mapping Table Creation* dialog box opens to inform you that the mapping table has been successfully saved and released.
4. Click *OK*.

Save and Release a New Version of an Existing Mapping Table into the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Choose date* dialog box opens.
2. Select a date from which the modified mapping table will be effective, and then click *OK*.
3. If the *Mapping Table Modification* dialog box opens to inform you that the mapping table has been successfully modified and released, click *OK*.

4. If a *Warning* dialog box opens to inform you that the mapping table has been modified since you opened it, you can:
    1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the mapping table.
    2. Click *Cancel* to come back to the mapping table. See later in this chapter how to save the mapping table as an XML file.

Save a Mapping Table as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the mapping table.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The mapping table is saved as an XML file in your selected directory.

1.14.5.4 Opening an Existing Mapping Table

You can open:

- An existing mapping table saved in the database
- A previous version of a mapping table saved in the database
- An existing mapping table saved in an XML file

Open an Existing Mapping Table Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Mapping Table*. The *Open Mapping Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. In the *Mapping Tables* folder of the catalog of the owner, select the mapping table. The *Mapping Table* window opens in display mode.

Open an Existing Mapping Table Saved in an XML File

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.

- In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.14.5.5 Modifying a Mapping Table

> ⚠ Caution
>
> Do not modify a mapping table created by a Customer Relationship Management or another provisioning system interfaced with SAP Convergent Charging. Refer to the caution about Mapping Tables [page 179].

You can modify a mapping table by adding, modifying, or deleting content, but you cannot modify the name or the mapping table class of a mapping table once you have saved it.

Modify a Mapping Table

1. From the menu bar in the main window, select *File* > *Open* > *Mapping Table*. The *Open Mapping Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. In the *Mapping Tables* folder of the catalog of the owner, select the mapping table.
4. Click *Open*. The *Mapping Table* window opens in display mode.
5. Set the edit mode by clicking the *Edit Mode* icon on the main toolbar. The mapping table may be reloaded if it has been modified since you have opened it.
6. In the Table Instance tab [page 189], set up the rows of the mapping table.

   > ⓘ Note
   >
   > When verifying the mapping table, you verify that the edited page contains no error. However, this check does not apply to the other pages of the mapping table. Therefore, when you save the mapping table, a message is displayed for each found error and asks you to correct this error.

7. Verify the mapping table structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
8. Save [page 194] the mapping table.

Related Information

Display/Edit Mode [page 46]
Setting Up a Mapping Table [page 188]
Saving a Mapping Table [page 194]

1.14.5.6 Deleting a Mapping Table

> ⚠ Caution
>
> Do not delete a mapping table created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about mapping tables [page 179].

> → Remember
>
> As of SAP Convergent Charging 2022 FPS 2, for performance reasons, the system does not check contract dependencies anymore. Remember that deleting a mapping table corrupts the provider contracts or allowances still using it. You must ensure that the mapping table is not referenced anywhere before deletion.

Before you delete a mapping table, you must first delete all master data objects which reference it.

To delete a mapping table

1. From the menu bar in the main window, select *File* > *Delete* > *Mapping Table*. The *Delete Mapping Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. In the *Mapping Tables* folder of the catalog of the owner, select the mapping table.
4. Click *Delete*. A deletion dialog box asks if you want to delete the mapping table
5. Confirm the deletion of the mapping table by clicking *Yes*:
    1. If a dialog box confirms your choice, the mapping table has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the mapping table is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> ⓘ Note
>
> If an error message appears after clicking *Yes* at step 5, you must delete the objects that reference the mapping table. The types of these master data objects appear in the message.

1.14.5.7 Exporting a Mapping Table

You can export mapping tables to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import [page 198] the mapping table back into SAP Convergent Charging 2023.

To export a mapping table

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the mapping table.
3. In the catalog of the owner, select the mapping table.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the mapping table.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.14.5.8 Importing a Mapping Table

You can import mapping tables in XML format from a directory in your database.

Note that only one version of a mapping table can exist in a database. You can import the unique version of a mapping table class into SAP CC *Core Tool*.

To import a mapping table

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the mapping table is saved, and then select the mapping table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected.
6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

> **Note**
>
> SAP CC *Core Tool* can only import an object which has previously been exported [page 198] and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).

This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.15 Range Table Classes

 A **range table class** in SAP Convergent Charging (SAP CC) allows you to define a **common data structure for some range tables** (in your pricing catalog) and subscriber range tables (in customer master data). When you model and design some pricing logic based on these data lookup tables, a range table class is required. A range table class defines the following characteristics for the corresponding data tables:

- Number (0 to 5) of input columns (string values as keys for comparisons)
- Number (1 to 15) and types [page 207] of output columns (string or number values)
- Some options to define if:
  - The upper bound of the ranges (intervals) is inclusive (default) or exclusive.
  - The last interval of the ranges is bounded (default) or not (+ Infinite).

These settings depend on your pricing modeling and design.

A range table class is included in the pricing catalog [page 109] of the service provider and is part of its master data.

See how to create [page 205], set up [page 207], save [page 209], and delete [page 212] a range table class.

> ⓘ Note
>
> Once you have saved a range table class, you can no longer modify its table schema when the class is referred by a range table or a subscriber range table. You can always change the description [page 207] and the additional information items [page 209].

Use

You use range table classes when you want to base some pricing logic on data available in range tables or subscriber range tables. A corresponding class is expected.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

SAP Solution

In an integrated *SAP Solution* scenario, the master data includes some range tables and corresponding classes. Refer to the integration guide that relates to your SAP system landscape.

Versions

SAP Convergent Charging 2023 does not manage several versions of a range table class.

Transport

You can use the Catalog and Data Transport feature to transport a consistent subset of data from a source system to target systems (as destinations). See Transporting Catalog and Data [page 109].

Otherwise, you can:

- Export [page 212] a range table class
- Import [page 213] a range table class

Maintenance

You can conditionally change [page 211] a range table class.

Related Information

Working with Range Table Classes [page 204]

1.15.1 Industry Examples of Range Table Classes

This topic presents some typical business use cases for different industries.

> **Note**
>
> This section details the range table classes that are necessary to set up the examples of corresponding range tables [page 217].

Telecommunications Industry

Example #1: Tiered Pricing

> **Example**
>
> A company sells voice services that apply decreasing tariffs according to the duration of the phone calls.
>
> You translate this marketing and pricing policy by modeling and designing some range tables and classes in SAP Convergent Charging.
>
> You design a range table class that defines range tables whose last ranges are unbounded. The class specifies an output column that is named Price for numerical values.
>
> The computation mode for the values of this column is "range size cumulative" to add up the prices of each selected range.
>
> As a pricing specialist, you design and configure the following range table class:
>
> - Upper bounds of ranges: inclusive
> - Upper bound of the last range: unbounded (+Infinite)
> - You set up the "Price" output column, type of number.
>
> The class defines no input columns.
>
> See next: Example of Tiered Pricing with a Range Table [page 218]

Example #2: Mutlidimensional Pricing

> **Example**
>
> A company sends threshold notifications to end customers depending on their consumption of services.
>
> You set up a range table class with input columns (prepaid or postpaid and service type), one output column (notification to be sent), and you define the Upper Bound of Range as inclusive and the Upper Bound of Last Range as Unbounded. You also define the type of the output column as String.
>
> See next: Example of Multidimensional Pricing with a Range Table [page 220]

Transports Industry

> **Example**
>
> A company sells postal services whose parcel postal tariffs are priced according to the weight of the parcel (weight in kilogram) that is limited to 20 kg.
>
> As a pricing specialist, you design and configure the following range table class:
>
> - Upper bounds of ranges: inclusive
> - Upper bound for the last range: bounded
> - Computation mode: single (none)

- You set up the 'Gold Price' output column, type of number.

See next: Transports Industry [page 220]

Example of Range Table Class Dedicated to Spending Status Monitoring

Example

You implement policy control in your system landscape to ensure fair usage and to optimize the network bandwidth allocation with predefined policy rules.

The appropriate 'DATA_FAIR_USAGE' spending status description [page 485] exists in the SAP CC system.

You must configure a range table and its class to be able to customize the necessary monitoring plans and set up the generation of a spending status that is based on this range table.

You configure the 'RTC dedicated to data monitoring' range table class that defines a particular structure that is relevant for configuring range tables that can be used to configure monitoring plans and to set up the generation of spending statuses.

You set up the following range table class:

- Codename: RTC dedicated to data monitoring
- Upper bounds of ranges: inclusive
- Upper bound for the last range: unbounded (+Infinite)
- You set up a unique **LABEL** output column; The type must be string [page 206].

Note

In this example, spending thresholds are included in the spending limits.

You can set up the "RT dedicated to data monitoring" range table in SAP CC. See in the Industry Examples of Range Tables [page 217].

See next: Example of a Range Table Dedicated to Spending Status Monitoring [page 223]

Related Information

Industry Examples of Range Tables [page 217]
Example of a Monitoring Plan [page 493]

1.15.1.1 Pricing for National and International Shipping

Example

A company sells postal services. The business model is based on the format of the letters and parcels, the weight, and shipping options. The marketing team defines some different pricing and discounting rules. For letters, the shipping options are economy, green, and priority mail. For parcel posts, the shipping options are the destination zones (national or international zone).

As the pricing specialist for this company, you translate this pricing policy to SAP Convergent Charging concepts and capabilities. You decide to implement range tables with two input columns for multidimensional pricing.

You design and configure the following range table class *Postal Services - National/International Letter and Parcel Pricing Structure*:

- Class Options:
  - Range upper bounds: included (aka inclusive)
  - Last range: bounded
- Input Columns:
  - **`Format`** (letter or parcel)
  - **`Shipment Class`** (economy/green/priority for letters and national/zoneA/zoneB/zoneC for parcels)
- Output Columns:
  - **`Number of Stamps`** (Number / no computation mode)
  - **`Total Price`** (Number / no computation mode)

Input columns:			**Output columns:**	
Format	**Shipment**	**Range**	**Total Stamps**	**Total Price**
"letter"	"economy"	] 0, 20]	1	0.8
"letter"	"economy"	] 20 , 100]	2	1.72
"letter"	"economy"	] 100 , 250]	4	3.44
"letter"	"green"	] 0 , 20]	1	0.88
"letter"	"green"	] 20 , 100]	2	1.76
"letter"	"green"	] 100 , 250]	4	3.52
"letter"	"green"	] 250 , 500]	6	5.28
"letter"	"green"	] 500 , 3,000]	8	7.04
"letter"	"priority"	] 0 , 100]	2	2.1
"letter"	"priority"	] 100 , 250]	4	4.2
"letter"	"priority"	] 250 , 500]	6	6.3
"letter"	"priority"	] 500 , 3,000]	8	8.4
"parcel"	"national"	] 0 , 250]	0	4.95

Input columns:			Output columns:	
Format	**Shipment**	**Range**	**Total Stamps**	**Total Price**
"parcel"	"national"	] 250 , 500]	0	6.25
"parcel"	"national"	] 500 , 750]	0	7.1
"parcel"	"national"	] 750 , 1,000]	0	7.8
"parcel"	"zoneA"	] 0 , 500]	0	12.3
"parcel"	"zoneA"	] 500 , 1,000]	0	15.2
"parcel"	"zoneA"	] 5,000 , 10,000]	0	36.3
"parcel"	"zoneA"	] 10,000 , 20,000]	0	60.3
"parcel"	"zoneA"	] 20,000 , 30,000]	0	60.3
"parcel"	"zoneB"	] 0 , 500]	0	16.65
"parcel"	"zoneB"	] 500 , 1,000]	0	19.9
"parcel"	"zoneB"	] 5,000 , 10,000]	0	46.25
"parcel"	"zoneB"	] 10,000 , 20,000]	0	72.15
"parcel"	"zoneC"	] 0 , 500]	0	24.35
"parcel"	"zoneC"	] 500 , 1,000]	0	27.1
"parcel"	"zoneC"	] 10,000 , 20,000]	0	164.6

1.15.2 Working with Range Table Classes

Related Information

Creating a Range Table Class [page 205]
Using a Range Table Class with the Spending Status Monitoring Function [page 206]
Opening an Existing Range Table Class [page 210]
Modifying a Range Table Class [page 211]
Setting Up a Range Table Class [page 207]
Saving a Range Table Class [page 209]
Deleting a Range Table Class [page 212]
Exporting a Range Table Class [page 212]
Importing a Range Table Class [page 213]

1.15.2.1 Creating a Range Table Class

Use one of these procedures to create a range table class.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Create a Range Table Class

1. From the menu bar in the main window, select *File* > *New* > *Range Table Class*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the range table class.
3. Click *OK*. The *New Range Table Class* window opens.
4. Set up [page 207] the range table class.
5. Verify the range table class structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 209] your new range table class.

> → Tip
>
> You can create a range table class by opening an existing range table class and copying it.

> ⓘ Note
>
> Once you have saved a range table class, you can no longer modify its table schema as soon as the class is referred by a Range Table [page 213] or a Subscriber Range Table [page 557]. You can always modify the description and the additional information.

Create a Range Table Class from an Existing Range Table Class

1. From the menu bar in the main window, select *File* > *Open* > *Range Table Class*. The *Open range table class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table class.
3. In the *Range Table Classes* folder of the catalog of the owner, select the range table class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding range table class window opens.
5. Modify [page 211] the range table class if necessary.
6. Verify the range table class structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.

7. Save [page 209] your new range table class.

Related Information

Setting Up a Range Table Class [page 207]
Saving a Range Table Class [page 209]
Modifying a Range Table Class [page 211]

1.15.2.2 Using a Range Table Class with the Spending Status Monitoring Function

> → Remember

When designing Monitoring Plans [page 491], you must design and configure a particular range table [page 227] and its class to set up the generation of spending statuses.

> Example
>
> See the topic Industry Examples of Range Table Classes [page 200].

To create a range table class for a monitoring plan

This procedure consists in creating a particular range table class for configuring dedicated range tables that are used in monitoring plans.

1. Create [page 205] a range table class.
2. Set up [page 207] the following:
   - Set *Upper Bound of Range* to *Inclusive* or *Exclusive* depending on the business requirements
   - Set *Upper Bound of Last Range* to *Unbounded* so that when the SAP CC system executes the related range table, this determines an output value.
   - Add a unique output column with the string type.

Next Task

You can configure a range table that relates to this class [page 227].

1.15.2.3 Setting Up a Range Table Class

Create [page 205] a range table class before setting it up. A *Range Table Class* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

The range table class allows you to create the structure of your range tables [page 213] (and subscriber range tables [page 557] for end customers), which can consist of 0 to 5 input columns maximum of type string and 15 output columns maximum.

> ⓘ Note
>
> Once you have saved a range table class, you can no longer modify its table schema if the class is referenced by a range table or a subscriber range table.
>
> You can always modify the description and the additional information.

Window: Range Table Class

To set up a range table class, you must fill in or modify the following tabs:

- Definition [page 207]
- Table Schema [page 208]
- Additional Information [page 209]

Related Information

Setting Up a Range Table [page 228]

1.15.2.3.1 Tab: Definition

Name: Type the unique identifier of the range table class [page 199].

Description: Describe the range table class. This description is mandatory.

1.15.2.3.2 Tab: Table Schema

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

You set up the number of input columns and output columns, and the characteristics of all the numerical intervals of all the business data tables (range tables, subscriber range tables) that are based on this class:

Upper Bound of Range: Set up if the upper bounds of the ranges are inclusive (default) or exclusive.

Upper Bound of Last Range: Set up if the upper bound of a last range of a table is bounded (default) or not.

Use the following icons to fill in the table *Input Columns* and *Output Columns* with at least one row:

 Add a new row to the table.

 Delete a row from the table.

 Move a row up.

 Move a row down.

Input Columns

- *Name*: Type the input column name. This name must be unique within the set of input columns.
- *Description*: Describe the input column. This description is mandatory.

Output Columns

- *Name*: Type the output column name. This name must be unique within the set of output columns.
- *Description*: Describe the output column. This description is mandatory.
- *Type*: Select the type of the output column and the associated computation mode (see the Industry Examples of Range Tables [page 217]):
  - *String*: The input property value determines a row of the range table. The output column values of this row are assigned to the generated properties.
  - *Single Number*: The input property value determines a row of the range table. The output column values of this row are assigned to the generated properties.
  - *Cumulative Number*: This type is useful when you want to design pricing based on the cumulativeness of prices defined by ranges. The input property value determines a row of the range table. The output column values of this row are assigned to the generated properties. For each output column, the output column value of this row and the values of the output column of the previous rows are added together. The resulting values of these sums are then assigned to another generated properties.
  - *Range Size Cumulative Number*: This type is useful when you want to design cumulative tiered pricing that is based on a quantity (time) multiplied by a unit price. The input property value determines a row of the range table. The output column values of this row are assigned to the generated properties. For each output column, the output column value of this row and the values in the output column of the previous rows are processed according to the linear formula "a*(b-c)" or "a*size" (where "a" is the output column value, "b" is the property value, and "c" is the lower boundary of the range that includes

the property value, and "size" is the size of the range (difference between the lowed boundary and the upper boundary) and added together. Each value is multiplied by the size of the range. The resulting values are then assigned to another generated properties.

> → Tip
>
> For more information, see Range Table Introducer (Comparator) [page 751].

Related Information

Range Table Introducer (Comparator) [page 751]

1.15.2.3.3 Tab: Additional Information

Use the configuration table and its icons to set up some additional information items for your range table class.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information item.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the CRM application or the external provisioning system that is connected to SAP Convergent Charging 2021.

1.15.2.4 Saving a Range Table Class

Once you have created a range table class and then verified its configuration, you are ready to save it. You can save the range table class either into the back-end database or to an XML file on your hard disk.

To save and release a new range table class into the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.

2. Select the name of the catalog.
3. Click *Save*. The *Range Table Class Creation* dialog box opens to inform you that the range table class has been successfully saved and released.
4. Click *OK*.

To save and release an existing range table class into the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Range Table Class Modification* dialog box opens to inform you that the range table class has been successfully modified and released, click *OK*. The new version of the range table class is saved as the only current version.
3. If a *Warning* dialog box opens to inform you that the range table class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the range table class.
   - Click *Cancel* to come back to the range table class. See below how to save the range table class as an XML file.
4. Click *Save*. The *Range Table Class Modification* dialog box opens to inform you that the range table class has been successfully created.
5. Click *OK*.

To save a range table class as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the range table class.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The range table class is saved as an XML file in your selected directory.

1.15.2.5 Opening an Existing Range Table Class

You can open:

- An existing range table class saved in the database
- An existing range table class saved in an XML file

To open an existing range table class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Range Table Class*. The *Open Range Table Class* dialog box opens.

2. In the *Owner* drop-down menu, select the owner of the range table class.
3. In the *Range Table Classes* folder of the catalog of the owner, select the range table class. The *Range Table Class* window opens in display mode.

To open an existing range table class saved in an XML file

- Use the procedure for Importing a Range Table Class [page 213]

Related Information

Display/Edit Mode [page 46]

1.15.2.6 Modifying a Range Table Class

You can modify a range table class by adding, modifying, or deleting content. The modified range table class replaces the previous class in the database. Only one version is stored.

- You cannot modify the name of a range table class.
- You can always modify the description and the additional information.
- You can modify the table schema (ranges, upper bound inclusion, input and output columns) as long as no range table [page 213] or subscriber range table [page 557] relates to this class.

To modify a range table class

1. From the menu bar in the main window, select *File* > *Open* > *Range Table Class*. The *Open Range Table Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table class.
3. In the *Range Table Classes* folder of the catalog of the owner, select the range table class.
4. Click *Open*. The *Range Table Class* window opens in display mode.
5. Set the editing mode by clicking the *Edit Mode* icon on the main toolbar. The range table class may be reloaded if it has been modified since you have opened it.
6. Set up [page 207] the range table class.
7. Verify the range table class structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
8. Save [page 209] the range table class.

Related Information

Display/Edit Mode [page 46]
Setting Up a Range Table Class [page 207]
Saving a Range Table Class [page 209]

1.15.2.7 Deleting a Range Table Class

Before you delete a range table class, you must first delete all objects which reference it.

To delete a range table class

1. From the menu bar in the main window, select *File* > *Delete* > *Range Table Class*. The *Delete Range Table Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table class.
3. In the *Range Table Classes* folder of the catalog of the owner, select the range table class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the range table class
5. Confirm the deletion of the range table class by clicking *Yes*:
   1. If a dialog box confirms your choice, the range table class has been successfully deleted.
   2. If a *Warning* dialog box opens to inform you that the range table class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the objects that reference the range table class. The identifiers of these objects appear in the message.

1.15.2.8 Exporting a Range Table Class

You can export range table classes to a directory of your hard disk, where they are saved as XML files. After exporting range table classes, you can import [page 213] them back into another SAP CC system.

To export a range table class

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table class.

3. In the catalog of the owner, select the range table class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the range table class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.15.2.9 Importing a Range Table Class

You can import range table classes in XML format from a directory to your database.

Note that there is only one version of a range table class in the database because a range table class cannot be modified once it has been referenced in a range table or a subscriber range table.

To import a range table class

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the range table class is saved, and then select the range table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected, as there is only one version for each range table class.
6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

1.16 Range Tables

> ⚠ Caution
>
> **Do not modify or delete range tables with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Modifying or deleting range tables may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. To modify a range table, you must use this CRM or provisioning system in your SAP system landscape.
>
> - Consult the Application Help for more information about the product modeling process.
> - Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this master data in the pricing catalog of a the service provider for its subscription- and usage-based services.
>
> The manual procedures detailed in this section are relevant in:

- A production system landscape with few contracts directly managed with the Core Tool user interface (no CRM interfaced with SAP CC)
- A test, demo, or training system landscape

 A **range table** in SAP Convergent Charging (SAP CC) is a multicolumn table of business/technical data that consists of a set of 0 to 5 input columns of type string, a range definition column, and a set of 1 to 15 output columns. A range table is a data lookup table that the pricing specialist. Each row of output columns is defined by a range (numerical interval) to which corresponds a set of values in the related output columns (string or number).

At runtime, SAP Convergent Charging uses range tables to compare a quantity of services with ranges (aka tiers) in a selected range table. When the range table contains some input columns, the system initially operates a preliminary comparison between information from the rating context with data in the input columns.

Range Table	Input Columns	Service Quantity Comparison	Multidimensional Pricing
Basic Table	None	Yes	No
Advanced Table	1 to 5 for comparison keys	Yes	Yes

The input columns contain comparison keys. The output columns represent prices, unit prices, or credits. You can also apply combinatorial rules (cumulative, range size cumulative) to numerical output columns to compute supplementary values. You must however configure the pricing logic that defines the computation algorithm of the final values. See the logic component Range Table Introducer (Comparator) [page 751].

A range table is included in the pricing catalog [page 109] of the service provider and is part of its master data. A range table allows you to create new *properties* (in the rating context [page 13]) that you can use in the price plan [page 299], in the monitoring plan [page 491], in the refill logic [page 395], or in the allowance logic [page 449].

You define the data structure of your range tables in a range table class [page 199].

Use

You use range tables to configure the pricing logic that includes tiered pricing with convergent charging services that are based on provider contracts. If the tiered pricing is based on customer data, you must use subscriber range tables [page 557] that belong to a subscriber account [page 511].

For industry-specific examples, see Industry Examples of Range Tables [page 217].

→ Recommendation

For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

> ⓘ Note
>
> You can easily use and directly select an existing range table or select the appropriate table in a charge plan [page 343]. It is also possible to redefine this selection in each provider contract [page 573] by referring to a range table or a subscriber range table.

Range Table Step by Step

Before creating a range table, you must create [page 205], set up [page 207], and save [page 209] a range table class. You perform the following steps to use the range tables:

1. Create [page 225] a range table.
2. Set up [page 228] the range table.
3. Save [page 235] the range table.
4. Use the range table in one or several price plans (or in the refill logic) by setting up a dedicated logic component: Range Table Introducer (Comparator) [page 751]. We recommend that you determine the appropriate customization level of the range table [page 224] that must be processed by the SAP CC system.
5. Delete [page 238] the range table.

> → Tip
>
> You can use the same range table class for configuring different range tables.

After you have created a range table, you can then add [page 330] it to the price plan of a charge [page 290] or to the refill logic. You can use the same range table more than once in any price plan or refill logic.

SAP Solution

In an integrated *SAP Solution* scenario, the master data includes some range tables and corresponding classes. Refer to the integration guide that relates to your SAP system landscape.

Agreement Tables

See Agreement Tables [page 216].

Integration

This master data may be created and maintained by an external system connected to the SAP CC system during the product modeling process.

> ✤ Example
>
> In an integrated *SAP Solution* scenario, SAP CRM manages some range tables that are stored in SAP Convergent Charging.

Versions

You can modify and save different versions of a range table [page 240].

Transport

You can use the Catalog and Data Transport feature to transport a consistent subset of data from a source system to target systems (as destinations). See Transporting Catalog and Data [page 109].

Otherwise, you can:

- Export [page 239] a range table and export [page 212] its class
- Import [page 239] a range table

Maintenance

You can modify and delete the rows of a range table, but you will save a new version of the range table. You cannot change the name of the range table or the name of the related range table class.

Related Information

Industry Examples of Range Tables [page 217]
Customized Use of Range Tables in the Charges and Refill Logic [page 224]
Working with Range Tables [page 225]

1.16.1 Agreement Tables

In an integrated *SAP Solution* scenario with SAP SOM (SAP Subscription Order Management) in Convergent Invoicing, an agreement table is a particular mapping table or a particular range table stored in SAP CC that allows sharing data between several subscriber accounts.

> → Remember
>
> Agreement tables are part of customer master data. For technical and performance reasons, you are able to manage them like mapping or range tables.
>
> - To create an agreement table, use the procedures for creating a mapping table or a range table.
> - To search for an existing agreement table, use the procedures for finding a mapping table or a range table and by filtering the `Agreement ID` information.

1.16.2 Industry Examples of Range Tables

This topic presents some typical business use cases based on range tables and classes to model tiered pricing in the pricing logic:

Example	SAP Industry	Boundary	Computation Mode
Example of Tiered Pricing with a Range Table [page 218]	*SAP for Telecommunications*	Last range is unbounded	Range Size Cumulative
Transports Industry [page 220]	*SAP for Transportation and Logistics*	Last range is bounded	None (aka single)
Loyalty Management [page 222]		Last range is bounded	Cumulative
Spending status monitoring [page 223]		Last range is unbounded	n/a

You can configure the pricing logic by creating a range table [page 225] or a set of range tables that is added to the pricing catalog [page 109] of a service provider or by creating subscriber range tables [page 557] that are assigned to a subscriber account [page 511] and are dedicated to some provider contracts [page 573] that belong to this account.

Related Information

Industry Examples of Range Table Classes [page 200]
Example of a Monitoring Plan [page 493]
Spending Status Monitoring [page 25]

1.16.2.1 Telecommunications Industry

Related Information

Example of Tiered Pricing with a Range Table [page 218]
Example of Multidimensional Pricing with a Range Table [page 220]

1.16.2.1.1 Example of Tiered Pricing with a Range Table

Example

A company sells voice services that apply decreasing tariffs according to the duration of the phone calls.

- The first 2 minutes are charged at €0.20 per minute.
- The following 4 minutes are charged at €0.1 per mn.
- The following 24 minutes are charged at €0.05 per mn.
- The following minutes are charged at €0.01 per mn.

The pricing specialist for this company decides to model this marketing and pricing policy by using range tables in SAP CC. The specialist designs and configures the following range table and its class:

Range (in mn)	Unit Price (defined per mn)
]0, 2]	0.2
]2, 6]	0.1
]6, 30]	0.05
]30, +∞[	0.01

The following configuration applies:

- Choice of a range table class that defines range tables whose last ranges are unbounded. The class specifies an output column that is named **`Unit Price`** for numerical values. The computation mode for the values of this column is **`range size cumulative`** to add up the prices of each selected range.
- Pricing logic concept, in which the final price is the sum of the cumulative price and the unit tariff multiplied by the position in the range that includes the input quantity value (call duration).
- Linear Function component [page 733] to set up the final price: Final price = cumulative price + unit price (input quantity value - lower bound of the current range)

In this example, the final prices are computed by the SAP CC system:

Call Duration	1 mn	2 mn	3 mn	4 mn	5 mn	10 mn	20 mn	40 mn
Generated properties								
Unit price	0.2	0.2	0.1	0.1	0.1	0.05	0.05	0.01
Price (cumulative)	0	0	= 0.2 (2-0) = 0.4	= 0.4	= 0.4	= 0.2 (2-0) + 0.1 (6-2) = 0.4 + 0.4 = 0.8	= 0.7	= 0.7 + 1.2 = 1.9
Introduced properties								

Call Duration	1 mn	2 mn	3 mn	4 mn	5 mn	10 mn	20 mn	40 mn
Lower bound of the range	0	0	2	2	2	6	6	30
Beyond lower bound	= 1 - 0 = 1	= 2-0 = 2	= 3 -2 = 1	= 4-2 = 2	= 5-2 = 3	= 10 - 6 = 4	= 20 - 6 = 14	= 40 - 3 = 10
Final prices	= sum of price (cumulative) and prorated unit price							
Computed by a component function added to the branch "Is in a range"	= 0 + 0.2 (1-0) = 0 + 0.2 = 0.2	= 0.4	= 0.4 + 0.1 (3-2) = 0.4 + 0.1 = 0.5	= 0.6	= 0.4 + 0.1 (5-2) = 0.4 + 0.3 = 0.7	= 0.8 + 0.05 (10-6) = 0.8 + 0.2 = 1.0	= 1.4	= 1.9 +0.1 = 2.0

Note

The previous range table replaces a long price list such as:

Call duration	Total price
1 mn	0.20
2 mn	0.40
3 mn	0.50
4 mn	0.60
5 mn	0.70
6 mn	0.80
7 mn	0.85
8 mn	0.90
9 mn	0.95
10 mn	1.00
11 mn	1.00
12 mn	1.00
13 mn	1.00
...	...

1.16.2.1.2 Example of Multidimensional Pricing with a Range Table

Example

This example demonstrates the use of **input columns** for range tables.

A company sends threshold notifications to end customers depending on their consumption of services.

The pricing specialist for this company translates the marketing and pricing policy and decides to model, design, and configure a range table and its associated class.

- Two input columns determine whether the service is:
  - Prepaid or postpaid
  - Of type LTE (Long-Term Evolution) or not
- The ranges (or tiers) define percentages of consumption.
- And an output column (of type string) determines whether a notification is sent to end customers depending on their consumption of services.

Input Columns		Ranges		Output Column
Key 1	**Key 2**	**Lower Bound**	**Upper Bound**	**Notification to Be Sent**
PREPAID	LTE	0	50	No
PREPAID	LTE	51	90	Yes
PREPAID	LTE	91	100	Yes
PREPAID	LTE	100	+Infinity	Yes
POSTPAID	LTE	0	80	No
POSTPAID	LTE	81	100	Yes
POSTPAID	LTE	100	+Infinity	Yes

1.16.2.2 Transports Industry

Example

A company sells postal services whose parcel postal tariffs are priced according to the weight of the parcel (weight in kilogram) limited to 20 kg.

Range	Gold Price
]0 , 2.5]	65
]2.5 , 8]	102

Range	Gold Price
]8 , 20]	139

The following configuration applies:

- Computation mode: none (aka single mode)
- Pricing logic: a Flat Function component [page 729] to set up the final price

In this example, the final computed prices for different weights are:

Parcel Weight	-1	0	2.5	3	8	10	50
Total Price	Denied	65	65	102	102	139	Denied

See also:

- Flat Function [page 729]
- Working with Logic Components [page 671]

1.16.2.3 Pricing for Pay as You Grow Offering - Monthly or Annually

Example

A company loans some builds to other companies and must charge them for the usage.

As the pricing specialist for this company, you translate this pricing policy to SAP Convergent Charging concepts and capabilities. You decide to implement range tables.

You design and configure the following range table class *"Pay as you Grow - Monthly and Annual Fee Structure"*:

- Class Options:
  - Range upper bounds: excluded (aka exclusive)
  - Last range: unbounded - The last bound is +Infinity (+∞).
- Input Columns:
  - None
- Output Columns:
  - `Pay as You Grow` (String)
  - `Annual Fees` (Number, no computation mode)
  - `Monthly Fees` (Number, no computation mode)

This class defines the structure for basic range tables with no input columns.

The pricing specialist configures the following range table *Pay as you Grow - Monthly and Annual Fees*:

Range	Pay as You Grow	Annual Fees	Monthly Fees
[0, 1,000 [	"Start"	12.3	15
[1,000, 2,500 [	"Plus"	20.5	25
[2,500, 10,000 [	"Max"	53.3	65
[10,000, +∞ [	"Ultimate"	95.5	110

1.16.2.4 Loyalty Management

Example

A company provides end customer value-added services and want to manage a loyalty program based on the usage of services, for which cumulative bonus points are attributed to some end customers according to the level of price that is computed for the consumption of the services.

The marketing team defines the following bonus rules:

- For silver customers, a bonus is credited each time the service price reaches another tier in the following table:

Price Range	Bonus Point
]0, 2]	1
]2, 4]	10
]4, 6]	20

  It is similar than tiered pricing.

- For gold customers, an extra bonus (+5 points) is credited when the upper bound for a price range is reached
- For platinum customers, a mega bonus (+50 points) is credited when the last price range is exceeded

The following example provides a second option, for which special bonus that are attributed under certain conditions are granted to end customers (the upper bound of a price range is reached or the last range is exceeded).

The following configuration applies:

- Computation mode: cumulative
- Pricing logic: Flat Function component [page 729] to add up the single price and the cumulative price

In this example, the following result table shows granted bonus points according to different service consumptions and silver/gold/platinum levels:

Service Consumption Level (Charged Price)	1	2	3	4	5	6	10
Total bonus points (silver)	0	0	1	1	11	11	31
Total bonus points (gold)	1	1	11	11	31	31	31
Total bonus points (platinum)	1	6	11	16	31	36	81

See also:

- Flat Function [page 729]
- Working with Logic Components [page 671]

1.16.2.5 Example of a Range Table Dedicated to Spending Status Monitoring

Example

You implement policy control in your system landscape to ensure fair usage and to optimize the network bandwidth allocation with predefined policy rules.

You must configure a range table and its class to be able to customize the necessary monitoring plans and set up the generation of a spending status that is based on this range table.

You configure the "RT dedicated to data monitoring" range table that defines the spending thresholds to be associated to a status label that is defined in the 'Data fair usage' spending status description [page 485]. The range table is based on the 'RTC dedicated to data monitoring' range table class [page 200].

The range table is defined by the numerical intervals (ranges):

Range (Spending Thresholds)	LABEL (Spending Status Labels)
]0 -- 2,000]	QUOTA_OK
]2,000 -- 2,500]	QUOTA_WARNING
]2,500 -- +∞[	QUOTA_REACHED

1.16.3 Customized Use of Range Tables in the Charges and Refill Logic

This topic allows you to create the charges or the refill logic that can be used with different range tables during the charging or refilling operations.

The procedure below describes how to create a charge [page 290] that contains a price plan with a Range Table Introducer [page 751] (Comparator component) without directly specifying the name of the range table to be used. The idea is to use a parameter as the name of a range table and set this parameter later on at charge plan [page 343] or provider contract [page 573] level. You can easily use different range tables or subscriber range tables [page 557].

> **Note**
>
> You can use the same procedure as that of the refill logic.

Configuration in the pricing catalog (master data related to a service provider)

Configuring at charge or refill logic level

1. Create a charge with a Range Table Introducer (Comparator) [page 751] component (See Working with Logic Components [page 671] for more information on this price plan component).
2. Add a parameter [page 312] with the type string to the charge.
3. Set the visibility to *External*.
4. In the *Range Table* field of the Range Table Introducer component, select this parameter as the name of the range table.

Configuring at charge plan level

1. Create a charge plan that contains the previously created charge. Note that the parameter is now assigned the type *Range Table ID*.
2. Add a parameter [page 354] to the charge plan:
    1. Set the type to *Range Table ID*
    2. Set the visibility to *External* if you want to set the name of the range table at provider contract level or from a connected CRM application
    3. Link this parameter to the parameter of the charge

The parameter that is created in the charge plan can be used to set the name of the range table or to make the parameter available at provider contract level (parameter of a contract item).

Configuration of the customer master data

Configuring at provider contract level

If the visibility of a parameter in a charge plan is set to *External*, a CRM application or an external provisioning system can redefine the value of this parameter in a contract item (this contract item enables to activate the charge plan in a provider contract).

The value of the parameter in the contract item [page 616] can be:

- The name of a range table from the pricing catalog of the service provider
- The name of a subscriber range table [page 557] of the subscriber account that represents an end customer of the service provider

1.16.4 Working with Range Tables

Related Information

Creating a Range Table [page 225]
Using a Range Table in Monitoring Plans [page 227]
Opening an Existing Range Table [page 236]
Changing the Version of a Range Table [page 240]
Modifying a Range Table [page 237]
Setting Up a Range Table [page 228]
Filling In a Range Table [page 232]
Saving a Range Table [page 235]
Deleting a Range Table [page 238]
Exporting a Range Table [page 239]
Importing a Range Table [page 239]

1.16.4.1 Creating a Range Table

Prerequisites

Before creating a range table, you must design and create a range table class [page 205] that suits your business requirements.

→ Remember

For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Create a Range Table

1. From the menu bar in the main window, select *File* > *New* > *Range Table*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. Click *OK*. The *New Range Table* window opens.
4. Set up [page 228] the range table.
5. Verify the range table structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 235] your new range table.

You can create a range table by opening an existing range table and copying it.

Note

Once you have saved a range table, you can no longer modify its name or the related range table class. You can only add rows or modify the rows of the range table.

Create a Range Table from an Existing Range Table

1. From the menu bar in the main window, select *File* > *Open* > *range table*. The *Open Range Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the *Range Tables* folder of the catalog of the owner, select the range table to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Range Table* window opens.
5. Modify [page 237] the range table if necessary.
6. Verify the range table structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
7. Save [page 235] your new range table.

Related Information

Industry Examples of Range Tables [page 217]
Setting Up a Range Table [page 228]
Saving a Range Table [page 235]

Modifying a Range Table [page 237]

1.16.4.2 Using a Range Table in Monitoring Plans

When designing monitoring plans [page 491], you must design and configure a particular range table (and its class [page 206]).

Example

You can configure the following range table:

Range	Output
]0, 10]	LOW
]0, 30]	MEDIUM
]30, +∞[	HIGH

Prerequisites

You must at least:

- Configure one spending status description [page 486] in the SAP CC system
- Configure a range table class to define the conversion logic between monitored counters and reported spending statuses [page 206]

Task

To create a range table for a monitoring plan

This procedure consists in creating a range table for configuring the logic to convert at runtime a counter value (in a contract) to a label value of a spending status (in a report).

1. Create [page 225] a range table
2. Set up [page 228] the following elements:
   - Select the appropriate range table class that is dedicated to the configuration of monitoring plans
   - In the *Table Instance* tab, add a row for each label value that is configured in the spending status description, which this range table relates to. You can configure more rows depending on your business requirements.
   - Set up the ranges and the output values for each threshold according to the configuration of the spending status description and according to your business requirements. An output value must strictly be equal to one label that is configured in the spending status description. If the upper bound of the last defined range is bounded (and not +∞), the SAP CC system cannot convert a counter value to a label value in the related spending status. The reported spending status is undefined.

> **Note**
>
> If the texts configured in these dedicated range tables are not the same as the labels that are configured in the related spending status descriptions, the SAP CC system will not report the spending statuses that are based on these range tables.

Next Task

You can configure a monitoring plan and set up the generation of some spending statuses [page 507] based on this range table.

1.16.4.3 Setting Up a Range Table

Create [page 225], open [page 236], or modify [page 237] a range table before setting it up. A *Range Table* window must be open.

> **→ Remember**
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Window: Range Table

To set up a range table, you must:

1. Type the identification name of the range table, select the appropriate range table class [page 199] that defines its data structure, and fill in the Definition [page 229] tab.
2. Depending on the selected class and the business requirements, fill in or modify the following tabs:
   - Table Instance [page 229]
   - Additional Information [page 232]

> **→ Tip**
>
> You can import already-existing data from a CSV file.

Related Information

Setting Up a Range Table Class [page 207]

1.16.4.3.1 Tab: Definition

Fill in or modify the *Definition* tab:

Name: Type the unique identifier of this range table [page 213].

Description: Describe the range table.

Range Table Class: Select the range table class that defines the data structure for this range table.

Agreement ID: Type the identification code of the related *master agreement*, if any. This ID is useful in an integrated *SAP Solution* scenario with **SAP S/4HANA for Customer Management 2.0**. When you set up in this field, the range table is an *agreement table* and belongs to the identified master agreement. Consult Subscription Order Management in the Product Assistance for SAP S/4HANA for Customer Management 2.0 for more information and user assistance about:

- Master data
- Master agreement
- Available functions

See also the integration guides at help.sap.com/brim.

1.16.4.3.2 Tab: Table Instance

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Setting Up the Rows of the Range Table

Once you have selected the range table class for your range table, you can add one or more rows [page 232] to the table, and then fill in the cells of each row with relevant data values (business, technical).

> ⓘ Note
>
> In the *Output Values* area, the rows are always sorted by intervals and you cannot change the order of these rows.

After adding a row, you have to set it up.

Input Values: Type the input values for each row as specified in the input columns of the related range table class [page 199].

Output Values: Type the output values for this row as specified in the output columns of the related range table class [page 199].

Range: Double-click to set up the upper bound of the range that is specified in this row. The boundary is inclusive or exclusive according to the configuration of the related range table class.

> **Note**
>
> Instead of manually typing all of the data in the columns, you can import existing data from external .CSV type files by clicking  (refer to the process described below).

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values

Rule	Verifications
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

1.16.4.3.3 Tab: Additional Information

Use the configuration table and its icons to create some additional information items.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the configuration table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the CRM application or the external provisioning system that is connected to SAP Convergent Charging 2021.

1.16.4.4 Filling In a Range Table

Use the icons below to fill in the range table

Using the Icons

Use the following icon...	To...
	Add a new row to the configuration table.
	Delete a row from the table.
	Add a new group to the table.
	Delete a group from the table.
	Move a row up.
	Move a row down.
	Import data into the table according to the CSV format.(*)
	Clear all the table.

Use the following icon...	To...
	Search for a numerical or string value in the table. The search stops when the first occurrence has been found. When the search is successful, the cell is highlighted. To find the next occurrence, use  or  . When you click  , a *Search in table* dialog box is displayed. In the *Search* field, type the value to search for or select a value in the list. If you want to do a case-sensitive search, click the *Match case* option. In the *Search in column(s)* check boxes, select the columns where you want to search for the numerical or string value.
	Search for the next occurrence towards the top of the table if the previous search with the icon  was successful. The search is performed from left to right, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the beginning of the table is reached, a message informs you that the value has not been found.
	Search for the next occurrence towards the bottom of the table if the previous search with the icon  was successful. The search is performed from right to left, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the end of the table is reached, a message informs you that the value has not been found.

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.

Rule	Verifications
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of `2016-5-6 15:1:2`, the CSV file must contain `2016-05-06 15:01:02`.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .

2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

1.16.4.5 Saving a Range Table

Once you have created or modified a range table, and then verified its structure, you are ready to save it. You can save the range table to either the database or as an XML file on your hard disk.

To save a new range table to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Range Table Creation* dialog box opens to inform you that the range table has been successfully saved.
4. Click *OK*.

To save a new version of an existing range table to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the range table has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the range table.
> 2. Click *Cancel* to come back to the range table. See later in this chapter how to save the range table as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the range table.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:

- *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
- *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.

4. Click *Save*. The *Range Table Modification* dialog box opens to inform you that the range table has been successfully created.
5. Click *OK*.

To save a range table as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the range table.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The range table is saved as an XML file in your selected directory.

1.16.4.6 Opening an Existing Range Table

You can open:

- An existing range table saved in the database
- A previous version of a range table saved in the database
- An existing range table saved in an XML file

To open an existing range table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Range Table*. The *Open Range Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the *Range Tables* folder of the catalog of the owner, select the range table. The *Range Table* window opens in display mode.

Range tables are managed chronologically. After you have saved a range table, you can access the previous versions of it that exist in the database. However, if you modify and save a previous version of a range table, all later versions are deleted.

To open a previous version of a range table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Range Table*. The *Open Range Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the *Range Table* folder of the catalog of the owner, select the range table, and then click *Open*. The *Range Table* window opens in display mode.
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the range table from the list.
6. Click *View*.

To open an existing range table saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.16.4.7 Modifying a Range Table

> ⚠ Caution
>
> Do not modify a range table created by a CRM application or a provisioning system interfaced with SAP CC. Refer to the caution about Range Tables [page 213].

You can modify a range table by adding, modifying, or deleting content. When you save it in the database, you can modify its versions [page 166].

You cannot modify the name of a range table or the related range table class once you have saved the range table.

To modify a range table

1. From the menu bar in the main window, select *File* > *Open* > *Range Table*. The *Open Range Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the *Range Tables* folder of the catalog of the owner, select the range table.
4. Click *Open*. The *Range Table* window opens in display mode.
5. Set the editing mode by clicking the *Edit Mode* icon  on the main toolbar. The range table may be reloaded if it has been modified since you have opened it.
6. Set up [page 228] the range table.
7. Verify the range table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. Save [page 235] the range table.

Related Information

Display/Edit Mode [page 46]
Setting Up a Range Table [page 228]
Saving a Range Table [page 235]

1.16.4.8 Deleting a Range Table

> ⚠ Caution
>
> Do not delete a range table created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about range tables [page 213].

> → Remember
>
> As of SAP Convergent Charging 2022 FPS 2, for performance reasons, the system does not check dependencies anymore. Remember that deleting a range table corrupts the provider contracts or allowances still using it. You must ensure that the range table is not referenced anywhere before deletion.

Before you delete a range table, you must first delete all master data objects which reference it.

To delete a range table

1. From the menu bar in the main window, select *File* > *Delete* > *Range Table*. The *Delete Range Table* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the *Range Tables* folder of the catalog of the owner, select the range table.
4. Click *Delete*. A deletion dialog box asks if you want to delete the range table
5. Confirm the deletion of the range table by clicking *Yes*:
    1. If a dialog box confirms your choice, the range table has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the range table is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the objects that reference the range table. The types of these master data objects appear in the message.

1.16.4.9 Exporting a Range Table

You can export range tables to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import [page 239] the range table back into SAP CC Core Tool.

To export a range table

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the range table.
3. In the catalog of the owner, select the range table.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the range table.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.16.4.10 Importing a Range Table

You can import range tables in XML format from a directory in your database.

Note that only one version of a range table can exist in a database. You can import the unique version of a range table class into SAP CC Core Tool.

To import a range table

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the range table is saved, and then select the range table file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected.
6. In the *Import* column, clear the checkboxes of objects you do not wish to import.
7. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported [page 239] and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.16.5 Using the Action Menu

Related Information

Changing the Version of a Range Table [page 240]
Deleting the Versions of a Range Table [page 241]

1.16.5.1 Changing the Version of a Range Table

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> ⓘ Note
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.16.5.2 Deleting the Versions of a Range Table

The procedure is the same as Deleting the Versions of a Catalog Object [page 118].

1.17 Chargeable Item Packages (Revoked)

> ⓘ Note
>
> From SAP Convergent Charging release 4.0 onwards:
>
> - You can no longer create, set up, or modify a *chargeable item package* and add some *chargeable item classes* to it. As an object, the chargeable item package is no longer displayed in the *Core Tool user interface* of SAP CC.
> - You must use and configure some chargeable item classes [page 241] to build your charge plan [page 343] to be able to price and monetize the usage (consumption) of a service by the end customers of the service provider company.

> → Tip
>
> If you have some chargeable item packages in your master data for the service pricing configuration, contact your application administrator in order to migrate this data with the Setup Tool user interface and the `migration generateCIPackageReport` and `migration executeCIPackageReport` commands. See Setup Tool.

1.18 Chargeable Item Classes

> ⚠ Caution
>
> Read the caution [page 281] about the *consumption item mapping* before deleting a *chargeable item class*.

> ⓘ Note
>
> Do not forget that you can configure a reusable charge [page 290] with or without a chargeable item class.

- If you create a charge without a chargeable item class, you cannot configure a usage rate [page 690] in the price plan of the charge because only the chargeable item classes allow rating and charging some usage for the chargeable service.

In SAP Convergent Charging (SAP CC), a **chargeable item class** represents the common characteristics of high-volume usage events that relate to the mass consumptions of a chargeable/marketable service by end customers and subscribers. SAP CC receives these events called *chargeable items*, rates these items to dynamically compute a price and charges these fees to determined postpaid/prepaid accounts and other balances or decreases allowances [page 575]. See also: Transactional Data [page 34]

Example

For example, a chargeable item class can represent the usage events for a mobile phone service, a telephone call service, or an Internet service. See other industry-specific examples [page 243].

A chargeable item class **consists of a set of properties [page 244] and characterizes the way a usage-based service is consumed by end customers**. See Properties (Default and Custom) [page 244]. In a price plan, you can describe the rating parameters by configuring the properties that make up a chargeable item class.

A chargeable item class belongs to the pricing catalog [page 109] for the subscription- and usage-based services to monetize. Creating [page 248] a chargeable item class is thus the first step to build your charge plan [page 343].

Converting Chargeable Items onto Consumption Items

If you work with SAP ERP/FI-CA, you may need that your SAP CC system is able to convert chargeable items into *consumption items* and load them to SAP Convergent Invoicing in its system. This is possible by using the configuration of the consumption item mapping [page 281]. To convert chargeable items onto consumption items, you must create [page 283] the consumption item mapping between a consumption item class defined in the SAP ERP/FI-CA reference system and a chargeable item class.

Refer to the Consumption Item Mapping [page 281] topic for more information when working with SAP ERP/FI-CA.

Managing Allowances When SAP CRM is Integrated with SAP CC

When SAP CRM is integrated with SAP Convergent Charging, SAP CRM creates allowances [page 585] in SAP CC on demand or automatically when the provider contract is activated or changed. To create allowances, SAP CRM uses a predefined chargeable item containing specific information on the allowance (activation date, for example). SAP CC uses this information when executing the related charge for creating and initializing the allowance. Therefore, you need to create a predefined chargeable item class in SAP CC as a prerequisite for processing the chargeable items of allowances. See Managing Allowances for Integrating SAP CRM with SAP CC [page 585].

Related Information

Working with Chargeable Item Classes [page 247]
Properties (Default and Custom) [page 244]

1.18.1 Examples

Example

A service provider *Learn.com* offers lessons through an Internet portal. The end customers of this company can consume the service by reading online or downloading a lesson.

To model and translate this marketing and pricing policy, two chargeable item classes can represent these services and how to price and monetize the service consumptions:

- The first chargeable item class is called **`Read Online`**.
- And the second one is called **`Download`**.

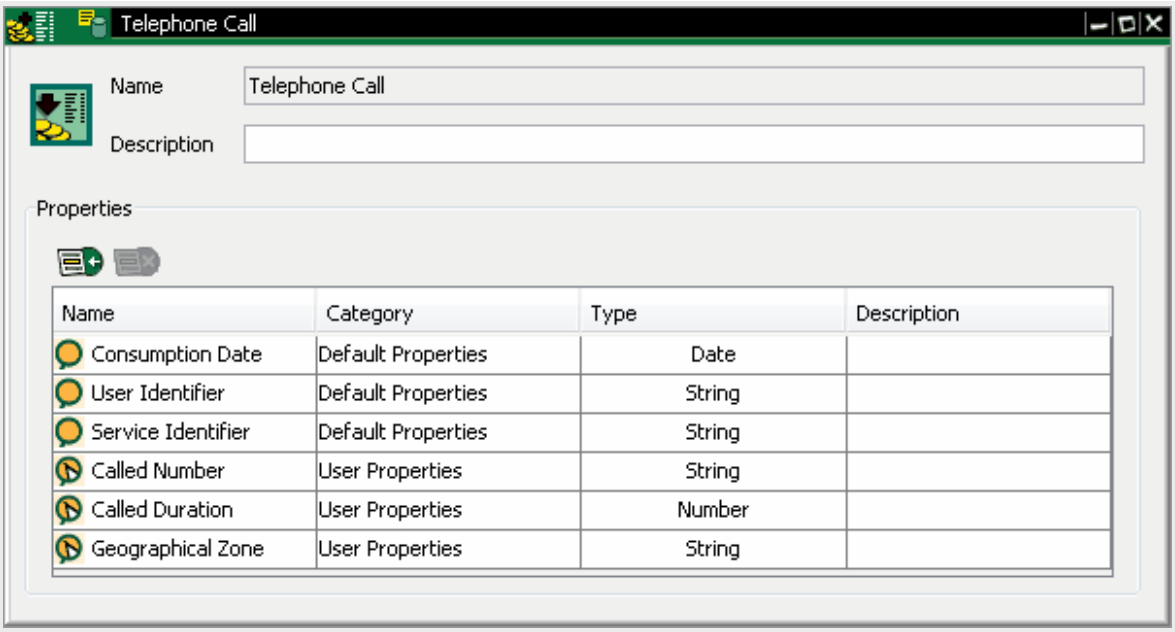
As a pricing specialist, you design, configure and maintain [page 248] these classes.

Example

The telephone operator *TelcoServ* sells telephone calls to its end customers. The company decides to bill these calls based on call duration but wants to apply different rates (prices) depending on geographical zones.

In SAP Convergent Charging, you can model and design the following related chargeable item class named **`Telephone Call`**. The class defines three custom properties:

- **`Called Number`** (String)
- **`Call Duration`** (Number)
- **`Geographical Zone`** (String)



As a pricing specialist, you design, configure and maintain [page 248] this class.

1.18.2 Properties (Default and Custom)

Properties of a Chargeable Item Class

A chargeable item class [page 241] contains several properties that describe each usage event for rating (dynamic pricing) purposes. All chargeable item classes define default properties, but you can define your own user properties (or custom properties) that are needed to design your price plan or that are needed by the final billing system.

Example

These basics examples illustrate the concept of chargeable item class:

- An Internet press service supplier prices article downloads depending on the document size. The chargeable item class **`Article Download`** will contain the custom property **`Document Size`**.
- A telecommunications operator chooses to bill telephone calls depending on the called number. The chargeable item class **`Telephone Call`** will contain the custom property **`Called Number`**.
- A press supplier charges the viewing of online articles depending on article categories such as daily newspaper, weekly review, and specialist review. The custom property **`Article Category`** is then added to the chargeable item class **`Viewed Article`**. The property type of this user property is defined as a character string.

As a pricing specialist, you design and configure this data in SAP CC.

Default Properties

Default properties provide basic information about the consumption of a chargeable item class for rating purposes. Each chargeable item class contains the same default properties:

- *Consumption Date*: this property describes the date and time when the service is consumed by the end customer
- *User Identifier*: this property describes the identifier of the end customer consuming the service
- *Service Identifier*: this property describes the identifier of the service consumed by the end customer

User-Defined Properties

In addition to default properties, you set up [page 249] some **custom properties** in a chargeable item class to be able to configure the price plans with calculation algorithms based on information included in these properties. At runtime, all the chargeable items sent by the connected mediation systems must include these custom properties with data values.

In your class, you set up these properties by assigning the following to them:

- Name
- Type (character string, number, or date/time)
- Description

> ⓘ Note
>
> You can identify some properties that are not used by the SAP CC system, but that are used by the billing system. Define these custom properties in your chargeable item class to be able to configure the output of this data in the generated charged items.

Related Information

Working with Chargeable Item Classes [page 247]

1.18.3 ALLO Chargeable Item Class

> → Remember
>
> These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your integrated *SAP Solution*, the SAP CRM system drives and triggers the creations for allowances [page 575] in *provider contracts* stored in SAP Convergent Charging (SAP CC) by sending charging requests with dedicated and specialized *chargeable items*. The SAP CC system (Core Server) handles and uses each chargeable item to identify the provider contract that will contain the new allowance.

These chargeable items must contain:

- A specific *user service identifier (USID)* that conforms to the syntax: **`ALLOW_*`** (where * matches 0 to 6 characters). For example, **`ALLOW_`** is a valid USID.
- The identifiers of the *allowance definition* used to initialize the allowance

As a prerequisite, a dedicated chargeable item class [page 241] must exist in SAP CC. This ALLO class contains the following predefined custom properties:

Custom Property	Property Description
`ALLOW_DEF_ID`	**`The identifier of the allowance definition used to create the allowance in the provider contract relating to the received chargeable items.`**
`ALLOW_START_DATE`	**`The date/time at which the allowance must be activated.`**

As an SAP CC pricing specialist, you configure and maintain [page 246] this class. Use these definitions to fill in the fields in the user interface.

You must then configure a particular *charge plan* and a *reusable charge* that use this chargeable item class and that define the allowance creation based on allowance definitions identified in the incoming chargeable items. The created allowance must be active at the specified date/time.

> ⓘ Note
>
> A corresponding **`ALLO consumption item class`** must be created in SAP Convergent Invoicing (in SAP ERP/FI-CA) to enable the loading of the *consumption items* into SAP Convergent Invoicing.

Related Information

Creating the Predefined ALLO Chargeable Item Class [page 246]
Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

1.18.3.1 Creating the Predefined ALLO Chargeable Item Class

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

Your Core Tool user interface provides you with a useful template to create an ALLO chargeable item class [page 245] in SAP CC.

To create the predefined ALLO chargeable item class

1. Make a copy of the `ALLO_chargeable_item_class.xml XML` file located in the installation directory of your Core Tool user interface in the templates folder.
2. Open the copied file and replace `${catalog}` by the name of your service pricing catalog.
3. Save and close the XML file.
4. Launch and log on [page 35] to your Core Tool user interface for SAP Convergent Charging.
5. Create the **ALLO** chargeable item class by importing [page 253] the modified file.
6. (Optional step) In the SAP ERP/FI-CA system, create the ALLO consumption item class for SAP Convergent Invoicing. For more information on SAP Convergent Invoicing, refer to the Help Resources [page 40].
7. (Optional step) Back in the Core Tool user interface, create [page 283] the corresponding consumption item mapping [page 281].

> **Note**
>
> You perform step 6 and step 7 only if the *Consumption Item Management* feature of SAP Convergent Invoicing is enabled for the storage of consumption items.

Next Step

Customizing the Allowance Interface for SAP CRM [page 94]

Related Information

Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

1.18.4 Working with Chargeable Item Classes

Related Information

Creating a Chargeable Item Class [page 248]

Opening an Existing Chargeable Item Class [page 251]
Modifying a Chargeable Item Class [page 252]
Setting Up a Chargeable Item Class [page 249]
Saving a Chargeable Item Class [page 249]
Deleting a Chargeable Item Class [page 252]
Exporting a Chargeable Item Class [page 253]
Importing a Chargeable Item Class [page 253]

1.18.4.1 Creating a Chargeable Item Class

Use one of these procedures to create a chargeable item class.

Create a Chargeable Item Class

1. From the menu bar in the main window, select *File* > *New* > *Chargeable Item Class*. The *Choose Owner* window opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. Click *OK*. The *New Chargeable Item Class* window opens.
4. Set up [page 249] the chargeable item class.
5. Verify the chargeable item class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save and release [page 249] your new chargeable item class.

> → Tip
>
> You can create a chargeable item class by opening an existing chargeable item class and copying it.

Create a Chargeable Item Class from an Existing Chargeable Item Class

1. From the menu bar in the main window, select *File* > *Open* > *Chargeable Item Class*. The *Open Chargeable Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. In the *Chargeable Item Classes* folder of the catalog of the owner, select the chargeable item class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding chargeable item class window opens.
5. **Rename the new chargeable item class** in the *Name* field, and then set up [page 249] it.
6. Verify the chargeable item class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.

7. Save and release [page 249] your chargeable item class.

Related Information

Setting Up a Chargeable Item Class [page 249]
Saving a Chargeable Item Class [page 249]

1.18.4.2 Setting Up a Chargeable Item Class

Create [page 248], open [page 251], or modify [page 252] a chargeable item class before setting it up. A *Chargeable Item Class* window must be open.

> **Note**
>
> - After saving a chargeable item class, you cannot modify the default properties, the existing user properties (custom properties), and the name of a chargeable item class. You can however add new user-defined properties to the chargeable item class.
> - If necessary, you can modify a chargeable item class at charge level when inserting a Usage Rate [page 302] component in the price plan of your charge.

To set up a chargeable item class, you first identify it, and then you add user properties to the table, or you modify the user properties of the table.

Name: Type the unique identifier of the chargeable item class.

Description: Describe the function of the chargeable item class.

 : Click to add a new row to the table.

 : Click to delete a row from the table.

Properties: Fill in or modify the columns of the table.

- *Name*: Type the identifier of the user property
- *Category*: Displays the category (default or user) of the property
- *Type*: Select the type of the property (number, string, or date)
- *Description*: Describe the function of the property

1.18.4.3 Saving a Chargeable Item Class

Once you have created a chargeable item class and then verified its configuration, you are ready to save it. You can save the chargeable item class to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving a chargeable item class, you cannot modify the default properties, the existing user properties, and the name of a chargeable item class. You can however add new user properties to the chargeable item class.

Save and Release a New Chargeable Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. Click *Save*. The *Chargeable Item Class Creation* dialog box opens to inform you that the chargeable item class has been successfully created and released.
4. Click *OK*.

Save and Release an Existing Chargeable Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Chargeable Item Class Modification* dialog box opens to inform you that the chargeable item class has been successfully modified, click *OK*. The new version of the chargeable item class is saved and released as the only current version.
3. If a *Warning* dialog box opens to inform you that the chargeable item class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the chargeable item class.
   - Click *Cancel* to come back to the chargeable item class. See below how to save the chargeable item class as an XML file.

Save a Chargeable Item Class as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the chargeable item class.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The chargeable item class is saved as an XML file in your selected directory.

1.18.4.4 Opening an Existing Chargeable Item Class

You can open:

- An existing chargeable item class saved in the database
- An existing chargeable item class saved in an XML file

Open an Existing Chargeable Item Class Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Chargeable Item Class*. The *Open Chargeable Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. In the *Chargeable Item Classes* folder of the catalog of the owner, select the chargeable item class.
4. Click the *Open* button. The *Chargeable Item Class* window opens in display mode.

Open an Existing Chargeable Item Class Saved in an XML File

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.18.4.5 Modifying a Chargeable Item Class

> **Note**
>
> You cannot modify the default properties, the existing user properties, and the name of a chargeable item class. You can however add new user properties to the chargeable item class.

Modify a Chargeable Item Class

1. Open [page 251] the chargeable item class.
2. Switch to the editing mode by clicking the *Edit Mode* icon on the main toolbar. The chargeable item class may be reloaded if it has been modified since you have opened it.
3. Set up [page 249] the chargeable item class.
4. Verify the structure of a chargeable item class by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Save and release [page 249] the chargeable item class.

1.18.4.6 Deleting a Chargeable Item Class

> **Caution**
>
> - Read the caution [page 281] about the consumption item mapping before deleting a chargeable item class.
> - There is no undo function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.

Before you delete a chargeable item class, you must first delete all objects which reference it.

For example, you can not delete a chargeable item class that is part of a charge plan in the database. You must first delete the charge plan and the charge that contain this chargeable item class, and then you can delete the chargeable item class.

Delete a Chargeable Item Class

1. From the menu bar in the main window, select *File* > *Delete* > *Chargeable Item Class*. The *Delete Chargeable Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. In the *Chargeable Item Classes* folder of the catalog of the owner, select the chargeable item class.

4. Click *Delete*. A deletion dialog box asks if you want to delete the chargeable item class
5. Confirm the deletion of the chargeable item class by clicking *Yes*:
    1. If a dialog box confirms your choice, the chargeable item class has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the chargeable item class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the chargeable item class. The identifiers of these objects appear in the message.

1.18.4.7 Exporting a Chargeable Item Class

You can export chargeable item classes to a directory of your hard disk, where they are saved as XML files.

> **Note**
>
> If you export a chargeable item class belonging to a chargeable item package [page 241], SAP Convergent Charging exports the chargeable item package with all its chargeable item classes.

Export a Chargeable Item Class to an XML File

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the chargeable item class.
3. In the catalog of the owner, select the chargeable item class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the chargeable item class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

Related Information

Importing a Chargeable Item Class [page 253]

1.18.4.8 Importing a Chargeable Item Class

You can import chargeable item classes in XML format from a directory of your hard disk.

> ⓘ Note
>
> If you import a chargeable item package [page 241], SAP CC Core Tool imports the chargeable item package with all its chargeable item classes. But you can individually use the chargeable item classes in your different price plans.

Import a Chargeable Item Class

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the chargeable item class is saved, and then select the file of the chargeable item class.
3. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a chargeable item class.
4. In the *Import* column, clear the checkboxes of objects you do not wish to import.
5. Click *Import*.

> ⓘ Note
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available for import.

1.19 Charged Item Classes

> ⚠ Caution
>
> As of SP 5 for release 5.0, the behavior of this master data is modified in case of an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition. **Every reference or operation related to prepaid accounts is no more available.** For more information and user assistance, refer to SAP Note 2772452.

> ⚠ Caution
>
> Read the caution [page 263] about the billable item mapping before deleting a charged item class.

In SAP Convergent Charging 2023, you use the **charged item classes** to configure the information included in the *charged items*, which are the resulting business transactions of the processing of the charging operations by the system. You also use the charged item classes to describe and filter the details of business transactions returned to the caller of the charging operation.

In your Core Tool user interface, a charged item class is graphically represented by a tree that consists of nodes representing custom data fields. Each node represents an output data field whose type can be a string  , a number  , a date  , or a boolean  . The tree allows you to configure how these fields are computed by the system in the high-volume charged items that will be based on your current class.

You can create and use charged item classes for different charges customized in a charge plan [page 343].

If you use an external billing system

If you want to send charged item classes to an external billing system or any other external system, you must create the charged item classes from scratch, and then set them up [page 257].

Setting up your charged item classes

Because you know the contents of the transactions resulting from the rating and charging processes, you can define and set up the data of your charged item classes accordingly to retrieve information from the transactions. Your system billing or another system receives and uses this information.

If you use an SAP ERP/FI-CA system

SAP Convergent Charging is integrated with SAP ERP/FI-CA for managing high-volume data. SAP Convergent Charging can:

- Produce billable items asynchronously from the converted charged items.
- Send the billable items to SAP Convergent Invoicing in the SAP ERP/FI-CA or SAP S/4HANA system.

Setting up your charged item classes

You must map charged item classes onto billable item classes defined in SAP Convergent Invoicing. SAP Convergent Invoicing generates automatically. To do that, you create a charged item class from the mapping of a billable item class. Refer to Billable Item Mapping [page 263] topic for more information when using SAP Convergent Invoicing and SAP ERP/FI-CA.

> **Note**
>
> For more information, consult the SAP CC Administration Guide

Related Information

Working with Charged Item Classes [page 256]

1.19.1 Working with Charged Item Classes

Related Information

Creating a Charged Item Class [page 256]
Opening an Existing Charged Item Class [page 260]
Modifying a Charged Item Class [page 261]
Setting Up a Charged Item Class [page 257]
Deleting a Charged Item Class [page 261]
Saving a Charged Item Class [page 259]
Exporting a Charged Item Class [page 262]
Importing a Charged Item Class [page 263]

1.19.1.1 Creating a Charged Item Class

Use this procedure to configure a new charged item class in the service pricing catalog.

Create a Charged Item Class

1. From the menu bar in the main window, select *File* > *New* > *Charged Item Class*. The *New Charged Item Class* window opens.
2. Set up [page 257] the charged item class.
3. Verify the charged item class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save and release [page 259] your new charged item class.

> → Tip
>
> You can create a charged item class by opening an existing charged item class and copying [page 257] it.

Related Information

Setting Up a Charged Item Class [page 257]
Saving a Charged Item Class [page 259]

1.19.1.2 Copying a Charged Item Class

Use this procedure to configure a new charged item class by copying an existing one.

Create a Charged Item Class from an Existing Charged Item Class

1. From the menu bar in the main window, select *File* > *Open* > *Charged Item Class*. The *Choose a Charged Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charged item class.
3. In the *Charged Item Classes* folder of the catalog of the owner, select the charged item class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding charged item class window opens.
5. Rename the charged item class in the *Name* field, and then set up [page 257] this new charged item class.
6. Verify the charged item class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save and release [page 259] your new charged item class.

Related Information

Setting Up a Charged Item Class [page 257]
Saving a Charged Item Class [page 259]

1.19.1.3 Setting Up a Charged Item Class

Create [page 256], open [page 260], or modify [page 261] a charged item class before setting it up. A *Charged Item Class* window must be open.

A charged item class is graphically represented by a tree that consists of nodes ( string,  number,  date/time,  boolean). Each node represents a data field that is generated by the system in corresponding charged items.

> **Note**
>
> If necessary, you can **redefine** the node values set in your charged item class when configuring a charge plan or an allowance plan. See:
>
> - Customized Charge: Output Item [page 372]
> - Allowance Logic: Output Item [page 479]

Set Up a Charged Item Class

1. In the *Name* field of the identification area, type the unique identifier of the charged item class. For example: **`Telephone Service`**.
2. If necessary, describe the charged item class in the *Description* field.
3. In the tree display area, click the  root node to activate the charged item class.
4. On the elements toolbar (left), click the icon relating to the node that you want to add to the charged item class.
5. In the *Name* field of the identification area, type the identifier of the node. For example: **`Call`**. It is the name for the data field that will be set at runtime in a charged item based on this class.
6. If necessary, describe the node in the *Description* field. It facilitates the redefinition of the data field generation.
7. In the *Value* list of the *Definition* tab, select one of the following list items:
   - *Type a value*: Allows you to type any constant value except an empty string (no spaces and no characters).
   - *Undefined*: Allows you to select a system property that has no assigned value. Selecting this property only means that you recommend changing this value at charge plan or allowance plan level.
   - *Empty*: Allows you to select a system property that has no assigned value. Selecting this property only means that you do not recommend changing this value at charge plan or allowance plan level.
   - Any other property of the list. The rest of the list depends on the type of the node (string, number, date/time, boolean).
8. If you want to define a new output data field in your class, continue the procedure at step 3 by **clicking the root node** in the displayed tree.

> ⓘ Note
>
> - If you set up a boolean node with a source property, the value is converted by SAP Convergent Charging during the charging process. It converts the strings "0" and "false" and the number 0 to `FALSE`. It converts the strings "1" and "true" and the number 1 to `TRUE`. Otherwise the charging process fails because it cannot convert the source value.
> - If you use SAP Convergent Charging with an SAP ERP system and if no value is assigned to the SUBPROCESS and BITTYPE nodes, you may need to assign these nodes the values which are defined in the related billable item mapping (Refer to the note describing these values [page 268]).

Change the Value of a Node

1. In the tree display area of the charged item class, click a node.
2. In the *Value* field of the *Definition* tab, type a new constant value or change the property.

1.19.1.4 Saving a Charged Item Class

Once you have created a charged item class and then verified its configuration, you are ready to save it. You can save the charged item class to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving a charged item class, you can no longer remove nodes and modify the node names. However, you can change the values of the existing nodes and add new nodes to the tree.

Save and Release a New Charged Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the Catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charged item class.
3. Click *Save*. The *Charged Item Class Creation* dialog box opens to inform you that the charged item class has been successfully created and released.
4. Click *OK*.

Save and Release an Existing Charged Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Charged Item Class Modification* dialog box opens to inform you that the charged item class has been successfully modified, click *OK*. The new version of the charged item class is saved and released as the only current version.
3. If a *Warning* dialog box opens to inform you that the charged item class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the charged item class.
   - Click *Cancel* to come back to the charged item class. See below how to save the charged item class as an XML file.

Save a Charged Item Class to an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the charged item class.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The charged item class is saved as an XML file in your selected directory.

→ Remember

You cannot use this file to import the charged item class in another SAP system landscape.

1.19.1.5 Opening an Existing Charged Item Class

You can open:

- An existing charged item class saved in the database
- An existing charged item class saved in an XML file

To open an existing charged item class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Charged Item Class*. The *Open Charged Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charged item class.
3. In the *Charged Item Classes* folder of the catalog of the owner, select the charged item class.
4. Click the *Open* button. The *Charged Item Class* window opens in display mode.

To open an existing charged item class saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

ⓘ Note

- If several versions of the object are already stored in XML files, you can open only the oldest version.
- In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.19.1.6 Modifying a Charged Item Class

When modifying a charged item class, you can:

- Add new nodes (string, number, date, boolean) to a charged item class.
- Assign a new property to a node.

The modified charged item class replaces the previous version in the back-end database.

> **Note**
>
> You cannot modify the name of the charged item class and the name of nodes.

Modify a Charged Item Class

1. Open [page 260] the charged item class.
2. Switch to the editing mode by clicking the *Edit Mode* icon on the main toolbar. The charged item class may be reloaded if it has been modified since you have opened it.
3. Set up [page 257] the charged item class.
4. Verify the structure of a charged item class by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Save and release [page 259] the charged item class.

1.19.1.7 Deleting a Charged Item Class

> **Caution**
>
> - Read the caution [page 263] about the billable item mapping before deleting a charged item class.
> - There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this operation is necessary.

Delete a Charged Item Class

1. From the menu bar in the main window, select *File* > *Delete* > *Charged Item Class*. The *Delete Charged Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charged item class.
3. In the *Charged Item Classes* folder of the catalog of the owner, select the charged item class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the charged item class.
5. Confirm the deletion of the charged item class by clicking *Yes*:
    1. If a dialog box confirms your choice, the charged item class has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the charged item class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the charged item class. The identifiers of these objects appear in the message.

1.19.1.8 Exporting a Charged Item Class

You can export charged item classes to a directory of your hard disk, where they are saved as XML files.

Export a Charged Item Class to an XML File

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charged item class.
3. In the catalog of the owner, select the charged item class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the charged item class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

Related Information

Importing a Charged Item Class [page 263]

1.19.1.9 Importing a Charged Item Class

You can import charged item classes in XML format from a directory of your hard disk.

> **Note**
>
> SAP CC Core Tool can only import an object that has previously been exported [page 262] (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available for import.

Import a Charged Item Class

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the charged item class is saved, and then select the file of the charged item class.
3. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a charged item class.
4. In the *Import* column, clear the checkboxes of objects you do not wish to import.
5. Click *Import*.
6. Work with the imported objects.
7. When necessary, save your data objects to the back-end database system.

Related Information

Exporting a Charged Item Class [page 262]

1.20 Billable Item Mapping

> **Caution**
>
> **Be careful if you want to delete the billable item mapping** and the related charged item classes [page 254] and refill record classes [page 414] in a production SAP system landscape where SAP Convergent Charging 2023 is installed and configured for working with SAP Convergent Invoicing.
>
> - Deleting these objects may prevent SAP Convergent Charging from converting charged items or refill records into billable items and bulk loading them to SAP Convergent Invoicing in its system (SAP ERP/FI-CA or SAP S/4HANA) or cloud (SAP S/4HANA Cloud Public Edition).

- Make sure that SAP Convergent Charging no longer uses these objects before deleting them.

This caution also applies to the consumption item mapping [page 281] and to the related chargeable item classes [page 241] that are configured in SAP CC.

→ Remember

These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP Convergent Invoicing. Several implementation scenarios and variants are available. For more information and user assistance, see the integration guides at help.sap.com/brim.

Billable Item Classes

Defined in the SAP ERP reference system, *billable item classes* are basically charged item classes or refill record classes with additional fields for control purposes.

Billable Item Mapping

SAP Convergent Charging creates business transactions (transactional data) called charged items in the case of the charging process or called refill records in the case of the refilling process. The SAP CC system can bulk load these transactions into one or more SAP ERP/FI-CA systems through billable items and according to the billable item mapping.

The billable item mapping consists in mapping charged item classes [page 254] or refill record classes [page 414] onto billable item classes defined in a particular SAP ERP/FI-CA system that is the ERP reference system in the SAP system landscape. At runtime, SAP CC uses the settings in the billable item mapping to convert generated charged items or refill records into billable items, and then loads the billable items into the relevant SAP ERP/FI-CA system that is declared in the identified subscriber account [page 511]. See the Customer Management Area Information [page 530].

ⓘ Note

The Core Tool user interface can automatically create the charged item class [page 266] (or refill record class) to facilitate the configuration of the charge plans [page 343] and refill plans [page 421].

See: Creating the Billable Item Mapping [page 266]

Data Maintenance

Because the SAP ERP reference system (or the SAP S/4HANA reference system) defines the billable item classes and modifies their contents if necessary, you can refresh [page 271] the mapping of the billable item classes to avoid any error relating to the creation and validation of the billable item classes.

See: Refreshing the Billable Item Mapping [page 271]

> **Note**
>
> In an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition, this refreshing mechanism is not available.

> **→ Remember**
>
> If a billable item class has been changed in the SAP ERP reference system (or in the SAP S/4HANA reference system), you must modify the existing billable item mapping defined in SAP Convergent Charging.
>
> For more information, refer to your administrator or refer to the maintenance procedure described in the SAP CC Administration Guide of SAP Convergent Charging.

The billable item mapping is not part of the master data and is not included in a pricing catalog [page 109]. The billable item mapping is shared by all catalogs in the system. It is a business data that is configured for an SAP CC system.

Tree Representation in Core Tool

In your Core Tool user interface, the billable item mapping is graphically represented by a tree that consists of nodes. Each node has a type which can be a string  , a number  , a date/time  , or a boolean  .

A node defines the format of the data generated in the billable items and specifies a source field from the charged items or the refill records.

See: Working with the Billable Item Mapping [page 266]

Additional Information

- For more information about the relationships between billable item mapping and charged item classes, refer to the topic Charged Item Classes [page 254].
- For the relationships between billable item mapping and refill record classes, refer to the topic Refill Record Classes [page 414].

Related Information

Consumption Item Mapping [page 281]

1.20.1 Working with the Billable Item Mapping

Related Information

Creating the Billable Item Mapping [page 266]
Opening the Billable Item Mapping [page 268]
Modifying the Billable Item Mapping [page 272]
Saving the Billable Item Mapping [page 267]
Deleting the Billable Item Mapping [page 273]

1.20.1.1 Creating the Billable Item Mapping

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP Convergent Invoicing. Several implementation scenarios and variants are available. For more information and user assistance, see the integration guides at help.sap.com/brim.

Use this procedure to map:

- A charged item class [page 254] onto a billable item class
- A refill record class [page 414] onto a billable item class

> ⓘ Note
>
> If you map an already-existing charged item class (or an already-existing refill record class) onto a billable item class, make sure that the structure of the charged item class (or the refill record class) is the same as the one of the billable item class; otherwise, you must modify the structure of the charged item class (or the refill record class) so that you can save the billable item mapping.

Create the Billable Item Mapping and its Associated Charged Item Class (or Refill Record Class)

As a pricing specialist for SAP Convergent Charging:

1. From the menu bar in the main window, select *File* > *New* > *Billable Item Mapping*. The *Choose a Billable Item Class by Use* dialog box opens.
2. In the dialog box, select one of the following types:
   - *Charging of Postpaid Account*: Allows you to map a charged item class onto a billable item class if you work with a charge [page 290] dedicated to external postpaid accounts [page 520].

- *Charging of Prepaid or Postpaid Account*: Allows you to map a charged item class onto a billable item class if you work with a charge dedicated to prepaid accounts [page 514] and external postpaid accounts.
- *Refilling of Prepaid Account*: Allows you to map a refill record class onto a billable item class if you work with a refill logic [page 395] dedicated to prepaid accounts.

3. Click *OK*. Depending on the selected option, the *Choose a Billable Item Class and a Charged Item Class* dialog box (or the *Choose a Billable Item Class and a Refill Item Class* dialog box) opens.
4. In the *Charged Item Class* list of the dialog box (or in the *Refill Item Class* list), select an existing charged item class (or an existing refill record class), or select *New Class* (recommended) to create a new charged item class (or a new refill record class), and then type its name in the next field. You can reuse this class later on.
5. Click *OK*. The window of the new billable item mapping is displayed[26].

> **Note**
>
> The *Function Module* field displays the name of the API called by SAP CC for the mapping. This API is provided by SAP Convergent Invoicing according to the billable item class.

6. If necessary, modify [page 272] the billable item class to reorganize the mapping between the billable item class and the charged item class (or the refill record class). In SAP Convergent Charging, SAP SE strongly recommends that you do not modify the structure of the billable item mapping.
7. Verify the structure of the billable item mapping by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. Save [page 267] both the billable item mapping and its associated charged item class (or refill record class).

1.20.1.2 Saving the Billable Item Mapping

Once you have created, and then verified its structure, you can save the billable item mapping to either the database or as an XML file on your hard disk.

If the charged item class does not exist, it is automatically created when you create the billable item mapping. In this case, you save the new charged item class in a catalog.

Save the New Billable Item Mapping and the Charged Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Billable Item Mapping and Charged Item Class Confirmation* dialog box opens to ask if you want to create the billable item mapping and save the new charged item class in a catalog.

[26] Depending on the selected type (step 2), the right panel of the window displays "Charged Item Class" or "Refill Record Class" as the *Source* of the billable item mapping. This means that in a charge plan, the *Charged Items* tab of a charge only provides a list of charged item classes while in a refill plan the Refill Records [page 395] tab of the refill logic only provides a list of refill record classes.

2. Click *Yes*. The *Save in Catalog* opens to save the charged item class.
3. In the *Owner* drop-down menu, select the owner of the charged item class.
4. Click *Save*. The *Billable Item Mapping and Charged Item Class Creation* dialog box opens to inform you that the billable item mapping and the charged item class have been successfully created[27].
5. Close the dialog box.

If the charged item class already exists (in a catalog), only the billable item mapping is created.

Save the New Billable Item Mapping to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Billable Item Mapping Creation* dialog box opens to ask if you want to create the billable item mapping.
2. Click *OK*. The billable item mapping is created.

Save the Billable Item Mapping as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the billable item mapping.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The billable item mapping is saved as an XML file in your selected directory.

1.20.1.3 Opening the Billable Item Mapping

Use this procedure to open the billable item mapping.

> **Note**
>
> You cannot open the billable item mapping from a file.

Open the Billable Item Mapping Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Billable Item Mapping*. The *Open Billable Item Mapping* dialog box opens.
2. In the *Billable Item Class* list of the dialog box, select the billable item mapping that you want to open.

[27] If the creation of the billable item mapping has created a new charged item class, the same message displays if you want to open the related charged item class.

3. Click *OK*. The window of the billable item mapping opens.
4. View [page 269] the complete mapping.
5. If necessary, refresh [page 271] the mapping.

Related Information

Viewing the Billable Item Mapping [page 269]

1.20.1.4 Viewing the Billable Item Mapping

Use this procedure to view the billable item mapping configured in SAP Convergent Charging.

View the Billable Item Mapping

1. Open [page 268] the billable item mapping.
2. View the complete mapping:
   By naming convention, the mapping name is composed of the names of the classes, separated by a slash ('/'):
   - Name of the charged item class/name of the billable item class
   - Name of the refill record class/name of the billable item class

   The window of the billable item mapping consists of two panes:
   - The left pane displays the tree of nodes. Each node represents a mapping setting. The root represents the billable item mapping object.
   - The right pane displays the name of the billable item mapping or the name of the selected node, and the *Definition* tab.

   The *Definition* tab displays:
   - The mapping information if you selected the tree root.
   - The target (billable item class) and the source (charged item class) of the billable item mapping (if you selected a node).
3. View the root:
   If you click the tree root, the *Definition* tab displays the following information:
   - *Target*:

     The icon  allows you to refresh [page 271] the billable item mapping (resynchronization with the SAP ERP reference system or the SAP S/4HAHA reference system). See Refreshing the Billable Item Mapping [page 271].

     > → Remember
     >
     > This icon is not available in case of an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition .

- *Billable Item Class*: The name of the billable item class (BIT class) configured in SAP Convergent Invoicing
- *Function Module*: The name of the API called by SAP Convergent Charging for the mapping. This API is provided by SAP Convergent Invoicing according to the billable item class.
  For example: **`/1FE/ALLO_BIT_CREATE_API`**

- *Source*:

  The icon  allows you to open [page 260] the charged item class, and then modify it if necessary.
  - *Charged Item Class*: The name of the class configured in SAP Convergent Charging

4. View a node representing some elementary mapping:
   If you click a tree node:
   - The node name is the target field (in the billable item class associated to the billable item mappping).
   - The *Definition* tab displays the following information:
     - *Source*: The source property in the charged item class (or refill record class). See Setting Up a Charged Item Class [page 257] and Setting Up a Refill Item Class [page 390] for more information.
     - *Type* aka *Data Type* (ex.: **`C`**, **`P`**, or **`D`**)
     - *Field Type*: The field type (aka subtype) (ex: BOOLEAN)
     - *Length*: The length
     - *Decimal Length*: The length of the decimal (when relevant)
     - *Size (Bytes)*: The size in bytes

   If you click the SUBPROCESS and BITTYPE nodes, the *Definition* tab displays respectively:
   - The value of the SUBPROCESS node
   - The values of the BITTYPE node which are associated to the SUPROCESS node

   > **Note**
   >
   > If you use SAP Convergent Charging with an SAP ERP or an SAP S/4HANA system, you may need to assign the values of the SUBPROCESS and BITTYPE nodes to the respective nodes of the charged item class associated to the billable item mapping. Refer to Setting Up a Charged Item Class [page 257].

5. If necessary, refresh [page 271] the mapping.

Similar

Viewing the Consumption Item Mapping [page 286]

Related Information

Refreshing the Billable Item Mapping [page 271]

1.20.1.5 Refreshing the Billable Item Mapping

> **Note**
>
> In an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition, this refreshing mechanism is not available.

Use this procedure to refresh the billable item mapping manually in order to resynchronize the configuration data between the two systems: SAP ERP (or SAP S/4HANA) with SAP Convergent Charging.

Refresh the Billable Item Mapping

As a pricing specialist for SAP Convergent Charging:

1. Open [page 268] the billable item mapping [page 263] in the Core Tool user interface.
2. Click the button icon  to refresh the billable item mapping (resynchronization with the SAP ERP reference system or the SAP S/4HAHA reference system).
   You maintain the billable item mapping (BIT mapping) and the associated charged item class when changes occurred in the structure of the billable item class (BIT class) that is used for this billable item mapping.

   > **Note**
   >
   > SAP Convergent Charging detects the new fields but ignores the changed/removed fields. For example, if the format of a field has changed in the BIT class, the refresh does not detect the new format.
   > You need to configure a new BIT mapping object and a new associated charged item class (or refill record class), and adapt the configurations of all the allowance plans, charge plans, and refill plans that previously used the replaced class.

   As of SAP CC 2022 FPS 2, if the type of the ALLOWANCE field changes from Boolean to String, the Core Tool user interface allows you to configure such a pair of new elements (a BIT mapping object and its associated charged item class or refill record class).
3. Verify the changes.
4. Save [page 267] the refreshed billable item mapping.
5. Maintain the charged item classes (and refill record classes) that are associated to the refreshed billable item mapping. See:
   - Modifying a Charged Item Class [page 261]
   - Modifying a Refill Record Class [page 419]

Related Information

Refreshing the Consumption Item Mapping [page 288]

1.20.1.6 Modifying the Billable Item Mapping

> ⚠ Caution
>
> In SAP Convergent Charging, SAP SE strongly recommends that you do not modify the structure of the billable item mapping because the SAP CC Core Server system is configured to map charged item classes onto billable item classes automatically.
>
> The procedure to modify the billable item mapping is given for information only. Refer to your system administrator for more information.
>
> First, consider Refreshing the Billable Item Mapping [page 271].

Because the charged item class (or refill record class) is mapped onto a billable item class, you can:

- Change the original mapping between these objects by reassigning the properties of the charged item class to the billable item class.
- Add one or more new properties to the tree of the charged item class, and then reorganise the mapping between the charged item class and the billable item class.

Change the Mapping

1. In the tree display area, select a node.
2. In the *Definition* tab of the tabs area, in the *Source* field, select the property of the charged item class (or refill record) to assign to the property of the billable item class.
3. Make sure that two or more properties have not already the same source; otherwise, you cannot save the billable item mapping, and a message informs you that the billable item mapping contains duplicates.

Add New Properties to a Charged Item Class

1. In the tree display area, click  to activate the billable item mapping.
2. In the *Definition* tab of the tabs area, click  to open the related charged item class. The charged item class window opens.
3. In the tree display area of the charged item class window, add one or more new properties to the charged item class and set them up [page 257].
4. Save [page 259] the charged item class.
5. Close and reopen the billable item mapping.
6. Reassign the new fields of the charged item class to the fields of the billable item class.

Related Information

Refreshing the Billable Item Mapping [page 271]

1.20.1.7 Deleting the Billable Item Mapping

> ⚠ Caution
>
> Read the caution [page 263] before deleting the billable item mapping [page 263] from your pricing catalog.

Use this procedure to delete the billable item mapping.

Delete the Billable Item Mapping

1. From the menu bar in the main window, select *File* > *Delete* > *Billable Item Mapping*. The *Delete Billable Item Mapping to Delete* dialog box opens.
2. In the *Billable Item Mapping* list of the dialog box, select the billable item mapping that you want to delete.
3. Click *OK*. A dialog box asks if you want to delete the consumption item mapping.
4. Click *Yes*. A dialog box opens to inform you that the billable item mapping has been successfully deleted.
5. Click *OK* to close the dialog box.

1.21 Aggregation Policies

In SAP Convergent Charging 2023, an **aggregation policy** defines the rules that are applied to the different properties of a charged item during a charging operation performed in the session-based execution mode [page 20]. The aggregation policies force the SAP CC system to merge multiple charged items into a single one.

> → Remember
>
> In an integrated *SAP Solution* scenario, these business settings decrease the number of consumption items that SAP Convergent Charging sends (bulk loads) to SAP Convergent Invoicing.

An aggregation policy consists of:

- Aggregation rules that can be applied to one or more properties of the charged item class associated to the aggregation policy
- Expiration parameters used to configure the duration of an aggregation period

Before creating [page 274] and setting up [page 275] an aggregation policy, you must be familiar with charged item classes [page 254] because the aggregation rules are related to the properties of these classes.

For more information about aggregation policies, consult the technical guides. efer to the chapters *Charged Item Aggregation* and *Charging Processes*. See Help Resources [page 40].

1.21.1 Working with Aggregation Policies

Related Information

Creating an Aggregation Policy [page 274]
Opening an Existing Aggregation Policy [page 278]
Modifying an Aggregation Policy [page 279]
Setting Up an Aggregation Policy [page 275]
Deleting an Aggregation Policy [page 280]
Saving an Aggregation Policy [page 277]
Exporting an Aggregation Policy [page 280]
Importing an Aggregation Policy [page 281]

1.21.1.1 Creating an Aggregation Policy

Use one of these procedures to create a aggregation policy.

Create an Aggregation Policy

To configure a new aggregation policy from scratch:

1. From the menu bar in the main window of the user interface, select *File* > *New* > *Aggregation Policy*. The *Choose Owner* dialog box opens.
2. In the *Owner of Object* drop-down menu, select the owner of the aggregation policy.
3. Click *OK*. The *New Aggregation Policy* window opens.
4. Set up [page 275] the aggregation policy.
5. Verify the aggregation policy structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 277] the aggregation policy.

→ Tip

You can create a aggregation policy by opening an existing aggregation policy and copying it.

To create a aggregation policy from an existing aggregation policy

1. From the menu bar in the main window, select *File* > *Open* > *Aggregation Policy*. The *Select an Aggregation Policy* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the aggregation policy.
3. In the *Aggregation Policies* folder of the catalog of the owner, select the aggregation policy to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding aggregation policy window opens.
5. **Rename the aggregation policy** in the *Name* field.
6. Set up [page 275] the new aggregation policy.
7. Verify the aggregation policy structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
8. Save [page 277] the aggregation policy.

Related Information

Setting Up an Aggregation Policy [page 275]
Saving an Aggregation Policy [page 277]

1.21.1.2 Setting Up an Aggregation Policy

You must create [page 274], open [page 278], or modify [page 279] an aggregation policy before setting it up. An *Aggregation Policy* window must be open.

Set Up an Aggregation Policy

1. In the upper part of the *Aggregation Policy* window, identify the aggregation policy:
   - *Name*: Type the unique identifier of the aggregation policy if you create the aggregation policy. You cannot modify this field.
   - *Description*: Describe the function of the aggregation policy if necessary.
   - *Charged Item Class*: Select a charged item class [page 254] which must correspond to the class of charged items to aggregate. The properties of the selected class are used to fill the table located in the *Aggregation Rules* tab.

2. In the lower part of the *Aggregation Policy* window, you must fill in or modify the following tabs:
   - Aggregation Rules [page 276]
   - Aggregation Period [page 277]

Related Information

Aggregation Period (Tab) [page 277]
Aggregation Rules (Tab) [page 276]

1.21.1.2.1 Aggregation Rules (Tab)

The *Aggregation Rules* tab allows you to define the aggregation rules for each property of the charged item class that has been selected.

Each rule is based on a function which is applied to a property of the charged item class. The following functions are available:

- *Constant*: The default function is used to aggregate the charged items which have the same value and the same property type.
- *First*: The aggregated value corresponds to the non-zero value of the first charged item.
- *Last*: The aggregated value corresponds to the non-zero value of the last charged item.
- *Lowest*: The aggregated value corresponds to the lowest value of the pending and the current charged items. The null value is considered a positive infinity value, except for the date properties for which the null value is not taken into account.
- *Greatest*: The aggregated value corresponds to the highest value of the pending and the current charged items. The null value is considered a negative infinity value, except for the date properties for which the null value is not taken into account.
- *Sum*: The aggregated value corresponds to the sum of the values of the pending and the current charged items. The null value is considered the value 0.

The following table shows the types of property to which the functions apply (X for applicable and n/a for not applicable):

	String	Number	Date	Boolean
Constant	X	X	X	X
First	X	X	X	X
Last	X	X	X	X
Lowest	n/a	X	X	n/a
Greatest	n/a	X	X	n/a
Sum	n/a	X	n/a	n/a

Note

- You can specify a reporting threshold for each function related to a Number property. For each Number property, the *Threshold* enables the reporting of the pending charged item each time the threshold value is exceeded.
- The Constant function also enables the reporting of the pending charged item if the condition is not respected (the value of the current charged item is not equal to the value of the pending charged item for the same property type).

1.21.1.2.2 Aggregation Period (Tab)

The *Aggregation Period* tab allows you to define a period for aggregating charged items that are related to confirmations of quotas. When this period expires, the pending charged item is considered reportable, and no other aggregation can be performed on it by the system. A new aggregation period is then started to aggregate the future charged items.

To define the aggregation period, you set up the following options:

- *Activate*: Select the checkbox to activate or deactivate the aggregation period.
- *Maximum Duration (in Hours)*: Type the number of hours of the aggregation period.

1.21.1.3 Saving an Aggregation Policy

Once you have created a aggregation policy and then verified its configuration, you are ready to save it. You can save the aggregation policy to either the database or as an XML file on your hard disk.

To save a new aggregation policy to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the aggregation policy.
3. Click *Save*. The *Aggregation Policy Creation* dialog box opens to inform you that the aggregation policy has been successfully saved.
4. Click *OK*.

To save an existing aggregation policy to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.

2. If the *Aggregation Policy Modification* dialog box opens to inform you that the aggregation policy has been successfully modified, click *OK*. The new version of the aggregation policy is saved as the only current version.
3. If a *Warning* dialog box opens to inform you that the aggregation policy has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the aggregation policy.
   - Click *Cancel* to come back to the aggregation policy. See below how to save the aggregation policy as an XML file.

To save a aggregation policy as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the aggregation policy.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The aggregation policy is saved as an XML file in your selected directory.

1.21.1.4 Opening an Existing Aggregation Policy

You can open:

- An existing aggregation policy saved in the database
- An existing aggregation policy saved in an XML file

To open an existing aggregation policy saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Aggregation Policy*. The *Open Aggregation Policy* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the aggregation policy.
3. In the *Aggregation Policies* folder of the catalog of the owner, select the aggregation policy.
4. Click the *Open* button. The *Aggregation Policy* window opens in display mode.

To open an existing aggregation policy saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.21.1.5 Modifying an Aggregation Policy

The modifications that you can make to an aggregation policy differ depending on whether it has been saved or not. You cannot modify the name of the aggregation policy and its related charged item class once you have saved it.

Modify an Aggregation Policy

1. Open [page 278] the aggregation policy.
2. Set the editing mode by clicking the *Edit Mode* icon on the main toolbar. The aggregation policy may be reloaded if it has been modified since you have opened it.
3. Set up [page 275] the aggregation policy.
4. Verify the structure of the aggregation policy by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Save [page 277] the aggregation policy.

1.21.1.6 Deleting an Aggregation Policy

Before you delete an aggregation policy, you must first delete all objects which reference it or remove the references in the objects.

Delete an Aggregation Policy

1. From the menu bar in the main window, select *File* > *Delete* > *Aggregation Policy*. The *Delete Aggregation Policy* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the aggregation policy.
3. In the *Aggregation Policies* folder of the catalog of the owner, select the aggregation policy.
4. Click *Delete*. A deletion dialog box asks if you want to delete the aggregation policy.
5. Confirm the deletion of the aggregation policy by clicking *Yes*:
    1. If a dialog box confirms your choice, the aggregation policy has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the aggregation policy is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the aggregation policy. The identifiers of these objects appear in the message.

1.21.1.7 Exporting an Aggregation Policy

You can export aggregation policyes to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import the aggregation policy back into SAP Convergent Charging.

To export a aggregation policy

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the aggregation policy.
3. In the catalog of the owner, select the aggregation policy.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the aggregation policy.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.21.1.8 Importing an Aggregation Policy

You can import aggregation policies in XML format from a directory of your hard disk.

Several versions of an aggregation policy can exist in a database, corresponding to their stages of modification. You can import all the versions of an aggregation policy into SAP CC Core Tool or select only the versions you want.

To import a aggregation policy

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the aggregation policy is saved, and then select the aggregation policy file.
3. Click *Open*. The *Import Objects* dialog box opens.
4. Leave the default *All versions* option selected, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as* of option and choose the date on which you save the imported version.
5. In the *Import* column, clear the checkboxes of objects you do not wish to import.
6. Click *Import*.

> ⓘ Note
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available for import.

1.22 Consumption Item Mapping

> ⚠ Caution
>
> Read the caution [page 263] about the billable item mapping before deleting the consumption item mapping.

> → Remember
>
> These topics are relevant only if you use SAP Convergent Charging 2023 in an integrated *SAP Solution* scenario with SAP Convergent Invoicing with the business function for managing consumption items. Refer to the *Data Storage of Consumption Items* feature.

For more information about the storage of consumption data, refer to the help.sap.com/brim.

The feature is not available with SAP S/4HANA Cloud Public Edition.

In your integrated *SAP Solution*, you store the original service consumption information in SAP Convergent Invoicing in the SAP ERP/FI-CA system or in the SAP S/4HANA system.

> **Note**
>
> The management of consumption items from SAP Convergent Charging is not available in SAP S/4HANA Cloud Public Edition.

Consumption Item Classes

Defined in the SAP ERP/FI-CA reference system, *consumption item classes* are basically chargeable item classes with additional data fields for control purposes.

Consumption Item Mapping

The consumption item mapping consists in mapping chargeable item classes onto consumption item classes that are already configured in the SAP ERP/FI-CA reference system. At runtime, SAP Convergent Charging uses the settings in the consumption item mapping to convert chargeable items into consumption items, and then loads the consumption items into SAP ERP/FI-CA for data storage.

> **Note**
>
> If a consumption item class has been changed in the SAP ERP/FI-CA reference system, you must refresh [page 288] the existing consumption item mapping defined in SAP Convergent Charging. For more information, refer to your system administrator or refer to the maintenance procedure described in the SAP CC Administration Guide.

The consumption item mapping is not part of the master data and is not included in a pricing catalog [page 109]. The consumption item mapping represents business data configured for an SAP Convergent Charging system.

See: Creating the Consumption Item Mapping [page 283]

Data Maintenance

See: Refreshing the Consumption Item Mapping [page 288]

Representation in Core Tool

In your Core Tool user interface, the consumption item mapping is graphically represented by a table that displays the names of the consumption item class, the chargeable item class, and all the properties of the mapping.

Note that the first column of the table displays the property name of the consumption item class while the second column displays the property name of the chargeable item class.

At the top of the table, the *Module Function* field displays the name of the module function relating to the structure of the consumption item class (information only relevant for SAP ERP/FI-CA).

See: Working with the Consumption Item Mapping [page 283]

Related Information

Billable Item Mapping [page 263]

1.22.1 Working with the Consumption Item Mapping

Related Information

Creating the Consumption Item Mapping [page 283]
Opening the Consumption Item Mapping [page 285]
Modifying the Consumption Item Mapping [page 289]
Saving the Consumption Item Mapping [page 284]
Deleting the Consumption Item Mapping [page 290]

1.22.1.1 Creating the Consumption Item Mapping

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP Convergent Invoicing. Several implementation scenarios and variants are available. For more information and user assistance, see the integration guides at help.sap.com/brim. In a scenario or variant based on SAP S/4HANA Cloud Public Edition, this topic is not relevant.

> ⚠ Caution
>
> The feature is not available when SAP Convergent Charging is integrated and interfaced with SAP S/4HANA Cloud Public Edition.

Use this procedure to map a consumption item class onto a chargeable item class [page 241].

> ⓘ Note
>
> When you create the consumption item mapping in your pricing catalog, the chargeable item class is automatically created if it does not exist.

Create the Consumption Item Mapping

As a pricing specialist for SAP Convergent Charging:

1. From the menu bar in the main window, select *File* > *New* > *Consumption Item Mapping*. The *Choose Consumption Item Class* dialog box opens.
2. In the *Consumption Item Class* list of the dialog box, select a consumption item class.
3. Click *OK* to create the consumption item mapping. The window of the new consumption item mapping is displayed [28].
4. Verify the structure of the consumption item mapping by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Save [page 284] both the consumption item mapping and its associated chargeable item class.

1.22.1.2 Saving the Consumption Item Mapping

Once you have created, and then verified its structure, you can save the consumption item mapping to either the database or as an XML file on your hard disk.

If the chargeable item class does not exist, it is automatically created when you create the consumption item mapping. In this case, you save the new chargeable item class in a catalog.

---

[28] You cannot manually modify the elements of the window of the consumption item mapping in the Core Tool user interface.

You can only click:

- The icon  to refresh [page 288] the consumption item mapping (synchronization with SAP ERP/FI-CA)
- The icon  to open the related chargeable item class

The *Function Module* field displays the name of the API called by SAP CC for the mapping. This API is provided by SAP Convergent Invoicing according to the consumption item class.

Remember that you can do these operations only if you have first saved the consumption item mapping in the database.

Save a New Consumption Item Mapping and the Chargeable Item Class to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Consumption Item Mapping and Chargeable Item Class Confirmation* dialog box opens to ask if you want to create the consumption item mapping and save the new chargeable item class in a catalog.
2. Click *Yes*. The *Save in Catalog* opens to save the chargeable item class.
3. In the *Owner* drop-down menu, select the owner of the chargeable item class.
4. Click *Save*. The *Consumption Item Mapping and chargeable Item Class Creation* dialog box opens to inform you that the consumption item mapping and the chargeable item class have been successfully created.
5. Close the dialog box.

If the chargeable item class already exists (in a pricing catalog), only the consumption item mapping is created.

Save the New Consumption Item Mapping to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Consumption Item Mapping Creation* dialog box opens to ask if you want to create the consumption item mapping.
2. Click *OK*. The consumption item mapping is created.

Save the Consumption Item Mapping as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the consumption item mapping.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The consumption item mapping is saved as an XML file in your selected directory.

1.22.1.3 Opening the Consumption Item Mapping

Use this procedure to open the consumption item mapping.

> **Note**
>
> You cannot open the consumption item mapping from a file.

Open the Consumption Item Mapping Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Consumption Item Mapping*. The *Open Consumption Item Mapping* dialog box opens.
2. In the *Consumption Item Mapping* list of the dialog box, select the consumption item mapping that you want to open.
3. Click *OK*. The window of the consumption item mapping opens.
4. View [page 286] the complete mapping.
5. If necessary, refresh [page 288] the mapping.

Related Information

Viewing the Consumption Item Mapping [page 286]

1.22.1.4 Viewing the Consumption Item Mapping

Use this procedure to view the consumption item mapping configured in SAP Convergent Charging.

View the Consumption Item Mapping

1. Open [page 285] the consumption item mapping.
2. View the complete mapping:

Element	Description and Recommendations
Consumption Item Class	Displays the identification name of the consumption item class that the consumption item mapping relates to. This class is configured in SAP Convergent Invoicing. The SAP CC system generates consumption items by converting received chargeable items into consumption items.
Function Module	Displays (for information only) the name of the function module that is called by the SAP CC system to transfer a consumption item to SAP Convergent Invoicing in its system (ERP or S/4HANA).
Chargeable Item Class	Displays the identification name of the chargeable item class that the consumption item mapping relates to.

Element	Description and Recommendations
Mapping	This section includes a nonmodifiable configuration table. Each row represents the mapping for a property onto a field (in the given consumption item class). **Note** The columns *Basic Type*, *Length*,*Decimal Length*, and *Size (Bytes)* contains the format of the target fields so SAP CC can convert properties into fields. **Column Name** / **Description** *Name* — Displays the identification name of the field in the consumption item class that is mapped onto the given chargeable item class. *Property Name* — Displays the name of the mapped property, which is: • A context property in SAP CC • Or a custom property defined in the chargeable item class. *Type* — Displays the type of the property in the chargeable item class: • *Boolean* • *Date* (date and time) • *Number* • *String* *Basic Type (aka Data Type)* — Displays the target type *Length* — Displays the target length *Decimal Length* *Size (Bytes)* — Displays the size in bytes **Note** The displayed mapping does not correspond to the order of fields in the consumption item class. The configuration table contains two subsets: • First rows correspond to unknown fields. The property-to-field mapping defines both the conversion format (data type, length, size) and a custom property that must be configured in the chargeable item class associated with the consumption item class.

Element	Description and Recommendations
	• Second group of rows corresponds to known fields. SAP CC sets up the property-to-field mapping by reusing rating context properties and the relevant conversion format. **Example** The field CONSUMDATE is mapped onto the property *Rating Consumption Date*. The field GPART is mapped onto the property *External Account Code*. The field VTREF is mapped onto the property *Contract Identifier*.

3. If necessary, refresh [page 288] the mapping.

Similar

Viewing the Billable Item Mapping [page 269]

Related Information

Refreshing the Consumption Item Mapping [page 288]

1.22.1.5 Refreshing the Consumption Item Mapping

Note

In an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition, this refreshing mechanism is not available.

Use this procedure to refresh the consumption item mapping manually in order to resynchronize the configuration data between the two systems: SAP ERP (or SAP S/4HANA) with SAP Convergent Charging.

Refresh the Consumption Item Mapping

As a pricing specialist for SAP Convergent Charging:

1. Open [page 285] the consumption item mapping [page 281] in the Core Tool user interface.
2. Click the refresh button  to maintain the consumption item mapping and the associated chargeable item class when changes occurred in the structure of the consumption item class.
   The refresh operation enables the SAP CC Core Server system to retrieve the consumption item class from the SAP ERP/FI-CA reference system and synchronizes the changes in SAP CC.

   > ⓘ Note
   >
   > However, refreshing the consumption item mapping may fail if:
   > - The name of the related chargeable item class [page 241] has changed.
   > - A field of the consumption item class has been deleted.
   > - The field type of the consumption item class has changed.

3. Verify the changes.
4. Save [page 284] the refreshed consumption item mapping.
5. Maintain the chargeable item class that is associated to the refreshed mapping. See Modifying a Chargeable Item Class [page 252].

Related Information

Refreshing the Billable Item Mapping [page 271]

1.22.1.6 Modifying the Consumption Item Mapping

You cannot modify the consumption item mapping, but you can open [page 285] it, and then refresh [page 288] it by clicking the icon  to synchronize any change relating to the consumption item mapping (synchronization with the SAP ERP/FI-CA reference system).

Related Information

Refreshing the Consumption Item Mapping [page 288]

1.22.1.7 Deleting the Consumption Item Mapping

> ⚠ Caution
>
> Read the caution [page 263] about the billable item mapping before deleting the consumption item mapping.

Use this procedure to delete the consumption item mapping [page 281] from your pricing catalog.

Delete the Consumption Item Mapping

1. From the menu bar in the main window, select *File* > *Delete* > *Consumption Item Mapping*. The *Delete Consumption Item Mapping to Delete* dialog box opens.
2. In the *Consumption Item Mapping* list of the dialog box, select the consumption item mapping that you want to delete.
3. Click *OK*. A dialog box asks if you want to delete the consumption item mapping.
4. Click *Yes*. A dialog box opens to inform you that the consumption item mapping has been successfully deleted.
5. Click *OK* to close the dialog box.

1.23 Charges

> ⚠ Caution
>
> As of FPS 0 for release 2020, some charge plans are managed by Cockpit, which means that their related charges are also managed by Cockpit.
>
> These charges and charge plans are visible from Core Tool. Managing them using Core Tool is possible, however, know the following restrictions:
>
> - the system automatically converts these objects into objects manageable by Core Tool
> - you cannot roll back this operation: these objects can no longer be managed by Cockpit
> - you cannot handle charges managed by Cockpit in charge plans managed by Core Tool: the system prevents you from creating hybrid objects
> - you can convert a charge plan managed by Cockpit in a charge plan managed by Core Tool by using the *Open from a File* feature
> - you can import charge plans managed by Cockpit in Core Tool, subject to confirmation according to potential warning messages
>
> A pop-up message always warns you when you handle a charge or a charge plan managed by Cockpit.
>
> Once a charge plan and its related charges managed by Cockpit are converted to be managed by Core Tool, rules and limitations of Core Tool apply.

→ Remember

- As of SP 5 for release 5.0, the behavior of this master data is modified in case of an integrated *SAP Solution* based on SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts [page 514] (in subscriber accounts) is no more available. For further information, refer to SAP Note 2772452.
- In an integrated *SAP Solution* scenario with the SAP CRM and SAP ERP/FI-CA components of SAP Business Suite, if you need to manage payments to a business partner, SAP SE recommends that you only configure appropriate master charges in SAP Convergent Charging (SAP CC) with the *Revenue Sharing and Partner Settlement* feature enabled in SAP ERP/FI-CA. Refer to the O2C Integration Guide [page 40] documentation for more information about partner settlement, customer billable item classes, and partner billable item classes.
- Dependent charges cannot be used in stateless mode (refer to the section below if needed) even if menus and commands are available in the Core Tool user interface.

Stateless and Stateful Modes

Stateful and stateless are adjectives that describe whether a computer or computer program is designed to note and remember one or more preceding events in a given sequence of interactions with a user, another computer or program, a device, or other outside element. Stateful means that the computer or the program keeps track of interactions, usually by setting values in a storage field designated for that purpose. Stateless means that there is no record of previous interactions. Each interaction request is entirely processed from information that comes with it.

Stateless Mode

The stateless rating is a service provided by SAP Convergent Charging to your client application (online mediation system). The stateless rating feature means that the SAP CC system does not charge any account for a service usage but only calculates the corresponding price amount (rating process). During the stateless rating operations, the state of SAP Convergent Charging does not change. In this mode, SAP Convergent Charging does not modify or keep any data in its database after rating (balances, credit limits, counters in contracts).

For example, states of counters are stored and maintained in an external system outside SAP Convergent Charging. SAP CC Core Server runs as a basic 'calculator'. This rating process corresponds to the 3GPP "Class A rating definition".

ⓘ Note

In stateless mode, you cannot used objects such as charge plans, provider contracts, and dependent charges.

Stateful Mode

SAP Convergent Charging maintains the states of counters, for example, and updates counters when the rating is confirmed. In this mode, SAP Convergent Charging modifies or keeps data in its back-end database after charging. Stateful charging corresponds to the 3GPP "Class B rating definition".

In stateful mode, you can use any of SAP Convergent Charging's objects.

In SAP Convergent Charging 2023, a charging business scenario consists of two stages:

- A rating process (aka dynamic pricing)
- And a charging process (determination of invoice recipient)

For example, when an end customer consumes a usage-based service to monetize, the rating process dynamically calculates a priced amount while the charging process defines which prepaid or external account will be charged or refilled with the amount.

A reusable charge allows you to configure this charging business scenario through a decision tree made up of two parts:

- The *price plan* that models the rating process (dynamic pricing)
- The *charging plan* that models the charging process (account assignment)

Charges consist of two types: master [page 293] and dependent [page 294] charges.

Price Plan

A reusable charge always includes a single price plan. The price plan [page 299] is a decision tree representing the algorithm to calculate an amount linked to a usage event, a recurring event, or a one-shot event.

For example, an amount can be linked to the usage of an Internet service through a chargeable item class [page 241] if the price plan contains a usage rate component [page 302].

You can think of charges as small pricing blocks that are related to chargeable item classes (*) and exist independently of offers. And you can even use charges in several offers that are related to the same chargeable item class.

Charging Plan

A reusable charge always includes a single charging plan. A charging plan [page 305] is a decision tree that defines which accounts will be charged or refilled with the amount of the service calculated by the related price plan.

To charge or refill an account, the charging plan is linked to a subscriber account [page 511]. When an end customer subscribes to different charges, the charging references of the related charging plan are connected to one or more accounts defined in the related subscriber account [page 511].

Before Starting

→ Remember

- It is strongly recommended to read the caution [page 299] about what you can do or not in price plans

- Read the note [page 316] about making retroactive changes to a charge or a charge plan
- Read the note [page 297] about the persistent counters

Before creating [page 309] and setting up [page 310] a reusable charge, you must be familiar with the following:

- Charge Window Areas [page 295]
- Type of charges: master [page 293] and dependent [page 294] charges.
- Counters [page 297]
- Price Plans [page 299]
- Charging Plans [page 305]
- Version change of a charge [page 334]

Related Information

Customized Use of Mapping Tables in the Charges and Refill Logic [page 185]
Customized Use of Range Tables in the Charges and Refill Logic [page 224]
Using SAP CRM Prices in Recurring Charges [page 319]
Synchronizing Recurring Fees with Billing Cycles in SAP Convergent Invoicing [page 321]

1.23.1 Master Charges

A **master charge** is a charge [page 290]:

- That is **not synchronized** with other charges customized in a charge plan
- That is **independent** of other charges concerning its trigger mode and its calculation mode

A charge plan [page 343] contains at least a master charge, and the master charge contains the price plan. A master charge can be used to credit or debit a client or partner depending on its role in the charge plan.

This charge represents a main amount that is agreed upon between the end customer and the service provider. It does not represent an associated amount such as sponsorship, the commission paid on the sale of a service, or a derived amount such as a discount on a basic price (which varies depending on usage).

Trigger Mode

Contrary to the dependent charge [page 294], a master charge is triggered independently of other charges. For example, if an end customer decides to subscribe to a document consultation service, the corresponding subscription charge is triggered without being dependent on another event. If the master charge is triggered periodically, the periodicity is not linked to another charge.

Calculation Mode

A master charge contains the main price plan [page 299] of an amount. The master charge can function using solely its rating context property values and does not require data calculated in another price plan. You can share transient counters set up in the price plan of the master charge with the price plan of dependent charges.

Characteristics

You need to set up charges if you use them in charge plans. Setting up a master charge means redefining its existing characteristics.

> ⓘ Note
>
> You define a reusable charge as a master charge on the Definition [page 311] tab.

1.23.2 Dependent Charges

> ⓘ Note
>
> - In an integrated *SAP Solution* scenario with the SAP CRM and the SAP ERP/FI-CA components of SAP Business Suite, if you need to manage payments to a business partner, SAP SE recommends that you do not configure any dependent charges. Refer to the note [page 290].
> - Dependent charges cannot be used in stateless mode (refer to the section below if needed) even if menus and commands are available in the graphical user interface.

Unlike a master charge [page 293], a **dependent charge**:

- Is synchronized with another charge customized in the charge plan.
- Uses either the trigger mode or the calculation mode of the other charge or uses both modes simultaneously.
- May depend upon another dependent charge. In this case, the links between charges are translated into a series of descending dependencies.
- Is assigned a role [page 346] when it is customized in a charge plan.

> ✤ Example
>
> Private individuals can subscribe to *Learn.com*, a company providing a range of online documents. Learn.com buys the documents from specialized authors to whom it pays royalties in the form of commissions on received income.
>
> Commissions are represented by a **`Sales Commission`** dependent charge linked to a **`Document Subscription`** master charge.

Trigger Mode

A dependent charge can only be triggered when the charge it depends on is triggered.

> Example
>
> For example, when a customer pays subscription fees, a commission is to be paid to a third party. In this way, the **`Document Subscription`** charge triggers the dependent **`Sales Commission`** charge.

Calculation Mode

The calculation of a dependent charge can be linked to the amount calculated by another charge. The price plan of the dependent charge depends on the result of another price plan that is integrated in the other charge. This dependency means an amount will first be calculated in the main price plan before it can be used by the dependent plan.

The property that you use to link a master charge amount to a dependent charge is the base amount property  . The link is determined by a common chargeable item class, a recurring rate, or a one-shot rate component. The base amount property can be used in cases such as commissions where a 10% commission is calculated from a master charge amount.

> Example
>
> End customers can download reference documents from *Learn.com*. Downloading costs vary according to the module and document volume. Learn.com must pay the document authors a commission equal to 5% of sales. You need to know the download amount to determine the amount of commission to be paid.
>
> The **`Document Download`** charge calculates the amount to be paid to Learn.com by its customers. The **`Document Commission`** charge calculates an amount to be paid to the authors, and this amount depends on the amount paid by the end customers.

Characteristics

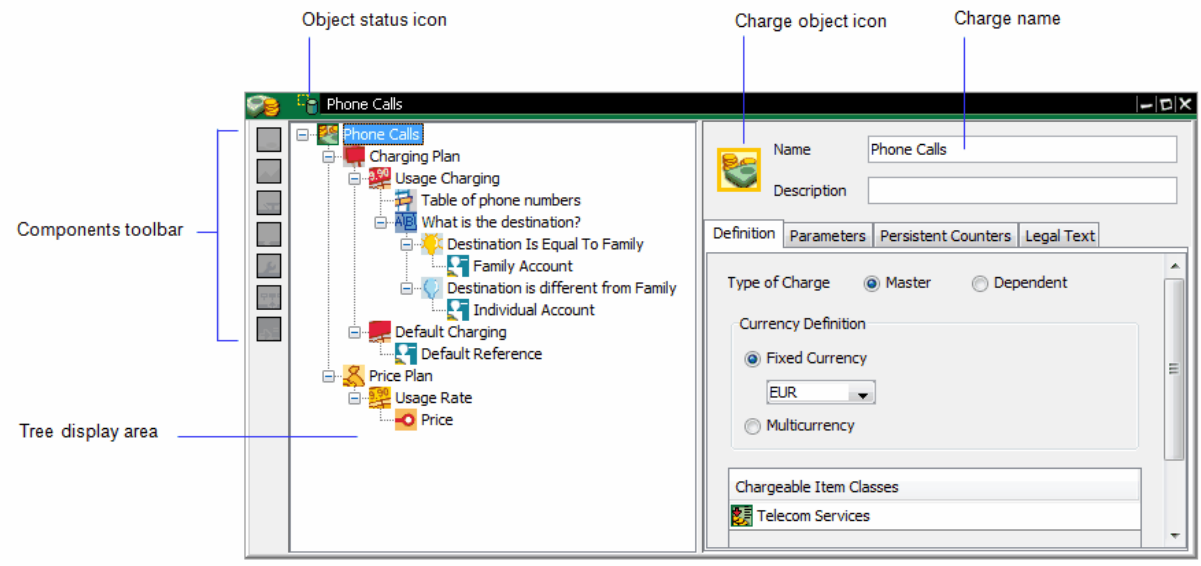
You need to set up charges if you use them in charge plans. Setting up a dependent charge means redefining its existing characteristics if necessary.

> Note
>
> Dependent charges are set up on the Definition [page 352] tab of charges included in charge plans .

1.23.3 Charge Window Areas

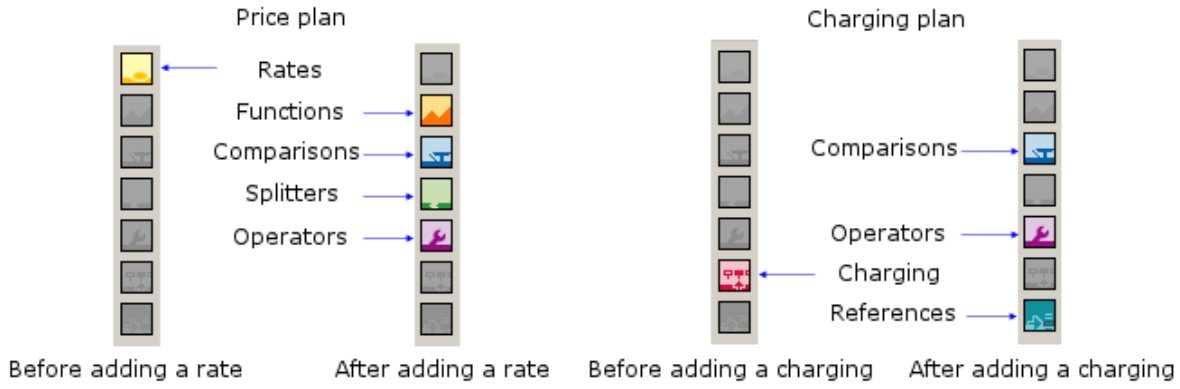
The *Charge* window opens when you open a charge or create a charge and link it to a chargeable item class.

The following figure shows a *Charge* window in which an SAP CC user has built a charging plan and a price plan. The charge "Phone Calls" is defined as a master charge and is linked to the chargeable item class “Telecom Services”.



The *Components toolbar* is available when you click the *Price Plan* icon  or the *Charging Plan* icon  in the tree display area. The toolbar contains shortcut buttons to add calculation tools (called components) that are categorized into seven families according to their main functions.

If you have not yet added a component to your tree, only the *Rates* family icon or the *Charging* family icon is available. After adding a Rate or a Charging component, the other component families are available depending on the type:



The object status icon [page 53] shows you if the open object is being created, is being modified, or has been saved.

The *tree display area* shows a graphical representation of the tree, which is made up of rating and charging components that appear as icons. By clicking on different elements in the tree, the information related to the selected element is displayed in the *tab area* and different component families are available on the *Components toolbar*.

1.23.4 Counters

In SAP Convergent Charging 2023, a *counter* is a numeric variable you can define and declare in a reusable charge. Each counter has a name and an initial value that you assign when you create the charge. A counter can be used in all rates, and you can change its value at any time in the charge. There is no limit to the number of counters you can create. During the implementation project phase, you check the performances.

You can declare two types of counters:

- Persistent [page 297]
- Transient [page 298]

Related Information

Sharing Transient Counters with Dependent Charges in Charge Plans [page 386]

1.23.4.1 Persistent Counters

> **Note**
>
> Persistent counters are declared at the root of the reusable charge (see the Definition [page 311] tab). You can redefine persistent counters in a price plan but you cannot redefine persistent counters in a charging plan. This means that the persistent counters are only used with their initial value in a charging plan.

When you declare a persistent counter  in a charge, SAP CC will create a counter in each provider contract that activates this charge. The counter is updated and saved to the back-end database during the charging process and refilling process.

Persistent counters declared in a charge are visible and can be redefined in charge plans including the charge.

In charge plans, you can configure persistent counters which can then be shared across some or all charges. These are called shareable counters.

Persistent counters introduce a new numeric rating property dedicated to a specific provider contract. You can retrieve the rating property via the calculation system.

> **Example**
>
> As the telephone operator TelcoServ, you offer a package called Advantage. End customers pay a subscription fee of $35 and receive special rates for weeknight and weekend calls. Anyone subscribing to this charge plan receives 30 free minutes of call time. In order to include these minutes of usage as free of charge in your price plan, you create a counter that decrements for each minute used and signals SAP CC Core Server when 30 minutes have been used up. In this way, it can then start charging the service usage.

1.23.4.2 Transient Counters

> **Note**
>
> Transient counters are only available in the price plan of a reusable charge (see the Definition [page 323] tab).

Contrary to a persistent counter, a transient counter  is not saved to the back-end database system but stored in internal memory during the charging process. The transient counters declared in a reusable charge are not visible and cannot be redefined in charge plans including the charge; However, transient counters can be shared [page 386] by master and dependent charges.

The life cycle of transient counters is linked to a charging session because the transient counters are created when the corresponding session-based charging process starts, and they are deleted when the session-based charging process stops.

Transient counters are mainly used in tariff plans intended for session-based charging [page 20].

> **Note**
>
> Transient counters can be shared with a master charge and its dependent charges once the reusable charges are customized in a charge plan. Refer to the procedure to enable this share.

> **Example**
>
> In the chargeable item class "Telephone Services", you charge the end customer $0.20 per call minute. To compute the price of a telephone call, you create a price plan using the transient counter "call_duration" and the user properties [page 241] "duration_quota" and "method". The user property "method" specifies if the duration quota is confirmed (updated) or reserved (not updated). The following table shows the session-based rating of a 3-minute call:

Session step	Amount to cancel	Amount to confirm	Amount to reserve
Start (method = reservation) (duration_quota = 3 minutes)			3 x $0.20 = $0.60 call_duration = 0
Update (method = confirmation) (duration_quota = 2 minutes) --- (method = reservation) (duration_quota = 3 minutes	$0.60 (3 x $0.20) call_duration = 0 ---	2 x $0.20 = $0.40 call_duration = 2 minutes ---	--- 3 x $0.20 = $0.60 call_duration = 2 minutes

Session step	Amount to cancel	Amount to confirm	Amount to reserve
Stop (method = confirmation) duration_quota = 1 minute	$0.60 (3 x $0.20) call_duration = 2 minutes	1 x $0.20 = $0.20 call_duration = 3 minutes	

1.23.4.3 Using a Charge to Configure the Initialization of a Shared Counter

You configure a reusable charge that defines the initialization of a shared persistent counter when it is created in a provider contract [page 573].

This charge must include:

- The declaration of the persistent counter you want to share
- A parameter dedicated to the counter initialization
- A *One-Shot Rate* component[29] that configures the initialization of a counter with the parameter value

> ⚠ Caution
>
> Do not initialize the same counter twice.

To create a reusable charge dedicated to the settings initializing the shared counter

1. Create and set up a reusable charge.
2. Add a one-shot rate component, a persistent counter, and a parameter to the charge. Note that the parameter enables you to initialize the counter value at provider contract level.
3. Set the *Event Name* to *Activation* in the *Definition* tab of the *One-Shot Rate* component.
4. Add a Counter Operator [page 302] component to the *One-Shot Rate* branch tree.
5. Set the counter to the value of the parameter in this *Counter Operator* component[30]

1.23.5 Price Plans

A price plan is a set of instructions that computes the amount of a charge [page 290]. A price plan is graphically represented by a rating tree.

[29] See the primary help of logic components [page 671] for more information about *One-Shot Rates* and *Counter Operators*
[30] See the primary help of logic components [page 671] for more information about *One-Shot Rates* and *Counter Operators*

At runtime, calculations in a price plan are only performed when the charge is triggered by:

- The arrival of service usage information or a ticket after a service[31] has been consumed.
- A provider contract event: activation, suspension, or resumption
- A periodic event in the case of a recurring amount such as an annual subscription fee
- The trigger of a master charge for a dependent change

> → Remember
>
> Read the topic [page 523] about taxes because you can include or not the taxes in price plans.

Before Starting

> ⚠ Caution
>
> We strongly recommend that you do not modify the structure of price plans in existing charges that are already customized in a released charge plan [page 352].
>
> You can however modify existing charges already inserted in a released charge plan by adding to or removing from price plans any component with the exception of RECURRING RATES or ONE-SHOT RATES components.
>
> If you want to add RECURRING RATES or ONE-SHOT RATES components to an existing charge already customized in a released charge plan, you must do the following:
>
> - Create a new charge in which you insert the one or more new components
> - Create and release a new charge plan by adding the new charge
> - Update the provider contracts to add a contract item referring to the new charge plan

Before setting up [page 323] a price plan, you must be familiar with the following:

- Rating Tree [page 301]
- Rate Components [page 302]
- Counters [page 297]
- Parameters [page 303]
- Rating Context [page 13]
- Session-Based Charging and Reservation [page 20]

Related Information

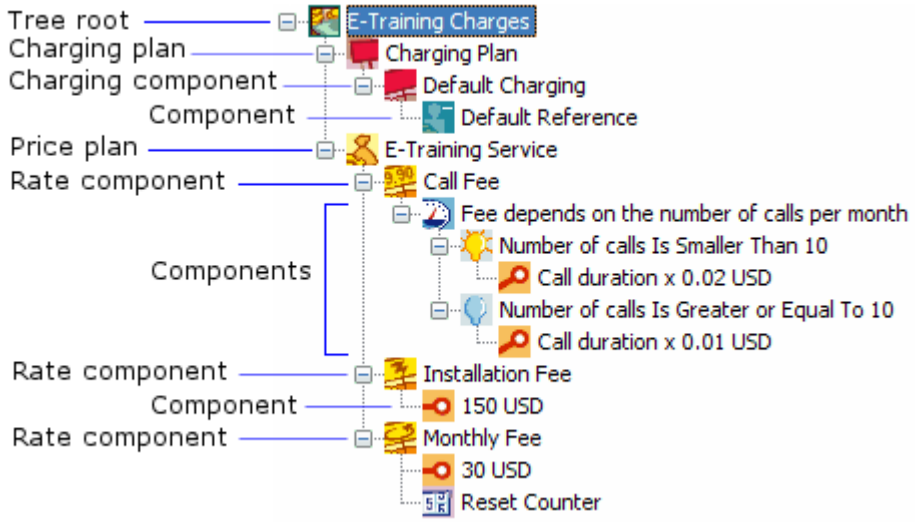
Working with Price Plans [page 323]

[31] Represented in SAP CC by a chargeable item class [page 241]

1.23.5.1 Rating Tree

A price plan [page 299] is graphically represented by a tree structure called a rating tree. A price plan component represents each instruction inside a price plan. A rating tree is a decision tree that contains the price plan. This structure consists of a root from which rate components [page 302] create branches. At the ends of each branch, a Function component describes an amount.

As shown in the illustration below, the tree contains the root  representing the charge and the price plan  with branches for the different rate components:



-
-
-
-

In addition to rate components, you can create the following items that can be redefined at charge plan or provider contract level:

- Counters [page 297]
- Parameters [page 303]

Several price plans can reuse pricing macros and translation tables.

Test of a price plan

Once your price plan is complete, SAP Convergent Charging provides you with the Rating Trial Run [page 335] tool to test your price plan.

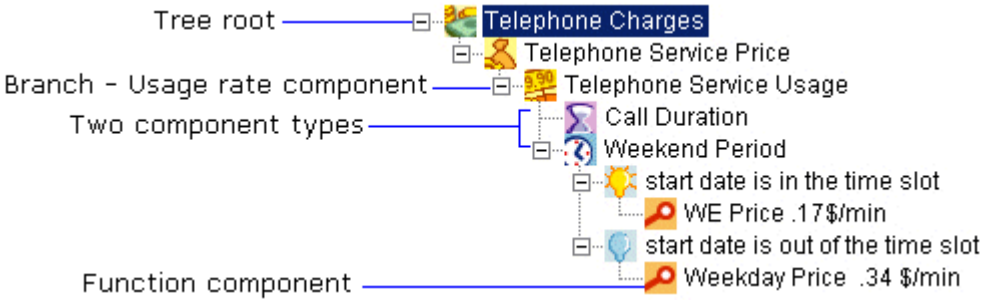
1.23.5.2 Example of a Price Plan

You are the telephone operator TelcoServ who sells telephone communications. You invoice your customers' telephone calls on a call-length basis. If the call is made during the weekend, the price is $0.17 cents/minute. If the call is made during the week, the price is $0.34 cents/minute.

Your Price Plan will contain a Usage rate branch. This branch will include several calculation or test actions which answer the following questions:

- How long was the call?
- Was the call made during the weekend period?
- What is the price to be applied?

You carry out these actions and decisions by using components that belong to several families. Your rating tree looks like this:



1.23.5.3 Rate Components

Rate components are calculation tools you use to create a price plan [page 299] or a pricing macro [page 121]. Together the components make up a rating tree. You use the rating tree in a price plan to calculate the amount of a charge [page 290].

Each main branch of the price plan begins with a Rate component that is linked to a rate activation method. There are three types of rate linked to three methods of activation:

Type of rate	Activation
Usage	Each usage
Recurring	Periodic
One-Shot	Once only (One-Off)

The values generated higher up a rating branch become available to all parts of the branch further toward the ends of the branch. When a property is generated on one rating branch, it cannot be used on another rating branch.

Once you have selected the price plan icon  in the rating tree display area of your charge, you can start building a rating tree.

Price plan components

The components of a rating tree are grouped into five families depending on their main function:

- Rate components  always begin the main branches of a rating tree.
- Comparator  , Splitter  and Operator  components help to calculate a rate.
- Function components  end each branch of a rating tree with a final calculated amount.

> **Note**
>
> For more comprehensive detail on using each component, refer to the primary help relating to components: Working with Logic Components [page 671]

1.23.5.4 Parameters

In SAP Convergent Charging 2023, a *parameter*  is a variable defined in a reusable charge [page 290] to customize the pricing logic for the subscription- and usage-based service to monetize. You set up a name, a type, and a default value to the parameter when you declare it in the reusable charge. This declaration leads to the automated creation of parameters in the subscribed provider contract by using the default value or a new and redefined value. The advantage of this parameter is that you can redefine its value in the pricing catalog or in the customer master data.

Use

At runtime, a parameter introduces a new numeric, a date, or a string-type property to the rating context [page 13]. You can configure a price plan [page 299] or a charging plan [page 305] that uses the parameter value.

> **Note**
>
> Particular string types are available in a reusable charge: *Mapping Table ID*, *Mapping Table Key*, and *Range Table ID*. You need these types to use mapping tables [page 179] or range tables [page 213] in your pricing logic or to use SAP CRM prices in recurring charges [page 319].

Redefinition

A parameter **can be redefined later** on to highly customize the pricing configuration of your chargeable service or product:

- In the pricing catalog, you can redefine the value of a parameter when you add the charge that contains it to a charge plan [page 343] (see the parameters of a charge plan [page 354]).
- In customer master data, a parameter is automatically added to a provider contract [page 573] that is subscribed to by an end customer. The parameter value is redefined when this charge is activated (contracted) as part of a provider contract. For example, see the parameters of contract items [page 616].

Example

See Example of Parameters [page 304].

Configuration

Parameters are set up on the Parameters [page 312] tab of reusable charges.

1.23.5.4.1 Example of Parameters

TelcoServ company proposes a certain number of call time (hours) for a given subscription. The configured price plan calculates the corresponding subscription fee, which can also be a parameter. The call plan has two parameters **`Subscription`** and **`Call Time`**. These parameters can vary as shown below:

Subscription Parameter	Call Time Parameter
$20	2 hours
$40	5 hours
$60	10 hours

Using these parameters, you can design and configure specialized packages (call plans) where no other parameters are visible.

For example, you can model, design, and configure a **`2-hour Subscription`** package in which the amount of call time is set to two hours and the amount of the subscription is $20.

1.23.6 Charging Plans

A charging plan is a set of instructions that determines a reference to:

- Charge the calculated amount of a price plan to an account.
- Refill the account with the calculated amount of the refill logic.

A charging plan is represented by a charging tree whose leaves are References components [page 307] (only Internal References are available).

Charging References

When working with charges and charge plans [page 343] for provider contracts, you create **charging references which are typed**.

In a charging plan, you can only use **internal charging references**. Each Internal Reference component [page 307] contains a charging reference to a prepaid or postpaid account. You create the charging references in the internal reference dictionary [page 327]. When the charging references are assigned to the accounts defined in a charge plan [page 343][32], SAP CC can charge or refill the accounts with the calculated amounts of the price plan or refill logic.

Before Starting

Before setting up [page 326] a charging plan, you must be familiar with the following:

- Charging Tree [page 305]
- Charging Components [page 307]
- Charging Plan rules (refer to the primary help relating to components here [page 671].)
- Account Assignments [page 593] in provider contracts

Related Information

Working with Charging Plans [page 326]

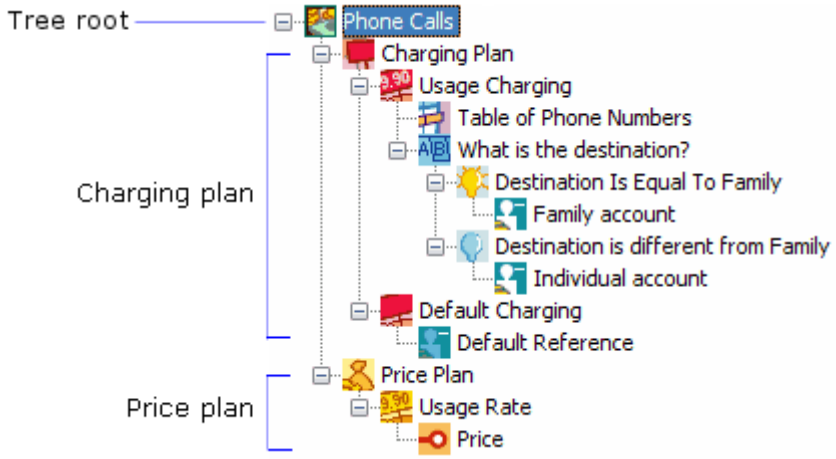
1.23.6.1 Charging Tree

A charging plan is graphically represented in the user interface by a tree structure called a charging tree. Each instruction inside a charging plan is represented by either a charging component or a pricing component[33]. A

[32] In a charge plan, you assign the charging references to accounts defined in the charge plan by configuring the Account Assignments [page 385] tabs accordingly.

charging tree is a decision tree that contains the charging plan. This structure consists of a root from which charging components create branches. At the ends of each branch, a Reference component describes which account must be debited or credited.

As shown in the illustration below, the tree contains the root  representing the charge, the charging plan [page 305]  , and the charging tree with the branches for the different charging components [page 307] or rate components [page 302]:



In a charging plan, you can use translation tables [page 132] or parameters [page 303][34]. The parameters can be redefined at charge plan or provider contract level. Translation tables can be reused in several charges.

1.23.6.2 Example of a Charging Plan

TelcoServ offers a Family package allowing to charge calls to a family or individual account depending on call destinations. The family Smith subscribed to the Family package allowing Mr and Mrs Smith's calls to be charged to an external account (postpaid account) while their son's calls are charged to a prepaid account. When John calls his parents, calls are charged to the external account

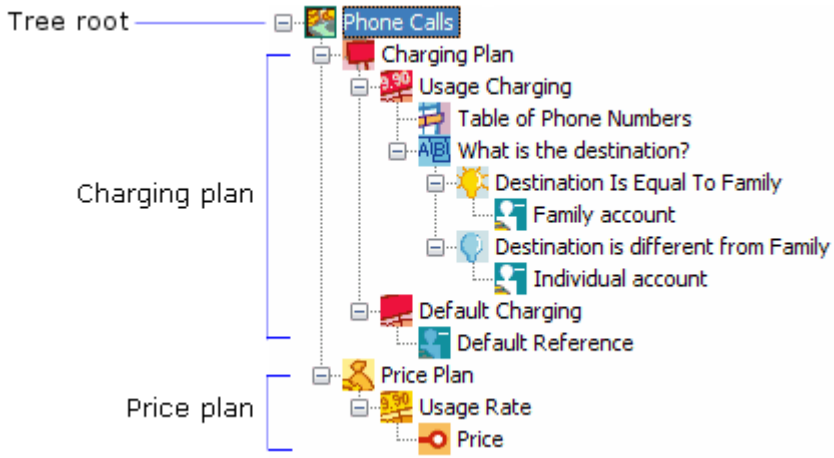
According to the illustration below, the price plan calculates the price of each call. After calculation, the amount of the price plan is charged to the Family account (external account) or the Individual account (prepaid account).

The charging plan contains a *Usage charging* branch. This branch includes test actions which answer the following questions:

- What is the phone number?
- What is the call destination?
- Which account should be charged?

These actions and decisions are carried out by using components that belong to a several families. The charging tree looks like this:

[33] A charging plan can also contains some rate components such as comparators and operators.
[34] Parameters are created in the charge.



1.23.6.3 Charging Components

Charging components are tools you use to create a charging plan [page 305]. Together the components make up a charging tree [page 305]. You use the charging tree in a charging plan to debit or refill an account.

Each main branch of the charging plan begins with a Charging component that is linked to a charging activation method. There are three types of charging linked to three methods of activation:

Type of charging	Activation
Usage	Each usage
Recurring	Periodic
One-Shot	Once only

The values generated higher up a charging branch become available to all parts of the branch further toward the ends of the branch. When a property is generated on one charging branch, it cannot be used on another charging branch.

Once you have selected the charging plan icon in the charging tree display area of your charge, you can start building a charging tree.

Charging plan components

The components of a charging tree are grouped into three families depending on their main function:

- Charging components  always begin the main branches of a charging tree.
- Comparator  and Operator  components[35] help to make comparisons, create, or modify properties.
- Reference components  end each branch of a charging tree.

> ⓘ Note
>
> For more comprehensive detail on using each component, refer to the primary help relating to components here [page 671].

1.23.7 Working with Reusable Charges

> → Remember
>
> Before working with reusable charges, you must have read the Caution note [page 290] about charges and charge plans.

Related Information

Creating a Reusable Charge [page 309]
Opening an Existing Charge [page 314]
Modifying a Charge [page 316]
Setting Up a Reusable Charge [page 310]
Adding a Component [page 330]
Saving a Charge [page 313]
Reading a Price Plan [page 335]
Doing a Trial Run of a Price Plan [page 335]
Deleting a Charge [page 317]
Deleting the Versions of a Charge [page 335]
Exporting a Charge [page 318]
Importing a Charge [page 318]
Using SAP CRM Prices in Recurring Charges [page 319]
Synchronizing Recurring Fees with Billing Cycles in SAP Convergent Invoicing [page 321]

[35] These components are also used in a price plan [page 299] to calculate a price.

1.23.7.1 Creating a Reusable Charge

Use one of these procedures to create a reusable charge [page 290].

To create a charge

1. From the menu bar in the main window, select *File* > *New* > *Charge*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the new charge.
3. Click *OK*. The *New Charge* window opens.
4. Set up [page 310] the reusable charge.
5. Verify the charge structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 313] the reusable charge.

> → Tip
>
> You can create a charge by opening an existing charge and copying it.

To create a charge from an existing charge

1. From the menu bar in the main window, select *File* > *Open* > *Charge*. The *Open Charge* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the new reusable charge.
3. In the *Charges* folder of the catalog of the owner, select the charge to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Charge* window opens.
5. Modify [page 316] the reusable charge if necessary.
6. Verify the charge structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 313] the reusable charge.

> ⓘ Note
>
> - A master charge [page 293] is independent of all other charges.
> - A dependent charge [page 294] depends on another charge in order to be calculated or triggered.
> - Each Usage Rate component [page 690] configured in a price plan can only refer to an exclusive chargeable item class [page 241], which means two usage rates cannot share the same chargeable item class (this is true for a Usage Charging component in a charging plan).
> - For more information about working with rating or charging components, see the topic Working with Logic Components [page 671].

Related Information

Setting Up a Reusable Charge [page 310]
Saving a Charge [page 313]

1.23.7.2 Setting Up a Reusable Charge

Create [page 309], open [page 314], or modify [page 316] a charge before setting it up. A *Charge* window must be open.

The elements of the Charge window [page 295] enable you to create:

- A price plan [page 299] that calculates the amount of a charge
- A charging plan [page 305] that charges the calculated amount of the price plan to an account

Set Up a Reusable Charge

1. In the tree display area of the *Charge* window, click the charge object icon  .
2. In the right pane of the charge window, identify the reusable charge. In the *Name* field, type the unique identifier of the charge. In the *Description* field, describe the function of the charge if necessary.
3. Set up the following tabs of the charge:
   - Definition [page 311]
   - Parameters [page 312]
   - Legal Text [page 313]
4. Click the price plan  in the right pane of the charge window, and then set up [page 323] the price plan of the charge.
5. Click the charging plan  in the right pane of the charge window, and then set up [page 326] the charging plan of the charge.

1.23.7.2.1 Charge: Definition

Definition (Tab)

The *Definition* tab allows you to define the type and currency of the reusable charge and to select the chargeable item classes relevant for the configuration of *Usage Rate* components.

Type of Charge: Select the charge type, Master [page 293] or Dependent [page 294]. A master charge is independent of all other charges while a dependent charge depends on another charge in order to be calculated or triggered.

> ⓘ Note
>
> About currencies:
>
> - At charge plan [page 370] level, you can redefine the currency if you assign the *Multicurrency* option to the charge.
> - The currency cannot be modified once the charge has been created.

Currency Definition: This area allows you to select the currency that you want to associate to the charge.

- *Fixed currency*: Provides the list of currencies. Select a currency.
- *Multicurrency*: Select this option if you do not want to associate a specific currency to the charge. This option allows you to create one only charge that can be reused in different charge plans or refill plans. This option avoid creating the same charge many times and associating each charge to a different currency.

Chargeable Item Classes: This area displays the chargeable item classes [page 241] related to the charge. Clicking a chargeable item class opens its window. Do not forget that a chargeable item class is associated to a charge when you add a usage rate to the price plan of the charge. If the chargeable item class belongs to a chargeable item package [page 241] coming from a version earlier than SAP CC 4.0, the chargeable item class is identified by a two-part name, one of which is in parentheses. For example, the chargeable item class "SMS (E2E_TELCOMOB_CIP)" is part of the chargeable item package "E2E_TELCOMOB_CIP".

> ⓘ Note
>
> Persistent counters can be reused in price plans and charging plans but cannot be modified in the price and charging plans.

Persistent counters: Click  to add a row to the table. Click  if you want to delete a persistent counter from the table. Fill in the following columns of each row that you add to the table:

- *Name*: Type the persistent counter name
- *Description*: Describe the function of the persistent counter if necessary
- *Initial Value*: Type a numerical value to initialize the persistent counter

1.23.7.2.2 Charge: Parameters

Create [page 309] or modify [page 316] a charge before setting it up [page 310]. In the left pane of the *Charge* window, click the charge root  to view, and then open the *Parameters* tab.

The *Parameters* tab allows you to add parameters to the charge [page 290].

Parameters (Tab)

Fill in or modify the *Parameters* tab.

 : Click to add a parameter [page 303] to the configuration table.

 : Click to delete a parameter from the table.

Name: Type the parameter name.

Description: Describe the function of the parameter if necessary.

Type: Provides the list of parameter types. Select the parameter type: *Number*, *String* or *Date*, *Mapping Table ID*, *Mapping Table Key*, or *Range Table ID*[36].

Value: Type a numerical value, a string value, or a date to initialize the parameter depending on its type. For the types *Mapping Table ID* and *Range Table ID*, a window is displayed. Select the appropriate class and the mapping table [page 179] or range table [page 213].

Visibility: Select one of the following option:

- *Internal*: The parameters are only available inside the reusable charge [page 290].
- *External*: The parameters can be modified outside the charge, in a charge plan [page 343], for example.

Recurring Charges Configuration

> ⓘ Note

In an integrated *SAP Solution* scenario, if your charge contains recurring rates [page 693], you can create special parameters to:

1. Use the prices [page 319] defined for SAP CRM products.
2. Synchronize your recurring rates [page 321] with the billing cycles of SAP Convergent Invoicing.

[36] Refer to Parameters [page 303] in price plans.

1.23.7.2.3 Charge: Legal Text

Create [page 309] or modify [page 316] a charge before setting it up [page 310]. In the left pane of the *Charge* window, click the charge root  to view, and then open the *Legal Text* tab.

The *Legal text* tab allows you to type text, legal or not, regarding the charge. You can save the text to a file or open a file to import text in the *Legal Text* tab.

Legal Text (Tab)

Fill in or modify the *Legal Text* tab.

 : Click to clear all the text in the *Legal Text* tab.

 : Click to import text in the *Legal Text* tab. The *Open* dialog box opens. Open the directory in which the text is saved, select the text file, and then click *Open*.

 : Click to save the text to a file. The *Save* dialog box opens. Select a directory to save the text to. In the *File name* field, type the name of the file, and then click *Save*.

1.23.7.3 Saving a Charge

Once you have created or modified a reusable charge, and then verified its structure, you are ready to save it. You can save the reusable charge either into the back-end database or as an XML file on your hard disk.

Save and Release a New Reusable Charge into the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the pricing catalog that includes the service pricing configuration for the subscription- and usage-based services to monetize.
3. Click *Save*. The *Charge Creation* dialog box opens to inform you that the charge has been successfully saved and released.
4. Click *OK*.

Save and Release a New Version of an Existing Reusable Charge into the Database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the charge has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the charge.
> 2. Click *Cancel* to come back to the charge. See later in this chapter how to save the charge as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the charge.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date/time while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates/times by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date/time in the past while *No End Date* represents an undetermined date/time in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Charge Modification* dialog box opens to inform you that the charge has been successfully saved and released.
5. Click *OK*.

> **Note**

Save a Reusable Charge as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the charge.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The charge is saved as an XML file in your selected directory.

1.23.7.4 Opening an Existing Charge

You can open:

- An existing charge saved in the database
- A previous version of a charge saved in the database
- An existing charge saved in an XML file

To open an existing charge saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Charge*. The *Open Charge* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge.
3. In the *Charges* folder of the catalog of the owner, select the charge. The *charge* window opens in display mode.

Charges are managed chronologically. After you save a charge, you can access previous versions [page 334] of it that exist in the database.

To open a previous version of a charge saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Charge*. The *Open Charge* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge.
3. In the *Charges* folder of the catalog of the owner, select the charge, and then click *Open*. The *Charge* window opens in display mode.
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the charge from the list.
6. Click *View*.

To open an existing charge saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.23.7.5 Managing the Versions

See:

- Changing the Version of a Charge [page 334]
- Deleting the Versions of a Charge [page 335]

1.23.7.6 Modifying a Charge

You can modify a price plan [page 299] or a charging plan [page 305] and add to or remove counter declarations or parameters from a previously saved charge. When you save the charge in the database, you can modify its versions (see the process below).

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

To modify a charge

1. From the menu bar in the main window, select *File* > *Open* > *Charge*. The *Open Charge* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge.
3. In the *Charges* folder of the pricing catalog [page 109], select the charge.
4. Click *Open*. The *Charge* window opens in display mode.
5. Set up the edit mode by clicking the *Edit Mode* icon on the main toolbar. The charge may be reloaded if it has been modified since you have opened it.
6. Set up [page 310] the charge.
7. Verify the charge structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
8. Save [page 313] the charge.

> **Note**
>
> You cannot modify a charge type once you have defined it.

Related Information

Display/Edit Mode [page 46]
Setting Up a Reusable Charge [page 310]
Saving a Charge [page 313]

1.23.7.7 Deleting a Charge

Before you delete a charge, you must first delete all master data objects which reference it.

> **Caution**
>
> There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.

To delete a charge

1. From the menu bar in the main window, select *File* > *Delete* > *Charge*. The *Delete Charge* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge.

3. In the *Charges* folder of the catalog of the owner, select the charge.
4. Click *Delete*. A deletion dialog box asks if you want to delete the charge
5. Confirm the deletion of the charge by clicking *Yes*:
    1. If a dialog box confirms your choice, the charge has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the charge is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> **Note**
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the charge. The identifiers of these objects appear in the message.

1.23.7.8 Exporting a Charge

You can export charges to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated objects to a file, allowing you to import [page 239] the charge back into SAP CC Core Tool.

> **Note**
>
> If you export a charge using chargeable item classes belonging to a chargeable item package, SAP Convergent Charging exports the chargeable item package [page 241] with all its chargeable item classes.

To export a charge

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge.
3. In the catalog of the owner, select the charge.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the charge.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.23.7.9 Importing a Charge

You can import charges in XML format from a directory in your database.

Several versions of a charge can exist in files, corresponding to their stages of modification. You can import all versions of a charge into SAP Convergent Charging, or select only the versions you want.

> **Note**
>
> You cannot import charges which are not compatible with charges plans.

To import a charge

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the charge is saved, and then select the charge file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected if you want to import all the versions of the imported objects, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option, and then select the date in the *Version as Of* field.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
6. In the *Import* column, clear the checkboxes of objects you do not want to import. You can remove the full list of chargeable item packages [page 241] if the chargeable item classes belonging to these chargeable item packages are not used in existing charges of the database. SAP CC Core Tool allows this operation only if you select the options *All versions* or *Replace All Versions* because the chargeable item packages are not arranged in chronological order.
7. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported [page 239] and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.23.7.10 Using SAP CRM Prices in Recurring Charges

> **→ Remember**
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

When working with SAP CRM, you may want to use the prices defined in the SAP CRM system to manage the some recurring fees in SAP Convergent Charging. You use a specific tariff: Recurring Rate [page 693]. The pricing configuration is based on this logic component.

Synchronizing SAP CC recurring rates with SAP CRM prices is done by using the special mapping table class [page 167] called "price table [page 171]".

To use SAP CRM prices in recurring charges, you must complete the following tasks:

- Import [page 171] the special mapping table class called "price table".
- Create a charge with a recurring rate.
- Use the charge in a charge plan.

This topic details these tasks.

To import the price table

> ⚠ Caution
>
> Price table instances [page 184] are automatically created and maintained in an SAP CRM system. Creating, modifying, or deleting price table instances may result in inconsistencies between these two systems. We strongly recommend that you do not modify or delete existing price table instances except under support supervision.

After you have created your pricing catalog and imported the price table, the price table instances[37] will then be automatically created from the SAP CRM system in your SAP system landscape.

1. Create [page 112] a catalog, if necessary.
2. Import [page 171] the special mapping table class called "price table".

To create a charge with a recurring rate

This procedure consists in creating a charge for managing recurring fees.

1. Create [page 309] a charge.
2. Add a parameter [page 312] to the charge, select the string type, and then type **`TABLE_NAME`** as the name of the parameter. This parameter allows storing the name of the price table instance.
3. Add a parameter [page 312] to the charge, select the string type, and then type **`PRICE_KEY`** as the name of the parameter. This parameter allows storing the key identifying the price in the price table instance.
4. Insert a recurring rate component [page 693] in the price plan of the charge.
5. Add a logic component Mapping Table Introducer (Comparator) [page 744] to the recurring rate component[38].
6. In the *Definition* tab of the Mapping Table Comparator, do the following:

[37] Mapping tables which have the price table as the mapping table class.

[38] For more information about the Mapping Table Comparator, refer to the primary help relating to components here [page 671]

- Set up the *Mapping Table Class* field to **`Price Table`**.
- Set up the *Mapping Table* field to **`TABLE_NAME`**.
- Set up the *Reference Date* to *Recurring Event Date*.
- Set up the Mapping Property of the **`price.key`** column to **`PRICE_KEY`** in the table Mapping of Rating Context.

To use the charge in a charge plan

> ⚠ Caution
>
> You must define at most one parameter with the type *Mapping Table ID (Price Table)* and one parameter with the type *Mapping Table Key* per charge plan. Otherwise, recurring prices would not be managed properly.

1. Add [page 366] the charge to the charge plan. Note that the parameter `TABLE_NAME` is now assigned the type *Mapping Table ID (Price Table)* and the parameter `PRICE_KEY` is assigned the type *Mapping Table Key*.
2. Create [page 354] two parameters in the charge plan with the visibility *External (Mandatory)* to make them visible at provider contract level.
   - Set the type of one of the parameters to *Mapping Table ID (price table)*. In the *Parameters* tab of the charge, link the parameter `TABLE_NAME` to this parameter.
   - Set the type of the other parameter to *Mapping Table Key*. In the *Parameters* tab of the charge, link the parameter `PRICE_KEY` to this parameter.

The SAP CRM system will set the names of the price table instance and the price key through the `TABLE_NAME` and `PRICE_KEY` parameters at provider contract level.

Related Information

Synchronizing Recurring Fees with Billing Cycles in SAP Convergent Invoicing [page 321]
Importing a Price Table as a Mapping Table Class [page 171]

1.23.7.11 Synchronizing Recurring Fees with Billing Cycles in SAP Convergent Invoicing

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging in an integrated *SAP Solution* scenario with SAP Convergent Invoicing.
>
> For more information, refer to the integration guides.

You may want to synchronize or correlate recurring fees computed in SAP Convergent Charging with billing cycles defined in the SAP ERP/FI-CA systems. For example, you want to charge monthly fees to an end customer for equipment renting but creating invoices every two months.

Synchronizing SAP CC and SAP Convergent Invoicing is done by using the `SAP_BILL_CYCLE_DATE` parameter in a particular charge plan that includes a dedicated recurring charge.

The `SAP_BILL_CYCLE_DATE` parameter gives the date/time when the first billing cycle starts in SAP Convergent Invoicing and is used as the reference date/time for the recurring charges in SAP CC.

When a provider contract is created, the `SAP_BILL_CYCLE_DATE` parameter is set in each contract item that activates the charge plan including the dedicated recurring charge.

To synchronize SAP CC and SAP Convergent Invoicing, you must:

- Create and configure a charge to synchronize recurring fees. A recurring charge is a charge that includes a unique rate component of recurring rate type.
- Insert the charge in a charge plan that you configure accordingly. This charge plan is activated in provider contracts as contract items that include the `SAP_BILL_CYCLE_DATE` parameter set to the start date/time of the first billing cycle.

Configure a Charge

This procedure consists in creating a reusable charge to synchronize recurring fees.

1. Create [page 309] a charge or use an existing charge.
2. Add a parameter [page 312] to the charge, select the date type, and then type **`RECURRING_REFERENCE_DATE`** as the name of the parameter. This parameter allows storing the start date/time of the first billing cycle defined in SAP Convergent Invoicing.
3. Insert a recurring rate [page 693] in the price plan of the charge.
4. In the *General* tab of the inserted recurring rate, set up the *Reference Date* field to `RECURRING_REFERENCE_DATE`.

You can configure a charge plan with this charge.

Configure a Charge Plan

After creating the reusable charge, you must insert it in a charge plan and link the `RECURRING_REFERENCE_DATE` parameter with the `SAP_BILL_CYCLE_DATE` of the parameter.

1. Add [page 366] the charge to the charge plan.
2. In the *Parameters* tab of the charge, set up the status of the `RECURRING_REFERENCE_DATE` parameter to *Linked* and its value to **`SAP_BILL_CYCLE_DATE`**.

> **Note**
>
> You can use the same procedure to synchronize recurring refills with billing cycles managed in SAP Convergent Invoicing.

Create first a dedicated parameter in the refill logic, and then link it to the `SAP_BILL_CYCLE_DATE` parameter of a refill plan.

Related Information

Recurring Rate [page 693]
Using SAP CRM Prices in Recurring Charges [page 319]

1.23.8 Working with Price Plans

Related Information

Setting Up the Price Plan of a Charge [page 323]
Changing the Sequence of Components [page 324]
Inserting an Existing Price Plan into a Charge [page 325]

1.23.8.1 Setting Up the Price Plan of a Charge

Create [page 309] or modify [page 316] a charge, and then set up [page 310] the charge before setting up the price plan. A *Charge* window must be open.

> ⓘ Note
>
> Before modifying a price plan, it is strongly recommended to read the caution [page 299] about what you can do or not.

In a *Charge* window, a price plan [page 299] is graphically represented by a rating tree, which can consist of price plan components using counters, parameters, pricing macros, mapping tables, or range tables depending on the requirements of the plan.

Set Up the Price Plan (Policy) of a Charge

1. In the rating tree display area of the *Charge* window, make sure that the price plan icon  is selected; otherwise, select it.

2. In the right pane of the charge window, identify the price plan. In the *Name* field, type the unique identifier of the price plan. In the *Description* field, describe the function of the price plan if necessary.
3. In the *Transient Counters* area, you can add transient counters [page 297] to the price plan. Click  to add a row to the table. Click  if you want to delete a transient counter from the table. Fill in the columns of each row that you add to the table: in the *Name* column, Type the transient counter name; In the *Description* column, describe the function of the transient counter if necessary; In the *Initial Value* column, type a numerical value to initialize the transient counter. If the counter name already exists in another price plan in a charge and if you create a charge plan by setting up a master and a dependent charge, then the transient counters are shared. At runtime, the SAP CC system the value of the counter from the master charge to the dependent charge. See Sharing Transient Counters with Dependent Charges in Charge Plans [page 386] for more information.
4. Add [page 331] rate components to the price plan, and then set up the rate components[39].
5. Add [page 330] functions, comparators, splitters or operators components to the price plan, and then set up the components[40].

> **Note**
>
> - You can remove [page 333], cut, copy and paste components or sections of a rating tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy* or *Paste*.
> - When you click the price plan icon  , the *Rates* icon  on the *Components* toolbar becomes available.

1.23.8.2 Changing the Sequence of Components

In some price plans, the order of logic components is important. For example, if two rates of the same component family are triggered by the same event, the first rate in the price plan will be calculated first.

You can change the sequence of Rate  components and Operator  components to create different calculation conditions in a price plan.

Moving Rate components

When you move a Rate component, you also move the rating branch attached to it.

Rate components are arranged in a price plan in the following order:

-  Usage rates

[39] For more information about setting up rate components, refer to the primary help relating to components here [page 671].
[40] For more information about setting up rate components, refer to the primary help relating to components here [page 671].

-  One-Shot rates
-  Recurring rates

You can only rearrange a rate within its section of the price plan. For example, you can only move a Recurring rate within the Recurring section - you cannot move a Recurring rate to the top of the price plan which is reserved for Usage rates.

Moving Operator components

Operator components can only be moved within their rating branch, and only at the same level of the branch.

To change the sequence of components in a price plan

1. In the rating tree display area of a charge, right-click the component you want to move.
2. Select one of these commands:
   - Move Up Component [page 333]
   - Move Down Component [page 334]

> ⓘ Note
>
> - You can remove [page 333], cut, copy and paste components or sections of a rating tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy* or *Paste*.

1.23.8.3 Inserting an Existing Price Plan into a Charge

> ⓘ Note
>
> Inserting a price plan into a charge also inserts the related charging plan.

If you have previously saved a charge as an XML file to your hard disk, you can insert its price plan into a new or an existing charge; however, if you want to replace a price plan in an existing charge, make sure that this action is necessary because the price plan will completely be replaced by the new one. Keep in mind that a charge window must be open if you want to insert a price plan into the charge.

> → Remember
>
> You can insert a price plan into an existing charge only if the price plan was correctly verified before you saved the charge [page 313] as an XML file; otherwise, the price plan does not appear in the *Open Parameters* window. To verify the price plan of a charge saved as an XML file, open the charge from the *Open Parameters* window by selecting the *Open as New Object* option, and then click the *Verify* icon  on the main window toolbar. Once the *Verify Charge* window is open, correct all elements underlined in red, and then save the charge again as an XML file.

To insert an existing price plan (and the related charging plan) into a charge

1. In the rating tree display area of the charge window, click the Price Plan icon  .
2. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
3. Open the directory in which the charge is saved, and then select the charge file.
4. Click *Open*. The *Open Parameters* dialog box opens.
5. Select the *Open as Sub-Element of Current Object* option. The price plans  in the list become available if they have correctly been verified before being saved as an XML file. Refer to the note above.
6. Select the price plan you that want to insert.
7. Click *Open*. The existing price plan is inserted into the charge.

1.23.9 Working with Charging Plans

Related Information

Setting Up the Charging Plan of a Charge [page 326]
Changing the Sequence of Components [page 328]
Inserting an Existing Charging Plan in a Charge [page 329]

1.23.9.1 Setting Up the Charging Plan of a Charge

Create [page 309] or modify [page 316] a charge, and then set up [page 310] the charge before setting up the price plan. A charge window must be open.

In a *Charge* window, a charging plan [page 305] is graphically represented by a charging tree, which can consist of charging plan components using counters, parameters, translation tables, tier tables, or mapping tables depending on the requirements of the plan.

Set Up the Charging Plan (Policy) of a Charge

1. In the charging tree display area of the charge window, make sure that the charging plan icon  is selected; otherwise, select it.

2. In the right pane of the charge window, identify the charging plan. In the *Name* field, type the unique identifier of the charging plan. In the *Description* field, describe the function of the charging plan if necessary.
3. Create and set up [page 327] the internal reference dictionary.
4. In the left pane, and if necessary[41], add [page 332] charging components to the charging plan, and then set up the charging components[42].
5. If you have added charging components, add [page 330] comparators, operators or references components to the charging plan, and then set up the components[43].

> **Note**
>
> - You can remove [page 333], cut, copy and paste components or sections of a rating tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy* or *Paste*.
> - When you click the charging plan icon  , the *Charging* icon  on the *Components* toolbar becomes available.

1.23.9.1.1 Internal Reference Dictionary

To access to an internal reference dictionary, you must click the charging plan [page 305] of a charge to display the related tab area.

A charging plan consists of one or more charging components [page 307] (usage charging, recurring charging, and one-shot charging). Each charging component that you add to a charging plan must refer to a prepaid or a postpaid account [page 511] which is charged after the charging plan has been triggered. Therefore, the internal reference dictionary allows you to create references which are used in the charging components to refer to an account. These references are then used at charge [page 385] level (in the charge plan) where they are mapped to account assignments [page 356] of the charge plan.

> **→ Remember**
>
> Do not forget that once you have created and saved the charge, you can non longer modify the dictionary of internal references. Therefore, you must create all the references you need to use in the charging plan before saving the charge.

You can only assign the *Internal Reference* component to each charging component inserted in the charging tree. For each *Internal Reference* component, you select an internal reference of the dictionary. If you want to make a refill, you use a refill plan [page 421].

---

41 Charging components are not mandatory because the charging plan already contains a  Default component. You can remove the  Default Reference, and then add any comparators or operators before adding any other reference to the Default component.

42 For more information about setting up charging components, refer to the primary help relating to componentshere [page 671]

43 For more information about setting up charging components, refer to the primary help relating to componentshere [page 671]

Internal reference dictionary tab

This tab allows you to create a dictionary of internal references.

Use the table and its icons to create charging references.

 : Add a new row to the table.

 : Delete a row from the table

Charging reference name: Type the name of the charging reference. You can reuse this charging reference in any branches of the charging tree.

Type: Select one of the following:

- *Prepaid*: Assigns the charging reference to a prepaid account
- *Postpaid*: Assigns the charging reference to a postpaid account

1.23.9.2 Changing the Sequence of Components

In some charging plans, the order of logic components is important. For example, if two Operator components are triggered by the same event, the first Operator component will be triggered first.

You can change the sequence of Operator  components to create different charging conditions in a charging plan.

Moving Charging components

When you move a Charging component, you also move the charging branch attached to it.

Charging components are arranged in a charging plan in the following order:

-  Usage charging
-  One-Shot charging
-  Recurring charging
-  Default component

You can only rearrange a charging component within its section of the charging plan. For example, you can only move a Usage charging within the Usage section, but you cannot move a Recurring rate to the top of the charging plan which is reserved for Usage charging.

Moving Operator components

Operator components can only be moved within their rating branch, and only at the same level of the branch.

To change the sequence of components in a charging plan

1. In the rating tree display area of a charge, right-click the component you want to move.
2. Select one of these commands:
   - Move Up Component [page 333]
   - Move Down Component [page 334]

> Note
>
> - You can remove [page 333], cut, copy and paste components or sections of a rating tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy* or *Paste*.

1.23.9.3 Inserting an Existing Charging Plan in a Charge

Refer to the topic Inserting an Existing Price Plan into a Charge [page 325].

1.23.10 Using the Action Menu

Related Information

Adding a Rate Component [page 331]
Adding a Charging Component [page 332]
Adding a Component [page 330]
Removing a Component [page 333]
Moving Up a Component [page 333]
Moving Down a Component [page 334]
Changing the Version of a Charge [page 334]
Deleting the Versions of a Charge [page 335]
Reading a Price Plan [page 335]
Doing a Trial Run of a Price Plan [page 335]

1.23.10.1 Adding a Component

You can add one or more components to the rating tree or the charging tree.

For adding a component to the rating tree

You build each branch of the rating tree by using several combinations of the following components:  Functions,  Comparators,  Splitters or  Operators. You must end each branch of the rating tree with a function component. Note that function components are the leaves of the tree.

> **Note**
>
> - Because you cannot insert a pricing macro [page 121] in a price plan, you need to insert components such as a macro function or a macro operator in a price plan to reuse an existing pricing macro.
> - Because you cannot insert a translation table [page 132] in a price plan, you need to insert a table operator component in a price plan to reuse an existing translation table.
> - Because you cannot insert a tier table [page 147] in a price plan, you need to insert a table operator component in a price plan to reuse an existing tier table.

To add and set up a component

1. In the rating tree, click the component under which you want to add a component.
2. On the *Components* toolbar, click the component that you need to add to the rating tree. You can select a component that belongs to one of these families:  Functions,  Comparators,  Splitters or  Operators.
3. In the *Name* field of the right pane, type the name of the component.
4. In the *Description* field, describe the function of the component if necessary.
5. Set up the component[44].

> **Note**
>
> If necessary, you can change the order of components. Place the cursor on the component to move, right-click to open the shortcut menu, and then select *Move Up Component* or *Move Down Component*.

[44] For more information about setting up components, refer to the primary help relating to components here [page 671].

To add a component to the charging tree

The rules and procedures are the same as those for the rating tree except that the  Functions, the  Splitters components and the pricing macros are not available for the charging tree. Furthermore,  References replace the  Functions as the leaves of the tree.

> ⓘ Note
>
> If you want to add components to the  Default component, you must first remove the  Default Reference, and then add any Comparators or Operators before adding any other reference.

1.23.10.2 Adding a Rate Component

You can add one or more rate components to the rating tree.

The rate branch is a mandatory base for your price plan. You cannot create a price plan without a rate branch that contains at least one of the following rate components:  Usage Rate,  Recurring Rate or  One-Shot Rate.

> → Remember
>
> Do not forget that you cannot add a usage rate component to a price plan whose charge is not associated to a chargeable item class [page 241].

To add and set up a rate component

1. In the rating tree, click the price plan icon  .
2. On the *Components* toolbar, move the pointer over  Rates, and then click one of the following rate components:  Usage Rate,  Recurring Rate or  One-Shot Rate.
3. In the *Name* field of the right pane, type the name of the rate component.
4. In the *Description* field of the right pane, describe the function of the rate component if necessary.
5. Set up the rate component[45].

> **Note**
>
> When a price plan contains several rate branches such as Usage Rates, Recurring Rates, and One-Shot Rates, the rate branches are automatically arranged in this order in the rating tree:
>
> -  Usage Rates
> -  One-Shot Rates
> -  Recurring Rates
>
> For example, if you add a new usage rate to the rating tree, it is automatically added after the last usage rate branch.

1.23.10.3 Adding a Charging Component

You can add one or more charging components to the charging tree.

The charging branch is not a mandatory base because the charging plan already contains a  Default component; however, if you create a charging branch, it must at least contain one of the following rate components:  Usage Charging,  Recurring Charging or  One-Shot Charging.

To add and set up a charging component

1. In the charging tree, click the charging plan icon  .
2. On the *Components* toolbar, move the pointer over  Charging, and then click one of the following rate components:  Usage Charging,  Recurring Charging or  One-Shot Charging.
3. In the *Name* field of the right pane, type the name of the charging component.
4. In the *Description* field of the right pane, describe the function of the charging component if necessary.
5. Set up the charging component[46].

[45] For more information about setting up rate components, refer to the primary help relating to components here [page 671]
[46] For more information about setting up charging components, refer to the primary help relating to components here [page 671].

> **Note**
>
> When a charging plan contains several charging branches such as Usage charging, Recurring charging, and One-Shot charging, the charging branches are automatically arranged in this order in the charging tree:
>
> -  Usage charging
> -  One-Shot charging
> -  Recurring charging
> -  Default component
>
> For example, if you add a new usage charging to the charging tree, it is automatically added after the last usage charging branch.

1.23.10.4 Removing a Component

Use this procedure to remove a component from a tree.

To remove a component from a tree

1. In the tree, click the component that you want to remove.
2. From the menu bar in the main window, select *Action* > *Remove Component*. The component is removed.

> **Note**
>
> - To remove an entire tree branch, right-click a component and select *Remove Component*.
> - Some components in the middle of a tree branch allow you to remove them and all other components attached to them towards the end of the branch. These branch sections can be removed only when the *Remove Component* command is available.
> - You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.23.10.5 Moving Up a Component

Use this procedure to move up a component.

To move up a component

1. In the tree, click the component that you want to move up.
2. From the menu bar in the main window, select *Action* > *Move Up Component*.

> **Note**
>
> You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.23.10.6 Moving Down a Component

Use this procedure to move down a component.

To move down a component

1. In the tree, click the component that you want to move down.
2. From the menu bar in the main window, select *Action* > *Move Down Component*.

> **Note**
>
> You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.23.10.7 Changing the Version of a Charge

Refer to the process below:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.23.10.8 Deleting the Versions of a Charge

The procedure is the same as Deleting the Versions of a Catalog Object [page 118].

1.23.10.9 Reading a Price Plan

Once you have created a price plan, you can generate a descriptive representation of it. This process is called reading a price plan.

To read a price plan

1. Open [page 314] a charge.
2. On the toolbar in the main window, click the *Read Price Plan* icon  . The *Read Price Plan* dialog box opens.
3. Check the price plan description for accuracy, and then click *Close*.

1.23.10.10 Doing a Trial Run of a Price Plan

Once you have built and verified a price plan, you can test the logic of its rating process to ensure that the correct results have been achieved. To do this, you use the *Rating Trial Run* tool which is a testing tool provided with SAP Convergent Charging.

To test a price plan

1. Make sure that the price plan that you want to test is open; otherwise, open [page 314] the price plan.
2. Open [page 339] the *Rating Trial Run* tool.
3. Set up [page 340] the *Rating Trial Run* tool.
4. Run [page 340] the *Rating Trial Run* tool.

1.23.11 Rating Trial Runs

Once you have built and verified a price plan or pricing macro, you can test the logic of their rating process to ensure that the correct results have been achieved. To do this, you can use the *Rating Trial Run* tool which is a testing tool provided with SAP Convergent Charging.

The *Rating Trial Run* tool allows you to manipulate and change the values of specific rates that are used in the rating process and locate the related components that do not work correctly.

When adjusting and finalizing a charge plan, the *Rating Trial Run* tool allows you to view the rating context [page 13] in detail.

> ⚠ Restriction
>
> Rating Trial Runs do not consider allowance creation.

Trial run procedure

The *Rating Trial Run* tool tests only one rating branch at a time; therefore, the trial run procedure consists in testing each branch of the rating tree of your price plan or pricing macro. For each rating branch, the trial run procedure is split into three main steps:

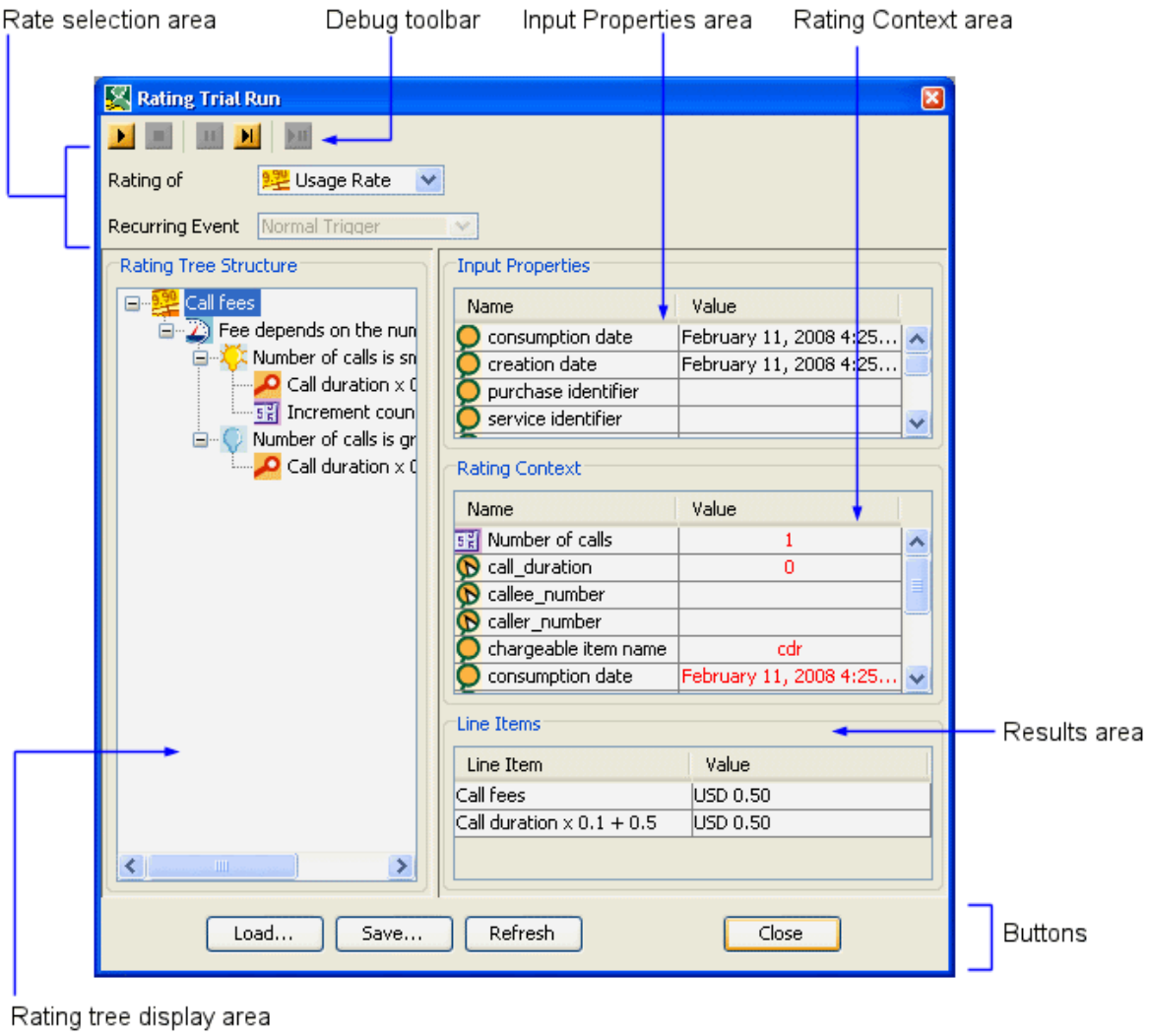
1. You need to set or change the values of the properties that are used in your price plan or pricing macro.
2. You start the *Rating Trial Run* tool. The tool uses the algorithm of the price plan or pricing macro, which is based on the values of the properties and provides a set of actual calculation results.
3. Once you have performed a rating trial run, the results area displays the final calculated price. You can compare the calculation results with the expected test results.

Testing a price plan or pricing macro

1. Open the price plan [page 299]or pricing macro [page 121] that you want to test.
2. Open [page 339] the *Rating Trial Run* tool.
3. Set up [page 340] the *Rating Trial Run* tool.
4. Run [page 340] the *Rating Trial Run* tool.

1.23.11.1 Rating Trial Run Window

Once you have opened the *Rating Trial Run* tool to test your price plan or pricing macro, you are presented with the *Rating Trial Run* window.



Rate Selection area

The *Rate Selection* area allows you to select one rate component of the price plan or pricing macro that you want to test. When the window opens initially, the first rate of the price plan or pricing macro is displayed by default.

Debug toolbar

Five buttons represent the modes to test your price plan or pricing macro. Each button corresponds to a specific test action. The buttons are the following:

Button	Mode	Description
	Start	Starts the test procedure.
	Stop	Stops the test procedure immediately.
	Set Breakpoint	Allows you to specify where the execution must be paused so that you can control the rating process.
	Step by Step	Allows you to control the execution of the program by moving from the current rating component to the next available component.
	Run to	Allows you to run the test procedure to the current cursor location. This is similar to a breakpoint; however, once the program runs a second time, this breakpoint is ignored.

Input Properties area

The *Input Properties* area displays the properties and values that are predefined in the price plan or pricing macro. These properties and values relate to the selected rate, the counters and parameters. This area also allows you to set or change the values of the properties.

Rating Context area

The Rating Context [page 13] area displays all of the properties that are available during the rating process. It also allows you to modify the values of these properties during the test procedure.

Rating tree display area

The *Rating Tree Display* area displays the rating tree that corresponds to the selected rate in the *Rate Selection* area.

Results area

The *Results* area displays the test results of the rating process. The first line displays the total calculated price while the remaining lines represent the individual items, component by component, that make up this total price.

Buttons

Load: Allows you to restore the input properties previously saved in a file. Click this button to open the directory in which the file is saved. Select the file, and then open it. The values saved in the file are assigned to the input properties. An error message is displayed if the input properties previously saved are not the same as the current input properties.

Save: Allows you to save the input properties in a file. Click this button to open the directory in which you want to save the file. Type the name of the file, and then save the file.

Refresh: Click to cancel any changes and restore the *Rating Trial Run* window to its original state.

Close: Click to close the *Rating Trial Run* window.

1.23.11.2 Working with Rating Trial Runs

Related Information

Opening the Rating Trial Run Tool [page 339]
Setting Up the Rating Trial Run Tool [page 340]
Running the Rating Trial Run Tool [page 340]

1.23.11.2.1 Opening the Rating Trial Run Tool

You must verify your price plan or pricing macro before debugging it. Once you have fully verified the price plan or pricing macro, you can open the *Rating Trial Run* tool.

To open the Rating Trial Run tool

1. Open the window of the price plan or pricing macro that you want to test.
2. On the main window of SAP Convergent Charging, click  . The *Rating Trial Run* window is displayed. The rating tree of the first rate in the price plan or pricing macro is displayed in the left pane of the window (*Rating Tree Display* area) and its corresponding properties in the right pane (*Input Properties* area).

1.23.11.2.2 Setting Up the Rating Trial Run Tool

Once you have opened the *Rating Trial Run* tool, you first need to choose the component to test if your price plan or pricing macro is made up of several components, and then you must set the property values of the price plan or pricing macro depending on your needs.

To set up the Rating Trial Run tool

1. In the Rating of list of the Rating Trial Run window, select the rate component that you want to test: Usage Rate, Recurring Rate or One-Shot Rate. The Rating Trial Run tool tests only one rating branch at a time.
2. [47]If you have selected  *Recurring Rate* as the component to test, you must also select the event that triggers the consumption of this rate in the *Recurring Event* field: *Normal Trigger*, *Subscription activation* or *Subscription cancellation*.
3. In the *Input Properties* area, change or set the values of the properties that are used by your price plan or pricing macro in the rating process. If necessary, use the vertical scroll bar to move the properties up or down. The values that you type or change are displayed in red.

When the *Rating Trial Run* tool is set up, you can run [page 340] it.

1.23.11.2.3 Running the Rating Trial Run Tool

Before running the *Rating Trial Run* tool, make sure that you have verified the price plan or pricing macro that you want to test; otherwise, the *Rating Trial Run* tool displays an error message.

Once you have set up [page 340] the *Rating Trial Run* tool, you can run it by selecting the mode in which you want to do the test. You can choose one the main modes to perform a trial run:

- Click  to quickly perform the trial run (with or without breakpoints)
- Click  to perform the trial run step by step and display the calculation results at each step
- Click  to perform the trial run that uses at least one breakpoint

[47] This step does not apply to  *Usage Rate* and  *One-Shot Rate*.

Button	Mode	Description
	Start	This mode performs the trial run of the rate component that you select in the *Rating of* field. When the trial run is complete, you can verify the values of the Rating Context [page 13] properties and the expected results in the *Line Items* area. **Note**: If the rating tree contains some breakpoints, you must click  ,  or  to continue the trial run.
	Stop	This mode stops the trial run permanently. You must click  ,  or  to start a new trial run. **Note**: This button is available when you select the *Step by Step* mode or the *Run to* mode.
	Set Breakpoint	This mode allows you to set or remove breakpoints that are used to stop the trial run temporarily. You can continue the trial run by clicking  ,  or  . To set a breakpoint, click a component in the *Rating Tree Structure* area, and then click  . The component label appears in red. To remove a breakpoint, click a component in the *Rating Tree Structure* area, and then click  . The component label appears in black. **Note**: You can use this mode in combination with the *Start* mode, the *Step by Step* mode or the *Run to* mode.

Button	Mode	Description
	Step by Step	This mode performs the trial run of the rate component that you select in the *Rating of* field, but performs one component at a time. You must click  to perform the next component. This mode allows you to verify the values of the *Rating Context* properties and the calculation results in the *Line Items* area at each step. **Note**: This button is available when you select the component just below the rate component of the tree.
	Run to	This mode performs the trial run of the rate component that you select in the *Rating of* field only if you have set at least one breakpoint. Once you have set a breakpoint, click  . The trial run stops at the breakpoint. To continue the trial run, click  . The breakpoint is automatically removed, but the *Run to* mode is not available any more.

Results

Once you have performed a rating trial run, the results are displayed in the *Line Items* area:

- The first line represents the final price of the rating branch
- The remaining lines represent the individual items, component by component, that make it up

Example

For example, you test the usage rate "Call fee" of your price plan. This usage rate is made up of two branches: Call duration x 0.02 USD and Call duration x 0.01 USD, which are triggered depending on the number of calls. If the number of calls is smaller than 10, the price is calculated according to Call duration x 0.02 USD; otherwise, the price is calculated in the second branch according to Call duration x 0.01 USD.

Before running the *Rating Trial Run* tool, you set the property Call duration to 100 and the property Number of calls to 10 in the *Input Properties* area.

You start the *Rating Trial Run* tool. Once the trial run is complete, the *Line Items* area displays the following:

Line Item	Value
Call fee	USD 1.00
Call Duration x 0.01 USD	USD 1.00

The first line of the *Line Items* area displays the final price of the usage rate "Call fee". The second line displays the result of the branch "Call duration x 0.01 USD". In this example, the results are the same, but results and number of line items depend on the complexity of your price plan.

1.24 Charge Plans

⚠ Caution

As of FPS 0 for release 2020, some charge plans are managed by Cockpit, which means that their related charges are also managed by Cockpit.

These charges and charge plans are visible from Core Tool. Managing them using Core Tool is possible, however, know the following restrictions:

- the system automatically converts these objects into objects manageable by Core Tool
- you cannot roll back this operation: these objects can no longer be managed by Cockpit
- you cannot handle charges managed by Cockpit in charge plans managed by Core Tool: the system prevents you from creating hybrid objects
- you can convert a charge plan managed by Cockpit in a charge plan managed by Core Tool by using the *Open from a File* feature
- you can import charge plans managed by Cockpit in Core Tool, subject to confirmation according to potential warning messages

A pop-up message always warns you when you handle a charge or a charge plan managed by Cockpit.

Once a charge plan and its related charges managed by Cockpit are converted to be managed by Core Tool, rules and limitations of Core Tool apply.

→ Remember

As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

In SAP Convergent Charging 2023, a **charge plan** defines the pricing and charging conditions to use a chargeable service or product. A charge plan consists of one or more customized charges [page 290] that refine these conditions.

Charge plans are referenced in contract items [page 592] about the pricing and charging conditions. A charge plan must be primarily combined as part of a commercial product or offer in a CRM application or another provisioning system. They are then activated (aka subscribed to) in a contract item for the ordered provider contract.

In the pricing catalog, the charge plan consists of:

- One or more customized charges [page 367] (also called charge plan items)
- A set of parameters to configure these charges

- A release status [page 352] that can be open, released, or obsolete

Before Starting

> → Remember
>
> - Read the note [page 316] about making retroactive changes to a charge or a charge plan.
> - Read the topic [page 523] about taxes because you can include or not the taxes in price plans.

Before creating [page 350] and setting up [page 351] a charge plan, you must be familiar with the following:

- Reusable Charges [page 290]
- Charged Item Classes [page 254]
- Chargeable Item Classes [page 241]
- Preview [page 358] of a charge plan
- Version change of a charge plan [page 360]

Transport

To transport a plan between different system landscapes (DEV to TST, DEV to PROD), you can:

- Create a change list [page 662] and manage a transport request.
- Export a charge plan [page 363] to an XML file and import a charge plan [page 364].

1.24.1 Characteristics of a Customized Charge

Related Information

Characteristics [page 345]
Relationship Types [page 346]
Operation Types [page 346]
Roles [page 346]

1.24.1.1 Characteristics

Charges can have main and secondary characteristics. You define main characteristics when you create [page 309] a reusable charge. When you customize charges [page 367] inside a charge plan, you give them secondary characteristics and you can redefine some main characteristics.

Once you have selected the charges that you want to add to a charge plan, you can redefine some of their main characteristics and assign secondary characteristics to them. The setup choices made for a particular charge plan only apply to this charge plan.

Main characteristics

When you create it, the charge can have the following characteristics:

- Charge type (master [page 293] or dependent [page 294])
- Parameters [page 303] if used
- Counters [page 297] if used

Redefinition

You cannot change the charge type when the charge is included in a charge plan, but you can modify initial values of parameters.

When you incorporate a charge into a charge plan, you can redefine its parameters if you have set the *Visibility* field to "External".

You cannot add further content to a translation table from within a charge plan.

> ⓘ Note
>
> Main characteristics are redefined on the Parameters [page 370] tab and the Counters [page 371] tab of the customized charge.

Secondary characteristics

A charge has several secondary characteristics which you define when you customize a charge in a charge plan:

- Relationship Types [page 346]
- Operation Types [page 346]
- Roles [page 346] for dependent charges

Secondary characteristics are only valid in the charge plan to which they have been assigned. If you reuse this charge in another charge plan, you can redefine secondary characteristics for the new charge plan.

1.24.1.2 Relationship Types

The relationship type allows you to choose whether the relationship with the service provider will be with a client or with a partner.

> **Note**
>
> Relationship types are set up on the Definition [page 368] tab of charges included in charge plans.

1.24.1.3 Operation Types

The operation type allows you to specify whether to credit or debit the client or the partner involved.

> **Note**
>
> Operation types are set up on the Definition [page 368] tab of charges included in charge plans.

> **Example**
>
> Learn.com offers its customers a range of online training materials. It buys tutorials from specialized authors, converts them into electronic format, and adapts them to e-learning. Customers can buy these training materials in modules and download them. Costs may vary from module to module and can be determined by a module code. The Purchase Module charge is a master charge for the following reasons:
>
> - When a customer decides to buy a module, the cost self-triggers because it does not depend on another event.
> - The price plan of the Purchase Module charge can calculate the price to be paid without requiring the intermediary calculation of another price plan. Customers choose their module when executing a download command, which means the module code is part of the rating context. Once the Module Code user property has been created in the chargeable item class, this data may be used in the price plan of the Purchase Module charge to determine the price.

1.24.1.4 Roles

> **Note**
>
> - In an integrated *SAP Solution* scenario with theSAP CRM and theSAP ERP/FI-CA components of SAP Business Suite, if you need to manage payments to a business partner, SAP SE recommends that you do not configure any dependent charges. Refer to the note [page 290].
> - Roles are set up on the Definition [page 368] tab of dependent charges customized in charge plans.

You choose the roles of dependent charges when configuring a charge plan.

A reusable charge is assigned its role only when it is customized in a charge plan and not when it is created. This means the same reusable charge may play different roles in different charge plans:

- Commission
- Third-party billed commission
- Sponsorship
- Discount
- Other

The role of a charge is only valid for charge plans for which it is created, which means you can use a single charge to calculate commission in one charge plan and sponsorship in another.

Commission

Commission, such as a sales commission, is an amount paid to a third party. A commission is charged to an account payable and is paid to a single agent.

Third-party Billed Commission

A third-party billed commission is an amount paid to a third party. This type of commission can be paid to a single third party (refer to the note above).

Sponsorship

Sponsorship is an amount paid by a third party to promote the sale of a service (through a chargeable item class). Because the costs of service consumption are shared between consumers and sponsors, the service provider is partly paid by its customers (consumers) and partly paid by its sponsors. Sponsorship may be paid by a single sponsor designated as "client" or "service provider" in the subscriber account (refer to the note above).

Discount

A discount is an amount to lower the price amount paid by an end customer.

Other

A dependent charge can also be assigned a customized role, known as “Other.” In this case, you need to configure certain parameters when setting up the charge. You must decide whether:

- The client or partner will be credited or debited.
- A new transaction will be created: For example, a commission generates a new transaction because there is an actor but a discount does not create a new transaction.
- The relationship type is client or partner, which means transactions will be charged to accounts receivable or accounts payable.
- The master transaction will be adjusted: For example, a sponsorship adjusts a master transaction because the dependent charges do not affect the revenue for the service provider. The revenue corresponds to the amount originally calculated by the master charge, and the final amount to be paid by the customer will be adjusted accordingly.

Dependent charges summary

The following table summarizes the parameters and fields for the different predefined roles that are assigned to a given dependent charge. Note that the only parameters and fields that the SAP CC user is free to configure are those for the "Other" role:

	Operation	Create new transaction	Relationship type	Adjust master transaction
Commission	Credit	Yes	Partner	No
Third-party billed commission	Debit	Yes	Client	No
Sponsorship	Debit	Yes	Client	Yes
Discount	Credit	No	n/a	n/a
Other	Free	Free	Free	Free

1.24.2 Sharing Counters with Charge Plans

Depending on your business requirements, you can configure charge plans or refill plans that share a counter.

> → Remember
>
> You must set up an additional charge plan that enables to configure the initialization.

To share a counter

In each charge plan or refill plan that will be part of the final provider contracts, you do the following.

1. Declare the same counter that you choose in the counter dictionary [page 86].
2. Set the visibility of the counter to *External*[48].

1.24.2.1 Using a Charge Plan to Configure the Initialization of a Shared Counter

You configure a charge plan that is dedicated to the configuration of the initialization of a shared persistent counter that is created in provider contracts [page 573].

This charge plan must include:

- The declaration of the persistent counter you want to share
- A *One-Shot Rate*[49]
- A parameter that is used to

> ⚠ Caution
>
> Do not initialize the same counter twice.

Prerequisites

A particular charge [page 299] is dedicated to this charge plan and is available in SAP CC.

Procedure

To create a charge plan containing this charge

1. Create a charge plan.
2. Add the dedicated charge to the charge plan.
3. Declare the counter to be shared with the visibility option set to *External* in the charge plan. The counter name is the same a in the charge.
4. Link the counter to the counter declared in the charge.
5. Define a parameter
6. Link the parameter with the visibility *External (Mandatory)* to the parameter defined in the charge.

1.24.3 Working with Charge Plans

> → Remember
>
> Before working with charge plans, you must have read the Caution note [page 290] about charges and charge plans.

[48] The CRM application or external provisioning system can then create provider contracts that share counters with contract items or linked contracts.

[49] See the primary help of logic components [page 671] for more information about *One-Shot Rates*

Related Information

Creating a Charge Plan [page 350]
Opening an Existing Charge Plan [page 360]
Modifying a Charge Plan [page 362]
Setting Up a Charge Plan [page 351]
Saving a Charge Plan [page 359]
Deleting a Charge Plan [page 363]
Exporting a Charge Plan [page 363]
Importing a Charge Plan [page 364]

1.24.3.1 Creating a Charge Plan

Use one of these procedures to create a charge plan [page 343].

Create a Charge Plan

1. From the menu bar in the main window, select *File* > *New* > *Charge Plan*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the charge plan.
3. Click *OK*. The *New Charge Plan* window opens.
4. Set up [page 351] the charge plan.
5. Verify the charge plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Check the *Preview* of the charge plan. See Previewing a Charge Plan [page 358].
7. Save [page 359] the charge plan.

> → Tip
>
> You can create a charge plan by opening an existing charge plan and copying it.

Create a Charge Plan from an Existing Charge Plan

1. From the menu bar in the main window, select *File* > *Open* > *Charge Plan*. The *Choose Charge Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge plan.
3. In the *Charge Plans* folder of the catalog of the owner, select the charge plan.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Charge Plan* window opens.

5. Rename the charge plan in the *Name* field, and then *Set up* the charge plan as necessary. the charge plan if necessary.
6. Verify the charge plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Check the *Preview* of the charge plan. See Previewing a Charge Plan [page 358].
8. Save [page 359] the charge plan.

Related Information

Setting Up a Charge Plan [page 351]
Previewing a Charge Plan [page 358]
Saving a Charge Plan [page 359]

1.24.3.2 Setting Up a Charge Plan

In the *Charge Plan* window, a charge plan is graphically represented by a tree, which consists of the following:

- 
- 

You must create [page 350], open [page 360], or modify [page 362] a charge plan before setting it up. A charge plan window must be open.

Set Up a Charge Plan

1. In the left pane of the *Charge Plan* window, click the charge plan root  .
2. In the right pane of the charge plan window, identify the charge plan. In the *Name* field, type the unique identifier of the charge plan if you create the charge plan. You cannot modify this field.
3. Set up the following tabs of the charge plan:
   - Definition [page 352]
   - Counters [page 353]
   - Parameters [page 354]
   - Additional Information [page 355]
   - Account Assignments [page 356]
   - Technical Data [page 356]
   - And Dependencies [page 357]

4. Add [page 366] one or more master and dependent charges by using reusable charges, and then set up [page 367] the charges.
5. If necessary, remove [page 366] one or more charges.

1.24.3.2.1 Charge Plan: Definition

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Definition* tab.

The *Definition* tab allows you to set up the status of the charge plan and gives a detailed description of the charge plan.

Definition (Tab)

Fill in the *Definition* tab.

Status: Provides the list of release statuses. Select one of the following statuses:

- *Open*: Select this status if you are working on the charge plan. This status allows you to modify the charge plan and save it in the database. However, the charge plan cannot be used by the SAP ERP/FI-CA system or by an external provisioning system. You cannot save several versions of the charge plan.
- *Released*: Select this status when the charge plan is ready to be in production. The charge plan can be used by the SAP ERP/FI-CA system or an external provisioning system. When you modify and save the charge plan in the database, the old versions of the charge plan are always stored in the database. According to this status, you are only allowed to do the following:
  - Add or remove an internal parameter [page 354] or counter [page 353] and change the description of counters and parameters.
  - Change some information about tax [page 374].
  - Change the charged item class and the charged item mapping (Output Item [page 372] and Response Item [page 374] tabs).
- *Obsolete*: Select this status when this charge plan must not be subscribed to anymore. The charge plan can be used by the SAP ERP/FI-CA system or an external provisioning system. You are allowed to do the same changes as with the *Released* status. When you modify and save the charge plan in the database, the old versions of the charge plan are always stored in the database. The charge plan cannot be subscribed in an existing provider contract [page 573] and cannot be subscribed in a new provider contract.

> **Note**
>
> - When you create a charge plan, its status is automatically set to *Open*. You must save the charge plan once before changing its status.
> - You can set the status to *Released* only if the current status of the charge plan is *Open*.

Description: Describe the charge plan by typing a long text if necessary. This field is mandatory. The SAP ERP/FI-CA system or an external provisioning system can display the description. Note that you cannot change the description of the charge plan if its status is *Released* or *Obsolete*.

1.24.3.2.2 Charge Plan: Counters (Declaration)

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Counters* tab.

The *Counters* tab allows you to declare counters to:

- Redefine the initial value of a counter defined in a reusable charge that is customized in this charge plan
- Share a counter among customized charges [page 371]
- Enable to share a counter between several contract items [page 596] belonging to the same provider contract
- Enable to share a counter between several provider contracts [page 607] belonging to the same subscriber account

Counters (Tab)

Use the configuration table and its icons to declare counters.

 : Click to add a counter row to the configuration table.

 : Click to delete a counter row from the table.

Name: Type the name of the counter if the *Visibility Level* option is set to *Internal* or select the name of the counter in the counter dictionary [page 86] if the *Visibility Level* option is set to *External*.

Visibility Level: Provides the list of *Visibility Level* options. The *Visibility Level* option allows you to share the counter value at provider contract level. Select one of the following visibility options:

- *External*: The counter is available for sharing its value with other counters at provider contract level. Note that the sharing is managed by an interfaced CRM or provisioning system when creating provider contracts in SAP CC (by using the SAP CC Web Services technical interface).
- *Internal*: The counter is not available for sharing its value with other counters at provider contract level. If you create a counter, you must link it to a counter at charge level [page 371].

Description: Describe the function of the counter if necessary.

Initial Value: Type a numerical value to initialize the counter. The value overwrites the initial values previously defined at charge level. This column is disabled if the *Visibility Level* option is set to *External*. In this case, the counter is initialized with the value 0.

Related Information

Setting Up the Generation of a Spending Status [page 507]

1.24.3.2.3 Charge Plan: Parameters

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Parameters* tab.

The *Parameters* tab allows you to create parameters that can be shared with the customized charges [page 370]. Depending on some conditions, you can redefine the parameter values when configuring provider contracts.

Parameters (Tab)

Use the configuration table and its icons to create shared parameters.

 : Click to add a parameter row to the configuration table.

 : Click to delete a parameter row from the table.

Name: Type the name of the shared parameter.

Type: Provides the list of parameter types. Select one of the following types: Number, String, Date, Currency, Invoice Text, Mapping Table ID, Mapping Table Key, or Range Table ID [page 303].

Visibility Level: Provides the list of options for modifying the parameter value at provider contract level. Select one of the following options:

- `Internal`: Enables the shared parameter to be modified in this charge plan. If you create a parameter, you must link it to a parameter of a Customized Charge: Parameters [page 370] in this charge plan.
- `External (Mandatory)`: Enables the shared parameter to be modified later in a provider contract througfh an interfaced CRM or a provisioning system. At provider contract level, you must enter the value of this parameter in the CRM system to define the parameter [page 616] in the contract item activating this charge plan.
- `External (Mandatory unmodifiable)`: Disables the shared parameter to be modified later in a provider contract through an interfaced CRM or a provisioning system. At provider contract level, you must enter the value of this parameter in the CRM system to define the parameter [page 616] in the contract item activating this charge plan.
- `External (Optional)`: Enables the shared parameter to be modified later in a provider contract through an interfaced CRM or a provisioning system. At provider contract level, you can redefine the value of this parameter in the CRM system but the default value is the value defined in this charge plan.

Description: Describe the function of the shared parameter. If the *Visibility Level* of the parameter is `External (Mandatory)`, `External (Mandatory unmodifiable)` or `External (Optional)`, the description of the parameter is mandatory.

Value: Type the default value of the shared parameter. If you have selected `External (Mandatory)` as the visibility level, you cannot type a value because this value is only mandatory at provider contract level.

Recurring Fees Synchronization

> **Note**
>
> This configuration is used when integrated with SAP CRM or SAP Convergent Invoicing in the SAP ERP/FI-CA or SAP S/4HANA system. By default, the SAP CC Core Tool user interface inserts a particular parameter `SAP_BILL_CYCLE_DATE` in the charge plan. You cannot modify or delete this parameter.

If your charge plan contains charges with recurring rates, you must create special parameters if you want to:

1. Use the prices defined commercial products configured in the SAP CRM system.
2. Synchronize your recurring rates with the billing cycles of SAP Convergent Invoicing in the SAP ERP/FI-CA system or the SAP S/4HANA system.

1.24.3.2.4 Charge Plan: Additional Information

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Additional Information* tab.

The *Additional Information* tab allows you to create information that is only descriptive and not reused by SAP Convergent Charging at any stage of the process. For example, you can create a table with the author's name of the charge plan, the author's code, the creation date, and another information. If necessary, you can retrieve this information by using an external system.

Additional Information (Tab)

Use the configuration table and its icons to create additional information.

 : Click to add a row to the configuration table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: Number, String, or Date.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the SAP ERP/FI-CA system or an external provisioning system.

1.24.3.2.5 Charge Plan: Account Assignments (Expected)

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Account Assignments* tab.

The *Account Assignments* tab allows you to declare which account references are expected when the charge plan is activated in a provider contract [page 593]. Through the provider contract [page 573], SAP Convergent Charging links the account assignment name to one prepaid or external account of a subscriber account [page 511] so that the activated charge plan can charge this account. The *Account Assignments* tab is also used to link the charge to the charge plan (refer to the charge account assignments [page 385]).

Account Assignments (Tab)

Use the configuration table and its icons to create account assignments.

 : Click to add a row to the configuration table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the account assignment. This name must be unique.

Type: Provides the list of the account types. Select one of the following types: *Prepaid* or *Postpaid*.

Description: Type the description of the account assignment. This description is visible from the provisioning system.

Related Information

Customized Charge: Account Assignments [page 385]

1.24.3.2.6 Charge Plan: Technical Data (Expected)

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Technical Data* tab.

The *Technical Data* tab allows you to define **custom types of technical identifiers** under which an end customer or the subscriber consumes a service. This identifier is used by the SAP CC system to rate and charge the usages of this end-customer service. Note that the same identifier can be used for rating and charging the usages of different services that are specified for each charge customized in this charge plan [page 385].

The identifiers of the end customer have to be specified by the provisioning system.

> ### Example
>
> Phone number, email address, carplate.

Technical Data (Tab)

Use the configuration table and its icons to create account assignments.

 : Click to add a row to the configuration table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

User Technical Identifier Name: Type the name of the technical data. This name has to be unique in the *Technical Data* tab. This name is visible from the SAP CRM system or another external provisioning system that is connected to the SAP CC system. For example, "`phoneNumber`" or "`Phone Number`".

Description: Type the detailed description of the technical data. This description is visible from the CRM application or provisioning system and facilitates the configuration of provider contracts in contract items [page 617].

Related Information

Customized Charge: Technical Data [page 385]

1.24.3.2.7 Charge Plan: Dependencies

Create [page 350] or modify [page 362] a charge plan before setting it up [page 351]. In the right pane of the *Charge Plan* window, open the *Dependencies* tab.

The *Dependencies* tab allows you to:

- Link master charges [page 293] to dependent charges [page 294].
- Prioritize the dependency links.

> ### → Remember
>
> You can use dependent charges as master charges except in the following cases:
>
> - A dependent charge with a sponsor or a discount role cannot be used as a master charge
> - A dependent charge with a commission role can only be master of a dependent charge with a commission role

> **Note**
>
> The *Dependencies* tab is unavailable if your charge plan contains only master charges.

Dependencies (Tab)

Use the configuration table and its icons to assign master charges to dependent charges. Be careful with the order in which the dependencies are listed in the table.

 : Click to add a dependency row to the configuration table.

 : Click to delete a dependency row from the table.

 : Click to move a row up. This changes the priority order of dependencies.

 : Click to move a row down. This changes the priority order of dependencies.

The charge: Provides the list of dependent charges customized in the charge plan. From the drop-down list, select a dependent charge.

Depends on: Provides the list of master or dependent charges customized in the charge plan. See the note about restrictions, earlier in this topic. From the drop-down list, select the charge that you want to link to the dependent charge.

1.24.3.3 Previewing a Charge Plan

This procedure allows you to preview a charge plan and check its signature that will be seen by SAP CRM or by another provisioning system interfaced with the SAP CC system.

To perform this procedure, you must open [page 360] a charge plan.

Preview a Charge Plan

1. From the menu bar in the main window, select *Action* > *Preview*, or click the *Preview* icon  on the toolbar in the main window.
2. The *Preview* window opens and displays a summary of the counters, parameters, account assignments, and technical data that will be seen by an external system. Read and check this summary.
3. Click *Close*.

> **Note**
>
> The *Preview* window displays only the counters and parameters which are assigned the external visibility option [page 370].

1.24.3.4 Saving a Charge Plan

Once you have created or modified a charge plan, and then verified its structure, you are ready to save it. You can save the charge plan to either the back-end database or as an XML file on your hard disk.

Save a New Charge Plan to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Charge Plan Creation* dialog box opens to inform you that the charge plan has been successfully saved.
4. Click *OK*.

Save a New Version of an Existing Charge Plan to the Database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the charge plan has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the charge plan.
> 2. Click *Cancel* to come back to the charge plan. See later in this chapter how to save the charge plan as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the charge plan.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Charge Plan Modification* dialog box opens to inform you that the charge plan has been successfully created.

5. Click *OK*.

Save a Charge Plan as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the charge plan.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The charge plan is saved as an XML file in your selected directory.

> **Note**
>
> The XML file is basic and does not include multiple versions of the charge plan.

1.24.3.5 Changing the Version of a Charge Plan

The procedure is the same as changing the version of an object explained below:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.24.3.6 Opening an Existing Charge Plan

You can open:

- An existing charge plan saved in the back-end database
- A previous version of a charge plan saved in the database
- An existing charge plan saved in an XML file

Open an Existing Charge Plan Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Charge Plan*. The *Open Charge Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge plan.
3. In the *Charge lans* folder of the catalog of the owner, select the charge plan. The *Charge Plan* window opens in display mode [page 46].

Charge plans are managed chronologically. After you have saved a charge plan, you can access previous versions of it that exist in the database. However, if you modify and save a previous version of a charge plan, all later versions of it are deleted.

Open a Previous Version of a Charge Plan Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Charge Plan*. The *Open Charge Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge plan.
3. In the *Charge Plans* folder of the catalog of the owner, select the charge plan, and then click *Open*. The *Charge Plan* window opens in display mode [page 46].
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the charge plan from the list.
6. Click *View*.

> **Note**
>
> You can only delete a previous version of a charge plan by modifying it, specifying an effective date earlier than the effective date of the original version, and then saving the charge plan.

Open an Existing Charge Plan Saved in an XML File

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.24.3.7 Modifying a Charge Plan

The modifications that you can make to a charge plan differ depending on whether it has been saved or not.

> → Remember
>
> Read the note [page 316] about making retroactive changes to a charge or a charge plan.

> ⓘ Note
>
> - When you make changes to charge plans, the system updates the related provider contracts depending on the changes applying to counters or parameters which are assigned the state *Not Redefined* in the related provider contracts.
> - You can modify and save different versions of the charge plan if its status is set to Released [page 352].

Modify a Charge Plan

1. Open [page 360] the charge plan.
2. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The charge plan may be reloaded if it has been modified since you have opened it.
3. *Set up* the charge plan.
4. Verify the charge plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Check the preview [page 358] of the charge plan.
6. Save [page 359] the charge plan.

To preserve the integrity of the database, you cannot remove elements from the structure of a saved charge plan, and many of the fields in the *Definition* tab of a charge are unavailable.

Related Information

Setting Up a Charge Plan [page 351]

Saving a Charge Plan [page 359]

1.24.3.8 Deleting the Versions of a Charge Plan

The procedure is the same as deleting the versions of a catalog object [page 118].

1.24.3.9 Deleting a Charge Plan

Before you delete a charge plan, you must first delete all master data objects which reference it.

For example, you can not delete a charge plan that is part of a contract in the back-end database. You must first delete the contract that uses this charge plan, and then you can delete the charge plan.

> ⚠ Caution
>
> There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.

Delete a Charge Plan

1. From the menu bar in the main window, select *File* > *Delete* > *Charge Plan*. The *Delete Charge Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge plan.
3. In the *Charge Plans* folder of the catalog of the owner, select the charge plan.
4. Click *Delete*. A deletion dialog box asks if you want to delete the charge plan.
5. Confirm the deletion of the charge plan by clicking *Yes*:
    1. If a dialog box confirms your choice, the charge plan has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the charge plan is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> ⓘ Note
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the charge plan. The identifiers of these objects appear in the message.

1.24.3.10 Exporting a Charge Plan

You can export charge plans to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to an XML file, the export operation saves all versions and associated objects to a single XML file, allowing you to import [page 364] this charge plan and its associated objects back into SAP CC system.

You can use this function to transport some consistent data from an SAP CC system to another SAP CC system.

> → Tip
>
> An alternative solution is to create a change list and use a transport request. See Change Lists and Transport Requests [page 660].

Export a Charge Plan to an XML File

As an SAP CC pricing specialist, to export a charge plan and its consistent subset of data objects, you complete the following procedure:

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the charge plan.
3. In the catalog of the owner, select the charge plan.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the charge plan.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

Related Information

Importing a Charge Plan [page 364]

1.24.3.11 Importing a Charge Plan

You can import charge plans in XML format from a directory to the database of another SAP CC system.

Several versions of a charge plan can exist in files, corresponding to their stages of modification. You can import all versions of a charge plan into SAP Convergent Charging, or select only the versions you want.

> ⓘ Note
>
> You cannot import charge plans that are not compatible with charge plans.

Import a Charge Plan

As an SAP CC pricing specialist, to import a previously-exported charge plan and its consistent subset of data objects, you complete the following procedure:

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the charge plan is saved, and then select the charge plan file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected if you want to import all the versions of the imported objects, or do the following:
   - To import only one version as of a certain date, select the *One version per object* option, and then select the date in the *Version as Of* field.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
6. In the *Import* column, clear the checkboxes of objects you do not want to import.
7. Click *Import*.

> ⓘ Note
>
> SAP Convergent Charging can only import an object that has previously been exported to an XML file (*File* > *Export*), rather than saved directly to an XML file (*File* > *Save* > *In a File*).
>
> The file contents differ.
>
> This is because the export operation saves all versions of an object and the objects associated to it, rather than just one version that lacks associated objects. The import operation requires all versions and associated objects to be available.

Related Information

Exporting a Charge Plan [page 363]

1.24.4 Working with Charges Customized in a Charge Plan

> → Remember
>
> Before working with charges customized in a charge plan, you must have read the Caution note [page 290] about charges and charge plans.

Related Information

Adding a Charge to a Charge Plan [page 366]
Removing a Charge from a Charge Plan [page 366]
Setting Up a Charge in a Charge Plan [page 367]
Sharing Transient Counters with Dependent Charges in Charge Plans [page 386]

1.24.4.1 Adding a Charge to a Charge Plan

This procedure allows you to add a charge to the charge plan.

Prerequisites

To perform this procedure, you must open [page 360] a charge plan.

To add a charge to a charge plan

1. From the menu bar in the main window, select *Action* > *Add Charge*, or click the icon  on the toolbar in the main window. You can also right-click the charge plan root  and click *Add Charge*. The *Choose a Charge* dialog box opens.
2. In the *Charges* folder, select the reusable charge [page 290] that you have prepared.
3. Click *OK*. The charge is added to the charge plan.
4. Select the charge  in the charge plan tree, and then set up [page 367] the tabs of the charge according to your business requirements.

1.24.4.2 Removing a Charge from a Charge Plan

If you have added a charge to a charge plan and the charge plan is not yet saved, you can remove the charge from the charge plan.

To preserve the integrity of the database, SAP Convergent Charging does not allow you to remove a charge from a charge plan after you have saved the charge inside the charge plan. If you need to remove a charge after it has been saved in the charge plan, you must delete the entire charge plan and build it again.

To perform this procedure, you must open [page 360] a charge plan.

To remove a charge from a charge plan

1. In an open charge plan, right-click the charge that you want to remove.
2. Select *Remove*. The charge is removed from the charge plan.

1.24.4.3 Setting Up a Charge in a Charge Plan

Customizing a charge in a charge plan means configuring aspects of it to meet the requirements of the charge plan. You can set up a charge in the charge plan root.

You must set up [page 351] a charge plan before setting up a customized charge. A *Charge Plan* window must be open.

Customize a Charge in a Charge Plan

1. In the left pane of the *Charge Plan* window, make sure that the charge  is selected; otherwise, select it.
2. In the right pane of the charge plan window, you can change the name of the customized charge. Note that two charges cannot have the same name. If several charges have the same name, the names are followed by a number to uniquely identify each charge.

   > **Example**
   >
   > A charge is named "SMS monthly fee". In the same charge plan, you add another charge with the same name, then its name will automatically be "SMS monthly fee#1". If you add a third charge with the same name, its name will be "SMS monthly fee#2" and so on.

3. In the right pane of the *Charge Plan* window, set up the following tabs of the customized charge:
   - Definition [page 368]
   - Parameters [page 370]
   - Counters [page 371]
   - Output Items [page 372]
   - Response Items [page 374]
   - Tax [page 374]
   - Account Assignments [page 385]
   - Technical Data [page 385]

When you set up a customized charge, only the *Definition* tab is mandatory. The other tabs are optional.

> **Note**
>
> - If you save the charge plan before defining the customized charge inside it, you cannot adjust the charge values later. If a charge is incorrectly defined once the charge plan is saved, you must delete the charge plan, and then re-create it.

- If the *Counters* or *Parameters* tabs of a charge are unavailable (dimmed), this means a counter or a parameter was not used at charge level.

1.24.4.3.1 Customized Charge: Definition

> ⓘ Note
>
> In an integrated *SAP Solution* scenario with the SAP CRM and the SAP ERP/FI-CA components of SAP Business Suite, if you need to manage payments to a business partner, SAP SE recommends that you do not configure any dependent charges. Refer to the note [page 290].

Create [page 350] or modify [page 362] a charge plan before setting up and customizing a charge [page 367] in this charge plan. In the left pane of the *Charge Plan* window, click a charge to view, and then open the *Definition* tab.

The *Definition* tab allows you to define how the master [page 293] or dependent [page 294] charge is used in the provider contracts and the role [page 346] it plays. The charge type is defined previously when you create [page 309] a reusable charge.

> ⚠ Caution
>
> If you save a charge plan before defining a customized charge, the charge will be saved using default settings. You cannot change the default settings in a charge that it is saved as part of a charge plan. If you need to change the definition settings of a charge after it has been saved in a charge plan, you must delete the charge plan, create it again, add the charge, and define it.

The *Definiton* tab displays different configuration fields depending on the type (master, dependent) of the charge that you want to set up and customize.

Definition (Tab)

If you set up a master charge

Fill in or modify the fields below:

Charge: Displays the name of the customized charge. If you want to modify the reusable charge before defining it in the charge plan, you can do so by clicking  , modifying [page 310] the charge, saving your changes, and closing its window.

Relationship Type: Select [page 346] one of the following options:

- *Client*: The subscriber of the charge plan consumes the service related to the charge, resulting in the generation of a client business transaction (or charged item).
- *Partner*: The subscriber of the charge plan provides the service related to the charge, resulting in the generation of a partner business transaction (or charged item).

Operation: Depends [page 346] on the option that you select in the *Relationship Type* field. Select one of the following options:

- *Debit Client*: The client pays a service provider for the consumption (aka usage) of the service related to the charge.
- *Credit Client*: The client is paid by a service provider after the service is consumed.
- *Debit Partner*: The partner pays for the consumption of the service related to the charge.
- *Credit Partner*: The partner is paid for the consumption of the service related to the charge.

If you set up a dependent charge

Fill in or modify the fields below:

Charge: Displays the name of the customized charge. If you want to modify the reusable charge before defining it in the charge plan, you can do so by clicking  , modifying [page 310] the charge, saving your changes and closing its window.

Role: Select [page 346] one of the following roles that the charge plays in the charge plan:

- *Commission*: Is an amount paid to an agent (third party) such as a sales representative.
- *Sponsorship*: Is an amount paid by a sponsor (third party) to a service provider.
- *Discount*: Is a deduction in price or amount of a bill paid by the client.
- *Third-Party Billed Commission*: Is an amount paid to a third party after being billed by the third party.
- *Other*: Select this option if you want to customize the role of the dependent charge.

Operation: Is only available [page 346] if you select the *Other* option in the *Role* field above. Select one of the following options:

- *Debit Client/Partner*: The client or partner is debited
- *Credit Client/Partner*: The client or partner is credited

Create New Transaction[50]: If selected, a business transaction is created which includes the price amount calculated by the dependent charge. Some fields are available or not depending on which option you select in the *Role* field. Selecting an option other than *Other* in the *Role* field only makes the *Client/Partner* and *Service Provider* fields available while selecting the *Other* option allows you to select or not the *Create New Transaction* checkbox. If this checkbox is selected, all the fields in this area are available; otherwise, they are unavailable.

> → Remember
>
> You must keep in mind that once a master charge has been triggered, the related dependent charges are triggered depending on the priority order defined in the Dependencies [page 357] tab. This is the case with the *Use previous discounted amount* option. When you select this option, a discount is calculated on the initial amount of the master charge from which the previous discounts have already been subtracted; therefore, the resulting amount of the discount depends on the order of the dependent charges in the dependencies table.

- *Relationship Type*: If you select the *Other* option in the *Role* field and the *Create New Transaction* checkbox, the Relationship Type [page 346] option allows you to tag generated transactions as either *Client* or *Partner*.

[50] If you choose not to create a new transaction, all the transactions created by the master charge and related dependent charges are merged into one single transaction; however, all the transaction details resulting from the dependent charges will not appear in the final transaction. Therefore, the transaction details of the dependent charges will be lost permanently.

- *Adjust Master Transaction*: If you select the *Other* option in the *Role* field and the *Create New Transaction* checkbox, the *Adjust Master Transaction* checkbox allows you to enable the amount calculated by the dependent charge to modify the master transaction amount. This will split the amount calculated by the master charge into two transactions: (i) Amount calculated by the dependent charge, and (ii) Amount calculated by the master charge minus the amount calculated by the dependent charge. Leaving the box unchecked will result in the creation of a transaction completely independent of a master charge.
- *Use previous discounted amount (if any)*: Is available if the *Discount* role and the *Create New Transaction* option are selected. If selected, this option allows you to apply the discount of the dependent charge to the amount of the master charge from which the previous discounts have already been subtracted. For example, the master charge amount "MA" is of $200. The master charge "MA" is linked to three dependent charges having a role of discount "D1" (10%), "D2" (20%), and "D3" (50%). The dependent charges and triggered in the order "D1", "D2", and "D3". When the master charge "MA" is triggered, a transaction of $200 is created. A second transaction is created for the dependent charge "D1": $200 x 10% = $20. A third transaction is created for the dependent charge "D2": ($200 - $20) x 20% = $36. A fourth transaction is created for the dependent charge "D3": ($200 - ($20 + $36) x 50% = $72.

1.24.4.3.2 Customized Charge: Parameters

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367] in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Parameters* tab.

The *Parameters* tab allows you to:

- Redefine or not the parameter of a charge.
- Link the parameter of a charge to a shared parameter of the charge plan.

> **Note**
>
> If the *Parameters* tab is unavailable (dimmed), this means that no parameter was created at charge level. To open the original charge, click the *Definition* tab of the charge plan, and then click  next to the *Charge* field.

Parameters (Tab)

Fill in or modify the *Parameters* tab.

> **Note**
>
> If you have assigned the Multicurrency [page 311] option to the charge inserted in the charge plan, SAP Convergent Charging adds a currency type parameter to the table. You can redefine this parameter or link this parameter to a shared parameter of the charge plan.

Name: Displays the name of the charge parameter.

Type: Displays the type of the parameter.

Status: Provides the list of parameter status. Select one of the following status:

- *Not Redefined*: Select this status if you do not want to link the parameter to a shared parameter of the charge plan.
- *Redefined*: Select this status if you want to redefine the parameter of a charge. You can redefine a parameter only if in the charge its visibility is set to *External*.
- *Linked*: Select this status if you want to link the parameter to a shared parameter of the charge plan [page 354].

Value: This field varies depending on the option that you select in the *Status* field:

- *State set to Not Redefined*: The value of the parameter is the value initially entered in the charge. The value of the parameter can not be changed.
- *State set to Redefined*: Type the new value of the parameter.
- *State set to Linked*: Provides the list of shared parameters having the same type as this parameter. Select the shared parameter that you want to link to your charge parameter. The value of the charge parameter is now the same as that of the shared parameter.

Recurring Charges Configuration

> ⓘ Note
>
> This configuration is used when implementing the data configuration for an *SAP Solution* scenario with SAP CRM or SAP Convergent Invoicing.

If your charge contains recurring rates, you may have to link special parameters to:

1. Use the prices [page 319] defined for SAP CRM products.
2. Synchronize your recurring rates [page 321] with the billing cycles of SAP Convergent Invoicing.

1.24.4.3.3 Customized Charge: Counters

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367].

Charge Plan window, click a charge  in the charge plan. In the to view, and then open the *Counters* tab.

The *Counters* tab allows you to link the counter of a charge to a counter of the charge plan. In this tab, a table displays the list of all counters created at charge level. Each row in the table represents a counter.

> ⓘ Note
>
> If the *Counters* tab is unavailable (dimmed), this means that no persistent counter was created in the original charge. To open the original charge, click the *Definition* tab of the charge plan, and then click  next to the *Charge* field.

Counters (Tab)

Fill in or modify the *Counters* tab.

Name: Displays the name of the charge counter.

Status: Provides the list of counter status. Select one of the following status:

- *Not Redefined*: Select this status if you do not want to link the counter to a counter of the charge plan [page 353]
- *Linked*: Select this status if you want to link the counter to a counter of the charge plan [page 353]. This *Status* enables you to redefine the initial value of the counter at charge plan level or to share this counter with other charges included in this charge plan.

Initial Value: Provides the initial value of the counter if the *Status* of the counter is *Not Redefined* or the list of the counters defined at charge plan level if the *Status* is *Linked*. In this case, select the counter that you want to link to your charge counter. The value of the charge counter is now the same as the value of the counter defined in the charge plan.

1.24.4.3.4 Customized Charge: Output Item

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367] in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Output Item* tab.

> → Remember
>
> Do not forget that you can create a charged item class:
>
> - From scratch [page 254] if you use an external billing system or any other external system.
> - From the billable item mapping [page 263] if you work with an SAP ERP/FI-CA system. In this case, when creating a charged item class, you must select the type of the charge item class which depends on the account type linked to the charge plan.

You must map charged item classes onto billable item classes generated automatically by SAP Convergent Charging. To do that, you create [page 266] a charged item class from the mapping of a billable item class. Refer to Billable Item Mapping [page 263] for more information when using an SAP ERP/FI-CA system.

The *Output Item* tab allows you to:

- Select the charged item class [page 254] to associate with the customized charge in your charge plan.
- Configure the generation of charged items (see charged item mapping) by the SAP CC system when processing the charging operations:
  - Change the constant values or properties of the fields predefined in the charged item class.
  - Assign values to the fields not set in the charged item class.
  - Select the aggregation policy (only relevant for the charging operations performed in the session-based execution mode [page 20]).

Output Item (Tab)

Fill in or modify the *Output Item* tab to define the generation of charged items by the system.

Charged Item Based On: Provides the list of the charged item classes. Select a charged item class if you want to replace the charged item class associated to the charge. Click  to open the charged item class, and then modify it if necessary. Click  to refresh the charged item mapping.

Charged Item Mapping: Provides the list of the fields of the selected charged item class, the respective values or properties of these fields, and the default values of these fields. To display less or more columns, click  , and then select the columns you want to display or not.

- *Charged Item Field*: Displays the field names of the charged item class.
- *Property/Value*: Allows you to change the values or properties of the fields depending on the following:
  - If a property or value is dimmed, it is not recommended changing this property by another property or value.
  - If a property or value is not dimmed, it is recommended changing this value by another property or value.
- *Default Value*: Displays the field values defined at class level [page 257].
- *Redefine*: Allows you to redefine any value of the *Property/Value* column. Refer to the recommendations listed above and the note below. By default, this column is not visible.

> **Note**
>
> - It is not recommended changing a property or value that is dimmed. For more information about the *Undefined* and *Empty* properties, see Setting Up a Charged Item Class [page 257].
> - If you set up a boolean field, the value is converted by SAP CC during the charging process. Strings "0" and "false" and number 0 are converted to FALSE. Strings "1" and "true" and number 1 are converted to TRUE; otherwise, the charging process fails because it cannot convert the value.

Session-Based Options: Provides the list of options which are specific to charging operations performed in the session-based rating/charging:

- *Aggregation Policy*: Provides the list of aggregation policies which are associated to the selected charged item class. Select one of the available aggregation policies or the *None* element of the list if you do not want to aggregate the generated charged items. Click  to open the selected aggregation policy, and then modify it if necessary.
- *Reporting Scope* (*): Select the reporting scope to use during the aggregation of charged items:
  - When using the *Charged Item* value, a reporting condition fulfilled for a given charged item triggers the reporting of the related charged item only.
  - When using the *Session* value, a reporting condition fulfilled for a given charged item triggers the reporting of all the charged items of the session.

> **Note**
>
> (*) The *Reporting Scope* option is not available for dependent charges and allowances.

1.24.4.3.5 Customized Charge: Response Item

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367] in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Reponse Item* tab.

> → Remember
>
> Do not forget that you can create a charged item class from:
>
> - Scratch [page 254] if you use an external billing system or any other external system.
> - The billable item mapping [page 263] if you work with an SAP ERP/FI-CA system. In this case, when creating a charged item class, you must select the type of the charge item class which depends on the account type linked to the charge plan.

You must map charged item classes onto billable item classes generated automatically by SAP Convergent Charging. To do that, you create [page 266] a charged item class from the mapping of a billable item class. Refer to Billable Item Mapping [page 263] for more information when using an SAP ERP/FI-CA system.

The *Response Item* tab allows you to:

- Select the charged item class [page 254] used to filter the content of the generated transaction.
- Configure the generation of transactions (see charged item mapping) returned by the charging process:
  - Change the values or properties of the fields predefined in the charged item class.
  - Assign values to the fields not set in the charged item class.

Response Item (Tab)

Charged Item Based On: Provides the list of the charged item classes. Select a charged item class if you want to replace the charged item class associated to the charge. Click  to open the charged item class, and then modify it if necessary. Click  to refresh any change relating to the charged item mapping.

Charged Item Mapping: Provides the list of the fields of the selected charged item class and the respective values or properties of these fields. For more information, consult the Charged Item Mapping of the Output Item [page 372] tab.

1.24.4.3.6 Customized Charge: Tax

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367] in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Tax* tab.

The *Tax* tab allows you to set up the tax settings for the charges customized in a charge plan. This information is used by the SAP CC system to calculate the tax data and determine who must pay the tax. Consult the Application Help for more information about the tax calculation feature and process.

> **Note**
>
> A charge in a charge plan represents a customer service of the service provider. If you want to configure the pricing logic for several services with different tax settings, you must customize several charges in your charge plan.

Tax (Tab)

> **Note**
>
> Depending on your SAP CC system configuration, the *Tax* tab displays the settings for the EU VAT tax system, the flat tax system, or the Avalara AvaTax for Communications (AFC) system. If the *Tax* tab is unavailable (dimmed), the SAP CC system is not configured for computing taxes. For more information, refer to your administrator and consult the SAP CC Administration Guide.

Fill in or modify the *Tax* tab.

Name: Depending on your SAP CC system configuration, select one of the following tax system:

- *Disabled*: No tax system applies to the charge.
- *EU VAT*: The European Union Value Added Tax (EU VAT) invoicing tax system applies to the charge.
- *Flat Tax*: A constant tax rate applies to all the amounts calculated in the charge.
- *US* (Avalara AvaTax for Communications (AFC)): The Avalara AvaTax for Communications (AFC) system applies to the charge.

> **Note**
>
> The tax system set in the dependent charges [page 294] of a charge plan must be the same than the tax system set in their related master charge [page 293].

With the EU VAT System [page 376]

With a Flat Tax System [page 379]

With the Avalara AvaTax for Communications (AFC) System/Engine for US Telecommunications Taxes [page 379]

1.24.4.3.6.1 With the EU VAT System

If you use the EU VAT system...

> **Note**
>
> About the *Choose a Property* option.
>
> If you need to apply more than one tax rate level to a customized charge, you can create a numerical property in the corresponding price plan, and then type different values into it depending on the branch of the rating tree.

The *European Union Value Added Tax* area displays the following:

- *Base Amount*: Select if the prices defined in the price plan of the charge are net or gross:
  - *Net Price*: The tax is calculated and added to the base amount (default configuration for charge plans)
  - *Gross Price*: The tax is calculated and removed from the base amount.

    > **Note**
    >
    > This configuration is not compliant with the *Account Holder Subject to Pay* or the *Account Holder Exempted* taxation mode that is configured in the subscriber accounts of end customers who have subscribed to a contract activating this charge plan. See the tax settings in subscriber accounts [page 534].

- *Tax Rate Level*: Select if you want to set another predefined tax rate, and then do one of the following:
  - Select the *Normal Tax Rate* option, and then select another predefined tax rate level from the list.
  - Select the *Choose a Property* option, and then select a numerical rating context property that represents the tax rate level (see the example below) at runtime.
    This property must be numerical and contain the following values during the tax calculations:
    - 0: Normal
    - 1: Intermediary
    - 2: Reduced
    - 3: Super Reduced
    - 4: Zero
    - 5: Special
    - -1: No Tax
- *VAT Rule*: Select the VAT rule that is relevant for your business if you want to support the *2010 EU VAT Invoicing Directive*. By default the following rules are configured and may differ from your settings:

VAT Rule	Service Provider Zone	Place of Taxation for B2B Services	Place of Taxation for B2C Services
EUVAT2010 - General rule for services	Any Country	Customer's place of establishment: The holder of the prepaid account [page 514] or external postpaid account [page 520]totally or partially impacted by the service consumption. The fees related to the consumption can be charged to several accounts in the subscriber account of the end customer.	Service provider's place of establishment: Its country is specified in this charge customized in a charge plan [page 343]. See the *Service Provider Country Code* option.
EUVAT2010 - Digital services	European Union	Customer's place of establishment: This country information is set in the tax settings [page 534] of the subscriber accounts [page 511].	Service provider's place of establishment
EUVAT2010 - Radio, TV and telecommunication services	Third Country	Customer's place of establishment	Customer's place of residence: The country of the holder of the prepaid account [page 514] or external postpaid account [page 520] in the subscriber accounts [page 511].
	European Union	Customer's place of establishment	Service provider's place of establishment
EUVAT2015 - Digital services	Third Country	Customer's place of establishment	Place of effective service consumption: The B2C taxation place is dynamically set by the SAP CC system according to the tax settings of this customized charge.
	Any Country	Customer's place of establishment	Customer's place of residence
EUVAT2015 - Radio, TV and telecommunication services	Any Country	Customer's place of establishment	Customer's place of residence

> **Note**
>
> If no VAT rule is selected, the SAP CC system will consider the tax settings [page 534] set in each subscriber account [page 511] hold by an end customer of the service provider:
>
> - To determine who is subject to pay the tax
> - And when the tax data must be computed

If a VAT rule is selected, do the following:

- *Description*: Display the information that describes the selected VAT rule.
- *Service Provider Country Code*: Select the place of establishment of the service provider that is considered by the VAT rule to compute tax data:
  - From a country code ISO 3166 (as configured in the SAP CC system)
  - From a string property of the context [page 13]

> **Note**
>
> The country codes set in the charges or retrieved from the rating context must be compliant with country codes known by SAP Convergent Charging.
>
> Refer to your administrator and consult the SAP CC Administration Guide to customize the country tax policies.

Depending on the selected VAT rule, select or view the taxation places and their country code.

- *B2B Taxation Place*: Select the place of taxation for B2B customers according to the selected VAT rule:
  - A country code ISO 3166 (as configured in the SAP CC system).
  - A string property of the context [page 13] that will contain at runtime the relevant value.
- When dimmed, view the settings from the VAT rule. The taxation place (country code) for B2B customers will be deducted from the:
  - Service provider's place of establishment (the owner of the pricing catalog including the charge plan of this charge) - See *Service Provider Country Code* above.
  - Account holder's place of establishment (the person who holds a prepaid account or an external postpaid account that is charged for the service consumption) - See the tax settings in subscriber accounts [page 534].
- *B2C Taxation Place*: Select the place of taxation for B2C customers according to the selected VAT rule:
  - A country code ISO 3166 (as configured in the SAP CC system)
  - From a string property of the context [page 13]
- When dimmed, view the settings from the VAT rule. The taxation place (country code) for B2C customer will be deducted from the:
  - Service provider's place of establishment (the owner of the pricing catalog including the charge plan of this charge) - See *Service Provider Country Code* above.
  - Account holder's place of residence (the person who holds a prepaid account or an external postpaid account that is charged for the service consumption) - See the tax settings in subscriber accounts [page 534].

> **Note**
>
> About the *B2C Taxation Place* option:
>
> If you select the *EUVAT2010 Radio, TV and telecommunication services*, the B2C taxation place must be the effective place of service consumption.
>
> Set up the related country code or select a property from the rating context and make sure that this property will have relevant values during the charging process. This can be a parameter, a computed property, or an input property from incoming chargeable items representing the service consumption.

1.24.4.3.6.2 With a Flat Tax System

If you use a flat tax system...

You first must select the rate value, and then select the identifier of the rate value.

The *Flat Tax* area displays the following:

- *Base Amount*: Choose if the prices defined in the price plan of the charge are net or gross:
  - *Net Price*: The tax is calculated and added to the base amount
  - *Gross Price*: The tax is calculated and removed from the base amount.

    > **Note**
    >
    > This configuration is not compliant with the *Account Holder Subject to Pay* or the *Account Holder Exempted* taxation modes that are configured in the subscriber accounts of end customers who have subscribed to a contract activating this charge plan.
    >
    > See the tax settings in subscriber accounts [page 534].
- *Flat Tax Rate*: To select the rate value, do one of the following:
  - Select the *%* option, and then type the value of the flat tax rate in the field.
  - Select the *Choose a Property* option, and then select a rating context property that represents the tax rate.
- *Identifier*: To select the identifier of the rate value, do one of the following:
  - Select the first option, and then type a name for the flat tax that will appear on invoices.
  - Select the *Choose a Property* option, and then select a rating context property that represents the tax rate identifier, the name of which will appear on invoices.

1.24.4.3.6.3 With the Avalara AvaTax for Communications (AFC) System/Engine for US Telecommunications Taxes

If you use the Avalara AvaTax for Communications (AFC) system...

> **Note**
>
> For more information about Avalara AvaTax for Communications (AFC) (solution, system, and engine), refer to the official product documentation that is provided by the vendor company.
>
> Check the user manual, wiki, and community for the Avalara AvaTax for Communications (AFC) software product. This software was known as *BillSoft EZTax*.

> **Example**
>
> For more information about EZTax, refer to the *EZTax® Telecom Users Manual* provided by *BillSoft®* company.

The *US Telco Tax* area displays the following tabs:

- Taxing Jurisdiction Info [page 381]
- Customer Info [page 382]
- Company Info [page 383]
- Transaction Info [page 383]
- Transaction Service Type [page 384]

As an SAP CC pricing specialist, you fill in these tabs with valid data that suits your business requirements. See also: Working with Charges Customized in a Charge Plan [page 365]

> **Note**
>
> As an application administrator (or system administrator), you regularly maintain some business data in the SAP CC system landscapes. This "application data" is provided by Avalara AvaTax for Communications (AFC) and contains the predefined definitions of the transaction types and service types.
>
> You use the Setup Tool user interface and the `eztax importTSTypes` command. Refer to the SAP documentation of this command line interface (CLI). See: SAP CC 2023 Setup Tool

1.24.4.3.6.3.1 Setting Up the Tabs of the Tax Tab

As an SAP CC pricing specialist, you fill in the tax settings for the customized charge (charge plan item) in your charge plan. For more information about Avalara AvaTax for Communications (AFC) or EZTax, refer to the official product documentation.

Avalara AvaTax for Communications (AFC) is the new brand name of: BillSoft EZTax

Taxing Jurisdiction Info [page 381]

Customer Info [page 382]

Company Info [page 383]

Transaction Info [page 383]

Transaction Service Type [page 384]

Related Information

Taxing Jurisdiction Info [page 381]
Customer Info [page 382]
Company Info [page 383]
Transaction Info [page 383]

Transaction Service Type [page 384]

1.24.4.3.6.3.1.1 Taxing Jurisdiction Info

On this tab, you assign the correct jurisdiction to the transaction for obtaining and returning the correct tax to the billing system.

Jurisdiction information can be supplied to the Avalara AvaTax for Communications (AFC) Engine/System in several ways. It can be supplied using a Permanent (location) Code (P-Code), a NPANXX code, or a State Code.

For example, a call can originate in Oregon, terminate in California and have a service address in Oregon.

Fill in or modify the *Taxing Jurisdiction Info* tab.

Origination: The origination location of the transaction to be taxed.

- *Type*: Provides the list of jurisdiction types. Select one of the following types:
  - *N (NPANXX)*
  - *P (P-Code)*
  - *S (State Code)*
- *Value*: Select or type the value assigned to the jurisdiction. Do one of the following:
  - Select the value defined in a property.
  - Type a constant value between "0000000000" and "9999999999" (10 digits).

Termination: The termination location of the transaction to be taxed.

- *Type*: Provides the list of jurisdiction types. Select one of the following types:
  - *N (NPANXX)*
  - *P (P-Code)*
  - *S (State Code)*
- *Value*: Select or type the value assigned to the jurisdiction. Do one of the following:
  - Select the value defined in a property.
  - Type a constant value between "0000000000" and "9999999999" (10 digits).

Service Address: The service address of the transaction to be taxed.

- *Type*: Provides the list of jurisdiction types. Select one of the following types:
  - *N (NPANXX)*
  - *P (P-Code)*
  - *S (State Code)*
- *Value*: Select or type the value assigned to the jurisdiction. Do one of the following:
  - Select the value defined in a property.
  - Type a constant value between "0000000000" and "9999999999" (10 digits).

1.24.4.3.6.3.1.2 Customer Info

On this tab, you define the customer information included in the transaction to be taxed.

Fill in or modify the *Customer Info* tab.

Customer Type: This option is used to specify the type of the end customer involved in the transaction. To select a type, do one of the following:

- Select one of the following types:
  - *Business*: If transactions are made at a place of business.
  - *Residential*: If transactions are made by a customer for home use.
  - *Industrial*: If transactions are made at an industrial business.
  - *Senior Citizen*: If transactions made by a customer meet the jurisdiction requirements.
- Select the type defined in a property.

Incorporated City Area: This option is used to specify whether the customer involved in the transaction is inside or outside of the incorporated area designated as their location. The tax may or may not be affected by this designator depending upon whether or not the unincorporated areas are taxed in the same manner as the incorporated areas. To select the option *inside* or *outside*, do one of the following:

- Select one of the following options:
  - *Inside*: The customer involved in the transaction is inside of the incorporated area.
  - *Outside*: The customer involved in the transaction is outside of the incorporated area.
- Select the option defined in a property.

Exemption: The exemption level is the jurisdictional level of the taxing authority that defines the tax. It is used to exempt taxes at specific federal, state, county and/or local taxes. Select the exemption levels:

- *Fed Exempt*: Is used to specify a Federal level tax exemption.
- *State Exempt*: Is used to specify a State level tax exemption.
- *County Exempt*: Is used to specify a County level tax exemption.
- *Local Exempt*: Is used to specify a Local level tax exemption.

Service Class: This option is provided to delineate the primary activity of an organization as either long distance or local service. To select a service class, do one of the following:

- Select one of the following service classes:
  - *Long Distance*: Primary long distance providers are carriers vending their services with over 50% of the gross business activities in long distance revenue.
  - *Local Primary*: Primary local service providers are carriers vending their services with over 50% of the gross business activities in local service revenue.
- Select the service class defined in a property.

Business Class: This option is used to specify if the business making the transaction is an Incumbent Local Exchange Company (ILEC) or a Competitive Local Exchange Company (CLEC). To select a business class, do one of the following:

- Select one of the following service classes:
  - *CLEC*: A CLEC company is engaged in selling services competing with an incumbent provider.
  - *ILEC*: An ILEC company is engaged in selling services over company-owned lines and equipment.

- Select the business class defined in a property

Lifeline: This option is used to indicate if a customer is a lifeline participant. Select the option defined in a property.

1.24.4.3.6.3.1.3 Company Info

On this tab, you define the transaction information that is contained in the transaction to be taxed.

Fill in or modify the *Company Info* tab.

Reseller: This option is used to specify if the telecommunication company and its services are regulated by the Public Utility Commission, the Federal Communication Commission, and/or other government authorities. To select an option, do one of the following:

- Select one of the following options:
  - *Regulated*: The telecommunication company and its services are regulated.
  - *Unregulated*: The telecommunication company and its services are not regulated.
- Select the option defined in a property.

Facilities Based: This option specifies the tangible facilities used by a company to deliver the service. If the carrier delivering the service has company-owned facilities used to provide the service, they are facilities based. If they do not own the facilities, they are non-facilities based.

- Select one of the following options:
  - *Facilities Based*: The carrier delivering the service has company-owned facilities used to provide the service.
  - *Non-Facilities Based*: The carrier delivering the service has not company-owned facilities used to provide the service.
- Select the option defined in a property.

Franchise: This option specifies if the company provides services sold according to a franchise agreement between the carrier and jurisdiction.

- Select one of the following options:
  - *Franchise*: The company provides services sold according to a franchise agreement.
  - *Non-Franchise*: The company does not provide services sold according to a franchise agreement.
- Select the option defined in a property.

1.24.4.3.6.3.1.4 Transaction Info

On this tab, you define the transaction information which is contained in the transaction to be taxed.

Fill in or modify the *Transaction Info* tab.

Number of Lines: When a local service is provided, a transaction must be generated with this option populated with the number of lines which the customer subscribes to. Avalara AvaTax for Communications (AFC)

uses this information for generating per line taxes usually associated with local E911 charges and local telecommunications relay service taxes and other assorted taxes. Do one of the following:

- Type the number of lines.
- Select the number of lines defined in a property.

Number of Locations: When a local service is provided, this option must be populated with the number of locations which the customer has. Avalara AvaTax for Communications (AFC) uses this information for generating local taxes that are mandated for location by the taxing jurisdiction. Most of these taxes are flat fees; however, the charge field must be specified to allow for taxation based upon percentages of revenue instead of per location.

- Type the number of locations.
- Select the number of locations defined in a property.

Client Resale: Telecommunication companies are taxed on transactions made by their clients, which in some cases can be passed on or "resold" to their customers in part or in total. This option is used to specify whether the service transaction is final or if it must be resold.

- Select one of the following types:
  - *Resale*: The service transaction must be resold.
  - *Sale*: The service transaction is final.
- Select the type defined in a property.

Telecommunication Type: The telecommunication type specifies the type of the telecommunication transaction. To select a type, select one of the following options:

- Select one of the following types:
  - *Calls*
  - *Debit Card Calls*
  - *Point of Sale*
- Select the type defined in a property.

Call Duration: The call duration specifies the length of phone call in minutes, with one tenth of a minute precision capability. Avalara AvaTax for Communications (AFC) uses the call duration for generating taxes that are specified as per minute flat fees in some taxing jurisdictions, and then stores the value in the Avalara AvaTax for Communications (AFC) log database. Select the call duration defined in a property.

1.24.4.3.6.3.1.5 Transaction Service Type

On this tab, you define the transaction and service types. Avalara AvaTax for Communications (AFC) stipulates a unique pair of numbers for each transaction and service type. The first number defines the transaction type and the second number defines the service type. Transaction types and service types are combined (or "paired") to uniquely describe a valid transaction / service pair for a transaction.

Fill in or modify the *Transaction Service Type* tab.

Types: Provides the list of transaction and service types. Select a type.

Description: Displays the description of the selected type.

1.24.4.3.7 Customized Charge: Account Assignments

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367]

in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Account Assignments* tab.

The *Account Assignments* tab allows you to link the charging references of the charge to account assignments [page 356] defined in the charge plan. The charging references of the charge are first defined in the Internal Reference Dictionary [page 327] of the charge.

Account Assignments (Tab)

Fill in or modify the *Account Assignments* tab.

Name: Displays the name of the charging references defined in the *Internal Reference Dictionary* of the charging plan of the charge. Even if no charging reference has been defined in the dictionary, there is always a default charging reference.

Type: Provides the account types. Select one of the following types: *Prepaid* or *Postpaid*.

Linked Value: Provides the list of the account assignments of the charge plan that have the same *Type* as this account assignment. Select the account assignment of the charge plan that you want to link to your account assignment.

> ⓘ Note
>
> If the list of account assignments is empty, make sure that the Account Assignments [page 356] tab of the related charge plan is not empty and the defined assignments have the appropriate type.

1.24.4.3.8 Customized Charge: Technical Data

Create [page 350] or modify [page 362] a charge plan before setting up and customize a charge [page 367]

in the charge plan. In the left pane of the *Charge Plan* window, click a charge  to view, and then open the *Technical Data* tab.

The *Technical Data* tab allows you to define identifiers that the end customer needs to access the related services of the service provider.

> ✤ Example
>
> SMS, voice, data.

Technical Data (Tab)

Fill in or modify the *Technical Data* tab.

 : Click to add a row to the configuration table.

 : Click to delete a row from the table.

> ⓘ Note
>
> The possible values of the service identifiers depend on how SAP Convergent Charging is integrated with the online or offline mediation system. The values must be consistent with the information supplied in the chargeable items received by the SAP CC system for rating or charging purposes.

Service Identifier: Type the technical identifier of the customer service (SMS, voice, data). This identifier must be unique in the *Technical Data* tab. This identifier (SID) must be present in the incoming chargeable items to charge in the SAP CC system. The same SID can relate to different customer services.

User Technical Identifier: Provides the list of the technical data of the charge plan. Select the technical data of the charge plan that you want to link to your technical data.

1.24.4.4 Sharing Transient Counters with Dependent Charges in Charge Plans

You can share transient counters [page 297] with a master charge [page 293] and its dependent charges [page 294] once they are customized in a charge plan. Apply the following procedure when configuring the pricing catalog of a service provider:

- Setting up the reusable charges that will be assigned to this charge plan.

To share a transient counter with master and dependent charges in a charge plan

1. First set up your reusable charges [page 290]:
   - Set up [page 323] a transient counter in the charge that will be the master charge in a charge plan.
   - Set up a transient counter with the same name in the reusable charges that will be the dependent charges.
2. Customize [page 367] these charges in a charge plan.

1.24.4.5 Changing the Version of a Charge Plan

1.24.5 Using the Action Menu

Use the *Action* menu of the user interface to manage your charge plans.

Possible actions are:

- Adding a Charge to a Charge Plan [page 366]
- Removing a Charge from a Charge Plan [page 366]
- Previewing a Charge Plan [page 358]

Version management:

- Changing the Version of a Charge Plan [page 360]
- Deleting the Versions of a Charge Plan [page 363]

1.25 Refill Item Classes

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

In SAP Convergent Charging 2023, a refill item class describes a refill event of usage type [page 398] coming from an external system such as SAP ERP/FI-CA or a Voucher Management System. Refill item classes allow you to model the properties used to configure the refill logic [page 395] of a refill plan. These properties can be used to set up the generation of resulting refill records at refill plan level.

The properties consist of default and user-defined properties.

> ⓘ Note
>
> - Tax Settings [page 523] do not apply to refill item classes
> - The refill item class is similar to the chargeable item class [page 241] used by a charge

> → Recommendation
>
> If you want the SAP CC system to copy data information from the incoming refill items to the generated refill records [page 414] (resulting from the refilling process [page 421]), you must add the relevant user properties to your refill record class. You must then configure the generation of refill records by setting up your refill logic [page 395] to enable the copy of data information from the input items to the output records.

Properties

A refill item class contains several properties that describe each refill item for refilling purposes. A refill item class contains only one default property, but you can configure a refill item class by creating your own user properties.

Example

The following table is an example of a refill item class:

Name	Category	Type	Description
Refill Date	Default	Date	Date of the refill
PREPR	User	String	Reference number used for the refill
GPART	User	String	Business partner number
BTRW	User	String	Refill amount with +/- sign
WAERS	User	String	Currency key
PPRSN	User	String	Reason for balance change on prepaid account
CRDID	User	String	Payment card ID

Default property

Each refill item class contains only one default property:

- Refill date/time

User-Defined Properties

In addition to the default property, you can create some user-defined and custom properties for a refill item class. You define these custom properties by assigning the following to them:

- Name
- Category
- Type (character string, number, or date/time)
- Description

SAP Solution

The integrated *SAP Solution* scenario (with SAP CRM and SAP ERP/FI-CA from SAP Business Suite) needs to configure the refill item classes specifically.

Note

Consult the SAP CC Administration Guide about the last recommendations for configuring master data in SAP CC.

> **Example**
>
> A refill item class automatically created by SAP CC contains the following fields:
>
> - SRCTATYPE (default constant value CCCIT)
> - PREPAID, VKONT (generated by SAP CC)
> - VTPOS, BKVID, CNTUI1 (to be set up in a refill plan)

1.25.1 Working with Refill Item Classes

Related Information

Creating a Refill Item Class [page 389]
Opening an Existing Refill Item Class [page 392]
Modifying a Refill Item Class [page 393]
Setting Up a Refill Item Class [page 390]
Deleting a Refill Item Class [page 394]
Saving a Refill Item Class [page 391]
Exporting a Refill Item Class [page 394]
Importing a Refill Item Class [page 394]

1.25.1.1 Creating a Refill Item Class

Use one of these procedures to create a refill item class.

To create a refill item class

1. From the menu bar in the main window, select *File* > *New* > *Refill Item Class*. The *New Refill Item Class* window opens.
2. *Set up* the refill item class.
3. Verify the refill item class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. *Save* the refill item class.

You can create a refill item class by opening an existing refill item class and copying it.

To create a refill item class from an existing refill item class

1. From the menu bar in the main window, select *File* > *Open* > *Refill Item Class*. The *Open Refill Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill item class.
3. In the *Refill Item Classes* folder of the catalog of the owner, select the refill item class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding refill item class window opens.
5. Rename the refill item class in the *Name* field, and then *set up* the refill item class.
6. Verify the refill item class structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
7. *Save* the refill item class.

Related Information

Setting Up a Refill Item Class [page 390]
Saving a Refill Item Class [page 391]

1.25.1.2 Setting Up a Refill Item Class

Create [page 389], open [page 392], or modify [page 393] a refill item class before setting it up. A refill item class window must be open.

In the graphical user interface of SAP Convergent Charging, a refill item class is represented by a table that consists of rows representing the properties of the refill item class. A refill item class contains only one default property, but you can add user properties to the refill item class.

> **Note**
>
> If necessary, you can modify a refill item class at refill logic level (refer to the Refill Logic Definition [page 402] tab of the refill logic).

To set up a refill item class, you first identify it, and then you add user properties to the table, or you modify the user properties of the table.

Name: Type the unique identifier of the refill item class.

Description: Describe the function of the refill item class.

: Click to add a new row to the table.

: Click to delete a row from the table.

Properties: Fill in or modify the columns of the table.

- *Name*: Type the identifier of the user property
- *Category*: Displays the category (default or user) of the property
- *Type*: Select the type of the property (number, string, or date)
- *Description*: Describe the function of the property

1.25.1.3 Saving a Refill Item Class

Once you have created a refill item class and then verified its configuration, you are ready to save it. You can save the refill item class to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving a refill item class, you can no longer:
>
> - Modify the name of the refill item class
> - Remove a user property from the table
> - Change the type of a user property

To save a new refill item class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill item class.
3. Click *Save*. The *Refill Item Class Creation* dialog box opens to inform you that the refill item class has been successfully saved.
4. Click *OK*.

To save an existing refill item class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Refill Item Class Modification* dialog box opens to inform you that the refill item class has been successfully modified, click *OK*. The new version of the refill item class is saved as the only current version.
3. If a *Warning* dialog box opens to inform you that the refill item class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the refill item class.
   - Click *Cancel* to come back to the refill item class. See below how to save the refill item class as an XML file.

To save a refill item class as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the refill item class.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The refill item class is saved as an XML file in your selected directory.

1.25.1.4 Opening an Existing Refill Item Class

You can open:

- An existing refill item class saved in the database
- An existing refill item class saved in an XML file

To open an existing refill item class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Refill Item Class*. The *Open Refill Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill item class.
3. In the *Refill Item Classes* folder of the catalog of the owner, select the refill item class.
4. Click the *Open* button. The *Refill Item Class* window opens in display mode.

To open an existing refill item class saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.

4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.25.1.5 Modifying a Refill Item Class

When modifying a refill item class, you can only add a new property to the refill item class. The modified refill item class replaces the previous version in the database.

> **Note**
>
> You cannot:
>
> - Modify the name of the refill item class
> - Remove a property from the table
> - Change the type of a property

To modify a refill item class

1. Open [page 392] the refill item class.
2. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The refill item class may be reloaded if it has been modified since you have opened it.
3. Set up [page 390] the refill item class.
4. Verify the structure of a refill item class by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Save [page 391] the refill item class.

Related Information

Display/Edit Mode [page 46]

1.25.1.6 Deleting a Refill Item Class

Use this procedure to delete a refill item class.

To delete a refill item class

1. From the menu bar in the main window, select *File* > *Delete* > *Refill Item Class*. The *Delete Refill Item Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill item class.
3. In the *Refill Item Classes* folder of the catalog of the owner, select the refill item class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the refill item class
5. Confirm the deletion of the refill item class by clicking *Yes*:
    1. If a dialog box confirms your choice, the refill item class has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the refill item class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48]for more information.

1.25.1.7 Exporting a Refill Item Class

You can export refill item classes to a directory of your hard disk, where they are saved as XML files.

To export a refill item class

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill item class.
3. In the catalog of the owner, select the refill item class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the refill item class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.25.1.8 Importing a Refill Item Class

You can import refill item classes in XML format from a directory of your hard disk.

To import a refill item class

1. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
2. Open the directory in which the refill item class is saved, and then select the file of the refill item class.
3. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a refill item class
4. In the *Import* column, clear the checkboxes of objects you do not wish to import.
5. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available for import.

1.26 Refill Logic

> **Caution**
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

In SAP Convergent Charging 2023, the *refill logic* is a reusable master data object, which is part of the pricing catalog [page 109] for a service provider. The refill plan [page 421] uses the refill logic that you can customize. The refill logic includes decision trees that specify the computation for the different credits (money, units, and so on) of a refill.

The refill logic performs calculations only when the refill plan is triggered by:

- A usage refill event [page 387]
- A periodic refill event
- A one-shot refill event

To process a refill event, the refill logic consists of a set of refill components [page 398].

Before Starting

Before setting up [page 401] the refill logic, you must be familiar with the following:

- Refill plan [page 421]

- Refill tree [page 397]
- Refill components [page 398]
- Refill context [page 398]
- Version change of refill logic [page 413]

Related Information

Customized Use of Mapping Tables in the Charges and Refill Logic [page 185]

1.26.1 Example of Refill Logic

The company TelcoServ sells the product "Walk & Talk", a triple play service with the following offer details:

- $150 Prepaid Card
- HD service
- Over 150 digital cable channels
- High-speed internet service
- Unlimited nationwide calling

Customers can choose to refill their prepaid card automatically when the balance of their prepaid account goes below a threshold defined according to the service plan which the customer has subscribed to. To meet this service, TelcoServ uses the refill logic in their offer.

This refill logic contains a recurring refill which once a day verifies that the balance of the prepaid account is not less than a defined threshold. If it is the case, the prepaid account is refilled according to the following options (refill policy):

- Option 1: If the balance is less than the defined threshold, the prepaid account is automatically refilled with the difference between the refill amount and the current amount of the balance. The prepaid account balance is refilled up to the refill amount.
- Option 2: If the balance is less than the defined threshold, the prepaid account is automatically refilled with the refill amount. The balance may exceed the refill amount.

The refill logic needs four parameters:

- A parameter to enable the refill (`THRESHOLD_REFILL_ENABLED`)
- A parameter to test the threshold of the refill (`THRESHOLD_AMOUNT`)
- A parameter to define the refill amount (`RECURRING_REFILL_AMOUNT`)
- A parameter to test the refill policy (`RECURRING_REFILL_POLICY`)

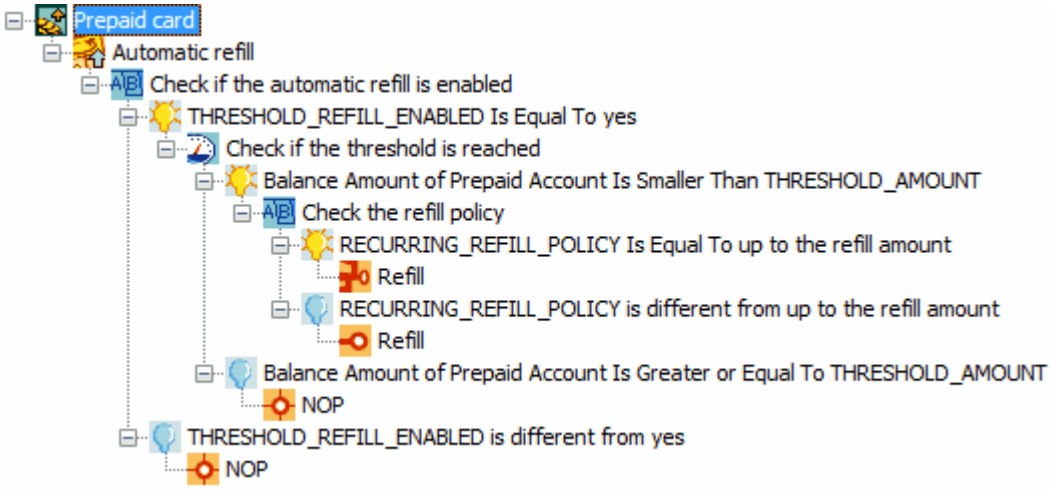
These parameters are defined according to the service plan which the customer has subscribed to.

In the refill tree:

- The first branch checks if the refill is enabled
- The second branch checks if the threshold is reached

- The third branch checks the refill policy

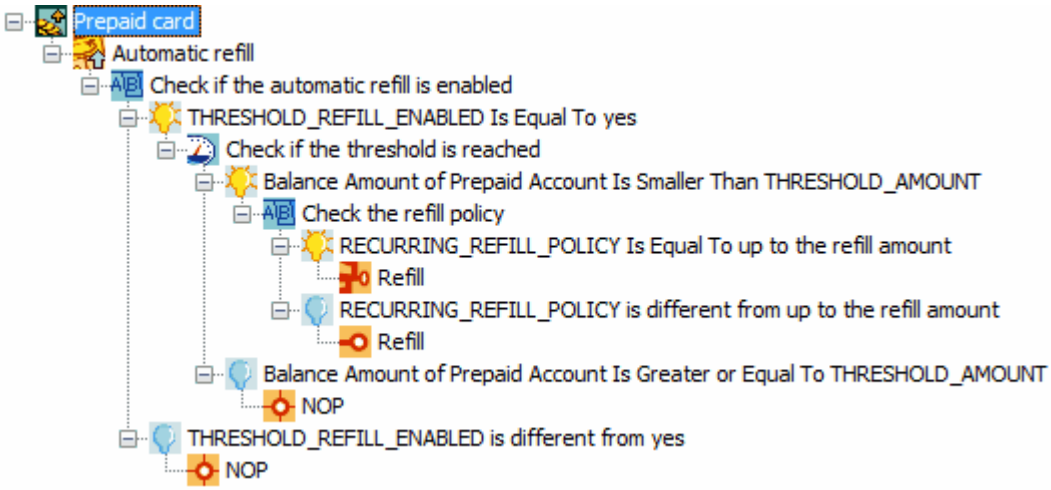
And the prepaid account is refilled or not depending on the test results.



1.26.2 Refill Tree

In your Core Tool user interface, the refill logic is graphically represented by a tree structure called a refill tree. A refill component represents each instruction inside the refill logic. This structure consists of a root from which the refill components create branches. At the end of each branch, a Function component describes a refill.

As shown in the illustration below, the tree contains the root  representing the refill logic and the branches for the different refill components [page 398]:



In addition to refill components, you can create the following items:

- Counters [page 297]
- Parameters [page 303]
- Pricing Macros [page 121]

- Translation Tables [page 132]
- Tier Tables [page 147]
- Mapping Tables [page 179]

Pricing macros, translation tables, tier tables, and mapping tables can be reused in any refill logic.

Test of the refill logic

Once your refill logic is complete, SAP Convergent Charging provides you with the Rating Trial Run [page 335] tool to test your refill logic.

1.26.3 Refill Context

The refill function of SAP Convergent Charging uses a set of data to complete its tasks at runtime. This data whose values vary over time is referred to as the refill context.

Data used to calculate a refill event comes directly from usage and is supplied by the properties of the received refill item or from the following sources:

- Counters
- Parameter values coming from:
  - The refill logic
  - The refill plan
  - The contract item of a provider contract
- Translation table [page 132], tier table [page 147], or mapping table [page 179] rows used in the refill logic

> **Note**
>
> When you are at the stage of adjusting and finalizing your refill logic, you can view the refill context in detail. You can also manipulate it in the GUI by performing a trial run [page 335] on the refill logic.

1.26.4 Refill Components

Refill components are calculation tools you use to create the . Together the components make up a refill tree [page 397] which computes the different credits (money, units, and so on) of a refill [page 290].

Each main branch of the refill logic begins with a Refill component that is linked to a refill activation method. There are three types of refill linked to three methods of activation:

Type of refill	Activation
 Usage	Each time a customer refills a prepaid account
 Recurring	Periodic Automatically triggered by SAP CC
 One-Shot	Once only Automatically triggered by SAP CC

Examples of activation

- Usage refill: Can be used each time a customer makes a refill on a prepaid account through a refill card or an ATM.
- Recurring refill: Can be used monthly to automatically add a fixed credit or to credit the balance of a prepaid account to a certain amount. Can be used daily to ensure a minimum balance amount.
- One-shot refill: Can be used when a bonus is charged on a prepaid account after a customer has just subscribed to a provider contract [page 573]

The values generated higher up a refill branch become available to all parts of the branch further toward the ends of the branch. When a property is generated on one refill branch, it cannot be used on another refill branch.

The components of a refill tree are grouped into five families depending on their main function:

- Refill components  always begin the main branches of a refill tree.
- Comparator  , Splitter  , and Operator  components help to calculate a refill.
- Function components  end each branch of a refill tree with a final calculated refill.

> ⓘ Note
>
> For more comprehensive detail on using each component, refer to the primary help relating to components here [page 671].

1.26.5 Working with the Refill Logic

Related Information

Creating the Refill Logic [page 400]

Opening the Existing Refill Logic [page 406]
Modifying the Refill Logic [page 407]
Setting Up the Refill Logic [page 401]
Saving the Refill Logic [page 404]
Deleting the Refill Logic [page 408]
Exporting the Refill Logic [page 408]
Importing the Refill Logic [page 408]
Setting Up the Tabs of the Refill Logic [page 401]

1.26.5.1 Creating the Refill Logic

Use one of these procedures to create the refill logic.

To create the refill logic

1. From the menu bar in the main window, select *File* > *New* > *Refill Logic*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the refill logic.
3. Click *OK*. The *New Refill Logic* window opens.
4. *Set up* the refill logic.
5. Verify the refill logic structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. *Save* the refill logic.

You can create the refill logic by opening the existing refill logic and copying it.

To create a refill logic from an existing refill logic

1. From the menu bar in the main window, select *File* > *Open* > *Refill Logic*. The *Open Refill Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill logic.
3. In the *Refill Logics* folder of the catalog of the owner, select the refill logic to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *refill logic* window opens.
5. Rename the refill logic in the *Name* field, and then *set up* the refill logic
6. Verify the refill logic structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. *Save* the refill logic.

Related Information

Setting Up the Refill Logic [page 401]
Saving the Refill Logic [page 404]

1.26.5.2 Setting Up the Refill Logic

Create [page 400], open [page 406], or modify [page 407] the refill logic before setting it up. The *Refill Logic* window must be open.

In a refill logic window, the refill tree consists of refill components (usage refill, recurring refill, one-shot refill) and other components (functions, comparators, and so on) using counters, parameters, pricing macros, and other objects which depend on the contents of the refill logic.

> → Remember
>
> Before modifying a price plan, it is strongly recommended to read the caution [page 299] about what you can do or not.

To set up the refill logic

1. In the right pane of the refill logic window, identify the refill logic. In the *Name* field, type the unique identifier of the refill logic. In the *Description* field, describe the function of the refill logic if necessary.
2. Set up the following tabs of the refill logic: Definition [page 402] tab, Parameters [page 403] tab, and Persistent Counters [page 404] tab.
3. Add [page 409] refill components to the refill logic, and then set up the refill components[51].
4. Add [page 410] functions, comparators, splitters, or operators components to the refill logic, and then set up the components[52].

> ⓘ Note
>
> You can remove [page 411], cut, copy, or paste components or sections of a refill tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.26.5.2.1 Setting Up the Tabs of the Refill Logic

[51] For more information about setting up components, refer to the primary help relating to components here [page 671]
[52] For more information about setting up components, refer to the primary help relating to components here [page 671]

Related Information

Refill Logic: Definition [page 402]
Refill Logic: Parameters [page 403]
Refill Logic: Persistent Counters [page 404]

1.26.5.2.1.1 Refill Logic: Definition

Create [page 400] or modify [page 407] the refill logic before setting it up [page 401]. In the right pane of the refill logic window, click the *Definition* tab to set it up.

Definition tab

The *Definition* tab allows you to define the currency of the refill logic and to select the refill item classes relevant for the configuration of *Usage Refill* components.

> ⓘ Note
>
> About currencies
>
> - Contrary to a charge plan [page 343], you cannot redefine the currency (fixed or multicurrency) at refill plan [page 421] level when customizing the refill logic.
> - The currency cannot be modified once the refill logic has been created.

Currency Definition: This area allows you to select the currency that you want to associate to the refill logic.

- *Fixed Currency*: Provides the list of currencies [page 81]. Select a currency.
- *Multicurrency*: Select this option if you do not want to associate a specific currency to the refill logic. This option allows you to create one only refill logic that can be reused in different refill plans. This option avoid creating the same refill logic many times and associating each refill logic to a different currency.

Refill Item Classes: Displays the refill item classes [page 387] used by the usage refills [page 398] configured in the refill logic. If necessary, click a refill item class to open it, and then modify [page 393] it.

Related Information

Setting Up the Refill Logic [page 401]

1.26.5.2.1.2 Refill Logic: Parameters

Create [page 400] or modify [page 407] the refill logic before setting it up [page 401]. In the right pane of the refill logic window, click the *Parameters* tab to set it up.

The *Parameters* tab allows you to add parameters to the refill logic.

Parameters tab

Fill in or change the *Parameters* tab.

 : Click to add a parameter to the refill logic.

 : Click to delete a parameter from the refill logic.

Name: Type the parameter name.

Description: Describe the function of the parameter if necessary.

Type: Provides the list of parameter types. Select one of the following types: Number, String, Date, Mapping Table ID, Mapping Table Key or Range Table ID [page 303].

Value: Type a numerical value, a string value, or a date to initialize the parameter depending on its type. For the types *Mapping Table ID* and *Range Table ID*, a window is displayed. Select the appropriate class and the mapping table [page 179] or range table [page 213].

Visibility: Select one of the following options:

- *Internal*: The parameters are only available inside the refill logic.
- *External*: The parameters can be modified in the related refill plan [page 421].

Synchronizing recurring rates with the billing cycles of SAP Convergent Invoicing

In your *SAP Solution* scenario, if your charge contains recurring refills [page 710], you may want to synchronize them with the billing cycles of SAP Convergent Invoicing. To synchronize these objects, you must create a dedicated parameter in the *Parameters* tab with the procedure described above. Then, you must:

- Set up the reference date of the recurring refill to this parameter (see the primary help of logic components [page 671])
- Link the previously created parameter [page 435] to the `SAP_BILL_CYLE_DATE` parameter of the refill plan

Related Information

Setting Up the Refill Logic [page 401]

1.26.5.2.1.3 Refill Logic: Persistent Counters

Create [page 400] or modify [page 407] the refill logic before setting it up [page 401]. In the right pane of the refill logic window, click the *Persistent Counters* tab to set it up.

The *Counters* tab allows you to create counters for the refill logic.

> ⓘ Note
>
> The counters created in the refill logic are available in the related refill plan [page 421] but cannot be modified.

Counters tab

Use the table and its icons to create counters.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

Name: Type the name of the counter.

Description: Describe the function of the counter if necessary.

Initial Value: Type a numerical value to initialize the counter.

Related Information

Setting Up the Refill Logic [page 401]

1.26.5.3 Saving the Refill Logic

Once you have created or modified the refill logic, and then verified its structure, you are ready to save it. You can save the refill logic to either the database or as an XML file on your hard disk.

> ⓘ Note
>
> After saving the refill logic, you can no longer modify:

- The name of the refill logic
- The currency definition

To save the new refill logic to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Refill Logic Creation* dialog box opens to inform you that the refill logic has been successfully saved.
4. Click *OK*.

To save a new version of an existing refill logic to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the refill logic has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the refill logic.
> 2. Click *Cancel* to come back to the refill logic. See later in this chapter how to save the refill logic as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the refill logic.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Refill Logic Modification* dialog box opens to inform you that the refill logic has been successfully created.
5. Click *OK*.

To save a refill logic as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the refill logic.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The refill logic is saved as an XML file in your selected directory.

1.26.5.4 Opening the Existing Refill Logic

You can open the existing refill logic:

- Saved in the back-end database
- Saved in an XML file

To open the existing refill logic saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Refill Logic* The *Open Refill Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill logic.
3. In the catalog of the owner, select the refill logic.
4. Click the *Open* button. The related window of the refill logic opens in display mode [page 46].

To open the existing refill logic saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.

4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.26.5.5 Modifying the Refill Logic

When you save the refill logic in the database, you can modify its versions (see the process below if necessary).

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

To modify the refill logic

1. Open [page 406] the refill logic.
2. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The refill logic may be reloaded if it has been modified since you have opened it.
3. Set up [page 401] the refill logic.
4. Verify the structure of the refill logic by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Save [page 404] the refill logic.

Related Information

Display/Edit Mode [page 46]

1.26.5.6 Deleting the Refill Logic

Use this procedure to delete the refill logic.

To delete a refill logic

1. From the menu bar in the main window, select *File* > *Delete* > *Refill Logic*. The *Delete Refill Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill logic.
3. In the *Refill Logics* folder of the catalog of the owner, select the refill logic.
4. Click *Delete*. A deletion dialog box asks if you want to delete the refill logic
5. Confirm the deletion of the refill logic by clicking *Yes*:
    1. If a dialog box confirms your choice, the refill logic has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the refill logic is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

1.26.5.7 Exporting the Refill Logic

You can export the refill logic to a directory of your hard disk, where it is saved as an XML file.

To export a refill logic

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill logic.
3. In the catalog of the owner, select the refill logic.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the refill logic.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.26.5.8 Importing the Refill Logic

You can import the refill logic in XML format from a directory of your hard disk.

To import a refill logic

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select ▶ *File* ▶ *Import* ▶. The *Open* dialog box opens.
3. Open the directory in which the refill logic is saved, and then select the file of the refill logic.
4. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of the refill logic
5. In the *Import* column, clear the checkboxes of objects you do not wish to import.
6. Click *Import*.

> ⓘ Note
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (▶ *File* ▶ *Export* ▶), rather than saved directly to a file (▶ *File* ▶ *Save* ▶ *In a File* ▶).

1.26.6 Using the Action Menu

Related Information

Adding a Refill Component [page 409]
Adding a Component [page 410]
Removing a Component [page 411]
Moving Up a Component [page 412]
Moving Down a Component [page 412]
Deleting the Versions of the Refill Logic [page 413]
Reading the Refill Logic [page 413]
Doing a Trial Run of the Refill Logic [page 414]

1.26.6.1 Adding a Refill Component

You can add one or more refill components to the refill tree.

The refill branch is a mandatory base for the refill logic. You cannot create the refill logic without a refill branch that contains at least one of the following refill components: 

Usage Refill,  Recurring Refill, or  One-Shot Refill.

To add and set up a refill component

1. In the left pane of the refill logic window, move the pointer over  Refills on the *Components* toolbar, and then click one of the following refill components:  Usage Refill,  Recurring Refill, or  One-Shot Refill.
2. In the *Name* field of the right pane, type the name of the refill component.
3. In the *Description* field of the right pane, describe the function of the refill component if necessary.
4. Set up the refill component[53].

> **Note**
>
> When a price plan contains several refill branches such as Usage Refills, Recurring Refills, and One-Shot Refills, the refill branches are automatically arranged in this order in the refill tree:
>
> -  Usage Refills
> -  One-Shot Refills
> -  Recurring Refills
>
> For example, if you add a new usage refill to the refill tree, it is automatically added after the last usage refill branch.

1.26.6.2 Adding a Component

You can add one or more components to the refill tree.

For adding a component to the refill tree

You build each branch of the refill tree by using several combinations of the following components:  Functions,  Comparators,  Splitters or  Operators. You must end each branch of the refill tree with a function component. Note that function components are the leaves of the tree.

> **Note**
>
> - Because you cannot insert a pricing macro [page 121] in a refill tree, you need to insert components such as a macro function or a macro operator in a refill tree to reuse an existing pricing macro.

[53] For more information about setting up refill components, refer to the primary help relating to components here [page 671].

- Because you cannot insert a translation table [page 132] in a refill tree, you need to insert a table operator component in a refill tree to reuse an existing translation table.
- Because you cannot insert a tier table [page 147] in a refill tree, you need to insert a table operator component in a refill tree to reuse an existing tier table.

To add and set up a component

1. In the refill tree, click the component under which you want to add a component.
2. On the *Components* toolbar, click the component that you need to add to the refill tree. You can select a component that belongs to one of these families:  Functions,  Comparators,  Splitters or  Operators.
3. In the *Name* field of the right pane, type the name of the component.
4. In the *Description* field, describe the function of the component if necessary.
5. Set up the component[54].

> **Note**
>
> If necessary, you can change the order of components. Place the cursor on the component to move, right-click to open the shortcut menu, and then select *Move Up Component* or *Move Down Component*.

1.26.6.3 Removing a Component

Use this procedure to remove a component from a tree.

To remove a component from a tree

1. In the tree, click the component that you want to remove.
2. From the menu bar in the main window, select *Action* > *Remove Component*. The component is removed.

> **Note**
>
> - To remove an entire tree branch, right-click a component and select *Remove Component*.
> - Some components in the middle of a tree branch allow you to remove them and all other components attached to them towards the end of the branch. These branch sections can be removed only when the *Remove Component* command is available.

[54] For more information about setting up components, refer to the primary help relating to components here [page 671].

- You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.26.6.4 Moving Up a Component

Use this procedure to move up a component.

To move up a component

1. In the tree, click the component that you want to move up.
2. From the menu bar in the main window, select *Action* > *Move Up Component*.

> **Note**
>
> You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.26.6.5 Moving Down a Component

Use this procedure to move down a component.

To move down a component

1. In the tree, click the component that you want to move down.
2. From the menu bar in the main window, select *Action* > *Move Down Component*.

> **Note**
>
> You can remove, cut, copy, or paste components or sections of a tree by right-clicking a component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.26.6.6 Changing the Version of the Refill Logic

Refer to the process below:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.26.6.7 Deleting the Versions of the Refill Logic

The procedure is the same as Deleting the Versions of a Catalog Object [page 118].

1.26.6.8 Reading the Refill Logic

Once you have created the refill logic, you can generate a descriptive representation of it. This process is called reading a refill logic.

To read a refill logic

1. Open [page 406] the refill logic..
2. On the toolbar in the main window, click the *Read Refill Logic* icon  . The *Read Refill Logic* dialog box opens.
3. Check the refill logic description for accuracy, and then click *Close*.

1.26.6.9 Doing a Trial Run of the Refill Logic

Once you have built and verified the refill logic, you can test it to ensure that the correct results have been achieved. To do this, you use the *Rating Trial Run* tool which is a testing tool provided with SAP Convergent Charging.

To test the refill logic

1. Make sure that the refill logic that you want to test is open; otherwise, open [page 406] the refill logic.
2. Open [page 339] the *Rating Trial Run* tool.
3. Set up [page 340] the *Rating Trial Run* tool.
4. Run [page 340] the *Rating Trial Run* tool.

1.27 Refill Record Classes

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

> ⚠ Caution
>
> Read the caution [page 263] about the billable item mapping before deleting a refill record class.

The refilling process [page 421] creates business transactions called refill records. With refill record classes, you can retrieve and configure the information included in the refill records. Through the mapping of a billable item class [page 263], SAP Convergent Charging then sends this information to a billing system such as an SAP ERP/FI-CA system or any other external system.

> ⓘ Note
>
> A refill record class is similar to a charged item class [page 254] that is used by a charge plan.

In SAP CC Core Tool, a table represents a refill record class. Each row in the table represents a property whose type can be a string, a number, a date, or a boolean. The table allows you to configure how the system generates these values.

You can create and use refill record classes for:

- An SAP ERP/FI-CA system
- Any other external system

If you use an external system

If you want to send refill record classes to an external billing system or any other external system, you must create [page 416] the refill record classes from scratch, and then set them up [page 416].

Setting up your refill record classes

Because you know the contents of the transactions resulting from the refilling processes, you can define and set up the data of your refill record classes accordingly to retrieve information from the transactions. Your system billing or another system receives and uses this information.

If you use an SAP ERP/FI-CA system

SAP Convergent Charging is integrated with SAP ERP/FI-CA for managing high-volume data. SAP Convergent Charging can:

- Produce billable items asynchronously from the refill records
- Send the billable items to one ore more SAP ERP/FI-CA system

Setting up your refill record classes

You must map refill record classes onto billable item classes SAP Convergent Charging generates automatically the billable item mapping. To do that, you create a refill record class from the mapping of a billable item class when you create [page 266] the billable item mapping. Refer to Billable Item Mapping [page 263] for more information when using an SAP ERP/FI-CA system.

> **Note**
>
> For more information, consult the SAP CC Administration Guide (Refer to Configuration for SAP ERP/FI-CA systems).

1.27.1 Working with Refill Record Classes

Related Information

Creating a Refill Record Class [page 416]
Opening an Existing Refill Record Class [page 418]
Modifying a Refill Record Class [page 419]
Setting Up a Refill Record Class [page 416]
Deleting a Refill Record Class [page 420]
Saving a Refill Record Class [page 417]

1.27.1.1 Creating a Refill Record Class

Use one of these procedures to create a refill record class.

To create a refill record class

1. From the menu bar in the main window, select *File* > *New* > *Refill Record Class*. The *New Refill Record Class* window opens.
2. *Set up* the refill record class.
3. Verify the refill record class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. *Save* the refill record class.

You can create a refill record class by opening an existing refill record class and copying it.

To create a refill record class from an existing refill record class

1. From the menu bar in the main window, select *File* > *Open* > *Refill Record Class*. The *Choose a Refill Record Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill record class.
3. In the *Refill Record Classes* folder of the catalog of the owner, select the refill record class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding refill record class window opens.
5. Rename the refill record class in the *Name* field, and then *set up* the refill record class.
6. Verify the refill record class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. *Save* the refill record class.

Related Information

Setting Up a Refill Record Class [page 416]
Saving a Refill Record Class [page 417]

1.27.1.2 Setting Up a Refill Record Class

Create [page 416], open [page 418], or modify [page 419] a refill record class before setting it up. A refill record class window must be open.

In the graphical user interface of SAP Convergent Charging, a refill record class is represented by a table that consists of rows which represent the properties of the refill record class.

To set up a refill record class, you first identify it, and then you add properties to the table, or you modify the properties of the table.

Name: Type the unique identifier of the refill record class.

Description: Describe the function of the refill record class.

 : Click to add a new row to the table.

 : Click to delete a row from the table.

Properties: Fill in or modify the columns of the table.

> ⓘ Note
>
> If you set up a boolean node with a source property, the value is converted by SAP Convergent Charging during the charging process. It converts the strings "0" and "false" and the number 0 to FALSE. It converts the strings "1" and "true" and the number 1 to TRUE. Otherwise the charging process fails because it cannot convert the source value.

- *Name*: Type the identifier of the property.
- *Type*: Select the type of the property (boolean, date, number, or string).
- *Value*: Select one of the following:
  - *Type a value*: Allows you to type any constant except an empty string (no space and no character).
  - *Undefined*: Allows you to select a system property that has no assigned value. Selecting this property only means that you recommend changing this value at refill plan level.
  - *Empty*: Allows you to select a system property that has no assigned value. Selecting this property only means that you do not recommend changing this value at refill plan level.
  - Any other property of the list.
- *Description*: Describe the function of the property.

1.27.1.3 Saving a Refill Record Class

Once you have created a refill record class and then verified its configuration, you are ready to save it. You can save the refill record class to either the database or as an XML file on your hard disk.

> ⓘ Note
>
> After saving a refill record class, you can no longer:
>
> - Modify the name of the refill record class
> - Remove a property from the table
> - Change the type of a property

To save a new refill record class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the Catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill record class.
3. Click *Save*. The *Refill Record Class Creation* dialog box opens to inform you that the refill record class has been successfully created.
4. Click *OK*.

To save an existing refill record class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Refill Record Class Modification* dialog box opens to inform you that the refill record class has been successfully modified, click *OK*. The new version of the refill record class is saved as the only current version.
3. If a *Warning* dialog box opens to inform you that the refill record class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the refill record class.
   - Click *Cancel* to come back to the refill record class. See below how to save the refill record class as an XML file.

To save a refill record class as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the refill record class.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The refill record class is saved as an XML file in your selected directory.

1.27.1.4 Opening an Existing Refill Record Class

You can open:

- An existing refill record class saved in the database
- An existing refill record class saved in an XML file

To open an existing refill record class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Refill Record Class*. The *Open Refill Record Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill record class.
3. In the *Refill Record Classes* folder of the catalog of the owner, select the refill record class.
4. Click the *Open* button. The *Refill Record Class* window opens in display mode.

To open an existing refill record class saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Display/Edit Mode [page 46]

1.27.1.5 Modifying a Refill Record Class

When modifying a refill record class, you can:

- Add new properties to the refill record class
- Assign a new value to an existing property

The modified refill record class replaces the previous version in the database.

> **Note**
>
> You cannot:
>
> - Modify the name of the refill record class
> - Remove a property from the table
> - Change the type of a property

To modify a refill record class

1. Open [page 418] the refill record class.
2. Set the edit mode [page 46] by clicking the *Edit Mode* icon on the main toolbar. The refill record class may be reloaded if it has been modified since you have opened it.
3. Set up [page 416] the refill record class.
4. Verify the structure of a refill record class by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Save [page 417] the refill record class.

1.27.1.6 Deleting a Refill Record Class

> **Caution**
>
> - Read the caution [page 263] about the billable item mapping before deleting a refill record class.

To delete a refill record class

1. From the menu bar in the main window, select *File* > *Delete* > *Refill Record Class*. The *Delete Refill Record Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill record class.
3. In the *Refill Record Classes* folder of the catalog of the owner, select the refill record class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the refill record class
5. Confirm the deletion of the refill record class by clicking *Yes*:
   1. If a dialog box confirms your choice, the refill record class has been successfully deleted.
   2. If a *Warning* dialog box opens to inform you that the refill record class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48]for more information.

1.28 Refill Plans

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

In SAP Convergent Charging 2023, the refilling business scenario enables to credit the prepaid account of a customer. For example, the customer can refill a prepaid account manually through a refill card or an ATM. In this case, a system such as a voucher management system sends a refill item to SAP Convergent Charging which performs the refill plan depending on the information contained in the refill item. SAP Convergent Charging can also refill a prepaid account automatically, for example, when the balance of a prepaid account goes below a defined threshold or if the refill of a prepaid account depends on a periodic event.

A refill plan defines the refilling conditions to credit the prepaid account of a customer. SAP CRM or an external provisioning system first combines a refill plan with charge plans, and then SAP CRM or the external system activates the refill plan in the provider contract [page 573] of the customer.

The refill plan consists of:

- The refill logic [page 395]
- A set of parameters to configure the refill logic (for example, the frequency of a periodic refill)
- A release status [page 424] which can be open, released, or obsolete

> ⓘ Note
>
> Tax Settings [page 523] do not apply to refill plans.

Before Starting

Before setting up [page 423] a refill plan, you must be familiar with the following:

- Refill item class [page 387]
- Refill Logic [page 395]
- Refill record class [page 414]
- Version change [page 440] of a refill plan
- Preview of a refill plan [page 439]

Transport

- Export [page 432]
- Import [page 432]

1.28.1 Working with Refill Plans

Related Information

Creating a Refill Plan [page 422]
Opening an Existing Refill Plan [page 430]
Modifying a Refill Plan [page 431]
Setting up a Refill Plan [page 423]
Deleting a Refill Plan [page 432]
Saving a Refill Plan [page 429]
Exporting a Refill Plan [page 432]
Importing a Refill Plan [page 432]
Setting Up the Tabs of a Refill Plan [page 423]

1.28.1.1 Creating a Refill Plan

Use one of these procedures to create a refill plan [page 421].

To create a refill plan

1. From the menu bar in the main window, select *File* > *New* > *Refill Plan*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the refill plan.
3. Click *OK*. The *New Refill Plan* window opens.
4. *Set up* the refill plan.
5. Verify the refill plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Check the *Preview* of the refill plan.
7. *Save* the refill plan.

You can create a refill plan by opening an existing refill plan and copying it.

To create a refill plan from an existing refill plan

1. From the menu bar in the main window, select *File* > *Open* > *Refill Plan*. The *Choose Refill Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill plan.
3. In the *Refill Plans* folder of the catalog of the owner, select the refill plan.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Refill Plan* window opens.
5. Rename the refill plan in the *Name* field, and then *Set up* the refill plan as necessary.
6. Verify the refill plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Check the *Preview* of the refill plan.
8. *Save* the refill plan.

Related Information

Setting up a Refill Plan [page 423]
Previewing a Refill Plan [page 439]
Saving a Refill Plan [page 429]

1.28.1.2 Setting up a Refill Plan

Create [page 422], open [page 430], or modify [page 431] a refill plan before setting it up. A refill plan window must be open.

To set up a refill plan

1. In the right pane of the refill plan window, identify the refill plan. In the *Name* field, type the unique identifier of the refill plan. You cannot modify this field after the refill plan is saved.
2. Set up the following tabs of the refill plan: Definition [page 424], Counters [page 425], Parameters [page 425], Additional Information [page 427], Account Assignments [page 428], Technical Data [page 427].
3. Add [page 439] the refill logic, and then set up [page 433] the refill logic.
4. If necessary, remove [page 440] the refill logic.

1.28.1.2.1 Setting Up the Tabs of a Refill Plan

Related Information

Refill Plan: Definition [page 424]
Refill Plan: Counters [page 425]
Refill Plan: Parameters [page 425]
Refill Plan: Additional Information [page 427]
Refill Plan: Technical Data [page 427]
Refill Plan: Account Assignments (Expected) [page 428]

1.28.1.2.1.1 Refill Plan: Definition

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Definition* tab.

The *Definition* tab allows you to set up the status of the refill plan and gives a detailed description of the refill plan.

Definition tab

Fill in the *Definition* tab.

Status: Provides the list of release statuses. Select one of the following statuses:

- *Open*: Select this status if you are working on the refill plan. This status allows you to modify the refill plan and save it in the database. However, the refill plan cannot be used by the SAP ERP/FI-CA system or by an external provisioning system. You cannot save several versions of the refill plan.
- *Released*: Select this status when the refill plan is ready to be in production. The refill plan can be used by the SAP ERP/FI-CA system or an external provisioning system. When you modify and save the refill plan in the database, the old versions of the refill plan are always stored in the database. According to this status, you are only allowed to do the following:
  - Change the status of counters [page 434] or parameters [page 435] of the refill logic depending on the counters [page 425] and parameters [page 425] declared at refill plan level
  - Change the refill record class and the refill record mapping (Refill Records [page 436] tab)
- *Obsolete*: Select this status when you want to archive the refill plan. The refill plan can be used by the SAP ERP/FI-CA system or an external provisioning system. You are allowed to do the same changes as with the *Released* status. When you modify and save the refill plan in the database, the old versions of the refill plan are always stored in the database. The refill plan can be subscribed in an existing provider contract [page 573] but cannot be subscribed in a new provider contract.

> **Note**
>
> - When you create a refill plan, its status is automatically set to *Open*. You must save the refill plan once before changing its status.
> - You can set the status to *Released* only if the current status of the refill plan is *Open*.

Description: Describe the refill plan by typing a long text if necessary. This field is mandatory. The SAP ERP/ FI-CA system or an external provisioning system can display the description. Note that you cannot change the description of the refill plan if its status is *Released* or *Obsolete*.

1.28.1.2.1.2 Refill Plan: Counters

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Counters* tab.

The *Counters* tab allows you to share counters which are declared in the counter dictionary [page 86]. After you have added a counter to the table, you can initialize the counter and link it to a counter of the refill logic [page 434].

> → Remember
>
> A counter declared in the counter dictionary can be used in several objects. This means that a counter used in an object can be changed in another object. For example, the counter dictionary includes the counter `FREE_SMS`. If you share this counter both in the refill plan `RefillWS` and in the charge plan `ChargeWS`, the value of the counter `FREE_SMS` will vary over time depending on its use in the related refill and charge plans.

Counters tab

Use the table and its icons to share counters declared in the counter dictionary. You cannot change the name of the counter and its description.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

Name: Displays the name of the counter.

Description: Displays the function of the counter.

> ⓘ Note
>
> The initial value of the counter is automatically set to zero.

1.28.1.2.1.3 Refill Plan: Parameters

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Parameters* tab.

The *Parameters* tab allows you to create parameters that can be shared with the refill logic [page 435]. Depending on some conditions, you can redefine the parameter values when configuring provider contracts.

> **Note**
>
> All parameters created at refill plan level must be linked to parameters of the refill logic.

Parameters tab

Use the table and its icons to create shared parameters.

 : Click to add a parameter row to the table.

 : Click to delete a parameter row from the table.

Name: Type the name of the parameter.

Type: Provides the list of parameter types. Select one of the following types: Number, String, Date, Mapping Table ID, Mapping Table Key or Range Table ID [page 303].

Condition: Provides the list of options for modifying the parameter value at provider contract level. Select one of the following options:: Provides the list of options for modifying the parameter value at provider contract level. Select one of the following options:

- *Optional*: Enables the shared parameter to be modified later in the provider contract of an interfaced CRM or a provisioning system. At provider contract level, you can redefine the value of this parameter in the CRM system but the default value is the value defined in this refill plan.
- *Mandatory*: Enables the shared parameter to be modified later in the provider contract of an interfaced CRM application or a provisioning system. At provider contract level, you must enter the value of this parameter in the CRM system to define the parameter [page 624] in the contract item activating this refill plan.

Description: Describe the function of the parameter. The description is mandatory.

Default Value: Type the value of the parameter. If you have selected *Mandatory* as the condition, you cannot type a value because this value is only mandatory at provider contract level (see the Mandatory option).

Synchronizing Recurring Refills with the Billing Cycles of SAP Convergent Invoicing

In your *SAP Solution* scenario, if SAP Convergent Invoicing is installed in your SAP system landscape, each refill plan contains a parameter called `SAP_BILL_CYCLE_DATE`. This parameter can be used for synchronizing the recurring refills of refill logic objects with the billing cycles of SAP Convergent Invoicing. You cannot modify or delete this parameter. If you want to synchronize the billing cycles of SAP Convergent Invoicing and the SAP CC recurring refills, you must link this parameter [page 370] with a parameter of the refill logic containing the recurring refill.

1.28.1.2.1.4 Refill Plan: Additional Information

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Additional Information* tab.

The *Additional Information* tab allows you to create information that is only descriptive and not reused by SAP Convergent Charging at any stage of the process. For example, you can create a table with the author's name of the refill plan, the author's code, the creation date, and another information. If necessary, you can retrieve this information by using an external system.

Additional Information tab

Use the table and its icons to create additional information.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: Number, String, or Date.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the SAP ERP/ FI-CA system or an external provisioning system.

1.28.1.2.1.5 Refill Plan: Technical Data

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Technical Data* tab.

The *Technical Data* tab allows you to define the technical identifier under which a customer can make a refill. If you provide different refill methods or services, you can use the same identifier.

SAP CRM system or the provisioning system must specify the identifier of the customer at provider contract level.

> ⓘ Note
>
> In most cases, you do not need to set up technical data in a refill plan.

> ✤ Example
>
> Examples of technical data: SMS, voice, data.

Technical Data tab

Use the table and its icons to create technical data.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in or modify each row that you add to the table.

User Technical Identifier Name: Type the name of the technical data. This name must be unique in the table. This name is visible from the SAP CRM system or the external provisioning system.

Description: Type the description of the technical data. This description is visible from the SAP CRM system or an external provisioning system.

1.28.1.2.1.6 Refill Plan: Account Assignments (Expected)

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Technical Data* tab.

The *Account Assignments* tab allows you to declare the unique prepaid account that is expected in a provider contract [page 573] that subscribes to the refill plan. Through the provider contract, the SAP CRM system[55] fills in the account assignment with one prepaid account of a subscriber account [page 511] so that the activated refill plan can charge this account. The *Account Assignments* tab is also used to link the refill logic to the refill plan (refer to the refill logic account assignments [page 437]).

> ⓘ Note
>
> - For a refill plan, you must declare one and only one expected account assignment for a prepaid account.
> - For a charge plan, you can declare several expected account assignments for both prepaid and postpaid accounts.

Tab: Account Assignments

Use the configuration table and its icons to create a prepaid account assignment.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in the row that you add to the table.

[55] Or an external provisioning system. In this case, the assignment mechanism may be different.

Name: Type the name of the account assignment which is expected in the provider contract. This name is visible from the SAP CRM system or an external provisioning system.

Type: Displays that the expected account is prepaid. You cannot change the type of the account, and only a prepaid account is used for the refill plan.

Description: Type the description of the account assignment. This description is visible from the SAP CRM system or an external provisioning system.

1.28.1.3 Saving a Refill Plan

Once you have created or modified a refill plan, and then verified its structure, you are ready to save it. You can save the refill plan to either the database or as an XML file on your hard disk.

To save a new refill plan to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Refill Plan Creation* dialog box opens to inform you that the refill plan has been successfully saved.
4. Click *OK*.

To save a new version of an existing refill plan to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the refill plan has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the refill plan.
> 2. Click *Cancel* to come back to the refill plan. See later in this chapter how to save the refill plan as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the refill plan.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No*

End Date represents an undetermined date in the future. A message informs you when you overwrite versions.

- *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.

4. Click *Save*. The *Refill Plan Modification* dialog box opens to inform you that the refill plan has been successfully created.
5. Click *OK*.

To save a refill plan as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the refill plan.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The refill plan is saved as an XML file in your selected directory.

1.28.1.4 Opening an Existing Refill Plan

You can open the existing refill plan:

- Saved in the back-end database
- Saved in an XML file

To open an existing refill plan saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Refill Plan* The *Open Refill Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill plan.
3. In the catalog of the owner, select the refill plan.
4. Click the *Open* button. The related window of the refill plan opens in display mode [page 46].

To open the existing refill plan saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.28.1.5 Modifying a Refill Plan

When modifying a refill plan, you can only add a new property to it.

> **Note**
>
> - You cannot modify the name of the refill plan
> - You can modify and save different versions of the refill plan if its status is set to Released [page 424]

To modify a refill plan

1. Open [page 430] the refill plan
2. Set the edit mode [page 46] by clicking the *Edit Mode* icon on the main toolbar. The refill plan may be reloaded if it has been modified since you have opened it.
3. Set up [page 423] the refill plan
4. Verify the structure of the refill plan by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Check the preview [page 439] of the refill plan
6. Save [page 429] the refill plan

1.28.1.6 Deleting a Refill Plan

To delete a refill plan

1. From the menu bar in the main window, select *File* > *Delete* > *Refill Plan*. The *Delete Refill Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill plan.
3. In the *Refill Plans* folder of the catalog of the owner, select the refill plan.
4. Click *Delete*. A deletion dialog box asks if you want to delete the refill plan
5. Confirm the deletion of the refill plan by clicking *Yes*:
    1. If a dialog box confirms your choice, the refill plan has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the refill plan is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

1.28.1.7 Exporting a Refill Plan

You can export refill plans to a directory of your hard disk, where they are saved as XML files.

To export a refill plan

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the refill plan.
3. In the catalog of the owner, select the refill plan.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the refill plan.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.28.1.8 Importing a Refill Plan

You can import refill plans in XML format from a directory of your hard disk.

To import a refill plan

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the refill plan is saved, and then select the refill plan file.
4. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a refill plan.
5. In the *Import* column, clear the checkboxes of objects you do not want to import.
6. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.28.2 Working with the Refill Logic Customized in a Refill Plan

Related Information

Setting Up the Refill Logic in a Refill Plan [page 433]
Setting Up the Tabs of the Customized Refill Logic [page 434]

1.28.2.1 Setting Up the Refill Logic in a Refill Plan

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. A *Refill Plan* window must be open.

To set up the refill logic

1. In the right pane of the refill plan window, identify the refill logic. In the *Name* field, type the unique identifier of the refill logic. You cannot modify this field after the refill logic is saved.

2. Set up the following tabs of the refill logic: Definition [page 434], Counters [page 434], Parameters [page 435], Refill Records [page 436], Account Assignments [page 437], Technical Data [page 438].

1.28.2.2 Setting Up the Tabs of the Customized Refill Logic

Related Information

Customized Refill Logic: Definition [page 434]
Customized Refill Logic: Counters [page 434]
Customized Refill Logic: Parameters [page 435]
Customized Refill Logic: Refill Records [page 436]
Customized Refill Logic: Account Assignments [page 437]
Customized Refill Logic: Technical Data [page 438]

1.28.2.2.1 Customized Refill Logic: Definition

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Definition* tab.

The *Definition* tab allows you to modify the refill logic if necessary.

Definition tab

Refill Logic: Displays the name of the refill logic used in the refill plan. Click  to display or modify the refill logic.

1.28.2.2.2 Customized Refill Logic: Counters

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Counters* tab.

The *Counters* tab allows you to redefine the counters of the refill logic by linking them to the counters of the refill plan [page 425]. Do not forget that the counters of the refill plan are counters declared in the counter dictionary [page 86].

Counters tab

Redefine the counters of the refill logic, if necessary.

Name: Displays the name of the counter.

Status: Provides the list of counter statuses. Select one of the following statuses:

- *Not Redefined*: Select this status if you do not want to link the counter to a counter of the refill plan
- *Linked*: Select this status if you want to link the counter to a counter of the refill plan

Initial Value: Provides the list of the counters of the refill plan. Select the counter that you want to link to the counter of the refill logic. The value of the counter is now the same as that of the counter defined in the refill plan.

1.28.2.2.3 Customized Refill Logic: Parameters

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Parameters* tab.

On the *Parameters* tab, you redefine the parameters of the refill logic by changing their values or by linking them to the parameters of the refill plan.

Parameters tab

Redefine the parameters of the refill logic if necessary.

Name: Displays the name of the parameter.

Type: Displays the parameter type.

Status: Provides the list of the parameter status. Select one of the following status:

- *Not Redefined*: Select this status if you do not want to link the parameter to a parameter of the refill plan
- *Redefined*: Select this status if you want to redefine the parameter of the refill logic
- *Linked*: Select this status if you want to link the parameter to a parameter of the refill plan

Value: Type a value or select a parameter of the refill plan to redefine the parameter. This option is not available for the status *Not Redefined*.

Synchronizing recurring refills with the billing cycles of SAP Convergent Invoicing

If you want to synchronize the recurring refills of refill logic objects with the billing cycles of SAP Convergent Invoicing, you must link a dedicated parameter with the `SAP_BILL_CYLE_DATE` parameter of the refill plan.

As a prerequisite, you must:

- Create a dedicated parameter [page 403] when setting up the refill logic containing the recurring rate
- Set the reference date of the recurring refill to this parameter (see the primary help of logic components [page 671])
- Link this parameter with the `SAP_BILL_CYLE_DATE` parameter of the refill plan

In the *Parameters* tab, you must then link the previously created parameter to the `SAP_BILL_CYLE_DATE` parameter of the refill plan.

1.28.2.2.4 Customized Refill Logic: Refill Records

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Refill Records* tab.

> → Remember
>
> Do not forget that you can create a refill record class from:
>
> - Scratch [page 414] if you use an external billing system or any other external system.
> - The billable item mapping [page 254] if you work with the SAP ERP/FI-CA

On the *Refill Records* tab, you can configure the generation of refill records by SAP CC Core Tool. To set up the refill record mapping, you can:

- Select the refill record class [page 414] used as the base for the refill record mapping system. In this case, when creating a refill record class, you must select a specific type.
- Change the values of the properties to change the contents of the generated data
- Add to or remove properties from the refill record mapping to change the structure of the refill records generated by SAP CC

Refill Records tab

Modify the *Refill Record* tab if necessary.

Refill Record Class: Provides the list of the refill records. Select a refill record if you want to replace that of the refill plan. Click  to open the refill record, and then modify [page 419] the refill record if necessary.

Refill Record Mapping system. In this case, when creating a refill record: Displays the list of refill record properties and their associated property or value. You can set up each refill record property with properties. To display less or more columns, click  , and then select the columns you want to display or not.

- *Charged Item Field*: Displays the field names of the charged item class.
- *Property / Value*: Allows you to change the values or properties of the fields depending on the following:
  - If a property or value is dimmed, it is not recommended changing this property by another property or value.

- If a property or value is not dimmed, it is recommended changing this value by another property or value.

- *Default Value*: Displays the field values defined at class level [page 416].
- *Redefine*: Allows you to redefine any value of the *Property / Value* column. Refer to the recommendations listed above and the note below.

> **Note**
>
> - It is not recommended changing a property or value that is dimmed. For more information about the *Undefined* and *Empty* properties, consult this topic [page 416].
> - If you set up a boolean field, the value is converted by SAP CC during the charging process. Strings "0" and "false" and number 0 are converted to FALSE. Strings "1" and "true" and number 1 are converted to TRUE; otherwise, the charging process fails because it cannot convert the value.

Integration

You need to specifically set up the refill record generation in the integrated end-to-end scenario (with SAP CRM and SAP ERP/FI-CA from SAP Business Suite)

> **Note**
>
> Consult the SAP CCAdministration Guide about the last recommendations for configuring master data in SAP Convergent Charging.

> **Example**
>
> A refill record mapping contains the following items:
>
> - CRDID (Source: CRDID user defined property of the refill item class)
> - CNTUI1 (Source: computed by the refill logic)
>
> SAP Convergent Charging generates billable items that include a CRDID field filled in with the value of the CRDID field from the received refill items.
>
> The billable items include a field (CNTUI1) computed by SAP CC and calculated according to the appropriate refill logic.

1.28.2.2.5 Customized Refill Logic: Account Assignments

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Account Assignments* tab.

The *Account Assignments* tab allows you to link the default charging reference of the refill logic to the expected account assignment [page 428] declared in the refill plan.

Account Assignments tab

Fill in or modify the *Account Assignments* tab.

Charging Reference: Displays "Default" because the refill logic contains one and only one charging reference.

Type: Displays "Prepaid" which is the default type of the charging reference.

Linked Value: Provides the name of the account assignment of the refill plan. Select this name.

1.28.2.2.6 Customized Refill Logic: Technical Data

Create [page 422] or modify [page 431] a refill plan before setting it up [page 423]. In the right pane of the *Refill Plan* window, open the *Technical Data* tab.

The *Technical Data* tab allows you to:

- Set up the identifiers of the service (refill from an ATM, refill from a call center) used by customers to refill their prepaid account [page 514].
- Link a service identifier of the refill logic to a user technical identifier of the refill plan. For more information, refer to the Refill Plan: Technical Data [page 427] tab.

> ⓘ Note
>
> In most of the cases, you do not need to set up technical data in the refill logic.

> ✤ Example
>
> Examples of technical data: Refill, refill from an ATM, or refill from a call center.

Technical Data tab

Use the table and its icons to create technical data.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in or modify each row that you add to the table.

Service Identifier: Type the identifier of the service. This identifier must be unique in the table.

User Technical Identifier: Type the description of the technical data. This description is visible from the SAP CRM system or an external provisioning system.

1.28.3 Using the Action Menu

Related Information

Previewing a Refill Plan [page 439]
Adding an Item [page 439]
Removing an Item [page 440]
Changing the Version of a Refill Plan [page 440]
Deleting the Version of a Refill Plan [page 441]

1.28.3.1 Previewing a Refill Plan

This procedure allows you to preview a refill plan and check its signature that will be seen by SAP CRM or by another provisioning system.

To perform this procedure, you must open [page 430] a refill plan.

To preview a refill plan

1. From the menu bar in the main window, select *Action* > *Preview*, or click the *Preview* icon  on the toolbar in the main window.
2. The *Preview* window opens and displays a summary of the counters, parameters, account assignments, and technical data that will be seen by the external system. Read and check this summary.
3. Click *Close*.

> **Note**
>
> The *Preview* window displays only the counters and parameters which are assigned the external visibility option [page 403].

1.28.3.2 Adding an Item

This procedure allows you to add a refill logic object to the refill plan. To perform this procedure, you must create [page 422] a refill plan.

To add the refill logic to a refill plan

1. From the menu bar in the main window, select *Action* > *Add Item*, or click the icon  on the toolbar in the main window. The *Choose the Refill Logic* dialog box opens.
2. In the *Refill Logic* folder, select the refill logic.
3. Click *OK*. The refill logic is added to the refill plan.
4. Select the refill logic  in the refill plan, and then set up [page 433] the tabs of the refill logic according to your requirements.

1.28.3.3 Removing an Item

These procedures allow you to remove the refill logic from a refill plan.

To remove the refill logic from a refill plan

1. In the *Refill Plan* window, right-click the refill logic.
2. Select *Remove*. The refill logic is removed from the refill plan.

or

1. In the *Refill Plan* window, right-click the refill logic.
2. From the menu bar in the main window, select *Action* > *Remove Item*, or click the icon  on the toolbar in the main window.
3. Click *OK*. The refill logic is removed from the refill plan.

1.28.3.4 Changing the Version of a Refill Plan

The procedure is the same as for changing the version of an object:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.28.3.5 Deleting the Version of a Refill Plan

The procedure is the same as for deleting the versions of a catalog object [page 118].

1.29 Allowance Event Classes

An *allowance event class* in SAP Convergent Charging (SAP CC) is a pricing element that **defines and describes the data structure of the allowance events [page 442] with a set of properties**. To each property in the class corresponds a data field in the calculated allowance events.

The allowance event class is similar to the chargeable item class [page 241] or the refill item class [page 387]. However, the allowance event class has no default properties but only custom properties (aka user-defined properties).

> **Note**
>
> The **name of the allowance event class determines the type of the allowance events** sent by the execution of the *Allowance Event Sender* component (see the example below) that you configure in the allowance logic.

Properties

A user-defined property is defined by:

- A *Name*
- A *Type* (character string, number, or date/time)
- And a *Description*

In the Core Tool user interface, the *Category* is not editable and is shown as a reminder.

Use

An allowance event class is used when you want to use the following logic components:

- Allowance Event Sender [page 777]
- Event-Based Trigger [page 717]

An allowance event class belongs to a pricing catalog [page 109] for the subscription- and usage-based services or products to monetize.

> **Example**
>
> The allowance event class named `ALLOWANCE_DEBIT` specifies the allowance events relating to the debit of an allowance in its provider contract. This allowance event class defines the following properties:
>
> - A number type property named `SERVICE_QTY`
> - A date type property named `SERVICE_TYPE`
>
> At runtime, the system calculates the allowance events that are based on this class. Each allowance event includes the fields `SERVICE_QTY` and `SERVICE_TYPE` with their data values.

Before Starting

Before setting up [page 444] an allowance event class, you must be familiar with the following:

- Basic Charging Concepts [page 18]
- Configuration of Master Data Step by Step [page 104]
- Provider Contracts [page 573]
- Allowances [page 575]
- Allowance Logic [page 449]
- Allowance Plans [page 466]
- Allowance Interface [page 91]

Related Information

Working with Allowance Event Classes [page 443]

1.29.1 Allowance Events

An **allowance event** is a modifiable message composed of a list of properties and their values and is managed by the system when running allowances (in provider contracts) and executing the allowance logic that is set in their contracted allowance plan. Allowances in provider contracts receive and process allowance events.

For more information on the allowance event, refer to the Application Help guide.

In SAP CC, the pricing specialist configures a class (allowance event class) that defines the data structure of the allowance events.

→ Tip

Consult the topic Allowance Event Updater (Operator) [page 782] about the modification of an allowance event.

Related Information

Allowances [page 575]

1.29.2 Working with Allowance Event Classes

Related Information

Creating an Allowance Event Class [page 443]
Opening an Existing Allowance Event Class [page 446]
Modifying an Allowance Event Class [page 447]
Setting Up an Allowance Event Class [page 444]
Saving an Allowance Event Class [page 445]
Deleting an Allowance Event Class [page 447]
Exporting an Allowance Event Class [page 448]
Importing an Allowance Event Class [page 448]

1.29.2.1 Creating an Allowance Event Class

Use one of these procedures to create an allowance event class [page 441].

To create a allowance event class

1. From the menu bar in the main window, select *File* > *New* > *Allowance Event Class*. The *New charged Item Class* window opens.
2. Set up [page 444] the allowance event class.
3. Verify the allowance event class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.

4. Save [page 445] the allowance event class.

You can create a allowance event class by opening an existing allowance event class and copying it.

To create a allowance event class from an existing allowance event class

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Event Class*. The *Choose a Allowance Event Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance event class.
3. In the *Allowance Event Classes* folder of the catalog of the owner, select the allowance event class to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding allowance event class window opens.
5. Rename the allowance event class in the *Name* field, and then set up [page 444] the allowance event class.
6. Verify the allowance event class structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 445] the allowance event class.

Related Information

Setting Up an Allowance Event Class [page 444]
Saving an Allowance Event Class [page 445]

1.29.2.2 Setting Up an Allowance Event Class

Create [page 443], open [page 446], or modify [page 447] an allowance event class before setting it up. An *Allowance Event Class* window must be open.

In your Core Tool user interface, an allowance event class is graphically represented by a data table that consists of rows representing the custom properties of the allowance event class.

To set up an allowance event class [page 441], you first identify it, and then you define or change some custom properties.

Name: Type the unique identifier of the allowance event class.

Description: Describe the function of the allowance event class.

 : Click to add a new row to the configuration table.

 : Click to delete a row from the configuration table.

Properties: Fill in or modify the columns of the configuration table.

- *Name*: Type the identifier of the custom property that characterizes the allowance event.
- *Category*: Displays the category of properties (User properties not modifiable)
- *Type*: Select the type of the property (date, number, or string).
- *Description*: Describe the function of the property.

1.29.2.3 Saving an Allowance Event Class

Once you have created a allowance event class and then verified its configuration, you are ready to save it. You can save the allowance event class to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving an allowance event class, you can no longer:
>
> - Modify the name of the allowance event class
> - Rename a property
> - Remove a property from the table
> - Change the type of a property

To save a new allowance event class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the Catalog* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance event class.
3. Click *Save*. The *Allowance Event Class Creation* dialog box opens to inform you that the allowance event class has been successfully created.
4. Click *OK*.

To save an existing allowance event class to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. If the *Allowance Event Class Modification* dialog box opens to inform you that the allowance event class has been successfully modified, click *OK*. The new version of the allowance event class is saved as the only current version.
3. If a *Warning* dialog box opens to inform you that the allowance event class has been modified since you opened it, you can:
   - Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the allowance event class.
   - Click *Cancel* to come back to the allowance event class. See below how to save the allowance event class as an XML file.

To save a allowance event class as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the allowance event class.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The allowance event class is saved as an XML file in your selected directory.

1.29.2.4 Opening an Existing Allowance Event Class

You can open:

- An existing allowance event class saved in the database
- An existing allowance event class saved in an XML file

To open an existing allowance event class saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Event Class*. The *Open Allowance Event Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance event class.
3. In the *Allowance Event Classes* folder of the catalog of the owner, select the allowance event class.
4. Click the *Open* button. The *Allowance Event Class* window opens in display mode [page 46].

To open an existing allowance event class saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.

4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.29.2.5 Modifying an Allowance Event Class

When modifying a allowance event class, you can:

- Change the description of the class
- Change the description or the properties
- Remove a property if the allowance event class is not already referenced
- Change the property type if the allowance event class is not already referenced
- Add new properties to the allowance event class if the allowance event class is not referenced or only referenced in the *Event Allowance* components (refer to the primary help relating to logic components here [page 671]).

The modified allowance event class replaces the previous version in the database.

To modify a allowance event class

1. Open [page 446] the allowance event class.
2. Set the edit mode [page 46] by clicking the *Edit Mode* icon  on the main toolbar. The allowance event class may be reloaded if it has been modified since you have opened it.
3. Set up [page 444] the allowance event class.
4. Verify the structure of a allowance event class by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Save [page 445] the allowance event class.

1.29.2.6 Deleting an Allowance Event Class

> **Note**
>
> You cannot delete an allowance event class used by the allowance logic [page 449] or by an *Allowance Event Sender* component.

To delete an allowance event class

1. From the menu bar in the main window, select *File* > *Delete* > *Allowance Event Class*. The *Delete Allowance Event Class* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance event class.
3. In the *Allowance Event Classes* folder of the catalog of the owner, select the allowance event class.
4. Click *Delete*. A deletion dialog box asks if you want to delete the allowance event class
5. Confirm the deletion of the allowance event class by clicking *Yes*:
    1. If a dialog box confirms your choice, the allowance event class has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the allowance event class is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48]for more information.

1.29.2.7 Exporting an Allowance Event Class

You can export allowance event classes to a directory of your hard disk, where they are saved as XML files.

To export an allowance event class

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance event class.
3. In the catalog of the owner, select the allowance event class.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the allowance event class.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.29.2.8 Importing an Allowance Event Class

You can import allowance event classs in XML format from a directory of your hard disk.

> **Note**
>
> Only one version of an allowance event class can exist in a database. You can import the unique version of an allowance event class into SAP CC Core Tool.

To import a allowance event class

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the allowance event class is saved, and then select the allowance event class file.
4. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a allowance event class.
5. In the *Import* column, clear the checkboxes of objects you do not want to import.
6. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.30 Allowance Logic


The **allowance logic** in SAP Convergent Charging (SAP CC) is a reusable object that belongs to the pricing catalog (in master data) of the chargeable service. The allowance logic includes decision trees that **define the rules and conditions of use of a contract allowance [page 575] that is based on this logic**.

The allowance plan uses the allowance logic that you can customize.

The allowance logic performs calculations only when the allowance plan is triggered by:

- Any allowance event
- A periodic allowance event
- A one-shot allowance event

To process an allowance event, the allowance logic consists of a set of allowance components.

> **Note**
>
> About shared allowances [page 575].
>
> - The allowance logic must be shareable [page 456] to be used for shared allowances.
> - The shareable allowance logic implies some restrictions regarding the use of components (For more information, refer to the primary help relating to components here [page 671]).
> - You can use the allowance logic in shareable or non-shareable allowance plans.

Use

To be able to set up your allowance plan [page 466] for the allowance management in provider contracts, you must design and configure the necessary pricing logic objects.

Test of the Allowance Logic

Once your allowance logic is complete, SAP Convergent Charging provides you with the Rating Trial Run tool [page 335] to test your allowance logic and its functional correctness.

Before Starting

Before setting up [page 444] an allowance logic object, you must be familiar with the following:

- Basic Charging Concepts [page 18]
- Configuration of Master Data Step by Step [page 104]
- Provider Contracts [page 573]
- Allowances [page 575]
- Allowance Logic [page 449]
- Example of Allowance Logic [page 450]
- Allowance Tree [page 451]
- Allowance Plans [page 466]
- Allowance Interface [page 91]
- Allowance Context [page 453]
- Allowance Components [page 452]
- Version change of the allowance logic [page 465]

Related Information

Working with the Allowance Logic [page 454]

1.30.1 Example of Allowance Logic

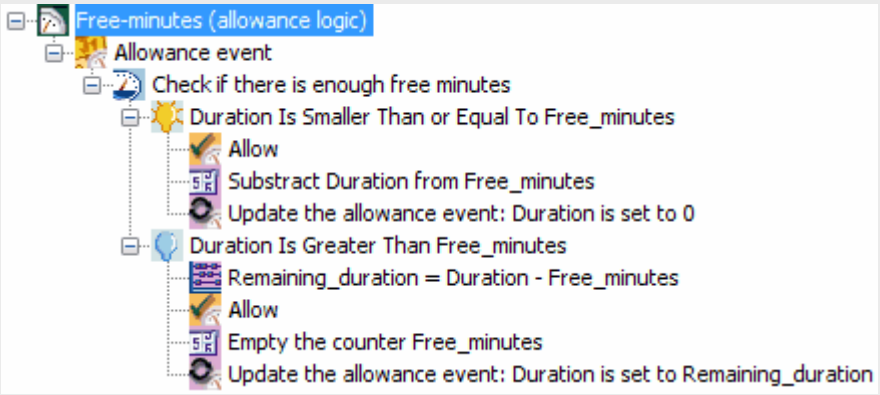
> ✤ Example
>
> The company TelcoServ sells the product Walk & Talk, a prepaid monthly plan including voice calls for 20¢/minute. End customers can choose to buy an add-on of 60 minutes for 1$ during one month. To meet this service, TelcoServ uses the allowance logic in their plan.

The allowance logic contains:

- An allowance component (*Event-Based Trigger* component) to decrement the counter `Free_minutes` (the *Event-Based Trigger* is activated when receiving an event related to the allowance and sent by an *Allowance Event Sender Operator* component)
- The persistent counter `Free_minutes` set to the initial value of 60

In the allowance tree:

- The first branch checks if there is enough remaining free minutes.
- The second branch allows the whole duration.
- The third branch allows part of the duration.

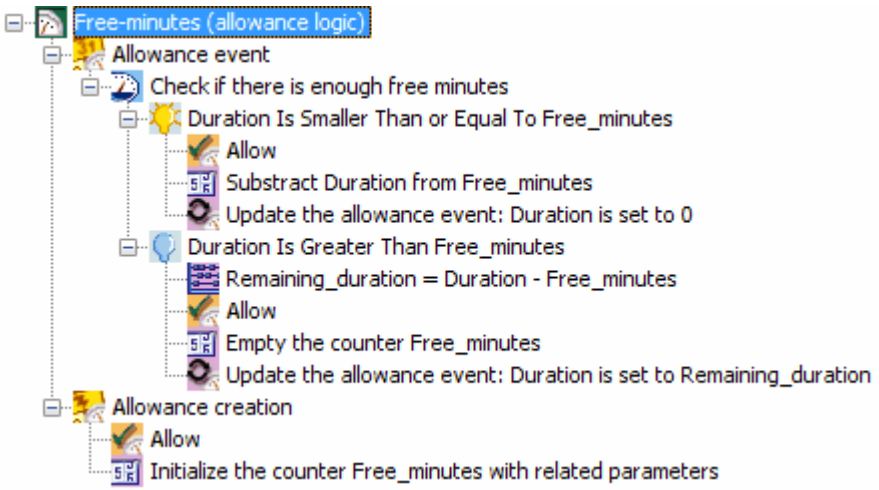


For more information on logic components, click here [page 671].

1.30.2 Allowance Tree

The allowance logic is graphically represented by a tree structure called an allowance tree. An allowance component [page 452] represents each instruction inside the allowance logic. This structure consists of a root from which the allowance components create branches. The *Allow Function* component marks the end of each branch.

As shown in the illustration below, the tree contains the root  representing the allowance logic and the branches for the different allowance components:



In addition to allowance components, you can create the following pricing elements:

- Counters [page 297]
- Parameters [page 303]
- Pricing Macros [page 121]
- Translation Tables [page 132]
- Tier Tables [page 147]
- Mapping Tables [page 179]
- Range Tables [page 213]

Pricing macros, translation tables, tier tables, mapping tables, and range tables can be reused in any allowance logic objects.

1.30.3 Allowance Components

Allowance components are calculation tools you use to create the allowance logic. Together the components make up an allowance tree [page 451] defining the algorithms, rules, and conditions of use of allowances [page 575].

Each main branch of the allowance logic begins with a *Trigger component* that is linked to an allowance activation method. There are three types of trigger linked to three methods of activation:

Type of trigger	Activation
Event-Based	Each time an event is triggered by the Allowance Event Sender component
Recurring	Periodic and linked to a reference date Automatically triggered by SAP CC

Type of trigger	Activation
 One-Shot	Once only at the allowance creation, activation, or expiration Automatically triggered by SAP CC

Example

Examples of activation

- *Event-Based Trigger*: Can be used to receive allowance events from the *Allowance Event Sender* components
- *Recurring Trigger*: Can be used to reset the allowance counter at the beginning of each month
- *One-shot Trigger*: Can be used to report the initial quantity at the allowance creation or to report the remaining quantity at the allowance expiration.

The values generated higher up a trigger branch become available to all parts of the branch further toward the ends of the branch. When a property is generated on one trigger branch, it cannot be used on another trigger branch.

The components of an allowance tree are grouped into four families depending on their main function:

- *Trigger components*  always begin the main branches of an allowance tree.
- *Comparator*  , and *Operator*  components help to update the allowance event.
- *Function components*  end each branch of an allowance tree and generate a charged item.

Note

For more comprehensive detail on using each component, refer to the primary help relating to components here [page 671]

1.30.4 Allowance Context

The allowance function of SAP Convergent Charging uses a set of data to complete its tasks at runtime. This data whose values vary over time is referred to as the allowance context.

Data used to calculate an allowance event comes directly from the *Allowance Event Sender* component[56] and is supplied by the properties of an Allowance Event Class [page 441] or from the following sources:

- Counters [page 297]
- Parameter values coming from:
  - The allowance logic [page 449]
  - The allowance plan [page 466]

[56] For more information on components, see the primary help of logic component [page 671]s.

- The allowance of a provider contract [page 575]
- Translation table [page 132], tier table [page 147], mapping table [page 179] or range table [page 213] rows used in the allowance logic

> **Note**
>
> When you are at the stage of adjusting and finalizing your allowance logic, you can view the allowance context in detail. You can also manipulate it in the GUI by performing a trial run [page 335] on the allowance logic.

1.30.5 Versions

Different versions of an allowance logic object can coexist in the pricing catalog for different periods.

Related Information

Changing the Version of the Allowance Logic [page 465]

1.30.6 Working with the Allowance Logic

Related Information

Creating the Allowance Logic [page 454]
Opening the Existing Allowance Logic [page 460]
Modifying the Allowance Logic [page 461]
Setting Up the Allowance Logic [page 455]
Deleting the Allowance Logic [page 462]
Saving the Allowance Logic [page 459]
Exporting the Allowance Logic [page 462]
Importing the Allowance Logic [page 463]

1.30.6.1 Creating the Allowance Logic

Use one of these procedures to create the allowance logic.

Create the Allowance Logic

1. From the menu bar in the main window, select *File* > *New* > *Allowance Logic*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the allowance logic.
3. Click *OK*. The *New Allowance Logic* window opens.
4. Set up [page 455] the allowance logic.
5. Verify the allowance logic structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 459] the allowance logic.

> → Tip
>
> You can create an allowance logic object by opening ane existing allowance logic object and copying it.

Create an Allowance Logic Object from an Existing Allowance Logic Object

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Logic*. The *Open Allowance Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance logic.
3. In the *Allowance Logics* folder of the catalog of the owner, select the allowance logic to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *allowance logic* window opens.
5. Rename the allowance logic in the *Name* field, and then set up [page 455] the allowance logic.
6. Verify the allowance logic structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 459] the new allowance logic.

Related Information

Setting Up the Allowance Logic [page 455]
Saving the Allowance Logic [page 459]

1.30.6.2 Setting Up the Allowance Logic

Create [page 400], open [page 406], or modify [page 461] the allowance logic before setting it up. The *Allowance Logic* window must be open.

In an allowance logic window, the allowance tree [page 451] consists of trigger components (event-based trigger, recurring trigger, or one-shot trigger) and other components (functions, comparators, and operators)

using counters, parameters, pricing macros, and other objects which depend on the content of the allowance logic.

Set Up the Allowance Logic

1. In the right pane of the *Allowance Logic* window, identify the allowance logic. In the *Name* field, type the unique identifier of the allowance logic. In the *Description* field, describe the function of the allowance logic if necessary.
2. Set up the following tabs of the allowance logic:
   - Definition [page 456]
   - Parameters [page 457]
   - Persistent Counters [page 457]
   - Transient Counters [page 458]
3. Add [page 464] trigger components to the allowance logic, and then set up the refill components[57].
4. Add [page 464] functions, comparators, or operators components to the allowance logic, and then set up the components[58].

> **Note**
>
> You can remove [page 464], cut, copy, or paste components or sections of a refill tree by right-clicking the component and selecting *Remove*, *Cut*, *Copy*, or *Paste*.

1.30.6.2.1 Allowance Logic: Definition

Create [page 400] or modify [page 407] the refill logic before setting it up [page 455]. In the right pane of the *Allowance Logic* window, click the *Definition* tab to set it up.

Definition (Tab)

Is Shareable: Select this option enables the allowance logic to be used in the context of shared allowances [page 575]. The shareable allowance logic can be used by shareable or non-shareable allowance plans [page 466]. When this option is selected, the usage of logic components is restricted depending on the context. For more information, see the Application Help and the primary help of logic components [page 671].

Allowance Event Class

Displays the allowance event classes [page 441] that are used by each allowance usage that is included in the allowance logic. If necessary, click an allowance event class to open it, and then modify it.

[57] For more information about setting up components, refer to the primary help relating to components here [page 671]
[58] For more information about setting up components, refer to the primary help relating to components here [page 671].

1.30.6.2.2 Allowance Logic: Parameters

Create [page 400] or modify [page 407] the refill logic before setting it up [page 455]. In the right pane of the *Allowance Logic* window, click the *Parameters* tab to set it up.

The *Parameters* tab allows you to add parameters to the allowance logic.

Parameters (Tab)

Fill in or modify the *Parameters* tab.

 : Click to add a parameter to the allowance logic.

 : Click to delete a parameter from the allowance logic.

Name: Type the parameter name.

Description: Describe the function of the parameter if necessary.

Type: Provides the list of parameter types. Select the parameter type:

- *Number*
- *String*
- *Date*
- *Mapping Table ID* (see Parameters [page 303] for details)
- *Mapping Table Key*
- *Range Table ID*

Value: Type a numerical value, a string value, or a date to initialize the parameter depending on its type[59].

Visibility: Select one of the following options:

- *Internal*: The parameters are only available inside the allowance logic.
- *External*: The parameters can be modified in the related allowance plan.

1.30.6.2.3 Allowance Logic: Persistent Counters

Create [page 454] or modify [page 407] the refill logic before setting it up [page 455]. In the right pane of the *Allowance Logic* window, click the Persistent Counters [page 297] tab to set it up (see also the persistent counters in session-based charging limitations [page 22]).

The *Persistent Counters* tab allows you to create counters for the allowance logic.

[59] For the types *Mapping Table ID* and *Range Table ID*, you cannot type a value but must select a mapping or a range table class and optionally select a mapping or a range table.

Persistent Counters (Tab)

Use the configuration table and its icons to create counters.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

Name: Type the name of the counter.

Description: Describe the function of the counter if necessary.

Initial Value: Type a numerical value to initialize the counter.

Operation:

- In case of a shareable [page 456] allowance logic, choose the operation applied to the counter: *Increment* or *Decrement*.
- In case of a non-shareable [page 456] allowance logic, the operation is set to *Update* by default.

> ⚠ Caution
>
> **About shared allowance**
>
> When a distributed counter is already used in an allowance plan, do not change its related operation, otherwise an exception will be raised in the rating process. SAP SE recommends adding a new counter instead of changing the related operation of an existing one.

1.30.6.2.4 Allowance Logic: Transient Counters

Create [page 454] or modify [page 407] the refill logic before setting it up [page 455]. In the right pane of the *Allowance Logic* window, click the Transient Counters [page 297] tab to set it up (see also the transient counters in session-based charging limitations [page 22]).

The *Transient Counters* tab allows you to create counters for the allowance logic.

Persistent Counters (Tab)

Use the configuration table and its icons to create counters.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

Name: Type the name of the counter.

Description: Describe the function of the counter if necessary.

Initial Value: Type a numerical value to initialize the counter.

1.30.6.3 Saving the Allowance Logic

Once you have created or modified the allowance logic, and then verified its structure, you are ready to save it. You can save the allowance logic to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving the allowance logic, you can no longer modify:
>
> - The name of the allowance logic
> - The *Is Shareable* option

To save the new allowance logic to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Allowance Logic Creation* dialog box opens to inform you that the allowance logic has been successfully saved.
4. Click *OK*.

To save a new version of an existing allowance logic to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the allowance logic has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the allowance logic.
> 2. Click *Cancel* to come back to the allowance logic. See later in this chapter how to save the allowance logic as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the allowance logic.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.

- *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.

4. Click *Save*. The *Allowance Logic Modification* dialog box opens to inform you that the allowance logic has been successfully created.
5. Click *OK*.

To save a allowance logic as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the allowance logic.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The allowance logic is saved as an XML file in your selected directory.

1.30.6.4 Opening the Existing Allowance Logic

You can open the existing allowance logic:

- Saved in the back-end database
- Saved in an XML file

To open the existing allowance logic saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Logic* The *Open Allowance Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance logic.
3. In the catalog of the owner, select the allowance logic.
4. Click the *Open* button. The related window of the allowance logic opens in display mode [page 46].

To open the existing allowance logic saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.30.6.5 Modifying the Allowance Logic

When you save the allowance logic in the database, you can modify its versions (see the process below if necessary).

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

Modify the Allowance Logic

1. Open [page 460] the allowance logic.
2. In the *Owner* drop-down menu, select the owner of the allowance logic to be modified.
3. In the *Allowance Logic* folder of the catalog of the owner, select the allowance logic.
4. Click *Open*. The allowance logic window opens.
5. Switch to the editing mode by clicking the *Edit Mode* icon  on the main toolbar. The allowance logic may be reloaded if it has been modified since you have opened it.

6. Set up [page 455] the allowance logic.
7. Verify the structure of the allowance logic by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. Save [page 459] the allowance logic.

Related Information

Display/Edit Mode [page 46]

1.30.6.6 Deleting the Allowance Logic

Use this procedure to delete the allowance logic.

> **Note**
>
> You cannot delete the allowance logic activated in an allowance plan.

Delete the Allowance Logic

1. From the menu bar in the main window, select *File* > *Delete* > *Allowance Logic*. The *Delete Allowance Logic* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance logic.
3. In the *Allowance Logics* folder of the catalog of the owner, select the allowance logic.
4. Click *Delete*. A deletion dialog box asks if you want to delete the allowance logic
5. Confirm the deletion of the allowance logic by clicking *Yes*:
    1. If a dialog box confirms your choice, the allowance logic has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the allowance logic is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

1.30.6.7 Exporting the Allowance Logic

You can export the allowance logic to a directory of your hard disk, where it is saved as an XML file.

Export the Allowance Logic

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance logic.
3. In the catalog of the owner, select the allowance logic.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the allowance logic.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.30.6.8 Importing the Allowance Logic

You can import allowance logics in XML format from a directory in your database.

Import the Allowance Logic

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the allowance logic is saved, and then select the allowance logic file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected if you want to import all the versions of the imported objects, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option, and then select the date in the *Version as Of* field.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
6. In the *Import* column, clear the checkboxes of objects you do not want to import, if necessary.
7. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported [page 239] and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.30.7 Using the Action Menu

Related Information

Adding an Allowance Component [page 464]
Adding a Component [page 464]
Removing a Component [page 464]
Moving Up a Component [page 464]
Moving Down a Component [page 465]
Changing the Version of the Allowance Logic [page 465]
Deleting the Versions of the Allowance Logic [page 465]
Reading the Allowance Logic [page 465]
Doing a Trial Run of the Allowance Logic [page 466]

1.30.7.1 Adding an Allowance Component

The procedure is the same as for adding a refill component [page 409] to a refill tree.

1.30.7.2 Adding a Component

The procedure is the same as for adding a component to a refill tree [page 410].

1.30.7.3 Removing a Component

The procedure is the same as for removing a component from a refill tree [page 411].

1.30.7.4 Moving Up a Component

The procedure is the same as for moving up a component in a refill tree [page 412].

1.30.7.5 Moving Down a Component

The procedure is the same as for moving down a component in a refill tree [page 412].

1.30.7.6 Changing the Version of the Allowance Logic

Refer to the process below:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version* icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.30.7.7 Deleting the Versions of the Allowance Logic

The procedure is the same as Deleting the Versions of a Catalog Object [page 118].

1.30.7.8 Reading the Allowance Logic

Once you have created the allowance logic, you can generate a descriptive representation of it. This process is called reading a allowance logic.

To read a allowance logic

1. Open [page 460] the allowance logic..
2. On the toolbar in the main window, click the *Read Allowance Logic* icon. The *Read Allowance Logic* dialog box opens.

3. Check the allowance logic description for accuracy, and then click *Close*.

1.30.7.9 Doing a Trial Run of the Allowance Logic

Once you have built and verified the allowance logic, you can test it to ensure that the correct results have been achieved. To do this, you use the *Rating Trial Run* tool which is a testing tool provided with SAP Convergent Charging.

To test the allowance logic

1. Make sure that the allowance logic that you want to test is open; otherwise, open [page 406] the allowance logic.
2. Open [page 339] the *Rating Trial Run* tool.
3. Set up [page 340] the *Rating Trial Run* tool.
4. Run [page 340] the *Rating Trial Run* tool.

1.31 Allowance Plans

In SAP Convergent Charging 2023, the allowance plan defines the behavior of an allowance when the charging and refilling processes are executed. The allowance plan consists of:

- One or several allowance logic objects
- A set of counters linked to the allowance logic objects
- A set of parameters optionally linked to the allowance logic objects
- A status which can be open or released

> ⓘ Note
>
> About shared allowances [page 575]
>
> - To be used for shared allowances, an allowance plan must be shareable [page 456] and only contains shareable allowance logic objects.
> - You can however use a shareable allowance plan for non-shared allowances.

> → Remember
>
> About distributed counters
>
> - If the allowance plan is shareable, you cannot link an increment counter with a decrement counter of the allowance logic
> - If the allowance plan is not shareable, you can link an update counter with any type of counter in the allowance logic (increment or decrement)

- If the allowance plan is not shareable, the operation associated to the counters is "update"

Before Starting

Before setting up [page 469] an allowance plan, you must be familiar with the following:

- Basic Charging Concepts [page 18]
- Configuration of Master Data Step by Step [page 104]
- Provider Contract [page 573]
- Allowance [page 575]
- Allowance Event Class [page 441]
- Allowance Interface [page 91]
- Allowance Logic [page 449]
- Charged Item Classes [page 254]
- Version change of an allowance plan [page 482]

1.31.1 Versions

Different versions of an allowance plan can coexist in the pricing catalog for different periods.

Related Information

Changing the Version of an Allowance Plan [page 482]

1.31.2 Working with Allowance Plans

Related Information

Creating an Allowance Plan [page 468]
Opening an Existing Allowance Plan [page 474]
Modifying an Allowance Plan [page 475]
Setting Up an Allowance Plan [page 469]
Saving an Allowance Plan [page 472]

Deleting an Allowance Plan [page 475]
Exporting an Allowance Plan [page 476]
Importing an Allowance Plan [page 476]

1.31.2.1 Creating an Allowance Plan

Use one of these procedures to create a allowance plan [page 421].

Create an Allowance Plan

1. From the menu bar in the main window, select *File* > *New* > *Allowance Plan*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the allowance plan[60].
3. Click *OK*. The *New Allowance Plan* window opens.
4. Set up [page 469] the allowance plan.
5. Verify the allowance plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Check the preview [page 472] of your allowance plan.
7. Save [page 472] the allowance plan.

> → Tip
>
> You can create a allowance plan by opening an existing allowance plan and copying it.

Create an Allowance Plan from an Existing Allowance Plan

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Plan*. The *Choose Allowance Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance plan.
3. In the *Allowance Plans* folder of the catalog of the owner, select the allowance plan.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Allowance Plan* window opens.
5. Rename the allowance plan in the *Name* field, and then *Set up* the allowance plan as necessary.
6. Verify the allowance plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. *Save* the allowance plan.

[60] Steps 1 and 2 are optional if you are the owner of the catalog. In this case, the *New Allowance Plan* window opens when you select *File* > *New* > *Allowance Plan*.

Related Information

Previewing an Allowance Plan [page 472]
Setting Up an Allowance Plan [page 469]
Saving an Allowance Plan [page 472]

1.31.2.2 Setting Up an Allowance Plan

Create [page 468], open [page 474], or modify [page 475] an allowance plan before setting it up. An *Allowance Plan* window must be open.

Set Up an Allowance Plan

1. In the right pane of the *Allowance Plan* window, identify the allowance plan. In the *Name* field, type the unique identifier of the allowance plan. You cannot modify this field after the allowance plan is saved.
2. Set up the following tabs of the allowance plan:
   - Definition [page 469]
   - Counters [page 470]
   - Parameters [page 471]
3. Add [page 481] one or more allowance logic objects to the plan, and then set up [page 477] the allowance logic objects.
4. If necessary, remove [page 481] one or more allowance logic objects.

   Allowance Plan: Definition [page 469]

   Allowance Plan: Counters [page 470]

   Allowance Plan: Parameters [page 471]

Related Information

Working with the Allowance Logic Customized in an Allowance Plan [page 477]

1.31.2.2.1 Allowance Plan: Definition

Create [page 468], open [page 474], or modify [page 475] an allowance plan before setting it up [page 469]. In the right pane of the allowance plan window, open the *Definition* tab.

The *Definition* tab allows you to set the status of the allowance plan and gives a detailed description of the allowance plan.

Definition (Tab)

Fill in the *Definition* tab.

Status: Provides the list of statuses. Select one of the following statuses:

- *Open*: Select this status if you are working on the allowance plan. This status allows you to modify the allowance plan and save it in the database. Furthermore, you cannot save several versions of the allowance plan.
- *Released*: Select this status when the allowance plan is ready to be in production. Furthermore, you can modify and save different versions of the released allowance plan. According to this status, you are only allowed to do the following:
  - Change the status of counters [page 478] or parameters [page 479] of the allowance logic depending on the counters and parameters declared at allowance plan level
  - Change the charged item class and the charged item mapping (Output Item [page 479] tab)

> **Note**
>
> - When you create an allowance plan, its status is automatically set to *Open*. You must save the allowance plan once before changing its status.
> - You can set the status to *Released* only if the current status of the allowance plan is *Open*.
> - The *Create Allowance* component can only reference an allowance plan whose status is *Released*. For more information on this component, refer to the primary help here [page 671].

Is Shareable: Select this option enables the allowance plan to be used in the context of shared allowances [page 575]. A shareable allowance plan only consists of one or several shareable allowance logic [page 449] objects. For more information about shared allowances, see the Application Help and the primary help of logic components [page 671].

Description: Describe the allowance plan by typing a long text if necessary. This field is mandatory. Note that you cannot change the description of the allowance plan if its status is *Released*.

1.31.2.2.2 Allowance Plan: Counters

Create [page 468], open [page 474], or modify [page 475] an allowance plan before setting it up [page 469]. In the right pane of the *Allowance Plan* window, open the *Counters* tab.

Each time you add a new counter to the configuration table, you must initialize the counter and link a counter of the allowance logic [page 478] to the new counter of the allowance plan.

The counter of the allowance plan enables to:

- Share a counter between several allowance logic objects.
- Be read by an external system, a CRM application for example, through the Web Services (WS) [page 40]. Refer to the WS operation *Find the allowances associated to a given charging contract*.

> **Note**
>
> When you create an allowance plan, the table of counters may already contain some counters that are created in the allowance interface [page 91] component. If necessary, you can also link the counters of the allowance logic to these counters.

Counters (Tab)

Use the configuration table and its icons to add counters to the allowance plan.

 : Click to add a counter row to the table.

 : Click to delete a counter row from the table.

Name: Displays the name of the counter.

Description: Displays the function of the counter.

1.31.2.2.3 Allowance Plan: Parameters

Create [page 468], open [page 474], or modify [page 475] an allowance plan before setting it up [page 469]. In the right pane of the allowance plan window, open the *Parameters* tab.

You can link a parameter of the allowance logic to the new parameter of the allowance plan.

The parameter of the allowance plan enables to:

- Share a parameter between several allowance logic objects
- Be read by an external system, CRM for example, through the Web Services [page 40] (refer to the operation *Find the allowances associated to a given charging contract*)

All these parameters are configured at the allowance creation time when using the *Create Allowance* component (Refer to the primary help relating to components here [page 671]).

> **Note**
>
> When you create an allowance plan, the table of parameters may already contain some parameters that are created in the allowance interface [page 91] component. If necessary, you can also link the parameters of the allowance logic to these parameters.

Parameters (Tab)

Use the configuration table and its icons to create parameters.

 : Click to add a parameter row to the table.

: Click to delete a parameter row from the table.

Name: Type the name of the parameter.

Type: Provides the list of parameter types. Select one of the following types: *Number*, *String*, *Date*.

Description: Describe the function of the parameter.

1.31.2.3 Previewing an Allowance Plan

This procedure allows you to preview a allowance plan and check its signature that will be seen by SAP CRM or by another provisioning system.

To perform this procedure, you must open [page 474] a allowance plan.

Preview an Allowance Plan

1. From the menu bar in the main window, select *Action* > *Preview*, or click the *Preview* icon on the toolbar in the main window.
2. The *Preview* window opens and displays a summary of the counters, parameters, account assignments, and technical data that will be seen by the external system. Read and check this summary.
3. Click *Close*.

> **Note**
>
> The *Preview* window displays only the counters and parameters which are assigned the external visibility option [page 403].

1.31.2.4 Saving an Allowance Plan

Once you have created or modified a allowance plan, and then verified its structure, you are ready to save it. You can save the allowance plan to either the database or as an XML file on your hard disk.

> **Note**
>
> After saving an allowance plan to the database, you can no longer modify:
>
> - The name of the allowance logic
> - The *Is Shareable* option

To save a new allowance plan to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Allowance Plan Creation* dialog box opens to inform you that the allowance plan has been successfully saved.
4. Click *OK*.

To save a new version of an existing allowance plan to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the allowance plan has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the allowance plan.
> 2. Click *Cancel* to come back to the allowance plan. See later in this chapter how to save the allowance plan as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the allowance plan.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Allowance Plan Modification* dialog box opens to inform you that the allowance plan has been successfully created.
5. Click *OK*.

To save a allowance plan as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.

3. Select a directory to save the allowance plan.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The allowance plan is saved as an XML file in your selected directory.

1.31.2.5 Opening an Existing Allowance Plan

You can open the existing allowance plan:

- Saved in the back-end database
- Saved in an XML file

To open an existing allowance plan saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Allowance Plan* The *Open Allowance Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance plan.
3. In the catalog of the owner, select the allowance plan.
4. Click the *Open* button. The related window of the allowance plan opens in display mode [page 46].

To open the existing allowance plan saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File* . The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.31.2.6 Modifying an Allowance Plan

> **Note**
>
> - You cannot modify the name of the allowance plan.
> - You can modify and save different versions of the allowance plan if its status is set to Released [page 469].

To modify a allowance plan

1. Open [page 474] the allowance plan.
2. Switch to the editing mode [page 46] by clicking the *Edit Mode* icon on the main toolbar. The allowance plan may be reloaded if it has been modified since you have opened it.
3. Set up [page 469] the allowance plan.
4. Verify the structure of the allowance plan by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
5. Check the preview [page 472] of your allowance plan.
6. Save [page 472] the allowance plan.

1.31.2.7 Deleting an Allowance Plan

Before you delete an allowance plan, you must first delete all objects which reference it.

For example:

- You cannot delete an allowance plan that is part of a contract
- You cannot delete an allowance plan that is used in a *Create Allowance component*

To delete a allowance plan

1. From the menu bar in the main window, select *File* > *Delete* > *Allowance Plan*. The *Delete Allowance Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance plan.
3. In the *Allowance Plans* folder of the catalog of the owner, select the allowance plan.
4. Click *Delete*. A deletion dialog box asks if you want to delete the allowance plan
5. Confirm the deletion of the allowance plan by clicking *Yes*:
    1. If a dialog box confirms your choice, the allowance plan has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the allowance plan is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

1.31.2.8 Exporting an Allowance Plan

You can export allowance plans to a directory of your hard disk, where they are saved as XML files.

To export a allowance plan

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the allowance plan.
3. In the catalog of the owner, select the allowance plan.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the allowance plan.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

1.31.2.9 Importing an Allowance Plan

You can import allowance plans in XML format from a directory of your hard disk.

Import an Allowance Plan

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the allowance plan is saved, and then select the allowance plan file.
4. Click *Open*. The *Import Objects* dialog box opens. The options for importing versions of an object do not apply to objects because there is only one version of a allowance plan.
5. In the *Import* column, clear the checkboxes of objects you do not want to import.
6. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).

1.31.3 Working with the Allowance Logic Customized in an Allowance Plan

Related Information

Setting Up the Allowance Logic in an Allowance Plan [page 477]

1.31.3.1 Setting Up the Allowance Logic in an Allowance Plan

Setting up the allowance logic consists in meeting the requirements of the allowance plan.

You must set up [page 469] an allowance plan before setting up the allowance logic. An allowance plan window must be open.

Set Up the Allowance Logic in an Allowance Plan

1. In the left pane of the allowance plan window, make sure that the allowance logic is selected; otherwise, select it.
2. In the right pane of the allowance plan window, you can change the name of the allowance logic. Note that two allowance logic objects cannot have the same name. If several allowance logic objects have the same name, the names are followed by a number to identify each allowance logic object.
3. In the right pane of the allowance plan window, set up the following tabs of the allowance logic:
   - Definition [page 477]
   - Counters [page 478]
   - Parameters [page 479]
   - Output Item [page 479]

> ⓘ Note
>
> The *Counters* or *Parameters* tabs of the allowance logic are unavailable (dimmed) if the original allowance logic contains no counters or no parameters.

1.31.3.1.1 Customized Allowance Logic: Definition

The *Definition* tab displays the allowance logic.

Definition (Tab)

Allowance Logic: Displays the name of the allowance logic used in the allowance plan. Click  to open the allowance logic.

1.31.3.1.2 Customized Allowance Logic: Counters

Create [page 468] or modify [page 475] an allowance plan before setting up the allowance logic in the allowance plan. In the left pane of the allowance plan window, click the allowance logic  to view, and then open the *Counters* tab.

The *Counters* tab allows you to link the counter of the allowance logic to a counter of the allowance plan. In this tab, a table displays the list of all counters created at allowance logic level. Each row in the table represents a counter.

> ⓘ Note
>
> The *Counters* tab is unavailable (dimmed) if the original allowance logic contains no counter.

Counters (Tab)

Fill in or modify the *Counters* tab.

Name: Displays the name of the allowance logic counter.

Status: Provides the list of counter statuses. Select one of the following statuses:

- *Not Redefined*: Select this status if you do not want to link the counter to a counter of the allowance plan.
- *Redefined*: Select this status if you want to redefine the value of the counter.
- *Linked*: Select this status if you want to link the counter to a counter of the allowance plan. This status enables to share this counter with other allowance logic objects included in the related allowance plan.

Initial Value: Provides the initial value of the counter if the status of the counter is *Redefined*. If the Status of the counter is *Not Redefined*, its initial value is the one defined in the allowance logic. If the status of the counter is *Linked*, its initial value is the same as that of the shared counter.

1.31.3.1.3 Customized Allowance Logic: Parameters

Create [page 468] or modify [page 475] an allowance plan before setting up the allowance logic in the allowance plan. In the left pane of the allowance plan window, click the allowance logic  to view, and then open the *Parameters* tab.

The *Parameters* tab allows you:

- To redefine or not the parameter of the allowance logic
- To link the parameter of the allowance logic to a shared parameter of the allowance plan

> **Note**
>
> The *Parameters* tab is unavailable (dimmed) if the original allowance logic contains no parameter.

Parameters (Tab)

Fill in or modify the *Parameters* tab.

Name: Displays the name of the allowance logic parameter.

Type: Displays the type of the parameter.

Status: Provides the list of parameter statuses. Select one of the following statuses:

- *Not Redefined*: Select this status if you do not want to link the parameter to a shared parameter of the allowance plan.
- *Redefined*: Select this status if you want to redefine the parameter of the allowance logic. You can redefine a parameter only if in the allowance logic its visibility is set to External.
- *Linked*: Select this status if you want to link the parameter to a shared parameter of the allowance plan.

Value: This field varies depending on the option that you select in the *Status* field:

- *Not Redefined*: The value of the parameter is the value initially entered in the allowance logic. The value of the parameter cannot be changed.
- *Redefined*: Type the new value of the parameter.
- *Linked*: Provides the list of shared parameters of the same type. Select the shared parameter that you want to link to your parameter so that its value is the same as that of the shared parameter.

1.31.3.1.4 Allowance Logic: Output Item

Create [page 468] or modify [page 475] an allowance plan before setting up the allowance logic in the allowance plan. In the left pane of the allowance plan window, click the allowance logic  to view, and then open the *Output Item* tab.

The *Output Item* tab allows you to:

- Select the charged item class [page 254] associated to the allowance logic in the allowance plan
- Change the values or properties of the charged item class
- Assign values to the fields of the charged item class
- Select the aggregation policy [page 273] (only relevant for the charging operations performed in the session-based execution mode [page 20])

Output Item (Tab)

Fill in or modify the *Output Item* tab.

Charged Item Based On: Provides the list of the charged item classes. Select a charged item class if you want to replace the charged item class associated to the allowance logic. Click  to open the charged item class, and then modify it if necessary. Click  to refresh the charged item mapping.

> **Note**
>
> This field is not mandatory.
>
> If no charged item class is provided, the verify panel shows that `no value` is set in *Charged Item Class* and *Aggregation Policy*.

Charged Item Mapping: Provides the list of the fields of the selected charged item class and the respective values or properties of these fields. To display less or more columns, click  , and then select the columns you want to display or not.

- *Charged Item Field*: Displays the field names of the charged item class.
- *Property / Value*: Allows you to change the values or properties of the fields depending on the following:
  - If a property or value is dimmed, it is not recommended changing this property by another property or value.
  - If a property or value is not dimmed, it is recommended changing this value by another property or value.
- *Default Value*: Displays the field values defined at class level [page 257].
- *Redefine*: Allows you to redefine any value of the *Property / Value* column. Refer to the recommendations listed above and the note below.

  > **Note**
  >
  > - It is not recommended changing a property or value that is dimmed. For more information about the *Undefined* and *Empty* properties, consult this topic [page 257].
  > - If you set up a boolean field, the value is converted by SAP CC during the charging process. Strings "0" and "false" and number 0 are converted to FALSE. Strings "1" and "true" and number 1 are converted to TRUE; otherwise, the charging process fails because it cannot convert the value.

Session-Based Options: Provides the list of options which are specific to charging operations performed in the session-based rating:

- *Aggregation Policy*: Provides the list of aggregation policies which are associated to the selected charged item class. Select one of the available aggregation policies or the *None* element of the list if you do not want to aggregate the generated charged items. Click  to open the selected aggregation policy, and then modify it if necessary.

1.31.4 Using the Action Menu

Adding an Item [page 481]

Removing an Item [page 481]

Changing the Version of an Allowance Plan [page 482]

Deleting the Versions of an Allowance Plan [page 482]

Previewing an Allowance Plan [page 482]

1.31.4.1 Adding an Item

This procedure allows you to add the allowance logic to the allowance plan. To perform this procedure, you must create [page 468] an allowance plan.

To add the allowance logic to an allowance plan

1. From the menu bar in the main window, select *Action* > *Add Item*, or click the icon  on the toolbar in the main window. The *Select the Allowance Logic* dialog box opens.
2. In the *Allowance Logic* folder, select the allowance logic.
3. Click *OK*. The allowance logic is added to the allowance plan.
4. Select the allowance logic in the allowance plan, and then set up [page 477] the tabs of the allowance logic according to your requirements.

1.31.4.2 Removing an Item

These procedures allow you to remove the allowance logic from a allowance plan.

To remove the allowance logic from a allowance plan

1. In the *Allowance Plan* window, right-click the allowance logic.
2. Select *Remove*. The allowance logic is removed from the allowance plan.

or

1. In the *Allowance Plan* window, right-click the allowance logic.
2. From the menu bar in the main window, select *Action* > *Remove Item*, or click the icon  on the toolbar in the main window.
3. Click *OK*. The allowance logic is removed from the allowance plan.

1.31.4.3 Changing the Version of an Allowance Plan

The procedure is the same as for changing the version of an object:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.31.4.4 Deleting the Versions of an Allowance Plan

The procedure is the same as for deleting the versions of a catalog object [page 118].

1.31.4.5 Previewing an Allowance Plan

From the *Action* menu, you can preview your allowance plan. See Previewing an Allowance Plan [page 472].

1.32 Spending Status Description

> ⚠ Caution
>
> When SAP Convergent Charging is integrated with a policy or monitoring system, check the settings configured in this system before modifying or deleting spending status descriptions with Core Tool [page 7]. Modifying or deleting spending status descriptions may result in inconsistencies between the two systems because there is no replication or synchronization mechanism.

> ⓘ Note
>
> Spending status descriptions allow to configure the monitoring plans [page 491]. A spending status description is a business data available in the complete SAP CC system. You create spending status descriptions only when SAP CC is connected to a policy or a monitoring system and when the spending status monitoring function is implemented in your system landscape.

In SAP Convergent Charging 2023, a *spending status description*  is a list of labels that represent information exposed by SAP CC to an external system (policy or monitoring) to convey valuable information relating to the end customer spending. It is part of the business data configuration. The SAP CC system uses these labels to convert a counter value of a provider contract into a spending status label that can be reported to a policy system or a monitoring system. The policy system can apply some rules depending on the label value of this reported spending status.

A spending status description consists of:

- A name that becomes the identifier of a spending status that can be exposed by the SAP CC system; you select this name when configuring monitoring plans and setting up the generation of spending statuses [page 507].
- A comprehensive description.
- Several string labels that represent different spending limits (inclusive or exclusive); you assign the necessary thresholds to these labels when configuring a particular range table dedicated to the configuration of monitoring plans [page 227].

> ⓘ Note
>
> In the 3GPP environment, you can use a spending status description to define the values of policy counter status.

> ✤ Example
>
> You implement policy control in your system landscape to ensure fair usage and to optimize the network bandwidth allocation with predefined policy rules. You can configure a spending status description 'Data fair usage' [page 485] that defines different quota levels of fair usage for the data service.

Prerequisites

This configuration must be shared between the SAP CC system and the connected policy system. In your policy system, you may need to set up a policy counter identifier for each spending status description configured in SAP CC. You set up the policy counter statuses that are available for this policy counter.

Integration

The configuration of spending status descriptions in SAP CC must be known in the policy system or in the monitoring system. In the connected system, the related settings can have different forms according to the technology: Policy counter values, policy rules, and so on.

Transport

You can transport this business dat [page 32]a between SAP CC systems or manually maintain the configuration in both systems.

To transport the spending status descriptions

For each spending status description, you apply the following procedure:

1. Save a spending status description in a file. [page 488]
2. Log on to the target SAP CC system, or launch another Core Tool.
3. In the target system landscape, create a new spending status description by opening and copying the previous file [page 486].

Data Maintenance

You must manually synchronize the configuration of the SAP CC system and the related configuration settings in the connected policy system or in the monitoring system. When you change the configuration of spending status descriptions, you change the configuration in the policy system.

Related Information

Creating a Spending Status Description [page 486]
Setting Up a Spending Status Description [page 487]
Setting Up the Generation of a Spending Status [page 507]
Example of a Spending Status Description [page 485]
Spending Status Monitoring [page 25]
Monitoring Plans [page 491]

1.32.1 Example of a Spending Status Description

Example

An end customer subscribes to a mobile data plan allowing "unlimited data usage". To protect itself from customer abuse in network usage, the operator adds a clause to the small print of the contract stipulating that data usage with regular quality of service is unlimited until a specified consumption threshold.

For more information see the application example [page 25]

To model this business case in SAP CC, the service provider configures the following spending status description "Data fair usage":

- *Name*: **`Data Fair Usage`**
- *Description*: **`Policy Control / Quality of Service / Fair Use Policy`**
- *Labels*:
  - **`QUOTA_OK`**
  - **`QUOTA_WARNING`**
  - **`QUOTA_REACHED`**

Related Information

Example of a Monitoring Plan [page 493]
Spending Status Monitoring [page 25]

1.32.2 Working with Spending Status Descriptions

Related Information

Creating a Spending Status Description [page 486]
Opening an Existing Spending Status Description [page 489]
Modifying a Spending Status Description [page 490]
Setting Up a Spending Status Description [page 487]
Deleting a Spending Status Description [page 491]
Saving a Spending Status Description [page 488]
Finding a Spending Status Description [page 488]

1.32.2.1 Creating a Spending Status Description

Use one of these procedures to create a spending status description in SAP CC. After creating a spending status description, you can configure a monitoring plan [page 491] in the pricing catalog [page 109] (master data) for the customer service of its service provider.

> → Remember
>
> Once you save your changes, you can only change an existing spending status description by adding new labels.

To create a spending status description

1. From the menu bar in the main window, select *File* > *New* > *Spending Status Description*. The *New charged Item Class* window opens.
2. *Set up* the spending status description.
3. Verify the spending status description structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. *Save* the spending status description.

You can create a spending status description by opening an existing spending status description and copying it.

To create a spending status description from an existing spending status description

1. From the menu bar in the main window, select *File* > *Open* > *Spending Status Description*. The *Choose a Spending Status Description* dialog box opens.
2. Select the spending status description to copy.
3. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding spending status description window opens.
4. Rename the spending status description in the *Name* field, and then *set up* the spending status description.
5. Verify the spending status description structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. *Save* the spending status description.

Related Information

Setting Up a Spending Status Description [page 487]

Saving a Spending Status Description [page 488]
Modifying a Spending Status Description [page 490]

1.32.2.2 Setting Up a Spending Status Description

Create [page 486], open [page 489], or modify [page 490] a spending status description before setting it up. A *Spending Status Description* window must be open.

To set up a spending status description, you must:

1. Set up the identification name and the comprehensive description of the spending status description
2. Fill in the label values

> ⚠ Restriction
>
> The name and the labels of a spending status description cannot exceed 256 characters.

Spending Status Description window

Fill in or modify the spending status description window.

Name: Type the unique identifier of the spending status description. This is also the identifier of the spending status that can be reported to a registered downstream system (policy system or monitoring system). For example in the 3GPP environment, this is the "policy counter identifier" reported in the spending limit reports.

Description: Type a mandatory detailed description of the spending status description.

Label: Provide the list of label values that define the spending status description. A label value is mandatory at least. The label values are known by the policy or monitoring system that monitors the spending status reports (see the Integration section). A label does not include any space character.

Fill in or modify the *Label* table.

 : Click to add a row to the table.

 : Click to delete a row from the table.

 : Click to reset the added rows from the table.

Related Information

Example of a Spending Status Description [page 485]
Spending Status Monitoring [page 25]

1.32.2.3 Saving a Spending Status Description

Once you have created [page 486] or modified [page 490] a spending status description, and then verified its structure, you are ready to save it. You can save the spending status description to either the back-end database or as an XML file on your hard disk.

To save a new spending status description to the database

1. From the menu bar in the main window, select *File* > *Save* > *In The Database*. The *Spending Status Description Creation* dialog box opens.
2. Click *OK*

To save a spending status description as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the spending status description.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The spending status description is saved as an XML file in your selected directory.

1.32.2.4 Finding a Spending Status Description

To search for a spending status description in the business data [page 32] configured in SAP CC, you perform the operation in two halves of the *Choose a Spending Status Description* window:

- The top part of the window is the filter area, which allows you to find a spending status description by using filter criteria.
- The bottom area of the window is the results area, which displays the search results that you perform.

Tasks

You can find spending status descriptions with or without search criteria.

- If you do not use search criteria, click  to start the search and view the results.
- If you use search criteria, set up the *Basic* tab, and then click  to start the search. The spending status descriptions are displayed in the results area. To display more columns of information in the results area,

click  , and then select which types of information you want to display. Click  to reset the filter and delete the result list.

Filter area

Basic tab

The *Basic* tab allows you to search for spending status descriptions depending on the identification name in SAP CC.

Name: Is the identifier of the spending status description. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database. For example, if you want to find all spending status descriptions containing the letters **`eric`**, you select the *Contains* option, and then type **`eric`**.

Results area

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

1.32.2.5 Opening an Existing Spending Status Description

You can open:

- An existing spending status description saved in the database
- An existing spending status description saved in an XML file

To open an existing spending status description saved in the database

1. First find the relevant spending status description.
2. From the menu bar in the main window, select *File* > *Open* > *Spending Status Description* . The *Open Spending Status Description* dialog box opens.

3. Search [page 488] for the spending status description, select it, and then click *OK*.

To open an existing spending status description saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.32.2.6 Modifying a Spending Status Description

> ⚠ Caution
>
> When SAP Convergent Charging is integrated with a policy or monitoring system, check the settings configured in this system before modifying or deleting spending status descriptions with Core Tool [page 7]. Modifying or deleting spending status descriptions may result in inconsistencies between the two systems because there is no replication or synchronization mechanism.

Once you save your changes, you can only add new labels in an existing spending status description.

To modify a spending status description

1. Open [page 489] the spending status description.
2. Set up [page 487] the spending status description.
3. Verify the structure of a spending status description by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.

4. Save [page 488] the spending status description.

1.32.2.7 Deleting a Spending Status Description

> ⚠ Caution
>
> When SAP Convergent Charging is integrated with a policy or monitoring system, check the settings configured in this system before modifying or deleting spending status descriptions with Core Tool [page 7]. Modifying or deleting spending status descriptions may result in inconsistencies between the two systems because there is no replication or synchronization mechanism.

Use this procedure to delete a spending status description in SAP CC that is not referred by any monitoring plan [page 491] to configure the generation of a spending status [page 507].

To delete an spending status description

1. From the menu bar in the main window, select *File* > *Delete* > *Spending Status Description*. The *Delete Spending Status Description* dialog box opens.
2. Search [page 488] for the relevant spending status description, select it, and then click *OK*.
3. A deletion dialog box asks if you want to delete the spending status description.
4. Confirm the deletion of the spending status description by clicking *Yes*:
    1. If a dialog box confirms your choice, the spending status description has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the spending status description is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48]for more information.

1.33 Monitoring Plans

> ⓘ Note
>
> Monitoring plans are available in the pricing catalogs [page 109] in the master data [page 28]. You create monitoring plans only when SAP CC is connected to a policy control system and when the spending status monitoring [page 25] function is implemented in your system landscape.

A monitoring plan  defines the generation of spending statuses that you can monitor through a connected policy system (or monitoring system) during charging and refilling operations relating to an end customer. The plan defines:

- A list of spending status identifiers

- The monitored sources (counters)
- Some rules for determining their label values

> **Note**
>
> You configure the logic to convert the value of a counter stored in a contract into a label reported in a spending status by setting up a range table dedicated to this scope.

They are then combined and assigned to commercial products in a CRM application or a provisioning system and are finally activated in the provider contract [page 573] of an end customer of the service provider.

The monitoring plan consists of:

- One or several configurations of target spending statuses to be generated and reported by the SAP CC system.
- A set of parameters [page 499] to configure the monitoring plan. The default value is the name of a range table.
- A release status [page 497] that can be open, released, or obsolete.

> **Example**
>
> You implement policy control in your system landscape to ensure fair usage and to optimize the network bandwidth allocation with predefined policy rules. You can configure a monitoring plan 'Monitoring plan dedicated to data usage' [page 493] that defines the reporting of a spending status that indicates if the service quota is still valid.

Before Starting

Before creating [page 495] and setting up [page 496] a monitoring plan, you must be familiar with the following:

- Spending Status Description [page 483]
- Range tables [page 227] and range table classes [page 227] needed in the monitoring plans
- Version change [page 510] of a monitoring plan
- Preview and signature [page 510] of a monitoring plan

Prerequisites

You must at least:

- Configure one spending status description [page 486] in the SAP CC system
- Configure a range table and its class to define the conversion logic between monitored counters and reported spending statuses [page 227]

Integration

This master data is used in an interfaced CRM or provisioning system to model commercial products. Once the plan is released, its signature cannot not change anymore. For example, SAP CRM manages some monitoring plans as charge plans.

Versions

You can modify and save different versions of a monitoring plan [page 510].

Transport

To transport a monitoring plan between different system landscapes, you can:

- Manually maintain the counter name dictionary [page 86] in the target SAP CC system
- Transport the necessary spending status descriptions [page 483]
- Export a monitoring plan [page 505]
- Import a monitoring plan [page 506]

Related Information

Spending Status Monitoring [page 25]
Charge Plans [page 343]
Example of a Monitoring Plan [page 493]

1.33.1 Example of a Monitoring Plan

> ### Example
>
> An end customer subscribes to a mobile data plan allowing "unlimited data usage". To protect itself from customer abuse in network usage, the operator adds a clause to the small print of the contract stipulating that data usage with regular quality of service is unlimited until a specified consumption threshold.
>
> For more information see the application example [page 25]
>
> To model this business case in SAP CC, the service provider configures in the pricing catalog [page 109] of this customer service:
>
> - A charge plan [page 343] dedicated to define the pricing policy related to the usage of the data service:

    - You configure a reusable charge and customize it in the charge plan. The price plan of the reusable charge includes a usage rate that defines the pricing of the consumption of service data and that is based on a quantity of bytes
    - You declare and share a **DATA_USAGE** counter that is incremented by a quantity of bytes
- A monitoring plan "Monitoring plan dedicated to data usage" associated to the previously configured charge plan and that reuses the **DATA_USAGE** shared counter:
    - *Target spending status*: **Data fair usage**
    - *Counter in contract to monitor*: **DATA_USAGE**
    - *Conversion logic based on*: RT dedicated to monitoring [page 217] range table

To enable the monitoring of the **DATA_USAGE** counter, you create a monitoring plan and set up the generation of a spending status based on this counter as follows:

- Create [page 495] the monitoring plan
- Set up a parameter [page 499] and change its default value to the relevant range table:
    - Select the RTC dedicated to monitoring [page 200] range table class
    - Select the RT dedicated to monitoring [page 217] range table
- Set up the generation of a spending status from counter: [page 507]
    - Select the Data fair usage [page 485] spending status description
    - Select the **DATA_USAGE** counter name
    - Select the parameter previously defined and that refers to the RT dedicated to monitoring [page 217] range table
- Save the monitoring plan [page 501] to the back-end database
- Release [page 504] the monitoring plan

Related Information

Spending Status Monitoring [page 25]
Configuration of Master Data Step by Step [page 104]
Example of a Spending Status Description [page 485]

1.33.2 Working with Monitoring Plans

Related Information

Creating a Monitoring Plan [page 495]
Opening an Existing Monitoring Plan [page 502]
Modifying a Monitoring Plan [page 504]

Setting Up a Monitoring Plan [page 496]
Deleting a Monitoring Plan [page 505]
Saving a Monitoring Plan [page 501]
Releasing a Monitoring Plan [page 504]
Exporting a Monitoring Plan [page 505]
Importing a Monitoring Plan [page 506]
Setting Up the Tabs of a Monitoring Plan [page 497]

1.33.2.1 Creating a Monitoring Plan

Prerequisites

Before creating a monitoring plan [page 491], you must:

- Create the necessary spending status descriptions [page 486]
- Create the necessary range tables [page 227] and classes [page 206] dedicated to the configuration of monitoring plans

Use one of these procedures to create a monitoring plan [page 491].

To create a monitoring plan

1. From the menu bar in the main window, select *File* > *New* > *Monitoring Plan*. The *Choose Owner* window opens.
2. In the *Owner of Object* drop-down menu, select the owner of the monitoring plan.
3. Click *OK*. The *New Monitoring Plan* window opens.
4. *Set up* the monitoring plan. By default, its release status is set to *Open*.
5. Verify the monitoring plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Check the preview and signature [page 510] of the monitoring plan.
7. *Save* the monitoring plan.
8. In a development system landscape, you can test this plan.
9. Release [page 504] the monitoring plan.

You can create a monitoring plan by opening an existing monitoring plan and copying it.

To create a monitoring plan from an existing monitoring plan

1. From the menu bar in the main window, select *File* > *Open* > *Monitoring Plan*. The *Choose Monitoring Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the monitoring plan.
3. In the *Monitoring Plans* folder of the catalog of the owner, select the monitoring plan.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Monitoring Plan* window opens.
5. Rename the monitoring plan in the *Name* field, and then *Set up* the monitoring plan as necessary. By default, its release status is set to *Open*.
6. Verify the monitoring plan structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Check the preview and signature [page 510] of the monitoring plan.
8. *Save* the monitoring plan.
9. In a development system landscape, you can test this plan.
10. Release [page 504] the monitoring plan.

Related Information

Setting Up a Monitoring Plan [page 496]
Previewing a Monitoring Plan [page 510]
Releasing a Monitoring Plan [page 504]

1.33.2.2 Setting Up a Monitoring Plan

In the *Monitoring Plan* window, a monitoring plan is graphically represented by a tree, which consists of the following levels:

-  Plan root
- Counter monitoring node:
  -  Generated spending status from monitored source: counters in provider contracts that activate this monitoring plan in a contract item

You must create [page 495], open [page 502], or modify [page 504] a monitoring plan before setting it up. A *Monitoring Plan* window must be open.

To set up a monitoring plan

1. In the left pane of the *Monitoring Plan* window, click the monitoring plan root  .
2. In the right pane of the *Monitoring Plan* window, identify the monitoring plan. In the *Name* field, type the unique identifier of the monitoring plan if you create the monitoring plan. You cannot modify this field after saving the plan.
3. Set up the following tabs of the monitoring plan: Definition [page 497], Parameters [page 499], Additional Information [page 499], Technical Data [page 500].
4. If necessary, set up the generation of one or more spending statuses [page 507] from counters in the provider contract that relates to the end customer who is subject to be monitored.
5. If necessary, remove [page 366] one or more entries.

Related Information

Releasing a Monitoring Plan [page 504]
Setting Up the Generation of a Spending Status [page 507]

1.33.2.2.1 Setting Up the Tabs of a Monitoring Plan

Related Information

Monitoring Plan: Definition [page 497]
Monitoring Plan: Parameters [page 499]
Monitoring Plan: Additional Information [page 499]
Monitoring Plan: Technical Data [page 500]

1.33.2.2.1.1 Monitoring Plan: Definition

Create [page 495] or modify [page 504] a monitoring plan before setting it up [page 496]. In the right pane of the *Monitoring Plan* window, open the *Definition* tab.

The *Definition* tab allows you to set up the release status of the monitoring plan and gives a comprehensive description of the monitoring plan. You maintain a monitoring plan and manage its life cycle.

> **Note**
>
> - When you create a monitoring plan, its status is automatically set to *Open*. You must save the monitoring plan once before changing its status.
> - You can set the status to *Released* only if the current status of the monitoring plan is *Open*.
> - When the status of a monitoring plan is *Obsolete*, the status cannot be changed anymore; the status is final. The monitoring plan cannot be deleted and is kept for reference for provider contracts activating it in a contract item [page 632].

Definition tab

Fill in the *Definition* tab.

Status: Provides the list of release statuses. Select one of the following statuses:

- *Open*: Select this status if you are working on the monitoring plan. This status allows you to modify the monitoring plan and save it in the back-end database. However, the monitoring plan cannot be used by theSAP CRM system or by another interfaced provisioning system. You cannot save several versions [page 510] of the monitoring plan. An open plan cannot be activated in a contract item [page 632] of a provider contract subscribed to by an end customer.
- *Released*: Select this status when the monitoring plan is ready to be in production in the system landscape. The monitoring plan will not change anymore regarding to its data structure and to its signature [page 510]. The monitoring plan can be used by the SAP CRM system or another interfaced provisioning system. You are only allowed to change the value of parameters. When you modify and save the monitoring plan in the database, the old versions of the monitoring plan are always stored in the database.
- *Obsolete*: Select this status when the plan is still released but should not be used anymore or under restricted conditions. The monitoring plan can be used by the SAP CRM system or another interfaced provisioning system. You are allowed to do the same changes as with a released plan. When you modify and save the monitoring plan in the database, the old versions of the monitoring plan are always stored in the database. An obsolete plan can or cannot be combined in a commercial product depending on the integration that has been implemented between SAP CC and your SAP CRM or provisioning system. Refer to the documentation of this system.

> **→ Remember**
>
> When SAP CC is connected to SAP CRM, you cannot assign an obsolete plan to a new commercial product. Refer to the SAP CRM documentation for more information.

Description: Describe the monitoring plan and its scope by typing a detailed text if necessary. This comprehensive information is mandatory. The *SAP CRM* system or provisioning system can display the description during the configuration of commercial products based on this monitoring plan. Note that you cannot change the description of the monitoring plan if its status is *Released* or *Obsolete*.

Related Information

Setting Up the Generation of a Spending Status [page 507]

1.33.2.2.1.2 Monitoring Plan: Parameters

Create [page 495] or modify [page 504] a monitoring plan before setting it up [page 496]. In the right pane of the *Monitoring Plan* window, open the *Parameters* tab.

The *Parameters* tab allows you to create internal parameters that you use to configure the generation of spending statuses [page 507]. The parameter type is always Range Table ID [page 303]. Each parameter identifies a range table that includes the logic to convert a counter value to a spending status label.

> → Remember
>
> The default value cannot be redefined when configuring a provider contract.

Parameters tab

Use the table and its icons to create parameters.

 : Click to add a parameter row to the table.

 : Click to delete a parameter row from the table.

Name: Type the name of the parameter.

Description: Describe the function of the parameter.

Value: Click the cell of the *Value* column to open a *Choose range table* window. Select the relevant range table class [page 471] and select the appropriate range table [page 227]. This range table represents the logic used by the SAP CC system to convert a counter value into a spending status label.

Related Information

Setting Up the Generation of a Spending Status [page 507]
Using a Range Table in Monitoring Plans [page 227]
Parameters [page 303]

1.33.2.2.1.3 Monitoring Plan: Additional Information

Create [page 495] or modify [page 504] a monitoring plan before setting it up [page 496]. In the right pane of the *Monitoring Plan* window, open the *Additional Information* tab.

The *Additional Information* tab allows you to create information that is only descriptive and not reused by SAP CC at any stage of the business processes. You set up additional information to facilitate the data management in SAP CC or to configure data information relevant for external systems connected to the SAP CC system like a CRM, a provisioning, or a billing system.

> **Note**
>
> Once the monitoring plan is released [page 504], you cannot change the additional information settings anymore.

> **Example**
>
> For example, you can create a table with the author's name of the monitoring plan, the author's code, the creation date, and another information. If necessary, you can retrieve this information by using an external system.

Additional Information tab

Use the table and its icons to create additional information.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the SAP CRM system or an external provisioning system.

1.33.2.2.1.4 Monitoring Plan: Technical Data

Create [page 495] or modify [page 504] a monitoring plan before setting it up [page 496]. In the right pane of the *Monitoring Plan* window, open the *Technical Data* tab.

The *Technical Data* tab allows you to define types of technical identifiers under which an end customer consumes a service and under which a spending status monitoring can be requested. The appropriate identifiers of the end customer must be specified by the CRM or provisioning system. This identifier is used by the SAP CC system to manage a monitoring seance dedicated to both this end customer and this identifier. Note that the same identifier can be used for different monitoring seances.

> **Note**
>
> Once the monitoring plan is released [page 504], you cannot rename an existing technical data entry or change its comprehensive description.

> ### Example
>
> You can configure the following settings: phone number, email address, IP address, and carplate.

Technical Data tab

Use the table and its icons to create technical data.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in or modify each row that you add to the table.

User Technical Identifier Name: Type the name of the technical data. This name must be unique in the table. This name is visible from the SAP CRM system or the external provisioning system.

Description: Type the description of the technical data. This description is visible from the SAP CRM system or an external provisioning system.

1.33.2.3 Saving a Monitoring Plan

Once you have created [page 495] or modified [page 504] a monitoring plan, you are ready to save it.

Prerequisites

You have verified its data structure and checked its preview and signature [page 510].

Tasks

To save a new monitoring plan to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Monitoring Plan Creation* dialog box opens to inform you that the monitoring plan has been successfully saved.
4. Click *OK*.

To save a new version of an existing monitoring plan to the database

> **Note**
>
> If a *Warning* dialog box opens to inform you that the monitoring plan has been modified since you opened it, you can:
>
> 1. Click *Overwrite* and then *Yes* to overwrite the changes performed by another user. Clicking *No* enables you to come back to the monitoring plan.
> 2. Click *Cancel* to come back to the monitoring plan. See later in this chapter how to save the monitoring plan as an XML file.

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. A dialog box opens to ask if you want to save the monitoring plan.
2. Click *Yes*. The *Save Version* window opens.
3. In this window, select one of the following options:
   - *Create a new version*: The *Start Date* is automatically filled in with the current date while the *End Date* is automatically filled in with *No End Date*. You can however replace these dates by selecting them with the calendar or the arrow at the right of the field. Clicking the arrow displays the dates related to existing versions. In this list, *No Start Date* represents an undetermined date in the past while *No End Date* represents an undetermined date in the future. A message informs you when you overwrite versions.
   - *Replace Versions*: The existing versions are displayed from the latest to the oldest. If the list contains one version, this version is displayed as "Unique Version". You can replace one or more consecutive versions. A message informs you when you overwrite versions. Note that the list can display identical versions whose periods are different.
4. Click *Save*. The *Monitoring Plan Modification* dialog box opens to inform you that the monitoring plan has been successfully created.
5. Click *OK*.

Related Information

Exporting a Monitoring Plan [page 505]
Changing the Version of a Monitoring Plan [page 510]

1.33.2.4 Opening an Existing Monitoring Plan

You can open:

- An existing monitoring plan saved in the back-end database
- A previous version of a monitoring plan saved in the database
- An existing monitoring plan saved in an XML file

To open an existing monitoring plan saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Monitoring Plan* > The *Open Monitoring Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the monitoring plan.
3. In the *Monitoring Plans* folder of the pricing catalog of the owner, select the monitoring plan, and then click *Open*. The window of the corresponding monitoring plan opens in display mode [page 46].
4. You can view or modify the plan or change the version.

Once released [page 504], the monitoring plans are managed chronologically. After you have saved a monitoring plan, you can access its previous versions that exist in the database.

To open a previous version of a charge plan saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Monitoring Plan* > The *Open Monitoring Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the monitoring plan.
3. In the *Monitoring Plans* folder of the pricing catalog of the owner, select the monitoring plan, and then click *Open*. The window of the corresponding monitoring plan opens in display mode [page 46].
4. On the toolbar in the main window, click the *Change Version* icon  . The *Change Version* dialog box opens.
5. Select a version of the monitoring plan from the list.
6. Click *View*.

To open an existing monotoring plan saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> ⓘ Note
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File* >. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.

5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

Related Information

Changing the Version of a Monitoring Plan [page 510]
Modifying a Monitoring Plan [page 504]

1.33.2.5 Modifying a Monitoring Plan

The modifications that you can make to a monitoring plan differ depending on whether it has been saved or not.

To modify a monitoring plan

1. Open [page 490] the relevant monitoring plan.
2. Set the edit mode by clicking the *Edit Mode* icon  on the main toolbar. The monitoring plan may be reloaded if it has been modified since you have opened it.
3. Set up [page 496] this monitoring plan.
4. Verify its data structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
5. Check the preview and signature [page 510] of the monitoring plan.
6. Save [page 501] the changed monitoring plan to the back-end database.

To preserve the integrity of the back-end database, you cannot remove elements from the structure of a saved monitoring plan. You can only modify some parameter values.

1.33.2.6 Releasing a Monitoring Plan

Only released or obsolete monitoring plans can be combined in commercial products in the CRM or provisioning system.

> **Note**
>
> Refer to the notes [page 497] about the release status and the life cycle of a monitoring plan.

To release a monitoring plan

You change the release status by modifying a monitoring plan [page 504].

Related Information

Monitoring Plan: Definition [page 497]

1.33.2.7 Deleting a Monitoring Plan

Before you delete an monitoring plan, you must first delete all objects which reference it.

To delete a monitoring plan

1. From the menu bar in the main window, select *File* > *Delete* > *Monitoring Plan*. The *Delete Monitoring Plan* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the monitoring plan.
3. In the *Monitoring Plans* folder of the catalog of the owner, select the monitoring plan.
4. Click *Delete*. A deletion dialog box asks if you want to delete the monitoring plan
5. Confirm the deletion of the monitoring plan by clicking *Yes*:
    1. If a dialog box confirms your choice, the monitoring plan has been successfully deleted.
    2. If a *Warning* dialog box opens to inform you that the monitoring plan is currently being edited, you are not allowed to delete it. See the topic Concurrent Edition Management [page 48] for more information.

> ⓘ Note
>
> If an error message appears after clicking *Yes* at step 5, you must delete the object or objects that reference the monitoring plan. The identifiers of these objects appear in the message.

1.33.2.8 Exporting a Monitoring Plan

You can export monitoring plans to a directory of your hard disk, where they are saved as XML files.

Unlike the process of simply saving an object to a file, the export process saves all versions and associated data objects to a file, allowing you to transport it by importing the monitoring plan [page 506] back into another SAP CC system.

> **Note**
>
> This export operation does not export the necessary business data (counter names, spending status descriptions). Refer to the prerequisites [page 506] of the import operation for more information.

To export a monitoring plan

1. From the menu bar in the main window, select *File* > *Export*. The *Choose Object* dialog box opens.
2. In the *Owner* drop-down menu, select the owner of the monitoring plan.
3. In the catalog of the owner, select the monitoring plan.
4. Click *OK*. The *Save* dialog box opens.
5. Select a directory to export the monitoring plan.
6. In the *File name* field, type the name of the file.
7. Click *Save*.

Related Information

Spending Status Description [page 483]

1.33.2.9 Importing a Monitoring Plan

To transport data, you can import monitoring plans exported in XML format from a directory to the back-end database of another SAP CC system or to another SAP CC system landscape.

Several versions of a monitoring plan can exist in files corresponding to their stages of modification. You can import all versions of a monitoring plan into SAP CC, or select only the version that you want.

Prerequisites

- The same pricing catalog exists in the target SAP CC system
- The necessary spending status descriptions [page 483] exist
- The necessary counter names exist in the dictionary [page 86]

To import a monitoring plan

1. Log on and connect to the target SAP CC system.
2. From the menu bar in the main window, select *File* > *Import*. The *Open* dialog box opens.
3. Open the directory in which the monitoring plan is saved, and then select the monitoring plan file.
4. Click *Open*. The *Import Objects* dialog box opens.
5. Leave the default *All versions* option selected if you want to import all the versions of the imported objects, or do the following:
   - To select only one version as of a certain date, select the *One version per object* option, and then select the date in the *Version as Of* field.
   - To replace all versions by the imported version, select the *Replace all versions* option.
   - To replace or not some existing versions by the imported version, select the *Use modification date as of* option and choose the date on which you save the imported version. Depending on this date, some existing versions will be replaced or not by the imported version.
6. In the *Import* column, clear the checkboxes of objects you do not want to import, if necessary.
7. Click *Import*.

> **Note**
>
> SAP CC Core Tool can only import an object which has previously been exported [page 462] and saved (*File* > *Export*), rather than saved directly to a file (*File* > *Save* > *In a File*).
>
> This is because the export process saves all versions of an object and the objects attached to it, rather than just one version that lacks associated objects. The import process requires all versions and associated objects to be available.

1.33.3 Working with Spending Status Generation

Related Information

Setting Up the Generation of a Spending Status [page 507]
Definition Tab [page 509]

1.33.3.1 Setting Up the Generation of a Spending Status

Setting up the generation of a spending status means determining the shared counters (information source) to be exposed in spending status reports as the form of spending statuses. At least a charge plan (or a refill plan) must share a counter and must be activated in the same provider contract as the activated monitoring

plan. You set up the target spending statuses, the source counters, and the logic that defines how the SAP CC system converts a label value (for spending statuses) from a counter value.

Prerequisites

You must set up [page 496] a monitoring plan before setting up a the generation of one or several spending statuses. A *Monitoring Plan* window must be open.

The following configuration must be available:

- A spending status description [page 483] in the SAP CC system
- A range table [page 227] and its class [page 206] to define the conversion logic between monitored counters and reported spending statuses

To set up the generation of a spending status

1. In the left pane of the *Monitoring Plan* window, make sure that the monitoring plan root  is selected; otherwise, select it.
2. Configure the generation of a new spending status. From the menu bar in the main window, select *Action* > *Add Spending Status From Counter*, or click the icon  on the toolbar in the main window. You can also right-click the monitoring plan root  and click Add Spending Status From Counter. A new leaf appears in the monitoring plan tree.
3. In the tree, select the target spending status  you added, and then set up the tab of the spending status generation according to your business requirements.

To remove the target spending status

1. In the left pane of the *Monitoring Plan* window, make sure that the appropriate generated spending status  is selected; otherwise, select it.
2. Remove its generation configuration. From the menu bar in the main window, select *Action* > *Remove*, or right-click the generated spending status  and click *Remove*.

1.33.3.2 Definition Tab

Create [page 495] or modify [page 504] a monitoring plan before setting up the generation of a spending status [page 507]. In the left pane of the *Monitoring Plan* window, make sure that the relevant target spending status  is selected and that the right pane displays the *Spending Status Generation* tab.

The *Spending Status Generation* tab allows you to set up the generation of a spending status in spending status reports: Source information (a shared counter in a provider contract), conversion logic, target spending status. At runtime, the SAP CC system converts and reports some counter values in the form of spending statuses with labels.

Definition tab

Fill in the *Definition* tab.

Spending Status Description: Provides the list of spending status descriptions [page 483] configured in SAP CC. Select the appropriate spending status description to identify a new generated spending status that can be reported when this monitoring plan is activated in a provider contract. Click  if you want to view the selected spending status description.

Counter Name: Provides the list of counter names configured in the counter dictionary [page 86]. Select the name of the counter [page 603] to be monitored. This counter must exist and be shared within the provider contracts that activate this monitoring plan.

> **Note**
>
> You must set up a charge plan or a refill plan that enables the counter sharing [page 353] of a counter with the same name.

Range Table: Provides the filtered list of parameters defined in the monitoring plan [page 499]. Select the appropriate parameter. The value of this parameter is the dedicated range table that includes the logic to convert counter values into spending status labels that are reported to a policy system or monitoring system.

Related Information

Setting Up a Monitoring Plan [page 496]

1.33.4 Using the Action Menu

Related Information

Changing the Version of a Monitoring Plan [page 510]
Deleting the Versions of a Monitoring Plan [page 510]
Previewing a Monitoring Plan [page 510]

1.33.4.1 Changing the Version of a Monitoring Plan

The procedure is the same as for changing the version of an object (as explained below).

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> ⓘ Note
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.33.4.2 Deleting the Versions of a Monitoring Plan

The procedure is the same as for deleting the versions of a catalog object [page 118].

1.33.4.3 Previewing a Monitoring Plan

This procedure allows you to preview a monitoring plan to check its signature. The CRM or provisioning system, interfaced with the SAP CC system, uses this signature to model commercial products.

To perform this procedure, you must open [page 502] a monitoring plan.

To preview a monitoring plan

1. From the menu bar in the main window, select *Action* > *Preview*, or click the *Preview* icon on the toolbar in the main window.
2. The *Preview* window opens and displays a summary of the monitoring plan: Target spending statuses, monitored counters, and expected technical data. This information will be seen by the interfaced CRM or provisioning system. Read and check this signature.
3. Click *Close*.

Integration

If you implement the *Web Services* technical interface between your CRM application (or a provisioning system) and SAP CC, the preview result is the same as the result of the *Find Monitoring Plan Classes* operation. The operation response includes the same information (see the `monitoringPlanClassFindResponse` message type).

Refer to the *Web Services Documentation* in the user assistance for more information about this service operation in the *Catalog Management* process component (Web service).

Related Information

Previewing a Charge Plan [page 358]
Previewing a Refill Plan [page 439]

1.34 Subscriber Accounts

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this master data is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. Every reference or operation related to prepaid accounts is no more available. For further information, refer to SAP Note 2772452.

> ⚠ Caution
>
> **Do not modify or delete subscriber accounts with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Creating, modifying, or deleting subscriber accounts **may result in inconsistencies** between the two systems because of data replication and synchronization mechanisms. To modify a subscriber account, you must use this CRM or provisioning system in your SAP system landscape.

- Consult the Application Help for more information about the *Customer Data Provisioning Process*.
- Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this customer master data.

The manual procedures detailed in this section are relevant in:

- A production system landscape with few contracts directly managed with Core Tool (no CRM interfaced with SAP CC)
- A test, demo, or training system landscape

Note

If you use provider contracts [page 573], the needed subscriber accounts are automatically created from outside SAP Convergent Charging. For test, demo, or training purpose you can configure master data related to end customers. You need the Customer Sales Representative (CSR) role [page 641] in your SAP CC user settings.



A **subscriber account** in SAP Convergent Charging (SAP CC) is a customer master data owned by a subscriber (party or individual) as the result of commercial agreements with a contracting service provider for the use of its chargeable services. A subscriber account contains all the prepaid accounts and external accounts that are debited when the subscriber or allowed end customers consume these digital services or credited when the end customers make refills. In the environment of SAP CC, a subscriber account is referenced by provider contracts and subscriber tables (data lookup tables) that contain various prices and other data for your business.

A subscriber account is part of the customer master data managed by SAP Convergent Charging as an extension of the *SAP Master Data* in your SAP system landscape.

In rare implementation scenarios, the subscriber accounts must be manually created by the SAP CC users granted the customer sales representative (CSR) role.

Before creating provider contracts [page 573], you must create at least one subscriber account. Each existing provider contract is related to one and only one subscriber account.

Caution

On a platform where SAP CC is integrated with SAP CRM or with another provisioning system, subscriber accounts, prepaid accounts, and external accounts are automatically created and maintained. Creating, modifying or deleting subscriber accounts may result in inconsistencies between these two systems. SAP SE strongly recommends that you do not modify or delete existing subscriber accounts except under support supervision.

A subscriber account is defined by:

- The **identifier of the subscriber account** (mandatory)
- The **identifier of the subscriber** who signed the business agreements with the service provider company for the chargeable services or physical goods. This subscriber and a group of end customers can consume the services or use a product whose pricing is configured in SAP CC.
- The **identifier of the pricing catalog and contracting service provider** (mandatory). This identifier is the name of the pricing catalog owned by the service provider. This catalog contains all the pricing elements that are combined together to form charge plans that are activated in subscribed provider contracts.

- The **customer management area (CMA) [page 87]** (optional) that is responsible for this subscriber account.
- A few **accounts**:
  - Prepaid accounts [page 514] (optional)
  - External (postpaid) accounts [page 520] (optional)
- A tax system and some taxation settings [page 523] (optional)
- Some additional information items (optional)

A subscriber account references a high volume of:

- Provider contracts [page 573] (optional), which belong to the subscriber account.
- Subscriber mapping tables [page 542] (optional)
- Subscriber range tables [page 557] (optional)

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

How an Account Is Debited or Credited

A subscriber account is linked to the charging plan (policy) [page 305] of a charge through an account assignment in provider contracts. When an end customer subscribes to a charge in a provider contract, the charging plan of the charge together with the charging mapping references specified in the account assignment in the provider contract allow the system to identify which account of the subscriber account will be debited or credited with the amount calculated in the price plan.

> Example
>
> In SAP Convergent Charging, the charging process consists usually of two stages: A dynamic pricing (aka rating) process and an account determination process.
>
> When an end customer consumes a service, the pricing process calculates the corresponding fees while the second process defines which prepaid and external accounts will be debited with the fees and calculates the distribution of the fees between these accounts.

Before Starting

Before creating [page 527] and setting up [page 528] a subscriber account, you should be familiar with:

- Tax Settings [page 523]
- Prepaid Accounts [page 514]
- External Accounts [page 520]
- Subscriber Mapping Tables [page 542]
- Subscriber Range Tables [page 557]

Integration

This customer master data must be created and maintained by an external system connected to the SAP CC system during the provisioning process. Depending on your business needs, replication, distribution, and synchronization processes may be required.

> → Remember
>
> In an integrated *SAP Solution* scenario, the subscriber accounts are replicated from another master data. Refer to the integration guide that relates to your implementation scenario.
>
> Several scenarios and scenario variants exist.
>
> In an integrated *SAP Solution* scenario based on Subscription Order Management (SOM) in SAP S/4HANA, subscriber accounts are replicated from business partners that are assigned the MKK role.
>
> In an integrated *SAP Solution* scenario with SAP ERP, the ERP system transfers the business partner, contract account, and prepaid account data to the SAP CC systems.

Transport

You cannot transport customer master data with the Core Tool user interface. It is a *Customizing* activity and need to be implemented in your productive SAP system landscape.

Depending on your SAP system landscape and business/technical requirements, master data distribution, replication, or synchronization processes must be implemented.

1.34.1 Prepaid Accounts

> ⚠ Caution
>
> **Do not modify or delete prepaid accounts with Core Tool** when SAP Convergent Charging is integrated with a CRM application or a provisioning system.
>
> Creating, modifying, or deleting prepaid accounts may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. Refer to the caution about subscriber accounts.



A **prepaid account** is an account managed and stored in SAP Convergent Charging. It is mainly characterized by a balance that represents a monetary counter associated to a currency. Funds are placed in the account before the service begins, and then the charges linked to the services consumed by the subscribers or agreed end customers are debited as much as necessary.

Balance

The prepaid account balance is a monetary counter associated to a currency. Funds are placed in the account before the service begins, and then the charges linked to the services consumed by the subscribers (or agreed end customers) are debited as much as necessary.

The balance of an account is defined by a minimal value (threshold). Depending on service providers' requirements, the threshold is usually close to the value of 0. Usage is rejected when the balance reaches the threshold. However, if overspending (refer to the note below) is authorized for specific usage, the threshold is ignored, and the usage can be debited from the account even though the balance of the account is less than the threshold.

Crediting a fee to a prepaid account consists in increasing the amount of its balance. Conversely, debiting a fee consists in decreasing the amount of its balance.

Prepaid Account Threshold

The prepaid account threshold is the minimal value of a balance ensuring that the account is sufficiently funded to pay an account fee, for example, or to reserve money in order to limit the processing error of the vendor. When the threshold is reached, usage or events are rejected by the system.

Alerts

A prepaid account can define some alerts based on times or amounts and that are automatically triggered by the system. A notification is sent to the dedicated subsystems or end customers. See Prepaid Account: Alerts [page 519].

Taxes

Depending on the ways you allow the end customers to make a refill (credit pre-purchased or cash advance), taxes may have been paid or not.

The holder of this account can be subject to pay the tax according to the tax settings configured in SAP Convergent Charging.

States of a Prepaid Account and Expiration Schedule

A prepaid account can be:

- *Active*: The prepaid account is charged depending on how you configure the default or internal reference components in the related charging plan.

- The active prepaid account can be debited until the threshold is reached.
- The prepaid account can be refilled.
- If overspending is authorized, debiting is possible.

You can block, lock, or close an active prepaid account.

- *Blocked*: The prepaid account is charged depending on how you configure the default or internal reference components in the related charging plan.
  - The blocked prepaid account cannot be debited (only "free" usage is accepted).
  - The blocked prepaid account can however be refilled.
  - If overspending is authorized, debiting is possible.

  You can reactivate, lock, or close a blocked prepaid account.
- *Locked*: The prepaid account is charged depending on how you configure the default or internal reference components in the related charging plan.
  - The locked prepaid account cannot be debited ("free" usage is not accepted).
  - Refills and overruns are authorized.
  - If overspending is authorized, debiting is possible.

  You can reactivate, block, or close a locked prepaid account.
- *Closed*: The prepaid account is charged depending on how you configure the default or internal reference components in the related charging plan.
  The closed prepaid account cannot be debited.
  Refills and overruns are forbidden.
  Once closed, you cannot change a prepaid account anymore.

You can define an expiration schedule [page 517] that specifies a workflow for the states of a prepaid account.

Similar Concept

Allowances [page 575]

See Also

Free Function [page 730]

Related Information

Setting Up a Prepaid Account [page 517]
Prepaid Account: Definition [page 517]
Prepaid Account: Alerts [page 519]
Free Function [page 730]

1.34.1.1 Working with Prepaid Accounts

Setting Up a Prepaid Account [page 517]

Refilling a Prepaid Account [page 519]

Related Information

Setting Up a Prepaid Account [page 517]

1.34.1.1.1 Setting Up a Prepaid Account

> ⚠ Caution
>
> Do not modify a prepaid account created by a CRM application or another provisioning system interfaced with SAP CC. Refer to the caution [page 511] about subscriber accounts.

In a prepaid environment scenario, use this procedure to set up a prepaid account [page 514] in the subscriber account for the belonging subscriber.

Set Up a Prepaid Account

1. In the left pane of the *Subscriber Account* window, click the prepaid account  to set up.
2. In the right pane of the subscriber account window, identify the prepaid account.
   - In the *Name* field, type the identifier of the prepaid account. You cannot modify this field.
   - In the *Description* field, describe the function of the prepaid account if necessary.
3. Set up the following tabs of the prepaid account:
   - Definition [page 517]
   - Alerts [page 519]
   - *Tax*

1.34.1.1.1.1 Prepaid Account: Definition

Create [page 527] or modify [page 540] a subscriber account and then set it up [page 528] before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Definition* tab, you define the amount of the prepaid account and its minimal amount. You define the currency and the account that will be debited when the prepaid account is empty. If necessary, you set up the schedule, which can be combined with alerts.

> **Note**
>
> The *Balance* and the *Currency* fields can be modified only when the subscriber account is being created. Once you have saved the subscriber account, these fields appears dimmed (unavailable). The balance is modified when it is charged for service consumption or when you refill it.

Definition (Tab)

Fill in or modify the *Definition* tab.

External Account Code: Type the code of the external account. In an integrated *SAP Solution* scenario with SAP Convergent Invoicing, it represents the business partner number (GPART).

External Account Name: Type the name of the external account. In an integrated *SAP Solution* scenario with SAP Convergent Invoicing, it represents the contract account number (VKONT).

Client: Type the name of the holder of the prepaid account, if necessary. This can be the subscriber of the subscriber account [page 530] (for example, the end customer of the service provider) or another person. This person can be subject to pay the tax according to the tax settings.

Balance: Type the initial amount of the prepaid account.

Threshold: Type the minimal amount of the prepaid account. Service consumption can be rejected by SAP Convergent Charging if the threshold is reached.

Currency: Select the currency of the prepaid account.

Overrun: Provides the list of the other prepaid or external accounts of the subscriber account. Select another account to which the prepaid account will be linked. The selected account will be debited when the balance of the prepaid account is not sufficient. For example, if the balance amounts to $50 with a threshold of $10 and if the amount to be debited from the balance is $70, the balance will be debited with $40 while the selected account is debited with the remaining amount of $30.

> **Note**
>
> The overrun function is deactivated when the prepaid account is closed. If no account is selected, service consumption is rejected by SAP Convergent Charging when the threshold is reached.

Schedule: Allows you to schedule the states of the prepaid account [page 514]. You can create alerts [page 519] that will be triggered depending on the schedule. A schedule can be postponed when making a refill [page 519].

- *Blocked on*: Click the calendar button , and then select a date on which the prepaid account will be blocked.
- *Locked on*: Click the calendar button , and then select a date on which the prepaid account will be locked.
- *Closed on*: Click the calendar button , and then select a date on which the prepaid account will be closed.

1.34.1.1.1.2 Prepaid Account: Alerts

Create [page 527] or modify [page 540] a subscriber account and then set it up [page 528] before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

You can define two types of alerts for a prepaid account:

- Alerts depending on an amount
- Alerts depending on time

Alerts (Tab)

Fill in or modify the *Alerts* tab.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Amount Alerts: They are displayed in a table. Each row of the table represents an amount alert defined by a code and an amount. A notification is sent to a dedicated system when the prepaid account is less than or equal to the threshold value of an alert. The currency used in alerts is that of the prepaid account.

- *Code*: Type a code for the alert. The code is used to mark the alert notification.
- *Threshold*: Type the amount triggering the alert.

Expiration Alerts: An alert is triggered a number of days before or after one of the dates defined in the schedule [page 517]. For example, the alert "alert1_prepaid" is triggered two days before the prepaid account is locked in order to warn the user that the account is due to expire soon; and therefore, the user must refill the account.

- *Code*: Type a code for the alert. The code is used to identify the alert notification.
- *Triggering time*: Type a number of days.
- *Triggering mode*: Select if the alert is triggered before or after the reference state changes.
- *State*: Select one of the following states: *Active*, *Blocked*, *Locked*, or *Closed*.

1.34.1.1.2 Refilling a Prepaid Account

You cannot manually refill a prepaid account from the SAP CC 2023 Core Tool user interface, but SAP Convergent Charging provides functions and operations that can be implemented in your SAP system landscape to provide refilling services on a prepaid account in a subscriber account.

Consult the SAP CC Administration Guide about the necessary *Customizing* activities: Implementation of dedicated service operations from the *Web Services* technical interface, dedicated configuration of pricing master data of a service provider (see refill plan [page 421]).

> **Note**
>
> When SAP CC is connected to SAP CRM or SAP ERP/FI-CA, refill functions are available. Refer to the documentation of these software components.

Prerequisites

Your convergent charging services are based on provider contracts.

Related Information

Prepaid Accounts [page 514]

1.34.2 External Accounts

> **Caution**
>
> **Do not modify or delete external accounts with Core Tool** when SAP Convergent Charging is integrated with a CRM application or a provisioning system.
>
> Creating, modifying, or deleting external accounts may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. Refer to the caution about subscriber accounts.


An **external account** is a reference to an account outside SAP Convergent Charging and managed by a third-party invoicing and billing system or accounting system. The external account contains the ID of this system.

For example, an external account represents a postpaid account owned by the subscriber (agreement signatory party), the end customer (service consumer), or another party. Invoicing and billing are also processed outside SAP CC.

The currency of the external account must be the same as the currency of the postpaid account or billing account.

Account Types

SAP CC verifies the compatibility between account types and business transactions in real time. To be checked in real time, an external account can be assigned one of the following types:

- **Payable**: An account payable is an account by which a service provider owes a partner money (and sends settlement reports to the partner).
- **Receivable**: An account receivable is an account by which a client owes a service provider money (and receives invoices from the service provider).

By default, no type is assigned to the external account and the system does not check the compatibility between account types and business transactions in real time.

Taxes

The holder of this account can be subject to pay the tax according to the tax settings configured in SAP Convergent Charging.

You can define the taxation policy to be applied to the external account. If you do not fill in the *Tax* tab of the account, the taxation policy [page 534] defined in the subscriber account applies.

Related Information

Setting Up an External Account [page 521]
External Account: Definition [page 522]

1.34.2.1 Working with External Accounts

Related Information

Setting Up an External Account [page 521]

1.34.2.1.1 Setting Up an External Account

> ⚠ Caution
>
> Do not modify an external account created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber accounts [page 511].

In a postpaid environment scenario, use this procedure to set up an external account [page 520] that references an account in an external billing system or accounting system.

Set Up an External Account

1. In the left pane of the *Subscriber Account* window, click the prepaid account  to set up.
2. In the right pane of the subscriber account window, identify the prepaid account.
   - In the *Name* field, type the identifier of the external account. You cannot modify this field.
   - In the *Description* field, describe the function of the external account if necessary.
3. Set up the following tabs:
   - Definition [page 522]
   - *Tax*

1.34.2.1.1.1 External Account: Definition

Create [page 527] or modify [page 540] a subscriber account and then set it up [page 528] before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Definition* tab, you define the amount of the external account and its minimal amount. You define the currency and the account that will be debited when the external account is empty. If necessary, you set up the schedule, which can be combined with alerts.

Definition (Tab)

Fill in or modify the *Definition* tab.

External Account Code: Type the identification code of the account in the external billing system (or accounting system). Note that this code is used by SAP Convergent Charging to mark the charged transactions that are assigned to this account.

> → Tip
>
> In an integrated *SAP Solution* scenario with SAP Convergent Invoicing, it represents the business partner number (GPART).

External Billing System: Type the ID of the external billing system that manages the billing account represented by this external account in the subscriber account.

Currency: Select the currency of the external account. It must be the currency of the referenced account in the external billing system or accounting system.

Control charged transactions compatibility: If you select this account option, SAP Convergent Charging checks in real time the compatibility between account types and transactions.

> ⓘ Note
>
> In the external billing system, the compatibility is based on the account type: receivable or payable.

In the definition of the charge customized in the charge plan [page 368], the charged transactions (aka charged items) are respectively marked *Client* for a receivable account and *Partner* for a payable account.

Therefore, the charged transactions are compatible with the external account if the account type is the same. If the charged transactions are not compatible, the charging request will not be completed, and the charged transaction will be rejected, triggering a charging exception at mediation level.

If you unselect this option, the type of the external account is set to *Undefined*, and no compatibility is checked by the system.

- *Account Type*: Select *Receivable* or *Payable* as the type [page 520] of the external account.

1.34.3 Tax Settings

SAP Convergent Charging provides tax computation. To do so:

- The SAP CC administrator (system or application) must set up the SAP CC system with required tax systems as a prerequisite for the configuration of the SAP Master Data.
- The SAP CC pricing specialist must set up some tax information in the pricing catalog [page 109] of the service provider.
- Some tax information must be set in the subscriber accounts [page 511]. In most implementation and integration scenarios, this data is replicated into SAP CC from an external system or application.

→ Remember

In a pricing catalog (in master data):

An SAP CC pricing specialist defined the tax information in the customized charges that represent the tax settings for the consumption of a customer service.

In the charge customized [page 374] of a charge plan, your SAP CC pricing specialist defined if the prices in its price plan are net or gross (the tax is removed from or added to the base amount). This specialist defined the invoicing tax system that applies to this customized charge and relevant options:

EU VAT System	• Base amount (net/gross) • Tax rate level • VAT rule and place of establishment of the service provider
Flat Tax System	• Base amount (net/gross) • Flat tax rate

US Telecommunications taxes • By Avalara AvaTax for Communications (AFC) Engine/System	• Taxing jurisdiction information (origination, termination, service address) [page 381] • Customer information (customer type, incorporated city area, exemption, service class, business class, lifeline) [page 382] • Company information (reseller, franchise) [page 383] • Transaction information (number of lines, of locations) [page 383] • Transaction types and service types (and pairs of transaction/service types) [page 384]

Tax Settings in Subscriber Accounts

- In customer master data:
  In most implementation scenarios, this master data (subscriber account with its prepaid/external accounts) is replicated into SAP CC from an interfaced system.
  If in your implementation scenario, the creation of subscriber accounts is manual, then, as a customer sales representative (CSR), you define the taxation type (VAT, US, or none) and the taxation mode (who pays the tax).
  See: Subscriber Account: Tax [page 534]

> **Note**
>
> - If you select a tax system (EU VAT or US), the tax is calculated after the rating process. For more information, consult the Application Help about the tax calculation process.
> - Taxes do not apply to refill plans and refill record classes [page 414].

The table below shows you the main settings regarding the computation of taxes. This table does not describe all the possible combinations that you can set up. If you choose a combination which is not authorized by SAP Convergent Charging, the charging process is not performed. For example, if you choose the VAT taxation type in the subscriber account and the US Avalara AvaTax for Communications (AFC) system in the charge of a charge plan, the charging process is not performed for the service consumption relating to this subscriber account.

Charge in a Charge plan				Subscriber Account			Postpaid Charged Item [page 254]/Prepaid Charged Item		
Tax system	**Base amount**	**VAT rule**	**Tax rate level**	**Taxation type**	**Taxation mode**	**Taxation zone**	**Amount**	**Tax amount**	**Total amount**
None	Net or Gross		From -1 to 5	None	None	NO_TAX	Base amount	0	Base amount
EU VAT	Net or Gross		From -1 to 5	VAT	None	NO_TAX	Base amount	0	Base amount

Charge in a Charge plan				Subscriber Account			Postpaid Charged Item [page 254]/Prepaid Charged Item		
EU VAT	Net or Gross		From -1 to 5	VAT	SPSP[61]	BY_RULE	Base amount	0	Base amount
EU VAT	Net or Gross		From -1 to 5	VAT	SPSP	Any country	Base amount	Tax	Base amount + tax
EU VAT	Net or Gross		From -1 to 5	VAT	SPSP	Any country	Base amount - tax	Tax	Base amount
Flat Tax	Net or Gross	N/A	N/A	VAT	SPSP	Any country (Information not used)			
US Telco (by Avalara AvaTax for Communications (AFC))	N/A	N/A	N/A	US	SPSP	Avalara AvaTax for Communications (AFC)-AREA	Base amount	Tax	Base amount + tax

Subscriber Account			Charge in a Charge Plan			Postpaid Charged Item [page 254]/ Prepaid Charged Item		
Taxation type	**Taxation mode**	**Taxation zone**	**Taxation type**	**Base amount**	**Tax rate level**	**Amount**	**Tax amount**	**Total amount**
None	None	NO_TAX	None	Net or Gross	From -1 to 5	Base amount	0	Base amount
VAT	None	NO_TAX	VAT	Net or Gross	From -1 to 5	Base amount	0	Base amount
VAT	SPSP	BY_RULE	VAT	Net or Gross	From -1 to 5	Base amount	0	Base amount
VAT	SPSP	Any country	VAT	Net	From -1 to 5	Base amount	Tax	Base amount + tax
VAT	SPSP	Any country	VAT	Gross	From -1 to 5	Base amount - tax	Tax	Base amount

[61] Service Provider Subject to Pay

Subscriber Account			Charge in a Charge Plan			Postpaid Charged Item [page 254]/ Prepaid Charged Item		
US	SPSP	Avalara AvaTax for Communications (AFC)-AREA	US	N/A	N/A	Base amount	Tax	Base amount + tax

Example

Your company offers a service called *Bilingual Dictionaries Online* which allows end customers to purchase a subscription to an online dictionary. When purchasing a subscription, the end customers receive an email to confirm that their subscription is ready for use. They are then be able to login to their subscription by going to the 'Bilingual Dictionaries Online' site.

Each bilingual dictionary in 'Bilingual Dictionaries Online' is available separately by annual subscription. A 3-month subscription to one language costs £15 + VAT and a 12-month subscription to one language costs £45 + VAT.

In a price plan called `Bilingual Dictionaries Online`, you use a Flat Function component to charge any customer £45 for each 12-month subscription to a bilingual dictionary. According to the table above, if you choose a VAT taxation type of 19,6% ("Normal rate") and 'GB' as the country, the results for a 12-month subscription are the following:

Subscriber Account			Charge in a Charge Plan			Price Plan	Postpaid Charged Item/Prepaid Charged Item		
Taxation type	**Taxation mode**	**Taxation zone**	**Taxation type**	**Base amount**	**Tax rate level**	**Base amount (*)**	**Amount**	**Tax amount**	**Total amount**
None	None	NO_TAX	None	Net or Gross	0	£45	£45	0	£45
VAT	None	NO_TAX	VAT	Net or Gross	0	£45	£45	0	£45
VAT	SPSP	BY_RULE	VAT	Net or Gross	0	£45	£45	0	£45
VAT	SPSP	GB	VAT	Net	0	£45	£45	£8,82	$53,82
VAT	SPSP	GB	VAT	Gross	0	£45	£37,63	£7,37	$45
US	SPSP	Avalara AvaTax for Communications (AFC)-AREA	US	N/A	N/A	£45	£45	Tax	$45 + tax

Note (*): In this example, **Base amount** is the price for a 12-month subscription that you type in the *Fixed Amount* field of the Flat Function component.

See also:

- Recurring Rate [page 693]
- Flat Function [page 729]

Additional Information

Consult the Tax Calculation section of the SAP CC 2023 Application Help documentation for more information about the tax calculation feature and process.

Consult the SAP CC 2023 Administration Guide for more information about the *Customizing* activities relevant for the tax management services.

1.34.4 Working with Subscriber Accounts

Related Information

Creating a Subscriber Account [page 527]
Finding a Subscriber Account [page 537]
Opening an Existing Subscriber Account [page 539]
Modifying a Subscriber Account [page 540]
Setting Up a Subscriber Account [page 528]
Saving a Subscriber Account [page 536]
Deleting a Subscriber Account [page 541]
Reporting a Subscriber Account [page 540]
Using the Action Menu [page 541]

1.34.4.1 Creating a Subscriber Account

> **Note**
>
> If you use provider contracts [page 573], the needed subscriber accounts are automatically created from outside SAP Convergent Charging.

Use this procedure to create a subscriber account.

To create a subscriber account

1. From the menu bar in the main window, select *File* > *New* > *Subscriber Account*. The *Choose Owner* dialog box opens.
2. In the *Owner of Object* drop-down menu, select the owner of the subscriber account.
3. Click *OK*. The *New Subscriber Account* window opens.
4. Set up [page 528] the subscriber account.
5. Verify the subscriber account structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 536] the subscriber account.

> ⚠ Caution
>
> Do not manually create a subscriber account if SAP Convergent Charging is already interfaced with a CRM or an external provisioning system responsible for the management of subscriber accounts.

1.34.4.2 Setting Up a Subscriber Account

Create [page 468], open [page 474], or modify [page 475] a subscriber account before setting it up. A *Subscriber Account* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

In a *Subscriber Account* window, a subscriber account is graphically represented by a tree, which can consist of prepaid accounts and external accounts depending on the business requirements of the subscriber account.

Set Up a Subscriber Account

1. In the tree display area of the *Subscriber Account* window, make sure that the subscriber account icon is selected; otherwise, select it.
2. In the right pane of the subscriber account window, identify the subscriber account.
   - In the *Name* field, type the unique identifier of the subscriber account.
   - In the *Description* field, describe the function of the subscriber account if necessary.
3. Set up the tabs of the subscriber account:
   - Definition [page 530]
   - Contracts [page 531]
   - Mapping Tables [page 531]

- Range Tables [page 532]
- Additional Information [page 533]
- Default Charging [page 534]
- Tax [page 534]

4. If necessary, add [page 529] one or more prepaid accounts, and then set them up [page 517].
5. If necessary, add [page 529] one or more external accounts, and then set them up [page 521].

> **Note**
>
> You can remove [page 530], cut, copy, and paste a component of a subscriber account tree by right-clicking the component and selecting *Cut*, *Copy*, or *Paste*.

1.34.4.2.1 Adding a Component to a Subscriber Account

You can add one or more  prepaid accounts,  external accounts, or  credit limit balances to the nodes of the subscriber account tree. Prepaid accounts and external accounts are the leaves of the tree.

Add a Component to a Subscriber Account

1. In the subscriber account tree, click the  root or the node which you want to add a component to.
2. On the *Components* toolbar, click the component that you need to add to the tree. You can select one of the following components:
   -  *Prepaid Account*
   -  *External Account*

   The component is automatically added to the corresponding node.
3. Set up the component.

> **Note**
>
> You can remove [page 530], cut, copy, and paste a component of a subscriber account tree by right-clicking the component and selecting *Cut*, *Copy*, or *Paste*.

Related Information

Setting Up a Subscriber Account [page 528]

1.34.4.2.2 Removing a Component

> ⚠ Caution
>
> Do not delete a prepaid account created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber accounts [page 511].

Use this procedure to remove a component from a subscriber account tree.

Remove a Component from a Subscriber Account

1. In the subscriber account tree, click the component that you want to remove.
2. From the menu bar in the main window, select *Action* > *Remove Component*[62]. The component is removed.

> ⓘ Note
>
> You can cut, copy, and paste a component of a subscriber account tree by right-clicking the component and selecting *Cut*, *Copy*, or *Paste*.

1.34.4.2.3 Subscriber Account: Definition

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Definition* tab, you identify the subscriber account [page 511] and view the identifier of the pricing catalog [page 109] owner.

Definition (Tab)

Fill in or modify the *Definition* tab.

Subscriber: Type an account identifier for the subscriber who contracts a commercial aggreement with the service provider company that offers chargeable services or products. The subscriber is a person or a party. A subscriber can use or not these chargeable services (or products).

Contracting Service Provider: Displays the identifier of the pricing catalog owner [page 109].

[62] You can also right-click to open the shortcut menu, and then select *Remove*.

Customer Management Area: Displays the name of the Customer Management Areas [page 87] to which this subscriber account is assigned to. This field is only visible in an SAP system landscape that includes one or more SAP ERP/FI-CA systems and SAP CRM systems in an integrated *SAP Solution* scenario.

1.34.4.2.4 Subscriber Account: Contracts

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Contracts* tab, you can view, create, modify, and open provider contracts [page 573] relating to the subscriber account.

Contracts (Tab)

Fill in or modify the *Contracts* tab.

> ⓘ Note
>
> You must save the new subscriber account before creating a new provider contract.

 : Select a provider contract in the list, and then click the icon to open the provider contract.

 : Click to create a new provider contract that will automatically be linked to the current subscriber account.

 : Click to refresh the list of provider contracts.

1.34.4.2.5 Subscriber Account: Mapping Tables

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Mapping Tables* tab, you can view, create, or open subscriber mapping tables [page 542] relating to the subscriber account.

Mapping Tables (Tab)

The *Mapping Tables* tab lists all the subscriber mapping tables that belongs to the displayed subscriber account.

Code: The identifier of a subscriber mapping table that belongs to the subscriber account.

Possible actions are:

Icon	Action
	Select a subscriber mapping table in the list and then click  or double-click the row to open and view it.
	Click  to configure a new subscriber mapping table which will automatically be linked to the current subscriber account. See: Creating a Subscriber Mapping Table [page 545]
	Click  to refresh the list of subscriber mapping tables. This icon is especially used after you create a new subscriber mapping table in the subscriber account window.

View a Subscriber Mapping Table

Select a subscriber mapping table in the list and then click  or double-click the row to open and view it.

Create a Subscriber Mapping Table

Click  to configure a new subscriber mapping table which will automatically be linked to the current subscriber account. See: Creating a Subscriber Mapping Table [page 545]

1.34.4.2.6 Subscriber Account: Range Tables

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

On the *Range Tables* tab, you can view, create, or open range tables [page 213] relating to the subscriber account.

Range Tables (Tab)

Fill in or modify the *Range Tables* tab.

: Select a subscriber mapping table in the list and then double-click to open it.

: Click to create a new subscriber mapping table which will automatically be linked to the current subscriber account.

: Click to refresh the list of subscriber range tables. This icon is especially used after you create a new subscriber mapping table in the subscriber account window.

1.34.4.2.7 Subscriber Account: Additional Information

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

The *Additional Information* tab allows you to define some custom information that is only descriptive and not reused by SAP Convergent Charging at a stage of the process.

> ### Example
>
> For example, additional information can be the author's name of the subscriber account, the author's code, the creation date/time, and another custom information or reference. If necessary, you can retrieve this information for use in an external system.

> ### Note
>
> In an integrated *SAP Solution* scenario with several SAP ERP or SAP CRM systems, the Core Tool user interface does not display the reserved **SAP_CMA** additional information items.
>
> Refer to the customer management area (CMA) [page 87] that is set in this subscriber account.
>
> Click the *Definition* tab.

Additional Information (Tab)

Fill in or modify the *Additional Information* tab.

Use the configuration table and its icons to add some additional information items.

: Click to add a row to the table.

: Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

1.34.4.2.8 Subscriber Account: Default Charging

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

The *Default Charging* tab allows you to define the default policy for the subscriber account. You choose an account to charge. Optionally, you choose credit limit balances if the related subscription specifies no charging mapping for the references identified in the charging plan.

Default Charging (Tab)

Fill in or modify the *Default Charging* tab.

Charged by Default On: Provides the list of prepaid [page 514] and external [page 520] accounts (postpaid) of the subscriber account. Select a default account that is assigned during charging operations.

Credit Limit Balances: This table displays the credit limit balances of the subscriber account. You can select no, one, or more default credit limit balances.

- *Code*: Displays the identification name of the credit limit balance.
- *Is Updated by Default*: Select the checkbox if you want the credit limit balance to be charged by default.

1.34.4.2.9 Subscriber Account: Tax

Create [page 527] or modify [page 540] a subscriber account before setting up this tab. In the tree display area of the *Subscriber Account* window, click  to view this tab.

> **Note**
>
> Depending on your SAP CC system configuration, the *Tax* tab of a subscriber account displays the settings for the Valued Added Tax (VAT) system, the Avalara AvaTax for Communications (AFC) system (for US Telecommunications taxes), or both systems.
>
> For more information, refer to your administrator and consult the SAP CC Administration Guide.

In SAP Convergent Charging, the tax is calculated by using information coming from both the customer data account holder (see subscriber account [page 511]) and the consumed service (see the charges customized in a charge plan [page 374]).

The subscriber account specifies tax information about the place of taxation (country) while the charge in the charge plan specifies information about the level of tax to be applied (for example, normal tax, reduced tax, and so on) in case of EU VAT system.

The *Tax* tab allows you to select the tax settings of the accounts (prepaid and postpaid) belonging to the subscriber account.

Tax (Tab)

Fill in or modify the *Tax* tab.

Taxation Type (or Tax System): Defines the invoicing tax system to be applied to the tax computation related to the service consumption for this subscriber account. Select one of the following types:

- *VAT*: The EU VAT system applies to the subscriber account (or Flat Tax system).
- *US*: The Avalara AvaTax for Communications (AFC) system applies to the subscriber account.
- *None*: No tax system applies to the subscriber account (the balances include no tax).

> **Note**
>
> - In the *Taxation Mode*, the options *Account Holder Subject to Pay* and *Account Holder Exempted* are reserved for managing the tax of external (postpaid) accounts [page 520]. Do not use the modes in a prepaid environment where prepaid accounts are charged.
> - These options are not compliant with a charge customized in a charge plan that specifies that the prices defined in the price plan are gross [page 374].

Taxation Mode: Defines who pays the tax. Select one of the following modes:

- *Service Provider Subject to Pay*: The tax is paid by the service provider.
- *Account Holder Subject to Pay*: The tax is paid by the holder of an external postpaid account. The holder can be different from the subscriber of the subscriber account. The holder must pay the tax in his or her country. Refer to the note above.
- *Account Holder Exempted*: The tax is calculated for information but not applied to billing. Refer to the note above.
- *None*: The tax calculation is not applied.

You set up the following fields depending on the selected *Taxation Type*.

If You Use the EU VAT System

The *Value Added Tax* area displays the following:

- *Tax Zone*: Allows you to select one of the following:
  - The country code ISO 3166 of the taxation place of the holders of prepaid [page 514] or external [page 520] accounts in the subscriber account. These holders must have the same country code as in the version of SAP Convergent Charging.
  - *No Tax*: This means that no tax applies to the accounts (prepaid and postpaid) of the subscriber account.

- *Determined by a VAT Rule*: This means that the VAT rule specified in the charge customized in a charge plan [page 374] applies if any. If no rule is specified in this charge, no tax applies to the accounts (prepaid and postpaid) of the subscriber account.

When *Determined by a VAT Rule* is selected, the *Customer Country* and *Business Category* fields display the related values.

- *Customer Country*: Select or view the country code ISO 3166 (FR, DE, IT, GB, ES, and so on) from the holders of the prepaid accounts and external accounts in the subscriber account. These holders must have the same country code as in the version of SAP Convergent Charging.
- *Business Category*: Select or view the business category (B2B services or B2C services):
  - *B2B*: The holder of the subscriber account is a company.
  - *B2C*: The holder of the subscriber account is a private individual.

If You Use the AvaTax System for US Telecommunications taxes

> **Note**
>
> For more information about AvaTax, refer to the AvaTax® Telecom Users Manual provided by Avalara®.

The US Telco Tax area is available on the tab if you select the US *Taxation Type* and if you select a *Taxation Mode* (service provider or account holder).

The options of this area are the same as those of the Tax tab of the customized charge in a charge plan [page 374]:

- *Origination*: Type the origination location of the transaction to be taxed.
- *Termination*: Type the termination location of the transaction to be taxed.
- *Service Address*: Type the service address of the transaction to be taxed.
- *Customer Type*: Select one of the types of customers involved in the transaction.
- *Incorporated Code*: Select if the customer involved in the transaction is inside or outside of the incorporated area.
- *Client Resale*: Select if the service transaction is final (Sale) or must be resold (Resale).

1.34.4.3 Saving a Subscriber Account

Once you have created or modified a subscriber account, and then verified its structure, you are ready to save it. You can save the subscriber account to either the database or as an XML file on your hard disk.

To save a new subscriber account to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. The *Subscriber Account Creation* dialog box opens to inform you that the subscriber account has been successfully saved.
3. Click *OK*.

To save an existing subscriber account to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. The *Subscriber Account Modification* dialog box opens to inform you that the subscriber account has been successfully modified.
3. Click *OK*. The new version of the subscriber account is saved as the only current version.

To save a subscriber account as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the subscriber account.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The subscriber account is saved as an XML file in your selected directory.

1.34.4.4 Finding a Subscriber Account

To search for a subscriber account in the customer master data [page 30] configured in SAP CC, you perform the operation in two halves of the *Choose a Subscriber Account* window:

- The top part of the window is the filter area, which allows you to find a subscriber account by using filter criteria.
- The bottom area of the window is the results area, which displays the search results that you perform.

Tasks

You can find subscriber accounts with or without search criteria.

- If you do not use search criteria, click  to start the search and view the results.
- If you use search criteria, set up the *Basic* tab, and then click  to start the search. The subscriber accounts are displayed in the results area. To display more columns of information in the results area, click  , and then select which types of information you want to display. Click  to reset the filter and delete the result list.

Filter Area

Basic Tab

The *Basic* tab allows you to search for subscriber accounts depending on an identification code, a subscriber, or a service provider.

Code: Is the identifier of the subscriber account. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database. For example, if you want to find all subscriber accounts containing the letters **`eric`**, you select the *Contains* option, and then type **`eric`**.

Subscriber: Is the end customer that holds an account. The drop-down list is described just above. Type the string that you want to search for in the database.

Service Provider: Is the owner of the pricing catalog [page 109]. The drop-down list is described just above. If you are associated to a catalog, this field is automatically filled in with the name of the related catalog and cannot be modified. Otherwise, you type the string that you want to search for in the database.

Customer Management Area: Is the customer management area [page 87] that is responsible for a subset of subscriber accounts when the system landscape includes several SAP ERP/FI-CA or SAP CRM systems. Select *Any* or the appropriate name.

Advanced Tab

To search for some subscriber accounts, the *Advanced* tab allows you to create a table with one or more search criteria based on additional information items that are set in the subscriber accounts (see Subscriber Account: Additional Information [page 533]).

Actions: To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> **Note**
>
> - If the criteria table comprises several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table consists of three fields. To set up a row, do the following:

1. In the first field, select a search criterion such as *Add. Info (Date)*, *Add. Info (Number)*, or *Add. Info (String)* in the list of identifiers.
2. In the second field, select a search comparator such as *Equals*, *Is Like*, *Starts With*, for example.
3. In the third field, select or type the string that you want to search in the database.

Results Area

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

Related Information

Opening an Existing Subscriber Account [page 539]

1.34.4.5 Opening an Existing Subscriber Account

You can open an existing subscriber account:

- Saved in the back-end database
- Saved in an XML file

To open the existing subscriber account saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Subscriber Account* > The *Open Subscriber Account* dialog box opens.
2. Search [page 537] for the subscriber account, select it, and then click *OK*. The subscriber account window opens.

To open the existing subscriber account saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.34.4.6 Modifying a Subscriber Account

> ⚠ Caution
>
> Do not modify a subscriber account created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber accounts [page 511].

You can modify, add, or remove accounts in a previously saved subscriber account.

To modify a subscriber account

1. Open [page 539] the subscriber account.
2. Set up [page 528] the subscriber account.
3. Verify the subscriber account structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save [page 536] the subscriber account

1.34.4.7 Reporting a Subscriber Account

You can view all information relating to the current subscriber account such as:

- Subscriber account: Code, Subscriber, and Service provider
- Prepaid account: Code, Amount, and Threshold
- External account: Code, External account code, and External billing system
- Credit limit balance: Code, Amount, and Reset credit value

Report a Subscriber Account

1. Open [page 539] a subscriber account.

2. On the toolbar in the main window, click the *Subscriber Account Report* icon  or use the *Action* menu[63]. The *Subscriber Account Report* dialog box opens to display a summary of the subscriber account.
3. Check the report, and then click *Close*.

Related Information

Working with Subscriber Accounts [page 527]

1.34.4.8 Deleting a Subscriber Account

> ⚠ Caution
>
> Do not delete a subscriber account created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber accounts [page 511].

> ⚠ Caution
>
> There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.

To delete a subscriber account

1. From the menu bar in the main window, select *File* > *Delete* > *Subscriber Account*. The *Delete Subscriber Account* dialog box opens.
2. Search [page 537] for the subscriber account to delete.
3. Select the subscriber account.
4. Click *Delete*. A deletion dialog box asks if you want to delete the subscriber account.
5. Confirm the charge deletion by clicking *Yes*. A dialog box confirms the successful deletion.

1.34.5 Using the Action Menu

Use the *Action* menu of the user interface to manage your subscriber accounts.

Possible configuration tasks:

- Adding a Component to a Subscriber Account [page 529]

[63] From the menu bar in the main window, you can also select *Action* > *Subscriber account report*.

- Removing a Component [page 530]
- Reporting a Subscriber Account [page 540]

Possible business operations and tasks:

- Refilling a Prepaid Account [page 519]

1.35 Subscriber Mapping Tables

> ⚠ Caution
>
> **Do not modify or delete subscriber mapping tables with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Creating, modifying, or deleting subscriber mapping tables may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. To modify a subscriber mapping table, you must use this CRM or provisioning system in your SAP system landscape.
>
> - Consult the Application Help for more information about the customer data provisioning process.
> - Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this customer master data.
>
> The manual procedures detailed in this section are relevant in:
>
> - A productive SAP system landscape with few contracts directly managed with Core Tool (no CRM interfaced with SAP CC)
> - A test, demo, or training system landscape

> → Remember
>
> In an integrated *SAP Solution* scenario based on SAP Billing and Revenue Innovation Management (BRIM), SAP CRM creates and maintains this data. Do not modify any data with Core Tool in a productive SAP system landscape.

 A **subscriber mapping table** in SAP Convergent Charging (SAP CC) is a multicolumn data table that is similar to a mapping table [page 179] except that it is a master data related to end customers or subscribers. A subscriber mapping table is attached to a subscriber account [page 511] instead of being defined in a pricing catalog [page 109] for the chargeable service or product of a provider company. It is used to replace a mapping table in the pricing catalog with a custom table that defines new values for a specific end customer or a group of end customers.

Using subscriber mapping tables in the pricing logic, allows you to refine the implementation of the pricing policy as defined by the marketing team.

You define the structure of a subscriber mapping table in a mapping table class [page 167].

Remind:

- A subscriber mapping table is a data-lookup multicolumn table composed of business or technical data values.
- Each row of the subscriber mapping table consists of a set of values of input columns to which corresponds a set of values of output columns. For each row, you have to specify the period during which the row is valid.
- A subscriber mapping table is part of the customer master data managed by SAP Convergent Charging. It is not part of a pricing catalog.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Subscriber Mapping Table Step by Step

Before creating a subscriber mapping table, an SAP CC pricing specialist must create [page 173], set up [page 174], and save [page 176] a mapping table class that defines its structure.

As a customer sales representative, do the following steps to use subscriber mapping tables:

1. Create [page 545] a subscriber mapping table. The subscriber mapping table is then attached to a subscriber account.
2. Set up [page 546] the subscriber mapping table.
3. Save [page 552] the subscriber mapping table.
4. Use the subscriber mapping table in a provider contract [page 544] related to the subscriber account the subscriber mapping table is attached to. You have to set a parameter with the type Mapping Tables ID at contract item level.
5. Delete [page 556] the subscriber mapping table.

You can use the same mapping table class for configuring different mapping tables and subscriber mapping tables.

Integration

This customer master data must be created and maintained by an external system connected to the SAP CC system during the provisioning process. Depending on your business needs, replication, distribution, and synchronization processes may be required.

Versions

You can modify and save different versions of a subscriber range table.

Transport

You cannot transport customer master data with the Core Tool user interface. It is a *Customizing* activity and need to be implemented in your productive SAP system landscape.

Depending on your SAP system landscape and business/technical requirements, master data distribution, replication, or synchronization processes must be implemented.

See Also

Subscriber Range Tables [page 557]

Related Information

Using Subscriber Mapping Tables in Provider Contracts [page 544]
Working with Subscriber Mapping Tables [page 545]

1.35.1 Using Subscriber Mapping Tables in Provider Contracts

The procedure below describes how to use a subscriber mapping table in a provider contract created manually.

> ⚠ Caution
>
> You use this procedure only when you want to create a provider contract manually for test, demo, or training purposes. Do not use this procedure if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts. Refer to the caution caution [page 573] for provider contracts.

To use a subscriber mapping table in a provider contract

1. Identify the following objects:
   - The provider contract [page 573]
   - The subscriber account [page 511] containing the provider contract
   - The mapping table class [page 167] used as a schema for the subscriber mapping table that you want to create and use in your contract
2. Create [page 545] a subscriber mapping table from the subscriber account containing the provider contract. The subscriber mapping table must be based on the mapping table class identified during the previous step.

3. Configure a charge or the refill logic, and then a charge plan or a refill plan to enable the dynamic use of mapping tables [page 185]. In the charge or the refill logic, use the mapping table class identified during the first step of the procedure.
4. Activate the charge plan or the refill plan [page 632] in the provider contract or create a provider contract with an item [page 594].
5. In the charge plan or the refill plan activated in the appropriate contract item, set the parameter of type Mapping Table ID [page 185] to the name of the subscriber mapping table to be used.

1.35.2 Working with Subscriber Mapping Tables

Related Information

Creating a Subscriber Mapping Table [page 545]
Finding a Subscriber Mapping Table [page 553]
Opening an Existing Subscriber Mapping Table [page 555]
Modifying a Subscriber Mapping Table [page 556]
Setting Up a Subscriber Mapping Table [page 546]
Saving a Subscriber Mapping Table [page 552]
Deleting a Subscriber Mapping Table [page 556]

1.35.2.1 Creating a Subscriber Mapping Table

Before creating a subscriber mapping table for demo or training purposes, you must:

- As a pricing specialist, create a mapping table class [page 173] in the pricing catalog for the chargeable service or product of your service/product provider company.
- As a customer sales representative, create a subscriber account [page 527] assigned to this service provider.

Use one of these procedures to create a subscriber mapping table.

> ⚠ Restriction
>
> For a matter of performances, do not exceed the recommended maximum values explained in SAP Note 24778958.

Create a Subscriber Mapping Table

1. From the menu bar in the main window, select *File* > *New* > *Subscriber Mapping Table*. The *Choose a Subscriber Account* window opens.
2. Search [page 537] for the subscriber account [page 511] which is linked to the subscriber mapping table.
3. Select the subscriber account.
4. Click *OK*. The *New Subscriber Mapping Table* window opens.
5. Set up [page 546] the subscriber mapping table.
6. Verify the mapping table structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 552] the new subscriber mapping table.

> ⓘ Note
>
> Once you have saved a subscriber mapping table, you can no longer modify its name or its mapping table class [page 167]. You can only modify the rows of the subscriber mapping table.

Create a Subscriber Mapping Table from a Subscriber Account

You can create a subscriber mapping table from the subscriber account it is related to. See the tab *Mapping Tables* of the *Subscriber Account* window.

> ⓘ Note
>
> You cannot create a subscriber mapping table by opening an existing subscriber mapping table and copying it.

1.35.2.2 Setting Up a Subscriber Mapping Table

Create [page 545], open [page 555], or modify [page 556] a subscriber mapping table before setting it up. A *Subscriber Mapping Table* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

To set up a subscriber mapping table, you must:

1. In the *Definition* tab, set up the identification name and select the mapping table class [page 167] that defines the structure of your subscriber mapping table.
2. Fill in or modify the *Table Instance* tab and the *Additional Information* tab.

1.35.2.2.1 Tab: Definition

Definition (Tab)

Name: Type the unique identifier of this subscriber mapping table.

Description: Describe the subscriber mapping table.

Subscriber Mapping Table Class: Select the mapping table class [page 167] that defines the data structure of this subscriber mapping table.

Subscriber Account

- *Contracting Service Provider*: Displays the identifier of the pricing catalog owner [page 109].
- *Code*: Displays the identifier of the related subscriber account [page 511].

Related Information

Mapping Table Classes [page 167]

1.35.2.2.2 Tab: Table Instance

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Tab: Table Instance

The *Table Instance* tab consists of two parts:

- The top part of the tab is the filter area, which allows you to find rows in the subscriber mapping table by using filter criteria.
- The bottom part of the tab is the results area, which displays the search results that you perform.

> ⓘ Note
>
> Use the arrows ▲▼ if you want to hide or show the filter area or the results area.

Setting up the filter

> **Note**
>
> When you add, remove, or change one or more rows in the results area, the filter area is no more available (dimmed) until you save your changes.

Use the following icons:

 : To start a new search. Every time you add, remove, or change one or more criteria, this button allows you to start a new search and display the results.

 : To reset the filter. After one or more searches, this button allows you to re-initialize the filter and display all the rows of the subscriber mapping table according to the number of results per page and the selected order.

 : To move to the previous page. This button is available when the results list is large.

 : To move to the next page. This button is available when the results list is large.

 : To add a criterion to the filter.

 : To remove a criterion from the filter.

Use the following fields:

Results per Page: Select the number of results that are displayed in the results area.

Identifier Sorted By: Select the order in which you want to display the results.

Found Rows: Displays the number of rows that are found by the search.

Total Number of Rows: Displays the total number of rows of the subscriber mapping table that is stored in the database.

Creating, modifying, or removing rows

Use the following icons:

 : To add a row to the table. The new row is added after the selected row in the current page of the results area. Otherwise, the new row is added to the end of the table in the current page of the results area. When you click this button, the filter area is no more available (dimmed) until you save your changes.

 : To remove a row from the table. Select a row in the current page of the results area, and then click this button to remove the row from the table. When you click this button, the filter area is no more available (dimmed) until you save your changes.

 : To import data into the subscriber mapping table according to the CSV format(*).

 : To export data of the subscriber mapping table into a CSV file (**). This button is not available (dimmed):

- If you are modifying the subscriber mapping table
- If the subscriber mapping table contains no row

 : To remove the added and filtered rows from the subscriber mapping table. This button allows you to remove the rows that you have added to the subscriber mapping table and the rows that have been found by the last search. When you click this button, a message displays the number of added and filtered rows and asks if you want to remove the rows. The added and filtered rows are removed from the results area but are not deleted from the database until you save the subscriber mapping table.

Use the following fields:

Identifier: Type an identifier for the row. This identifier must be unique in the subscriber mapping table.

Input Columns: Type the input values for this row. You are allowed to enter duplicate input values. In that case, the selection of a row depends on the *Start Date* and the *End Date*. Note that the number of input columns is the same as the number of input columns defined in the related subscriber mapping table class.

Start Date: Click the cell of the *End Date* column to open a calendar, and then select a date.

End Date: Click the cell of the *End Date* column to open a calendar, and then select a date. The Start Date and the *End Date* define the period when the row is valid.

Output Columns: Type the output values for this row. Note that the number of output columns is the same as the number of output columns defined in the related subscriber mapping table class.

> **Note**
>
> Instead of manually typing all of the data in the columns, you can import existing data from external .CSV type files by clicking  (refer to the process described below) .

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.

Rule	Verifications
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of `2016-5-6 15:1:2`, the CSV file must contain `2016-05-06 15:01:02`.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .

2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

**

(**) Exporting Data to a CSV File

This procedure allows you to export data from mapping tables or subscriber mapping tables into a CSV file.

Export Data to a CSV File

1. Click the button  . The *CSV File Format Options* dialog box is displayed.
2. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
3. In the *Number Format Parameters* area, select the *Decimal Separator* type (*Point*, *Comma*) and the *Thousands Separator* type (*No*, *Quote*, *Comma*, *Space*, *Point*) to format numbers. If there is no *Thousands Separator*, select *No*.
4. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns (not editable for mapping tables).
5. Click *OK* to export data. The *Save* dialog box is displayed.
6. In the *Save in* field, select the directory where you want to export data.
7. In the *File name* field, type the name of the file, and then click *Save*. If the file already exists, a message asks you if you want to overwrite it.

Restrictions on input values and dates

You are not allowed to put the same input values, the same start date, and the same end date for two distinct rows.

Example

The following table shows you an example of duplicate input value:

Identifier	First input column	Start Date	End Date	First output column
Row 1	A	July, 18th	October, 4th	12
Row 2	A	July, 18th	October, 5th	19
Row 3	A	July, 17th	October, 4th	16

For example, an event occurs on the 20th of July with the input value A. You use the following rules to determine which row of the mapping table applies to the example .

Rule 1: When more than one period correspond to an input value (A) at a date (the 20th of July), the rows whose start date is closest to the reference date applies. In our example, Row 1 and Row 2 are selected because their start date (July, 18th) is closer to the reference date (July, 20th) than the start date of Row 3 (July, 17th).

Rule 2: If two rows have the same start date, the one that has the shortest period applies. In the previous example, Row 1 applies.

1.35.2.2.3 Tab: Additional Information

Tab: Additional Information

Use the table and its icons to create additional information.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the SAP ERP/FI-CA system or the external provisioning system that is connected to SAP Convergent Charging.

1.35.2.3 Saving a Subscriber Mapping Table

Once you have created or modified a subscriber mapping table, and then verified its structure, you are ready to save it. You can save the subscriber mapping table to either the database or as an XML file on your hard disk.

To save a new subscriber mapping table to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.

3. Click *Save*. The *Subscriber Mapping Table Creation* dialog box opens to inform you that the subscriber mapping table has been successfully saved.
4. Click *OK*.

To save a subscriber mapping table as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the subscriber mapping table.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The subscriber mapping table is saved as an XML file in your selected directory.

1.35.2.4 Finding a Subscriber Mapping Table

To search for a subscriber mapping table in the customer master data [page 30] configured in SAP CC, you perform the operation in two halves of the *Choose a Subscriber Mapping Table* window:

- The top part of the window is the filter area, which allows you to find a subscriber mapping table by using filter criteria. The information in the *Basic* and *Advanced* tabs is combined during the search operation.
- The bottom area of the window is the results area, which displays the search results that you perform.

Tasks

You can find subscriber mapping tables with or without search criteria.

- If you do not use search criteria, click  to start the search and view the results.
- If you use search criteria, set up the *Basic* and *Advanced* tabs, and then click  to start the search. The subscriber mapping tables are displayed in the results area. To display more columns of information in the results area, click  , and then select which types of information you want to display. Click  to reset the filter and delete the result list.

Filter Area

Basic Tab

The *Basic* tab allows you to search for subscriber mapping tables depending on the criteria described below.

Code: Is the identifier of the subscriber mapping table. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database.

> **Example**
>
> For example, if you want to find all subscriber mapping tables containing the letters **eric**, you select the *Contains* option, and then type **eric**.

Subscriber: Is the end customer who holds the subscriber account [page 511] that the searched subscriber mapping tables belong to. The drop-down list is described just above. Type the string that you want to search for in the database.

Service Provider: Is the owner of the pricing catalog [page 109]. The drop-down list is described just above. If you are associated to a catalog, this field is automatically filled in with the name of the related catalog and cannot be modified. Otherwise, you type the string that you want to search for in the database.

Advanced Tab

To search for subscriber mapping tables, the *Advanced* tab allows you to create a table with one or more search criteria. The possible search criteria are:

- Mapping table class [page 167] that defines the structure for the searched subscriber mapping tables
- Additional information items [page 552] set in the searched tables
- Description

Actions: To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> **Note**
>
> - If the criteria table comprises several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table consists of three fields. To set up a row, do the following:

1. In the first field, select a search criterion such as:
   - *Add. Info (Date)*
   - *Add. Info (Number)*
   - *Add. Info (String)*
   - *Description*
   - *Mapping Table Class*
2. In the second field, select a search comparator such as *After*, *Before*, *After or Equals*, *Before or Equals* for example.
3. In the third field, select or type the number, the date/time, or the string that you want to find in the database.

Results Area

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

1.35.2.5 Opening an Existing Subscriber Mapping Table

You can open:

- An existing subscriber mapping table saved in the back-end database
- An existing subscriber mapping table saved in an XML file

To open an existing subscriber mapping table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Subscriber Mapping Table*. The *Open Subscriber Mapping Table* window opens.
2. Search [page 553] for the subscriber mapping table, select it, and then click *OK*. The *Subscriber Mapping Table* window opens.

To open an existing subscriber mapping table saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.35.2.6 Modifying a Subscriber Mapping Table

> ⚠ Caution
>
> Do not modify a subscriber mapping table created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber mapping tables [page 542].

You can modify a subscriber mapping table by adding, modifying, or deleting content.

You cannot modify the name or the related mapping table class [page 167] of a subscriber mapping table once you have saved it.

To modify a subscriber mapping table

1. Open [page 555] the subscriber mapping table.
2. Set up [page 546] the rows of the subscriber mapping table.
3. Verify the subscriber mapping table by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save [page 552] the subscriber mapping table.

1.35.2.7 Deleting a Subscriber Mapping Table

> ⚠ Caution
>
> Do not delete a mapping table created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber mapping tables [page 542].

Before you delete a subscriber mapping table, you must first delete all objects which reference it.

To delete a subscriber mapping table

1. From the menu bar in the main window, select *File* > *Delete* > *Subscriber Mapping Table*. The *Delete Subscriber Mapping Table* dialog box opens.
2. Search [page 553] for the subscriber mapping table, select it, and then click *OK*. The subscriber mapping table window opens.
3. Click *Delete*. A deletion dialog box asks if you want to delete the subscriber mapping table
4. Confirm the deletion of the subscriber mapping table by clicking *Yes*. A dialog box opens to inform you that the subscriber mapping table has been successfully deleted.

> **Note**
>
> If an error message appears after clicking *Yes* at step 4, you must delete the objects that reference the subscriber mapping table. The identifiers of these objects appear in the message.

1.36 Subscriber Range Tables

> **Caution**
>
> **Do not modify or delete subscriber range tables with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Creating, modifying, or deleting subscriber range tables may result in inconsistencies between the two systems because of data replication and synchronization mechanisms. To modify a subscriber range table, you must use this CRM or provisioning system in your SAP system landscape.
>
> - Consult the Application Help for more information about the customer data provisioning process.
> - Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this customer master data.
>
> The manual procedures detailed in this section are relevant in:
>
> - A production system landscape with few contracts directly managed with Core Tool (no CRM interfaced with SAP CC)
> - A test, demo, or training system landscape

> **→ Remember**
>
> In an integrated *SAP Solution* scenario based on SAP Billing and Revenue Innovation Management (BRIM), SAP CRM creates and maintains this data. Do not modify any data with Core Tool in a productive SAP system landscape.

 A **subscriber range table** in SAP Convergent Charging (SAP CC) is a table of business/technical data that consists of a set of 0 to 5 input columns of string type and a set of 1 to 15 output columns. Each row of output columns is defined by a value range to which corresponds a set of values in the related output columns. Each output column is associated to a type [page 207] that you define at range table class [page 199] level.

The subscriber range table allows you to map ranges of positive numbers onto a set of output string or number values that represent prices, unit prices, or credits. You can also apply combinatorial rules to output columns to compute supplementary values. You must however configure the pricing logic that defines the computation algorithm of the final values. See some examples [page 217].

Similar to a range table [page 213], the subscriber range table is however only related to end customers of a service provider and is assigned to a subscriber account [page 511] instead of being defined in a pricing catalog [page 109] of a service provider. In addition, a subscriber range table is used to define or redefine the values of a range table for a specific end customer or a set of contracts [page 573]. A subscriber range table allows you

to create new properties that you can use in the pricing logic that is assigned to a provider contract (see active charge plans).

You define the structure of a subscriber range table in a range table class [page 199].

A subscriber range table is part of the customer master data managed by SAP Convergent Charging. It is not part of a pricing catalog.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Subscriber Range Table Step by Step

Before creating a subscriber range table, an SAP CC pricing specialist must create [page 205], set up [page 207], and save [page 209] a range table class.

As a customer sales representative, you perform the following steps to use the subscriber range tables:

1. Create [page 560] a subscriber range table. When it is created, the subscriber range table is attached to a subscriber account.
2. Set up [page 563] the subscriber range table.
3. Save [page 567] the subscriber range table.
4. Use the subscriber range table in a provider contract that is related to the subscriber account to which the subscriber mapping table is attached. You must set a parameter [page 303] with the type *Range Tables ID* at contract item level.
5. Delete [page 571] the subscriber range table.

You can use the same range table class for configuring different range tables and subscriber range tables.

Integration

This customer master data must be created and maintained by an external system connected to the SAP CC system during the provisioning process. Depending on your business needs, replication, distribution, and synchronization processes may be required.

Versions

You can modify and save different versions of a subscriber range table [page 572].

Transport

You cannot transport customer master data with the Core Tool user interface. It is a *Customizing* activity and need to be implemented in your productive SAP system landscape.

Depending on your SAP system landscape and business/technical requirements, master data distribution, replication, or synchronization processes must be implemented.

Maintenance

The only subscriber range table that is available is the last modified one. You can modify and delete the rows of a subscriber range table, but you cannot change the name of a subscriber range table or the related range table class.

Related Information

Subscriber Mapping Tables [page 542]
Changing the Version of a Subscriber Range Table [page 572]

1.36.1 Working with Subscriber Range Tables

Related Information

Creating a Subscriber Range Table [page 560]
Finding a Subscriber Range Table [page 568]
Opening an Existing Subscriber Range Table [page 570]
Changing the Version of a Subscriber Range Table [page 572]
Modifying a Subscriber Range Table [page 571]
Filling In a Subscriber Range Table [page 561]
Setting Up a Subscriber Range Table [page 563]
Saving a Subscriber Range Table [page 567]
Deleting a Subscriber Range Table [page 571]

1.36.1.1 Creating a Subscriber Range Table

Before creating a subscriber range table for demo or training purposes, you must:

- Create a range table class [page 199] in the catalog of the service provider
- Create a subscriber account [page 527] assigned to this service provider

Use one of these procedures to create a subscriber range table.

> ⚠ Restriction
>
> For a matter of performances, do not exceed the recommended maximum values explained in SAP Note 24778958.

To create a subscriber range table

1. From the menu bar in the main window, select *File* > *New* > *Subscriber Range Table*. The *Choose a Subscriber Account* window opens.
2. Search [page 537] for the subscriber account [page 511] which is linked to the subscriber range table.
3. Select the subscriber account.
4. Click *OK*. The *New Subscriber Range Table* window opens.
5. Set up [page 563] the subscriber range table.
6. Verify the range table structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
7. Save [page 567] the subscriber range table.

> ⓘ Note
>
> Once you have saved a subscriber range table, you can no longer modify its name or its range table class [page 199]. You can only modify the rows of the subscriber range table.

To create a subscriber range table from a subscriber account

You can create a subscriber range table from the related subscriber account. See the tab *Range Tables* of the *Subscriber Account* window.

> ⓘ Note
>
> You cannot create a subscriber range table by opening an existing subscriber range table and copying it.

1.36.1.2 Filling In a Subscriber Range Table

Use the icons below to fill in the subscriber range table

Using the Icons

Use the following icon...	To...
	Add a new row to the configuration table.
	Delete a row from the table.
	Add a new group to the table.
	Delete a group from the table.
	Move a row up.
	Move a row down.
	Import data into the table according to the CSV format.(*)
	Clear all the table.
	Search for a numerical or string value in the table. The search stops when the first occurrence has been found. When the search is successful, the cell is highlighted. To find the next occurrence, use  or  . When you click  , a *Search in table* dialog box is displayed. In the *Search* field, type the value to search for or select a value in the list. If you want to do a case-sensitive search, click the *Match case* option. In the *Search in column(s)* check boxes, select the columns where you want to search for the numerical or string value.
	Search for the next occurrence towards the top of the table if the previous search with the icon  was successful. The search is performed from left to right, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the beginning of the table is reached, a message informs you that the value has not been found.
	Search for the next occurrence towards the bottom of the table if the previous search with the icon  was successful. The search is performed from right to left, row by row. Each time that you click  , the cell is highlighted if the search is successful. When the end of the table is reached, a message informs you that the value has not been found.

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns

Rule	Verifications
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

1.36.1.3 Setting Up a Subscriber Range Table

Create [page 560], open [page 570], or modify [page 571] a subscriber range table before setting it up. A *Subscriber Range Table* window must be open.

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

To set up a subscriber range table, you must:

1. In the *Definition* tab, set up the identification name for the range table and then select the range table class [page 199] that defines its data structure.

2. Fill in or modify the *Table Instance* tab and the *Additional Information* tab.

1.36.1.3.1 Tab: Definition

Tab: Definition

Name: Type the unique identifier of this subscriber range table.

Description: Describe the subscriber range table.

Range Table Class: Select the range table class [page 199] that defines the data structure of this subscriber range table.

Related Information

Range Table Classes [page 199]

1.36.1.3.2 Tab: Table Instance

> → Remember
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Setting Up the Rows of the Subscriber Range Table

Once you have selected the range table class for your range table, you can add one or more rows [page 561] to the table, and then fill in the cells of each row with appropriate values.

> ⓘ Note
>
> In the *Output Values* area, the rows are always sorted by intervals (ranges) and you cannot change the order of these rows.

After adding a row, you have to set it up.

Input Values: Type the input values for each row as specified in the input columns of the related range table class [page 199].

Output Values: Type the output values for this row as specified in the output columns of the related range table class [page 199].

Range: Double-click to set up the upper bound of the range that is specified in this row. The boundary is inclusive or exclusive according to the configuration of the related range table class.

> **Note**
>
> Instead of manually typing all of the data in the columns, you can import existing data from CSV files by clicking  (refer to the process described below).

(*) Importing Data from a CSV File

Before Importing

Make sure that the CSV file to import is compliant with the following rules:

Rule	Verifications
CSV File Structure	**Note** Make sure that the order of the columns in the CSV file is the same as the order in the *Column Indexes in File area* of the dialog box when you import this file into the table. Otherwise, change the CSV file or change the order of the columns through the dialog box. • The CSV file has no header. • The input data file does not contain comment. • The format of the input data is compliant with the table.
Date/Time Format	In the CSV file to import, the items of the dates/times (months, days, minutes, and seconds) must have a fixed size. The format is the following: yyyy-MM-dd HH24:mm:ss. For example, instead of **`2016-5-6 15:1:2`**, the CSV file must contain **`2016-05-06 15:01:02`**.
Field Separator	By default, the field separator is a semicolon, a tabulation, a comma, a space, or a slash, but any printable character is suitable, except the double quote ("). In any case, a field separator must not exceed one character.
Specific Rules about Translation Tables	Each row in the CSV file consists of N inputs and M outputs: • Column 1 to column N: Input key • Column N+1 to column N+M: Output values

Rule	Verifications
Specific Rules about Tier Tables	**Note** An empty field represents the infinite. Each row in the CSV file consists of a name, two bounds, and N outputs: • Column 1: Group name • Column 2: Lower bound • Column 3: Lower bound is included or not (the expression "true" means that the bound is included. Any other expression than "true" means that the bound is excluded. The expression "true" is case-insensitive.) • Column 4: Upper bound • Column 5: Upper bound is included or not (see above) • Column 6 to column 6+N+1: Output columns
Specific Rules about Range Tables	Only the upper bound must be represented by a CSV field because the lower bound is expected to be the previous upper bound of the range table (for the first range, 0 is expected). A bound can be null (0) but not negative. If the range table class is unbounded: • The last range in the CSV file can have an infinite upper bound. The infinite is represented by anything that is not a valid number. For example, an empty field, a space, or any string. • The last range in the CSV file can also have a finite upper bound. In this case, after you import the CSV file into the range table, you must manually define an unbounded range at the end of the range table to validate it. If the range table class is bounded, the last range in the CSV file must have a defined upper bound.

Import Data from a CSV File into a Table

To import some data into a table in your Core Tool user interface:

1. Click the button  .
2. The *Open* dialog box is displayed to search for a file. When the file has been found, select the file, and then click the *Open* button.
3. The *CSV File Format Options* dialog box is displayed. In the *Field Separator* area, click the separator that is used in your CSV file to separate the numerical or string values, or type a specific separator.
4. In the *Number Format Parameters* area, select the *Decimal Separator* type (Dot, Comma) and the *Thousands Separator* type (No, Quote, Comma, Space, Dot) to format numbers. If there is no *Thousands Separator*, select No.
5. In the *Column Indexes in File* area, you can type the order in which the numerical or string values of the CSV file are imported into the table columns.
6. Click *OK* to import data into the table.

1.36.1.3.3 Tab: Additional Information

Tab: Additional Information

Use the configuration table and its icons to add some additional information items.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This description is visible from the CRM application or the external provisioning system that is connected to SAP Convergent Charging 2021.

1.36.1.4 Saving a Subscriber Range Table

Once you have created or modified a subscriber range table, and then verified its structure, you are ready to save it. You can save the subscriber range table to either the database or as an XML file on your hard disk.

To save a new subscriber range table to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Save in the catalog* dialog box opens.
2. Select the name of the catalog.
3. Click *Save*. The *Subscriber Range Table Creation* dialog box opens to inform you that the subscriber range table has been successfully saved.
4. Click *OK*.

To save a new version of an existing subscriber range table to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Choose date* dialog box opens,
2. Select a date from which the subscriber range table modifications will be effective, and then click *OK*. The *Subscriber Range Table Modification* confirmation box opens to inform you that the subscriber range table has been successfully modified.
3. Click *OK*.

To save a subscriber range table as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the subscriber range table.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The subscriber range table is saved as an XML file in your selected directory.

1.36.1.5 Finding a Subscriber Range Table

To search for a subscriber range table in the customer master data [page 30] configured in SAP CC, you perform the operation in two halves of the *Choose a Subscriber Range Table* window:

- The top part of the window is the filter area, which allows you to find a subscriber range table by using filter criteria. The information in the *Basic* and *Advanced* tabs is combined during the search operation.
- The bottom area of the window is the results area, which displays the search results that you perform.

Tasks

You can find subscriber range tables with or without search criteria.

- If you do not use search criteria, click  to start the search and view the results.
- If you use search criteria, set up the *Basic* and *Advanced* tabs, and then click  to start the search. The subscriber range tables are displayed in the results area. To display more columns of information in the results area, click  , and then select which types of information you want to display. Click  to reset the filter and delete the result list.

Filter Area

Basic Tab

The *Basic* tab allows you to search for subscriber range tables depending on the criteria described below.

Code: Is the identifier of the subscriber range table. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database. For example, if you want to find all subscriber range tables containing the letters **`eric`**, you select the *Contains* option, and then type **`eric`**.

Subscriber: Is the end customer who holds the subscriber account [page 511] that the searched subscriber range tables belong to. The drop-down list is described just above. Type the string that you want to search for in the database.

Service Provider: Is the owner of the pricing catalog [page 109]. The drop-down list is described just above. If you are associated to a catalog, this field is automatically filled in with the name of the related catalog and cannot be modified. Otherwise, you type the string that you want to search for in the database.

Advanced Tab

To search for subscriber range tables, the *Advanced* tab allows you to create a table with one or more search criteria. The possible search criteria are:

- Range table class [page 199] that defines the structure for the searched subscriber range tables
- Additional information items [page 567] set in the searched tables
- Description

Actions: To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> **Note**
>
> - If the criteria table comprises several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table consists of three fields. To set up a row, do the following:

1. In the first field, select a search criterion such as:
   - *Add. Info (Date)*
   - *Add. Info (Number)*
   - *Add. Info (String)*
   - *Description*
   - *Range Table Class Identifier* (the name for the range table class that specifies your searched subscriber range tables)
2. In the second field, select a search comparator such as *After*, *Before*, *After or Equals*, *Before or Equals* for example.
3. In the third field, select or type the number, the date/time, or the string that you want to find in the database.

Results Area

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

Related Information

Opening an Existing Subscriber Range Table [page 570]

1.36.1.6 Opening an Existing Subscriber Range Table

You can open:

- An existing subscriber range table saved in the back-end database
- An existing subscriber range table saved in an XML file

To open an existing subscriber range table saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Subscriber Range Table*. The *Open Subscriber Range Table* window opens.
2. Search [page 568] for the subscriber range table, select it, and then click *OK*. The *Subscriber Range Table* window opens.

To open an existing subscriber range table saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.36.1.7 Modifying a Subscriber Range Table

> **Caution**
>
> Do not modify a subscriber range table created by a CRM application or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber range tables [page 557].

You can modify a subscriber range table by adding, modifying, or deleting content.

You cannot modify the name or the related range table class [page 199] of a subscriber range table once you have saved it.

To modify a subscriber range table

1. Open [page 570] the subscriber range table.
2. Set up [page 563] the rows of the subscriber range table.
3. Verify the subscriber range table by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save [page 567] the subscriber range table.

1.36.1.8 Deleting a Subscriber Range Table

> **Caution**
>
> Do not delete a mapping table created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution about subscriber range tables [page 557].

Before you delete a subscriber range table, you must first delete all objects which reference it.

To delete a subscriber range table

1. From the menu bar in the main window, select *File* > *Delete* > *Subscriber Range Table*. The *Delete Subscriber Range Table* dialog box opens.
2. Search [page 568] for the subscriber range table, select it, and then click *OK*. The subscriber range table window opens.
3. Click *Delete*. A deletion dialog box asks if you want to delete the subscriber range table
4. Confirm the deletion of the subscriber range table by clicking *Yes*. A dialog box opens to inform you that the subscriber range table has been successfully deleted.

> **Note**
>
> If an error message appears after clicking *Yes* at step 4, you must delete the objects that reference the subscriber range table. The identifiers of these objects appear in the message.

1.36.2 Using the Action Menu

Related Information

Changing the Version of a Subscriber Range Table [page 572]

1.36.2.1 Changing the Version of a Subscriber Range Table

The procedure is the same as for changing the version of an object.

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version* icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> **Note**
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.37 Provider Contracts

> **Caution**
>
> **Do not modify or delete provider contracts with Core Tool** when SAP Convergent Charging 2023 is integrated and interfaced with a Customer Relationship Management (CRM) application or a provisioning system.
>
> Creating, modifying, or deleting provider contracts **may result in inconsistencies** between the two systems because of data replication and synchronization mechanisms. To modify a provider contract, you must use this CRM or provisioning system in your SAP system landscape.
>
> - Consult the Application Help for more information about the *Customer Data Provisioning process*.
> - Consult the Web Services Documentation [page 40] if you need to integrate your CRM or provisioning system by implementing the *Web Services* technical interface provided by SAP Convergent Charging to manage this customer master data.
>
> The manual procedures detailed in this section are relevant in:
>
> - A production SAP system landscape with few contracts directly managed with the Core Tool user interface (no CRM interfaced with SAP CC)
> - A test, demo, or training system landscape


A **provider contract** in SAP Convergent Charging (SAP CC) is customer master data that represents a long-term business agreement between a *service/product provider company* and a *subscriber (individual or party)*. A provider contract mostly includes pricing and charging data that was agreed upon and allows one or many end customers to use the chargeable service or product. The subscriber must have a subscriber account in SAP CC in order to be charged for the service or product: subscription-based fees and usage-based fees.

The subscribed agreement is based on specific terms that have been negotiated beforehand and that defines how and how much this subscriber (and authorized end customers) is charged to:

- Have access to goods or services.
- Consume these goods or services.

The provider contract belongs to the subscriber and to the contracting service/product provider. Note that a provider contract is associated with the pricing catalog [page 109] that includes the pricing configuration (logic) for the service to monetize.

A provider contract contains:

- Contract items [page 592] that represent such a business term. For example, a term includes counters [page 615] that represent some free credits and that are used by the pricing logic relating to the term.

- Several allowances [page 575] that authorize end customers to consume services that are limited in quantity or in time; for example, a credit bucket, a spending limit, or a promotion.
- A provider contract can also include terms to refill the prepaid account of the subscriber.

> → Recommendation
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Structure

A provider contract relates to one and only one subscriber account [page 511] (for the end customer) and consists of several lines:

- One or more **allowances** (see Searching For and Viewing the Allowances of a Provider Contract [page 633])
- One or more **contract items**:
  - One or more contract items that "activate" and reference charge plans [page 343] that are subscribed by the subscriber to be charged
  - One unique contract item that activates and references at most a refill plan [page 421] that is subscribed by the subscriber
  - One or more contract items that activate and reference monitoring plans [page 491] that are subscribed by an end customer.
    When several contract items activate respective monitoring plans, the spending status reporting considers the sum of all the counters selected [page 629] in the activated plans.

  Every provider contract groups several contract items. During its lifecycle, its composition could vary. Note that every contract item can also change during the lifecycle or the belonging provider contract.

Parent Contracts

You can link a provider contract to a parent contract to:

- Share a pool of counters [page 607] with the related contracts (the pool of counters is stored in the parent contract)
- Access the allowances of the parent contract from the linked contracts through the logic components *Allowance Event Sender* and *Allowance Property Introducer* (click here [page 671])
- Define a simplified hierarchy between provider contracts

> ⓘ Note
>
> - There are close relationships between the charging process, the activation process, and the rerating process relating to the linked provider and parent contracts. Consult the Application Help for more information about these processes.
> - Remember that you trigger the rerating process [page 59] with SAP CC Core Tool.

Integration

This customer master data must be created and maintained by an external system connected to the SAP CC system during the provisioning process. Depending on your business needs, replication, distribution, and synchronization processes may be required.

Versions

You can modify and save different historical versions of a provider contract [page 632]. These versions are only possible for the contract items.

> → Remember
>
> There are no versions for the allowances that belong to a provider contract.

Transport

You cannot transport customer master data with the SAP CC 2023 Core Tool user interface. This is a *Customizing* activity that needs to be implemented in your SAP system landscape.

When SAP CC is integrated with a CRM application responsible for the management, the replication, and the synchronization of the customer master data, consult the transport function of this system.

Maintenance

You can conditionally change [page 595] a provider contract.

1.37.1 Allowances

An **allowance** in SAP Convergent Charging (SAP CC) is a credit for a service consumption (or use of a product) that a subscriber for a service can redeem for one or more services on a quantity basis or for a specific amount of time. For example, a credit of 3 GB, which must be used within 30 days. This allowance is part of a provider contract data [page 573]. It allows the subscriber and agreed end customers to consume one or use a product for a limited quantity and/or time.

An allowance mostly contains:

- A reference to an allowance plan [page 466] and its allowance logic in the pricing catalog [page 109],
- Parameters to parameterize the logic
- Counters that are used in the logic or that represent a spending limit, a promotion, or a prepaid service allowance

Allowances can be used in a single subscriber account (belonging to the contract) or in different subscriber accounts (see shared allowances [page 579]).

Allowances in contracts allow you to model, design, and configure:

Model	Characteristics	Example
A **credit bucket**, which is a credit given to an end customer or bought by an end customer to use one or several services.	A credit bucket is expressed as a monetary or nonmonetary amount and is valid for a certain period of time.	3GB of GPRS data valid for 30 days
A **spending limit**, which is the right for an end customer to use a service or a group of services limited in quantity or in time.	The end customer usually defines the limits that are expressed as monetary or nonmonetary amounts.	100 SMS at maximum per day
A **promotion**, which is a rule to extend the capacity for using a service.	Promotions can be limited in quantity or in time.	15% discount on national voice calls (valid for two weeks)

→ Recommendation

For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Content and Structure

In detail, an allowance is defined by:

- An (allowance) **identifier**, which is unique in the SAP CC system
- The **ID of the belonging provider contract**
- The **name of the allowance plan** in the pricing catalog that is associated with the given allowance. This reference plan specifies some logic to be executed by the system.
- An account assignment, which is **identifier of the account** (in the subscriber account) that is assigned to the allowance and that can be debited by SAP Convergent Charging.
- A **share ID**, which is a unique identifier that is used to group and access shared allowances
- A **status** (inactive/active/expired)
- A **validity period** made up with a start date/time and an end date/time
- The ID of the provider contract that created the allowance
- The ID and the external ID of the contract item that created the allowance
- A set of **parameters** (optional)
- A set of **counters** (optional) - a counter can relate to a credit bucket, a spending limit, or a promotion
- A currency (for monetary amounts)
- A set of **activated allowance logic objects** that relate to the allowance logic objects that are customized in the related allowance plan.
  The customized allowance logic defines the business/technical/pricing logic that is executed by the system when the activated allowance logic is triggered in an allowance.
- The last date/time when the shared counters of the allowance have been refreshed.

> **Note**
>
> An allowance does not contain any balance. The allowance for a service is debited by SAP CC by using the account that is declared in the allowance data as the account assignment.

An allowance is made up of **activated allowance logic objects**. Every **activated allowance logic object** relates to an allowance logic object that is configured in the allowance plan referenced by the allowance.

Every activated allowance logic object contains **persistent counters** and relates to an allowance logic object that is defined in the allowance plan.

See more here:

- Shared Allowances [page 579]
- Life Cycle and Validity Period Management [page 578]

> **→ Tip**
>
> To view all allowances of a provider contract [page 633], find and open the expected provider contract, and do one of the following:
>
> - Click the icon  on the main window toolbar.
> - Select the *View Allowances* submenu in the *Action* menu.
>
> See also:
>
> - Opening an Existing Provider Contract [page 611]
> - Viewing an Allowance [page 583]

Version

In SAP CC environment, there is no version management for an allowance.

Note that some lifecycle management is available. See Life Cycle and Validity Period Management [page 578].

Automated purges of allowance data are available.

Sizing

> **→ Recommendation**
>
> For optimal performance, do not exceed the recommended maximum sizes explained in SAP Note 2478958. Refer to the provided guidance.

Consider also the purge mechanisms.

Configuration

- How to Configure the Creation of Allowances [page 579]
- Managing Allowances for Integrating SAP CRM with SAP CC [page 585]
- Working with Allowances (in an Integrated SAP Solution Scenario) [page 587]

Related Information

Basic Charging Concepts [page 18]
Configuration of Master Data Step by Step [page 104]
Allowance Logic [page 449]
Allowance Plans [page 466]
Managing Allowances for Integrating SAP CRM with SAP CC [page 585]

1.37.1.1 Allowance Counters

An *allowance counter* is a real counter that belongs to an allowance (in a provider contract).

An allowance counter:

- Can be a variable that is used in the executed logic
- Can be used to reflect the amount/quantity of the service allowance

> **Note**
>
> There are no historical versions of a counter.

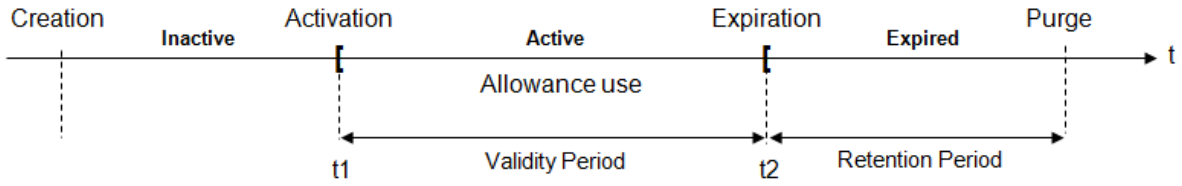
1.37.1.2 Life Cycle and Validity Period Management

Allowance Lifecycle

As part of a provider contract, an *allowance* is first created (*inactive* state) and then activated (*active* state) by a scheduler of the SAP CC system.

- The system creates the allowance data by executing the refill or charge plan identified in a contract item.
- The allowance can be used through the charging or refilling processes triggered internally or by an external system.

- Each allowance has a *validity period*. An allowance expires if the current date/time is after its validity end date/time (*expired* state). SAP CC then allows you to purge expired allowances from the system. The *retention period* is the period during which expired allowances are kept in the system until they are purged.



Allowance Validity Period Management

In the previous illustration, the validity period of an allowance is defined by a start date/time (t1) and an end date/time (t2). By convention, the start date/time is included in the validity period and the end date/time is excluded from the validity period.

Setting Up the Allowance Validity Period

You can set up the start date/time and the end date/time of the validity period of an allowance when configuring the creation of the allowances in the allowance logic [page 449].

Extending the Allowance Validity Period

When the allowance has been activated by the SAP CC system, you can extend the validity period of the allowance through the *Allowance Validity Period* component.

See Allowance Validity Period Updater (Operator) [page 787].

1.37.1.3 How to Configure the Creation of Allowances

At runtime, the SAP CC system creates allowances in provider contracts by executing a *Create Allowance Operator* component that is configured in a charge plan, a refill plan, or in an allowance plan. This logic component allows you to model, design, and configure the creation of allowances by selecting a released allowance plan [page 466] and defining the validity period [page 578] for each allowance.

As an SAP CC pricing specialist, you manage these configuration settings for the pricing of your subscription- and usage-based services.

For more information, see the logic component: Create Allowance Operator [page 792].

1.37.1.4 Shared Allowances

> ⚠ Caution
>
> Unlike the other cached structures that contain partitioned data, the shared allowances cache contains data that is shared among the running raters. To ensure consistency and synchronization between the

running raters, the cached structure containing the shared allowances must be regularly refreshed. For more information, refer to the:

- Configuring the Automated Refreshment of the Shared Allowance Caches section of the SAP CC 2023 Administration Guide documentation
- SAP Note 3200945

Shared allowances are the allowances that you create to be shared between provider contracts that belong to different subscriber accounts. When creating shared allowances, you assign the same unique identifier to a group of shared allowances. Provider contracts use this identifier to access the shared allowances of a group.

Example

As a pricing specialist for a service provider, you create a bucket of 100,000 SMS for a company. This bucket is shared by a subset of contracts that are subscribed by the employees of this company.

This bucket is decremented through a *Decrement Counter Operator* component every time an employee uses a SMS. When the value of the bucket is equal to 0, employees cannot sent SMS anymore.

See the logic component Decrement Counter Operator [page 795].

Example

A company wants to be notified when the number of SMS sent by its employees exceeds 50,000.

As a pricing specialist for the service provider of this company, you define a specific counter which is incremented through an *Increment Counter Operator* component each time an employee sends an SMS. Once this counter reaches 50,000, the company receives a notification stating that the threshold is reached.

See the logic component Increment Counter Operator [page 803].

Creation of Shared Allowances

Allowances are created from a charge plan or a refill plan that is referenced by a provider contract. Once created, a group of allowances can be used from provider contracts that belong to other subscriber accounts.

When creating an allowance in a *Create Allowance Operator* component, the provider contract owner specifies a unique share identifier that allows the other provider contracts to reference this shared allowance and those that have the same share identifier. To use this shared allowance, the other provider contracts send events through an *Allowance Event Sender Operator* component that is configured with the share identifier.

See:

- Create Allowance Operator [page 792]
- Allowance Event Sender (Operator) [page 777]

For more information on this process, see the Application Help. For more information on the logic components, click here [page 671].

Note

- Shared allowances can use the mapping tables [page 179] that are stored in the pricing catalog, but they cannot use the subscriber mapping tables [page 542].
- The configuration of shared allowances implies some restrictions regarding the use of logic components [page 671].

- Refer to the notes about the usage of shared allowances in the allowance logic [page 449] and the allowance plan [page 466].
- The persistent counters that are implemented by shared allowances are not visible in the allowance interface [page 91].

⚠ Restriction

A few components are forbidden within **shareable allowance logics** as inconsistencies in the validity periods may occur:

- Recurring Trigger [page 720]
- One-Shot Trigger [page 723]
- Allowance Event Sender (Operator) [page 777]
- Allowance Property Introducer (Operator) [page 784]
- Allowance Validity Period Updater (Operator) [page 787]
- Create Allowance Operator [page 792]

1.37.1.5 Working with Allowances

1. Searching For Allowances [page 581]
2. Viewing an Allowance [page 583]

In an integrated *SAP Solution*, see:

- Managing Allowances for Integrating SAP CRM with SAP CC [page 585]
- Working with Allowances (in an Integrated SAP Solution Scenario) [page 587]

1.37.1.5.1 Searching For Allowances

Find existing allowances that belong to a provider contract.

Search For Allowances

Find [page 609] and open [page 611] the main window of a provider contract, and then on the *Action* menu, select *View Allowances* to search for and list all the allowances of this provider contract. They are stored in the SAP CC system.

ⓘ Note

On the main window toolbar, you can click  to view all the allowances of a given provider contract.

When the window for viewing the allowances is displayed by the Core Tool user interface, the *Basic* tab allows you to find all the allowances of the belonging provider contract. Work with this result list to complete your tasks.

Use the following icons for your actions:

Icon	Action
	To search for and display the allowances.
	To move to the previous page. This button is available when the result list is large.
	To move to the next page. This button is available when the result list is large.
	To open and display an allowance. You must select an allowance in the list before clicking the icon. See Viewing an Allowance [page 583].

The result list contains the following columns:

- *Unique Identifier*: The unique identifier of the allowance in the SAP CC system
- *Allowance Plan*: The identifier of the allowance plan related to the allowance
- *Validity Start Date*: The date/time from which the allowance is valid and can be debited.
- *Validity End Date*: The date/time from which the allowance is no longer valid and can no longer be debited.
- *Account Code*: The code of the account assigned to the allowance and that can be debited by the SAP CC system. This account is a prepaid account [page 514] or an external account [page 520] in the subscriber account [page 511] that is identified by the system when processing a business operation.
- *Share Identifier*: The unique identifier to group and access shared allowances [page 579]. This share ID indicates that the allowance is shared with some provider contracts (member contracts) that can use and "decrease" the allowance.
- *Currency*: The currency of the allowance

To display more columns of information or to display the counters and parameters of the allowances, click  , and then select which types of information you want to display.

> **Note**
>
> If an allowance does not have a value for a parameter or a counter defined in the allowance interface, the corresponding value in the results list will be the default value of the parameter or counter as defined in the allowance interface.

Next

Next step: Viewing an Allowance [page 583]

Related Information

Viewing an Allowance [page 583]

1.37.1.5.2 Viewing an Allowance

View an Allowance

An allowance is a part of a provider contract.

To open and view a given allowance, select an allowance in the result table [page 581], and then:

- Click  .
- Or double-click the related row.

The window of an allowance displays detailed information about the allowance graphically represented by a tree.

In the tree, each node represents an activated allowance logic object. All these objects compose the displayed allowance. There is an activated allowance logic object for each allowance logic object that is configured in the allowance plan referenced by the allowance.

The icons of the allowance tree are the following:

- The icon  represents the allowance identified by a unique identifier.
- The icon  represents the activated allowance logic (in an allowance).
  Click the icon to view the activated allowance logic. [page 585]

> **Note**
>
> You cannot modify the allowance, but you can save the allowance in a file by clicking *File* > *Save* > *In a File* on the menu bar of the main window. The allowance is saved in the selected file according to the XML format.

Consult the tabs:

- Definition [page 584]
- Counters [page 584]
- Parameters [page 584]

Definition (Tab)

This tab displays the related provider contract, the related allowance plan, and the validity period of the allowance.

Provider Contract: Displays the provider contract of the allowance. Click  to open the provider contract.

Allowance Plan: Displays the allowance plan related to the allowance. Click  to open the allowance plan [page 466].

> **Note**
>
> The dates *Validity Start Date* and *Validity End Date* define the period `[Validity Start Date, Validity End Date[` where *Validity End Date* is excluded.

Validity Start Date: Displays the date/time from which the allowance is valid and can be debited.

Validity End Date: Displays the date/time from which the allowance is no longer valid and can no longer be debited.

Account Code: The code of the account assigned to the allowance for being debited by SAP CC.

Account Type: The type of the account assigned to the allowance (*Prepaid* or *Postpaid*).

Share Identifier: The unique identifier to group and access shared allowances.

Last Update to Counters: The moment (date and time) that corresponds to the last consolidated value of the counters that are used by the shared allowance.

Currency: The currency of the allowance.

Counters (Tab)

This tab displays the counters set in the allowance.

Name: Displays the name of the counter.

Value: Displays the current value of the counter in real time.

Parameters (Tab)

This tab displays the parameters set in the allowance.

Name: Displays the name of the parameter.

Type: Displays the type of the parameter (*Number*, *String*, or *Date*).

Value: Displays the value of the parameter.

1.37.1.5.2.1 Viewing the Activated Allowance Logic Objects

An allowance contains several activated allowance logic objects.

View the Activated Allowance Logic

In the left pane of the Allowance [page 583] window, click the activated allowance logic  that you want to view, and then open the *Definition* tab in the right pane. The *Definition* tab provides basic information on the activated allowance logic.

Definition (Tab)

Last Activation Date: Displays the date and time when the activated allowance logic was last activated by one of the following:

- A charging request
- The internal scheduler
- An external system

Counters (Tab)

This tab displays the counters defined in the activated allowance logic:

Name: Displays the name of the counter.

Value: Displays the current value of the counter in real time.

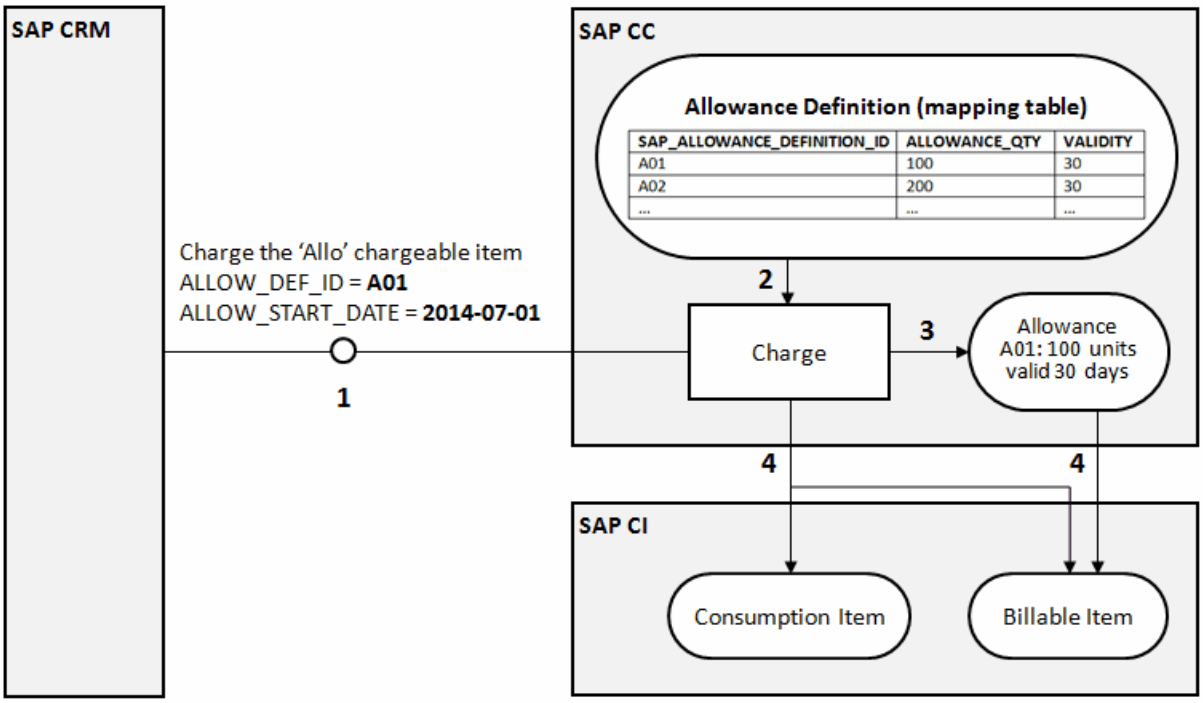
1.37.1.6 Managing Allowances for Integrating SAP CRM with SAP CC

> → Remember
>
> These topics are relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your *SAP Solution*, the *cross-catalog mapping* in the SAP CRM system defines which allowances [page 575] must be created when end customers sign contracts [page 573] for a commercial product or when changes are made to the customers' contract.

- SAP CRM creates allowances in the SAP CC system on demand or automatically when the provider contract is activated or changed.
- SAP CRM sends a charging operation request to SAP CC with an *ALLO chargeable item* (1). This specialized chargeable item contains the identifier of the allowance definition to be created and the date/time at which the allowance must be activated.
- SAP CC processes this charging request, identifies and executes the relating contracted charge [page 613] (2) in this contract.
- This charge retrieves the corresponding allowance definition from the allowance definition table, creates the corresponding allowance, and initializes the allowance according to the allowance definition (3).
- The charge can generate billable items [page 263] for reporting purposes (4), and the allowance can generate billable items for Deferred Revenue Recognition purposes (4).

> **Note**
>
> The numbers in parentheses refer to the steps describing the process of allowance creation in the following illustration.



As a pricing specialist you configure:

- ALLO Chargeable Item Class [page 245]
- Allowance Interface [page 91]
- Allowance Definition Table Class [page 169]
- Allowance Definition Tables [page 182]
- Allowance Event Classes [page 441]
- A charge [page 290]
- A charge plan [page 343]

To customize SAP CC, you are provided with the templates of the ALLO chargeable item class, the allowance definition table class, and the instance. The allowances tutorial provides also concrete examples of charges and charge plans. For more information, refer to the Allowances Tutorial guide [page 40].

Configuring the Master Data

As an SAP CC pricing specialist, you perform the following procedures for configuring the master data:

1. Create [page 246] the predefined ALLO chargeable item class.
2. Customize [page 94] the allowance interface.
3. Create [page 170] an allowance definition table class (as a specific mapping table class).
4. Create [page 184] an allowance definition table (as a specific mapping table).
5. Create [page 443] the allowance event class.
6. Configure [page 588] the allowance creation in an additional charge plan (or refill plan). This plan includes a charge (refill logic) that uses the ALLO chargeable item class.

Related Information

ALLO Chargeable Item Class [page 245]
Allowance Definition Table Class [page 169]
Allowance Definition Tables [page 182]

1.37.1.7 Working with Allowances (in an Integrated SAP Solution Scenario)

Related Information

Creating the Predefined ALLO Chargeable Item Class [page 246]
Customizing the Allowance Interface for SAP CRM [page 94]
Creating an Allowance Definition Table Class [page 170]
Creating an Allowance Definition Table [page 184]
Configuring the Creation of Allowances in Charge Plans [page 588]

1.37.1.7.1 Configuring the Creation of Allowances in Charge Plans

> → Remember
>
> This topic is relevant only if you use SAP Convergent Charging (SAP CC) in an integrated *SAP Solution* scenario with SAP CRM and with SAP Convergent Invoicing in SAP ERP/FI-CA. For more information and user assistance, see the integration guides at help.sap.com/brim.

In your SAP system landscape, to implement the automated creation of an allowance from an *allowance definition* known by the SAP CRM system, you must first design and configure in the pricing catalog:

- A specific reusable charge [page 290] that defines the creation of an allowance in a provider contract in response to the reception of an ALLO chargeable item
- A specific charge plan [page 343] that customizes this charge and can be activated and subscribed to in provider contracts.

Configure the Charge Plan

The charge plan item can have several user service identifiers, but one user service identifier (USID) must match the pattern below so that the SAP CRM system is able to know which user service identifier must be used to activate the charge creating the allowance from the allowance definition:

- `ALLOW_*`

Where * matches 0 to 6 characters. For example, ALLOW_ and ALLOW_TUT are valid user service identifiers.

> ⓘ Note
>
> The charge plan item must have one and only one user service identifier matching the previous pattern.

The charge plan must have at least one mapping table ID of parameter type initialized with a default allowance definition table instance.

To configure the creation of allowances in charge plans, refer to the relevant topics for modeling the charge [page 290] and the charge plans [page 343] accordingly.

Related Information

Working with Charge Plans [page 349]

1.37.1.7.2 Setting up Allowances for the Posting of Deferred Revenues

As a pricing specialist, you design and configure the necessary elements in the pricing catalog (billable item mapping, allowance logic, allowance plan).

In order to post deferred revenues in Contract Accounts Receivable and Payable (FI-CA), allowances should be modeled to generate a billable item when:

- An allowance is created, to report its initial quantity and the corresponding revenue to be recognized
- An allowance is consumed, to report the consumed quantity
- An allowance expires, to report the remaining quantity

When working with SAP Convergent Invoicing, SAP Convergent Charging (SAP CC) must be able to send billable items that include the following data fields:

Field Name (in Billable Items)	Field Description	In SAP CC
ALLOWANCE	*Allowance* It indicates whether a billable item (BIT) is created within the context of the processing of an allowance defined in a provider contract. • ' ' indicates that the billable item does not represent an allowance. • 'X' indicates that the billable item represents an allowance.	SAP CC manages the billable item mapping. When configuring the allowance logic, the default property `Produced by an Allowance` is available.
DEFREV_CAT	*Category of deferred revenues* The category can be time-based ("01") or event-based ("02"). **For allowances, only event-based revenue recognition is relevant.**	
DEFREV_PDATE	Transfer posting date for deferred revenues	
DEFREV_ACTION	*Action code for deferred revenues* • *"01": Create deferred revenue* It indicates that the billable item represents a newly created allowance. • *"02": Recognize revenue* It indicates that the billable item represents usage (consumption) of an allowance or expiration of an allowance.	

Field Name (in Billable Items)	Field Description	In SAP CC
DEFREV_CTYPE	Cost type or revenue type It determines the account and the deferred revenue calculation base (amount or quantity).	
DEFREV_ASSKY	*Assignment key* It is the allowance ID.	
BIT_QUANTITY	*Billing quantity of billable item* Depends on the value of the DEFREV_ACTION field. • If DEFREV_ACTION 01, the • If DEFREV_ACTION 02, the	
BIT_AMOUNT	Amount of billable item Depends on the value of the DEFREV_ACTION field. • If DEFREV_ACTION is set to 01, then the BIT_AMOUNT field is the deferred revenue. • If DEFREV_ACTION is set to 02, then the BIT_AMOUNT field is the amount of the revenue to be recognized.	
BIT_CURR	Currency of billable item Depends on the value of the DEFREV_ACTION field. • If DEFREV_ACTION 01, then the BIT_CURR field is the currency of the deferred revenue. • If DEFREV_ACTION 02, then the BIT_CURR field is the currency of the amount revenue to be recognized.	

Billable items generated to trigger the posting of deferred revenue require specific information from SAP CC.

> → Remember
>
> Deferred revenue is only relevant for allowances when the revenue corresponding to the consumed or lost credit can be recognized.

	Values in Billable Items (BITs)		
Field	Deferred Revenue Creation (Allowance Creation)	Revenue Activation (Allowance Usage)	Revenue Activation (Allowance Expiration)
ALLOWANCE	'X'	'X'	'X'

	Values in Billable Items (BITs)		
Field	**Deferred Revenue Creation (Allowance Creation)**	**Revenue Activation (Allowance Usage)**	**Revenue Activation (Allowance Expiration)**
DEFREV_ASSKY	The allowance ID	The allowance ID	The allowance ID
DEFREV_CAT	"02"	"02"	"02"
DEFREV_PDATE	The event date/time	The event date/time	The event date/time
DEFREV_ACTION	"01"	"02"	"02"
DEFREV_CTYPE	A custom value	A custom value	A custom value
BIT_QUANTITY	The initial quantity of the allowance	The consumed quantity of the allowance	The remaining quantity of the allowance
BIT_QTY_UNIT	The initial quantity of the allowance	The consumed quantity of the allowance	The remaining quantity of the allowance
BIT_AMOUNT	The cost of the allowance	0 for this allowance consumption	0 for this allowance expiration
BIT_CURR	The currency of the cost of the allowance		

An allowance that generates a billable item that triggers the posting of deferred revenue must feature the following:

- A trigger that the SAP Convergent Charging system executes when the allowance is created.
- Event-based triggers for debiting the allowance quantity.
- A trigger that the SAP Convergent Charging system executes when the allowance is expired.

1. Check the billable item class that you must use.
2. Create and update the corresponding billable item mapping in SAP CC.
3. Create and adjust the charged item mapping for the allowance plan in order to set up the fields that are imported for handling deferred revenue in SAP Convergent Invoicing.
4. Depending on your business scenario, create the allowance logic objects.
   - The first allowance logic defines two one-shot triggers [page 723]:
     - A first trigger configures how to report the initial quantity and cost to SAP FICA.
       Configure the business logic by using the Allow Function [page 728] and the Allowance Property Introducer (Operator) [page 784].
     - A second trigger configures how to report lost quantity to SAP FICA.
       Configure the logic.
   - Configure a second allowance logic object that specifies the behavior for the debit of a service credit when the service is consumed.
     Configure the business logic by using Trigger Components [page 716].

1.37.2 Contract Items

A **contract item** is a charging term of the provider contract [page 573] subscribed to by a party known as the subscriber with a service/product provider company. The contract item refers to and activates a charge plan [page 343] or a refill plan [page 421] (or a monitoring plan [page 491]) in the pricing catalog [page 109]. These plans specify the pricing configurations for the subscription- and usage-based service to monetize.

The contract item defines:

- Some identification information
- A validity period for the contract item
- A reference of the charge plan [page 343], refill plan [page 421], or monitoring plan [page 491] that is activated in this contract item by a CRM application or a provisioning system (or by an individual SAP CC user in rare cases)
- Some contract configurations:
  - Some **parameter values** (depending on the referenced plan)
    - Provider Contract>Charge Plan: Parameters [page 616]
    - Provider Contract>Refill Plan: Parameters [page 624]
  - Some account assignments [page 593] specifying the accounts to be charged (debit or credit) for the use of the chargeable service (or product)
  - Some **technical data** elements that specify the technical identifiers under which the subscriber and agreed end customers who consume the usage-based service. The subscriber is charged for the service consumptions.
    - Provider Contract>Charge Plan: Technical Data [page 617]
    - Provider Contract>Refill Plan: Technical Data [page 625]
    - Provider Contract>Monitoring Plan: Technical Data [page 630]
- **Contracted charges** that were subscribed to and that are triggered to compute fees.
  See: contracted charges [page 613] and their counters
- The contracted refill logic [page 621]
- **Real counters** (defined at contract item level) that are updated by the system in real time
  - Provider Contract>Charge Plan: Counters [page 615]
  - Provider Contract>Refill Plan: Counters [page 623]
  - Provider Contract>Monitoring Plan: Counters [page 629]

> **Note**
>
> Counters and contracted charges are not managed with versions.

Structure

A contract item is made up with settings

- Historical versions:
  - Parameters, account assignments, technical data

- No versioning:
  - Real counters
  - Contracted charges

Versions

In SAP CC, a contract item support a partial version management.

A version is partial and concerns only the parameter values, the account assignments, and the technical data.

A contract item is made up with several consecutive versions.

There if no history of neither the contracted charges nor the decreased/increased counters.

Related Tasks

You can complete the following business configuration tasks:

- Activating [page 632] a charge plan, a refill plan, or a monitoring plan
- Sharing counters [page 596] with contract items
- Sharing counters [page 596] with contract items and provider contracts

Related Information

Provider Contract>Charge Plan: Counters [page 615]
Provider Contract>Charge Plan: Parameters [page 616]
Provider Contract>Charge Plan: Account Assignments [page 617]
Provider Contract>Charge Plan: Technical Data [page 617]

1.37.2.1 Account Assignments

An **account assignment** is a **link** between a contract item [page 592] and an account defined in a subscriber account [page 511]. The SAP CC system uses this information during the charging and refilling operations **to determine the relevant accounts to credit or debit**.

The CRM application or the provisioning system must specify the relevant account assignments when creating or maintaining the provider contracts (customer master data): Each contract item must include the account assignments as expected in the activated charge plan or refill plan:

- Account assignments configured in a contract item activating a charge plan [page 617]
- Account assignments configured in a contract item activating a refill plan [page 624]

Otherwise, you manually set up the necessary account assignments in the contract item.

→ Remember

In the service pricing catalog [page 109] (master data), the expected account assignments [page 356] are declared in the charge plan and in the refill plan that are then activated in some contract items of the provider contract subscribed to by an end customer or another subscriber:

- Expected account assignments defined in a charge plan [page 356]
- Expected account assignments defined in a refill plan [page 428]

Related Information

Provider Contract>Charge Plan: Account Assignments [page 617]
Provider Contract>Refill Plan: Account Assignment [page 624]
Subscriber Accounts [page 511]

1.37.3 Provider Contract Creation Stages

⚠ Caution

Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts. Refer to the caution [page 573] for provider contracts.

This procedure describes the main steps for creating and setting up a provider contract [page 573] created manually.

Create a Provider Contract

1. Create [page 599] a new provider contract.
2. Create [page 599] additional information for the provider contract if needed.
3. Add [page 632] contract items by activating charge plans and at most one refill plan.
4. Set up [page 614] the provider contract items that activate a plan (set up the end date of the contract item, share counters, and set up the value of the parameters, accounts, and technical data).

ⓘ Note

The accesses [page 634] corresponding to the provider contract are automatically created by the SAP CC system.

1.37.4 Provider Contract Changes

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts in customer master data.
>
> Refer to the caution [page 573] for provider contracts.

After you have set up and saved a provider contract, you can further modify it to suit your business needs.

Structure Changes

You can modify the composition of a provider contract by activating a plan or deactivating a plan [page 595] and creating new contract items [page 592].

> ⓘ Note
>
> This change creates a new version [page 573] of the provider contract. You must select a date when saving the provider contract to indicate when the change is effective.

The creations/purges of allowances do not change the structure of a provider contract.

Attribute Changes

You can add, remove, or modify some additional information items.

Changes in the Contract Items

At contract item level, you are allowed to modify:

- The external identifier of the contract item[64]
- The end date/time of the contract item
- The share of counters [page 573]
- The values of the parameters
- The account in the account assignment
- The user service identifier of the technical data
- The technical data

The SAP CC system updates the counter values.

[64] This modification does not create a new version of the provider contract.

1.37.5 Sharing Counters with Contract Items

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts. Refer to the caution [page 573] for provider contracts.

SAP Convergent Charging enables you to share a unique counter with several contract items belonging to the same provider contract. This means that one unique counter is created, and then debited or credited by the subscribed charges and the contracted refill logic of several contract items at the same time.

> Example
>
> - When creating a provider contract in SAP CRM, you can share some counters declared in a charge plan or in a refill plan. Consult the configuration documentation of SAP CRM for more information about the product maintenance, the charge plan assignments, and the counter sharing settings.
> - When implementing the spending status monitoring feature for policy control, the contract items that activates a monitoring plan must be configured with counters shared with other contract items of the provider contract.

Namespaces

In addition, you can define a namespace for a shared counter. The namespace enables you to use the same counter name from the counter dictionary in different contexts. The pair (counter name, namespace) is unique and is used to identify the shared counter.

> Example
>
> A counter called **`FREE_UNITS`** can be used in the namespace SMS and in the namespace VOICE. In that case, two different counters with the same name will be created and used separately inside the same provider contract (see also Sharing Counters with Contract Items [page 596]).

Prerequisites

The following prerequisites must be fulfilled to share a counter:

- In each charge plan or refill plan that will be activated in a provider contract, the SAP CC pricing specialist must:
  - Declare the counter with the same name. Note that you must choose this name in the counter dictionary [page 86] configured by the application administrator.
  - Set the visibility of the counter to *External*[65]

[65] Only in charge plans. The visibility of counters is always external in refill plans.

- A particular charge plan is dedicated to configuration of the initialization of shared counters [page 349]

Share a Counter

Apply the following procedure. This manual procedure is only for test, demo, or training purposes.

To share a counter inside a provider contract:

1. As a customer sales representative (CSR), create or modify the provider contract with which you want to share counters.
2. Verify in each charge plan and refill plan activated in the provider contract that the prerequisites are fulfilled.
3. Share the counter in a first contract item by displaying the *Counters* tab and selecting the status *Shared Within Contract*:
   - In a contract item activating a charge plan, set up the Counters [page 615] tab.
   - In a contract item activating a refill plan, set up the Counters [page 623] tab.
4. Share this counter with the required contract items. For example, if you want to be able to monitor this counter in the form of a spending status, you can set up a monitoring plan [page 491].

Initialize the Shared Counter

The procedure to initialize the shared counter consists in creating in the pricing catalog, an additional charge plan that contains a dedicated parameter.

Then in the provider contract containing the shared counter, the customer sales representative creates a contract item that "activates" this new charge plan. This contract item includes a parameter that is dedicated to the counter initialization.

The procedure consists in the following steps:

1. The SAP CC pricing specialist maintains the pricing catalogs by creating:
   1. A reusable charge dedicated to initializing the shared counter. See Using a Charge to Configure the Initialization of a Shared Counter [page 299].
   2. A dedicated charge plan containing this new charge. See Using a Charge Plan to Configure the Initialization of a Shared Counter [page 349].
2. A customer sales representative (CSR) changes the provider contract by.
   1. Creating a new contract item referencing the charge plan dedicated to initializing the shared counter.
   2. Setting up the status of the counter to *Shared*.
   3. Specifying the same namespace as for the other contract items, if needed.
   4. Setting up the parameter linked to that of the charge to the value needed to initialize the shared counter.

> → Remember
>
> Do not initialize the same counter twice.

1.37.6 Spending Statuses

Once the spending status monitoring function is implemented, the SAP CC system can report some business indicators to a registered policy or monitoring system able to make policy decisions. For each provider contract, the label values of these spending statuses reflect the current context of an end customer regarding to the customer service use and to monitored counters in the provider contract.

This data is not persistent. The SAP CC never stores this data in the back-end database system.

The possible label values of each type of spending status are configured in spending status descriptions [page 483] and must be synchronously configured in the policy or monitoring system.

During online charging and refilling operations, the SAP CC system manages these spending statuses according to the configuration set in one or more contract items that activate a monitoring plan [page 628].

> ### Example
>
> **High-Tech and Media Industry**
>
> The SAP CC system reports spending statuses to a registered policy system. These reports include the "Data fair usage" spending status. The possible labels are configured both in the SAP CC (in a spending status description [page 483]) and in the policy system.

> ### Example
>
> **Telecommunications Industry**
>
> In the 3GPP environment, the SAP CC system generates spending limit reports (named spending status reports) that include different policy counter statuses and identifiers to a registered PCRF system.

Related Information

Spending Status Description [page 483]
Setting Up a Monitoring Plan in a Provider Contract [page 628]
Provider Contract>Monitoring Plan: Counters [page 629]
Setting Up the Generation of a Spending Status [page 507]
Setting Up a Spending Status Description [page 487]

1.37.7 Working with Provider Contracts

Related Information

Creating a Provider Contract [page 599]
Finding a Provider Contract [page 609]
Opening an Existing Provider Contract [page 611]
Changing the Version of a Provider Contract [page 632]
Modifying a Provider Contract [page 612]
Setting Up a Provider Contract [page 600]
Activating a Plan When Adding a Contract Item [page 632]
Saving a Provider Contract [page 606]
Deleting a Provider Contract [page 612]
Sharing Counters with Provider Contracts [page 607]
Searching For and Viewing the Allowances of a Provider Contract [page 633]

1.37.7.1 Creating a Provider Contract

⚠ Caution

Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts and customer master data.

Refer to the caution [page 573] for provider contracts.

→ Recommendation

You can manually create a provider contract to:

- Set up end-to-end tests of your pricing modeling.
- Simulate the results of the implementation of the provisioning process between your CRM application and SAP Convergent Charging. This simulation can help you in validating the design of this implementation activity.
- Configure this customer master data to conform to your pricing modeling and design. During the implementation phase, the project team creates this data.

Create a Provider Contract for Test, Demo, or Training Purposes

→ Remember

As prerequisites:

- A subscriber account must exist in customer master data because the provider contract belongs to it.

- A pricing catalog must exist and include the necessary pricing elements for the pricing configuration of the chargeable service or product to monetize.

1. From the menu bar in the main window, select *File* > *New* > *Provider Contract*.
2. Click *OK*. The *Choose a Subscriber Account* window opens.
3. Search for [page 537] the subscriber account [page 511] that must refer to the new provider contract.
4. Select the relevant subscriber account, and then click *OK*. The *New Provider Contract* window opens.
5. Type the name of the provider contract. Choose a name which is not currently used for an existing provider contract.
6. Set up [page 600] the provider contract.
7. Verify that the provider contract is correctly set up by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
8. Save [page 606] the provider contract.

Related Information

Setting Up a Provider Contract [page 600]
Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]

1.37.7.2 Setting Up a Provider Contract

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts and customer master data. Refer to the caution [page 573] for provider contracts.

Create [page 599], open [page 611], or modify [page 612] a provider contract before setting it up. A *Provider Contract* window must be open.

A provider contract  is graphically represented in the *Provider Contract* window by a tree. In this tree, each node is a contract item [page 592] that activates a charge plan, a refill plan, or a monitoring plan from the pricing catalog [page 109] for the subscription- and usage-based services to monetize. The contract items have the same structure as the original plans.

In the provider contract tree, the icons are the following:

 : Represents the provider contract.

 : Represents a contract item that references a charge plan in the service pricing catalog.

 : Represents a contract item that references a refill plan.

 : Represents a contracted charge originally contained in the charge plan or represents the contracted refill logic originally contained in the refill plan.

 : Represents a contract item that references a monitoring plan.

Set Up a Provider Contract

1. In the left pane of the *Provider Contract* window, click the provider contract root  .
2. If necessary, link the contract to a parent contract: Definition tab [page 601].
3. Set up the following tab of the provider contract: Additional Information tab [page 605].
4. If necessary, set up [page 614] the contract items that activate a charge plan in the provider contract.

   For each contract item, click  in the left pane, and then set up the contract item.
5. If necessary, set up [page 614] the unique contract item that activates a refill plan in the provider contract.

   Click  in the left pane, and then set up the contract item.
6. If necessary, share a pool of counters [page 607] with provider contracts.
7. If necessary, share counters [page 596] with the contract items of the provider contract.
8. If necessary, activate or deactivate [page 632] charge plans or a refill plan.

> ⓘ Note
>
> If the *Counters*, *Parameters*, or *Additional Information* tabs of a contract item are unavailable (dimmed), this means no external counter, no external parameter or no additional information was created in the original charge plan or refill plan activated in this contract item.

Related Information

Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]

1.37.7.2.1 Provider Contract: Definition

In the left pane of the provider contract window, click the contract root node  to view the provider contract, and then open the *Definition* tab.

The *Definition* tab provides information on the subscriber account [page 511] and the batch rating group [page 59] of the provider contract. If the provider contract is linked to another provider contract, some information related to this provider contract is displayed (see the Counters [page 603] tab).

Definition (Tab)

Subscriber Account: Displays the service provider and the code of the subscriber account to which the provider contract is related or the information from the parent contract. These fields are automatically filled when you specify the subscriber account to create the provider contract [page 599].

- *Contracting Service Provider*: Displays the identification information of the contracting service provider that manages this provider contract. This service provider is also the owner information [page 111] of the pricing catalog that includes charge plans and refill plan activated in this provider contract. The name of the contracting service provider is the name of its catalog.
- *Code*: Displays the identifier (or name) of the related subscriber account [page 511]. Click  if you want to view and modify the subscriber account.

Linked to Provider Contract: Select the parent contract of a linked contract that reuses a pool of counters and all the allowances shared by this provider contract with all the linked contracts. Click  if you want to select a parent contract to reuse its counters and allowances. Click  if you want to view the selected parent contract, its shared counters, and its allowances. Once saved, you cannot unlink contracts. Refer to the notes to save a provider contract.

Batch Rating Group: Displays the code name of the batch rating group assigned to the provider contract if your system landscape includes SAP CC BART Database to provide offline charging services. The SAP CC system sets up the selected group in the background. If the provider contract is linked to a parent contract, the batch rating group is the same. Consult the SAP CC Administration Guide for more information about the necessary *Customizing* activities.

Operational Status: Displays the status of the provider contract. SAP Convergent Charging allows you to manage an operational status for each provider contract when transferring provider contracts from an SAP CC system to another. The values of this status are the following:

- *Active*: The provider contract is fully operational.
- *Locked*: The provider contract is temporarily locked and the processes of SAP CC related to the provider contract are not available. You cannot modify the provider contract.
- *Closed*: The provider contract is no longer operational and is deactivated. You can no longer modify the provider contract.

> **Note**
>
> For more information on the operational status, consult the technical documentation (Integration & Analytics Information) [page 40] about SAP CC Web Services on the service operation `Change the Operational Status of a Charging Contract`.

Related Information

Setting Up a Provider Contract [page 600]

1.37.7.2.2 Provider Contract: Counters

The *Counters* tab allows you to view the counters of the provider contract that are shared with several contract items inside the provider contract and their current value. If the provider contract is a parent contract or a linked contract, the *Counters* tab displays the pool of counters shared with these provider contracts. The tab is dimmed when no counters are shared inside or outside the provider contract.

The fields of the *Counters* tab cannot be edited. Use the Counters [page 615] tab of each contract item.

Tab: Counters

Name: Displays the name of the shared counter.

Status: Displays the status of the shared counters:

- *Shared Within Contract*: The counter is shared with all the contract items of the provider contract. See the counters of a contract item [page 615] for more details.
- *Shared With Linked Contracts*: The counter is shared with all the contract items of the provider contract and is also shared with all the linked contracts. This status is only available with a parent contract.
- *Shared With Other Contracts*: The counter is shared with all the contract items of the provider contract and is also shared with all the contracts that have the same parent. This status is only available with a provider contract linked to a parent contract to share a pool of counters.

Namespace: Displays the namespace [page 596] of the shared counter. If there is no namespace, this field is set to *None*.

Value: Displays the current value of the shared counter. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. This means that no usage event (end-customer service consumption) and no activation has occurred yet. If the provider contract is linked to a provider contract, the shared counters are stored in a pool in the parent contract.

Related Information

Setting Up a Provider Contract [page 600]

1.37.7.2.3 Provider Contract: Rerating State

The *Rerating State* tab allows you to view information on the rerating process of the provider contract.

Note that the fields of the *Rerating State* tab cannot be edited.

Tab: Rerating State

> **Note**
>
> The fields *Last Rerating Status* and *Suggested Rerating Date* are available when SAP Convergent Charging performs the rerating process with BART; otherwise, these fields are not available. This configuration depends on how SAP Convergent Charging is installed and configured to work with SAP ERP/FI-CA.
>
> For more information, consult the Administration Guide and the Installation and Maintenance Guide.

Rerating Lock Code: Displays the lock code of the provider contract when the provider contract is being rerated. After the provider contract is rerated, the code is deleted from the field. The code remains displayed if the provider contract has not been unlocked yet or if an error has occurred during the rerating process.

Last Rerating Lock Code: Displays the lock code of the last rerating process.

Last Rerating Status: Displays the status of the last rerating process. You can use the code associated to the status to find the provider contracts [page 609] that have failed during the rerating process. The statuses are the following:

- `alreadyLockedFailure`: The rerating process failed because the provider contract was locked by the last rating operation or by another rerating operation.
- `counterRestorationFailure`: The restoration process failed during the last rerating operation because provider contract counters were not correctly restored.
- `billingFailure`: The rerating process has failed because of a billing issue.
- `restorationSuccess`: The restoration process was successful during the last rerating operation.
- `unlockfailure`: The provider contract has not been unlocked.
- `prepaidAccountFailure`: The rerating process has failed because the account for the provider contract was a prepaid account.
- `selectionSuccess`: The selection of the provider contract was successful.
- `reratingSuccess`: The rerating process was successful during the last rerating operation.

Suggested Rerating Date: Displays the rerating date and time that have been suggested by the external billing system during the last rerating session. This field is empty if the rerating date has not been modified by the external billing system.

Related Information

Setting Up a Provider Contract [page 600]

1.37.7.2.4 Provider Contract: Additional Information

The *Additional Information* tab allows you to view the additional information items set in a provider contract. This custom information data is only descriptive and is not reused by SAP Convergent Charging at any stage of the charging process. You cannot configure any pricing logic based on this data. This custom information facilitates the search in Core Tool (see Finding a Provider Contract [page 609]) or in an external system (billing, CRM, ERP) that is integrated with SAP Convergent Charging. This system reuses the configured additional information for further processing.

> ### Example
>
> For example, additional information can be the author's name of the provider contract, the author's code, the creation date/time, and other custom information or reference.

This information is set by the external system (ERP, CRM, or provisioning) that normally creates a provider contract. This system knows how to use this information.

If your implementation scenario and pricing modeling does not include such a system, you manually set up the relevant additional information items. Create [page 599] or modify [page 612] a provider contract before setting in up.

Tab: Additional Information

Fill in or modify the *Additional Information* tab.

Use the configuration table and its icons to add some additional information items.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Fill in each row that you add to the table:

Name: Type the name of the additional information.

Type: Provides the list of the additional information types. Select one of the following types: *Number*, *String*, or *Date*.

Value: Type the additional information according to the selected type.

Description: Type the description of the additional information. This field is optional.

Related Information

Setting Up a Provider Contract [page 600]
Finding a Provider Contract [page 609]

1.37.7.3 Saving a Provider Contract

Once you have created or modified a provider contract, and then verified that it is correctly set up, you are ready to save it. You can save the provider contract to either the database or as an XML file on your hard disk.

> **Note**
>
> - When a provider contract is saved, all the charges [page 290] of the subscribed charge plans and the refill logic [page 395] of the subscribed refill plan are automatically activated by the SAP CC system and may trigger one-off fees depending on their configuration.
> - After saving a provider contract to the database, you can no longer modify the relationship to a parent contract [page 601].
> - No new version of the provider contract is created if it is identical to the existing version valid at the date of the modification.

To save a new provider contract to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Create a Provider Contract* dialog box opens to inform you that the provider contract has been successfully created.
2. Click *OK*.

To save a new version of an existing provider contract to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Choose date* dialog box opens
2. Select a date from which the provider contract modifications will be effective, and then click *OK*. The *Provider Contract Modification* confirmation box opens to inform you that the provider contract has been successfully modified.
3. Click *OK*.

> **Caution**
>
> Refer to the notes for modifying [page 612] a provider contract.

To save a provider contract as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the provider contract.

4. In the *File name* field, type the name of the file.
5. Click *Save*. The provider contract is saved as an XML file in your selected directory.

1.37.7.4 Sharing Counters with Provider Contracts

> **Note**
>
> Do not modify a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts. Refer to the caution [page 573] for provider contracts.

SAP Convergent Charging enables you to share a counter with several provider contracts linked to the same parent provider contract. This means that only one counter is created and used by the charges and the refill logic of several provider contracts at the same time. A counter share is defined for a version of a provider contract. You can change the counter share level in another version of the provider contract.

You configure:

- A charge plan for the parent contract; It configures the reset of the shared counter.
- A charge plan for the linked contracts; It configures the pricing logic that is based on the counter shared by a parent contract.

> **Example**
>
> A central provider contract shares some counters for credit pooling. A linked contract reuses a counter of its parent contract during the first year. After this period, the linked contract uses its own counter.

Prerequisites

The following prerequisites must be fulfilled to share a counter with some provider contracts.

- You do not use the counter in contract items activating a monitoring plan set with this counter.
- In each charge plan or refill plan that will be part of the parent contract or the linked contracts, you must:
  - Declare the counter with the same name. Note that you must choose this name in the counter dictionary [page 86].
  - Set the visibility of the counter to *External*[66].

Sharing a counter with provider contracts

Apply the following procedure.

[66] Only in charge plans. The visibility of counters is always external in refill plans.

To share a counter

1. Create or modify the provider contract that will be the parent contract.
2. Share [page 615] the counter share with linked contracts.
3. Create or modify the provider contracts.
4. At charge plan and refill plan levels, verify that the prerequisites are fulfilled.
5. Share the counter[67]. For SAP Convergent Charging, this modifies:
   - The provider contract whose items reference the charge plans and the refill plan which contain the counter
   - The counter of each contract item which is assigned the *Shared* status

In addition, you can define a namespace for a shared counter. The namespace enables you to use the same counter name from the counter dictionary in different contexts. The pair (counter name, namespace) is unique and is used to identify the shared counter.

> ✤ Example
>
> A counter called F**REE_UNITS** can be used in the namespace SMS and in the namespace VOICE. In that case, two different counters with the same name will be created and used separately inside the same provider contract.

Initializing and resetting the counter

The procedure to initialize and reset the shared counter consists in creating an additional charge plan which contains a dedicated parameter. This charge plan is then referenced by an item of the parent provider contract containing the shared counter.

The procedure consists of the following steps:

1. Create a charge dedicated to initializing the shared counter.
2. Create a charge plan containing this charge.
3. Insert the charge plan in the provider contract.

> ⚠ Caution
>
> Do not initialize the same counter twice.

To create a charge dedicated to initializing the shared counter

1. Create a charge.
2. Add a one-shot rate, a persistent counter, and a parameter to the charge. Note that the parameter enables you to initialize the counter at provider contract level.
3. Set the *Event Name* to *Activation* in the *Definition* tab of the one-shot rate.
4. Add a Counter Operator component [page 302] to the one-shot rate.
5. Set the counter to the value of the parameter in the *Counter Operator* component[68].

[67] This step must be performed from a CRM application or an external system through SAP Convergent Charging Web Services.

[68] See the primary help of components [page 671] for more information about one-shot rates and counter operators.

To create a charge plan containing this charge

1. Create a charge plan.
2. Add the charge to the charge plan.
3. Create the counter to be shared with the visibility option set to *External* in the charge plan. Do not forget to use the same name as previously.
4. Link the counter to the counter declared in the charge.
5. Define a parameter, and then link the parameter with the visibility *External (Mandatory)* to the parameter declared in the charge.

Insert the charge plan in the provider contract

1. Create a new contract item referencing the charge plan dedicated to initializing the shared counter.
2. Set the status of the counter to *Shared*.
3. Specify the same namespace as for the other contract items, if needed.
4. Set the parameter linked to that of the charge to the value needed to initialize the shared counter.

1.37.7.5 Finding a Provider Contract

To search for a provider contract in the customer master data [page 30] configured in SAP CC, you perform the operation in two halves of the *Choose a Provider Contract* window:

- The top part of the window is the filter area, which allows you to find a provider contract by using filter criteria.
- The bottom area of the window is the results area, which displays the search results that you perform.

Tasks

You can find provider contracts with or without search criteria.

- If you do not use search criteria, click  to start the search and view the results.
- If you use search criteria, set up the *Basic* tab, and then click  to start the search. The provider contracts are displayed in the results area. To display more columns of information in the results area, click  , and then select which types of information you want to display. Click  to reset the filter and delete the result list.

Filter Area

Basic Tab

The *Basic* tab allows you to search for provider contracts depending on an identification code, a subscriber, or a service provider.

Code: Is the identifier of the provider contract. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database.

> ### Example
>
> For example, if you want to find all provider contracts containing the letters **`eric`**, you select the *Contains* option, and then type **`eric`**.

Subscriber Account: Is the identifier of the subscriber account [page 511] related to the provider contract. The drop-down list is described just above. Type the string that you want to search for in the database.

Service Provider: Is the owner information of the pricing catalog [page 109]. The drop-down list is described just above. If you are associated to a catalog, this field is automatically filled in with the name of the related catalog and cannot be modified. Otherwise, you type the string that you want to search for in the database.

Customer Management Area: Is the customer management area [page 87] that is responsible for a subset of provider contracts when the system landscape includes several SAP ERP/FI-CA or SAP CRM systems. Select *Any* or the appropriate name.

Exclude all the linked contracts: This option allows you to search for standard provider contracts and parent contracts [page 601] that share a pool of counters with linked contracts. To search for the linked contracts of a parent contract, use the *Advanced* tab.

Advanced Tab

To search for some provider contracts, the *Advanced* tab allows you to create a table with one or more search criteria based on configuration for searched subscriber accounts:

- Rerating information (see Provider Contract: Rerating State [page 604])
- Additional information items (see Provider Contract: Additional Information [page 605])
- Parent provider contract information

Actions: To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> ### Note
>
> - If the criteria table comprises several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table consists of three fields. To set up a row, do the following:

1. In the first field, select a search criterion such as:
   - *Add. Info (Date)*
   - *Add. Info (Number)*
   - *Add. Info (String)*
   - *Last Rerating Code*
   - *Last Rerating Status*
   - *Rerating Lock Code*
   - *Suggested Rerating Date*

- *Parent Provider Contract*

2. In the second field, select a search comparator such as *After*, *Before*, *After or Equals*, *Before or Equals* for example.
3. In the third field, select or type the number, the date/time, or the string that you want to find in the database.

Results Area

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

Related Information

Opening an Existing Provider Contract [page 611]

1.37.7.6 Opening an Existing Provider Contract

The *Choose a Provider Contract* window allows you to find the provider contract that you want to open from the database.

> **Note**
>
> - Because provider contracts are managed chronologically, you can open a previous version of a provider contract.
> - To open a previous version of a provider contract, open the provider contract as described in this topic.
>
> When the provider contract is open, click the *Change Version* icon  on the toolbar in the main window. The *Change Version* dialog box opens. Select a version of the provider contract from the list, and then click *View*.

To open an existing provider contract

1. From the menu bar in the main window, select *File* > *Open* > *Provider Contract*. The *Open Provider Contract* dialog box opens.
2. Search [page 609] for the provider contract.
3. Select the provider contract, and then click *OK*. The provider contract window opens.

1.37.7.7 Modifying a Provider Contract

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts and customer master data. Refer to the caution [page 573] for provider contracts.

To modify a provider contract

1. Open [page 611] a provider contract.
2. Set up [page 600] the provider contract.
3. Verify that the provider contract is correctly set by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. Save [page 606] the provider contract.

1.37.7.8 Deleting a Provider Contract

> ⓘ Note
>
> - There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.
> - Do not delete a provider contract created by a CRM or another provisioning system interfaced with SAP CC. Refer to the caution [page 573] about provider contracts.

To delete a provider contract

1. From the menu bar in the main window, select *File* > *Delete* > *Provider Contract*. The *Delete Provider Contract* window opens.

2. Search [page 609] for the provider contract.
3. Once you have located the provider contract that you want to delete, select the provider contract row, and click *Delete*. A deletion dialog box asks if you want to delete the provider contract.
4. Click *Yes* to delete the provider contract. A dialog box confirms that the provider contract has been successfully deleted.

1.37.8 Working with Contract Items Activating Charge Plans

In a provider contract, you work with a contract item [page 592] that activated a charge plan [page 343] from a corresponding pricing catalog.

The contract item includes one or more contracted charges (aka subscribed charges) [page 613].

Related Information

Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]
Contracted Charges [page 613]
Working with Contracted Charges [page 618]

1.37.8.1 Contracted Charges

Contracted charges are charges recopied from the charge plan (in the pricing catalog for the chargeable service) that has been subscribed to and are included in a contract item [page 592] as result of the subscription (aka contract ordering) operations.

In a provider contract, each contracted charge of a contract item includes the following data and information:

- Last activation/trigerring date and time [page 619]
- Real counters [page 620]

This contracted charge relates to a customized charge in the corresponding charge plan. See Characteristics of a Customized Charge [page 344].

> → Remember
>
> The **technical name** is `Charge Activation`.
>
> Keep in mind this terminology when working with the SAP software product, logs, and APIs.

Related Information

Working with Contracted Charges [page 618]

1.37.8.2 Setting Up a Provider Contract Item That Activates a Charge Plan

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or a provisioning system responsible for managing provider contracts and customer master data. Refer to the caution [page 573] for provider contracts.

Create [page 599], open [page 611]modify [page 612] a provider contract before setting up a provider contract item activating a charge plan. A , or *Provider Contract*, or window must be open. A provider contract item is also known as a contract item.

> → Remember
>
> Each provider contract must include contract items [page 592] that activate [page 632] a charge plan or a refill plan from the pricing catalog of the contracting service provider.

Configure a Contract Item That Activates a Charge Plan

1. In the left pane of the *Provider Contract* window of your provider contract, click  .
2. Set up the tabs of the contract item that references a given charge plan:
    1. Definition [page 615]
    2. Counters [page 615]
    3. Parameters [page 616]
    4. Account Assignments [page 617]
    5. Technical Data [page 617]

> ⓘ Note
>
> If the *Counters* or the *Parameters* tab of the charge plan is not available (dimmed), it means no external counter or no external parameter was declared in the original charge plan [page 343] or refill plan [page 421].

Related Information

Setting Up a Provider Contract Item That Activates a Refill Plan [page 621]
Setting Up a Monitoring Plan in a Provider Contract [page 628]

1.37.8.2.1 Provider Contract>Charge Plan: Definition

The name of the charge plan activated by the contract item [page 592] is displayed on the *Definition* tab.

Definition (Tab)

External Code: Type the code to identify the contract item in the customer data of an external system. Used with SAP CRM and SAP ERP/FI-CA, this code identifies an item in the master data managed by SAP CRM and that must be present in the billable items sent to SAP Convergent Invoicing (in SAP ERP/FI-CA) or in SAP S/4HANA). See the configuration of the billable item mapping [page 263] and the output field CONTRACTITEMID.

If SAP CC is integrated with another CRM or provisioning system, the use of this code is customized during the project implementation phase.

Activated Plan: Displays the name of the original charge plan [page 343] from the pricing catalog [page 109] that is activated in this contract item. Click  if you want to view the charge plan in the pricing catalog for the chargeable service or product.

End Date: Select the date and time when this contract item is no longer valid.

Related Information

Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]

1.37.8.2.2 Provider Contract>Charge Plan: Counters

The list of the counters declared in the original charge plan [page 353] is displayed on the *Counters* tab. Each row in the list represents a real counter (*) and its current value as stored in the contract item (in the provider contract in customer master data).

If the contract is a linked contract, the counters are only stored in the parent contract [page 573]. You define the shared counter for the current version of the provider contract. Refer to the notes [page 596].

(*) Counters in Contracts

A counter is a numerical data stored in provider contracts in customer master data. During the charging and refilling operations, the SAP CC system uses the counter values to dynamically computes the price of a customer service usage and then updates these counters for further calculation.

A counter can represent a technical information or a business information like a service credit.

When provisioning the provider contracts, the CRM application or the provisioning system creates all the counters as declared in the charge plans and refill plans that are referenced by a contract item.

In the pricing catalog of the service provider, you configure the conditions of counter creations by setting up the charge plans and the refill plans that will be activated in the provider contracts of the end customers.

Counters (Tab)

Name: Displays the name of the counter.

Description: Displays the business or functional description of the counter.

Status: Select the status of the shared counters:

- In a provider contract, set up the status to *Shared Within Contract* if you want this counter to be shared with several contract items of the provider contract. Otherwise, select *Not Shared*.
- In a parent contract, select *Shared With Other Contracts* to share the counter with all the contract items and all the linked contracts.
- In a linked contract, select *Shared With Other Contracts* to reuse the counter shared by the parent contract with all its linked contracts. If you remove this status in another version of the linked contract, the counter is reset to zero.

Namespace: Select the namespace of the shared counter. Select *None* if there is no namespace.

Value: Displays the current value of the counter. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. It means that no usage event (customer service consumption) has been processed by the system and no activation has occurred yet.

Related Information

Provider Contract: Counters [page 603]
Charge Plan: Counters (Declaration) [page 353]

1.37.8.2.3 Provider Contract>Charge Plan: Parameters

The list of the shared parameters declared for the original charge plan is displayed on the *Parameters* tab. Each row in the list represents a **shared parameter and its current value**.

Parameters (Tab)

Name: Displays the name of the parameter.

Type: Displays the type of the parameter (Number, String, or Date/Time).

Value: Set up the value of the parameter if you want to refine the default value.

Related Information

Charge Plan: Parameters [page 354]

1.37.8.2.4 Provider Contract>Charge Plan: Account Assignments

The list of the account assignments [page 593] as expected in the original charge plan is displayed on the *Account Assignment* tab. Each row in the list represents an account assignment. The SAP CC system uses this information to credit or debit the relevant accounts during charging operations.

Account Assignment (Tab)

Account Assignment: Displays the name of the account assignment.

Account Type: Displays the type which is expected for the account (*Prepaid* or *Postpaid*).

Account: Select the account to be credited or debited. This account is a prepaid account [page 514] or an external account (postpaid) [page 520] defined in the subscriber account [page 511] that relates to the provider contract.

Related Information

Charge Plan: Account Assignments (Expected) [page 356]

1.37.8.2.5 Provider Contract>Charge Plan: Technical Data

In the contract item [page 592], the *Technical Data* tab allows you to view the technical identifier under which an end customer or a subscriber consumes a service. The interfaced CRM or provisioning system must specify

these technical identifiers depending on the settings in the charge plan (from the pricing catalog) that has been activated in this contract item.

> **Note**
>
> For a given service, the value of this technical identifier must be present in all the input *chargeable items* that represent the usage (consumption) of this service. These chargeable items contain a data field `USID` that is set with a value that match the value of the technical data that is set in the contract item.

Technical Data (Tab)

Each row in the configuration table is a technical data. A contract item can include a few pieces of technical data.

User Technical Identifier: Select the name of the user technical identifier (UTI) which is declared in the activated charge plan [page 356]. This is a "custom category" of user service identifiers (USIDs). For example, `phoneNumber`, `rechargeCardID`, `emailAddr`, or `logonToService`.

User Service Identifier (USID): Type in the technical identifier of the user (end customer) who consumes the chargeable service.

> **→ Remember**
>
> To be charged, a service consumption must relate to a chargeable item that contains a `USID` field whose value must match the value of the technical data that is set in this contract item.
>
> See also: Transactional Data [page 34]

> **Example**
>
> The User Technical Identifier (UTI) is set to "phoneNumber" and the User Service Identifier (USID) is set to "33432536445".

Related Information

Changing the Version of a Provider Contract [page 632]
Charge Plan: Technical Data (Expected) [page 356]

1.37.8.3 Working with Contracted Charges

Related Information

Opening a Contracted Charge [page 619]
Viewing a Contracted Charge [page 619]

1.37.8.3.1 Opening a Contracted Charge

Open a Contracted Charge

1. Open [page 611] a provider contract.
2. In the tree of the provider contract, click  to open a contracted charge.
3. View [page 619] the following tabs of the charge: Definition [page 619], Counters [page 620].

1.37.8.3.2 Viewing a Contracted Charge

Related Information

Provider Contract>Contract Item>Contracted Charge: Definition [page 619]
Provider Contract>Contract Item>Contracted Charge: Counters [page 620]

1.37.8.3.2.1 Provider Contract>Contract Item>Contracted Charge: Definition

The name of the contracted charge in the contract item [page 592] is displayed on the *Definition* tab.

Definition (Tab)

Last Activation Date: The latest event date/time on which the contracted charge has been triggered in SAP Convergent Charging. The event could be a recurring event or a one-shot event.

> **Note**
>
> - The usage event does not modify the last activation date
> - The event date/time is not necessarily the date/time this event was processed by the system

Related Information

Recurring Rate [page 693]
One-Shot Rate [page 698]

1.37.8.3.2.2 Provider Contract>Contract Item>Contracted Charge: Counters

The list of the counters declared in the original charge [page 290] (in the pricing catalog) is displayed on the *Counters* tab. Each row in the list represents a real counter and its current value as stored in the provider contract.

Counters (Tab)

Name: Displays the name of the counter.

Description: Displays the business or functional description of the counter. The description is set in the pricing catalog by the SAP CC pricing specialist. It facilitates the understanding of the counter.

Value: Displays the current value of the counter in real time. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. This means that no service usage event and no activation has occurred yet.

Related Information

Provider Contract>Charge Plan: Counters [page 615]

1.37.9 Working with Contract Items Activating a Refill Plan

In a provider contract, you work with a contract item [page 592] that activated a refill plan [page 421] from a corresponding pricing catalog.

Related Information

Setting Up a Provider Contract Item That Activates a Refill Plan [page 621]
Contracted Refill Logic [page 621]
Working with the Contracted Refill Logic [page 625]

1.37.9.1 Contracted Refill Logic

Contracted refill logic objects are refill logic objects from a refill plan (in a pricing catalog) that have been subscribed to and are included in a contract item as result of the subscription (aka contract ordering).

Each contracted refill logic in the contract item includes the following data and information:

- Last activation/trigerring date and time [page 626]
- Real counters [page 627]

Related Information

Working with the Contracted Refill Logic [page 625]

1.37.9.2 Setting Up a Provider Contract Item That Activates a Refill Plan

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or a provisioning system responsible for managing provider contracts and customer master data. Refer to the caution [page 573] for provider contracts.

Create [page 599], open [page 611], or modify [page 612] a provider contract before setting up a contract item activating a refill plan. A *Provider Contract* window must be open.

The provider contract must include some items that activate [page 632] a charge plan or a refill plan.

Configure a Contract Item That Activates a Refill Plan

1. In the left pane of the *Provider Contract* window, click  .

2. Set up the tabs of the contract item which references a refill plan:
    1. Definition [page 622]
    2. Counters [page 623]
    3. Parameters [page 624]
    4. Technical Data [page 625]
    5. Account Assignment [page 624]

> **Note**
>
> If the *Counters* or the *Parameters* tab of the refill plan is not available (dimmed), this means no external counter or no external parameter was configured in the original refill plan [page 421] (in the service pricing catalog).

Related Information

Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]
Setting Up a Monitoring Plan in a Provider Contract [page 628]

1.37.9.2.1 Provider Contract>Refill Plan: Definition

The name of the refill plan activated by the contract item [page 592] is displayed on the *Definition* tab.

Definition tab

External Code: Type the code to identify the contract item in the customer data of an external system. Used with SAP CRM and SAP ERP/FI-CA, this code identifies an item in the master data managed by SAP CRM and that must be present in the billable items sent to SAP Convergent Invoicing (in the SAP ERP/FI-CA or SAP S/4HANA system or cloud). See the configuration of the billable item mapping [page 263] and the output field **`CONTRACTITEMID`**. If SAP CC is integrated with another CRM application or provisioning system, the use of this code is customized during the project implementation phase.

Activated Plan: Displays the name of the original refill plan activated in this contract item. Click  if you want to view the refill plan from the pricing catalog [page 109] of the service provider.

End Date: Select the date/time when this contract item is no longer valid.

Related Information

Setting Up a Provider Contract Item That Activates a Refill Plan [page 621]

1.37.9.2.2 Provider Contract>Refill Plan: Counters

The list of the counters declared in the original refill plan [page 425] is displayed on the *Counters* tab. Each row in the list represents a counter (*) stored in the provider contract in customer master data. If the contract is a linked contract, the counters are only stored in the parent contract [page 573]. You define the shared counter for the current version of the provider contract. Refer to the notes [page 596].

(*) Counters in Contracts

A counter is a numerical data stored in provider contracts in customer master data. During the charging and refilling operations, the SAP CC system uses the counter values to dynamically computes the price of a customer service usage and then updates these counters for further calculation.

A counter can represent a technical information or a business information like a service credit.

When provisioning the provider contracts, the CRM application or the provisioning system creates all the counters as declared in the charge plans and refill plans that are referenced by a contract item.

In the pricing catalog of the service provider, you configure the conditions of counter creations by setting up the charge plans and the refill plans that will be activated in the provider contracts of the end customers.

Counters tab

Name: Displays the name of the counter.

Description: Displays the description of the counter.

Status: Select the status of the shared counters:

- In a provider contract, set the status to *Shared Within Contract* if you want this counter to be shared with several contract items of the provider contract. Otherwise, select *Not Shared*.
- In a parent contract, select *Shared With Other Contracts* to share the counter with all the contract items and all the linked contracts.
- In a linked contract, select *Shared With Other Contracts* to reuse the counter shared by the parent contract with all its linked contracts. If you remove this status in another version of the linked contract, the counter is reset to zero.

Namespace: Select the namespace of the shared counter. Select *None* if there is no namespace.

Value: Displays the current value of the counter. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. This means that no usage event (customer service consumption) and no activation has occurred yet.

Related Information

Provider Contract: Counters [page 603]
Activating a Plan When Adding a Contract Item [page 632]
Refill Plan: Counters [page 425]

1.37.9.2.3 Provider Contract>Refill Plan: Parameters

The list of the shared parameters created for the original refill plan is displayed on the *Parameters* tab. Each row in the list represents a shared parameter.

Parameters tab

Name: Displays the name of the parameter.

Type: Displays the type of the parameter (Number, String or Date).

Value: Set the value of the parameter if you want to refine the default value.

Related Information

Refill Plan: Parameters [page 425]

1.37.9.2.4 Provider Contract>Refill Plan: Account Assignment

The list of the account assignments [page 593] as expected in the original refill plan is displayed on the *Account Assignment* tab. It assigns the contract item to a prepaid account of the same subscriber account.

Account Assignment tab

Account Assignment: Displays the name of the account assignment as configured in the original refill plan [page 421].

Account Type: Displays *Prepaid* which is the sole type of the account for a contract item that activates a refill plan..

Account: Select the prepaid account [page 514] to be used. This prepaid account belongs to the subscriber account [page 511] that relates to the provider contract you are configuring. Note that a prepaid account can only be assigned by one contract item activating a refill plan.

Related Information

Refill Plan: Account Assignments (Expected) [page 428]

1.37.9.2.5 Provider Contract>Refill Plan: Technical Data

The *Technical Data* tab allows you to view the technical identifier (User Service Identifier) under which a customer makes refills.

Technical Data tab

User Technical Identifier: Select the name of the technical data (UTI) which is declared in the activated plan. This is a type of user service identifiers.

User Service Identifier: Type the technical identifier of the user (end customer) inside the customer service.

> ✤ Example
>
> "refillByPhoneNumber" is a user technical identifier and the phone number "3312355435" is the related user service identifier.

Related Information

Changing the Version of a Provider Contract [page 632]
Refill Plan: Technical Data [page 427]

1.37.9.3 Working with the Contracted Refill Logic

Related Information

Opening the Contracted Refill Logic [page 626]
Viewing the Contracted Refill Logic [page 626]

1.37.9.3.1 Opening the Contracted Refill Logic

Open the Contracted Refill Logic

1. Open [page 611] a provider contract.
2. In the tree of the provider contract, click  to open the contracted refill logic.
3. View the following tabs of the contracted refill logic: Definition [page 626], Counters [page 627].

1.37.9.3.2 Viewing the Contracted Refill Logic

Related Information

Provider Contract>Contract Item>Contracted Refill Logic: Definition [page 626]
Provider Contract>Contract Item>Contracted Refill Logic: Counters [page 627]

1.37.9.3.2.1 Provider Contract>Contract Item>Contracted Refill Logic: Definition

The name of the activated refill logic in the contract item is displayed on the *Definition* tab.

Definition (Tab)

Last Activation Date: The latest event date/time on which the refill logic has been triggered in SAP Convergent Charging. The event could be a recurring event or a one-shot event.

> **Note**
>
> - The usage event does not modify the last activation date
> - The event date/time is not necessarily the date/time this event was processed by the system

Related Information

Recurring Refill [page 710]
One-Shot Refill [page 713]

1.37.9.3.2.2 Provider Contract>Contract Item>Contracted Refill Logic: Counters

The list of the counters created in the original refill logic is displayed on the *Counters* tab. Each row in the list represents a counter.

Counters (Tab)

Name: Displays the name of the counter.

Description: Displays the description of the counter.

Value: Displays the current value of the counter. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. This means that no usage event and no activation has occurred yet.

Related Information

Provider Contract>Refill Plan: Counters [page 623]

1.37.10 Working with Monitoring Plans Activated in a Contract

Related Information

Setting Up a Monitoring Plan in a Provider Contract [page 628]
Setting Up the Tabs of a Monitoring Plan in a provider Contract [page 628]

1.37.10.1 Setting Up a Monitoring Plan in a Provider Contract

> ⚠ Caution
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or a provisioning system responsible for managing provider contracts and customer master data. Refer to the caution [page 573] for provider contracts.

Create [page 599], open [page 611], or modify [page 612] a provider contract before setting up a contract item activating a monitoring plan. A *Provider Contract* window must be open.

The provider contract must include some items that activate [page 632] a charge plan or a refill plan that shares the necessary counters to be monitored and reported as form of spending statuses.

To set up a contract item activating a monitoring plan in a provider contact

1. In the left pane of the *Provider Contract* window, click  .
2. Set up the tabs of the contract item which references a monitoring plan: Definition [page 629], Counters [page 629], Technical Data [page 630].

Related Information

Setting Up a Provider Contract Item That Activates a Charge Plan [page 614]
Setting Up a Provider Contract Item That Activates a Refill Plan [page 621]

1.37.10.2 Setting Up the Tabs of a Monitoring Plan in a provider Contract

Related Information

Provider Contract>Monitoring Plan: Definition [page 629]
Provider Contract>Monitoring Plan: Counters [page 629]
Provider Contract>Monitoring Plan: Technical Data [page 630]

1.37.10.2.1 Provider Contract>Monitoring Plan: Definition

The name of the monitoring plan activated by the contract item [page 592] is displayed on the *Definition* tab.

Definition tab

External Code: Type the code to identify the contract item in the customer data of an external system. Used with SAP CRM and SAP ERP/FI-CA, this code identifies an item in the master data managed by SAP CRM and that must be present in the billable items sent to SAP Convergent Invoicing in the SAP ERP/FI-CA or SAP S/4HANA system. See the configuration of the billable item mapping [page 263] and the output field CONTRACTITEMID. If SAP CC is integrated with another CRM application or provisioning system, the use of this code is customized during the project implementation phase.

Activated Plan: Displays the name of the original monitoring plan activated in this contract item. Click  if you want to view the monitoring plan from the pricing catalog [page 109] of the service provider.

End Date: Select the date when this contract item is no longer valid.

Related Information

Setting Up a Monitoring Plan in a Provider Contract [page 628]

1.37.10.2.2 Provider Contract>Monitoring Plan: Counters

The list of the counters as declared in the original monitoring plan is displayed on the *Counters* tab. Each row in the list represents a counter (*) stored in the provider contract. The SAP CC system uses this list as source of spending statuses [page 598] that muse be monitored and reported.

> → Remember
>
> The counter must be shared inside the provider contract with the appropriate namespace. The counters cannot be shared by a parent contract [page 573]. The spending status monitoring feature does not support parent contracts.

(*) Counters in Contracts

A counter is a numerical data stored in provider contracts in customer master data. During the charging and refilling operations, the SAP CC system uses the counter values to dynamically computes the price of a customer service usage and then updates these counters for further calculation.

A counter can represent a technical information or a business information like a service credit.

When provisioning the provider contracts, the CRM application or the provisioning system creates all the counters as declared in the charge plans and refill plans that are referenced by a contract item.

In the pricing catalog of the service provider, you configure the conditions of counter creations by setting up the charge plans and the refill plans that will be activated in the provider contracts of the end customers.

Counters tab

Name: Displays the name of the counter.

Description: Displays the description of the counter.

Status: Select the status of the shared counters:

- In a provider contract, set the status to *Shared Within Contract* if you want this counter to be shared with several contract items of the provider contract. Otherwise, select *Not Shared*.

Namespace: Select the namespace of the shared counter. Select *None* if there is no namespace.

Value: Displays the current value of the counter. If the counter has no value because it has not been initialized yet, the value is set to *Not Yet Initialized*. This means that no usage event (customer service consumption) and no activation has occurred yet.

Related Information

Provider Contract: Counters [page 603]
Activating a Plan When Adding a Contract Item [page 632]
Sharing Counters with Provider Contracts [page 607]

1.37.10.2.3 Provider Contract>Monitoring Plan: Technical Data

The *Technical Data* tab allows you to view the technical identifier under which an end customer consumes a service or under which the end customer is identified in the system landscape. The CRM or provisioning system must specify this identifier of the end customer. It can be the same technical ID used during the charging/refilling operations or it can be another type of identifier. For example, this ID can be stored in

telephone directory or email directory accessible via directory services based on LDAP (Lightweight Directory Access Protocol).

The policy/monitoring system and the CRM/provisioning system must share the knowledge related to this identifier.

Technical Data tab

User Technical Identifier: Select the name of the user technical identifier (UTI) which is declared in the activated monitoring plan [page 500]. This is a type of user service identifiers (USIDs).

User Service Identifier: Type the technical identifier of the user (end customer) inside the customer service. The SAP CC system uses this USID information to manage the operation requests for spending status monitoring from the registered policy or monitoring system. By default, the SAP CC system generates spending status reports that include spending statuses relating to the selected counters [page 629].

> → Remember
>
> When a provider contract includes several contract items activating monitoring plans, the spending status monitoring feature extends the reporting of counters by considering all the counters selected in all the activated monitoring plans. The SAP CC system takes into account all these counters. The registered downstream system (policy or monitoring) can access to the sum of the spending statuses by using each USID.

> ⁂ Example
>
> The technical data includes:
>
> - The User Technical Identifier (UTI) is set to "phoneNumber" and the User Service Identifier (USID) is set to "33432536445".
> - The UTI is set to "SUBSCRIBER_IN_GENERAL_DIRECTORY" and the USID is set to "SBSCRBRUVWXYZ".
>
> The registered policy/monitoring system specifies one of these identifiers to manage the spending status monitoring of the end customer who owns the 33432536445 phone number and who is identified by SBSCRBRUVWXYZ in the telephone directory.

Related Information

Monitoring Plan: Technical Data [page 500]

1.37.11 Using the Action Menu

Related Information

Changing the Version of a Provider Contract [page 632]
Activating a Plan When Adding a Contract Item [page 632]
Searching For and Viewing the Allowances of a Provider Contract [page 633]

1.37.11.1 Changing the Version of a Provider Contract

The procedure is the same as for changing the version of an object:

Change the Version of an Object

When you open an object, you can access its versions by clicking the *Change Version*  icon on the toolbar.

Modifying and saving an object modifies the *Change Version* list by adding a new version, deleting one or more existing versions, or overwriting some existing versions.

> ⓘ Note
>
> If you save an object with a specified date/time in the past, the system deletes all its versions whose date/time is later than the specified date/time.

1.37.11.2 Activating a Plan When Adding a Contract Item

> ⚠ Caution
>
> You activate a charge plan [page 343], a refill plan [page 421], or a monitoring plan [page 491] only if you create a provider contract manually and want to add a contract item [page 599].
>
> Do not manually create a provider contract if SAP Convergent Charging is already interfaced with a CRM application or an external provisioning system responsible for managing provider contracts. Refer to the caution [page 573] for provider contracts.

In a provider contract [page 573], you can activate a charge plan [page 343], a refill plan [page 421], or a monitoring plan [page 491] by adding it to the provider contract as a new contract item [page 592]. You must then:

- Set up [page 614] the contract item for the activated charge plan
- Or set up [page 621] the contract item for the activated refill plan
- Or set up [page 628] the contract item for the activated monitoring plan

Prerequisites

Before performing this task, you must:

- Open [page 611] an existing provider contract
- Or create [page 599] a new provider contract

Tasks

To activate a plan in a provider contract

1. In the provider contract tree, right-click the provider contract icon  .
2. Click *Activate Charge Plan*, or *Activate Refill Plan*, or *Activate Monitoring Plan*.
3. Select the plan to be activated to the provider contract.
4. A new contract item is added. You must set up this contract item:
   - Set up [page 614] the contract item for the activated charge plan
   - Set up [page 621] the contract item for the activated refill plan
   - Set up [page 628] the contract item for the activated monitoring plan

To deactivate an activated charge plan, refill plan, or monitoring plan

1. Right-click the activated plan.
2. Click *Remove Plan*.

> **Note**
>
> To activate or deactivate a charge plan (or a refill plan or a monitoring plan), you can also:
>
> - Use the *Action* menu in Core Tool: *Activate Charge Plan...*, *Activate Refill Plan...*, *Activate Monitoring Plan...*, or *Remove Plan...*.
> - Click the following icons on the toolbar: *Activate Charge Plan...*, *Activate Refill Plan...*, or  *Activate Monitoring Plan...*.

1.37.11.3 Searching For and Viewing the Allowances of a Provider Contract

You can search for and view all the allowances that belong to a provider contract.

Open [page 611] the main window of a provider contract, and then on the *Action* menu, select *View Allowances* to search for and view all the allowances of the provider contract.

1. Searching For Allowances [page 581]
2. Viewing an Allowance [page 583]

1.38 Accesses

An **access** in SAP Convergent Charging (SAP CC) is a customer master data that grants a license to an end customer of the service provider to access and use its customer services. In return for this service usage, the access identifies the business agreement that is charged for this service consumption or product use.

An access specifies:

- Technical data that includes:
  - A technical identifier of the customer service (Service ID [page 636])
  - A technical identifier under which an end customer consumes the service (Setting Up an Access [page 636])
- Business agreement data: Each charge of a provider contract [page 614] that is configured to price usage rates corresponding to the consumption of the customer service.

At runtime, the SAP CC system uses the access information to determine the fees to charge and to identify the provider contract [page 573].

> ⚠ Caution
>
> Accesses related to provider contracts are automatically created and maintained by the SAP CC system. Creating, modifying, or deleting accesses may result in inconsistencies between these two systems. We strongly recommend that you do not modify or delete existing accesses except under support supervision.

An access consists of a user (service) identifier, a service identifier, and a master charge from the provider contract. The combination of the user and service identifiers must be unique for a given period of time within the SAP CC system (this combination can however be simultaneously defined for several distinct periods of time).

> ⓘ Note
>
> If you are associated to a pricing catalog [page 109], you can only open accesses relating to this catalog.

> → Remember
>
> A "No access" message returned by SAP CC indicates:
>
> - Either that the ID pair is not assigned to any access at all
> - Or that the ID pair exists but none of its assignment periods include the consumption date/time.

Access Version and Chronology

The pair User Service Identifier (USID)/Service Identifier identifies each access in the back-end database. The same access can be divided chronologically into time periods that do not overlap. You can assign each time period of the same access to a different or same provider contract. For example, you can divide the access USI1/SI1 into two time periods:

- First time period: From 01 January to 01 February
- Second time period: From 02 February to an unspecified date

You can then assign each time period of the access USI1/SI1 to a different provider contract.

Because accesses are arranged into a chronological order, the time periods cannot overlap but can be discontinuous. For example, you can divide the access USI2/SI2 into third periods:

- First time period: From 01 May to 01 June
- Second time period: From 01 August to 01 September
- Third time period: From 01 November to 01 December

> **Note**
>
> The access chronology is optional. This means that the access can be assigned to the provider contract for an unlimited time period.

> **Example**
>
> A company wants to change a 5 free MMS plan into a 20 free MMS plan on January 1st. In this example, the price plan includes the same service (MMS). Processing possible delayed MMS messages requires to divide the same access into time periods. This case requires January 1st as the end date/time of the first access while the start date/time of the second access is after January 1st.

Where to create or modify accesses

You can create [page 636] or modify [page 640] accesses from the *File* menu.

1.38.1 Working with Accesses

Related Information

Creating an Access [page 636]
Opening an Existing Access [page 637]
Selecting an Access [page 638]
Modifying an Access [page 640]
Setting Up an Access [page 636]
Deleting an Access [page 640]
Saving an Access [page 638]

1.38.1.1 Creating an Access

You can create or modify an access from the *File* menu.

> → Remember
>
> You can assign accesses to master charges only.

From the File menu

Create an Access from the File Menu

1. From the menu bar in the main window, select *File* > *New* > *Access*. The *Choose owner* dialog box opens.
2. In the *Owner of Object* drop-down menu, select the owner of the access.
3. Click *OK*. The *New Access* window opens.
4. Set up [page 636] the user access.
5. Verify the user access structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
6. Save [page 638] the user access.

1.38.1.2 Setting Up an Access

Create [page 636], open [page 637], or modify [page 640] an access before setting it up. An *Access Window* must be open.

To set up an access, you first identify it, and then if necessary, you define time periods during which the access is assigned to one or more charges relating to provider contracts. You perform the operation in two halves of this window.

> ⓘ Note
>
> The fields *User Service Identifier* and *Service Identifier* are unavailable (dimmed) when you modify an access.

Top Half of the Access Window

Fill in or modify the fields below.

User Service Identifier: Type the identifier of the end customer (aka user) who will use the usage-based service. For example, a technical ID such as an MSISDN number or an IP address.

Service Identifier: Type the identification code of the service consumed by the end customer. For example, Telecommunications or Internet services.

Bottom Half of the Access Window

Fill in or modify the table to define time periods.

 : Click to add a new row to the table.

 : Click to delete a row from the table.

An access can be assigned to one or more charges relating to provider contracts during a period delimited by a start date/time and an end date/time. The start date/time is mandatory except for the first period. The end date/time is mandatory except for the last period.

> ⓘ Note
>
> You can:
>
> - Leave the table empty. This means that the access is assigned to no charge. This option allows you to create accesses for later use.
> - Add only a row to the table, and then select a provider contract and a charge without selecting a time period. This option means that the access is assigned to a charge for an unlimited time period.

Start Date: Click  to select the beginning of the period during which the access is assigned to the charge.

End Date: Click to select the end of the time period during which the access is assigned to the charge.

Provider Contract: Click  to choose a provider contract [page 573] to which you want to assign the access.

After you select the provider contract, click  to view it if necessary. See Opening an Existing Provider Contract [page 611].

Charge: Once you have selected the provider contract, click  to display the charges relating to it, and then click the master charge to which you want to assign the access.

1.38.1.3 Opening an Existing Access

You can open an existing user access from the *File* menu.

Open an Existing Access from the File Menu

1. From the menu bar in the main window, select *File* > *Open* > *Access*. The *Open Access* dialog box opens.
2. Select [page 638] the access that you want to open. The access window is displayed.

1.38.1.4 Saving an Access

Once you have created or modified a user access, you are ready to save it. You can save the user access to either the database or as an XML file on your hard disk.

To save a user access to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Access Creation* dialog box opens to inform you that the user access has been successfully saved.
2. Click *OK*.

To save a user access as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. The *Save* dialog box opens.
2. Select a directory to save the user access.
3. In the *File name* field, type the name of the file.
4. Click *Save*. The user access is saved as an XML file in your selected directory.

1.38.1.5 Selecting an Access

You must open [page 637] or modify [page 640] an existing user access to open the *Select an Access* window.

Select an Access window

The *Select an Access* window consists of two areas:

- The top half is the filter area, which allows you to select a user access by using filter criteria.
- The bottom half of the window is the results area, which displays the search results that you perform.

Once a search is complete and the results are displayed, select a user access, and then click *OK*.

> **Note**
>
> You can select a user access with or without using a filter. Use the arrows  if you want to hide or show the filter area or the results area.

Filter Area

The *Basic* tab allows you to find user accesses depending on search criteria. Each pair (User Service Identifier, Service Identifier) identifies a unique access in the database.

User Service Identifier: Provides a list of search comparators such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. Select a search comparator, and then type in the last field the string that you want to search in the database. If you choose the *Is Like* comparator, you can type "*****" to represent a number of characters and "**?**" to represent a single character.

> **Example**
>
> For example, if you want to find all user service identifiers containing the string `John`, you select the *Contains* option, and then type **John**.

Service Identifier: Provides a list of search comparators such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. Select a search comparator, and then type in the last field the string that you want to search in the database.

> **Example**
>
> For example, if you want to find all service identifiers containing the string `John`, you select the *Contains* option, and then type **John**.

Results Area

This area displays the results search. Click the icon  to display the type of information that you want to view.

 : Click to start a search.

 : Click to delete the results list.

 : Click to move to the previous page if the results list is large.

 : Click to move to the next page if the results list is large.

1.38.1.6 Modifying an Access

You can modify an access from the *File* menu.

Modify an Existing Access from the File Menu

1. Open [page 637] the access.
2. Set up [page 636] the access.
3. Verify the user access structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
4. *Save* the access.

1.38.1.7 Deleting an Access

Use this procedure to delete a user access.

To delete a user access

1. From the menu bar in the main window, select *File* > *Delete* > *Access*. The *Delete Access* dialog box opens.
2. Select [page 638] the user access that you want to delete.
3. Click *Delete*. A deletion dialog box asks if you want to delete the user access.
4. Confirm the user access deletion by clicking *Yes*. A dialog box confirms the successful deletion.

1.39 Users

SAP CC Core Tool uses your logon and password to give you a unique user ID. Only user administrators can configure new users [page 645] for SAP Convergent Charging 2023.

 An **SAP CC user** in SAP Convergent Charging (SAP CC) is:

- A **person**, an *individual user* who works with the user interfaces delivered with the SAP Convergent Charging software product.
- A **machine**, a *service user* that connects to and communicates with the SAP Convergent Charging systems through different technical interfaces.

SAP Convergent Charging manages the security for the SAP CC systems and gradually controls the accesses for the individual users and service users.

This machine or this person is:

- Authorized to work with the SAP Convergent Charging systems.
- Granted with one or more roles [page 641] in the organization.
- Assigned or not to a unique pricing catalog [page 109] to define some restrictions on the following master data:
  - Data that belongs to this service pricing catalog, which is owned by the service provider for the services to monetize
  - Data that is related to the end customers (subscriber accounts, for example) and that is associated to the pricing catalog for these services. It belongs to the end customers of the service provider company.

> **Note**
>
> To restrict access and control rights to users, you can also associate a list of authorized accesses [page 646] with each user.

When the *Core Tool* user interface is running, you cannot log on to it by using another user name/ID. To change the user ID, you must close Core Tool and relaunch it with the user ID and password of the new user.

More Information

Consult the SAP CC Security Guide for more information and user assistance.

Related Information

User Roles [page 641]
Working with SAP CC Users [page 644]

1.39.1 User Roles

In SAP Convergent Charging, **roles are predefined levels of authorizations** that grant SAP Convergent Charging users specific access and control rights in the SAP CC systems. Permitted operations depend on these levels.

The various roles that are available and the level of authority associated with each role is listed in the table below.

As a user administrator (or system administrator), you manage [page 644] the users and their granted roles to conform to your global security policy.

→ Remember

- If you are assigned a user administrator role, inform your system administrator before changing your password. The system administrator must update a specified configuration file (refer to the SAP CC 2023 Installation and Maintenance Guide) each time you change your password.
- You are not allowed to modify provider contracts in Core Tool in a production system landscape. You can only view the provider contract. To modify a provider contract, you must use a CRM application or an external provisioning system connected to SAP Convergent Charging.

Role	Authorized Domain	Authorized Action
Administrator	All	- User management Operations: - Audit management - Bulk operation management - Rerating management - Refresh rating instances - Specific tasks when using administrative tools of SAP CC (refer to the SAP CC 2023 Administration and Operation Guide)
Authorization Delegator	None	This role gives the possibility to use the authorization delegation mechanism for the SAML 2.0 authentication protocol.
User Administrator	Users	- Create and delete SAP CC users - Modify password or user roles - Lock and unlock a user
Customer Sales Representative (CSR)	Provider contracts, accesses, subscriber mapping tables, subscriber range tables	**Customer master data** - Create, modify, and delete customer master data [page 30]. **Master data** Display these objects in the pricing catalogs (master data [page 28]) and in business data [page 32].
Marketing	Charge plans, Refill plans	Create, modify, and delete objects in the pricing catalogs (master data [page 28]) and in business data [page 32].

Role	Authorized Domain	Authorized Action
Pricing Specialist - Price Data	Mapping tables, range tables	Read, edit, import, and export mapping tables and range tables of a catalog, or agreement type, with the Cockpit user interface. **⚠ Caution** Once this role is assigned to a user, the concerned user can log in to Core Tool only for password management purpose, as this role is strictly limited to Cockpit users.
Process Manager	Business functions	Use the operations via the Web Services technical interface to trigger business and technical operations such as: - Provider contract activation - Charged item bulk loading
Batch Rating Administrator	Batch rating and charging	- Log on to the *BART* user interfaces: BART Tool and BART+ - Perform all actions related to batch operations (refer to the SAP CC BART Tool documentation) - Connect the SAP CC BART Server system to the SAP CC Core Server system (refer to the SAP CC 2023 Administration Guide)
Remote Support	All	Read-only rights relating to the data and configuration of SAP CC (refer to the SAP CC 2023 Security Guide).
Message Charging Client	The Message TCP technical interface	Allow a service user to establish an authenticated connection to the Message TCP technical interface of SAP CC (refer to the SAP CC SAP CC 2023 Security Guide). The login and the password are required only if the MESSAGE_API_AUTHENTICATION_REQUIRED parameter is set to true (refer to the SAP CC 2023 System Parameter Reference).
System Manager	Business functions	This role contains the rights required for **System Configuration** and **System Information** Cockpit apps.

1.39.2 Changing Your Password

You use the *Changing Password* window:

- When you want to change your password
- When a new password is required[69]

To change your password, you access the *Changing Password* window from the menu bar in the main window. On the menu bar, select *File* > *Change Password* to open the *Changing Password* window.

> **Note**
>
> All the rules regarding passwords are determined and set up by your SAP CC system administrator. The administrator configures the system to apply the global security policy of your company. For more information, refer to the administrator for SAP Convergent Charging 2021 in your SAP system landscape.

Changing Password (Window)

Fill in the *Changing Password* window.

Old Password: Type the old password that you want to change.

New Password: Type your new password.

Retype New Password: Retype your new password.

> **→ Tip**
>
> You can contact your user administrator to reset your password.

Related Information

Concepts
Password Management Policy

1.39.3 Working with SAP CC Users

As a user, you cannot neither view nor change any user settings. Only the user administrator and the system administrator can manage these settings and grant some roles to you for example. Note that they can reset your user password.

[69] The *Changing Password* window can be displayed depending on the rules defined by your administrator.

- Changing Your Password [page 644]

As a user administrator or a system administrator, you manage your SAP CC users with the Core Tool user interface.

New User Configuration	User Operations	User Maintenance
• Creating a User for SAP CC [page 645] • Setting Up a User [page 646] • Saving a User [page 648]	Basic: • Granting Roles to a User [page 650] • Assigning a User to a Unique Pricing Catalog [page 650] • Locking a User [page 651] • Unlocking a User [page 652] User's password: • Forcing a Password Change for an Individual User [page 652] • Changing or Resetting Passwords for Your Users [page 653]	• Opening an Existing User [page 648] • Modifying a User [page 649] • Deleting a User [page 653]

1.39.3.1 Creating a User for SAP CC

> ⓘ Note
>
> Only system administrators and user administrators are authorized to create users for SAP Convergent Charging.

When you create an SAP CC user, you must do the following:

- Choose whether or not to add password protection.
- Grant [page 650] at least one role to the new SAP CC user.
- Assign or not [page 650] this new user to a pricing catalog.

> → Tip
>
> If you create a user for the *Web Services* technical interface of SAP Convergent Charging, refer to the SAP CC 2023 Security Guide for more information and user assistance.

Create an SAP CC User

1. From the menu bar in the main window, select *File* > *New* > *User*. The *New User* window is displayed.
2. Set up [page 646] the new user for SAP CC.
3. Save [page 648] the user.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

Related Information

User Roles [page 641]
Users [page 640]

1.39.3.2 Setting Up a User

Create [page 645] or open [page 648] an SAP CC user before setting it up. A *User* window must be open.

User (Window)

Fill in or modify the *User* window.

Log On: Type the unique identifier of the new user for your SAP Convergent Charging system landscape. This identifier will be used by the user to log on to the SAP CC user interfaces (Cockpit, Core Tool) or to connect to SAP CC systems. SAP CC systems authenticate users by checking their user credentials (ID/name, password).

With Password: This field is available only when you create a user for the first time. Click this option to create the password authentication for the new user, and then type the initial password in the *Password* field.

Password: Type a legible password for the user if you have clicked the *With Password* option.

> ⓘ Note
>
> SAP CC systems authenticate users by checking their user credentials (ID/name, password).

Catalog: Only available when the user administrator that creates a new user or modifies an existing one is not associated to a catalog. Select the unique pricing catalog [page 109] to assign to the SAP CC user for whom you want to define some restrictions on master data related to a service provider (pricing catalog) or related to its end customers (subscriber accounts, provider contracts, and so on).

Modify Password: This field is available after an SAP CC user is created and saved in the back-end database. Click this option to modify the user's password, and then type the password in the *Password* field.

Force Password Modification: This field[70] is only relevant for individual users and is used to force these typical users to change their passwords next time they log on to an SAP CC user interface. This field is automatically checked:

- After an individual user is created
- Every time the password of an individual user has been changed

Security Profile:

- *Individual user*: This option applies a strong password policy for the user. We recommend using this option for individual users. The strong password policy includes:
  - Password expiration and locking policy if the system is configured according to the security policy for user passwords
  - High complexity for password encryption. You can configure it at system level
- *Service User*: This option deactivates the strong password policy for the user. Password expiration and locking policy are not applied. The complexity of password encryption is adapted to service (technical) users. You can configure it at system level. We recommend not using this option for individual users.

> **Note**
>
> - If you change the security profile of the user, the complexity of password encryption is not modified immediately. It changes when the user changes the password.
> - If the user isolation feature is enabled, you cannot change your own security profile. For further information about this feature, refer to the SAP CC Security Guide.

Locking:

- *Locked*: This checkbox[71] is available only after a user is created and saved in the database. Click this checkbox if you want to lock the user. When locked, the user cannot log on anymore. Only an administrator or a user administrator can unlock a user.

  > **Note**
  >
  > This checkbox is clicked if the system date is out of the validity period.

- *Valid From*: Click to select the date from which the user can log on to SAP Convergent Charging
- *Valid To*: Click to select the date after which the user can no longer log on to SAP Convergent Charging

> **Note**
>
> The fields *Valid From* and *Valid To* are optional. Leaving one of them or both of them empty means an endless validity period.

Roles: Click one or more checkboxes corresponding to the roles [page 641] that you want to assign to the user for SAP Convergent Charging. The roles granted to the user define the subset of authorizations to perform certain operations.

Creation Date: Displays the date on which the user was created

Last Modification Date: Displays the date on which the last change has been made for this user

> **Note**
>
> *Creation Date* and *Last Modification Date* fields are not editable.

---

70 This field must be configured by your administrator to be displayed in the window. For more information, refer to your administrator.

71 This checkbox must be configured by your administrator to be displayed in the window. For more information, refer to your administrator.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

Related Information

Users [page 640]

1.39.3.3 Saving a User

> ⓘ Note
>
> Only administrators and user administrators are authorized to save users for SAP Convergent Charging.

Use this procedure to save an SAP CC user into the back-end database.

Save a User into the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database* or click the *Save into Database* button on the toolbar. A confirmation message is displayed.
2. Click *OK*.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.4 Opening an Existing User

> ⓘ Note
>
> Only administrators and user administrators are authorized to view users.

You can open an existing SAP CC user saved in the back-end database system.

Open an Existing User Saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *User*. The *Open User* dialog box is displayed.
2. Select the SAP CC user that you want to open, and then click *OK*. The window of the selected user is displayed.
3. View or edit [page 646] the user settings.

Related Information

Modifying a User [page 649]
Deleting a User [page 653]

1.39.3.5 Modifying a User

> **Note**
>
> Only administrators and user administrators are authorized to view/modify users for SAP Convergent Charging.

Modify an Existing User

1. Open [page 648] an SAP CC user in the *Core Tool* user interface.
2. Set up [page 646] this user.
3. Save [page 648] the SAP CC user.

> **Restriction**
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

Related Information

Granting Roles to a User [page 650]
Assigning a User to a Unique Pricing Catalog [page 650]

1.39.3.6 Granting Roles to a User

As a user administrator (or system administrator), you must assign or reassign roles [page 641] to an SAP CC user.

Grant Roles to an SAP CC User

1. Create [page 645] or open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this user by clicking the necessary *Roles* checkboxes. This selection defines the associated authorizations and permitted operations.
3. Save [page 648] the SAP CC user.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.7 Assigning a User to a Unique Pricing Catalog

As a user administrator, you can assign an SAP CC user to a unique pricing catalog [page 109].

> Example
>
> Set up this assignment to a service user that is dedicated to the communications through the *Web Services (WS)* technical interface.
>
> Set up the assignment to an individual user who must be associated and restricted to a single catalog and must not see any other data.

Assign an SAP CC User to a Pricing Catalog

1. Create [page 645] or open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this user by selecting the relevant pricing catalog in the *Catalog* field to define the association and restriction.
3. Save [page 648] the SAP CC user.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.8 Locking a User

As a user administrator (or system administrator), you can lock or unlock your users for SAP Convergent Charging.

Lock an SAP CC User

1. Open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this user by selecting the *Locked* checkbox.
3. Save [page 648] the SAP CC user.

> ⓘ Note
>
> The access to SAP Convergent Charging is granted to a user only if both of the following conditions are verified:
>
> - The current date/time matches with the validity period defined for the SAP CC user.
> - The *Locked* checkbox is not selected in the user settings.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.9 Unlocking a User

As a user administrator (or system administrator), you can lock or unlock your users for SAP Convergent Charging.

For troubleshooting purposes, you want to unlock an existing SAP CC user. This user is currently locked for different possible reasons. Check this reason.

Unlock an SAP CC User

1. Open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this user by unselecting the *Locked* checkbox.
3. Save [page 648] the SAP CC user.

> ⓘ Note
>
> The access to SAP Convergent Charging is granted to a user only if both of the following conditions are verified:
>
> - The current date/time matches with the validity period defined for the SAP CC user.
> - The *Locked* checkbox is not selected in the user settings.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.10 Forcing a Password Change for an Individual User

As a user administrator (or system administrator), you can force an individual user to change the user's password next time they log on to an SAP CC user interface.

Force the Password Change

1. Open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this user by clicking the *Force Password Modification* checkbox.
3. Save [page 648] the SAP CC user.

⚠ Restriction

- Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
- If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.11 Changing or Resetting Passwords for Your Users

As a user administrator (or system administrator), you can change or reset a user password for SAP Convergent Charging.

For security reasons or for troubleshooting purposes (forgotten password), you want to reset a user password.

Reset a User Password

1. Open [page 648] an SAP CC user in the *Core Tool* user interface.
2. In the *User* window, set up [page 646] this SAP CC user by clicking the *Modify Password* checkbox and typing a new password or a reinitialization password.
3. Save [page 648] the SAP CC user.
4. Inform your individual SAP CC user about the changes or contact the relevant skateholder about changes for a service user that is used by interfaced systems or applications.

⚠ Restriction

- Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
- If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.39.3.12 Deleting a User

ⓘ Note

Only administrators and user administrators are authorized to delete users.

To delete an existing user, complete the following steps.

Delete an Existing User

1. From the menu bar in the main window, select *File* > *Delete* > *User*. The *Delete User* dialog box is displayed.
2. Select the user you want to delete, and then click *Delete*. A message is displayed and asks you to confirm that you want to delete the selected user.
3. Verify the selected user and choose one of the following actions:
   - To abandon the deletion, click *No*.
   - To delete the selected user, click *Yes*. If you click *Yes*, the user is deleted and the *User Deletion* dialog box opens to inform you that the user has been successfully deleted.
4. Click *OK*.

> ⚠ Caution
>
> You cannot delete your own user.

> ⚠ Restriction
>
> - Only a system administrator can create, modify, lock/unlock, or delete a system administrator account.
> - If a system administrator is assigned to a catalog, this administrator can only manage user accounts assigned to the same catalog.

1.40 Audits

 An **audit** in SAP Convergent Charging (SAP CC) is an administration object that facilitates your manual inspections by configuring a custom audit filter. You can create [page 655] an audit object to search efficiently for audited user operations in the back-end database. According to the criteria of the audit object, you can search for [page 99] and view [page 99] all user operations that can be audited by the SAP CC system.
User operations are treatments that are processed by the SAP CC system as requested by the SAP CC users (service users, individual users).

For example, creating an object is a user operation.

Audit data recordings are stored in the database if the audit functions are enabled in the SAP CC Core Server system. For more information, refer to your administrator and consult the SAP CC Administration Guide.

The elements of the *Audit* window enable you to create an audit object. Using specific filters, an audit allows administrators to query the database of the SAP CC system for a user operation search. You can retrieve specific user operations by using a filter. The filter consists of criteria rows that specify the characteristics of the audited user operations you want to retrieve.

You can audit:

- Any operation that modifies the back-end database

- Any operation that modifies the state of the SAP CC system
- Any operation that relates to security (SAP CC user authentication)
- Any unsuccessful operation

> → Remember
>
> - The operations that do not modify the database can be audited only if they have failed.
> - You can audit bulk operations.
> - You can audit mass and bundle operations as sets of single operations. Multiple audited operations relate to the same mass (or bundle) operation.
> - You cannot audit the single charging operations, but you can audit the rerating operations.

Refer to the SAP CC Application Help for more information and user assistance about:

- The audit concept, features, and functions
- The specificities and conditions
- The notes and recommendations

Related Information

Viewing Operations [page 99]

1.40.1 Working with Audits

Related Information

Creating an Audit [page 655]
Opening an Existing Audit [page 658]
Modifying an Audit [page 659]
Setting Up an Audit [page 656]
Deleting an Audit [page 659]
Saving an Audit [page 658]

1.40.1.1 Creating an Audit

Use one of these procedures to create an audit. After creating an audit, you can search [page 99] for audited operations based on this audit object.

To create an audit

1. From the menu bar in the main window, select *File* > *New* > *Audit*. The *New Audit* window opens.
2. Set up [page 656] the audit.
3. Verify the audit structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
4. *Save* the audit.

You can create an audit by opening an existing audit and copying it.

To create an audit from an existing audit

1. From the menu bar in the main window, select *File* > *Open* > *Audit*. The *Choose an Audit* dialog box opens.
2. Select the audit to copy.
3. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding audit window opens.
4. Rename the audit in the *Code* field, and then make other changes as necessary - see modifying an audit [page 659] for details.
5. Verify the audit structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
6. Save [page 658] the audit.

1.40.1.2 Setting Up an Audit

Create [page 655], open [page 658], or modify [page 659] an audit before setting it up. An *Audit* window must be open.

Audits are triggered cyclically to extract user operations from the database; therefore, an audit is defined by:

- The start date of the first audit
- The frequence of the next audits
- The set of filters that are used for extracting specific user operations

Audit window

Fill in or modify the *General* tab and the *Filters* tab in the audit window.

General tab

Code: Type the unique identifier of the audit.

Description: Type a description of the audit if necessary.

Original Date: Click to select the start date of the first audit. By default, the current date is displayed.

Frequency: Provides the list of time periods to trigger the next audits. Select one of the following frequencies: *Daily*, *Weekly*, *Biweekly*, *Monthly*, *Bimonthly*, *Quarterly*, *Semiannual*, or *Yearly*.

Depending on the frequency that you choose, the three fields below display the dates of the last audit and the next audit.

Beginning of Last Audit: Displays the date on which the last audit started.

End of Last Audit: Displays the date on which the last audit stopped or will stop.

Next Audit Date: Displays the date on which the next audit starts.

Filters tab

The *Filters* tab allows you to create audit criteria for extracting user operations from the database.

Fill in or modify the table.

 : Click to add a row to the table.

 : Click to delete a row from the table.

Operation: Select the type of the user operation to be audited.

User: Select the name of the user to be audited.

Type: Select "successful operations" or "failed operations".

Operand: Select *any value* to audit all the operands, or select *literal value* to audit the operands that match a particular value. Operands are the external or internal references of objects to be audited.

> ⓘ Note
>
> About operand definitions.
>
> - The external reference of an object is its user-defined name such as the name of the chargeable item class or the name of the charge plan.
> - The internal reference of an object is its key in the database.

1.40.1.3 Saving an Audit

Once you have created or modified an audit, and then verified its structure, you are ready to save it. You can save the audit to either the database or as an XML file on your hard disk.

To save a new audit to the database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*. The *Audit Creation* dialog box opens.
2. Click *OK*.

To save an audit as an XML file

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the audit.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The audit is saved as an XML file in your selected directory.

1.40.1.4 Opening an Existing Audit

You can open:

- An existing audit saved in the database
- An existing audit saved in an XML file

To open an existing audit saved in the database

1. From the menu bar in the main window, select *File* > *Open* > *Audit*. The *Open Audit* dialog box opens.
2. Select an audit.
3. Click *Open*. The corresponding audit window opens.

To open an existing audit saved in an XML file

Opening an existing object saved in an XML file

When you open an XML file, the window displays the object as a new object, to which you must assign a new identifier if you want to save the object to the back-end database.

> **Note**
>
> - If several versions of the object are already stored in XML files, you can open only the oldest version.
> - In the *Open Parameters* dialog box, the *Open as Sub-Element of Current Object* option is used only when you want to insert the price plan [page 325] of a charge into an existing charge.

Open an Existing Object Saved in an XML File

1. From the menu bar in the main window, select *File* > *Open* > *From File*. The *Open* window opens.
2. Open the directory in which the object is saved, and then select the object file.
3. Click *Open*. The *Open Parameters* dialog box opens.
4. Select the *Open as New Object* option.
5. Select the object that you want to insert.
6. Click *Open*. A window displays the object as a new object.

1.40.1.5 Modifying an Audit

You can modify the date on which the audit starts, its description, and all fields on the *Filters* tab. Once you save your changes, you cannot access the previous versions of an audit.

To modify an audit

1. Open [page 658] an audit.
2. Set up [page 656] the audit.
3. Verify the audit structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
4. Save [page 658] the audit.

1.40.1.6 Deleting an Audit

Use this procedure to delete an audit.

To delete an audit

1. From the menu bar in the main window, select *File* > *Delete* > *Audit*. The *Delete Audit* dialog box opens.
2. Select the audit to be deleted.
3. Click *Delete*. A dialog box opens asking you to confirm your choice.
4. Confirm the audit deletion by clicking *Yes*. A dialog box confirms that the audit has been successfully deleted.

1.41 Change Lists and Transport Requests

You create a change list when you want to transport a consistent subset of catalog objects to target SAP CC systems (as destinations) by using:

- A transport request triggered with SAP CC Core Tool
- Or a transport operation in SAP Solution Manager (with *Transport Organizer Web UI* and an SAP CTS+ plugin)

The content of a change list export can also be reimported into SAP Convergent Charging using the *Import* feature.

> → Recommendation
>
> Whenever several related catalog objects must be retrieved together, exporting the whole change list is strongly recommended: it is easier than exporting the objects separately, and it also helps you to preserve the integrity of your catalog, as each export or import operation handles the entirety of the change list's object set.

> → Remember
>
> In SAP Convergent Charging, a change list is not just a "list of changes". A change list is a real object. It includes many cross-referenced objects and versions.

A **change list** is a "container" including some master data that the SAP CC user selects in order to transport data between systems. The selectable data is:

- Pricing macro
- Translation table
- Tier table
- Mapping table class
- Mapping table
- Range table class
- Range table
- Chargeable item class
- Charged item class

- Charged item aggregation policy
- Charge
- Charge plan
- Refill item class
- Refill record class
- Refill logic
- Refill plan
- Allowance event class
- Allowance logic
- Allowance plan
- Monitoring plan

> **Note**
>
> A change list contains all the versions of the data objects. Any change to these versions after creating the change list has no impact on its content.

A change list can also include dependent objects such as:

- Billable Item Mapping
- Consumption Item Mapping
- Allowance Interface
- Counter Name Dictionary
- Refill Record Mapping
- Spending Status Description

> **Note**
>
> Dependent objects are included only if the user ticks the corresponding checkbox when creating the change list.

Working with Change Lists [page 661]

Using the Action Menu [page 667]
Use the *Action* menu of the user interface to manage your change lists and transport requests.

Managing the Transport Requests [page 668]

Related Information

Catalog Transport

1.41.1 Working with Change Lists

Creating a Change List [page 662]

Setting Up a Change List [page 663]

Saving a Change List [page 664]

Finding a Change List [page 664]

Opening an Existing Change List [page 666]

Exporting a Change List and Included Data [page 666]

Use this procedure to be able to transport consistent objects (master data, business data) with SAP CTS in SAP Solution Manager to the target SAP CC systems in your landscapes.

Deleting a Change List [page 667]

Related Information

Creating a Change List [page 662]
Setting Up a Change List [page 663]
Saving a Change List [page 664]
Finding a Change List [page 664]
Opening an Existing Change List [page 666]
Exporting a Change List and Included Data [page 666]
Deleting a Change List [page 667]

1.41.1.1 Creating a Change List

Use the following procedure to create a change list.

> **Note**
>
> Once you have saved a change list, you can no longer modify it.

Create a Change List

1. From the menu bar in the main window, select *File* > *New* > *Change List*. The *New Change List* window opens.
2. Set up [page 663] the change list.
3. Verify the change list structure by clicking the *Verify* icon on the main window toolbar. Correct all elements underlined in red.
4. Save [page 664] the change list. If you save the change list to the database, you are asked to configure and process [page 668] the transport requests that you need.

→ Tip

You can create an change list by opening an existing change list and copying it.

Create a Change List from an Existing Change List

1. From the menu bar in the main window, select *File* > *Open* > *Change List*. The *Choose a Change List* dialog box opens.
2. Search for [page 664] the change list.
3. Select the change list to copy.
4. Select the *Open a Copy* checkbox, and then click *Open*. The corresponding *Change List* window opens.
5. Set up [page 663] the change list.
6. Verify the change list structure by clicking the *Verify* icon  on the main window toolbar. Correct all elements underlined in red.
7. Save [page 664] the change list.

1.41.1.2 Setting Up a Change List

Create [page 662] a new change list before setting it up. A *Change List* window must be open.

To set up a change list, you first identify it, and then you add the data objects to the list.

Name: Type the unique identifier of the change list.

Description: Type a description of the change list if necessary.

User Login: The SAP CC user who created the change list.

Creation Date: The date and time at which you have saved the change list. Note that this date/time is only displayed after saving the change list.

Objects

The *Objects* area allows you to add objects to the change list from a catalog, remove any object from the change list, and view the snapshot of objects after saving the change list.

 : Click to add a new object to the list. This button opens the *Choose Object* window, in which you select the owner of the catalog and the object to add to the list. You can also select the *Include the Dependent Objects* option that adds all the dependent objects of the selected object to the list.

 : Click to remove the selected objects from the list.

> **Note**
>
> You can only remove objects that are not dependent on other objects.

 : Click to open the selected object and view its snapshot. You can click this button only after saving the change list.

Code: The identifier of the object.

Type: The type of the object.

1.41.1.3 Saving a Change List

Save a New Change List to the Database

1. From the menu bar in the main window, select *File* > *Save* > *In the Database*.
2. The *Change List Creation* dialog box opens to inform you that the change list has been successfully created and that you can configure and process [page 668] the transport requests that you need.
3. Click *Configure* if you want to configure transport requests. Otherwise, click *No*.

Save a Change List as an XML File

1. From the menu bar in the main window, select *File* > *Save* > *In a File*. A confirmation dialog box opens.
2. Click *OK*. The *Save* dialog box opens.
3. Select a directory to save the change list.
4. In the *File name* field, type the name of the file.
5. Click *Save*. The change list is saved as an XML file in your selected directory.

> **Caution**
>
> This XML file is not compatible with the *Transport Organizer Web UI* user interface of the SAP CTS system that is integrated with your SAP CC system. Use the XML file created by the Export [page 666] procedure instead.

1.41.1.4 Finding a Change List

To search for a change list, you perform the operation in two halves of the *Choose a Change List* dialog:

- The top part of the dialog is the filter area, which allows you to find a change list by using filter criteria. The information in the *Basic* and *Advanced* tabs is combined during the search operation.
- The bottom area of the dialog is the results area, which displays the search results that you perform.

You can find change lists with or without using search criteria. If you do not use search criteria, click  to start the search. If you use search criteria, set up the *Basic* and *Advanced* tabs, and then click  to start the search. The change lists are displayed in the results area. To display more columns of information in the results area, click  , and then select the types of information you want to display.

The Choose a Change List dialog

Use the following icons:

 : To start a new search whether you use search criteria or not.

 : To delete the results list and reset the filter.

 : To move to the next page. This button is available when the results list is large.

 : To move to the previous page. This button is available when the results list is large.

Basic Tab

The *Basic* tab allows you to search for change lists depending on a name.

Name: Is the identifier of the change list. In the drop-down list, select a search comparator such as *Equals*, *Is Like*, *Starts With*, *Ends With*, or *Contains*. In the next field, type the string that you want to search in the back-end database. For example, if you want to find all provider contracts containing the letters **`eric`**, you select the *Contains* option, and then type **`eric`**.

Advanced Tab

To find change lists, the *Advanced* tab allows you to create a table with one or more search criteria based on additional information. To add a row to the table, click the *More* button. To delete a row from the table, click the *Less* button.

> **Note**
>
> - If the criteria table comprises several rows and if you want to delete a row, the *Remove* dialog box opens to display the identifier names. Select the row that you want to delete from the table, and then click *OK*.
> - If you choose the *Is Like* comparator when building criteria rows, you can type "*" in the right-hand field to represent a number of characters and "?" to represent a single character.

Each row of the table consists of three fields. To set up a row, do the following:

1. In the first field, select a search criterion such as *Add. Info (String)*, *Add. Info (Number)*, or *Add. Info (Date)* in the list of identifiers.

2. In the second field, select a search comparator such as *Equals*, *Is Like*, or *Starts With* for example.
3. In the third field, select or type the string that you want to find in the database.

1.41.1.5 Opening an Existing Change List

The *Choose a Change List* dialog allows you to find the change list that you want to open from the back-end database.

Open a Change List saved in the Database

1. From the menu bar in the main window, select *File* > *Open* > *Change List*. The *Open Change List* dialog box opens.
2. Search for [page 664] the change list that you want.
3. Select the change list, and then click *Open*. The *Change List* window opens.

1.41.1.6 Exporting a Change List and Included Data

Use this procedure to be able to transport consistent objects (master data, business data) with SAP CTS in SAP Solution Manager to the target SAP CC systems in your landscapes.

Create [page 662] or open [page 666] a change list before exporting it. A *Change List* window must be open.

Export a Change List Object to a File

1. From the menu bar in the main window, select *Action* > *Export Change List*. The *Save* dialog box opens.
2. Select a directory to export the change list as a file.

> ⚠ Caution
>
> The file includes some business information and data from the pricing catalog and from the system.

3. In the *File name* field, type the name of the file.
4. Click *Save*. The change list is exported as an XML file in your selected directory.

Next Steps

To transport the master data, use this exported XML file with the *Transport Organizer Web UI* user interface in SAP Solution Manager when the SAP CTS system is integrated with your SAP CC 2023 Core Server system.

Otherwise, contact the individual who is responsible for the transport operations with SAP CTS.

> → Recommendation
>
> To ease the attach operation, choose a file name as descriptive and as unique as possible to give SAP CTS users the possibility to attach the adequate change lists to the transport operations. Do not hesitate to use timestamps or any other information which can identify a file and avoid errors during the selection.

1.41.1.7 Deleting a Change List

> ⚠ Caution
>
> There is no “undo” function in SAP Convergent Charging. Before deleting an object, make sure that this action is necessary.

Use the following procedure to delete a change list.

Delete a Change List

1. From the menu bar in the main window, select *File* > *Delete* > *Change List*. The *Delete Change List* dialog box opens.
2. Search [page 664] for the change list to delete.
3. Select the change list to be deleted.
4. Click *Delete*. A dialog box opens asking you to confirm your choice.
5. Confirm the change list deletion by clicking *Yes*. A dialog box confirms that the change list has been successfully deleted.

1.41.2 Using the Action Menu

Use the *Action* menu of the user interface to manage your change lists and transport requests.

Configure Transport Requests (Menu) [page 668]

Export Change List (Menu) [page 668]

Related Information

Managing the Transport Requests [page 668]
Exporting a Change List and Included Data [page 666]

1.41.2.1 Configure Transport Requests (Menu)

This menu is available in the *Action* primary menu.

See: Managing the Transport Requests [page 668]

1.41.2.2 Export Change List (Menu)

This menu is available in the *Action* primary menu.

See: Exporting a Change List and Included Data [page 666]

1.41.3 Managing the Transport Requests

Prerequisites

- You have saved the change list [page 660] in the back-end database system.
- The system administrator has defined and configured the list of transport destinations for sending the subset of object versions to a remote SAP CC system. If this operation is not already done, refer to theAdministration Guide for more information.
- Make sure that the system scheduler is configured if you want to schedule the execution of transport requests. For more information, refer to the Administration Guide.

Manage the Transport Requests

You can configure, run, schedule, monitor, or delete a transport request when:

- Saving [page 664] a change list to the back-end database.
- Opening [page 666] the main window of an existing change list by selecting *Configure Transport Requests* in the *Action* menu.

> **Note**
>
> On the main window toolbar, you can click  to configure the transport requests related to the change list.

Transport Requests (Window)

The *Transport Requests* window allows you to execute the transport requests. If necessary, you can schedule the execution of the transport requests.

The buttons *Schedule* and *Execute* are available only if you select an item in the *Transport Destination Alias* column and if the scheduler and the list of transport destinations are correctly configured.

Change List: Displays the change list [page 660] to be sent to the transport destination. Click  to open the change list.

Use the following buttons:

Schedule: Schedules the transport requests for the selected transport destinations. This button is not available if the status of the selected transport destinations is “Scheduled” or “In process”.

Unschedule: Unschedules the transport requests of the selected transport destinations. This button is available only if the status of the selected transport destinations is “Scheduled”.

Execute: Executes the transport request according to the selected transport destinations. This button is not available if the status of the selected transport destinations is “In process”.

The transport requests table contains the following columns:

Transport Destination Alias: Displays the alias of the transport destination to which the change list (and related master data) is sent.

Creation Date: Displays the creation date and time of the transport request.

Execution Status: Displays the execution status of the transport request for the selected transport destination. The execution statuses mean the following:

- *None*: The change list has not been yet sent to the transport destination.
- *Scheduled*: The change list will be sent when the scheduler is triggered.
- *In Process*: The change list has been sent to the transport destination asynchronously, and the transport destination has not yet responded (or the response has not been received).
- *Successful*: The change list has been successfully sent to the transport destination.
- *Failed*: An error has occurred while sending the change list to the transport destination. Verify the error information (*Error Category*, *Error Description*) and retry the operation or troubleshoot the issue.

Error Category: Displays the error category if the transport request failed to execute the change list according to the selected transport destination. The categories mean the following:

- *Authentication Failure*: A failure such as an invalid password, a wrong user ID, and so on, has occurred.
- *Communication Failure*: A failure such as a network issue has prevented the connection to the target SAP CC system.
- *Invalid*: A master data included in the change list is not configured correctly.

- *Incompatible Configuration*: The configuration of the target SAP CC system is not compatible with a master data included in the change list.
- *Illegal State or Temporary Illegal State*: The change list has not been sent to the target SAP CC system because of its state.

Error Description: Displays the error message if the transport request failed to execute the change list according to the selected transport destination. You can click the message to display a dialog with more information about the error, which is often an error code. See Troubleshooting Transport Requests [page 670].

Last Send Date: The date and time when the transport request was last sent.

Response Date: The date and time when the last response of the transport destination was received.

Transport Requests window buttons

Use the following buttons:

Save To File: To save the transport request and its associated destination statuses to a file. Click this button to open the directory in which you want to save the file. Type the name of the file, and then save the file. The execution history of the transport requests is stored in this saved file.

Refresh: To view the last status of the transport requests. Click this button when you want to refresh the status.

Close: To close the *Transport Status* window in the user interface.

> **Note**
>
> - Refreshing a transport request whose status is `In Process` re-sends the transport request and modifies the related `Last Send Date` information. A transport request is executed only once even it is sent many times.
> - If a change list has been sent to a transport destination that does no more exist in the SAP system landscape, the associated transport request is dimmed (not available) and can no more be selected.

Related Information

Change Lists and Transport Requests [page 660]

1.41.3.1 Troubleshooting Transport Requests

As a system administrator or a pricing specialist, you troubleshoot some issues by investigating the discovered symptoms.

See the following situations:

Failure	Possible Reasons or Investigations
Authentication Failure	*Authentication Failure* can concern failures such as invalid password, wrong logon, and so on.

Failure	Possible Reasons or Investigations
Communication Failure	*Communication Failure* can be due for example to a network issue preventing the connection to the target SAP CC system.
Invalid	*Invalid* means that a master data included in the change list is invalid and cannot be applied in the target SAP CC system.
Incompatible Configuration	*Incompatible Configuration* means that the configuration of the target SAP CC system is not compatible with a master data included in the change list.
Illegal State	*Illegal State* means that the change has not been applied because of the overall state of the target SAP CC system itself. Contact the system administrators.
Temporary Illegal State	*Temporary Illegal State* means that the change has not been applied because of the overall state of the target SAP CC system itself. This system state is temporary, please try again later.

1.42 Working with Logic Components

Logic Components

> ⚠ Caution
>
> If you are designing and configuring a price plan for a prepaid environment (session-based charging with credit control and quota reservation), do not forget that the confirmed amount must not be greater than the reserved amount.
>
> For example, if 5 minutes are reserved at the price of $2.50, the confirmed amount must not be greater than $12.50.
>
> For more information about session-based charging, refer to the primary help [page 7] of Core Tool.

SAP Convergent Charging consists of the following business scenarios:

- Process of charging: Allows to rate a usage event and debit the price amount from an account or credit it to an account. The SAP CC system performs this process through the execution of a price plan and a charging plan configured in a charge and subscribed in a provider contract.
- Process of refilling: Allows a customer to credit prepaid accounts manually or automatically. This process is performed through the execution of a refill plan activated in a provider contract item.
- Process of creating billable items: Allows to convert charged items and refill records into billable items that are sent to SAP Convergent Invoicing in the SAP ERP/FI-CA or in the SAP S/4HANA system. This process

is performed through the execution of the billable item mapping. For more information, see Help resources [page 686].

- Process of creating consumption items: Allows to convert chargeable items into consumption items that are sent to SAP Convergent Invoicing in SAP ERP/FI-CA system for charging or deferred charging purposes, for example. For more information, see Help resources [page 686].
- Process of using allowances: Allows to create allowances [page 792], send events [page 777] to these allowances from the charging or refilling process, modify [page 782] these events, and redefine [page 782] the validity period of allowances. For revenue recognition purposes, SAP CC can inform SAP Convergent Invoicing through billable items when an allowance is created, used, and has expired. The billable items contain information so that SAP ERP/FI-CA can recognize revenues. For more information, see Help resources [page 686]. About the share of allowances (shared allowances) between subscriber accounts, refer to the primary help of Core Tool and the Application Help for more information.

> ⓘ Note
>
> The processes of creating consumption items and billable items are available in the end-to-end *SAP Solution* scenarios that include the SAP ERP/FI-CA component of the SAP Business Suite. Consult the SAP CC Getting Started Guide for more information.

Process Configuration

You configure the charging process and the refilling process by:

- Designing and specifying the pricing logic.
- Configuring the relevant master data included in the pricing catalog of a service provider.

This logic is configured through the decision trees of a charge, the refill logic, or the allowance logic. These trees consist of logic components[72]:

- Basic Components [page 726]
- Logic Components for a Price Plan [page 688]
- Logic Components for a Charging Plan [page 700]
- Logic Components in the Refill Logic [page 706]
- Logic Components in the Allowance Logic [page 715]

Price Plan

A price plan specifies the business logic to calculate an amount linked to a usage event [page 690], a recurring event [page 693], or a one-shot event [page 698]. The amount is calculated by using a set of instructions described in a price plan that is contained within a charge. A charge always includes a unique price plan.

> → Remember
>
> **About Taxes**
>
> Remember that the taxes are defined at the following levels in SAP CC:
>
> - Subscriber account
> - Charge in a charge plan (charging based on provider contracts)

[72] Except comparators and operators which are basic components [page 726] used both by the price plan, the charging plan, the refill logic, and the allowance logic.

Tax computation only applies to price plan components but does not apply to charging and refill logic components. For more information, consult the primary help of Core Tool [page 12] (see the Tax Settings [page 523] topic).

Charging Plan

A charging plan specifies which accounts will be charged or refilled with the amount of a service calculated by the related price plan.

Refill Logic

Part of a refill plan, the refill logic specifies the logic to compute the different credits (money, units, and so on) of a refill. At runtime, calculations in the refill logic are only performed when the subscribed refill plan is triggered by:

- A usage event [page 708]
- A periodic event [page 710]
- A one-off event [page 713]

To compute a refill event, the refill logic consists of a set of refill logic components.

Allowance Logic

Part of an allowance plan, the allowance logic defines the rules and conditions of use of an allowance and can model an algorithm based on several types of triggers: event-based, one-shot, and recurring.

At runtime, calculations in the allowance logic are only performed when the subscribed allowance plan is triggered by:

- Any event [page 717]
- A periodic event [page 720]
- A one-shot event [page 723]

To compute an allowance event, the allowance logic consists of a set of allowance logic components.

Logic Components

To set up the charge logic [page 290] (in price plan and in the charging plan) and the refill logic [page 395], SAP CC provides logic components to define the logic of these objects. Components are grouped into families according to their functional usage.

> ⓘ Note
>
> For more comprehensive detail on using each component, refer to the primary help relating to logic components:Working with Logic Components [page 671]

Family	Description	Price Plan	Charging Plan	Refill Logic
 Rates	Defines the tariff type for pricing (Usage, Recurring or One-Shot)	Yes	-	-
 Charging	Defines the charging type (Usage, One-Shot, Recurring or Default)	-	Yes	-
 Refill	Defines the refill type (Usage, Recurring or One-Shot)	-	-	Yes
 Functions	Calculates the final amount	Yes	-	Yes
 References	Identifies and charges or refills an account Example: Internal Reference	-	Yes	-
 Comparators	- Generates two or more branches in order to evaluate properties - Selects one branch according to the result Example: - Mapping Table Comparator - Range Table Comparator - Public Holidays Comparator	Yes	Yes	Yes
 Splitters	- Splits a number or duration - Generates and rates multiple branches Example: Public Holidays Splitter	Yes	-	Yes
 Operators	Adds properties to the calculation context Example: Tier Table Operator	Yes	Yes	Yes

1.42.1 What's New?

What's New in SAP CC 2020 FPS 1

As of SAP CC 2020 FPS 1:

- The Allowance Event Sender Operator [page 777] component allows the pricing specialist to send events to shared allowances directly from the price plan of a charge or a refill plan. So far, the only possibility to do so was to create non-shared allowances and use the Allowance Event Sender Operator component in the logic of these allowances.

  The enhancement of the Allowance Event Sender Operator component consists in enabling the Only Shared Allowances [page 779] option of the *Definition* tab of the component when this component is used in a price plan [page 299] of a charge [page 343] or a refill logic [page 395].
- The Property Introducer Operator [page 808] can manage empty strings.

1.42.2 Master Data

Master data

Logic components are used to design and set up the pricing logic of customer services of a service provider. These logic components are configured and combined in pricing objects that are stored in the master data related to this service provider.

Master Data Related to the Service Provider	Customer Master Data
SAP Convergent Charging manages master data related to the pricing logic for the business of a service provider.	SAP Convergent Charging manages master data related to the end customers of a service provider.
This pricing logic is configured and stored in a pricing catalog dedicated to this service provider.	The service provider remains the final owner of this master data.

Master Data Related to the Service Provider	Customer Master Data
The pricing catalog includes mostly: • Reusable charges that includes: • A price plan [page 299] and its logic components [page 688] • A charging plan [page 305] and its components [page 700] • The refill logic [page 395] and some components [page 706] • The allowance logic [page 449] and Logic Components in the Allowance Logic [page 715] • Pricing macros [page 121] • Charge plans [page 343], refill plans [page 421] and allowance plan [page 466] • Business data tables: • Mapping tables [page 179] and classes [page 167] • Range tables [page 213] and classes [page 199] • Translation [page 132] and tier [page 147] tables (rarely)	• Subscriber accounts [page 511] • Business data tables: • Subscriber mapping tables [page 542] • Subscriber range tables [page 557] • Provider contracts [page 573]

1.42.3 Matrix of Components

The matrix shows you in which master data objects (for example, price plan, charging plan) you can use the logic components. Each cell that contains 'X' means you can use the component in the related master data object. Because the usage of logic components can depend on specific contexts, refer to the following notes for X (1), X (2), X (3), X (4), X (5), X (6), and X (7):

> ⓘ Note
>
> - X (1): Persistent counters cannot be referenced or used in this context.
> - X (2): You can only use a One-Shot Trigger component with a Creation event or an Expiration event (no shift).
> - X (3): You can only use a Create Allowance Operator in a One-Shot Trigger component with an Expiration event (no shift). The renewal of shared allowances is not allowed in this context.
> - X (4): If the scope is limited to shared allowances in the Allowance Property Introducer Operator, the property that contains the Share Identifier must refer to a property of the rating context.
> - X (5): The usage of the Allowance Event Sender Operator depends on its scope:
> - If the scope is limited to shared allowances, you can only use the Allowance Event Sender Operator component in an Event-Based Trigger component of the allowance logic that is not shareable.
> - If the scope is limited to the allowances of the contract or the allowances of the contract and its parent, you can only use the Allowance Event Sender Operator component in the price plan and the refill plan (refill logic).
> - X (6): Persistent counters can only be referenced and used in a One-Shot Trigger (Creation event).
> - X (7): You can only use the Allowance Validity Period Updater Operator in a One-Shot Trigger component with a Creation event.

Rate Components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Usage Rate [page 690]	X					
Recurring Rate [page 693]	X					
One-Shot Rate [page 698]	X					

Charging components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Usage Charging [page 701]		X				
Recurring Charging [page 703]		X				
One-Shot Charging [page 703]		X				
Default Charging [page 704]		X				

References	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Default Reference		X				
Internal Reference [page 705]		X				
External Reference		X				
Refill		X				

Refill components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Usage Refill [page 708]				X		

Refill components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Recurring Refill [page 710]				X		
One-Shot Refill [page 713]				X		
Account Event Refill [page 714]				X		

Trigger Components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Event-Based Trigger [page 717]	X				X	X
Recurring Trigger [page 720]	X				X	
One-Shot Trigger [page 723]	X				X	X (2)

Function components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Allow Function [page 728]					X	X
Flat Function [page 729]	X		X	X		
Free Function [page 730]	X		X	X		
Generic Function [page 731]	X		X	X		
Linear Function [page 733]	X		X	X		
Macro Function [page 734]	X		X	X		
No Access Function [page 735]	X				X	X (1)
Polynomial Function [page 736]	X		X	X		

Comparison components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Date/Time Slot Comparator [page 738]	X	X	X	X	X	X (1)
Dates Comparator [page 740]	X	X	X	X	X	X (1)
Interval Switching Comparator [page 742]	X	X	X	X	X	X (1)
Mapping Table Introducer (Comparator) [page 744]	X	X	X	X	X	X (1)
Numbers Comparator [page 747]	X	X	X	X	X	X (1)
Prefix Switching Comparator [page 748]	X	X	X	X	X	X (1)
Public Holidays Comparator [page 750]	X	X	X	X	X	X (1)
Range Table Introducer (Comparator) [page 751]	X	X	X	X	X	X (1)
Strings Comparator [page 758]	X	X	X	X	X	X (1)
Switching Comparator [page 759]	X	X	X	X	X	X (1)

Splitters components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Amounts Aggregator Splitter [page 761]	X		X	X		
Numbers Splitter [page 763]	X		X	X		

Splitters components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Public Holidays Splitter [page 765]	X		X	X		
Time Slot Splitter [page 769]	X		X	X		

Operators components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Alert Triggering Operator [page 776]	X	X	X	X	X	X (1)
Allowance Event Sender (Operator) [page 777]	X (5)			X (5)	X (5)	
Allowance Event Updater (Operator) [page 782]					X	X (1)
Allowance Property Introducer (Operator) [page 784]	X (4)			X (4)	X (4)	
Allowance Validity Period Updater (Operator) [page 787]					X	X (7)
Arithmetic Computation Operator [page 789]	X	X	X	X	X	X (1)
Concatenate Operator [page 791]	X	X	X	X	X	X (1)
Create Allowance Operator [page 792]	X			X	X (3)	
Date Shift Operator [page 794]	X	X	X	X	X	X (1)

Operators components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Decrement Counter Operator [page 795]	X		X	X	X	X
Duration Operator [page 797]	X	X	X	X	X	X (1)
Duration Adapter Operator [page 798]	X	X	X	X	X	X (1)
Filter Operator [page 800]	X	X	X	X	X	X (1)
Increment Counter Operator [page 803]					X	X
Macro Introducer (Operator) [page 804]	X		X	X	X	X (1)
Multi-Valued Evaluator (Operator) [page 805]	X	X	X	X	X	X (1)
Output Property Update Operator [page 807]			X			
Property Introducer (Operator) [page 808]	X	X	X	X	X	X (1)
Property Type Convertor (Operator) [page 810]	X	X	X	X	X	X (1)
Prorata Operator [page 813]	X	X	X	X	X	X (1)
State Schedule Updater (Operator) [page 817]				X		
Switch Table Operator [page 819]	X	X	X	X	X	X (1)
Tier Table Introducer (Operator) [page 822]	X	X	X	X	X	X (1)

Operators components	Price plan	Charging plan	Pricing macro	Refill plan / Refill logic	Allowance plan / Allowance logic (not shareable)	Allowance plan / Allowance logic (shareable)
Translation Table Introducer (Operator) [page 823]	X	X	X	X	X	X (1)
Update Counter Operator [page 826]	X		X	X	X	X (6)

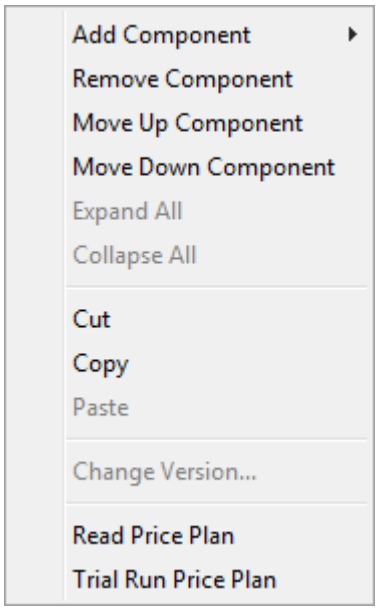
1.42.4 Working with Components

Related Information

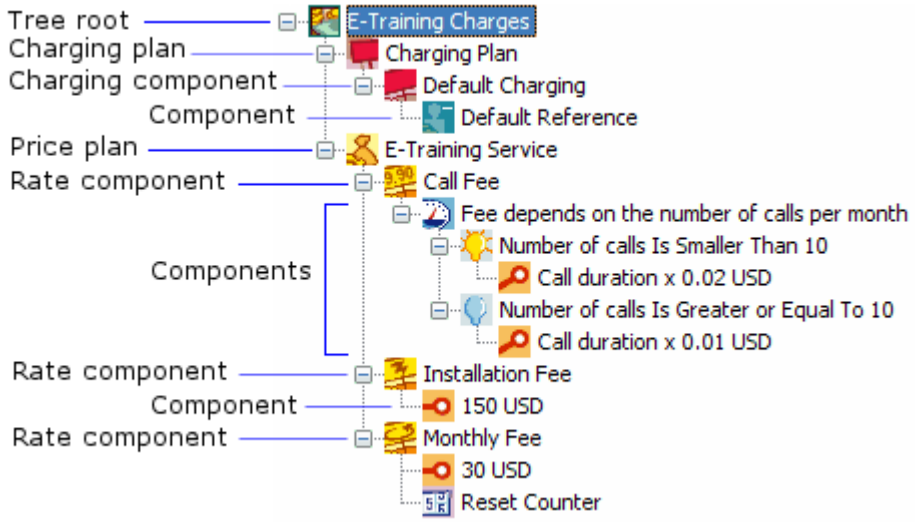
Arranging Components [page 682]
Setting Up a Component [page 685]

1.42.4.1 Arranging Components

Commands to arrange the components are grouped on the shortcut menu, which is displayed when you right-click the root of the tree, a rate component, a charging component, a refill component, or an allowance component:



The command features depend on the level at which you use them:



Root level

If you click the root of the tree, you can only carry out the commands *Add*, *Paste*, and *Copy*.

Add: Allows you to add rate, charging, allowance, or refill components, which are automatically arranged in this order:

- Usage Rates (or Usage Charging, Usage Refills, Event-Based Triggers)
- One-Shot Rates (or One-Shot Charging, One-Shot Refills, One-Shot Triggers)
- Recurring Rates (or Recurring Charging, Recurring Refills, Recurring Triggers)

For each component family of a price plan, a charging plan, a refill plan, or an allowance plan, you can add as many components as you want; however, each component is automatically added after the last component of its family. For example, if you add a usage rate, it is added after the last component of the usage rate family.

In a charging plan, you can add one One-Shot Charging and one Recurring at the most.

Copy: Allows you to retrieve the XML version of the tree. Click the root of the tree, right-click to open the shortcut menu, click the *Copy* command, and then open a text editor to paste the XML version of the tree.

Paste: Allows you to paste a part of the tree that you have first cut or copied from the tree. The part of the tree is added after the last component of its family. For example, if you have cut a One-Shot Rate component, this component is added after the last component of the One-Shot Rate family when you paste it. The *Paste* command is available only if you have first cut or copied a component (usage, one-shot, or recurring).

Rate, charging, refill, or allowance component level

If you click a rate, a charging, a refill, or an allowance component, you can carry out all the commands of the shortcut menu; however, the commands are available depending on the context. For example, you cannot move up a rate component if this component is the first component of the tree.

Add: If you click a rate, charging, refill, or an allowance branch, you can add these components:

- Functions[73]
- Comparators
- Splitters[74]
- Operators
- References[75]

Components are available on the shortcut menu depending on the components already in the rate, charging, refill, or allowance branch. For example, if the rate branch contains a function component, you can only add operator components. If the charging branch contains a reference component, you can only add operator components.

> **Note**
>
> You remove, move up or down, cut, copy, or paste a rate, charging, refill, or an allowance component along with its branch.

Remove: Allows you to remove a rate, charging, a refill, or an allowance component from the tree permanently. You cannot paste a rate, charging, refill, or an allowance component that you have removed with this command.

Move Up and *Down*: Allows you to change the order of the rate, charging, refill, or allowance components in their family. For example, if the price plan tree contains several usage rates and recurring rates, you can change the order of usage rates in the usage rate family, but you cannot move a usage rate after the last component of the recurring rate family.

Cut: Allows you to remove a rate, charging, refill, or an allowance component from the tree. To cut a component, click the component, and then click the *Cut* command on the shortcut menu.

Copy: Allows you to copy a rate, charging, refill, or an allowance component. This command also allows you to retrieve the XML version of a rate, charging, or refill branch. Click a rate, charging, refill, or an allowance component, right-click to open the shortcut menu, click the *Copy* command, and then open a text editor to paste the XML version of the rate, charging, or refill branch.

Paste: Allows you to paste a rate, charging, refill, or an allowance component that you have first cut or copied from the tree. For example, you must click the price plan  or the charging plan  before respectively pasting a rate or a charging component. The component is automatically added after the last component of its family.

Component level

If you click a component, you can carry out all the commands of the shortcut menu; however, the commands are available depending on the context. For example, you cannot move up a component if this component is the first component of the branch.

Add: Allows you to add components in the branches of comparators or splitters only.

---

73 Not available in a charging branch
74 Not available in a charging branch
75 Not available in a price plan or in the refill logic

> **Note**
>
> You remove, move up or down, cut, copy, or paste a component along with its branch.

Remove: Allows you to remove a component from a branch permanently. You cannot paste a component that you have removed with this command.

Move Up and *Down*: Allows you to change the order of the components in a branch. This command is available depending on the components already in the branch. For example, if a rate branch contains an operator and a function, you cannot move up or down either of them. In a charging plan, this command is only available for operator components.

Cut: Allows you to remove a component from a branch. You can paste this component at another location in the tree.

Copy: Allows you to copy a component and paste it at another location in the tree. This command also allows you to retrieve the XML version of a component. Click a component, right-click to open the shortcut menu, click the *Copy* command, and then open a text editor to paste the XML version of the component.

Paste: Allows you to paste a component that you have first cut or copied. You can paste a component at rate, charging, refill, or allowance component level or in the branch of comparators or splitters. For example, to paste a component in the branch of a usage charging component, click the usage charging, and then paste the component.

1.42.4.2 Setting Up a Component

After you add or insert a component in the decision tree of a price plan [page 299], a charging plan [page 305], a refill plan [page 421], an allowance plan [page 466], or a pricing macro [page 121], you set it up.

To set up a component

1. In the tree display area of the plan window, select the component.
2. On the right side of the plan window, type the *Name* and the *Description* in the *Identification* area.
3. In the *Tabs* area, select or type the appropriate parameters that are needed to set up the component. For more information, refer to the specific component that you need to set up.

1.42.5 Getting Help

Related Information

Help Resources [page 686]
Typographical Conventions [page 688]

1.42.5.1 Help Resources

To help you get the most out of SAP Convergent Charging, SAP SE provides you with an extensive list of resources you can use for support and guidance.

Integrated SAP Solution Scenarios

SAP Convergent Charging is integrated with different software components of the SAP Business Suite: SAP CRM, SAP ERP/FI-CA, and SAP Convergent Invoicing in the SAP ERP system.

The primary help provides you with specific information and recommendations for configuring data in SAP Convergent Charging for the integrated scenarios.

Consult the SAP Help Portal

On the SAP Help Portal, you find extensive documentation from the SAP Library on SAP products and solutions, along with additional information about the documentation, information design at SAP, educations services, and links to the SAP Community Network (SCN).

> ⓘ Note
>
> SAP Convergent Charging is part of SAP Billing and Revenue Innovation Management. When your SAP CC system is connected to the SAP CRM system or to SAP Convergent Invoicing in the SAP ERP system, the following SAP libraries are relevant in SAP Help Portal at:
>
> - SAP EHP 4 for SAP CRM 7.0 > Application Help >SAP Library > SAP CRM for Industries > Telecommunications > Functions for Provider Sales and Order Management
>   - Products > Cross Catalog Mapping
>   - Business Agreement
>   - Provider Order and Provider Contract
>   - Sharing Contract
>
> and at:
>
> - SAP EHP 8 for SAP ERP 6.0 > Application Help > SAP Library > Industries in SAP ERP > SAP Telecommunications > Contract Accounts Receivable and Payable > SAP Convergent Invoicing
> - • Master Data

- Rating
- Billing
- Discounts and Charges on Billable Items
- Partner Settlement
- Invoicing

Access the Reference Guides of SAP Convergent Charging

These guides are available in PDF format and provided on the DVD-ROM with the release of this product. They contain comprehensive information about each functional aspect of SAP Convergent Charging. The guides are also available on SAP Help Portal for online consultation or download.

- Getting Started Guide
- Administration Guide
- Installation and Maintenance Guide

To find a key word once a guide is open, press CTRL + F (Windows users), type the word in the *Find* field, and then click *Find Next*.

Access the Configuration Guide of SAP Convergent Charging

If you need more information about configuring and maintaining the master data for a service provider, we recommend that you consult the SAP Convergent Charging Administration Guide on SAP Help Portal.

Access the Primary help for Subscription-Based Business

In case of a system landscape with customer master data including subscriptions, you can set up the SAP CC Core Tool user interface to work with all kinds of data related to the charging process based either on subscriptions or provider contracts.

Consult the Primary Help of Core Tool [page 12] and the Logic Component Reference [page 671] Reference on the SAP site.

Access the Application Help Documentation

The Application Help document is provided in the Software Development Kit (SDK) of SAP Convergent Charging. A version is available on SAP Help Portal for online consultation.

Access the Technical Documentation

The Web Services Documentation, the SAP CC Java/XML API References, and the SAP CC Error Code Reference documents are provided in the Software Development Kit (SDK) of SAP Convergent Charging. A version is available on SAP Help Portal for online consultation,

- Web Services Documentation
- Core APIs
- Error Code Reference

> → Tip
>
> You can also download these product documents on SAP Help Portal.

Access the SAP Notes

SAP Notes and SAP Knowledge Base Articles (KBA) help you find answers to questions that arise from your daily work with SAP Convergent Charging:

- Searching for 'cockpit' in SAP Notes and Knowledge Base Articles (KBAs)

1.42.5.2 Typographical Conventions

The following typographical conventions are used in the primary help of SAP Convergent Charging 2021:

- Text or a button that appears on the graphical user interface is written like *this*.
- System commands, names of folders and files, and codes used in APIs are written in `Code` font.
- Menu choices are shown in this format: *menu name* > *submenu name* > *menu itme name*.
- Text to be typed into a field is shown as follows: **`userinput`**.
- Blue text indicates an active hyperlink to another part of the primary help or a pop-up window that provides you with extra information on a task.
- UI names or features are displayed in purple.

1.42.6 Logic Components for a Price Plan

> ⚠ Caution
>
> Read the caution about components [page 671] carefully.

Each main branch of a price plan [page 299] in a charge [page 299] begins with a rate component [page 690] (usage, recurring, or one-shot) linked to a rate activation method.

To create a decision tree, you must:

1. Select the *Price Plan* node in the charge to set up.
2. In the left bar, select a rate component [page 690] in the *Rates* pane.
3. Add components from other families according to your price plan requirements.
4. End each branch of the tree with a function component [page 727].

Component Families

In SAP Convergent Charging, logic components are grouped into families according to their functional usage. Each component is represented by a specific icon. In the user interface, these families are represented by a specific color and an icon, and they are visible in the toolbars of the *Charge* and *Pricing Macro* windows. A logic component is of the same color as the one of its family.

The following table presents the component families available in a price plan:

Family icon	Name of the family	Function of the component family	Location of the component family
	Rates [page 690]	Defines the rating type (Usage [page 690], Recurring [page 693] or One-Shot Rate [page 698])	Always at the root of the decision tree. **Note** Unavailable in a pricing macro.
	Functions [page 727]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Comparators [page 737]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Splitters [page 760]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Operators [page 773]	Refer to the basic components [page 726]	Refer to the basic components [page 726]

Related Information

Logic Components for a Charging Plan [page 700]
Logic Components in the Refill Logic [page 706]

1.42.6.1 Rate Components

Rate components define tariffs (prices) for the chargeable services or products. They are always found at the root of the decision tree of a price plan.

The different types of rate components are available depending on the triggering type that the SAP CC pricing specialist needs to implement.

The following table shows the different types and the relevant triggering:

Type of Rate Component	Triggering	Examples
Usage	Each consumption of a chargeable service or use of a product to monetize	A telephone call, a video download
Recurring	Periodic	Yearly fees or monthly fees
One-Shot	Once only	One-off fees (subscription, contract re-sumption) or administrative fees

Rate Components in SAP CC

The following rate components are available in SAP Convergent Charging:

Icon	Name	Action/Use
	Usage Rate [page 690]	Calculates the price of a usage event (service consumption or product use). This Rate is activated each time a customer service is used (or consumed).
	Recurring Rate [page 693]	Calculates a price on a periodic basis: Day, week, month, and year. For example, the Recurring Rate can be used in the case of contracts or line rental.
	One-Shot Rate [page 698]	Calculates the price of a one-shot event such as the activation, the suspen-sion, or the resumption of a contract.

Related Information

Usage Rate [page 690]
Recurring Rate [page 693]
One-Shot Rate [page 698]

1.42.6.1.1 Usage Rate


The **Usage Rate** component creates in the decision tree a main branch that specifies the tariffs (prices) for the use of a product of for the consumption of a service by end customers. Each product use or service

consumption leads to corresponding transactional events. These events are named *chargeable items* in SAP Convergent Charging (SAP CC). Each time an end customer consumes a service or use a product, some chargeable items are sent to the SAP CC systems for being real-time or batch processed.

The configuration of a Usage Rate component requires the selection of a chargeable item classes [page 241] that specifies the properties of the incoming chargeable items (input data) to be charged by SAP Convergent Charging.

> ⓘ Note
>
> Only the *Usage Rate* component uses chargeable item classes [page 241]. The Recurring Rate [page 693] and One-Shot Rate [page 698] components do not use chargeable item classes.

Rules

The rules are the following:

- A price plan can contain several usage rates.
- You must at least assign a chargeable item class to a usage rate.
- If the *Chargeable Item Class* list of the *Definition* tab contains many chargeable item classes, you assign each chargeable item class to a different usage rate, or you assign all the chargeable item classes to the same uage rate. You cannot however assign a chargeable item class to several usage rates.
- If a chargeable item class is assigned to a usage rate, the default properties and the user-defined properties of this chargeable item class are available only in the branch using this usage rate.
- The *Charge Effective Date* property, which represents the start date/time of the related provider contract item, is available in the *Usage Refill* branch.
- If all the chargeable item classes are assigned to a usage rate, the default properties and the user properties of these chargeable item classes are not available in the branch using this usage rate. Only the properties related to the charge are available.
  For example, the *Charge Effective Date*, the *Currency Code*, and the *ISO Currency Code* properties are available in the related branch.

Use

You use the *Usage Rate* component to trigger and rate chargeable item classes.

> ✤ Example
>
> The following examples shows how to use the *Usage Rate* component:
>
> - Phone call fees that are based on the **`Phone Call`** chargeable item class and charged each time that the customer makes a call
> - Online news fees that are based on the **`Open Document`** chargeable item class and charged each time that the customer opens a document
>
> To use a chargeable item class called **`Phone Communication`** a customer is charged a fee for each phone call. The rate is $0.10 per minute. The following table shows the rating process information:

Data	Name/Content
Chargeable item class	Name: Phone Communication. Type of chargeable item class: Service. User property name: Call duration.
Branch of the price plan	The chargeable item class “Phone Communication” is assigned to the Usage Rate component. The branch contains a linear function component to compute the call price with the numerical property “Call duration”.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Type the business description of the component (optional).

Definition (Tab)

> ⓘ Note
>
> If the chargeable item class belongs to a **chargeable item package coming from a version earlier than SAP CC release 4.0**, the chargeable item class is identified by a two-part name, one of which is in parentheses. For example, the chargeable item class "SMS (E2E_TELCOMOB_CIP)" is part of the chargeable item package "E2E_TELCOMOB_CIP".
>
> Remember that you cannot assign a chargeable item class to several Usage Rates in the same charge. For example, you cannot assign the chargeable item class "SMS" to a Usage Rate and the chargeable item class "SMS (E2E_TELCOMOB_CIP)" to another Usage Rate because SAP CC (from 4.0 onwards) considers the two chargeable item classes as the same one.
>
> For more information about chargeable item packages, consult the primary help of Core Tool [page 7].

Chargeable Item Class: Provides the list of chargeable item classes [page 241] that define the data structure of the incoming chargeable items (input data) to be rated and charged by the system. Select a chargeable item class or select *All Chargeable Item Classes*.

If you select the *All Chargeable Item Classes* option, you assign all the chargeable item classes to a *Usage Rate* component. The *All Chargeable Item Classes* option is available if no chargeable item is already assigned to another *Usage Rate*.

→ Tip

To open and modify the chargeable item class, click  . Note that this icon is dimmed if you have selected the *All Chargeable Item Classes* option.

1.42.6.1.2 Recurring Rate

 The **Recurring Rate** component creates in the decision tree a main branch that is charged on a periodic basis. This logic component is characterized by:

- A **triggering rule** expressed by a:
  - Frequency (day, week, month, year)
  - Starting date and time
- **Payment rules** expressed by a:
  - Charging mode at the beginning of the period (prepaid periods)
  - Prorated payment based on the start date/time and end date/time of the provider contract item

→ Remember

When you insert a *Recurring Rate* in the decision tree of a dependent charge, you cannot set up and use the prorata parameters of the *Advanced* tab. Calculations made on a prorata basis are not allowed in a dependent charge; therefore, if your master charges make calculations on a prorata basis, SAP SE recommends that you use the property *Base Amount* in the related dependent charges to make calculations triggered by the master charges.

For example, if a refund is made on a prorata basis in a master charge, use the property *Base Amount* in the related dependent charges to make calculations relative to the master charge.

Rules

The rules are the following:

- A price plan can contain several recurring rates.
- When you insert a recurring rate in a price plan, some properties are available in the *Recurring Rate* branch:

Property	Description
Charge Effective Date	The creation date/time of the provider contract item that subscribes to a charge plan containing this recurring rate.
Recurring Event Date	The date and time when the recurring rate is triggered by the system.

Property	Description
Prorata Computation	This string property takes one of the two values `"prorated"` or `"not prorated"` depending on the result of the prorated calculation. In your logic, you can test this property to check if a transaction has been prorated or not.
Refund Computation	*Refund Computation*: This string property takes one of the two values `"refunded"` or `"not refunded"` depending on changes made to the *End Date* in provider contract items. In your logic, you can test this property to check if a transaction has been refunded or not. In a provider contract item, if you change the *End Date* to an earlier date, a refund is automatically generated by the system depending on the periodic basis set up to trigger the recurring rate. For example, a provider contract item includes a recurring rate that triggers a subscription fee of $5 the 1st of every month. In the provider contract item, the *End Date* is the 5th of July. You decide to change the *End Date* to the 5th of April. Thus, the total refund for May, June, and July will be an amount of $15. **Note** A refund is not possible for the components usage rate [page 690] and one-shot rate [page 698]. .
Reference Start for Recurring Period (aka *Recurring Period Reference Start*)	The start date/time of the reference period related to a recurring rate
Reference Stop for Recurring Period (aka *Recurring Period Reference Stop*	The stop date/time of the reference period related to a recurring rate
Recurring Period Start	The period start related to a recurring rate. It may be different from the *Reference Start for Recurring Period* and can be used to compute the prorator factor.
Recurring Period Stop	The period stop related to a recurring rate. It may be different from the *Reference Stop for Recurring Period* and can be used to compute the prorator factor.

- If you insert a Recurring Rate component in a dependent charge[76], the related master charge always triggers the dependent charge.

[76] A dependent charge is synchronized with or triggered by another charge. For example, a dependent charge is triggered when a provider contract is activated. A dependent charge can be a commission, a sponsorship, or a discount. Contrary to the dependent charge, the master charge is not triggered by another charge.

Use

You use the *Recurring Rate* component:

- To trigger and calculate a charge that is based on a time period such as equipment rental costs that are billed on an annual basis
- To calculate a commission, a sponsorship, or a discount through a dependent charge that is triggered by a master charge
- To describe specific recurring actions such as resetting a counter to zero

> ### Example
>
> You may use a *Recurring Rate* component in a master charge to define recurring charges for a monthly telephone service subscription. If a commission, which is also generated on a monthly basis, is paid to a third party, you need to use a *Recurring Rate* component in the dependent charge that includes this commission.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label for the pricing transactions.

Description: Type the business description of the component (optional).

General (Tab)

This tab defines the frequency and the starting date for triggering the *Recurring Rate* component.

> ### → Remember
>
> Once you have set up a recurring rate, SAP SE recommends not to change the frequency or the starting date/time. A change can cause serious errors.

> ### ⓘ Note
>
> In a dependent charge, you cannot configure the settings of the *General* Tab.

Reference Date: Provides the list of dates/times for triggering the *Recurring Rate* component. Select a date and time that you can combine with the settings chosen in the *This recurring rate is triggered* area to trigger the *Recurring Rate* component.

> ### ⓘ Note
>
> The *Reference Date* is used only if you select the following options:
>
> - *The same day as the reference date*

- *The same day as the reference date or first day of next month*
- *The same time as the reference date*

If you select another option, the *Reference Date* is ignored.

Calendar: Provides the list of available calendars on which the recurring event is based.

> ⚠ Caution
>
> Once the calendar is set, SAP SE highly recommends not to change it. A change can cause inconsistencies.

This recurring rate is triggered: Provides the list of recurrencies or frequencies:

- For *Gregorian Calendar*, the available periodicities are the following:
  - *Every*, *Every other*, *Every 3rd*, and other items associated with the following list: *day*, *week*, *month*, *year*. For example, you can set up the triggering of a recurring rate that is "Every 3nd month on Tuesday". The different options to select the starting date are the following:
    - *The same day as the reference date*: If you click this option, the recurring rate is triggered depending on the date selected in the *Reference Date* list. See the note below.
    - *The same day as the reference date or first day of next month*: If you click this option, the recurring rate is triggered depending on the date selected in the *Reference Date* list or the first day of the month following that of the selected date in the *Reference Date* list. See the note below.
    - *the*: If you have selected the frequency *Every* (3rd, 4th, and so on) *month* or *year*, this option triggers the *Recurring Rate* depending on the day of the month that you select. This option is combined with the list of numbers (from 1 to 31), the list of items such as Last, First, Second, Third and Fourth and the list of weekdays and months. Examples: the *Recurring Rate* is triggered Every 5th month the 11 of the month; the recurring rate is triggered Every year The Second Sunday of January.
    - *on*: If you have selected the frequency *Every* (3rd, 4th, and so on) *week* or *year*, this option triggers the *Recurring Rate* depending on the day of the week or the month that you select. Examples: the recurring rate is triggered Every 4th week on Wednesday; the recurring rate is triggered Every year on the 1st of January
    - *at*: Allows you to select the starting hour if you have selected any of the previous options. This option includes hours and minutes combined with AM (ante meridiem) or PM (post meridiem). For example: the Recurring Rate is triggered Every day at 07:25 PM. You can select the option *The same time as the reference date* that is the date selected in the *Reference Date* list.

> ⓘ Note
>
> About *The same day as the reference date* and *The same day as the reference date or first day of next month* options.
>
> If you select both of the *Every month* and *The same day as the reference date* options, the *Recurring Rate* will be triggered every month on the *Reference Date*; however, if you select the 31st of the month as the *Reference Date*, the *Recurring Rate* will be triggered the 31st of every odd month but the 30th of every even month (or the 28th of February, for example, if February has 28 days).
>
> If you select both of the *Every month* and *The same day as the reference date or first day of next month* options, the *Recurring Rate* will be triggered every month on the *Reference Date* or the first day of the next month depending on how many days the month has. For example, if you select the 31st of the month as the *Reference Date*, the *Recurring Rate* will be triggered the 31st of every odd month but

the 1st of the month following every even month; however, as February can have 28 or 29 days, the *Recurring Rate* will always be triggered the 1st of March if the *Reference Date* is the 31st of the month.

The table below shows you the dates on which the *Recurring Rate* will be triggered if you select the 31st of March as the *Reference Date*:

In the month of	The recurring rate will be triggered
January	The 31st of January
February	The 1st of March
March	The 31st of March
April	The 1st of May
May	The 31st of May
June	The 1st of July
July	The 31st of July
August	The 31st of August
September	The 1st of October
October	The 31st of October
November	The 1st of December
December	The 31st of December

- For *4-4-5 Calendar*, the available periodicities are the following:
  - *Every month*, *Every quarter*, *Every semester* and *Every year*.

> **Note**
>
> The other options, available with the Gregorian Calendar, do not appear any longer. They are useless with the 4-4-5 Calendar.

Advanced (Tab)

This tab defines the charging mode and the settings of the prorated calculation.

> **Note**
>
> In a dependent charge, you cannot configure these options.

Charged at the beginning of the period: Click to start the charge at the beginning of the period. You must specify the number of prepaid periods.

- *Prepaid Period Number*: The value of 0 means that one period will be prepaid; the value of 1 means that two periods will be prepaid; the value of 2 means that three periods will be prepaid, and so on.
  Default option: 0

Proportional during the period: Click to prorate the calculation of the recurring charge according to the the start date/time and the end date/time of the provider contract item.

You must specify the calculation parameters in *Prorata Parameters*:

- *Refund*: Define how transactions generated for the last period are refunded at the end of the contract item.
  - *Partial*: Issue a transaction with a negative prorated amount to correct the initial transaction. For example, a monthly recurring rate generates a transaction with an amount of 30€ for a given month. The end date of the related contract item is set to the 20th of the month. On the end date, a transaction with a -10€ amount is generated (assuming there are 30 days in the given month).
  - *Complete*: Issue a first transaction with a negative full amount to correct the initial transaction and then a second transaction with the correct prorated amount. For example, a monthly recurring rate generates a transaction with an amount of 30€ for a given month. The end date of the related contract item is set to the 20th of the month. On the end date, a first transaction with a -30€ amount is generated (assuming there are 30 days in the given month) and a second one with a 20€ amount is generated as well.
- *Rounding step*: Adjust the step rule for rounding the duration.

  > ✤ Example
  >
  > For example, if the rounding step value is 5 seconds and the rounding mode is *To the next greater value*, a duration of 47 seconds is rounded to 50 seconds. In this example, therefore, the rounding step consists of 5-second intervals. The time unit of the rounding step is expressed in *second*, *minute*, *day*, or *year*.

  Default option: *seconds*
- *Rounding mode*: Select one of the rounding mode of the calculation result: *To the next greater value*, *To the previous value* or *To the nearest value*. Default option: *To the nearest value*.
- *Use typical period of*: If selected, this option allows you to turn a variable period into a constant period. Once you have selected the *Use typical period* option, you can either select the value of a property or select *Numerical Value*, and then type a number.

  > ✤ Example
  >
  > For example, to turn each month into a constant period of 30 days, click *Use typical period of*, select *Numerical Value* in the list, and then type 30. If you clear *Use typical period of*, the reference period is 28 days in February, 31 days in January, and so on.

1.42.6.1.3 One-Shot Rate


The **One-Shot Rate** component creates in the decision tree a main branch that is charged only when precise lifecycle events related to provider contracts and their contract items occur. These events are the following:

- The **activation** of a the charge plan and creation of the contract item in a provider contract
- The **suspension** of a contract item in a provider contract
- The **resumption** of a contract item in a provider contract

Rules

The rules are the following:

- A price plan can contain several one-shot rates. For example, a one-shot rate for each type of the event such as *Activation*, *Suspension*, *Resumption*.
- When you insert a *One-Shot Rate* component in a price plan, the *Charge Effective Date/Time*, the *Suspension Date/Time* and the *Resumption Date/Time* of the provider contract item are available in the *One-Shot Rate* branch depending on the selected type of the event.

Use

You use the *One-Shot Rate* component to configure one-off fees in SAP Convergent Charging. A one-shot rate specifies the triggering dates/times of fees relating to events linked to contract items in provider contracts. You can use this component to charge an end customer for the equipment installation fees, the creation fees of a contract, and so on.

> ### Example
>
> For the pricing of your chargeable service **`Telephone Service`**, every new customer has to pay a one-time installation fee, which is paid when the provider contract is signed. This charge can be rated with a *One-Shot Rate* component, which is triggered by the *Activation* event at runtime.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Type the business description of the component (optional).

Definition (Tab)

Event Name: Provides the list of lifecycle events that trigger the one-shot rate component in a contracted charge at runtime. Select one of the following events:

- *Activation* (aka **creation of a contract item** in a provider contract that subscribes to the charge plan containing the one-shot rate)
- *Suspension*
- *Resumption*

1.42.7 Logic Components for a Charging Plan

Charging components are used to create a charging plan in a charge [page 290]. Together the components make up a charging tree. You create a charging plan to configure the determination of the prepaid account [page 514] or the external postpaid account [page 520] to be debited or credited with the calculated amount of the executed price plan [page 299].

To create a decision tree, you must:

1. Select the *Charging Plan* node in the charge to set up.
2. In the left bar, select a charging component [page 701] in the *Charging* pane.
3. Add components from other families to the charging tree according to your charging plan requirements.
4. End each branch of the decision tree with a *Reference* component.

Component Families

In SAP Convergent Charging, logic components are grouped into families according to their functional usage. Each logic component is represented by a specific icon. In the user interface, these families are represented by a specific color and icon, and they are visible in the toolbar of the *Charge* window. A logic component is of the same color as the family which it belongs to.

The following table presents the component families available in a charging plan:

Family Icon	Name of the Family	Function of the Component Family	Location of the Component Family
	Charging [page 701]	Defines the charging type (Usage, One-Shot, Recurring or Default).	Always at the root of the charging tree.
	Reference Component [page 705]	Identifies and charges or refills an account.	Always at the end of the branches. References are the leaves of the tree.
	Comparators [page 737]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Operators [page 773]	Refer to the basic components [page 726]	Refer to the basic components [page 726]

Related Information

Logic Components for a Price Plan [page 688]
Logic Components in the Refill Logic [page 706]

1.42.7.1 Charging Components

Charging components are always found at the root of the charging tree. A charging component is activated by one or more rate components [page 690].

The following table shows the different types and their relevant activation:

Type of Charging Component	Activation
Usage	Usage rate
Recurring	Any recurring rate
One-Shot	Any one-shot rate
Default	Any rate component

Charging Components in SAP CC

The following charging components are available in SAP Convergent Charging:

Icon	Name	Action/Use
	Usage Charging [page 701]	Creates in the charging tree a main branch that charges to an account each calculated amount of one or more usage rates [page 690] defined in the price plan.
	Recurring Charging [page 703]	Creates in the charging tree a main branch that charges to an account each calculated amount of all recurring rates [page 693] defined in the price plan.
	One-Shot Charging [page 703]	Creates in the charging tree a main branch that charges to an account each calculated amount of all one-shot rates [page 698] defined in the price plan.
	Default Charging [page 704]	Is in the charging tree a main branch that charges to an account each calculated amount of rate components [page 690] when no specified charging is defined.

Related Information

Logic Components for a Charging Plan [page 700]

1.42.7.1.1 Usage Charging


The *Usage Charging* component creates in the charging tree a main branch that charges to an account the calculated amount of each Usage Rate [page 690] defined in the price plan.

Rules

> **Note**
>
> If the price plan contains one or more Usage Rates and if the charging plan contains no *Usage Charging*, the Default Charging [page 704] is activated by each Usage Rate.

The following rules apply to the *Usage Charging* component:

- You must assign at least a chargeable item class to a Usage Charging.
- If the Chargeable Item Class list of the Definition tab contains many chargeable item classes, you assign each chargeable item class to a different Usage Charging, or you assign all the chargeable item classes to the same Usage Charging. You cannot however assign a chargeable item class to several Usage Charging
- See the rules below relating to the price and charging plans:

If the price plan contains	And if the charging plan contains	Then
n (≥ 1) Usage Rate(s)	One Usage Charging	The Usage Charging is activated: • EITHER by the Usage Rate set up with the same chargeable item class as the Usage Charging[77] • OR by all the Usage rates if the Usage Charging is set up with the option *All Chargeable Item Classes*
n (> 1) Usage Rates	m (> 1) Usage Charging	• n < m: Impossible[78] • n ≥ m: Each Usage Charging is activated by a different Usage Rate set up with the same chargeable item class as the related Usage Charging[79]

Identification area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of charging transactions.

Description: Allows you to type the description of the component (optional).

Definition tab

Chargeable Item Class: Provides the list of chargeable item classes related to chargeable item packages. Select a chargeable item class or select All Chargeable Item Classes. If you select the *All Chargeable Item Classes* option, you assign all the chargeable item classes of one or more chargeable item packages to one Usage

[77] The Usage Rates which are not linked to a Usage Charging activate the Default component
[78] The number of Usage Rate cannot be less than the number of Usage Charging
[79] The Usage Rates which are not linked to a Usage Charging activate the Default component

Charging component; however, user properties or default properties are not available in the Usage Charging branch except the charge effective date property. The *All Chargeable Item Classes* option is available if no chargeable item class is already assigned to another Usage Charging. To open and modify the chargeable item class, click  . Note that this icon is dimmed if you selected the *All Chargeable Item Classes* option.

1.42.7.1.2 Recurring Charging

 The *Recurring Charging* component creates in the charging tree a main branch that charges to an account the calculated amount of each Recurring Rate [page 693] defined in the price plan.

Rules

The following rules apply to the *Recurring Charging* component:

- A charging plan contains one Recurring Charging at the most
- If the price plan contains n (≥ 1) Recurring Rate(s), the Recurring Charging is activated by each Recurring Rate
- If the charging plan contains no Recurring Charging, the Default Charging [page 704] is activated by each Recurring Rate

Identification area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of charging transactions.

Description: Allows you to type the description of the component (optional).

1.42.7.1.3 One-Shot Charging

 The *One-Shot Charging* component creates in the charging tree a main branch that charges to an account the calculated amount of each One-Shot Rate [page 698] defined in the price plan.

Rules

The following rules apply to the One-Shot Charging component:

- A charging plan contains one One-Shot Charging at the most
- If the price plan contains n (≥ 1) One-Shot Rate(s), the One-Shot Charging is activated by each One-Shot Rate

- If the charging plan contains no One-Shot Charging, the Default Charging [page 704] is activated by each One-Shot Rate

Identification area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of charging transactions.

Description: Allows you to type the description of the component (optional).

1.42.7.1.4 Default Charging


The *Default Charging* component is in the charging tree a main branch that charges to an account the calculated amount of each Rate component [page 690] when no specific charging is defined. Unlike the other charging components, the *Default Charging* component cannot be removed from the charging tree.

> **Note**
>
> When you create a charge, the *Charge* window displays a default charging plan with a *Default Charging* and a *Default Reference*. If necessary, you can remove the *Default Reference* that you replace by any other reference.

Rules

The following rules apply to the *Default Charging* component:

If the price plan contains	And if the charging plan contains	Then
 n (≥ 1) Usage Rate(s)	No Usage Charging	The Default Charging component is activated by each Usage Rate
 n (≥ 1) Usage Rate(s)	One Usage Charging	The Default Charging component is activated by the Usage Rates which are not linked to a Usage Charging
 n (> 1) Usage Rates	m (> 1) Usage Charging	The Default Charging component is activated by the Usage Rates which are not linked to a Usage Charging
 n (≥ 1) Recurring Rate(s)	No Recurring Charging	The Default Charging component is activated by each Recurring Rate
 n (≥ 1) One-Shot Rate(s)	No One-Shot Charging	The Default Charging component is activated by each One-Shot Rate

Identification area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of charging transactions.

Description: Allows you to type the description of the component (optional).

1.42.7.2 Reference Component

In the charging plan of a charge, a reference defines which account of a subscriber will be charged or refilled with a calculated amount of the price plan. The *Reference* component makes up the leave of the decision tree.

Reference Component

The following reference component is available in SAP Convergent Charging:

Icon	Name	Action/Use
	Internal Reference [page 705]	Charges one account through a reference of the Internal Reference Dictionary [page 86].

Related Information

Logic Components for a Charging Plan [page 700]

1.42.7.2.1 Internal Reference

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, the behavior of this component is modified in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. For further information, refer to SAP Note 2772452.

 The *Internal Reference* component charges one account through a reference of the Internal Reference Dictionary [page 86].

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition Tab

Internal Reference: Provides the list of the references of the Internal Reference Dictionary. Select the internal reference to assign to the *Internal Reference* component.

Prepaid Account Charging Depending on Account State: Allows you to define the conditions for charging the prepaid accounts depending on their state and the component types used in the pricing branch. You are provided with two sliders, one for the Free Function component and one for the other components.

- *Charge Free Component Rates on*: This slider defines the conditions for charging the prepaid accounts when using Free Function [page 730] components in the pricing branch. The calculated amount of the price plan is charged to the prepaid accounts depending on both the slider position and the current state of the prepaid accounts. The table below shows you the expected charging: Yes (the prepaid accounts can be charged) or No (the prepaid accounts cannot be charged). For example, if you set the slider to *Blocked*, the prepaid accounts can be charged only if their state is *Active* or *Blocked*. But if the slider is set to *Closed*, the prepaid accounts can be charged whatever their state. Select one of the slider positions:

	State of the prepaid accounts			
Slider position	**Active**	**Blocked**	**Locked**	**Closed**
Closed	Yes	Yes	Yes	Yes
Locked	Yes	Yes	Yes	No
Blocked	Yes	Yes	No	No
Active	Yes	No	No	No

- *Charge Rates Other Than Free on*: This slider defines the conditions for charging the prepaid accounts when using any other components other than Free Function components in the pricing branch. Refer to the description above.

Authorize Overspending on Prepaid Account: Click this option to allow the prepaid account to be overspent.

1.42.8 Logic Components in the Refill Logic

Each main branch of the refill logic [page 395] begins with a refill component [page 707] (usage [page 708], recurring [page 710], or one-shot [page 713]) linked to a refill activation method.

To create a refill tree, you must:

1. Select a refill component [page 707].

2. Add components from other families according to your refill plan requirements.
3. End each branch of the tree with a function component [page 727].

Component Families for the Refill Logic

In SAP Convergent Charging, logic components are grouped into families according to their functional usage. Each component is represented by a specific icon. In the graphical user interface, these families are represented by a specific color and an icon, and they are visible in the toolbar of the *Refill* window. A logic component is of the same color as the family which it belongs to.

The following table presents the component families available in the refill logic:

Family Icon	Name of the Family	Function of the Component Family	Location of the Component Family
	Refills [page 707]	Defines the refill type (Usage [page 708], Recurring [page 710], One-Shot [page 713], or Account Event [page 714])	Always at the root of the refill tree. **Note** Unavailable in a pricing macro.
	Functions [page 727]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Comparators [page 737]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Splitters [page 760]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Operators [page 773]	Refer to the basic components [page 726]	Refer to the basic components [page 726]

Related Information

Logic Components for a Price Plan [page 688]
Logic Components for a Charging Plan [page 700]

1.42.8.1 Refill Components

Caution

As of SAP CC 5.0 SP 5, these logic components are not available in case of an integrated *SAP Solution* scenario with SAP S/4HANA Cloud Public Edition.

For further information, refer to SAP Note 2772452.

Refill components are always found at the root of the refill tree in the refill logic [page 395]. The different types of refill components are available depending on the triggering type that the pricing specialist needs to implement.

The following table shows the different types and the relevant triggering:

Type of Refill Component	Triggering	Examples
Usage	Each usage	Refill through a cash machine
Recurring	Periodic	Refill depending on the threshold of a balance
One-Shot	Once only	Refill at the contract activation
Account Event	Once only	Automatic refill when the prepaid balance goes under a specified threshold

Refill Components in SAP CC

The following refill components are available in SAP Convergent Charging:

Icon	Name	Action/Use
	Usage Refill [page 708]	Computes refill events
	Recurring Refill [page 710]	Computes refills on a periodic basis
	One-Shot Refill [page 713]	Computes refills only when precise events related to provider contracts occur
	Account Event Refill [page 714]	Computes refills only when precise events related to the account occur

Related Information

Logic Components in the Refill Logic [page 706]

1.42.8.1.1 Usage Refill


The **Usage Refill** component creates in the refill tree a main branch that computes refill events. When executing the *Usage Refill* component at runtime, the system uses a *refill item class* to compute the refills.

Rules

The rules are the following:

- The refill logic can contain several *Usage Refill* components.
- You must at least assign a refill item class to a usage refill.
- If a refill item class is assigned to a usage refill, the default properties and the user properties of this refill item class are available only in the branch that uses this usage refill.
- You can assign the same refill item class to different usage refills so that they use the properties of the same refill item class. But you must assign a *Usage Identifier* to each usage refill to identify them.
- The *Charge Effective Date* property which represents the start date of the related provider contract item is available in the *Usage Refill* branch

Use

As a pricing specialist, you can use the *Usage Refill* component to refill a prepaid account through a card or voucher.

Example

A service provider offers bundled services in one card. End customers use this card as a calling card and also recharge their prepaid account from it. The following table shows the refill process information:

Data	Name/Content
Refill Item Class	Name: **`Calling Card`**
Properties	*Refill Date*: Date of the refill **PREPR**: Reference number used for the refill **GPART**: Business partner number **BTRW**: Refill amount with +/- sign **WAERS**: Currency key **PPRSN**: Reason for balance change on prepaid account **PINUM**: PIN Number
Refill branch of the refill plan	The Refill Item Class **`Calling Card`** is assigned to the Usage Refill component. The branch contains a linear function component to compute the recharge.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of the resulting transactions (refill records) that the system generates.

Description: Type the business description of the component (optional).

Definition (Tab)

> **Note**
>
> If SAP Convergent Charging is connected to SAP ERP/FI-CA from SAP Business Suite, you must add two Usage Refill components in the refill logic, a component for the refill requests and another one for the reverse refills.
>
> You must respectively set up the *Usage Identifier* option of these components to **`_refill_`** and to **`_reverse_refill_`**.

Refill Item Class: Provides the list of refill item classes. Select a refill item class. A refill item class consists of default and user properties that describe a refill event also called a refill item coming from an external system such as SAP CC or a Voucher Management System. The properties of the refill item class are used to perform a refill plan through the refill logic.

Usage Identifier: Type a unique usage identifier to identify each usage refill used in the refill logic. The SAP CC system uses the usage identifier to perform the related branch of the refill logic. To open and modify the refill item class, click  .

1.42.8.1.2 Recurring Refill

 The **Recurring Refill** component creates in the refill tree a main branch that computes refills on a periodic basis. This component is characterized by an activation rule expressed by a:

- Frequency (day, week, month, year)
- Starting date and hour

Rules

The rules are the following:

- The refill logic can contain several recurring refills.

- When you insert a *Recurring Refill* in the refill logic, some properties are available in the Recurring Refill branch:
  - *Charge Effective Date*: Start date of the related provider contract item
  - *Recurring Event Date*: Date on which the *Recurring Refill* is activated
  - *Reference Start for Recurring Period*: The start date of the reference period related to a recurring refill
  - *Reference Stop for Recurring Period*: The stop date of the reference period related to a recurring refill
  - *Recurring Period Start*: The period start related to a recurring refill. It may be different from the "Recurring Period Reference Start" and can be used to compute the prorator factor.
  - *Recurring Period Stop*: The period stop related to a recurring refill. It may be different from the "Recurring Period Reference Stop" and can be used to compute the prorator factor.

Use

You use the *Recurring Refill* component to refill a prepaid account depending on periodic usage events.

> ### Example
>
> You may use a *Recurring Refill* component to set up automatic refills so that a service is automatically refilled when the balance of an account reaches a defined threshold or when a recharge from a bank account has been scheduled monthly.

Identification area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Allows you to type the description of the component (optional).

General tab

This tab defines the frequency and the starting date for triggering the *Recurring Refill*.

> ### Note
>
> The *Reference Date* is used only if you select the options *The same day as the reference date*, *The same day as the reference date or first day of next month* or *The same time as the reference date*. If you select another option, the *Reference Date* is ignored.

Reference Date: Provides the list of dates for triggering the *Recurring Refill* component. Select a date that you can combine with the parameters chosen in the *Periodicity* area to trigger the *Recurring Refill* component.

Calendar: Provides the list of available calendars on which the recurring event is based.

> ⚠ Caution
>
> Once the calendar is set, SAP SE highly recommends not to change it. A change can cause inconsistencies.

Periodicity: Provides the list of frequencies:

- For *Gregorian Calendar*, the available periodicities are the following:
  - *Every*, *Every other*, *Every 3rd*, and other items associated with the following list: *day*, *week*, *month*, *year*. For example, you can trigger the Recurring rate Every 3nd month on Tuesday. The different options to select the starting date are the following:
    - *The same day as the reference date*: If you click this option, the *Recurring Refill* is triggered depending on the date selected in the *Reference Date* list. See the note below.
    - *The same day as the reference date or first day of next month*: If you click this option, the *Recurring Refill* is triggered depending on the date selected in the *Reference Date* list or the first day of the month following that of the selected date in the *Reference Date* list. See the note below.
    - *the*: If you have selected the frequency *Every* (3rd, 4th, and so on) *month* or *year*, this option triggers the *Recurring Refill* depending on the day of the month that you select. This option is combined with the list of numbers (from 1 to 31), the list of items such as Last, First, Second, Third and Fourth and the list of weekdays and months. Examples: the *Recurring Refill* is triggered Every 5th month The 11 of the month; the *Recurring Refill* is triggered Every year The Second Sunday of January.
    - *on*: If you have selected the frequency *Every* (3rd, 4th, and so on) *week* or *year*, this option triggers the *Recurring Refill* depending on the day of the week or the month that you select. Examples: the *Recurring Refill* is triggered Every 4th week on Wednesday; the *Recurring Refill* is triggered Every year on the 1st of January
    - *at*: Allows you to select the starting hour if you have selected any of the previous options. This option includes hours and minutes combined with AM (ante meridiem) or PM (post meridiem). For example: the Recurring Refill is triggered Every day at 07:25 PM. You can select the *same time as the reference date* that is the date selected in the *Reference Date* list.

> ⓘ Note
>
> About *The same day as the reference date* and *The same day as the reference date or first day of next month* options.
>
> If you select both of the *Every month* and *The same day as the reference date* options, the *Recurring Refill* will be triggered every month on the *Reference Date*; however, if you select the 31st of the month as the *Reference Date*, the *Recurring Refill* will be triggered the 31st of every odd month but the 30th of every even month (or the 28th of February, for example, if February has 28 days).
>
> If you select both of the *Every month* and *The same day as the reference date or first day of next month* options, the *Recurring Refill* will be triggered every month on the *Reference Date* or the first day of the next month depending on how many days the month has. For example, if you select the 31st of the month as the *Reference Date*, the *Recurring Refill* will be triggered the 31st of every odd month but the 1st of the month following every even month; however, as February can have 28 or 29 days, the *Recurring Refill* will always be triggered the 1st of March if the *Reference Date* is the 31st of the month.
>
> The table below shows you the dates on which the *Recurring Refill* will be triggered if you select the 31st of March as the *Reference Date*:

In the month of	The recurring refill will be triggered
January	The 31st of January
February	The 1st of March
March	The 31st of March
April	The 1st of May
May	The 31st of May
June	The 1st of July
July	The 31st of July
August	The 31st of August
September	The 1st of October
October	The 31st of October
November	The 1st of December
December	The 31st of December

- For *4-4-5 Calendar*, the available periodicities are the following:
  - *Every month*, *Every quarter*, *Every semester* and *Every year*.

> **Note**
>
> The other options, available with the Gregorian Calendar, do not appear any longer. They are useless with the 4-4-5 Calendar.

1.42.8.1.3 One-Shot Refill


The **One-Shot Refill** component creates in the decision tree a main branch that computes refills only when precise events related to provider contracts and contract items occur. There are three types of events:

- The **activation (creation)** of a contract item in a provider contract
- The **suspension** of a contract item
- The **resumption** of a contract item

Rules

The rules are the following:

- The refill logic can contain several One-Shot Refill components. For example, a One-Shot Refill for each type of event such as *Activation*, *Suspension*, and *Resumption*.

- When you insert a One-Shot Refill in the refill logic, the *Charge Effective Date* is available in the *One-Shot Refill* branch.

Use

You can use the One-Shot Refill component to compute refills (bonuses, free SMS, and so on) linked to the activation of a contract.

> ### Example
>
> In TelcoServ, the end customers who have subscribed to a mobile phone service receive bonuses when their contract is activated.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Event Name: Provides the list of events triggering the One-Shot Refill component. Select one of the following events:

- *Activation*
- *Suspension*
- *Resumption*

1.42.8.1.4 Account Event Refill


The **Account Event Refill** component creates in the refill tree a main branch that computes refills only when the prepaid balance of a subscriber account goes under a specified threshold.

Contrary to the *Recurring refill* component that is triggered by a scheduler at a later time, the *Account Event Refill* component is triggered immediately after a usage event when the related balance goes under a specified threshold.

Rules

The rules are the following:

- The refill logic can contain several Account Event Refill components. For example, several Account Event Refill components are needed to refill an account automatically with amounts that depend on different thresholds.
- When you insert an Account Event Refill component in the refill logic, you can set the threshold by using a fixed amount or a numeric parameter defined at charge component level.
- Every Account Event Refill component introduces the `Threshold Amount` and `Account Event Date` properties for the *Account Event Refill* branch.

Use

You use the *Account Event Refill* component to refill a prepaid account automatically when its balance goes under a specified threshold.

> ✤ Example
>
> You can use the *Account Event Refill* component to execute a 10$ automatic refill on a prepaid account when its balance goes under a 12$ threshold.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

Balance falls under threshold: Type the amount under which an immediate refill is triggered. The threshold amount can be a numerical value or a numeric property inherited from the parent charge component.

1.42.9 Logic Components in the Allowance Logic

Each main branch of the allowance logic [page 449] begins with a trigger component [page 716] (event-based, recurring, or one-shot) linked to an allowance activation method.

To create an allowance tree, you must:

1. Select a trigger component [page 716].
2. Add components from other families according to your allowance plan requirements.
3. End each branch of the tree with a function component [page 727].

Component Families for the Allowance Logic

In SAP Convergent Charging, logic components are grouped into families according to their functional usage. Each component is represented by a specific icon. In the user interface, these families are represented by a specific color and an icon, and they are visible in the toolbar of the *Allowance* window. A logic component is of the same color as the family which it belongs to.

The following table presents the component families available in the allowance logic:

Family Icon	Name of the Family	Function of the Component Family	Location of the Component Family
	Triggers [page 716]	Defines the trigger type (Event-Based, Recurring, or One-Shot)	Always at the root of the allowance tree. **Note** Unavailable in a pricing macro.
	Functions [page 727]	Refer to the basic components [page 726] (only the *No Access* and *Allow* components)	Refer to the basic components [page 726]
	Comparators [page 737]	Refer to the basic components [page 726]	Refer to the basic components [page 726]
	Operators [page 773]	Refer to the basic components [page 726]	Refer to the basic components [page 726]

Related Information

Logic Components for a Price Plan [page 688]

1.42.9.1 Trigger Components

Trigger components are always found at the root of the allowance tree of the allowance logic [page 449]. The different types of trigger components are available depending on the activation type.

The following table shows the different types and the relevant activation:

Type of Trigger Component	Triggering (Activation)	Example
Event-Based	Each allowance event	A new allowance is created for a carry-over, or an allowance is extended for a stacking.
Recurring	Periodic	The allowance counter is reset at the beginning of each month.
One-Shot	Once only	The initial quantity is reported at the allowance activation, or the remaining quantity is reported at the allowance expiration.

Trigger Components in SAP CC

The following trigger components are available in SAP Convergent Charging:

Icon	Name	Action/Use
	Event-Based Trigger [page 717]	Activates and executes a branch when an allowance event triggers the logic component.
	Recurring Trigger [page 720]	Activates and executes a branch when a periodic event triggers the logic component.
	One-Shot Trigger [page 723]	Activates and executes a branch when a one-shot event triggers the logic component.

Related Information

Logic Components in the Allowance Logic [page 715]

1.42.9.1.1 Event-Based Trigger

 The **Event-Based Trigger** component creates in the decision tree a main branch that is executed at runtime each time an *allowance event* is sent by the SAP system to contract allowances in customer master data through an Allowance Event Sender (Operator) [page 777] component.

The configuration of an Event-Based Trigger component requires the selection of an allowance event class [page 441] that specifies the type of the allowance event.

Rules

The rules are the following:

- The allowance logic [page 449] can contain several *Event-Based Trigger* components but only one per allowance event class.
- You must at least assign an allowance event class to an Event-Based Trigger component.
- If an allowance event class is assigned to an Event-Based Trigger component, the user-defined properties of this allowance event class are available only in the branch that uses this Event-Based Trigger component.
- When you insert an Event-Based Trigger component in the allowance logic, some properties (see below) are available in the *Event-Based Trigger* branch.
  These properties are the following:

Property	Description
Event Date	The date/time of the allowance event
Validity Start Date	The start date/time of the validity period of the allowance
Validity End Date	The end date/time of the validity period of the allowance
Allowance Unique Identifier	The unique identifier of the allowance
Currency Code	The currency code of the allowance
ISO Currency Code	The currency code of the allowance in the ISO format

Use

As a pricing specialist, you use the Event-Based Trigger component to configure the interactions with allowances created in several decision trees: Price Plans [page 299], Refill Logic [page 395], and Allowance Logic [page 449].

Example

Example with an event update

As an operator, you create an offer for your end customers so that they can add recharges to a base plan. Each time an end customer buys a recharge, the new recharge (or the new allowance) is created, and the current allowance is extended to the expiration date of the new one.

Furthermore, you want to declare the revenues related to allowances by reporting the exact revenue at the creation, the usage, and the expiration of the allowances.

You can then create in the allowance logic of your allowance plan:

- An *Event-Based Trigger* branch to report the revenues related to the usage of allowances
- A One-Shot Trigger branch to report the revenues related to the creation of allowances
- A One-Shot Trigger branch to report the revenues related to the expiration of allowances

Each time the allowance is used with an Allowance Event Sender (Operator) [page 777] component in the price plan of a charge, the Event-Based Trigger component computes the properties to report or updates

the allowance event. The Event-Based Trigger branch needs a Property Introducer (Operator) [page 808] component to define the reason of revenue recognition. This Event-Based Trigger branch enables the generation of a charged item.

Example

Example with a validity period

As an operator, you create an offer for your customers so that they can add recharges to a base plan. Each time the customer buys a recharge, the new recharge (or the new allowance) is created, and the current allowance is extended to the expiration date of the new one.

This offer contains an allowance plan, and its allowance logic includes an Event-Based Trigger component whose branch checks whether the validity period of the allowance can be extended or not. This branch contains a Numbers Comparator [page 747] component and an Allowance Validity Period Updater (Operator) [page 787] component.

The *Numbers Comparator* component checks that:

- The allowance start date/time is before the start date/time of the new allowance
- The allowance end date/time is after the start date/time of the new allowance
- The allowance end date/time is before the end date/time of the new allowance

If the previous rules are true, the *Allowance Validity Period Updater Operator* component can extend the validity period of the allowance.

Note

The *Event-Based Trigger* component needs an allowance event class [page 441] with related properties describing the event.

In this example, the allowance event class can be named **`Carryover`** and contains the validity period (start date/time and end date/time) of the new allowance.

Important: The name of the allowance event class determines the type of the event.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Allowance Event Class: Provides the list of allowance event classes. Select an allowance event class. If necessary, click  to open the selected allowance event class.

Related Information

Allowance Event Sender (Operator) [page 777]

1.42.9.1.2 Recurring Trigger

 The **Recurring Trigger** component creates in the decision tree a main branch that is activated on a periodic basis. This component is characterized by an activation rule expressed by a:

- Frequency (day, week, month, year)
- Starting date and time

Rules

The rules are the following:

- The *Recurring Trigger* component is activated only during the validity period of the allowance.
- The Recurring Trigger component is executed at the beginning of the recurring periods (for example, if the recurring trigger period starts from the first day to the last day of the month, the recurring trigger is executed the first of the month), except for the first recurring period where it is executed at the activation date of the allowance.
- The allowance logic can contain several Recurring Trigger components.
- When you insert a Recurring Trigger component in the refill logic, some properties are available in the *Recurring Trigger* branch:

Property	Description
Event Date	The date and time of the recurring event
Validity Start Date	The start date/time of the validity period of the allowance
Validity End Date	The end date/time of the validity period of the allowance
Allowance Unique Identifier	The unique identifier of the allowance
Currency Code	The currency code of the allowance

Property	Description
ISO Currency Code	The currency code of the allowance in the ISO format
Reference Start for Recurring Period	The start date/time of the reference period related to a recurring trigger
Reference Stop for Recurring Period	The stop date/time of the reference period related to a recurring trigger
Recurring Period Start	The period start related to a recurring trigger. It may be different from the *Recurring Period Reference Start* setting and can be used to compute the prorator factor.
Recurring Period Stop	The period stop related to a recurring trigger. It may be different from the *Recurring Period Reference Stop* setting and can be used to compute the prorator factor.

Use

As a pricing specialist, you use the Recurring Trigger component to configure the reset of contract allowances [page 575] on a periodic basis.

> ⚠ Restriction
>
> The *Recurring Trigger* component is forbidden within a **shareable allowance logic**.

> ✤ Example
>
> As an operator, your company proposes a recharge providing 6GB 4G data for a period of 90 days to your new end customers. At the end of the period, the allowances are reset.
>
> As a pricing specialist, you model, design, and configure some allowance logic that contains a *Recurring Trigger* component. It is dedicated to the allowance reset.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Type the business description of the component (optional).

General (Tab)

This tab defines the frequency and the starting date for triggering the recurring trigger.

> **Note**
>
> The *Reference Date* is used only if you select the options:
>
> - *The same day as the reference date*
> - *The same day as the reference date or first day of next month*
> - Or *The same time as the reference date*
>
> If you select another option, the *Reference Date* is ignored.

Reference Date: Provides the list of dates/times for triggering the *Recurring Trigger* component at runtime.

Select a date that you can combine with the parameters chosen in the *Periodicity* area to trigger the *Recurring Trigger* component.

Calendar: Provides the list of available calendars which the recurring event is based on.

> **Caution**
>
> Once the calendar is set, SAP SE highly recommends not to change it. A change can cause inconsistencies.

Periodicity: Provides the list of frequencies:

- For *Gregorian Calendar*, the available periodicities are the following:
  - *Every*, *Every other*, *Every 3rd*, and other items associated with the following list: *day*, *week*, *month*, *year*. For example, you can trigger the Recurring rate Every 3nd month on Tuesday. The different options to select the starting date are the following:
    - *The same day as the reference date*: If you click this option, the *Recurring Trigger* is triggered depending on the date selected in the *Reference Date* list. See the note below.
    - *The same day as the reference date or first day of next month*: If you click this option, the *Recurring Trigger* is triggered depending on the date selected in the *Reference Date* list or the first day of the month following that of the selected date in the *Reference Date* list. See the note below.
    - *the*: If you have selected the frequency *Every* (3rd, 4th, and so on) *month* or *year*, this option triggers the *Recurring Trigger* depending on the day of the month that you select. This option is combined with the list of numbers (from 1 to 31), the list of items such as Last, First, Second, Third and Fourth and the list of weekdays and months. Examples: the *Recurring Trigger* is triggered Every 5th month The 11 of the month; the *Recurring Trigger* is triggered Every year The Second Sunday of January.
    - *on*: If you have selected the frequency *Every* (3rd, 4th, and so on) *week* or *year*, this option triggers the *Recurring Trigger* depending on the day of the week or the month that you select. Examples: the *Recurring Trigger* is triggered Every 4th week on Wednesday; the *Recurring Trigger* is triggered Every year on the 1st of January
    - *at*: Allows you to select the starting hour if you have selected any of the previous options. This option includes hours and minutes combined with AM (ante meridiem) or PM (post meridiem). For example: the Recurring Trigger is triggered Every day at 07:25 PM. You can select the *same time as the reference date* that is the date selected in the *Reference Date* list.

> **Note**
>
> About *The same day as the reference date* and *The same day as the reference date or first day of next month* options.

If you select both of the *Every month* and *The same day as the reference date* options, the *Recurring Trigger* will be triggered every month on the *Reference Date*; however, if you select the 31st of the month as the *Reference Date*, the *Recurring Trigger* will be triggered the 31st of every odd month but the 30th of every even month (or the 28th of February, for example, if February has 28 days).

If you select both of the *Every month* and *The same day as the reference date or first day of next month* options, the *Recurring Trigger* will be triggered every month on the *Reference Date* or the first day of the next month depending on how many days the month has. For example, if you select the 31st of the month as the *Reference Date*, the *Recurring Trigger* will be triggered the 31st of every odd month but the 1st of the month following every even month; however, as February can have 28 or 29 days, the *Recurring Trigger* will always be triggered the 1st of March if the *Reference Date* is the 31st of the month.

The table below shows you the dates on which the *Recurring Trigger* will be triggered if you select the 31st of March as the *Reference Date*:

In the month of	The recurring trigger will be triggered
January	The 31st of January
February	The 1st of March
March	The 31st of March
April	The 1st of May
May	The 31st of May
June	The 1st of July
July	The 31st of July
August	The 31st of August
September	The 1st of October
October	The 31st of October
November	The 1st of December
December	The 31st of December

- For *4-4-5 Calendar*, the available periodicities are the following:
  - *Every month*, *Every quarter*, *Every semester* and *Every year*.

> **Note**
>
> The other options, available with the Gregorian Calendar, do not appear any longer. They are useless with the 4-4-5 Calendar.

1.42.9.1.3 One-Shot Trigger



The **One-Shot Trigger component** creates in the decision tree a main branch that is activated when precise events related to the allowance occur:

- The **creation of the allowance** in a provider contract
- The **activation** of the allowance
- The **expiration** of the allowance

In addition, you can activate the One-Shot Trigger component within the allowance validity period (after the activation of the allowance or before the expiration of the allowance).

Rules

The rules are the following:

- The allowance logic can contain several *One-Shot Trigger* components. For example, you can set up a one-shot trigger for each type of event (allowance creation, activation, and expiration).
- When you insert a one-shot trigger in the allowance logic, some properties are available in the *One-Shot Trigger* branch. These properties are the following:

Property	Description
Event Date	The date and time of the one-shot event
Validity Start Date	The start date/time of the validity period of the allowance
Validity End Date	The end date/time of the validity period of the allowance
Allowance Unique Identifier	The unique identifier of the allowance
Currency Code	The currency code of the allowance
ISO Currency Code	The currency code of the allowance in the ISO format

Use

As a pricing specialist, you use the *One-Shot Trigger* component to configure the recognition for the revenues of allowances.

⚠ Restriction

The *One-Shot Trigger* component is forbidden for an allowance activation within a **shareable allowance logic**.

✤ Example

As an operator, you create an offer for your end customers so that they can add recharges to a base plan. Each time the end customer buys a recharge, the new recharge (or the new allowance) is created, and the current allowance is extended to the expiration date of the new one.

Furthermore, you want to declare the revenues related to allowances by reporting the exact revenue at the creation, the activation, or the expiration of the allowances. You can then create in the allowance logic of your allowance plan:

- An *Event-Based Trigger* branch to report the revenues related to the event of allowances.
- Three *One-Shot Trigger* branches to report the revenues related to the creation (first branch), the activation (second branch), and the expiration of allowances (third branch) in contracts.

The *One-Shot Trigger* components can check and report the lost credit of end customers when allowances expire. Each One-Shot Trigger branch needs a Property Introducer (Operator) [page 808] component to define the reason of revenue recognition. These *One-Shot Trigger* branches enable the generation of a charged item by the system.

Identification Area

Name: Type the name of the component (mandatory). Choose an appropriate name because it is reused as the label of pricing transactions.

Description: Type the business description of the component (optional).

Definition (Tab)

Event Name: Provides the list of events activating the One-Shot Trigger component. Select one of the following allowance events:

- *Creation*
- *Activation*
- *Expiration*

Shifted By: This option allows activating the One-Shot Trigger component after the activation of the allowance or before the expiration of the allowance. The time shift is defined as a quantity and a unit of time (second, minute, hour, and day).

By default, the time shift is 0.

Select the quantity and the time of unit depending on the selected *Event Name*:

- *Creation*: No shift is possible.
- *Activation*: Only a positive shift is possible.
- *Expiration*: Only a negative shift is possible.

Note

- The shifted One-Shot Trigger components are always activated before the Recurring Trigger [page 720] components.
- The shifted One-Shot Trigger components are activated only if their computed date is included in the validity period of the allowance.
- A shifted One-Shot Trigger component can be activated several times if the expiration date of the allowance is postponed so that the date activating the One-Shot Trigger component is not only included in the validity period of the allowance but is also greater than the last activation date of the allowance.

1.42.10 Basic Components

The basic components are used in the trees and branches of the following:

- Rate Components [page 690]
- Charging Components [page 701]
- Refill Components [page 707]
- Trigger Components [page 716]
- Pricing Macros [page 121]

> **Note**
>
> The matrix of components [page 676] shows which basic components you can use in the master data objects (price plans, charging plans, pricing macros, refill plans, and allowance plans).

Component Families

In SAP Convergent Charging, logic components are grouped into families according to their functional usage. Each logic component is represented by a specific icon. In the user interface, these families are represented by a specific color and an icon, and they are visible in the toolbar of the *Charge*, *Pricing Macro*, *Refill*, or *Allowance* window. A logic component is of the same color as the family which it belongs to. The following table presents these component families:

Family icon	Name of the family	Function of the component family	Location of the component family
	Functions [page 727]	Calculates the final amount.	Always at the end of the branches. Functions are the leaves of the tree.
	Comparators [page 737]	• Generates two or more branches to evaluate properties. • Selects one branch according to the result.	Middle area of the decision tree: • After rates, charging, refills, triggers • Before functions
	Splitters [page 760]	• Splits a number or duration. • Generates and rates multiple branches.	Middle area of the decision tree: • After rates and refills • Before functions

Family icon	Name of the family	Function of the component family	Location of the component family
	Operators [page 773]	Adds properties to the calculation context.	Middle area of the decision tree: • After rates, charging, refills, triggers • Before functions **Note** The Update Counter Operator [page 826] component is always located after a Function [page 727] component.

1.42.10.1 Function Components

A function component calculates an amount from context properties. Function components make up the leaves of the decision tree and are used for calculations.

The *Flat Function* component, which calculates a fixed amount, is one of the simplest. The *Linear Function* component calculates an amount by using a numerical property. For example, the *Linear Function* component calculates the price of a telephone call that is charged 0.34 cents per minute.

Function Components

The following function components are available with SAP Convergent Charging:

Icon	Name	Action/Usage
	Allow [page 728]	Indicates that no additional logic is needed.
	Flat [page 729]	Provides a priced amount based on a constant value or a numerical property expressed in the currency of the plan.
	Free [page 730]	Represents a free service in the decision tree.

Icon	Name	Action/Usage
	Generic [page 731]	Computes an amount P with one of these three formulae: P = A.X + B P = A.X.Y + B P = A.X + B.Y + C
	Linear [page 733]	Computes an amount P with the formula P = A.X + B
	Macro [page 734]	Allows to use an existing pricing macro to calculate a final amount.
	No Access [page 735]	Indicates an unauthorized use of a service, an invalid access, or an unexpected case when usage events are processed.
	Polynomial [page 736]	Calculates an amount by computing the sum of n number of terms in the form of A.X.Y such as: A1.X1.Y1 + A2.X2.Y2 + ... + An.Xn.Yn

1.42.10.1.1 Allow Function

 The **Allow Function** component indicates that no additional logic is needed.

Use

As a pricing specialist, you use the *Allow Function* component to terminate each Trigger [page 716] component (event-based, recurring, and one-shot) that you configured in your allowance logic.

> ### Example
>
> You use a One-Shot Trigger [page 723] component triggered by a creation event to calculate the validity period of the allowance through an Allowance Validity Period Updater (Operator) [page 787] component. Because you do not need to keep track of this event, you clear the *Generate Charged Item* option.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

Generate Charged Item: If selected, a charged item is created by the SAP CC system at runtime.

> → Remember
>
> An Output Item [page 479] has to be defined in the Allowance Plans [page 466] in order to have a charged item creation. The charged item content will follow the output item configuration. If there is no output item setting, then the charged item will not be created (even if the property *Generate Charged Item* is selected).

Related Information

Trigger Components [page 716]

1.42.10.1.2 Flat Function


The *Flat Function* component provides a constant amount based on a constant value or a numerical property expressed in the currency of the plan.

Use

In a price plan, the *Flat Function* component allows you to assign a constant price to a consumed service. This constant price is represented by a value or a numerical property and is expressed in the currency of the plan.

> ⓘ Note
>
> The usage of the *Flat Function* component with a zero amount generates transactions while the Free Function [page 730] component never generates any transactions.

In a pricing macro, the component returns a priced amount with no currency.

> ✤ Example
>
> You propose an Internet subscription that consists of a monthly connection plan for $15 per month. Using a Flat Function component, you can bill this subscription by assigning the value of $15 to the monthly subscription price.
>
> Instead of indicating a numerical value, you can assign a price by using a numerical property. For example, if the subscription corresponds to several plans whose price varies according to the number of hours, you

can create a **`fixed price`** parameter that takes different values according to the number of hours. Using the Flat Function component, you assign the `fixed price` to the monthly subscription price.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the description of the component (optional).

Definition (Tab)

Fixed Amount: Provides the list of numerical properties or allows you to type a numerical constant value. Select a numerical property or select *Numerical Value*, and then type a numerical constant value.

On the right, a code is displayed and expresses:

- The currency defined in the charge component (for example, USD for United States Dollar)
- Any currency (symbol "¤") when the flat function is used in a pricing macro

1.42.10.1.3 Free Function


The *Free Function* component represents a free service in the decision tree. It neither calculates any amount nor creates any transactions (charged items) for billing systems. It must be used only if you do not need to produce a transaction. Because no transaction is created, the branch including the *Free Function* component is not rated at runtime.

Use

The *Free Function* component is used only when:

- The consumed service is free.
- It is not necessary to keep a record of the transaction.

> ⚠ Caution
>
> When the related price plan uses a *Free Function* component, no accounting transactions are created; therefore, you cannot rerate transactions. To create transactions, use the Flat Function [page 729] component with a zero amount instead of the *Free Function* component.

Example

You propose a special offer to try out an Internet connection for one month free of charge. At the end of one month, this offer becomes invalid. If the customer wants to continue the service, you propose a different offer. Because you do not need keep a record of connections during the limited free offer period, you can use a *Free Function* component.

Caution

SAP SE recommends that you respect the following rules:

- Use the *Free Function* component in a Recurring Rate [page 693], in a Recurring Refill [page 710], in a One-Shot Rate [page 698], or in a One-Shot Refill [page 713].
- Do not use the *Free Function* component in a Usage Rate [page 690] or in a Usage Refill [page 708], but use a Flat Function [page 729] component with a zero amount to set up a free service and create transactions. The service is free, but a transaction is created and can therefore be saved.
- Do not use a *Free Function* component in a master charge [page 293] linked to dependent charges [page 294] because this component stops the rating of dependent charges.

Identification Area

Name: Type the name of the logic component (mandatory).

Description: Type the description of the component (optional).

Definition (Tab)

The *Definition* tab is empty because the *Free Function* component need not be set up.

1.42.10.1.4 Generic Function


The *Generic Function* component computes an amount P with one of these three formulas:

- P = A.X + B
- P = A.X.Y + B
- P = A.X + B.Y + C

Where:

- A, B and C are numerical values expressed in the currency of the plan
- X and Y are mathematical expressions which combine a numerical property, an operator, and a numerical value. For example, X = Duration multiplied by 5, or Y = Duration plus 10, in which Duration is a numerical property.

You can apply the following operations to the numerical properties:

- Addition
- Subtraction
- Multiplication
- Division

Use

You use a *Generic Function* component when you want to calculate a price that uses numerical properties.

> **Example**
>
> In a chargeable item class called **`Video Conference`**, you charge the customer $0.10 per videoconference minute. This price is multiplied by the number of members, and then added to a fixed amount of $15. To compute the price of a **`Video Conference`**, you use a *Generic Function* component according to the formula AXY + B, in which A, X, Y, and B have the following values:

A	X	Y	B
$0.10	Call duration property (in minutes)	Number of members property	$15

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Type: Click one of the following formulas:

- *AX + B*: If this option is selected, the *Set the Y parameters* area and the *Money Amount C* box are unavailable
- *AXY + B*: If this option is selected, the *Money Amount C* box is unavailable
- *AX + BY + C*: If this option is selected, all is available

Money Amount A: Type the value that represents A in the selected formula. This box is required: Type 1 if A = 1. The displayed currency is the currency of the plan.

Money Amount B: Type the value that represents B in the selected formula. This box is required: Type 0 if B = 0. The displayed currency is the currency of the plan.

Money Amount C: Type the value that represents C in the formula AX + BY + C. This box is required: Type 0 if C = 0. The displayed currency is the currency of the plan.

Set the X parameters: Allows you to define the calculation of X in the selected formula. This area is made up of three boxes. From left to right:

- First box: Provides the list of numerical properties. Select a numerical property.
- Second box: Provides the list of operators to be applied to the numerical property that you have selected. Select an operator. Selecting the *-Operator-* item in the list allows you to not apply an operation to the numerical property.
- Third box: Once you have selected a property and an operator, type a numerical value. This box is required if you select an operator in the second box.

Set the Y parameters: Allows you to define the calculation of Y in the formulas AXY + B or AX + BY + C. Apply the same rules as in the Set the X parameters area.

1.42.10.1.5 Linear Function


The *Linear Function* component computes an amount P with the formula P = A.X + B where:

- A and B are numerical values or numerical properties
- X is a numerical property

Use

You can use the *Linear Function* when you want to calculate a price with numerical properties of the plan.

> Example
>
> In a chargeable item class called **`Telephone Services`**, you charge the customer $0.10 per call minute. To compute the price of a telephone call, you use a *Linear Function* component with the following values:

A	X	B
$0.10	Call duration property (in minutes)	0

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of numerical properties. Select the numerical property that represents X in the formula A.X + B.

Scaled Amount: Type the value that represents A in the formula A.X + B. This box is required: Type 1 if A = 1. The displayed currency is the currency of the plan.

Fixed Amount: Type the value that represents B in the formula A.X + B. This box is required: Type 0 if B = 0. The displayed currency is the currency of the plan.

1.42.10.1.6 Macro Function

 The *Macro Function* component allows you to use an existing pricing macro to calculate a final amount. Because a pricing macro cannot be inserted in a decision tree, you must use a *Macro Function*:

- To select a pricing macro
- To provide values to the pricing macro

The *Macro Function* uses the properties of the plan to provide the values to the pricing macro.

Use

You use the *Macro Function* to integrate an existing pricing macro at the end of a decision tree branch. The amount calculated in the pricing macro is the final amount of the branch.

> ### Example
>
> In a chargeable item class called **`Communication`**, you charge the customer according to the type of communication. The pricing rule is described in a unique Pricing Macro that completely calculates the final price. You can use the Macro Function component to integrate this Pricing Macro into each price plan that relates to the **`Communication`** chargeable item class.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Pricing Macro: Provides the list of available pricing macros. Select a pricing macro.

Mapping of Rating Context: Displays the rating context properties of the pricing macro and allows you to assign a property of the plan to each rating context property. The pricing macro runs with the values of the plan properties.

- *Name*: Displays the name of the rating context property.
- *Description*: Displays the description of the rating context property.
- *Mapping Property*: Provides the list of plan properties. Select the property that initializes the rating context property of the pricing macro.

1.42.10.1.7 No Access Function

 The *No Access Function* component indicates an unauthorized use of a service, an invalid access, or an unexpected case when usage events are processed. When it is inserted in a branch, the No Access Function rejects the usage tickets and returns an information message to the process.

Use

The *No Access Function* allows you to control certain access rights such as:

- Exceeding the consumption limit
- Prohibiting international telephone calls

> ⓘ Note
>
> The *No Access Function*:
>
> - Can be used only in Usage Rate-type branches
> - Does not calculate prices and cancels counter updates
> - Appears in the decision tree with the default Name “Access Denied”

> ✤ Example
>
> In a chargeable item class called **`Telephone Service`**, every consumption is billed according to the dialing codes of the called number. The dialing codes are tested in the price plan which includes a *No Access* component in a branch. When no value corresponds to a dialing code of the called number, the *No Access* component is activated.

> → Recommendation
>
> SAP SE recommends that you respect the following rules:
>
> - Use the *No Access* component only in a Usage Rate-type branch
> - Fill in the information message field, and choose properties so that you can easily locate and correct the unexpected incident

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Message: Allows you to type an information message to locate an incident.

Properties: Provides the list of properties. When the incident occurs, the values of the properties are displayed below the message. Select the properties which you want to know the values when the incident occurs. Clear the boxes of the properties that you do not need.

Trigger Best Effort Reservation: Clear this check box if you do not want that the mechanism of best effort reservation computes and reserves the maximum of an available service quota. This option is not available when a No Access Function component is inserted in the refill logic.

1.42.10.1.8 Polynomial Function


The *Polynomial Function* component calculates an amount by computing the sum of n numbers of terms in the form of AXY such as: A1X1Y1 + A2X2Y2 + ... + AnXnYn where:

- A and n are numerical values
- X and Y are numerical properties

You can select different numerical properties for each AXY term or reuse the same properties several times.

Use

The *Polynomial Function* is used when an amount is calculated with two or more numerical properties. It allows you to multiply numerical properties two by two for each term, and then add up the results.

Example

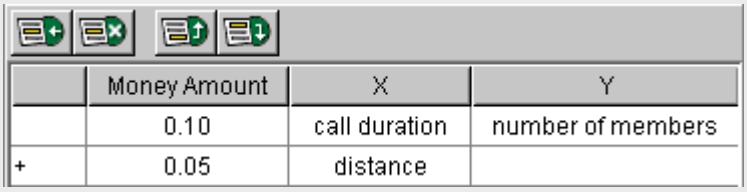
In a chargeable item class called **`Video Conference`**, you charge the customer $0.10 per videoconference minute. This price is multiplied by the number of members, and then added to a price that varies according to the distance ($0.05 cents per kilometer). To compute the price of a **`Video Conference`**, you use a *Polynomial Function* component with the following values:

Term of the polynomial sum	A	X	Y
1	$0.10	call duration (in minutes)	number of members
2	$0.05	distance (in kilometers)	None

The polynomial sum is made up of two terms and equal to:

- 0.10 * call duration * number of members + 0.05 * distance (with * as the multiplier sign)

Using the table in the *Definition* tab, you can express this polynomial expression as follows:



	Money Amount	X	Y
	0.10	call duration	number of members
+	0.05	distance	

Definition tab

This tab includes a table that allows you to calculate a polynomial expression in the form of AXY. Each row in the table represents one of the terms AXY of the polynomial expression. For example, if the polynomial expression to calculate is A1X1Y1 + A2X2Y2 + A3X3Y3, you must add three rows to the table.

> **Note**
>
> The most left column does not contain entries to be filled in. This column displays the sign "+" so that you can better visualize the polynomial formula.

 : Adds a row to the table. Each row in the table allows you to define a term AXY in the polynomial formula.

 : Deletes a row from the table and causes the deletion of a term from the polynomial formula.

 : Moves a row up in the table to change the order of the terms.

 : Moves a row down in the table to changes the order of the terms.

Money Amount: Allows you to type the value that represents A in the term AXY. The currency is that of the plan. This box is required.

X: Provides the list of numerical properties that represent X in the term AXY. This box is optional. If necessary, select a numerical property.

Y: Provides the list of numerical properties that represent Y in the term AXY. This box is optional. If necessary, select a numerical property.

1.42.10.2 Comparison Components

A comparator component compares properties of the same type. The comparator component can compare the properties of chargeable item classes, parameters, counters, or properties generated by another logic component. When you insert it in the decision tree, the comparator component creates one or more branches depending on the type of comparison.

Several types of comparator component are available depending on the type of the property used for comparison. For example, you can use:

- A Numerical Comparator component to compare the value of a numerical property with another value
- A Date Comparator component to compare a date property type with time slots

Comparator Components

The following comparator components are available with SAP Convergent Charging:

Icon	Name	Action/Usage
	Date / Time Slot [page 738]	Compares two date/time properties with five comparison operators.
	Dates [page 740]	Compares the value of a date/time property with one or several criteria.
	Interval Switching [page 742]	Compares a numerical property with intervals or values.
	Mapping Table Introducer [page 744]	Allows to look up for data in a mapping table (or subscriber mapping table) in a plan.
	Numbers [page 747]	Compares the value of a numerical property with a value or another numerical property.
	Prefix Switching [page 748]	Compares a string property with prefixes of string type.
	Public Holidays [page 750]	Compares a date property with one or more public holidays.
	Range Table Introducer [page 751]	Allows to look up for data in a range table (or subscriber range table) in a plan.
	Strings [page 758]	Compares the value of a string property with a value or with another string property.
	Switching [page 759]	Compares the state of a prepaid account with one or more predefined values (Active, Blocked, Locked, or Closed).

1.42.10.2.1 Date/Time Slot Comparator

 The *Date / Time Slot Comparator* compares a date property with one of the following time slots:

- Daily
- Weekly
- Monthly
- Yearly

The comparison is true when the date property is inside the selected time slot.

When you insert it in the decision tree, the *Date / Time Slot Comparator* component creates two separate branches:

- A branch that is activated when the comparison is true

- A branch that is activated when the comparison is false

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices which vary from one time period to another time period, you may need to compare a date with a time slot. To use the *Date / Time Slot Comparator* inside your price plan, do the following:

1. After you add the component, choose the date and the time slot.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> **Example**
>
> Within the context of a promotional offer that contains a calling plan, you want to implement a special price of $0.02 per minute for telephone calls that start during the weekend. The standard price is $0.04 per minute. The following table presents the information that you need to set up the Date / Time Slot component inside your price plan:

Data	Settings
Property date	Start date of the phone call
Time slot parameters	Type: Weekly. From Wednesday at 10:00 PM To Monday at 6:00 AM.
Pricing rules	Price that is applied to the "is in the time slot" branch: $0.02 per minute. Price that is applied to the "is out of the time slot" branch: $0.04 per minute.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of date properties. Select a date property.

Set the Time Slot parameters: Allows you to compare the date property with a time slot.

- *Time Slot Type*: Provides the list of time slots. Select one of the following time slots:
  - *daily*: Represents a part of the day or a whole day
  - *weekly*: Represents a part of the week or a whole week

    - *monthly*: Represents a part of the month or a whole month
    - *yearly*: Represents a part of the year or a whole year
- *From*: Allows you to select or type a start date or time (hour and minute) of the time slot.
- *at*: Allows you to select or type a time (hour and minute).
- *To*: Allows you to select or type an end date or time (hour and minute) of the time slot.

> ⓘ Note
>
> The following examples show different formats of time slots:
>
> - *daily*: *From* 05.00 AM *To* 07.00 AM
> - *weekly*: *From* Monday at 05.00 AM *To* Wednesday at 07.00 AM
> - *monthly*: *From* 10 *at* 05.00 AM *To* 12 *at* 07.00 AM
> - *yearly*: *From* 22 May *at* 05.00 AM *To* 18 June *at* 07.00 AM

1.42.10.2.2 Dates Comparator


The *Dates Comparator* component compares two date properties. Six comparison operators are available:

- Has the same
- Is equal to
- Is after
- Is strictly after
- Is before
- Is strictly before

The comparison is true when a date property matches a reference date or defined constraints.

When you insert it in the decision tree, the *Dates Comparator* component creates two separate branches:

- A branch that is activated when the comparison is true
- A branch that is activated when the comparison is false

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches, you may need to compare a date property with a reference date. To use the *Dates Comparator* component in the price plan, do the following:

1. After you add the component, choose a date property, a reference date or define constraints.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> ✤ Example
>
> In the context of a special "end of the year" promotional offer, a telephone operator wants to charge a special price ($0.15 per minute) for all the calls that begin on Sundays during the month of December.

All the other calls are rated at the regular price. The following table presents the information that the telephone operator needs to set up the *Dates Comparator* component inside the price plan:

Data	Settings
Property	`Call start date`
First constraint	Select *Sunday* as the *Day of week*
Second constraint	Select *December* as the *Month*
Pricing rules	Price that is applied to the *is* branch: $0.15 per minute. Price that is applied to the *is not* branch: usual price.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of date properties. Select the first date property.

Operator: Provides the list of operators. The operator *Has the same* is used with the option *Set the Constraints to Apply to This Date*. The other operators are used only to compare the first date property with a reference date. Select one of the following operators:

- *has the same*
- *is equal to*
- *is after*
- *is strictly after*
- *is before*
- *is strictly before*

Reference Date: Provides the list of date properties. Select the reference date that you want to compare with the first date property.

→ Remember

The option *Set the Constraints to Apply to This Date* allows you to compare:

- A date property with a reference date: You first select the date property and the reference date. You then use the set of constraints to select which part of the date property you want to compare with the reference date. For example, you select `agreement_creation_date` as the date property and `contract_date` as the reference date. The set of constraints allows you to select the different parts of the `agreement_creation_date` you compare with the `contract_date`: Day of week (contract_date's one), Day of month (contract_date's one), Month (contract_date's one), and so on.

- A date property with a set of defined constraints: You first select the date property, and you select*Choose a property* in the *Reference Date* box. You then define the constraints that you want to compare with the date property. For example, you select Day of week (Monday), Day of month (5), Month (January), and so on.

Set the Constraints to Apply to This Date: Allows you to set up the constraints on the date property. This option is available only if you have selected *Has the same* in the Operator list. You can select one or more criteria:

- *Day of week*: Allows you to compare the day of the date property with any day of the week or with the day of the reference date. Click this option, and then select the day in the list.
- *Day of month*: Allows you to compare the day of the date property with any day of the month or with the day of the month of the reference date. Click this option, and then select the day of month in the list.
- *Month*: Allows you to compare the month of the date property with any month or with the month of the reference date. Click this option, and then select the month in the list.
- *Year*: Allows you to compare the year of the date property with any year or with the year of the reference date. Click this option, and then select the year in the list.

1.42.10.2.3 Interval Switching Comparator


The *Interval Switching Comparator* component compares a numerical property with intervals or values of the same type. This component allows you to compare a numerical property with:

- One or several values
- One or several intervals
- One or several values and one or several intervals

When inserted in the decision tree, the *Interval Switching Comparator* component creates:

- A branch for each interval or value, which is activated if the property matches the related interval or value.
- An additional branch that is activated if the property does not match an interval or a value.

> ⓘ Note
>
> The test stops when a branch matches the comparison; therefore, the order of intervals and values is important.

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches according to the value of a property, you may need to compare a property with a set of intervals. To use the *Interval Switching Comparator* inside your price plan, do the following:

1. After you add the component, choose the property and the limits of the intervals or values.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> ✤ Example
>
> A software editor wants to determine the license price of their chargeable item class according to the number of users, which can vary from 10 users to 50 users. An interval represents a group of 10 users. For

more than 50 users, the price is $700. To determine the license price, the software editor needs to create a numerical property called **`number of users`** and test if the property matches one of the intervals. The following table presents the information that the software editor needs to set up the *Interval Switching Comparator* component inside the price plan:

Data	Settings
Property	Number of users
Value "0"	0
Interval I1	From 0 to 10 users:]0, 10]
Interval I2	From 10 to 20 users:]10,20]
Interval I3	From 20 to 30 users:]20, 30]
Interval I4	From 30 to 40 users:]30, 40]
Interval I5	From 40 to 50 users:]40,50]
Pricing rules	Branch value 0: error message. Branch I1: $150. Branch I2: $280. Branch I3: $400. Branch I4: $500. Branch I5: $600. Branch "outside of interval": $700.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of numerical properties. Select the first numerical property.

 : Adds a row to the table. Click this button to add an interval or a value to the table.

 : Deletes a row from the table. To remove an interval, select the corresponding interval row, and then click this button.

 : Moves a row up in the table. To move an interval up, select the corresponding interval row, and then click this button.

 : Moves a row down in the table. To move an interval down, select the corresponding interval row, and then click this button.

 : Opens the Open dialog box to search for a file and import data into the table. Click this button when you want to create intervals with data from a CSV file instead of manually entering the data.

 : Deletes all the rows in the table. Click this button to delete the list of intervals.

Parameters: Allows you to set up each row of the table. Each row represents an interval or a value to be compared with the first numerical property. Each interval row also represents a branch in the decision tree.

- *One Bound Interval*: Click this option to compare the first numerical property with a value instead of an interval, and then type the value or select a property in the *Between* box.
- *Between*: If the *One Bound Interval* option is selected, the *Between* option represents a value to compare with the first numerical property; otherwise, the *Between* option is the lowest limit of the interval to be compared with the first numerical property. Type a value or select a property to compare with the first numerical property. In the case of an interval, click the Included option to include the limit or clear it to exclude the limit.
- *And*: If the *One Bound Interval* option is not selected, the *And* option is the highest limit of the interval to be compared with the first numerical property. Type a value or select a property to compare with the first numerical property. Click the *Included* option to include the limit or clear it to exclude the limit. This box is not available if you have clicked the *One Bound Interval* option.

1.42.10.2.4 Mapping Table Introducer (Comparator)

 The **Mapping Table Introducer Comparator** component allows you to use an existing mapping table in a plan and set up the logic that defines the computation of the final values.

At runtime (or during trial run), SAP Convergent Charging proceeds as follows when executing the logic component *Mapping Table Introducer Comparator*:

1. The SAP CC system compares row by row the input columns of a mapping table with the properties of the plan.
2. The comparison stops when the properties match at least one row of the input columns.
3. The output columns related to the input columns are retrieved in generated properties that you can use in the plan.
4. If no property matches the comparison, the mapping table returns a default value for each generated property.

> **Note**
>
> The number of properties must be equal to the number of input columns of the mapping table.

When it is inserted in the decision tree, the Mapping Table Introducer Comparator component creates two separate branches:

- A branch (*Row found*) that is activated when the comparison is true

- A branch (*Row not found*) that is activated when the comparison is false

When the component is performed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches according to the value of a specific property, you may need to compare a property with a numerical value or another property.

To use the *Mapping Table Introducer* component inside your price plan, do the following:

1. After you add the component, choose a mapping table class [page 167], a mapping table [page 179], a date/time reference, and two properties to compare with the input columns of the mapping table.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> **Example**
>
> An operator needs a price table to retrieve the prices of offers expressed in many currencies depending on the validity periods of the offers. To do that in a price plan, the operator uses a *Mapping Table Introducer* which contains:
>
> - Mapping table input
> - **`OfferCode`**: The offer code
> - **`Currency`**: The currency for the offer code
> - **`ValidityPeriod`**: The validity period of the offer code
> - Mapping table output
> - **`OfferPrice`**: The price of the offer expressed in the selected currency

Identification Area

Name: Type the name of the component (mandatory).

Description: Describe the scope of the component (optional).

Definition (Tab)

Mapping Table Class: Provides the list of available mapping table classes. Select the mapping table class that you want to use with the *Mapping Table Introducer*. To open and modify the mapping table class, click  .

Mapping Table: Provides the list of available mapping tables that relate to the selected class. Select the mapping table that you want to use with the *Mapping Table Introducer* component, or select a parameter or a property that will include the identifier of a mapping table.

Reference Date: Provides the list of date/time properties. Select a date/time property. Each row in the mapping table includes a validity period. Therefore, the *Reference Date* setting enables to select the row whose time period includes the reference date and time. If several rows contain the same reference date/time:

- Rule 1: The SAP CC system selects the row whose start date/time is closest to the reference date/time set in the *Mapping Table Introducer* component.
- Rule 2: And if two time periods have the same start date/time, the shortest time period is first selected.

Column for Longest Prefix Match: Provides the list of input column names of the mapping table. Select an input column name that you define as the input column for containing the prefixes to be compared with the value of the related *Mapping Property*. When comparing the prefixes of this input column with the value of the related Mapping Property, the *Mapping Table Introducer* component finds a row only if the two conditions are true:

- The value of the input column (prefix) that matches the value of the related Mapping Property is the longest prefix match of all rows of the mapping table
- The values of the other input columns of the mapping table match the values of the related Mapping Properties

The Longest Prefix Match (LPM) algorithm is available as of SAP CC 4.1 SP 2.

Mapping of Rating Context: Displays the input columns of the mapping table, and allows you to select the properties that you want to compare with the input columns.

- *Column Name*: Displays the input column name of the mapping table.
- *Column Description*: Displays the description of the input column of the mapping table.
- *Type*: Provides the list of two property types: *String* or *Currency*. Select a property type.
- *Mapping Property*: Provides the list of properties whose type is defined in the *Type* column. Select the property that you want to compare with the input column of the mapping table.

Mapping of Generated Properties: Displays the output columns of the mapping table and allows you to rename the generated properties. When a comparison is true between properties, *Reference Date*, and input columns, the generated properties retrieve the values of the output column entries.

- *Column Name*: Displays the output column name of the mapping table.
- *Column Description*: Displays the description of the output column of the mapping table.
- *Type*: Displays the type of the generated property: *Number* or *String*.
- *Generated Property Name*: Displays the default generated property that retrieves the output column value of the mapping table. To change the name of the default generated property, double-click the entry, and then type a new name. You can use this new generated property in the plan.

Related Information

Mapping Tables [page 179]

1.42.10.2.5 Numbers Comparator


The *Numbers Comparator* compares a numerical property with a numerical value or another numerical property. Five comparison operators are available:

- Is equal to
- Is greater than
- Is less than
- Is greater than or equal to
- Is less than or equal to

When inserted in the decision tree, the *Numbers Comparator* component creates two separate branches:

- A branch that is activated when the comparison is true
- A branch that is activated when the comparison is false

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches according to the value of a specific property, you may need to compare a property with a numerical value or another property. To use the *Numbers Comparator* inside your price plan, do the following:

1. After you add the component, choose the first property, the value or the second property to compare with the first one, and then the Comparator operator.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

Example

In a monthly telephone plan, calls are free if the total duration of all the calls per month does not exceed the amount that is specified in the plan. If the total duration exceeds this amount, each call costs $0.34 per minute. The following table presents information that you need to set up the *Numbers Comparator* component inside the price plan:

Data	Settings
First property	Total duration of all calls per month
Comparator operator	*is greater than*
Second property	Duration of the monthly plan
Total duration of all calls per month	Price that is applied to the *is greater than* branch: $0.34 per minute. Price that is applied to the *is less than or equal to* branch: $0.00 per minute (free).

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of numerical properties. Select the first numerical property.

Operator: Provides the list of comparison operators. Select one of the following operators:

- *is equal to*
- *is greater than*
- *is less than*
- *is greater than or equal to*
- *is less than or equal to*

Comparison value: Provides the list of numerical properties or allows you to type a numerical value. Do one of the following:

- Select the second property to compare with the first property
- Type the numerical value to compare with the first property

1.42.10.2.6 Prefix Switching Comparator



The *Prefix Switching Comparator* component compares a string property with prefixes of string type.

When you insert it in the decision tree, the *Prefix Switching Comparator* component creates:

- A branch for each prefix, the branch of which is activated if the string property matches the related prefix.
- An additional branch that is activated if the string property does not match a prefix.

> **Note**
>
> When several prefixes contain identical characters, the *Prefix Switching Comparator* selects the prefix which contains the most characters in common with the string property.

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches, you may need to compare a property with a set of prefixes. To use the *Prefix Switching Comparator* inside your price plan, do the following:

1. After you add the component, choose the property and the prefixes.

2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

Example

An operator applies different prices to phone calls depending on the called number. The operator wants to determine the price for one minute of call time by analyzing the prefix of the called number. Using the *Prefix Switching Comparator* component, the operator needs to create a numerical property **`called number`** and compares it with other prefixes. The following table presents the information that the operator needs to set up the *Prefix Switching Comparator* component inside the price plan:

Data	Settings
Property	Called number
Prefix P1	02
Prefix P2	06
Prefix P3	089
Pricing rules	P1 Branch: $0.17 per minute. P2 Branch: $0.34 per minute. P3 Branch: $0.06 per minute. “Other prefix” branch: $0.25 per minute.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of string properties. Select a string property.

Prefixes: Each row in the table represents a prefix to be compared with the property. Each prefix row also represents a branch in the decision tree. Once you have added a row to the table, double-click the row, and then type a prefix.

 : Adds a row to the table. Click the button to add a prefix row to the table.

 : Deletes a row from the table. To remove a prefix, select the corresponding prefix row, and then click this button.

 : Moves a row up in the table. To move a prefix up, select the corresponding prefix row, and then click this button.

 : Moves a row down in the table. To move a prefix down, select the corresponding prefix row, and then click this button.

 : Opens the Open dialog box to search for a file and import data into the table. Click this button when you want to create prefixes with data from a CSV file instead of manually entering data.

 : Deletes all the rows in the table. Click this button to delete the list of prefixes.

1.42.10.2.7 Public Holidays Comparator

 The *Public Holidays Comparator* component compares a date property with one or more public holidays.

The comparison is true when the date property matches at least one public holiday.

When you insert it in the decision tree, the *Public Holidays Comparator* creates two separate branches:

- A branch that is activated when the comparison is true
- A branch that is activated when the comparison is false

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches, you may need to compare a property with a set of public holidays. To use the *Public Holidays Comparator* inside your price plan, do the following:

1. After you add the component, choose the date property and the public holidays.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> ### Example
>
> An operator charges a special price for telephone calls when they begin on a holiday. To do this, the operator compares the **`start date of the start of the call`** property to the holidays in the United States by using the *Public Holidays Comparator* component. The following table presents the information that is needed to set up the *Public Holidays Comparator* component inside the price plan:
>
Data	Settings
Date property	Start date of the start of the call
Public holidays	US holidays
Pricing rules	Price applied to the *is* branch: $0.15 per minute. Price applied to the *is not* branch: usual price.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of date properties. Select a date property.

Set the Public Holidays: Allows you to select the public holidays that you want to compare with the date property.

- *Group*: Select an item in the list[80], and then click  to select the item-related public holidays. Repeat this action for others items if necessary. To clear all the selected public holidays, click  .

> **Note**
>
> Do not pay attention to the public holidays that appear dimmed (unavailable) in the list because they are not longer recognized by the SAP CC system and are not longer active in the charges where they are used. For more information, consult the important note in the topic describing how to import public holidays. For more information on the import and export functions, refer to the dedicated topic [page 73] of primary help of Core Tool.

1.42.10.2.8 Range Table Introducer (Comparator)

 The logic component *Range Table Introducer Comparator* allows you to use an existing range table in a plan and set up the logic that defines the computation of the final values. You can also use a parameter or a property of the rating context to identify the range table. If you use a parameter, its value can be redefined:

- In a charge plan [page 343] to refer to another range table [page 213] in the pricing catalog [page 109] of the service provider
- In a provider contract [page 573] to refer to different subscriber range tables [page 557] that are dedicated to each end customer of the service provider

The comparison in the *Range Table Introducer Comparator* component mainly involves two steps:

- The comparison of input column values in the selected range table with properties of the rating context (see *Mapping of Input Properties*)
- The comparison of a quantity property of the rating context with the ranges (numerical intervals) in the selected range table (see *Range Input Property*)

[80] An item can be a region, a country, or anything else. The list of public holidays is created by your administrator. See your application administrator or consult the Administration Guide for more information

The comparison algorithm depends on the characteristics for the selected range table. The processing depends on the options defined for the ranges (numerical intervals):

- Upper bounds are inclusive or not.
- Last upper bound is +Infinity or is a business value.

> **Note**
>
> The comparison depends on the number of input columns in the selected range table:
>
> - If the range table contains one or more input columns, the *Range Table Introducer Comparator* performs the two steps.
> - If the range table contains no input columns, the *Range Table Introducer Comparator* only performs the second step.
>
> For more information about range table classes and range tables, refer to the dedicated [page 199] topics [page 213].

At runtime (or during trial run), SAP Convergent Charging proceeds as follows when executing the logic component *Range Table Introducer Comparator*:

- The system first compares row by row the set of input column values of the range table with a subset of properties in the rating context. The comparison stops when the subset of properties matches one entire row of input values in the range table. If no set of input values matches the subset of properties in the rating context, the fourth branch is (*Input Values not Found*) is executed.
- The system then compares row by row the ranges of the *Range* column in the range table with a given property of the rating context. The comparison stops when the property matches one range of the *Range* column. The output columns that relate to the selected range are retrieved in generated properties, which you can use in the plan. If no property matches the comparison, an appropriate branch (*Is Below the First Bound* or *Is Above the Last Defined Bound*) is executed and some default additional values are computed (*Computed Properties*).

When you insert the *Range Table Introducer Comparator* component in the decision tree for your plan, it creates three or four separate branches:

- A first branch (*Is Below the First Bound*) that is processed when the value of the range input property is negative.
- A second branch (*Is in a Range*) that is processed when the value of the range input property belongs to a range that is defined in the range table.
- A third branch that depends on the related range table class [page 199] that is specified in the component:

Third branch	Condition (in the range table, if the upper bound for last range is...)
A branch (*Is Above the Last Defined Bound*) that is processed when the value of the range input property is outside the last bunded range.	Bounded
A branch (*Found in Last Unbounded Range*) that is processed when the value of the range input property is in the last infinite and unbounded range.	Unbounded (+Infinity)

- A fourth branch (*Input Values not Found*) that is processed when no combination of input values has been found in the range table. This branch is available only if at least one input value has been defined in the

related range table class. All the properties that relate to the range table are not available in the context of this branch (for example, output values and computed values).

> **Note**
>
> When the component is executed by the SAP CC system at runtime, only one branch is executed depending on the comparison result.

When you set up the component, you can use the following generated properties in the logic of a branch:

- All output column values are usable in the rating or refilling context.
- When a column represent a number and the column is marked as cumulative, the cumulated value is generated in a context property.
- Range boundaries are available as context properties: Lower bound of the range and upper bound of the range.
- The size of the range is available: Value of the upper bound of the range minus the value of the lower bound of the range.
- When the value of the range input property is included in a range, the proration within the range is available (the value of the range input property minus the value of the lower bound of the range) / (the value of the upper bound of the range minus the value of the lower bound of the range).
- The difference between the value of the range input property and the lower bound of the range (the upper bound in case of an unbounded range) is also provided.

> **Note**
>
> If the value of the range input property is equal to 0, the SAP CC system executes different branches according to the configuration of the range table class that is specified in this component. The system takes into account the upper bounds of the intervals that are defined as inclusive or exclusive in the related class:
>
> - If the upper bounds of the intervals are inclusive, the system selects the first branch (*Is Below the First Bound*).
> - If the upper bounds of the intervals are exclusive, the system selects the second branch (*Is in a Range*).

Use

In a price plan [page 299], if you want to apply different prices to branches according to the value of a specific property, you may need to compare a property with a numerical value or another property. To use the *Range Table Introducer* inside your price plan, do the following:

1. After you add the component, choose a range table class, a range table, a date/time reference, and a property to compare with the ranges of the range table.
2. Choose input properties and add them to the mapping of input properties if the range table contains input columns.
3. Describe the pricing logic of each decision tree branch by using logic components and end each branch with a Function [page 727] component. In each branch, different new properties are available.

> **Example**
>
> In the telecommunication industry, a company sells voice services and has configured a range table class and a unique range table in its pricing catalog. The table specifies a unit price per minute regarding the call

duration. To use this range table in the pricing logic, you set up a charge with a price plan that includes a *Range Table Introducer Comparator* component.

In the pricing catalog [page 109] of this service provider, you must configure:

- A range table class [page 199] (where the last range is unbounded and an output column is a unit price per minute)
- A range table [page 213]

You create a new reusable charge and set up its price plan.

- You insert the *Range Table Introducer Comparator*.
- You set up the component and its branches by using Function [page 727] components and other logic components[81]:
  - In the first branch (*Is Below the First Bound*): Add a No Access [page 735] function.
  - In the second branch (*Is in a Range*): Add a Linear [page 733] function where A is the unit price that is set for the selected range, X is the call duration since the beginning of the range (see *Value Beyond Lower Bound*), and B is the cumulative price since the beginning of the phone call.
  - In the third branch (*Found in Last Unbounded Range*): Add a Linear [page 733] function.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Range Table Class: Provides the list of available range table classes. Select the range table class that you want to use with the *Range Table Introducer*. Note that the *Range Table Introducer Comparator* only creates the fourth branch (*Input Values not Found*) if the range table class contains input columns. To open and modify the range table class, click  .

Range Table: Provides the list of available range tables that relate to the selected class with the list of parameters and properties of type string. Select the range table that you want to use with the *Range Table Introducer*, or select a parameter or a property that must include at runtime the identifier of a range table or a subscriber range table.

> ⓘ Note
>
> If you select a parameter, the value of this parameter can be redefined in:
>
> - A charge plan [page 343] or refill plan [page 421] to share a range table between the charges of a charge plan

[81] You can configure more complex pricing logic in these branches.

- A contract item of a provider contract [page 592] to define the subscriber range table [page 557] that must be used for the computation during the charging process or the refilling process
- Refer also to Customized Use of Range Tables in the Charges and Refill Logic [page 224].

Reference Date: Provides the list of date/time properties. Select a date property. This reference date/time information enables the SAP CC system to select the appropriate version of the range table.

Range Input Property: Displays the properties of type number from the rating context and allows you to select the property that you want to compare with the ranges (numerical intervals) in the selected range table. Select the property as a quantity that you want to compare with the ranges of positive numbers in the range table.

Mapping of Input Properties: Displays the input columns of type string in the range table. When a range table contains input columns, the mapping of input properties allows you to determine which row of input columns you want to compare with properties of the rating context. The mapping of input properties allows you to map the properties of the rating context onto the input values of the range table:

- *Column Name*: Displays the name of the input column of the range table.
- *Column Description*: Displays the description of the input column of the range table.
- *Input Property Name*: Allows you to assign a property of the rating context to the input value of the range table.

Mapping of Generated Properties: Displays the output columns of the range table and allows you to rename the generated properties. When a comparison is true between properties, the *Reference Date*, and the *Range Input Property*, the generated properties retrieve the values of the output column entries.

- *Column Name*: Displays the output column name of the range table.
- *Column Description*: Displays the description of the output column of the range table.
- *Type*: Displays the type of the output column as configured in the selected range table class. The type are: *String*, *Single Number*, *Cumulative Number*, or *Range Size Cumulative Number*. Depending on this type, the SAP CC system generates one or two new properties for each output column.
- *Generated Property Name*: Displays the name of the generated property that retrieves the value of the cell in the output column and that relates to the row that is selected by the range comparison. The generated property retrieves the value of the cell in the output column that relates to the row that is selected by the range comparison. To change the name of the generated property, double-click the entry, and then type a new name. You can use this new generated property in the branches of this *Range Table Introducer* component. The default name is the name of the output column.
- *Generated Cumulative Property Name*: Displays the name of the generated cumulative property that retrieves the cumulative value or the range size cumulative value that is computed by the system. You can set up this name when the output column is of type *Cumulative Number* or *Range Size Cumulative Number*. To change the name of the generated cumulative property, double-click the entry, and then type a new name. You can use this new generated property in the branches of the component. The generated cumulative property retrieves the value of the cell in the output column that relates to the row that is selected by the range comparison.
  - *Cumulative value*: The generated property is of type number. Its value is computed by adding the output column values of all the rows until the row that is selected by the range comparison in the range table.
  - *Range size cumulative value*: The generated property is of type number. Its value is computed according to a formula (the output column value is multiplied by the range size) that is applied to the row just before the row that is selected by the range comparison in the range table.

The following table shows you the availability of generated properties:

Type of Generated Properties	The range input property is below the first bound	The range input property is in a range	The range input property is in the last unbounded range (+∞)	The range input property is after the last defined bound
String or Single Number	Yes It is the value of the column in the first range.	Yes It is the value of the column in the related range.	Yes It is the value of the column in the related range.	Yes It is the value of the column in the last range.
Cumulative value	Yes It is equal to 0.	Yes It is the sum of the values from the previous ranges excluding the selected range.	Yes It is the sum of the values from the previous ranges but excluding the last range.	Yes It is the sum of the column values from all the ranges.
Range size cumulative value	Yes It is equal to 0.	Yes It is the sum of the values that are computed according to the formula[82] from the previous ranges but excluding the selected range.	It is the sum of the values that are computed according to the formula[83] from the previous ranges but excluding the selected range.	Yes It is the sum of the values that are computed according to the formula[84] from all the ranges.

Computed Properties: Displays the computed properties that allows you to rename the generated properties that are visible in the rating contexts of the different branches.

- *Computed Value*: Displays the computed values that are available in the branch of the component. Some values are not available depending on the branch.
- *Computed Property Name*: Displays the names of the computed properties that provide additional information and can be used for the configuration of the pricing logic of the branches. To change the name of the default computed property, double-click the entry, and then type a new name. You can use this new computed property in the branches of the component according to their availability (see the table below).

The following table shows you the availability of computed properties:

Computed Values	The range input property is before the first bound	The range input property is in a range	The range input property is in the last unbounded range (+∞)	The range input property is after the last defined bound
Lower Bound of Range	Yes It is equal to 0.	Yes It is equal to the lower bound of the selected range.	Yes It is equal to the lower bound of the selected range.	Yes It is equal to the last upper bound.

82 The value is multiplied by the range size.
83 The value is multiplied by the range size.
84 The value is multiplied by the range size.

Computed Values	The range input property is before the first bound	The range input property is in a range	The range input property is in the last unbounded range (+∞)	The range input property is after the last defined bound
Upper Bound of Range	Yes It is equal to 0.	Yes It is equal to the upper bound of the selected range.	Not available	Yes It is equal to the last upper bound.
Range Size: Upper Bound of Range - Lower Bound of Range	Yes It is equal to 0.	Yes	No (N/A)	Yes It is equal to the last upper bound.
Prorata in Range: Value Beyond Lower Bound / Range Size	Not available	Yes	No (N/A)	Not available
Value Beyond Lower Bound: Difference between the lower bound of the range and the value of the input property that is used to select the appropriate range	Not available	Yes	Yes	Not available
Value Beyond Upper Bound: Difference between the upper bound of the last defined range and the value of the input property	Not available	Not available	Not available	Yes

Related Information

Range Tables [page 213]

1.42.10.2.9 Strings Comparator


The *Strings Comparator* component compares a string property with a string value or another string property. Four comparison operators are available:

- Is equal to
- Starts with
- Ends with
- Contains

When you insert it in the decision tree, the *Strings Comparator* component creates two separate branches:

- A branch that is activated when the comparison is true
- A branch that is activated when the comparison is false

When the component is executed, only one branch is activated depending on the test result.

Use

In a price plan [page 299], if you want to apply different prices to branches, you may need to compare a string property with a string value or another string property. To use the *Strings Comparator* inside your price plan, do the following:

1. After you add the component, choose the first property, the second property or the string value to be compared with the first one, and then the Comparator operator.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

> ### Example
>
> A telephone operator decides to apply a special rate ($0.06 per minute) when the called number begins with "089"; otherwise, the call is billed at the usual price. The following table presents the information that the telephone operator needs to set up the *Strings Comparator* component inside the price plan:

Data	Settings
First property	`Number called`
Comparator operator	*starts with*
Value	089
Pricing rules	Price that is applied to the “starts with” branch: $0.06 per minute. Price that is applied to the *does not start with* branch: Usual price.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of string properties. Select the first string property.

Operator: Provides the list of comparison operators. Select one of the following operators:

- *is equal to*
- *starts with*
- *ends with*
- *contains*

Comparison Value: Provides the list of string properties or the *String Value* option. Select a second property or select *String Value*, and then type a string.

1.42.10.2.10 Switching Comparator

> ⚠ Caution
>
> As of SAP Convergent Charging 2023 FPS 2, this component not available in case of an installation scenario integrated with SAP S/4HANA Cloud Public Edition. For further information, refer to SAP Note 2772452.


The *Switching Comparator* component compares the state of a prepaid account with one or more predefined values (*Active*, *Blocked*, *Locked*, or *Closed*).

The comparison is true when the state of the prepaid account matches one of the predefined values.

When you insert it in the decision tree, the *Switching Comparator* creates two to five separate branches:

- One to four branches corresponding to a predefined value (*Active*, *Blocked*, *Locked*, or *Closed*), but only one branch is activated if the comparison is true.
- One branch that is activated when the comparison is false.

When the component is executed, only one branch is activated depending on the test result.

Use

In the refill logic [page 395] of a refill plan [page 421], if you want to make a refill depending on the state of the prepaid account, you may need to compare the state of the prepaid account with one of the predefined states (*Active*, *Blocked*, *Locked*, or *Closed*). To use the *Switching Comparator* inside your price plan [page 299], do the following:

1. After you add the component, choose the "*Account State of Prepaid Account* as the property name.
2. Describe the refill of each decision tree branch by using a Function [page 727] component.

Definition tab

Property Name: Provides the list of properties. Because a refill plan is associate to one and only one prepaid account, the list provides only the property *Account State of Prepaid Account*. Select this property.

Values: Provides the list of the states of a prepaid account. Select a state, and then click  to add a value to the table.

- *Active*: Any operation is possible when the prepaid account is active
- *Blocked*: Some operations (refills, overruns, and overspending) are possible when the prepaid account is blocked
- *Locked*: Some operations (refills, overruns, and overspending) are possible when the prepaid account is locked
- *Closed*: No operation is possible when the prepaid account is closed

 : Adds a row to the table. Click the button to add a value to the table.

 : Deletes a row from the table. To remove a value, select the corresponding row, and then click this button.

 : Moves a row up in the table. To move a value up, select the corresponding row, and then click this button.

 : Moves a row down in the table. To move a value down, select the corresponding row, and then click this button.

 : Opens the Open dialog box to search for a file and import data into the table. Click this button when you want to create values with data from a CSV file instead of manually entering data.

 : Deletes all the rows in the table. Click this button to delete the list of values.

1.42.10.3 Splitter Components

A **splitter component** divides a numerical property or a date/time property into two parts (or more for the Amounts Aggregator Splitter [page 761] component). Each part is represented by a branch in the decision tree. At runtime, when a splitter component is executed by the system, all the branches are activated and provide interim amounts that are added up to compute the final amount that is the total amount. Note that the amounts aggregator splitter is particular; see the note below.

The interim amounts are aggregated and only one rated transaction is produced by the system.

> ⓘ Note
>
> In this component family, the amounts aggregator splitter component represents an exception.
>
> When it is configured to compute the sum value of several branches, the component complies with the definition for splitter components.

When it is configured to compute the minimum, maximum, or average value, the component represents an exception. Interim amounts are not added up to each other. The final amount is not the total amount of the executed branches as only one branch is executed by the system.

There are several types of splitter components. For example, the *Numbers Splitter* component splits an existing property related to a number, and the *Time Slot Splitter* component splits a property related to time slots.

Splitter Components

The following splitter components are available with SAP Convergent Charging:

Icon	Name	Action/Usage	Examples
	Amounts Aggregator Splitter [page 761]	Gets the sum (total amount), the minimum value, the maximum value, or the mean value of several prices computed in different ways.	1. Pricing based on the total of interim amounts/prices/fees 2. Pricing based on the most expensive amount 3. Pricing based on a computed average amount
	Numbers Splitter [page 763]	Splits a numeric value in two parts, according to a reference value, executes the business logic of both branches, and computes the total amount.	Different pricing under and above a limit value: Free upload under a limit (100 MB) and USD 1.00/MB above 100 MB
	Public Holidays Splitter [page 765]	Splits a duration according to a reference time slot that is made of public holidays, executes the business logic of both branches, and computes the total amount.	Different pricing during a public holiday
	Time Slot Splitter [page 769]	Splits a duration according to a reference time slot, executes the business logic of both branches, and computes the total amount.	Different pricing during night and day: The charging of an electric vehicle started during the day and ending during the night

1.42.10.3.1 Amounts Aggregator Splitter

 The *Amounts Aggregator Splitter* component creates two or more branches in the decision tree to calculate amounts and performs one of the following operations:

- Calculates the sum of the amounts of all the branches
- Compares the amounts of all the branches to activate either the lowest amount or the highest amount
- Calculates the average amount of all the branches

In each branch of the *Amounts Aggregator Splitter* component, you insert a series of logic components to calculate an amount. A Function [page 727] component ends each branch.

The *Amounts Aggregator Splitter* component allows you to describe independent rules in each branch. You can compare or combine the calculated amounts in the different branches to provide one single amount. The final amount depends on the type of the operation:

With	The final amount is
Sum	The total amount of all the amounts calculated in each branch
Minimum	The minimum of all the amounts calculated in each branch
Maximum	The maximum of all the amounts calculated in each branch
Mean	The average of all the amounts calculated in each branch

When the component is inserted in pricing macro, it calculates amounts. When the component is inserted in a price plan, it calculates prices.

Use

In a price plan [page 299], if you want to bill a chargeable item class according to two distinct criteria and charge the customer the lower price, do the following:

- In the branch of the price plan, insert the *Amounts Aggregator Splitter* component by selecting the operation *Minimum* and the number *2* as the number of amounts or branches.
- Describe the pricing, criterion by criterion, of each branch.

Example

You sell educational documents on the web. Your end customer is charged an opening fee according to the size and the category of documents. To attract new customers, you decide to charge only the size or the category that represents the lowest fee.

As the pricing specialist for the service provider, you design and configure a price plan that is executed by the SAP system at runtime to calculate the fees to be charged.

You perform this operation by inserting the *Amounts Aggregator Splitter* component at the suitable level of your price plan. You set up the component by choosing *minimum* as the operation and *2* as the number of operations. The component will calculate:

- The price according to the category in the first branch
- The price according to the size in the second branch

After calculation, the end customer is charged the lowest price.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Operation: Provides the list of operations to apply to the tree branches of the *Amounts Aggregator Splitter* component. *Sum* is the default operation. Select one of the following operations:

- *sum*: Calculates the sum of the amounts of all the executed branches. The final result is the aggregation of these amounts. All the branches are sequentially executed. A counter change in a branch may influence calculations in another branch.
- *minimum*: Activates the branch whose amount is the lowest
- *maximum*: Activates the branch whose amount is the highest
- *mean*: Calculates the average of the amounts of all the branches

Number of Amounts: Provides the list of the number of branches that you want to compare or aggregate. The number *2* is the default value. Select a number ranging from *2* to *20*.

1.42.10.3.2 Numbers Splitter


The *Numbers Splitter* component compares a numerical property P with a reference threshold T and divides the numerical property into two parts, p1 and p2, according to the following conditions:

- If P is less than or equal to T, then p1 = P and p2 = 0.
- If P is greater than T, then p1 = T and p2 = P - T.

> **Note**
>
> In all cases, P = p1 + p2.

When you insert it in the decision tree, the *Numbers Splitter* component divides the numerical property P into two parts and creates:

- Two separate branches in the decision tree:
  - A first branch *Up to* that represents the "p1" value
  - A second branch *Beyond* that represents the "p2" value
- Two numerical properties to retrieve the "p1" and "p2" values

At runtime, when the component is executed by the system:

- If p1 and p2 are not equal to 0, the two branches are activated and provide two amounts which are added up to compute the final amount (total amount).
- If p2 = 0, the branch *Beyond* is not activated.
- The computations are aggregated. Only one rated transaction is produced and corresponds to the final total amount.

Use

In a price plan [page 299], do the following to use the *Numbers Splitter*:

1. Insert the component in the decision tree to create the two branches *up to* and *beyond*. Choose a numerical property, define the expected reference threshold, and choose the name of the generated property.
2. Describe the pricing of each decision tree branch by using a Function [page 727] component.

Example

An end customer has a telephone service that contains a calling plan of two hours per month at the reduced price of $0.02 per minute. Beyond these two hours, the calls cost $0.04 per minute.

To calculate the call price, you must compare and divide the call duration according to the time that remains in the monthly calling plan. The remaining time can be a counter that is decreased each time that there is a call and reset to two hours at the beginning of each month.

The following table presents the information that is needed to set up the *Numbers Splitter* component inside the price plan:

Data	Settings
Numerical property	Duration of the phone call (numerical property)
Threshold (Split value)	Monthly remaining duration in the package (counter)
Generated property	Call duration according to the calling plan
Pricing rules	Price that is applied to the *up to* branch: $0.02 per minute Price that is applied to the *beyond* branch: $0.04 per minute

Note

In a price plan, you may need to apply different prices that vary depending on a numerical threshold or a reference value. To do this, you can divide a numerical property according to the numerical threshold. **A typical use** of the *Numbers Splitter* component is the calculation of a call within the framework of a calling plan. The calculation rule is the following:

- If the call duration is less than the time that remains in the calling plan, you must apply the calling plan price to the entire call duration.
- If the call duration is greater than the time that remains in the calling plan, you must do the following:
  - Apply the non-calling plan price to the part of the call duration that is outside of the calling plan limits
  - Apply the calling plan price to the part of the call that is within the calling plan limits
  - Add the two resulting amounts

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Property Name: Provides the list of numerical properties. Select a numerical property that contains the value to be compared with at runtime.

Split Value: Provides the threshold value list, or allows to type a threshold value. Select or type the expected split value that is used as reference at runtime.

Names of Computed Properties: Allows you to name two new numerical properties:

- *Up to Split Value*: Type the name of the new numerical property that retrieves the result p1. You can reuse this computed property in the branch *Up to* of the decision tree.
- *Beyond Split Value*: Type the name of the new numerical property that retrieves the result p2. You can reuse this computed property only in the branch *Beyond* of the decision tree.

1.42.10.3.3 Public Holidays Splitter

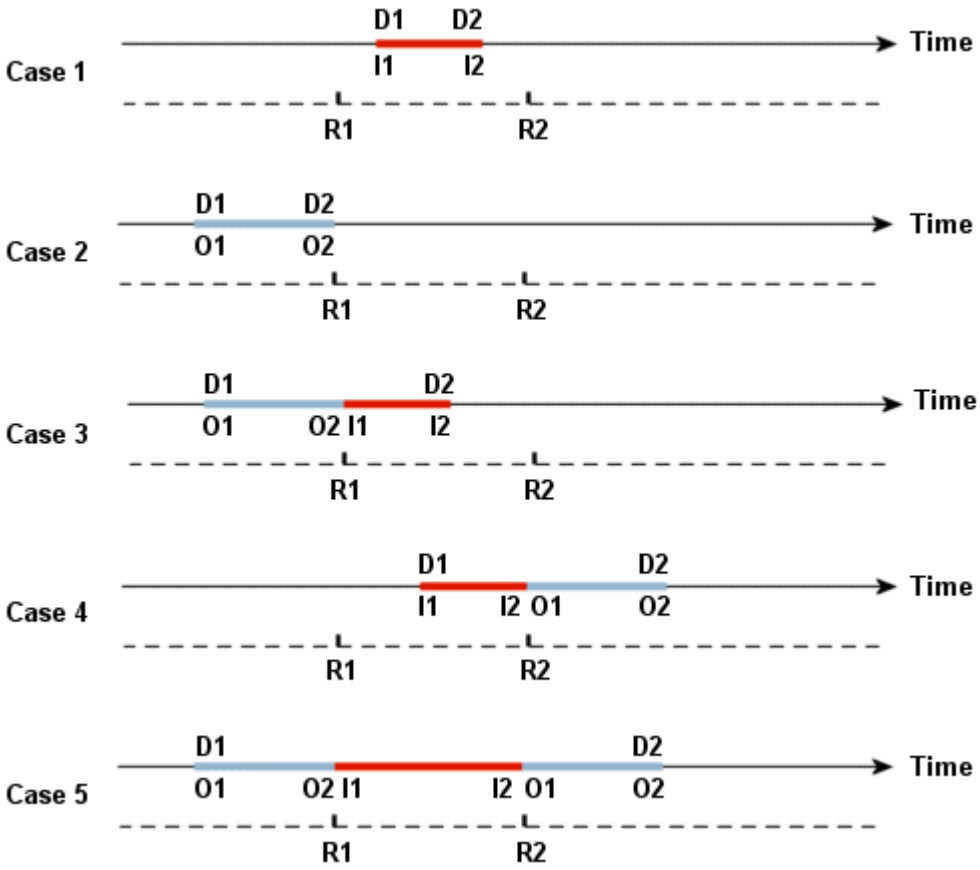

The *Public Holidays Splitter* component compares a time slot D with a reference time slot R and divides D into two time slots:

- Time slot I, which is inside the public holidays R
- Time slot O, which is outside the public holidays R

The reference time slot R is made up of public holidays. You can choose dates by selecting one or more countries or choose one or more dates by selecting individual holidays. The reference time can then be a discontinuous period or just one day.

The illustration below shows different cases you can meet when using a *Public Holidays Splitter*. The time slots are represented as follow:

- Time slot D, represented by [D1-D2]
- Time slot R, represented by [R1-R2]
- Time slot I, represented by [I1-I2]
- Time slot O, represented by [O1-O2]



> **Note**
>
> - If the time slot D is completely inside R, then part O = 0 (Case 1 in the previous illustration).
> - If the time slot D is completely outside R, then part I = 0 (Case 2 in the previous illustration).
> - In all cases, D = I + O (Refer to the previous illustration).
> - If time slot D extends beyond the reference time slot R on either side of R, the time slot results are coherent: I = R, and O represents the sum of the two time slots that extend outside of either end of R (Cases 3, 4 and 5 in the previous illustration).

Use

When you insert it in the decision tree, the *Public Holidays Splitter* component divides the time slot D into two time slots and creates:

- Two separate branches in the decision tree:
  - The first branch *During Public Holidays* represents the time slot I
  - The second branch *Not During Public Holidays* represents the time slot O
- Two date properties which are the start date and the end date of the time slots I and O:
  - The start date takes two values I1 and O1 in succession, depending on the branch

- The end date takes two values I2 and O2 in succession, depending on the branch

At runtime, when the component is executed by the system, the two branches are activated and provide two amounts that are added up to compute the final amount (total amount).

- If I = 0 or O = 0, the corresponding branch is not activated.
- The computations are aggregated. Only one rated transaction is produced and corresponds to the final total amount.

⚠ Caution

If D = 0, (case of D1 = D2), no branch is activated and no transaction is generated (like with a Free Function [page 730]).

If the *Public Holidays Splitter* component processes "case 5" in the previous illustration, it does the following in this order:

1. Process of the first time slot [O1 - O2] with Start Date (out of the slot) = O1 and End Date (out of the slot) = O2.
2. Process of the time slot [I1 - I2] with Start Date (in the slot) = I1 and End Date (in the slot) = I2.
3. Process of the last time slot [O1 - O2] with Start Date (out of the slot) = O1 and End Date (out of the slot) = O2.

Example

An end customer has subscribed to a telephone service that includes a reduced price of $0.02 per minute if the call takes place during a public holiday; otherwise, the call costs $0.04 per minute.

As the pricing specialist for the service provider, you design and configure a price plan that is executed by the SAP system at runtime to calculate the fees to be charged.

To calculate the cost of a call, you need to do the following:

1. Compare the call time slot with the reference time slot of public holidays.
2. Divide the call time slot into two time slots, **`During Public Holidays`** and **`Not During Public Holidays`**.
3. Calculate the duration of the two time slots and apply the corresponding price.

The following table presents the information that is needed to set up the *Public Holidays Splitter* component in the price plan:

Data	Settings
Time slot D	Start date of the phone call (D1). Stop date of the phone call (D2).
Reference time slot R	All USA public holidays
Generated property	Start Date and End Date (in the slot): [I1 - I2] Start Date and End Date (out of the slot): [O1 - O2]

Data	Settings
Pricing rules	Duration of each time slot: Difference between the two generated properties Start Date and End Date for each time slot. Price that is applied to the "During Public Holidays" branch: $0.02 per minute. Price that is applied to the "Not During Public Holidays" branch: $0.04 per minute.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Start Date Name: Provides the list of date properties. Select the date property that represents the value D1 of the [D1-D2] time slot.

End Date Name: Provides the list of date properties. Select the date property that represents the value D2 of the [D1-D2] time slot.

Public Holidays: Allows you to set up the reference time slot [R1-R2] that represents one or more Public Holidays.

- *Group*: Select an item in the list[85], and then click  to select the item-related public holidays. Repeat this action for others items if necessary. To clear all the selected public holidays, click  .

> **Note**
>
> Do not pay attention to the public holidays that appear dimmed (unavailable) in the list because they are not longer recognized by the SAP CC system and are not longer active in the charges where they are used. For more information, consult the important note in the topic describing how to import public holidays. For more information on the import and export functions, refer to the primary help of Core Tool [page 73].

Names of Computed Properties: Allows you to type the names of the new date properties that retrieve the resulting time slots. These properties can be reused in a Duration Operator [page 797] component to calculate a duration.

[85] An item can be a region, a country or anything else. The list of public holidays is created by your administrator. See your application administrator or consult the Administration Guide for more information

- *Start Date (in the slot)*: Allows you to type the name of the new date property that retrieves the start date of the resulting time slot [I1 - I2]. You can reuse this computed property in the branch *During Public Holidays* of the decision tree.
- *End Date (in the slot)*: Allows you to type the name of the new date property that retrieves the end date of the resulting time slot [I1 - I2]. You can reuse this computed property in the branch *During Public Holidays* of the decision tree.
- *Start Date (out of the slot)*: Allows you to type the name of the new date property that retrieves the start date of the resulting time slot [O1 - O2]. You can reuse this computed property in the branch *Not During Public Holidays* of the decision tree.
- *End Date (out of the slot)*: Allows you to type the name of the new date property that retrieves the end date of the resulting time slot [O1 - O2]. You can reuse this computed property in the branch *Not During Public Holidays* of the decision tree.

1.42.10.3.4 Time Slot Splitter

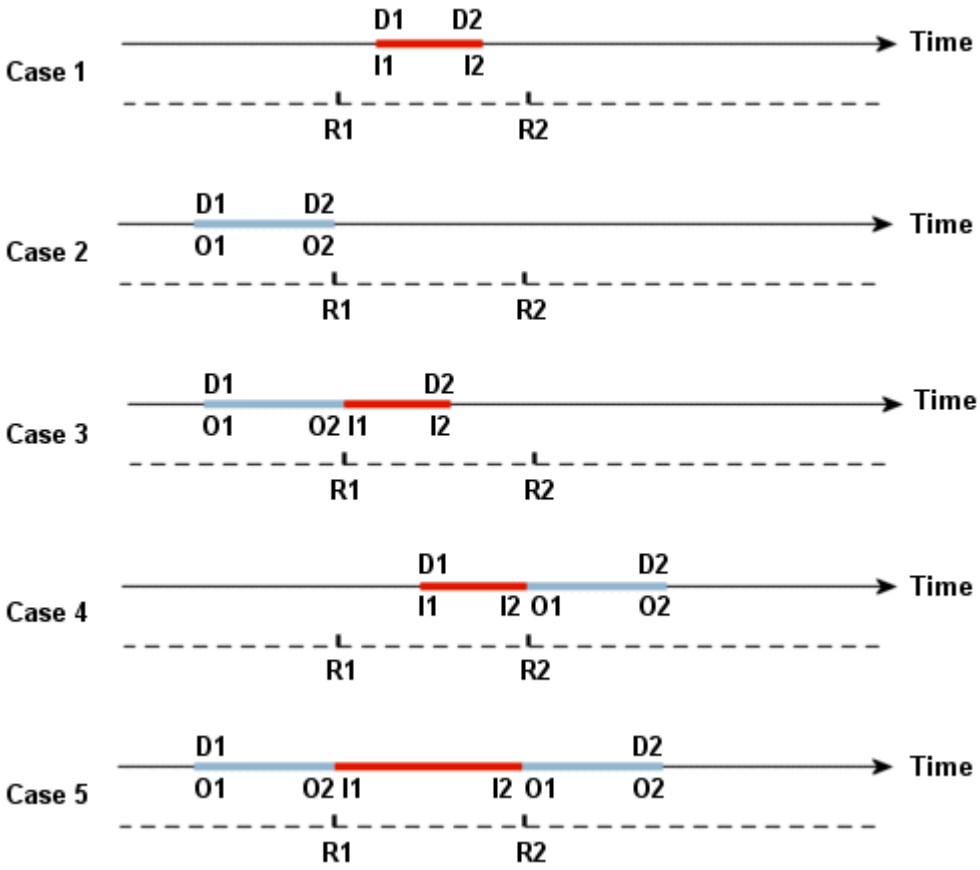
 The *Time Slot Splitter* component compares a time slot D with a reference time slot R and divides D into two time slots:

- Time slot I, which is inside the reference time slot R
- Time slot O, which is outside the reference time slot R

The reference time slot is a **recurring time slot**.

The illustration below shows different cases you can meet when using a *Time Slot Splitter*. The time slots are represented as follow:

- Time slot D, represented by [D1-D2]
- Time slot R, represented by [R1-R2]
- Time slot I, represented by [I1-I2]
- Time slot O, represented by [O1-O2]



> **Note**
>
> - If the time slot D is completely inside R, part O = 0 (Case 1 in the previous illustration).
> - If the time slot D is completely outside R, part I = 0 (Case 2 in the previous illustration).
> - If the time slot D extends beyond the reference time slot R on either side of R, the time slot results are coherent: I = R, and O represents the sum of the two time slots that extend outside of either end of R (Cases 3, 4, and 5 in the previous illustration).
> - In all cases, D = I + O.

Use

When you insert it in the decision tree, the *Time Slot Splitter* component divides the time slot D into two time slots and creates:

- Two separate branches in the decision tree:
  - The first branch *In the Time Slot*, which represents the time slot I
  - The second branch *Out of the Time Slot*, which represents the time slot O
- Two date properties which are the start date and the end date of the time slots I and O:
  - The start date takes two values I1 and O1 in succession, depending on the branch

- The end date takes two values I2 and O2 in succession, depending on the branch

Pricing rules:

At runtime, when the component is executed by the SAP CC system, the two branches are activated and provide two amounts that are added up to compute the final amount (total amount).

- The computations are aggregated. Only one rated transaction is produced and corresponds to the final total amount.
- If I = 0 (see Case 2) or O = 0 (see Case 1), the corresponding branch is not activated.
- If the SAP CC system processes the siutation of the Case 5 in the previous illustration, it does the following in this order:
  1. Process of the first time slot [O1 - O2] with Start Date (out of the slot) = O1 and End Date (out of the slot) = O2.
  2. Process of the time slot [I1 - I2] with Start Date (in the slot) = I1 and End Date (in the slot) = I2.
  3. Process of the last time slot [O1 - O2] with Start Date (out of the slot) = O1 and End Date (out of the slot) = O2.

→ Remember

If D = 0, (case of D1 = D2), no branch is activated and no transaction is generated by the SAP CC system (like with a Free Function [page 730] component).

Interfaced billing or reporting systems cannot view these calculations.

Example

An end customer has a telephone service that includes a reduced price of $0.02 per minute every day during the 10:00 PM to 6:00 AM time slot. Outside this time slot, the call costs $0.04 per minute. To calculate the cost of a call, you need to do the following:

1. Compare the call time slot with the reference time slot (10:00 PM to 6:00 AM).
2. Divide the call time slot into two time slots: *In the Time Slot* and *Out of the Time Slot*.
3. Calculate the duration of the two time slots and apply the corresponding price. For example, you insert a Duration Operator [page 797] component into the two branches to calculate the duration, and then insert a Linear Function [page 733] component to calculate the price of each branch.

The following table presents the information that is needed to set up the *Time Slot Splitter* component inside the price plan:

Data	Settings
Time slot D	Start date/time of the phone call (D1) Stop date/time of the phone call (D2)
Reference time slot R	Type of recurrence: Daily From: 10:00 PM To: 06:00 AM
Generated properties	Start date/time and end date/time (in the slot): [I1 - I2] Start date/time and end date/time (out of the slot): [O1 - O2]

Data	Settings
Pricing rules	Duration of each time slot: Difference between the two generated properties Start Date and End Date for each time slot. Price that is applied to the *In The Time Slot* branch: $0.02 per minute. Price that is applied to the *Out of The Time Slot* branch: $0.04 per minute.

> **Note**
>
> In a price plan [page 299], you may need to apply prices which vary according to different time slots. The Time Slot Splitter component is often used to calculate the cost of a telephone call, the price of which varies according to the cases that are shown in the following table:

If the call takes place...	You must apply the price to...	During...
Inside the reference time slot	*In the Time Slot*	The entire call duration
Outside the reference time slot	*Out of the Time Slot*	The entire call duration
Partly inside and partly outside the reference time slot	*In the Time Slot*	The call duration that is inside the time slot
	Out of the Time Slot	The call duration that is outside the time slot

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Start Date Name: Provides the list of date properties. Select the date property that represents the value D1 of the [D1-D2] time slot.

End Date Name: Provides the list of date properties. Select the date property that represents the value D2 of the [D1-D2] time slot.

Time Slot Parameters: Allows you to set up the reference time slot [R1-R2]. It is a recurring time slot.

- *Time Slot Type*: Provides the list of time slot types. Select one of the following recurrence types:
  - *daily*
  - *weekly*
  - *monthly*

- *yearly*

- *From*: Allows you to select or type a start date or time (hour and minute) of the time slot.
- *At*: Allows you to select or type a time (hour and minute).
- *To*: Allows you to select or type an end date or time (hour and minute) of the time slot.

> **Note**
>
> Depending on the selected time slot, the format varies as shown in the following examples:
>
> - *daily*: *From* 05.00 AM *To* 07.00 AM
> - *weekly*: *From* Monday *at* 05.00 AM *To* Wednesday *at* 07.00 AM
> - *monthly*: *From* 10 *at* 05.00 AM *To* 12 at 07.00 AM
> - *yearly*: *From* 22 May *at* 05.00 AM *To* 18 June *at* 07.00 AM

Names of Computed Properties: Allows you to type the names of the new date properties that retrieve the resulting time slots. These properties can be reused in a Duration Operator [page 797] component to calculate a duration.

- *Start Date (in the slot)*: Allows you to type the name of the new date property that retrieves the start date of the resulting time slot [I1 - I2].
  You can reuse this computed property in the branch *In the Time Slot* of the decision tree.
- *End Date (in the slot)*: Allows you to type the name of the new date property that retrieves the end date of the resulting time slot [I1 - I2].
  You can reuse this computed property in the branch *In the Time Slot* of the decision tree.
- *Start Date (out of the slot)*: Allows you to type the name of the new date property that retrieves the start date of the resulting time slot [O1 - O2].
  You can reuse this computed property in the branch *Out of the Time Slot* of the decision tree.
- *End Date (out of the slot)*: Allows you to type the name of the new date property that retrieves the end date of the resulting time slot [O1 - O2].
  You can reuse this computed property in the branch *Out of the Time Slot* of the decision tree.

1.42.10.4 Operator Components

> **Note**
>
> You can use the operator components in a price plan [page 299], in a pricing macro [page 121], in a charging plan [page 305], in a refill plan [page 421] (refill logic [page 395]), or in an allowance plan [page 466] (allowance logic [page 449]).
>
> But you cannot use:
>
> - The components *Counter Operator* and *Macro Operator* in a charging plan.
> - The component *State Schedule Updater* in a price plan, a pricing macro, in a charging plan, and in an allowance plan (allowance logic).

An operator component can do the following in the rating context:

- Modify counter values
- Create allowances
- Use allowances by sending events or introducing allowance properties
- Create new properties resulting from calculation

There are several types of operator component.

For example, the *Duration* operator component calculates the duration between two dates while the *Update Counter* operator component can update counters that are defined in the price plan, the charging plan, or the refill logic.

Operator Components

The following operator components are available with SAP Convergent Charging:

Icon	Name	Action/Usage
	Alert Triggering [page 776]	Configures an alert which is triggered if the branch containing the component is activated.
	Allowance Event Sender [page 777]	Sends an allowance event to allowances under certain conditions.
	Allowance Event Updater [page 782]	Modifies the event handled by the allowance.
	Allowance Property Introducer [page 784]	Retrieves the property values of a set of allowances and saves them in generated properties that are available for the next components of the decision tree.
	Allowance Validity Period Updater [page 787]	Extends the end date or redefines the start date and the end date of the validity period of the allowances.
	Arithmetic Computation [page 789]	Carries out an arithmetical operation between two numerical values or numerical properties.
	Concatenate [page 791]	Concatenates two or more string properties or character strings.
	Create Allowance [page 792]	Creates an allowance for the related provider contract through a refill plan, a charge plan, or an allowance plan.
	Date Shift [page 794]	Calculates a new date by adding a period of time to an existing date property or by subtracting a period of time from an existing date property.

Icon	Name	Action/Usage
	Decrement Counter [page 795]	Debits a counter
	Duration [page 797]	Calculates the period of time between two dates.
	Duration Adapter [page 798]	Converts a period of time into another time unit.
	Filter [page 800]	Compares each value of a multi-valued property of string type with one or more string values.
	Increment Counter [page 795]	Credits a counter
	Macro Introducer [page 804]	Allows to use an existing pricing macro in a plan.
	Multi-Valued Evaluator [page 805]	Counts the number of values which are in a multi-valued property.
	Output Property Update [page 807]	Modifies the generated properties of a pricing macro.
	Property Introducer [page 808]	Creates new properties for a plan.
	Property Type Convertor [page 810]	Creates a new property from an existing property by modifying the type of the existing property.
	Prorata [page 813]	Calculates the product Y = X * R, in which: - Y is the numerical result - X is a numerical value - R is a ratio ranging from 0 to 1
	State Schedule Updater [page 817]	Changes the dates on which a prepaid account is blocked, locked, or closed.
	Switch Table [page 819]	Compares the same property with one or more values or properties of the same type.
	Tier Table Introducer [page 822]	Allows to use an existing translation table in a plan.
	Translation Table Introducer [page 823]	Allows to use an existing tier table in a plan.

Icon	Name	Action/Usage
	Update Counter [page 826]	Updates a counter.

1.42.10.4.1 Alert Triggering Operator

 The **Alert Triggering Operator** component allows configuring an alert which is triggered if the branch containing the component is activated. When an alert is triggered and if the charging operation is successful, a notification is then published by SAP Convergent Charging to a dedicated system.

> ⓘ Note
>
> - The notification is published only if this function is activated. Refer to your administrator for more information or consult the SAP CC SAP CC 2023 Administration and Operation Guide.
> - The notification handling must be implemented in your external system by the implementation project team. If SAP Convergent Charging is connected to SAP ERP/FI-CA, the default integration does not take into account the notification based on the *Alert Triggering Operator* component and a specific implementation project is necessary.

> ✤ Example
>
> Using Wireless M2M communications, a vending machine can be installed virtually anywhere but should be connected to a support centre to signal when the vending machine is empty, full of change, or out of service. To do that, the price plan can include an Alert Triggering operator component to trigger an alert that notifies a dedicated system.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Code: Allows you to type the code identifying the alert, or select the string property whose value represents the alert code.

Message: Allows you to type an information message describing the alert.

Properties: Provides the list of rating context properties. The properties give additional information about the alert. Select none, one or more string properties. The values of the selected properties are included in the alert published by the dedicated system.

1.42.10.4.2 Allowance Event Sender (Operator)

> → Tip
>
> You cannot use this logic component after an Account Event Refill [page 714] component in the refill logic or after an Event-Based Trigger [page 717] component in the allowance logic.


The **Allowance Event Sender** component is an operator component that allows you to configure the sending of an *allowance event* to certain allowances [page 575] that exist in provider contracts. In order to receive this allowance event, the allowances must:

- Belong to the related provider contract or to the related provider contract and its parent contract.
- Be shared (see Shared Allowances [page 579] for more information).
- Be active at the date/time of the reception of the allowance event.
- "Listen for" the *allowance events* of the related allowance event class [page 441].

This operator component allows you to set up some filtering by using selection criteria based on the values of properties.

> ⓘ Note
>
> - You can trigger the automatic sending of the allowance event depending on the value of a monitored property.
> - At runtime, the allowance that sends the allowance event through the execution of this *Allowance Event Sender* component, does not receive this event.
> - The available scope of allowances depends on the usage of logic components. For more information, refer to the matrix of components [page 676].

Use

As a pricing specialist, you use the *Allowance Event Sender* component for sending allowance events to allowances in provider contracts by configuring this logic component in the price plan, the refill logic, or the allowance logic.

> ⚠ Restriction
>
> The *Allowance Event Sender (Operator)* component is forbidden within a **shareable allowance logic**.

> ✤ Example
>
> **Example case: The allowance event is modified.**

A pricing specialist designs and sets up the *Allowance Event Sender* component in a price plan to consume the allowances related to the duration of a videoconference. At runtime, when the system executes the *Allowance Event Sender* component, the system sends an allowance event based on the allowance event class `ALLOW_VIDEO_MINUTES` whose property is the duration of the videoconference in minutes.

The pricing specialist models the allowance logic through the `ALLOW_VIDEO_MINUTES` *Allowance Event Class* (the type of the "Event Allowance"):

- To authorize part of the duration based on the persistent counters of the allowances
- To update the allowance event based on the authorized part

Example

Example case: The allowance event is not modified.

An allowance is created through a refill plan. The allowance is valid six months and is part of a group of allowances named `SAMPLE_GROUP`. The allowances of this group have the same parameter value and extend their validity period each time a new allowance is added to the group.

At runtime, when an allowance is created for the group, the new allowance must inform the existing allowances of its group with an allowance event of the type `NEW_ALLOWANCE_CREATED`. The event includes the following fields:

- The allowance group ID
- The creation date/time
- Its validity period start date/time
- Its validity period end date/time

The other allowances of the group can then perform actions when receiving the allowance events.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

Allowance Event Class: Provides the list of allowance event classes [page 441]. If necessary, click  to open the selected allowance event class. Select an allowance event class, set up the values of the properties of the allowance event class, and then define the new output property if necessary:

- *Property Name*: Displays the property name.
- *Type*: Displays the property type.
- *Value*: Provides the list of available properties or allows you to type a value.
- *Output Property Name*: Type the name of the new property.

Scope: The scope defines the allowances to which the system sends an allowance event when executing *Allowance Event Sender* component.

Select one of the three options for the action perimeter that suits your business requirements:

- *Only Allowances of the Contract*: Allowance events are only sent to the allowances of the related provider contract.
- *Allowances of the Contract and its Parent*: Allowance events are only sent to the related provider contract and its parent contract.
- *Only Shared Allowances*: Allowance events are only sent to shared allowances.

Share Identifier: Select a property to identify the group of shared allowances [page 579] to which the system sends an allowance event when executing the *Allowance Event Sender* component at runtime.

Stop sending when the monitored property is lower than or equal to 0: Select this option if you want to stop sending the allowance event to subsequent allowances when the monitored property (typically a quantity to debit from allowances) reaches 0.

- *Monitored Property*: Provides the list of numerical properties of the selected allowance event class. Select a numerical property to monitor.
- *Revert allowance changes if condition is not fulfilled*: Click this option to apply the allowance modifications only when the condition *monitored property is lower than or equal to 0* is fulfilled.

> ⓘ Note
>
> The definition of output properties and the control of the sending of the allowance event require that the allowance logic modifies the monitored property of the event.

Selection (Tab)

This configuration table defines the conditions for selecting the allowances that will receive the allowance event.

 : Adds a row to the table.

 : Deletes a row from the table.

Property Name: Select the name of the allowance property, which can be:

- A parameter of the allowance interface [page 91]
- A counter of the allowance interface

  > → Remember
  >
  > The counters that are defined in the allowance interface for shared allowances are not available in the configuration table.

- A default allowance property such as:
  - The unique identifier of the allowance
  - The allowance activation date/time (aka *Validity Start Date*)

- The allowance expiration date/time (aka *Validity End Date*)

Type: Displays the type of the allowance property (*Date*, *Number*, or *String*)

Operator: Provides the list of comparison operators. Select an operator:

- *Date* (Equals, Before, After, Before or Equals, After or Equals)
- *Number* (Equals, Greater Than, Lower Than, Greater Than or Equal To, Less Than or Equal To, Not Equal)
- *String* (Equals, Starts With, Ends With, Contains)

Comparison Value: Type a value or select a rating context property (a parameter or a counter).

> **Note**
>
> - You must verify that the selection criteria are consistent because you can apply different selection criteria to the same allowance property.
> - By default, 2 selection criteria are available and correspond to the criteria that are used to select non-expired allowances. These 2 criteria cannot be modified or deleted from the table.

Sorting (Tab)

On the *Sorting* tab of the *Allowance Event Sender* component, you define your sorting criteria based on the two-level rules for sorting the allowances:

- The first level consists in ordering the criteria depending on their position (rank) in the table. This ranking defines the sending priorities.
- The second level consists in defining the ascending or descending order of each criterion.

The criteria are the following:

- The parameters defined in the allowance interface [page 91]
- The counters defined in the allowance interface
- The allowance default properties:
  - The unique identifier of the allowance
  - The allowance activation date/time (aka *Validity Start Date*)
  - The allowance expiration date/time (aka *Validity End Date*)

> **Note**
>
> - If the values of the properties, parameters, or counters of two or more allowances are equal, the sort system uses the unique identifier of the allowances as the sort criterion. For example, if the values of the properties, parameters, and counters of three allowances A, B, and C are equal, the system sorts allowances A, B, and C according to their unique identifier.
> - The criteria cannot be duplicated in the list.
> - The counters that are defined in the *Allowance Interface* for shared allowances are not available in the table.

Example

On the *Sorting* tab of the *Allowance Event Sender* component, the table represents a set of rules for sorting allowances. Each row of the table represents a rule, and the first column (#) of the table represents the priority order of the rules.

If we consider a set of allowances a1, a2,..., an, and the following rules for sorting the allowances:

#	Name	Type	Order
1	Property 1	number	Descending
2	Property 2	date	Descending
3	Property 3	number	Descending

The set of allowances is sorted according to the order of the following properties:

1. Property 1: From the greatest number to the smallest number.
2. Property 2: From the earliest date to the oldest date.
3. Property 3: From the greatest number to the smallest number.

For example, if we sort the following set of allowances according to the previous rules:

	Property 1	Property 2	Property 3
Allowance 1	4	26/09/2013	2
Allowance 2	4	26/09/2013	28
Allowance 3	4	26/09/2013	4

The sort result is the following:

	Property 1	Property 2	Property 3
Allowance 2	4	26/09/2013	28
Allowance 3	4	26/09/2013	4
Allowance 1	4	26/09/2013	2

The system first sorts the allowances according to this order: Property 1, Property 2, and Property 3 (position in the table). But if the values of the properties (Property 1 and Property 2) are equal like in this example, the system sorts the allowances according to the different values of Property 3 (in the descending order of the criterion).

If we keep the same rules as in the previous example and if the set of allowances is the following:

	Property 1	Property 2	Property 3
Allowance 1	4	26/09/2013	9
Allowance 2	8	26/09/2013	6
Allowance 3	12	26/09/2013	1

The sort result is the following:

	Property 1	Property 2	Property 3
Allowance 3	12	26/09/2013	1
Allowance 2	8	26/09/2013	6
Allowance 1	4	26/09/2013	9

Because the values of Property 1 are not equal, the system first sorts the allowances according to Property 1 (position in the table) and according to the different values of Property 1 (in the descending order of the criterion). And the system ignores Property 2 and Property 3.

The configuration table defines the criteria and the priorities for sending the allowance event.

Use the following icons for your actions:

Icon	Action
	Adds a row to the configuration table. You define a new sorting criterion.
	Deletes a row from the table.
	Moves a row up in the table. You change the execution priority for the criterion that is defined in the row.
	Moves a row down in the table.

Fill in the *Sorting* tab to define the priorities that suit your business requirements:

#: Displays the sort order (rank) of allowances.

Name: Provides the list of properties, parameters, and counters that the system uses at runtime to sort the allowances. Select a property, a parameter, or a counter.

Type: Displays the property, the parameter, or the counter type.

Order: Provides the list of sort orders. Select a sort order: *Ascending* or *Descending* that applies to the sorting criterion.

1.42.10.4.3 Allowance Event Updater (Operator)

→ Tip

You can only use this component after an Event-Based Trigger [page 717] component in the allowance logic.


The **Allowance Event Updater** component is an operator component that allows you to modify the *allowance events* handled by the allowances configured in provider contracts (in customer master data). These modifications are visible in the output properties of the charged item mapping, in the "allowance logic" of other allowances, and in the Allowance Event Sender [page 777] component (for more information on the Allowance Logic, refer to the primary help of Core Tool [page 449]).

Use

As a pricing specialist, you use the *Allowance Event Updater* component when you need to **change the properties of an allowance event** sent to allowances by the system at runtime.

> → Remember
>
> Use the allowance event updater after an event-based trigger.

> Example
>
> A telecommunication operator rates duration-based voice calls. Before calculating the price of each call, a **`Voice Call`** allowance event is sent to allowances in provider contracts.
>
> As a pricing specialist and in order to model this pricing policy by marketing, you design and configure your settings:
>
> - The allowance event class related to the allowance event contains the **`Duration (sec)`** property that is a number representing the duration of the call.
> - The allowance logic has an *event allowance* for the *allowance event class* called **`Voice Call`**. To indicate that the current allowance has authorized a part of the duration of the call, you must modify the event by setting up the remaining duration in the *Duration (sec)* field.
>
> The configuration of a Trigger Allowance component requires the selection of an Allowance Event Class [page 441] that specifies the type of the allowance event.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

This tab includes a configuration table that allows you to fill in the allowance event properties provided by the *allowance event class* defined in the related Event-Based Trigger [page 717] component.

Name: The name of the property.

Type: The type of the property: Date (and Time), Number, or String.

Value: Select a property (mandatory) or type a value.

Related Information

Allowance Event Sender (Operator) [page 777]

Event-Based Trigger [page 717]

1.42.10.4.4 Allowance Property Introducer (Operator)



The **Allowance Property Introducer** component is an operator component that allows you to retrieve the property values of a set of allowances and save them in generated properties that are available for the next logic components of the decision tree.

> ⓘ Note
>
> - The available scope of allowances depends on the usage of logic components. For more information, refer to the matrix of components [page 676].
> - Persistent counters that are used by shared allowances cannot be selected for creating generated properties. About the sharing of allowances between subscriber accounts, refer to the primary help of Core Tool [page 12] and the SAP CC 2023 Application Help for more information.

Use

As a pricing specialist, you use the *Allowance Property Introducer* component when:

- The properties of the available allowances in a provider contract are needed for the decision tree of the price plan of a charge, the refill logic, or the allowance logic.
- Some properties of allowances must be reported in charged items.
- The properties of allowances must be used in the Allowance Event Sender (Operator) [page 777] component for sending an allowance event to allowances.

> ⚠ Restriction
>
> The *Allowance Property Introducer* component is forbidden within the **shareable allowance logic**.

> ✤ Example
>
> You want to retrieve the lowest and greatest credit values from the following set of allowances but only through the “USER” scope:

Allowance ID	Start Date	End Date	Credit (Counter)	Scope (Parameter)
1234	20xx-01-01	20xx-02-01	10	USER
1235	20xx-01-15	20xx-02-15	250	COMPANY
1236	20xx-01-28	20xx-02-28	300	USER
1238	20xx-02-01	20xx-03-01	120	COMPANY
1240	20xx-03-15	20xx-03-15	50	USER

To do that, you configure an *Allowance Property Introducer* component to select the allowances whose property named 'Scope' is equal to 'USER', and you create two generated properties named 'LowestValue' and 'GreatestValue' to save the values of the 'Scope' properties.

From the previous set of allowance properties, the *Allowance Property Introducer Operator* thus selects the allowances whose ID are 1234, 1236, and 1240:

Allowance ID	Start Date	End Date	Credit (Counter)	Scope (Parameter)
1234	20xx-01-01	20xx-02-01	10	USER
1236	20xx-01-28	20xx-02-28	300	USER
1240	20xx-03-15	20xx-03-15	50	USER

The *Allowance Property Introducer Operator* then creates a generated property named 'LowestValue' with the value of 10 (the allowance whose ID is 1234) and a generated property named 'GreatestValue' with the value of 300 (the allowance whose ID is 1236).

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

This tab includes two configuration tables:

- The *Allowance Selection Criteria* table that defines the criteria for processing the selection of allowances according to the values of their properties
- The *Generated Properties* table to define generated properties in which you save the values of the allowance properties

Allowance Selection Criteria

This configuration table defines the conditions for selecting the allowances in a provider contract.

 : Adds a row to the table.

 : Deletes a row from the table.

Property Name: Select the name of the allowance property, which can be:

- A parameter of the *Allowance Interface*
- A counter of the allowance interface
- A default allowance property such as:
  - The unique identifier of the allowance

- The allowance activation date and time (*Validity Start Date*)
- The allowance expiration date and time (*Validity End Date*)

Type: Displays the type of the allowance property (Date, Number, or String)

Operator: Provides the list of comparison operators. Select an operator:

- *Date* (Equals, Before, After, Before or Equals, After or Equals)
- *Number* (Equals, Greater Than, Lower Than, Greater Than or Equal To, Less Than or Equal To, Not Equal)
- *String* (Equals, Starts With, Ends With, Contains)

Comparison Value: Type a value or select a rating context property (a parameter or a counter).

> ⓘ Note
>
> - Because you can apply different selection criteria to the same allowance property, you must verify that the selection criteria are coherent.
> - If you have defined no selection criteria for selecting allowances, this means that all the allowances of the provider contract must be selected.
> - Only non-expired allowances can be selected.

Scope: The scope defines the allowances from which you want to retrieve the property values. Select one of the three options:

- *Only Allowances of the Contract*: Retrieves the property values that only belong to the allowances of the related provider contract
- *Allowances of the Contract and its Parent*: Retrieves the property values that only belong to the allowances of the related provider contract and its parent
- *Only Shared Allowances*: Retrieves the property values that only belong to the shared allowances

Share Identifier: Select a property to identify the group of shared allowances from which you want to retrieve the property values.

Generated Properties

This configuration table allows you to specify generated properties which are initialized with the result of operators. You need to select an operator that is applied to the selected allowance properties. The generated properties are then available as computed properties in the rating context.

Allowance Count Property Name: Type the name of the generated property that represents the count of selected allowances.

 : Adds a row to the table.

 : Deletes a row from the table.

Property Name: Select the name of the allowance property.

Type: Displays the type of the allowance property: *Date* or *Number*

Operator: Select one of the following operators to initialize the generated properties:

- *Greatest*: Returns the greatest value of a given allowance property. If no allowance has been selected, the default value defined in the allowance interface is returned.

- *Lowest*: Returns the lowest value of a given allowance property. If no allowance has been selected, the default value defined in the allowance interface is returned.
- *Sum*: Returns the sum of the allowance property if exists. If no allowance has been selected this operator returns '0'.

Generated Property Name: Type the name of the generated property.

> ⓘ Note
>
> You can give the name of an allowance property to any generated property.

1.42.10.4.5 Allowance Validity Period Updater (Operator)

> → Remember
>
> In the case of automatic allowance renewal and at runtime, if an *Allowance Validity Period Updater*[86] component tries to set up the validity start date/time of an allowance to a date/time that is earlier than its current validity start date/time, the validity period of the allowance is automatically shifted to the current validity start date/time.
>
> See also the Create Allowance Operator [page 792] component about the allowance renewal.


The **Allowance Validity Period Updater** component is an operator component that enables you to redefine the end date/time or redefine the start date/time and the end date/time of the validity period of the allowances.

This component is available in the Allowance Logic [page 449] and can be inserted in the decision tree after an Event-Based Trigger [page 717], a Recurring Trigger [page 720], or a One-Shot Trigger [page 723] component.

Use

As a pricing specialist, you can use the *Allowance Validity Period Updater* component when you need to change the validity period of an allowance.

> ⚠ Restriction
>
> The *Allowance Validity Period Updater Operator* component is forbidden within a **shareable allowance logic**.

> ✤ Example
>
> As a service provider, you create an allowance valid for six months for a customer. The allowance extends its validity period when a new allowance is created (carry-over).

[86] This rule is only true if the *Allowance Validity Period Updater* is used in a one-shot trigger (creation event) of the allowance logic and if the allowance was created by a one-shot trigger at expiration time.

At creation time, the new allowance generates an allowance event of the **ALLOWANCE_CREATED** type including the following fields:

- Its validity period start date
- Its validity period end date

The old allowance that receives the allowance event first checks that its end date is before the end date of the new allowance. If it is the case, the old allowance extends its own validity end date.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the business description of the component (optional).

Definition (Tab)

An allowance has a validity period, which is the period of time during which the allowance can be debited by the charging process. The validity period is specified according to two dates, the start date and the end date. Note that the start date is included in the validity period and the end date is excluded from the validity period: [start date, end date[.

Update the validity period according to the allowance status

> ⓘ Note
>
> When selecting a date property, keep in mind the following:
>
> - *Keep*: Does not change the date
> - *Creation Date*: Is the date when the allowance is created (only available in a Creation One-Shot Trigger)
> - *Event Date*: Is the date of the current event (Event, Recurring or One-Shot)
> - *Validity Start Date*: Is the current start date of the validity period of the allowance
> - *Validity End Date*: Is the current end date of the validity period of the allowance
> - You can select your own date properties created in the allowance logic (context properties)

> ⓘ Note
>
> The values of the *Validity Start Date* and *Validity End Date* properties are updated according to the modifications performed by the *Allowance Validity Period Updater Operator* and are then available for the next logic components that can use these dates.

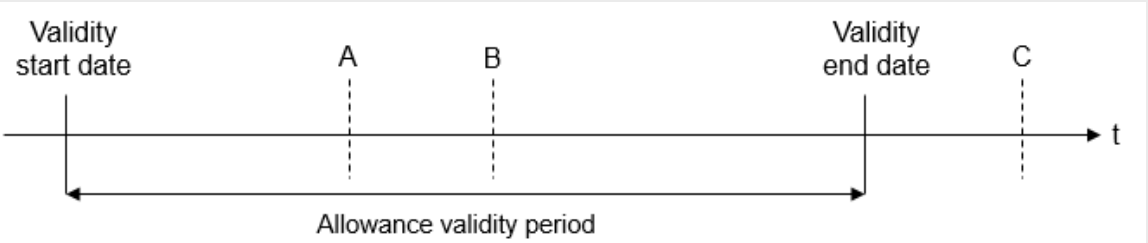
Inactive: The allowance has not been yet activated by SAP CC (activation process). You can then redefine the start and end dates of the validity period of the allowance:

- *Validity End Date*: Select a date property.
- *Allow to shorten the validity period*: If you select this option, the validity period of the allowance can be either shortened or extended; otherwise, the validity period of the allowance can only be extended.

Note

- If the *Allow to shorten the validity period* option is selected, make sure that the validity end date is greater than the following dates:
  - The date of the current event, which corresponds to the value of the **`Event Date`** property
  - The date when the last recurring or one-shot was triggered
  - The date of the latest reservation which may have accessed the allowance (so that such reservation is not impacted by the modification of the validity end date)
- If the specified validity end date is not greater than these dates, the validity end date is automatically set to the greatest of these dates, plus 1 second (because the end date is excluded from the validity period).
- If the *Allow to shorten the validity period* option is not selected, make sure that the new value of the validity end date is greater than its current value. Otherwise, the validity end date is not modified.

Example



- If the end date of the validity period of the allowance is modified to the date C, the modification is performed as configured whether the *Allow to shorten the validity period* option is selected or not.
- If the end date of the validity period is modified to the date A:
  - If the *Allow to shorten the validity period* option is not selected, the end date of the validity period of the allowance is not modified.
  - If the *Allow to shorten the validity period* is selected and there is no reservation in progress, the end date of the validity period of the allowance is modified to the date A as configured.
  - If the *Allow to shorten the validity period* option is selected and there is a pending reservation at the date B, the end date of the validity period of the allowance is modified to the date B plus 1 second instead of the date A.

Expired: The allowance is expired. You cannot update the validity period of the allowance.

1.42.10.4.6 Arithmetic Computation Operator


The **Arithmetic Computation Operator** component carries out an arithmetical operation between two numerical values or properties. The calculation result is saved in a new property. The Arithmetic Computation Operator component can carry out the following operations:

- Addition

- Subtraction
- Multiplication
- Division
- Modulo
- Power

Use

To calculate a price in a price plan [page 299], you can combine one of the Function [page 727] components with the *Arithmetic Computation Operator*. The *Arithmetic Computation Operator* allows you to calculate numerous intermediate results that are saved in new properties; therefore, you can reuse these properties to calculate the final price by using a function component.

Example

In a price plan, the price of a telephone call is calculated by using the following basic formula:

Call price = Constant x Base price x Call Duration

`Base price` is a parameter in the offer and represents a price per minute. *Call Duration* is a numerical property. The generic function component, which is a "AXY + B" type formula, allows you to calculate **`call price`**. However, for certain customers, a percentage-type discount that is saved as a parameter is applied to **`Base price`**. To calculate the discount, you can insert an *Arithmetic Computation Operator* component before the generic function component as shown in the following table:

Step	Component	Calculation
1	*Arithmetic Computation Operator*	Discounted Price = Base price x Customer discount
2	Generic Function [page 731]	Call price = Constant x Discounted Price x Call Duration

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the description of the component (optional).

Definition (Tab)

First Operand: Type a constant value or select a numerical property that contains the value.

Operator: Provides the list of arithmetical operators. Select one of the following operators:

- *Plus*
- *Minus*
- *Multiplied By*
- *Divided By*
- *Modulo*
- *Power*

Second Operand: Type a constant value or select a numerical property.

Computed Property: Type the name of the new computed property that contains the result of the operation.

Rounding Mode: Provides the list of modes for rounding the decimal portion of the result. The default option is *To the nearest value*. Select one of the following rounding modes:

- *To the next greater value*
- *To the previous value*
- *To the nearest value*

Scale: Provides the list of decimal places of the result. The default option is *12*.

1.42.10.4.7 Concatenate Operator


The *Concatenate Operator* component allows you to concatenate two or more string properties or character strings.

> ✤ Example
>
> For reporting purpose, an operator needs to create a CSV file which must contain some information such as customer name, service, consumption date, and duration. In a price plan [page 299], the operator uses a *Concatenate Operator* component to generate a computed property from the properties of the price plan. For example, the operator can build sentences like this one: "John Smith consumed the Internet service QuickAccess on Monday the 03 of October 2011 at 11:55:26 during 15 seconds".

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Computed Property Name: Type the name of the new string property that retrieves the concatenation result.

Fields Separator: Provides the set of separators that is used to separate the concatenated properties. Select a separator (*Semicolon*, *Tabulation*, and so on), or click *Other*, and then type a character that you want as a separator.

- *No Separator*: Select this option if you do not want to separate the concatenated properties
- *Semicolon*: Select this option if you want a semicolon as a separator
- *Tabulation*: Select this option if you want a tabulation as a separator
- *Comma*: Select this option if you want a comma as a separator
- *Space*: Select this option if you want a space as a separator
- *Slash*: Select this option if you want a slash as a separator
- *Other*: Type a character that you want as a separator

Elements to Concatenate: Allows you to choose the string properties or type the character strings that you want to concatenate. The Concatenate Operator concatenates the values of properties or character strings in the order of the table from the top to the bottom of the table. To add a property to the table, click  , and then select a property or type a new property.

 : Adds a row to the table.

 : Deletes a row from the table.

 : Moves a row up in the table to change the order of properties or character strings.

 : Moves a row down in the table to changes the order of properties or character strings.

1.42.10.4.8 Create Allowance Operator

→ Remember

- The *Create Allowance Operator* is executed at the end of the decision tree if there is one master charge or at the end of all the decision trees if there is one or more dependent charges. For more information on charge plans [page 343], charges [page 290], and dependencies between master [page 293] and dependent charges, [page 294] refer to the dedicated topics in the primary help of Core Tool.
- The configuration of a *Create Allowance Operator* component requires the selection of an Allowance Plan [page 466] that specifies the type of the allowance. If you want to create a shared allowance, you must select a shareable allowance plan. For more information on the allowance plan, refer to the dedicated topic in the primary help of Core Tool.
- In the Allowance Logic [page 449], you cannot use the *Create Allowance Operator* in the Event-Based Trigger [page 717] and Recurring Trigger [page 720] branches, but you can use the *Create Allowance Operator* in a One-Shot Trigger [page 723] branch if you configure the *One-Shot Trigger* with an expiration event and no time shift.
- In the case of automatic allowance renewal and at runtime, if a one-shot trigger (creation event) that contains a *Create Allowance Operator* is executed in the allowance logic, the duration between the validity start date of the current allowance and the validity start dates of the newly created allowances cannot be less than one day (calendar). If this duration is greater than one day, the validity period is

automatically shifted. The new validity start dates are automatically set to the effective date of the current allowance plus one day while the validity period of the allowances do not change. See also the Allowance Validity Period Updater (Operator) [page 787] about the allowance renewal.


The **Create Allowance Operator** creates an allowance for the related provider contract [page 573] through a refill plan [page 421], a charge plan [page 343], or in an allowance plan [page 466].

> ⚠ Restriction
>
> Rating Trial Runs do not consider allowance creation.

Use

As a pricing specialist, you can use the *Create Allowance Operator* component to create allowances for your provider contracts.

> ⚠ Restriction
>
> The *Create Allowance Operator* component is forbidden within a **shareable allowance logic**.

> ✤ Example
>
> An operator sells data plans to its customers. They can buy recharges including data buckets that are valid for a limited period of time and can be modeled as allowances. The *Create Allowance Operator* can be used to create such allowances. In this context, the initial value of the bucket can be represented as an allowance parameter.

Identification Area

Name: Type the name of the component (mandatory).

Description: Type the description of the component (optional).

Definition (Tab)

Allowance Plan: Provides the list of *Allowance Plans*. The *Create Allowance Operator* component must reference an allowance plan whose status is set to `Released`. Select an *Allowance Plan* (mandatory). If necessary, click


to open the selected allowance plan. On the right side of the field, a text displays if the *Allowance Plan* is shareable or not.

> **Note**
>
> You cannot create a shared allowance in the allowance logic because the renewal of allowances is not allowed in this context.
>
> You can only create a shared allowance in a charge or the refill logic. For more information, refer to the dedicated section in the primary help of Core Tool.

Create a Shared Allowance: Select this box if you want to create a shared allowance. The *Allowance Plan* must be shareable.

- *Share Identifier*: Type or select an identifier that identifies a group of shared allowances.

Validity Start Date: Provides the list of date properties. Select a start date property for the Allowance validity period (mandatory).

Validity End Date: Provides the list of date properties. Select an end date property for the Allowance validity period (mandatory).

Allowance Parameters: Displays the table of Allowance parameters provided by the related *Allowance Plan*. Fill in the values of the allowance parameters:

- *Name*: Displays the name of the parameter.
- *Type*: Displays the type of the parameter (*Date*, *Number*, *String*, *Counter*, *Mapping Table Key*, *Mapping Table Identifier*, or *Range Table Identifier*).
- *Value*: Select a property, or type a value.

1.42.10.4.9 Date Shift Operator


The *Date Shift Operator* component calculates a new date by adding a period of time to an existing date property or by subtracting a period of time from an existing date property. The new date is saved in a new date property. The time unit is expressed in seconds, minutes, hours, or days. You use this component to change an existing date into an earlier or a later date.

> **Example**
>
> In a chargeable item class called **`Telephone Service`**, you want to give a customer a time-credit of six seconds for each call. To define the payable period, you use the *Date Shift* component to recalculate the start date of the customer's call.
>
> > **Note**
> >
> > You can easily calculate the local time based on GMT time if you know the GMT offset. You enter the consumption date as the original date to which you add the numerical property `GMT Offset` representing the period of time in hours. The calculation result determines the local time that is saved in the computed property. You can also use the local time in the subbranch of the price plan.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Date Property Name: Provides the list of date properties. Select a date property.

Computed Property Name: Type the name of the new date property that retrieves the calculation result.

Operation: Provides the list of operations. The default option is *Subtraction*. Select one of the following operations:

- *Subtraction*: The period of time is subtracted from the date property
- *Addition*: The period of time is added to the date property

Duration: Allows you to type the period of time that is added to or subtracted from a numerical property, or allows you to select a numerical property that represents the period of time.

> **Note**
>
> Whether you type the period of time or select a numerical property, if the period of time is a decimal number, the fractional part of the decimal number is truncated. Only the integer part is added to or subtracted from the date property. For example, adding 1.5 hours to August 28, 2007 2:13:48 PM gives the result August 28, 2007 3:13:48 PM, or subtracting 2.10 hours to August 28, 2007 2:13:48 PM gives the result August 28, 2007 0:13:48 PM.

Duration Unit: Provides the list of time units. The default option is *seconds*. Select one of the following time units:

- *seconds*
- *minutes*
- *hours*
- *days*
- *months*
- *years*

1.42.10.4.10 Decrement Counter Operator


The **Decrement Counter Operator** component debits a counter (transient or persistent) without exceeding the current value of the counter.

Use

The *Decrement Counter Operator* is used to decrement counters defined in shared allowances (in provider contracts).

> ### Example
>
> As a service provider, you create a bucket of 100,000 SMS for a company. This bucket is shared by the employees of this company and implemented as a shared allowance that includes a persistent counter.
>
> Based on the previous example, a bucket of 100,000 SMS is debited through a *Debit Counter Operator* each time an employee uses a SMS. When the value of the bucket is equal to 0, employees cannot send SMS anymore.

See also: Increment Counter Operator [page 803]

> ### Note
>
> The *Decrement Counter* component was known as the *Debit Counter* component in former versions of SAP CC.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Counter Name: Provides the list of counters. Select a counter.

Decrement: Type or select the positive value to be withdrawn from the counter.

> ### Caution
>
> If you type or select a negative value, an exception is raised during the rating process.

Generated Properties: Allows you to name two new numerical properties.

- *Non Decremented*: Type the name of the new property that contains the value that could not be withdrawn from the counter. For example, the counter is equal to 15. If the decrement is equal to 16, the counter is debited of 15, and its new value is equal to 0 because the counter cannot be less than 0. The non decremented (decrement minus value of the counter) is equal to 1 and is the value that could not be debited from the counter
- *Decremented*: Type the name of the new property that contains the value that is withdrawn from the counter. For example, the counter is equal to 20. If the decrement is equal to 18, the decremented is equal to 18. But if the decrement is equal to 24, the decremented is equal to 20 because the counter cannot be less than 0

1.42.10.4.11 Duration Operator

 The *Duration Operator* component calculates the period of time between two dates. The period of time is expressed in seconds, minutes, hours, or days and saved in a new numerical property. You can specify the precision unit, the rounding mode, the rounding step, and the threshold of the period of time.

Use

In a price plan [page 299], you may need to calculate a price that is dependent of time. To compute the period of time, you use the *Duration Operator* that calculates the period of time between a start date and an end date.

> Example
>
> In a chargeable item class called **`Video Conference`**, you charge the customer $0.10 per video conference minute. After the start and stop date of the video conference, a *Duration* component calculates the duration of the video conference in the price plan.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

The *Duration Operator* calculates the period of time between two dates: The start date property and the end date property. The result is saved in a generated property.

Start Date Property Name: Provides the list of date properties. Select a start date property.

End Date Property Name: Provides the list of date properties. Select an end date property.

Name of the Generated Property: Type the name of the new numerical property that retrieves the calculation result.

Time Unit: Provides the list of time units. Select one of the following units:

- *seconds*
- *minutes*
- *hours*
- *days*

Advanced tab

You set up the parameters to make calculation.

Rounding Mode: Provides the list of modes for rounding the decimal portion of the period of time. The default option is *To the next greater value*. Select one of the following rounding modes:

- *To the next greater value*
- *To the previous value*
- *To the nearest value*

Precision Unit: Provides the list of time units. The default option is *seconds*. Select one of the following time units:

- *seconds*
- *minutes*
- *hours*
- *days*

Rounding Step: Allows you to type a step rule to round the period of time, or allows you to select a numerical property that represents the step rule. The default option is *1*. Example: If the rounding step value is 5 seconds and the rounding mode is *To the next greater value*, the period of 47 seconds is rounded to 50 seconds. In this example, the rounding step then consists of 5-second intervals.

Threshold: Allows you to type the minimum value of the period of time, or allows you to select a numerical property that represents the minimum value. The default option is *0*. Example: If the threshold is 30 seconds, a period less than 30 seconds is rounded to 30 seconds.

1.42.10.4.12 Duration Adapter Operator


The *Duration Adapter* operator component converts a period (of time) into another time unit. The calculation result is saved in a new numerical property. You can convert into days a numerical property expressed in seconds or in hours. You can specify the time unit of the source and the target, the rounding mode, the rounding step, and the threshold of the target.

Use

In a price plan [page 299], you may need to make calculations dependent of time and that you want to convert into a different time unit. To do that, you use the *Duration Adapter* component to convert a period (of time). The conversion result is saved in a new numerical property.

> Example
>
> In a chargeable item class called **`Telephone Service`**, a phone call duration is expressed in seconds. You need to convert the call duration into minutes because your price is defined in minutes. You use the *Duration Adapter Operator* to convert the call duration into minutes.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

You define the names and the time unit of the source, the new numerical property that retrieves the conversion result, and the time unit for converting the source.

Source Duration Name: Provides the list of numerical properties. Select a numerical property.

Source Time Unit: Provides the list of time units of the source. Select one of the following time units:

- *seconds*
- *minutes*
- *hours*
- *days*

Target Duration Name: Allows you to type the name of the new property that retrieves the conversion result.

Target Time Unit: Provides the list of time units for converting the source. Select one of the following time units:

- *seconds*
- *minutes*
- *hours*
- *days*

Calculation Parameters (Tab)

You set up the calculation parameters to convert the conversion result.

Rounding Unit: Provides the list of time units for rounding the conversion result. The default option is *seconds*. Select one of the following time units:

- *seconds*
- *minutes*
- *hours*
- *days*

Rounding Mode: Provides the list of modes for rounding the decimal portion of the period of time. The default option is *To the next greater value*. Select one of the following rounding modes:

- *To the next greater value*
- *To the previous value*
- *To the nearest value*

Rounding Step: Allows you to type a step rule to round the period of time, or allows you to select a numerical property that represents the step rule. The default option is *1*.

Example: If the rounding step value is 5 seconds and the rounding mode is *To the next greater value*, the period of 47 seconds is rounded to 50 seconds. In this example, the rounding step then consists of 5-second intervals.

Threshold: Allows you to type the minimum value of the period of time, or allows you to select a numerical property that represents the minimum value. The default option is *0*.

Example: If the threshold is 30 seconds, a period less than 30 seconds is rounded to 30 seconds.

1.42.10.4.13 Filter Operator


The *Filter Operator* compares each value of a multi-valued property of string type with one or more string values. Each time that the result of a comparison is true, the value of the multi-valued property is stored in a specified multi-valued property associated with the comparison. Each time that the result of a comparison is false, the value of the multi-valued property is rejected and stored in a specified multi-valued property.

> **Note**
>
> If a value of the multi-valued property matches several string values, the value is stored in the specified multi-valued properties associated with the comparisons for which the result is true.

> **Example**
>
> You want to filter a multi-valued property that contains the following values:
>
> - image/jpeg:2500
> - image/jpeg:4000
> - image/jpeg:5000
> - image/gif:4000
> - image/gif:10000
> - text/plain:100
> - audio/mp3:20000
> - Message:20
>
> Each value of this multi-valued property is compared with the comparison values that are shown in the next table. When the result of a comparison is true, the value is stored in the property associated to the comparison (`ImageFolder`, `TextFolder`, and so on). If a value of the multi-valued property does not match any string values, the value is stored in the property `RejectedValues`.

Property Name	Comparator	Comparison Value
`ImageFolder`	*starts with*	**`image`**
`TextFolder`	*starts with*	**`text`**

Property Name	Comparator	Comparison Value
`AudioFolder`	*starts with*	**`audio`**
`JpegImageFolder`	*contains*	**`jpeg`**
`GifImageFolder`	*contains*	**`gif`**

Once the *Filter Operator* have completed the comparisons, each property contains the following list:

`ImageFolder`	`TextFolder`	`AudioFolder`	`JpegImageFolder`	`GifImageFolder`
image/jpeg:2500	text/plain:100	audio/mp3:20000	image/jpeg:2500	image/gif:4000
image/jpeg:4000			image/jpeg:4000	image/gif:10000
image/jpeg:5000			image/jpeg:5000	
image/gif:4000				
image/gif:10000				

The value "Message:20" does not match any string values. This value is thus stored in the multi-valued property `RejectedValues`.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Source: Allows you to select a multi-valued property that you want to filter.

- *Property Name*: Provides the list of multi-valued properties. Select a multi-valued property.

Target: Allows you to create the table of comparisons that filters the selected multi-valued property.

- *Unmatched Values Property*: Allows you to name the property that retrieves all the values of the multi-valued source property that do not match the table of comparisons.
-  : Adds a row to the table. Click this button to add a comparison.
-  : Deletes a row from the table. To remove a comparison, select the corresponding comparison row, and then click this button.
- *Property Name*: Allows you to type the multi-valued source property name that retrieves all the values of the multi-valued source property that match the comparison.
- *Operator*: Provides the list of comparators. Select an operator. The comparators are the following:

- *is equal to*: All the values that are equal to the value of Comparison Value are saved in the Property Name column entry
- *starts with*: All the values that start with the value of Comparison Value are saved in the Property Name column entry
- *ends with*: All the values that end with the value of Comparison Value are saved in the Property Name column entry
- *contains*: All the values that contain at least the value of Comparison Value are saved in the Property Name column entry
- *matches*(*): All the values that match the regular expression of Comparison Value are saved in the Property Name column entry
- *matches (capturing group)*: All the values that match the regular expression containing parenthesis in Comparison Value are saved in the Property Name column entry. For example, in the following multi-valued source property containing the string "text/html;12000;text/html;06000;text/html;03000", you can retrieve the values "text/html;12000", "text/html;06000" and "text/html;03000" with a regular expression such as "text/.+;([0-9]+)".

- *Comparison Value*: Allows you to type a value or select a property that is compared with each value of the source property. You cannot type a regex or a capturing group.

(*) The last two comparisons use regular expressions that you should know. These regular expressions are compliant with the POSIX standard. For more information about regular expressions, consult the relevant J2SE websites.

The following table shows the regular expressions (Regex) that are equivalent to the most used wildcards:

Wildcard	Regex	Comment
*	*	Represents a number of characters (including zero or more characters). If you type "cd*" or "cd.*", you can retrieve "cda", "cdrom", "cdrecord" and anything that starts with "cd", "cd" itself included.
?	.	Represents a single character. If you type something like "hd?" or "hd.", you can retrieve hda, hdb, hdc and every other letter/number between a-z, 0-9.
[]	[]	Specifies a range. If you type m[aou]m, this expression can become: mam, mum, mom. If you type m[a-d]m, this expression can become anything that starts and ends with m and has a character a to d in-between. For example: mam, mbm, mcm, mdm. This kind of wildcard specifies an "or" relationship.
\	\	Represents an "escape" character, which role is to protect a subsequent special character. Thus, "\\" searches for a backslash.

1.42.10.4.14 Increment Counter Operator

 The **Increment Counter Operator** component credits a distributed counter.

Use

The *Increment Counter Operator* is used to increment counters defined in shared allowances (in provider contracts).

> ### Example
>
> A company wants to be notified when the number of SMS sent by its employees exceeds 50,000.
>
> As a pricing specialist, you define a specific counter which is incremented through an *Increment Counter Operator* component each time an employee sends an SMS. Once this counter reaches 50,000, the company receives a notification reporting that the threshold is reached.

See also: Decrement Counter Operator [page 795]

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Counter Name: Provides the list of counters. Select a counter.

Increment: Type or select the positive value to be added to the counter.

> ### Note
>
> If you type or select a negative value, an exception is raised during the rating process.

Generated Property: Displays the counter value after the increment operation.

1.42.10.4.15 Macro Introducer (Operator)


The **Macro Introducer** operator component allows you to use an existing pricing macro in a plan. This pricing macro contains some reusable logic (as a combination of pricing components). This component allows you:

- To select a pricing macro [page 121] in a pricing catalog
- To provide the values to the rating context properties that the pricing macro needs to make calculations
- To retrieve the calculation results from the generated properties of the pricing macro

The *Macro Introducer* component uses the price plan properties:

- To provide the values to the pricing macro
- To retrieve the calculation results from the pricing macro

> ### Example
>
> An operator bills telephone calls according to:
>
> - The duration of calls
> - The location of called numbers
> - The type of telephone such a fixed line or a mobile
>
> As the pricing specialist, you design and configure a pricing macro named **`MacroTel`** that computes the cost of calls that is based on:
>
> - The prefix of the called number (**`Prefix_Called`**)
> - The type of telephone (**`Phone_Type`**)
> - The duration of the call (**`Call_Duration`**)
>
> The *Macro Introducer* component provides this information to the pricing macro through price plan properties. To calculate the price of each call, the pricing macro needs the price code provided by a translation table. The pricing macro returns the total amount of calls and the price code, both of which are retrieved by the *Macro Introducer* through price plan properties. In the price plan, the *Macro Introducer* component shows the following information:
>
> - The pricing macro **`MacroTel`**
> - The price plan properties (**`Prefix_Called`**, **`Phone_Type`** and **`Phone_Type`**) that contain the values to initialize the rating context properties of the pricing macro
> - The price plan property **`Total_Amount`** that retrieves the computed amount of the pricing macro
> - The price plan property **`Code_Price`** that retrieves the price code of the generated property of the pricing macro

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Pricing Macro: Provides the list of available pricing macros [page 121]. Select the pricing macro that yo want to use or reuse and that suits your needs.

Mapping of Rating Context: Displays the rating context properties of the pricing macro, and allows you to assign a property of the plan to each rating context property. The pricing macro runs with the values of the plan properties.

- *Name*: Displays the name of the rating context property.
- *Description*: Displays the description of the rating context property.
- *Mapping Property*: Provides the list of the plan properties. Select the property that initializes the rating context property of the pricing macro.

Mapping of Generated Properties: Displays the generated properties of the pricing macro, and allows you to assign a new property to each generated property that retrieves the calculation results from the pricing macro.

- *Computed Amount Property Name*: Allows you to type the name of the new property that retrieves the computed amount of the pricing macro.
- *Name*: Displays the name of the generated property.
- *Description*: Displays the description of the generated property.
- *Generated Property Name*: Allows you to type the name of the new property that retrieves the value of the generated property of the pricing macro. The default property is the generated property that is displayed in the *Name* column. Double-click the entry to modify it. You can use this new property in the plan.

Related Information

Pricing Macros [page 121]
Working with Pricing Macros [page 122]

1.42.10.4.16 Multi-Valued Evaluator (Operator)


The *Multi-valued Evaluator Operator* counts the number of values which are in a multi-valued property. If the multi-valued property is of numerical type, the *Multi-valued Evaluator Operator* calculates the sum and the average of all the numerical values which are in the multi-valued property and determines the smallest and the greatest of these numerical values.

> **Note**
>
> Use the Property Type Convertor (Operator) [page 810] to convert an existing property into a multi-valued property.

Example

To calculate a price that depends on a number of recipients, the rating branch of a price plan [page 299] needs two components:

- A *Filter Operator* that tests the recipient addresses
- A Multi-valued Evaluator that counts the number of values which are in the multi-valued property **`RecipientsAddresses`**

To identify the recipients of the service providers, a Filter Operator [page 800] tests their address. The *Filter Operator* is configured to create one multi-valued property per service provider: **`MeteoRecipients`**, **`NewsRecipients`**, **`ExchangeRecipients`**, **`SpotsRecipients`**, **`TrafficRecipients`** and **`TaxiRecipients`** . All the addresses that do not match a comparison are added to the property **`RejectedAdresses`**.

The configuration of the *Filter Operator* relating to the service provider Meteo is the following:

Property Name	Comparator	Comparison Value
`MeteoRecipients`	*ends with*	@meteo.com

In this example, it is supposed that once the *Filter Operator* have completed the comparisons:

- The multi-valued string **`MeteoRecipients`** contains two values: "greatbritain.country@meteo.com" and "london.city@meteo.com". The multi-valued property **`RejectedAdresses`** contains "ysmith.messagestore@blue.uk".
- The multi-valued properties of the other recipients are empty.

Once the *Filter Operator* have completed its tasks, the *Multi-Valued Evaluator Operator* counts:

- The number of recipients per service provider, the number of which is six.
- The number of addresses that are retrieved in the multi-valued property of each recipient, which is two for MeteoRecipients and zero for the other recipients.

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Property Name: Provides the list of multi-valued properties of the plan. Select the multi-valued property that you want to evaluate.

Count Property Name: Allows you to name the property that retrieves the number of all the string values which are in a multi-valued property of string type.

Sum: This area is only available if you want to evaluate a multi-valued property of numerical type.

- *Property Name*: Allows you to name the property that retrieves the sum of all the numerical values which are in a multi-valued property.
- *Default Value*: Allows you to type a default value that is assigned to the previous property if the multi-valued property is empty. The default value is mandatory if the *Property Name* box is not empty.

Minimum: This area is only available if you want to evaluate a multi-valued property of numerical type.

- *Property Name*: Allows you to name the property that retrieves the smallest value of all the numerical values which are in a multi-valued property.
- *Default Value*: Allows you to type a default value that is assigned to the previous property if the multi-valued property is empty. The default value is mandatory if the *Property Name* box is not empty.

Maximum: This area is only available if you want to evaluate a multi-valued property of numerical type.

- *Property Name*: Allows you to name the property that retrieves the greatest value of all the numerical values which are in a multi-valued property.
- *Default Value*: Allows you to type a default value that is assigned to the previous property if the multi-valued property is empty. The default value is mandatory if the *Property Name* box is not empty.

Average: This area is only available if you want to evaluate a multi-valued property of numerical type.

- *Property Name*: Allows you to name the property that retrieves the average of all the numerical values which are in a multi-valued property.
- *Default Value*: Allows you to type a default value that is assigned to the previous property if the multi-valued property is empty. The default value is mandatory if the *Property Name* box is not empty.

1.42.10.4.17 Output Property Update Operator


The *Output Property Update Operator* component allows you to modify the generated properties of a pricing macro.

Use

In a pricing macro [page 121], you can create generated properties to which you assign default values. Before retrieving these default values in the price or charging plan through a Macro Operator [page 804] component, you can modify them by using the *Output Property Update Operator*.

> ### Example
>
> An operator bills telephone calls according to the duration and the geographical zone of called numbers. The price plan contains a pricing macro **`TelNumber`** that analyzes the telephone number and returns a generated property **`ZoneCode`**. This property is retrieved in the price plan to calculate the price of the call. In the pricing macro, an *Output Property Update Operator* is used to assign a specified value to **`ZoneCode`**". This value depends on the prefix of the telephone number. In the price plan, a *Macro Operator* is used to retrieve the calculation results from the pricing macro. The *Macro Operator* component shows the following information:
>
> - The **`TelNumber`** pricing macro
> - The **`NumberCalled`** price plan property that provides the telephone number to the pricing macro

- The **`GeographicZone`** price plan property that retrieves the value of the generated property **`ZoneCode`**

Identification area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition tab

Output Property Name: Provides the list of generated properties of the pricing macro. Select the generated property to which you want to assign a new value.

Value To Set: Provides the list of rating context properties of the pricing macro or allows you to type a value. Select a rating context property or select *Numerical Value*, and then type a numerical value to set the generated property. Values or properties must be the same type as the generated property.

> ⓘ Note
>
> You cannot type a value for a multi-valued string or a multi-valued number. You must select *Empty* or a property.

1.42.10.4.18 Property Introducer (Operator)


The **Property Introducer** operator component allows you to create new properties for a plan or to modify the properties that are created with an above Property Introducer component. This component can also be used to access properties that are not available for other components (for example, the contract identifier).

Use

In a pricing plan, you can insert a first *Property Introducer* operator component to create new properties available for the branch of a rate component. If you need to modify the values of these properties in the branch of a component, you can insert a second Property Introducer component in this branch to assign new values to the properties that are created with the first Property Introducer component.

> ✤ Example
>
> In a price plan [page 299], you need an intermediate property to save a result. You then insert a Property Introducer component into your price plan to create the property **`Intermediate result`**.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

This tab includes a table that allows you to create new properties.

 : Adds a row to the table

 : Deletes a row from the table

Property Name: Type the name of the new property.

Type: Select the type of the new property: *Date*, *Number*, or *String*.

Value: Type a value, or select a property, a parameter, or a counter.

> ⓘ Note
>
> An empty string is a possible value.

Attribute: You can assign one of the following attributes to the selected value:

- *Not Applicable*: No attribute is assigned to the selected value.
- *Current Value*: The selected value includes the modifications that result from ongoing reservations. This attribute only applies to properties of number type and values that are persistent counters.
- *Reservation-free Value*: The selected value does not include the modifications resulting from ongoing reservations. This attribute only applies to properties of number type and values that are persistent counters.
- *Length*: The property includes the length of the selected value. This attribute only applies to properties of number type and values of string type.

> ⓘ Note
>
> - The reservation-free value is not compatible with the cancellation of the previous transaction in the session-based charging. In case of cancellation, the reservation-free value will be the current value at the time of the canceled reservation.
> - The reservation-free value of the persistent counter is provided for information and reporting and must be carefully used in the decision tree.

> ✤ Example
>
> In the offer of a telephone operator, an **`SMS count`** persistent counter represents the number of SMS that the consumer can use. The reservation-free value is 3, which means that the consumer can use 3 SMS. And the current value of **`free SMS`** is 1 because there are 2 ongoing reservations for this counter.

You must not authorize the consumer to use more than 1 free SMS even if the reservation-free value is 3 because you cannot predict if the reservations will be confirmed or canceled.

1.42.10.4.19 Property Type Convertor (Operator)


The **Property Type Convertor** operator component creates a new property from an existing property by modifying the type of the existing property.

Use

In a plan, some calculations use numerical data which are included in string properties. The Property Type Convertor operator component allows you to convert a string property into a number property.

The table below shows the authorized and unauthorized conversions:

Source property type	Target property type	Comment
String	String	Authorized
String	Date	Authorized
String	Number	Authorized
String	Multi-valued string	Authorized
String	Multi-valued number	Authorized
Date	String	Authorized
Date	Number	Authorized
Date	Date	**Not authorized**
Number	String	Authorized
Number	Date	**Not authorized**
Number	Number	**Not authorized**
Multi-valued string	Multi-valued string	Authorized
Multi-valued string	String	Authorized
Multi-valued string	Multi-valued number	Authorized
Multi-valued number	String	Authorized
Multi-valued number	Multi-valued string	Authorized
Multi-valued number	Multi-valued number	**Not authorized**

> ### Example
>
> An operator uses a translation table providing a discount percentage from the number of units that are consumed by customers. The `number of units consumed` property is numerical.
>
> Using the Property Type Convertor operator component, you can convert this property into string before putting it into the translation table.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Source

- *Property Name*: Provides the list of the properties. Select a property that you want to convert.
- *Property Type*: Displays the type of the selected property. "Undefined" is the default value.

Target

- *Property Name*: Allows you to type the property name that retrieves the result of the conversion.
- *Property Type*: Provides the list of conversion types. The available types vary according to the source property type. Select one of the following conversion types:
  - *String*: Represents a character string. For example, "image", "text" and "price" are strings.
  - *Number*: Represents a decimal or an integer number. For example, 25.60, 6523 and 4.55 are numbers.
  - *Date*: Represents a combination of the day, the month, the year and the time. For example, August 12, 2008 12:00:00 AM is a date.
  - *Multi-valued String*: Represents a list of string values. For example, the multi-valued string `ImageType` contains the following list: image/jpeg:5000, image/gif:10000, image/eps:25000 and image/pic:8000.
  - *Multi-valued Number*: Represents a list of numerical values. For example, the multi-valued number `ImageSize` contains the following list: 5000, 10000, 15000, 20000.

Advanced (Tab)

This tab is separated in four areas. Each area corresponds to a type of property, but only one or two areas are available according to the conversion type that you select on the *Definition* tab.

For example, if you want to convert a string into a string, only the *String Parameters* area is available.

> **Note**
>
> About string parameters:
>
> - The values of *Start Offset* and *Stop Offset* can range from 1 to the size of the string. The value of *Stop Offset* must be equal to or greater than the value of *Start Offset*.
>
> Particular values:
>
> - If *Stop Offset* is set to -1, this value represents the size of the string.
> - If *Start Offset* is set to 1 and if *Stop Offset* is set to 0, the combination of these values represents an empty string.
> - If *Start Offset* is set to a value greater than 1 and if *Stop Offset* is set to 0, the system generates an error.

String Parameters: Specifies the positions of start and end of a character substring in an input character string or in each string of a multi-valued string.

- *Start Offset*: Specifies a start position in the string or the multi-valued string by typing a number value or selecting *Start of String*, a counter, a property of number type, or a parameter of number type.
- *Stop Offset*: Specifies an end position in the string or the multi-valued string by typing a number value or selecting *End of String*, a counter, a property of number type, or a parameter of number type.

Date Parameters: Specifies the date format that allows you to convert a string into a date, or specifies the date field that allows you to convert a date either into a number or into a string.

- *Date Format*: Allows you to type or select a date format.

  The date format conventions are the following:
  - The character 'm' stands for month, 'd' for day and 'y' for year. For example, the date format 'm d y' represents dates such as 03 16 2021 or 05 26 2021.
  - A lower-case letter represents numbers, and an upper-case letter represents letters. For example, the date format 'D d M y' represents the date Tuesday 05 June 2001, in which 'D' represents Tuesday and 'M' represents June.
  - The '/' character is used to separate each part of the date. For example, the date format 'm/d' represents dates such as 03/16 or 05/26. The date format 'y-m-d' represents dates such as 2021-03-16 or 2021-05-26. Any other characters can be used as a separator.
- *Field to use*: Converts the part of a date into a number. The date format is 'D d M y'. For example, if you select the *Month* option, you can convert the date Tuesday 05 June 2001 into the number 6 because June is the 6th month of the year. To convert a part of a date into a number, select one of the following parts of the date:
  - *Year*: Converts the date into a number of years. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 2001.
  - *Month*: Converts the date into a number between 1 and 12. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 6 because June is the 6th month of the year.
  - *Day of Year*: Converts the date into a number between 1 and 366. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 156 because 05 June is the 156th day of the year.
  - *Day of Month*: Converts the date into a number between 1 and 31. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 5 because 05 June is the 5th day of the month.
  - *Hour*: Converts the date into a number between 0 and 12. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 2 because the number of hours in 02:12:46 PM is 2. The particular values are the following: The time 12:00:00 AM is turned into 0, the time 12:00:00 PM is turned into 12.

- *Minute*: Converts the date into a number between 0 and 59. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 12 because the number of minutes in 02:12:46 PM is 12.
- *Second*: Converts the date into a number between 0 and 59. Example: The date Tuesday 05 June 2001 02:12:46 PM is converted into 46 because the number of seconds in 02:12:46 PM is 46.
- *Milliseconds since 1970*: Converts the source date/time into a timestamp that is equal to the number of elapsed milliseconds since the UNIX epoch time (1/1/1970 at midnight GMT). Source dates/times prior to this epoch time are converted into negative timestamps.

Number Parameters: Specifies the target number format that converts:

- A string into a number or multi-valued number
- A multi-valued number into a string or multi-valued string
- A multi-valued string into a multi-valued number

You must specify the decimal and hundreds separators.

- *Decimal Separator*: Select one of the following decimal separators:
  - *Point*: For example, 1234567.89
  - *Comma*: For example, 1234567,89
- *Thousand Separator*: Select one of the following thousand separators:
  - *No Separator*: For example, 1234567,89
  - *Space*: For example, 1 234 567,89
  - *Comma*: For example, 1,234,567,89
  - *Quote*: For example, 1'234'567,89

Multi-Valued Parameters: Specifies the field separator that converts:

- A string into a multi-valued string or multi-valued number
- A multi-valued string into a string
- A multi-valued number into a string

Click one of the available separators (*Semicolon*, *Tabulation*, *Comma*, *Space*, or *Slash*), or click *Other*, and then type a separator.

The default value is *Semicolon*.

For example, to convert the string `MeteoRecipients;NewsRecipients;ExchangeRecipients;SpotsRecipients;TrafficRecipients;TaxiRecipients` into a multi-valued string, you must click *Semicolon* so that the Property Type Convertor operator component removes all the semicolons for converting the string into a list of a multi-valued string.

If you want to convert the multi-valued number (2000 5000 10000 20000) into a string, you must select a field separator to separate the elements of the list in the string. If you select *Slash* as the field separator, the resulting string is `2000/5000/10000/20000`.

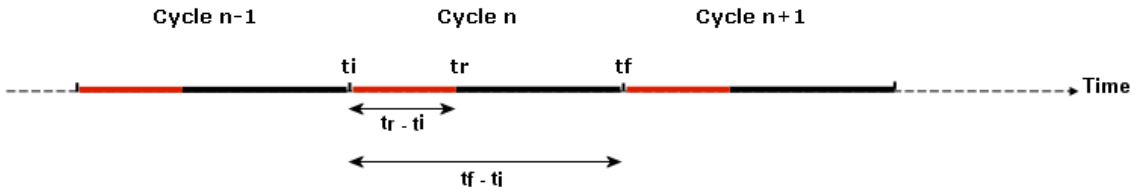
1.42.10.4.20 Prorata Operator

 The **Prorata Operator** calculates the product Y = X * R, in which:

- Y is the numerical result
- X is a numerical value
- R is a ratio ranging from 0 to 1 (R is a decimal number; for example, 0.45)

As shown in the next illustration, the *Prorata Operator* calculates R = (tr - ti) / (tf - ti) as the ratio of two time periods:

- tf - ti: Time period that is delimited by the initial time ti and the final time tf. The condition ti < tf is always true.
- tr - ti: Part of the time period tf - ti that is delimited by the initial time ti and the reference time tr. The condition ti <= tr <= tf is always true.



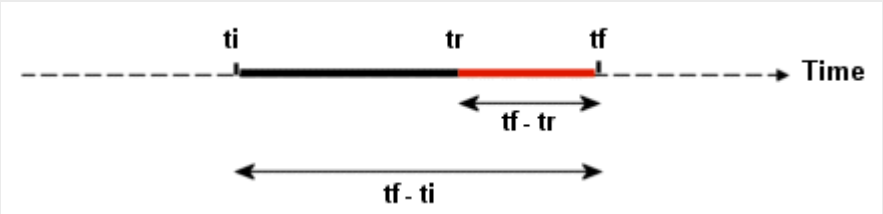
→ Remember

Keep in mind that tf - ti is a cycle defined by one of the following frequencies:

- A day or 24 hours
- A week
- A month
- A year

ⓘ Note

Instead of calculating R = (tr - ti) / (tf - ti), the *Prorata Operator* can calculate the ratio R = (tf - tr) / (tf - ti) as shown in the next illustration.



✤ Example

If you propose some services that include a monthly calling plan, you may want to adjust a new client's plan for the first month according to the subscription date of the contract. Likewise, you may want to adjust the plan for the last month when the plan is based on the cancellation date of the contract. In both cases, you use the *Prorata* component to calculate the calling plan.

A customer signs a contract on the 10th of the month for a service that includes two free hours per month. Because the customer signed on the 10th of the month, you want to prorate two free hours from the 10th to the end of the month. The following table presents the information that you need to set up the *Prorata* component:

Data	Name of properties/Settings	Comment
Property to prorate	Number of free hours	In the product Y = X * R, the numerical value X is Number of free hours (two hours).
Generated Property	Number of free hours during the first month	The generated property contains the resulting amount Y = X * R.
Prorata Date	Charge effective date	In the product Y = X * (tf - tr) / (tf - ti), the reference time tr is the Charge effective date (the 10th of the month).
Time period	Every month	In the product Y = X * (tf - tr) / (tf - ti), the initial time ti is the first of the month at 12:00 AM and the final time tf is the last day of the month.
Type of ratio	Calculation according to the *Fraction from prorata date* option	The *Fraction from prorata date* option calculates R = (tf - tr) / (tf - ti).

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Property to prorate: Provides the list of numerical properties. Select the property that represents the numerical value X of the product Y = X * R.

Generated Property Name: Allows you to type the name of the new property that retrieves the numerical result of Y = X * R.

Calendar: Provides the list of available calendars on which the recurring event is based.

Reference period: Allows you to set up the time period tf - ti and the initial time ti. To set up the time period tf - ti, this area provides you with a list of frequencies:

- For *Gregorian Calendar*, the available periodicities are the following:
  - *Every*, *Every other*, *Every 3rd*, and other items associated with the following list: *day*, *week*, *month*, and *year*. For example, you can choose a time period of two months by selecting *Every other month*, or you can choose a time period of one day by selecting *Every day*.

> **Note**
>
> The *Reference Date* is used ONLY if you select either or both of the options *The same day as the reference date* and *The same time as the reference date*. Instead of the *Reference Date*, you can use the other options.

- Once you have selected a frequency or time period, you can select the initial time ti:
  - *Reference Date*: Provides the list of date properties. Select the property that represents the initial time ti.
  - *The same day as the reference date*: If you click this option, the initial time ti is the date selected in the *Reference Date* field. Example: you have selected the Charge Effective Date. If you select Every month, the time period tf - ti is one month and the initial time ti starts at the *Charge Effective Date*.
  - *The same day as the reference date or first day of next month*: If you click this option, the *Recurring Rate* is triggered depending on the date selected in the *Reference Date* list or the first day of the month following that of the selected date in the *Reference Date* list. See the note below.
  - *the*: If you have selected the frequency *Every* (3rd, 4th, and so on) *month* or *year*, this option allows you to select the initial time ti as a day of the month. This option is combined with a list of numbers (from 1 to 31), a list of items such as Last day, First, Second, Third and Fourth and the list of weekdays and months. Example: selecting Every 5th month The 11 of the month at 04:00 PM means that the time period tf - ti is five months and that the initial time ti starts the 11 of the month at 04:00 PM.
  - *on*: If you have selected the frequency *Every* (3rd, 4th, and so on) *week* or *year*, this option allows you to select the initial time ti as a day of the week or the month. This option is combined with a list of numbers (from 1 to 31) and months and a list of weekdays. Example: selecting Every 4th week on Wednesday at 02:00 AM means that the time period tf - ti is four weeks and that the initial time ti starts on Wednesday at 02:00 PM.
  - *at*: Allows you to select the starting hour of the initial time ti if you have selected any of the previous options. This option includes hours and minutes combined with AM (ante meridiem) or PM (post meridiem). Example: selecting Every day at 07:25 PM means that the time period tf - ti is one day and that the initial time ti starts at 07:25 PM. You can select *The same time as the reference date* that is the date selected in the *Reference Date* field. Example: if you select the *Charge Effective Date*, the initial time of the time period tf - ti is that of the *Charge Effective Date*.
- For *4-4-5 Calendar*, the available periodicities are the following:
  - *Every month*, *Every quarter*, *Every semester* and *Every year*.

> **Note**
>
> The other options, available with the Gregorian Calendar, do not appear any longer. They are useless with the 4-4-5 Calendar.

Fraction of the reference period to prorate: Allows you to select the reference time tr and the type of ratio to calculate: R = (tr - ti) / (tf - ti) or R = (tf - tr) / (tf - ti).

- *Prorata date*: Provides the list of numerical properties. Select the reference time tr.
  - *Fraction up to prorata date*: Calculates the ratio R = (tr - ti) / (tf - ti)
  - *Fraction from prorata date*: Calculates the ratio R = (tf - tr) / (tf - ti)

Calculation Parameters

You set up the parameters for making the calculation.

Time Unit: Provides the list of time units to calculate the total part of the ratio R. The default option is seconds. Select one of the following units:

- *seconds*
- *minutes*
- *hours*
- *days*

Rounding Mode: Provides the list of modes for rounding the decimal portion of the ratio R. The default option is *To the next greater value*. Select one of the following rounding modes:

- *To the next greater value*
- *To the previous value*
- *To the nearest value*

Rounding Step: Allows you to type a step rule for rounding the ratio R, or allows you to select a numerical property that represents this step rule. The default option is *1*.

> ✤ Example
>
> If the rounding step value is 5 seconds and the rounding mode is *To the next greater value*, a period of tima of 47 seconds is rounded to 50 seconds. In this example, the rounding step consists of 5-second intervals.

Use typical period of: If selected, this option allows you to turn a variable period into a constant period. Once you have selected the *Use typical period of* option, you can either select the value of a property or select *Numerical Value*, and then type a number.

> ✤ Example
>
> For example, to turn each month into a constant period of 30 days, click *Use typical period of*, select *Numerical Value* in the list, and then type 30. If you clear *Use typical period of*, the reference period is 28 days in February, 31 days in January, and so on.

1.42.10.4.21 State Schedule Updater (Operator)


The **State Schedule Updater** operator component allows to change the dates/times on which a prepaid account is blocked, locked, or closed.

Remember that a prepaid account can be active, blocked, locked, or closed and these states are initially scheduled in the subscriber account including the prepaid account. For more information, refer to the primary help of Core Tool [page 514].

You can insert several *State Schedule Updater* components in a refill tree because each component is independent. Therefore, the dates scheduled in each component are effective when each component is performed.

> ✤ Example
>
> An operator can uses the *State Schedule Updater Operator* component in a price plan:
>
> - To block the prepaid accounts if the end customers or subscribers have not refilled their prepaid account

- To close the prepaid accounts if the end customers or subscribers have terminated their contract

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Blocking Date Change

- *Operation*: Provides the list of operations. Select an operation.
  - *None*: Does not change the date on which the prepaid account is blocked
  - *Clear*: Clears the date on which the prepaid account is blocked
  - *Set*: Sets or changes the date on which the prepaid account is blocked
  - *Add Duration*: Adds a duration value to the date on which the prepaid account is blocked
  - *Subtract Duration*: Subtracts a duration value from the date on which the prepaid account is blocked
- *Date*: Provides the list of date properties and date parameters. Select a date.
- *Duration*: Provides the list of numerical properties and counters and parameters. Select a date.
- *Duration Unit*: Provides the list of duration units (*Seconds*, *Minutes*, *Hours*, *Days*, *Months*, and *Years*). Select a duration unit.

Locking Date Change

- *Operation*: Provides the list of operations. Select an operation.
  - *None*: Does not change the date on which the prepaid account is locked
  - *Clear*: Clears the date on which the prepaid account is locked
  - *Set*: Sets or changes the date on which the prepaid account is locked
  - *Add Duration*: Adds a duration value to the date on which the prepaid account is locked
  - *Subtract Duration*: Subtracts a duration value from the date on which the prepaid account is locked
- *Date*: Provides the list of date properties and date parameters. Select a date.
- *Duration*: Provides the list of numerical properties and counters and parameters. Select a date.
- *Duration Unit*: Provides the list of duration units (*Seconds*, *Minutes*, *Hours*, *Days*, *Months*, and *Years*). Select a duration unit.

Closing Date Change

- *Operation*: Provides the list of operations. Select an operation.
  - *None*: Does not change the date on which the prepaid account is closed
  - *Clear*: Clears the date on which the prepaid account is closed
  - *Set*: Sets or changes the date on which the prepaid account is closed
  - *Add Duration*: Adds a duration value to the date on which the prepaid account is closed

- *Subtract Duration*: Subtracts a duration value from the date on which the prepaid account is closed
- *Date*: Provides the list of date properties and date parameters. Select a date.
- *Duration*: Provides the list of numerical properties and counters and parameters. Select a date.
- *Duration Unit*: Provides the list of duration units (*Seconds*, *Minutes*, *Hours*, *Days*, *Months*, and *Years*). Select a duration unit.

1.42.10.4.22 Switch Table Operator


The **Switch Table Operator** compares the same property with one or more values or properties of the same type. When a comparison is true, the test stops and a numerical or string value is assigned to a computed property.

If no comparison is true, a default value is assigned to the computed property.

Use

In a price plan, you may need to apply different prices that depend on the value of a property. To use the *Switch Table Operator* inside your price plan, do the following:

1. Choose the first property, a comparison operator, and the values or properties to compare to the first property.
2. Type the value of the computed property for each comparison.
3. Describe the pricing of each value of the computed property.

> ⓘ Note
>
> The *Switch Table Operator* stops the test when a comparison is true; therefore, the order of the comparisons in the table is important.

> ✤ Example
>
> An operator bills telephone calls that depend on the duration and the type of call (local, international and information). In the price plan, the property "CallType" is assigned one of these three values: "Loc", "Int" or "Inf". In the Switch Table Operator component, the computed property "Price" is expressed in cents/minute and is assigned three values as described in the following table:

Property	Comparator	Value	Price
CallType	equal to	Loc	$0.10
CallType	equal to	Int	$0.60
CallType	equal to	Int	$0.20

The table rows mean that:

- If *CallType* is equal to "loc", then *Price*=0.10
- If *CallType* is equal to "int", then *Price*=0.60
- If *CallType* is equal to "inf", then *Price*=0.20

The string property *CallType* is compared with the values "loc", "int" and "inf". *Price* is a new numerical property that is assigned a numerical value for each comparison. The *Switch Table Operator* stops the test when a comparison is true, and then *Price* is assigned the value relating to the comparison.

For example, if *CallType* is equal to "inf", the value of *Price* is 0.20. If no comparison is true, a default value is assigned to *Price*.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Property Name: Provides the list of properties. Select a property. The selected property is compared with the values or properties that are set up in the table. The property name appears in the heading of the first column *If ? ...* of the table. Example: If you select a property called Duration, the first column heading displays *If Duration ...*.

Parameters of the computed property: Allows to name and set up the computed property, which is assigned a value depending on the comparisons of the table.

- *Name*: Allows you to type the name of the computed property. This property is assigned a value depending on the comparisons that are set up in the table. The property name appears in the heading of the third column *.... ?=* of the table. Example: If you type Price, the third column heading in the table displays ...*Price=*.
- *Type*: Select *String* or *Number* as the type of the computed property.
- *Default Value*: Allows you to type the default value of the computed property, or allows you to select a numerical property that represents the default value. The computed property is assigned the default value if no comparison is true.

Table area

Each row added to the table allows you to make a comparison such as: If (condition) then (result). Example: If Duration... is equal to ... the value ...25...Price=100, where Duration is the property that you select in the Property Name list and Price is the computed property name.

 : Adds a row to the table. Click this button to add a comparison.

 : Deletes a row from the table. To remove a comparison, select the corresponding comparison row, and then click this button.

 : Moves a row up in the table. To move a comparison up, select the corresponding comparison row, and then click this button.

 : Moves a row down in the table. To move a comparison, select the corresponding comparison row, and then click this button.

 : Opens the *Open* dialog box to search for a file and import data into the table. Click this button when you want to create comparisons with data from a CSV file instead of manually entering the data.

 : Deletes all the rows in the table. Click this button to delete the list of comparisons.

- Case of a *String* type:
  - *If ? ...*: The "?" symbol is replaced by the property that you select in the *Property Name* list. Each row that you add to the table allows you to select one of the following operators:
    - *is equal to*: For example, if a property that contains "06526179" is compared with the second column entry that contains "06526179", the comparison is true; otherwise, the comparison is false.
    - *starts with*: For example, if a property that contains "06526179" is compared with the second column entry that contains "06", the comparison is true; otherwise, the comparison is false.
    - *ends with*: For example, if a property that contains "06526179" is compared with the second column entry that contains "79", the comparison is true; otherwise, the comparison is false.
    - *contains*: For example, if a property that contains "06526179" is compared with the second column entry that contains "261", the comparison is true; otherwise, the comparison is false.
  - *...the value...*: Each row added to the table allows you to type a value or select a property to compare with the first column.
  - *... ? =*: The "?" symbol is replaced by the computed property name. Each row added the table allows you to type a value or select a property, either of them is assigned to the computed property when the comparison is true.
- Case of a *Number* type:
  - *If ? ...*: The "?" symbol is replaced by the property that you select in the *Property Name* list. Each row that you add to the table allows you to select one of the following operators:
    - *Is Equal To*: For example, if a property that contains "12" is compared with the second column entry that contains "12", the comparison is true; otherwise, the comparison is false.
    - *Is Greater Than*: For example, if a property that contains "12" is compared with the second column entry that contains "10", the comparison is true; otherwise, the comparison is false.
    - *Is Smaller Than*: For example, if a property that contains "12" is compared with the second column entry that contains "30", the comparison is true; otherwise, the comparison is false.
    - *Is Greater Than or Equal To*: For example, if a property that contains "12" is compared with the second column entry that contains "10", the comparison is true; otherwise, the comparison is false.
    - *Is Smaller Than or Equal To*: For example, if a property that contains "12" is compared with the second column entry that contains "30", the comparison is true; otherwise, the comparison is false.
    - *Is Not Equal To*: For example, if a property that contains "12" is compared with the second column entry that contains "10", the comparison is true; otherwise, the comparison is false.

- *...the value...*: Each row added to the table allows you to type a value or select a property to compare with the first column.
- *... ? =*: The "?" symbol is replaced by the computed property name. Each row added the table allows you to type a value or select a property, either of them is assigned to the computed property when the comparison is true.

1.42.10.4.23 Tier Table Introducer (Operator)


The *Tier Table Introducer Operator* component allows you to use an existing tier table [page 147] in a plan.

In a tier table, each group consists of one or more numerical intervals. The output columns of the tier table are number or string values assigned to each numerical interval. The *Tier Table Introducer* component compares row by row each numerical interval of a group with a property value of the plan. The comparison stops when the property value matches a numerical interval of the group. The output columns related to the numerical interval are retrieved in generated properties that you can use in the plan.

With the *Tier Table Introducer Operator* component, you can:

- Select a tier table.
- Select one of the groups of the tier table.
- Select the property value to compare with the numerical intervals of the group.

Example

A service provider bills Internet services to its end-customers according to their contract length. To calculate the service price, a tier table provides the contract lengths with their price. The contract type (Gold prices or Silver prices) determines the service price. To use the tier table, a *Tier Table Introducer* must be inserted in the decision tree of the price plan. The tier table **`TieredTariff`** consists of:

- Numerical intervals representing the contract length in months (]-∞, 6],]6, 12],]12, 24],]24, +∞[)
- Output columns representing the related prices (Gold prices and silver prices) in USD

In the price plan [page 299], the *Tier Table Introducer* component shows the following information:

- The **`TieredTariff`** tier table
- The **`ContractLength`** price plan property that is compared to the numerical intervals (contract lengths) of the group
- The **`GoldenPrice`** price plan property that retrieves the value of the "Golden price" output column
- The **`SilverPrice`** price plan property that retrieves the value of the "Silver price" output column

In this example, the tier table **`TieredTariff`** uses the default group. Therefore, the *Group Name Property* is *Default Group*. If the tier table uses another group, the list of *Group Name Properties* must contain the property whose value is the group name.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Tier Table: Provides the list of available tier tables. Select the tier table that you want to use with the *Tier Table Introducer*. To open and modify the tier table, click  .

Mapping Property: Provides the list of properties. Choose the property to compare with the numerical intervals of the selected group.

Group Name Property: Provides the list of properties. Choose the property whose value is the group name, or choose the default group.

Mapping of Generated Properties: Displays the output columns of the tier table and allows you to rename the generated properties. When a comparison is true between a numerical interval and the property, the generated properties retrieve the output column values of the tier table.

- *Column Name*: Displays the output column name of the tier table.
- *Column Description*: Displays the description of the output column of the tier table.
- *Type*: Displays the type of the generated property.
- *Generated Property Name*: Displays the default generated property that retrieves the output column value of the tier table. To change the name of the default generated property, double-click the entry, and then type a new name. You can use the generated property in the plan.

Related Information

Tier Tables [page 147]

1.42.10.4.24 Translation Table Introducer (Operator)

 The *Translation Table Introducer Operator* component allows you to use an existing translation table in a plan.

The *Translation Table Introducer Operator* component compares row by row the input columns of a translation table [page 132] with properties of the plan. The comparison stops when the properties match at least one row of the input columns. The order of the rows in the translation table is therefore important. The output columns related to the input columns are retrieved in generated properties that you can use in the plan. If no property matches the comparison, the translation table returns a default value for each generated property.

> **Note**
>
> The number of properties must be equal to the number of input columns of the translation table.

With the *Translation Table Introducer Operator* component, you can:

- Select a translation table
- Select the properties to compare with the input columns of the translation table
- Select the type of comparison

> **Example**
>
> An operator bills telephone calls according to the geographic zone of the called numbers. To calculate the price of each call, a translation table provides the price plan with the geographic zone code that is assigned to each telephone number prefix. To use the translation table, a *Translation Table Introducer* must be inserted in the rating tree of the price plan. The translation table **`TableZone`** is made up of the input column **`Prefix`** and the output column **`ZoneCode`**. The input column contains the telephone number prefixes, and the output column contains the geographic zone codes.
>
> In the price plan [page 299], the *Translation Table Introducer* component shows the following information:
>
> - The **`TableZone`** translation table
> - The **`NumberCalled`** price plan property that provides the telephone number to the translation table
> - The **`GeographicZone`** price plan property that retrieves the value of the **`ZoneCode`** output column
> - The comparison test *starts with* that is used to compare the price plan properties with the input columns of the translation table

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Translation Table: Provides the list of available translation tables. Select the translation table that you want to use with the *Translation Table Introducer*. To open and modify the translation table, click  .

Comparison Operator: Provides the list of comparison operators. To compare the properties with the input columns of the translation table, select one of the following operators:

- *Is equal to*: The comparison is true when the property matches the value of the input column. For example, if the property that contains "06526179" is compared with an input column entry that contains "06526179", the comparison is true; otherwise, the comparison is false.
- *Starts with*: The comparison is true when the property starts with the value of the input column. For example, if the property that contains "06526179" is compared with an input column entry that contains "06", the comparison is true; otherwise, the comparison is false.

- *Contains*: The comparison is true when the property contains the value of the input column. For example, if the property that contains "06526179" is compared with an input column entry that contains "261", the comparison is true; otherwise, the comparison is false.
- *Ends with*: The comparison is true when the property ends with the value of the input column. For example, if the property that contains "06526179" is compared with an input column entry that contains "79", the comparison is true; otherwise, the comparison is false.

Reference Date: Is an additional criterion to compare a date with the *Start Date* and *End Date* columns of the translation table[87]. If you do not select a *Reference Date*, this option is ignored. If selected, the comparison result depends on the following rules:

- If a row includes a non-null *Start Date* and a non-null *End Date*, the comparison is true if the *Reference Date* is equal or greater than the *Start Date* and strictly less than the *End Date*; otherwise, the comparison is false.
- If a row includes a non-null *Start Date* and no *End Date*, the comparison is true if the *Reference Date* is greater than or equal the *Start Date*; otherwise, the comparison is false.
- If a row includes no *Start Date* and a non-null *End Date*, the comparison is true if the *Reference Date* is strictly less than the *End Date*; otherwise, the comparison is false.
- If a row includes no *Start Date* and no *End Date*, the comparison is always true.

Mapping of Rating Context: Displays the input columns of the translation table, and allows you to select the properties that you want to compare with the input columns.

> **Note**
>
> In the *Type* column, you are allowed to select only one multi-valued string as a property type.

- *Column Name*: Displays the input column name of the translation table.
- *Column Description*: Displays the description of the input column of the translation table.
- *Type*: Provides the list of two property types: *String* or *Multi-Valued String*. Select a property type.
- *Mapping Property*: Provides the list of properties relating to the selected type of the Type column. Select the property that you want to compare with the input column.

Mapping of Generated Properties: Displays the output columns of the translation table, and allows you to rename the generated properties. When a comparison is true between properties, *Reference Date* (if selected), and input columns, the generated properties retrieve the values of the output column entries.

- *Column Name*: Displays the output column name of the translation table.
- *Column Description*: Displays the description of the output column of the translation table.
- *Type*: Displays the type of the generated property.
- *Generated Property Name*: Displays the default generated property that retrieves the output column value of the translation table. To change the name of the default generated property, double-click the entry, and then type a new name. You can use this new generated property in the plan.

[87] *Reference Date* is used in combination with the *Start and end dates can be specified for each row* option of translation tables.

1.42.10.4.25 Update Counter Operator

> ⓘ Note
>
> If your configuration is part of session-based charging services, be careful when using the Update Counter component and read the note at the end of this topic.


The **Update Counter Operator** component updates a counter of a plan. To update a counter, you can:

- Reset the counter
- Assign a value such as a constant or numerical property to the counter
- Add a value such as a constant or numerical property to the counter
- Subtract a value such as a constant or numerical property from the counter

> → Remember
>
> This component is not compatible with counters defined in shared allowances (distributed counters). To update such counters, you have to use the Decrement Counter Operator [page 795] or the Increment Counter Operator [page 803] component.

Use

In a periodic-type telephone calling plan, the *Update Counter* operator component can allow you to upgrade the **`number of hours in the calling plan`** counter and reset this counter at the beginning of the period.

> ✤ Example
>
> A monthly calling plan includes two hours of free call time for end customers.
>
> This plan is billed using a recurring rate component (for monthly subscriptions) and a usage rate component (for calls).
>
> The price plan [page 299] contains a counter **`number of hours in the calling plan`**.
>
> - In the recurring rate branch, this counter is reset to the original value of two hours at the beginning of the month by using an *Update Counter Operator* component.
> - In the usage rate branch, an *Update Counter Operator* component subtracts the duration of each call from the counter **`number of hours in the calling plan`**.

Identification Area

Name: Type the name of the component (mandatory).

Description: Allows you to type the description of the component (optional).

Definition (Tab)

Counter Name: Allows you to type the name of the counter that you want to update.

Counter operation: Provides the list of counter updates. Select one of the following counter updates:

- *Reset to the initial value*[88]: The counter is set to its initial value.
- *Set to the value defined below*: The counter is set to the *Operation Value*
- *Add the value defined below*: The *Operation Value* is added to the counter value.
- *Subtract the value defined below*: The *Operation Value* is subtracted from the counter value.

Operation value: Allows you to type the value that updates the counter, or allows you to select a numerical property that represents the value.

Update During the Reservation for a Session: This option is available only when you select a transient counter. If selected, this option means that the transient counter is updated at reservation time within a session. This option does not change the behavior of a persistent counter.

1.43 Troubleshooting

As part of the technical operations, you may have to troubleshoot some problems when using the Core Tool user interface to configure or maintain the data or the system. Find afterwards a list of possible symptoms and related solutions that you can try to implement.

Related Information

Enabling the Traces for Troubleshooting [page 827]
Troubleshooting Connection Issues to Core Server [page 828]
Error Messages in Core Tool [page 828]

1.43.1 Enabling the Traces for Troubleshooting

For troubleshooting purposes, SAP SE recommends that you temporarily enable the logging and tracing functions in the graphical user interface.

For more information, refer to the Enabling the Traces for Troubleshooting procedure in the SAP CC 2023 Administration and Operation Guide documentation.

[88] Do not set or reset counters when working with session-based charging operations. Refer to the SAP CC 2023 Application Help documentation for more information about the process constraints and limitations relating to the charging process.

> **Note**
>
> In addition, you can consult the messages exchanged between your Core Tool user interface and the SAP CC system (XML verbose mode). See the Preferences [page 56] menu.

1.43.2 Troubleshooting Connection Issues to Core Server

> **Note**
>
> For more information about this connection issue, refer to your administrator and the related SAP Note 1846673.

Symptom	An issue occurs when the Core Tool user interface tries to connect to the distant multi-instance Core Server system. The user interface displays the following error message: *An error occurred during communication with the server.*
Reasons	Several permanent or temporary reasons are possible. The user interface must sequentially connect to several instances of the target SAP system.
Details	The dispatcher and the updater instances should be accessible through the firewall. For more information, refer to the Securing Communications with the Core Server System section of the SAP CC 2023 Installation and Maintenance Guide documentation about the communication interfaces that must be accessible through a firewall.
Possible Solutions	1. Verify with the system administrator that the SAP system is up and running. The dispatcher and updater instances must be started and ready. 2. Verify that the HCI URLs of the updater and dispatcher instances are directly accessible from the Core Tool host. For example, ping these instances to check the connections. Core Tool uses the dispatcher instance address to retrieve the adress of the running updater. Once connected to the updater instance, Core Tool only communicates with it. Refer to Launching Core Tool and Logging On [page 35] topic if necessary. 3. Make sure that you provide Core Tool with the IPv4 or IPv6 address of the dispatcher instance, **and not the updater instance's.** Once connected to the updater instance, Core Tool only communicates with it. 4. In addition, you should also check that the entered *Server URL* included the relevant protocol (HTTP vs HTTPS) in your SAP system landscape.

1.43.3 Error Messages in Core Tool

Core Tool provides you with some user guidance when an error occurs. To improve your user experience, the user interface interprets all the issues and displays an error message that must help you.

Find here some typical error messages, possible symptoms, and related solutions that you can try to implement.

Error message: "This database does not support to store strings which end with a space character (<ERRONEOUS_TEXT>)."

This error message means that the Core Tool user interface requested the SAP Convergent Charging system (Core Server) to save data into the back-end database system (Core Database). The operation request failed. This situation may be due to:

- The database technology for the back-end database does not support any values that end with a space character. Verify with your system administrator. For example, SAP ASE does not store string values ending with trailing spaces.
  In Core Tool, do not enter any string values that include trailing spaces when such a database is in your SAP system landscape.
- You may encounter this problem when trying to import some data to a target SAP CC system. The data was previously exported from another system. The back-end database for the target system does not support any values ending with a space character.
- This problem may relate to a database migration project. Contact your implementation and integration project team.

See also:

- Log message com.sap.SCC.infra.000003
- Error code SQL-59

Error message: "An error occurred during communication with the server."

This error message means that the Core Tool user interface cannot communicate with the target SAP CC system.

Several permanent or temporary reasons are possible in your environment.

See: Troubleshooting Connection Issues to Core Server [page 828]

1.43.3.1 Invalid Property VTPOS in the Consumption Item Class Z005

Error message: "Invalid Property <FIELD_NAME> in the consumption item class <CONSUMPTION_ITEM_CLASS_ID_OR_NAME>"

When creating or refreshing the consumption item mapping in the user interface

Several reasons are possible:

- The settings in SAP Convergent Invoicing are not correct or some additional tasks are expected.
  In SAP ERP:
  - Go to SPRO.
  - Search for "Transfer Consumption Items".
  - Under that, go to "Assign data type to standard fields" and correct the data type of the property shown as invalid in error message to correct one and then the consumption item class starts to replicate successfully.
- The default mapping known by SAP Convergent Charging is not correct. Refer to the Tuning Guide.

See also:

- Creating the Consumption Item Mapping [page 283]
- Refreshing the Consumption Item Mapping [page 288]

1.44 Administering the SAP CC System

Admin+ is a user interface of SAP Convergent Charging (SAP CC). It is a console program that allows you to manage local or remote instances of the SAP CC Core Server system and change their configuration parameters. The console program runs in interactive mode to get commands/arguments and to give you results or prompts for new options. It provides an IT administrator with some troubleshooting tools to resolve problems about SAP CC systems, data refreshing, and SAP CC users.

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